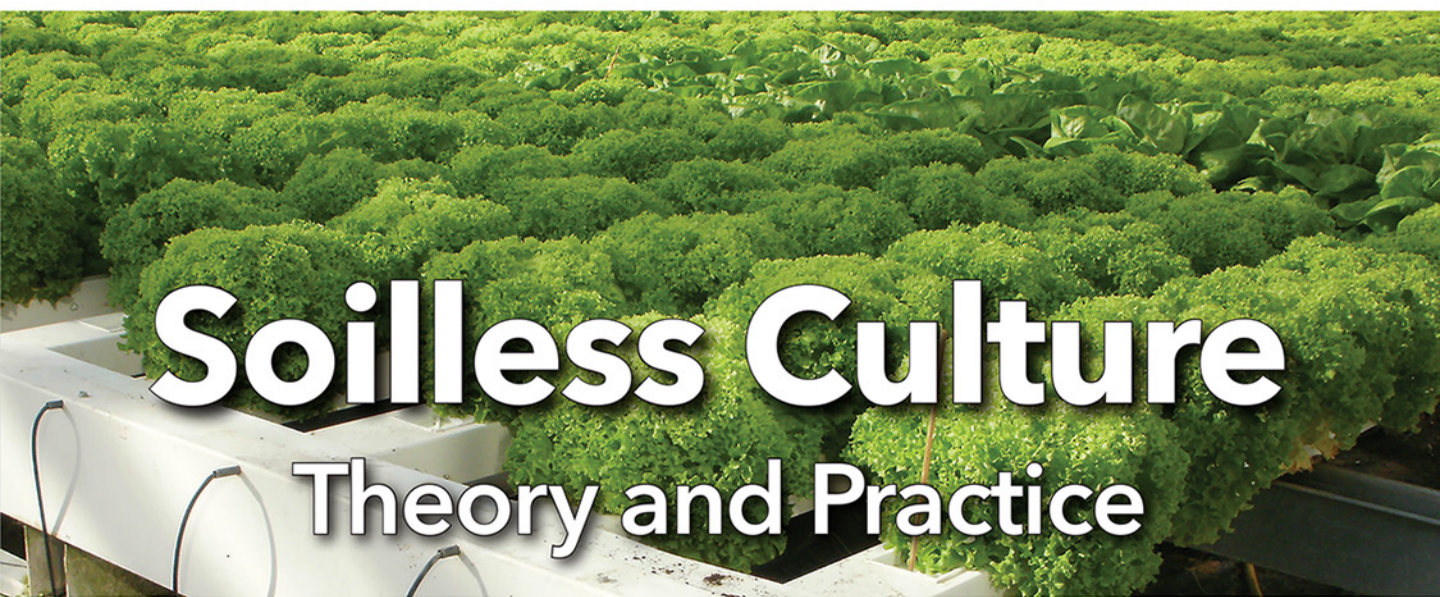




Second Edition



Soilless Culture

Theory and Practice



Edited by
**Michael Raviv, J. Heinrich Lieth
and Asher Bar-Tal**



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Edited by

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Preface

Since the onset of the commercial application of soilless culture, this production approach has evolved at a fast pace, gaining popularity among growers throughout the world. As a result, a lot of information has been developed by growers, advisors, researchers, and suppliers of equipment and substrate. With the rapid advancement of the field, an authoritative reference book is needed to describe the theoretical and practical aspects of this subject. Our goal for this book is to describe the state of the art in the area of soilless culture and to suggest directions in which the field could be moving. This book provides the reader with background information of the properties of the various soilless media, how these media are used in soilless production, and how this drives plant performance in relation to basic horticultural operations such as irrigation and fertilization.

As we assembled the first edition of this book, we were aware that many facets of the field are rapidly changing, so that the state of the art is continuing to advance. Several areas, in particular are in flux. Two such factors are (1) the advent of governmental pressures to force commercial soilless production systems to recirculate irrigation effluent and (2) a desire for society to use fewer agricultural chemicals in food production. The group of authors that have contributed to this book are all aware of these factors and their contributions to this book attempt to address the state of the art.

In the years between the first edition and this one, the advancements in the field have continued. Many facets that were just starting to be an issue in 2007 are now even more important. Examples of this are the changes in the domain where soilless culture is the only approach to growing plants indoors with artificial light. Some substrates are now even more mainstream (e.g., coir) and hydroponics is now much more prevalent, both in commercial as well as home horticulture. As such, all chapters have been updated from the first editions. The chapters on root function ([Chapter 2: Root Growth, Physiology, and Potential Impact of Soilless Culture on Their Functioning](#)) and the effects from microbes in the root zone ([Chapter 5: Root Zone Microbiology of Soilless Cropping Systems](#)) were completely replaced with chapters by authors who were not part of the first edition, whereas in some chapters, new coauthors joined authors of the first edition. A new chapter serving as introduction to hydroponics ([Chapter 12: Liquid Culture Hydroponic System Operation](#)) was added to serve practitioners who are new to the domain.

This book should serve as reference book or textbook for a wide readership including researchers, students, greenhouse, nursery, and indoor production managers and extension specialists; in short, all those who are involved in the production of plants and crops in systems where the root-zone consists of soilless media or no media at all. It provides information concerning the fundamental principles involved in plant production in soilless culture and, in addition, may serve as a manual that describes many of the useful techniques that are constantly emerging in this field.

In preparing this book, we were helped by many authorities in the various specialized fields that are covered. We would like to take this opportunity to thank the colleagues who contributed their time and expertise to improve the quality of the book.

For us editors, the assembly of this book has been an arduous task in which we have had numerous discussions about the myriad of facets that make up this field. This has served to stimulate for us a more in-depth respect for the field and a deeper appreciation for many of our colleagues throughout the world. We are very appreciative of all the work that our authors invested to make

this book the highest quality that we could achieve and hope that after all the repeated requests from us for various things, that they are still our friends.

We would also note that while no specific agency or company sponsored any of the effort to assemble this book, we are in debt to some extent to various funding sources that supported research of all the authors over decades-long research careers. Our own employers (The Agricultural Research Organization of Israel and the University of California), of course, supported our efforts to create this work, and for that we are deeply grateful. We are also indebted to the many discerning readers of the first edition who provided us with valuable feedback in the decade between 2008 and 2018.

We also thank our wives, Ayala Raviv, Smadar Bar-Tal, and Sharyn Lieth, for their tolerance toward our time constraints while editing this book.

PART

BACKGROUND, BASIC
INFORMATION,
PRINCIPLES

I

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SIGNIFICANCE OF SOILLESS CULTURE IN AGRICULTURE

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1.1 HISTORICAL FACETS OF SOILLESS PRODUCTION

Although we normally think about soilless culture as a modern practice, growing plants in containers aboveground has been tried at various times throughout the ages. The Egyptians did it almost 4000 years ago. Wall paintings found in the temple of Deir el Bahari (Naville, 1913) showed what appears to be the first documented case of container-grown plants (Fig. 1.1). They were used to transfer mature trees from their native countries of origin, to the king's palace and then to be grown this way when local soils were not suitable for the particular plant. It is not known what type of growing medium was used to fill the containers, but since they were shown as being carried by porters over large distances, it is possible that materials used were lighter than pure soil.

Starting in the 17th century, plants were moved around, especially from the Far and Middle East to Europe to be grown in orangeries, in order to supply esthetic value, and rare fruits and vegetables to wealthy people. An orangery is “a sheltered place, especially a greenhouse, used for the cultivation of orange trees in cool climates” (American heritage Dictionary) so it can be regarded as the first documented case of a greenhouse-container-grown system, although soil was mostly used to fill these containers. Orangeries can still be found today throughout Europe. An exquisite example from Moscow, Russia is shown in Fig. 1.2.

The orangery at Pillnitz Palace near Dresden Germany was used to protect container-grown citrus trees during the winter. Large doors at the east side allowed trees to be moved in and out so they could be grown outdoors during the summer and brought inside during the winter. Large floor-to-ceiling windows on the south side allowed for sunlight to enter.

As suggested by the name, the first plants to be grown in orangeries were different species of citrus. An artistic example can be seen in Fig. 1.3.

Two major steps were key to the advancement of the production of plants in containers. One was the understanding of plant nutritional requirements, pioneered by French and German scientists in the 19th century, and later perfected by American and English scientists during the first half of the 20th century. As late as 1946, British scientists still claimed that while it is possible to grow plants in silica sand using nutrient solutions, similarly treated soil-grown plants produced more

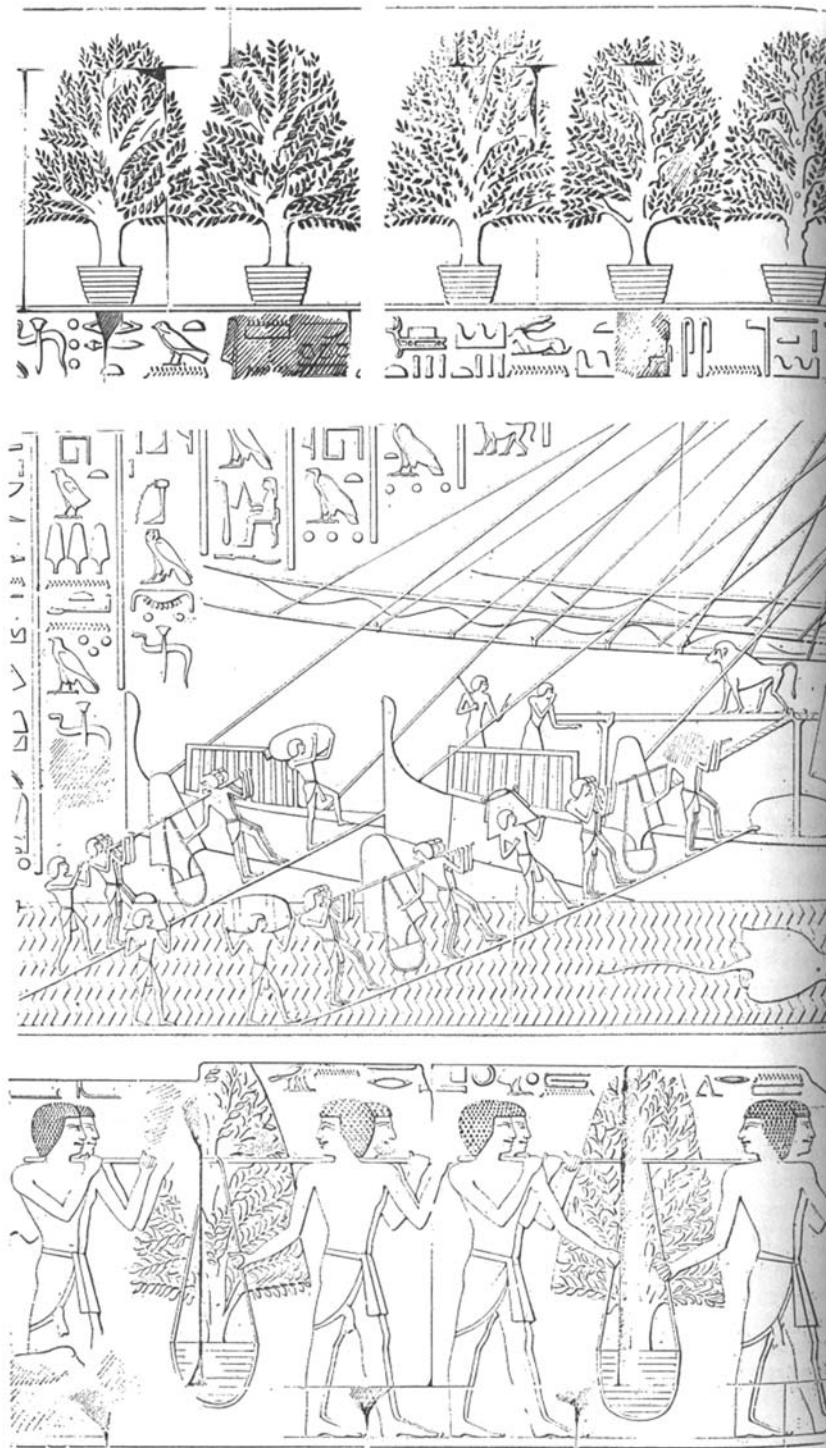


FIGURE 1.1

Early recorded instance of plant production and transportation, recorded in the temple of Hatshepsut, Deir el-Bahri, near Thebes, Egypt (Navelle, 1913; Matkin et al., 1957).



FIGURE 1.2

(A) Orangerie in Kuskovo, Moscow (1761–64). (B) The orangerie at Pillnitz Palace near Dresden, Germany.

(A) With kind permission of photographer Alexei Troshin and Wikipedia—The Free Encyclopedia. (http://en.wikipedia.org/wiki/Image:Kuskovo_orangerie.jpg).

yield and biomass (Woodman and Johnson, 1946a,b). It was not until the 1970s that researchers developed complete nutrient solutions, coupled their use to appropriate rooting media, and studied how to optimize the levels of nutrients, water, and oxygen to demonstrate the superiority of soilless media in terms of yield (Cooper, 1975; Verwer, 1977).



FIGURE 1.3

Orangery, from “The Nederlanze Hesperides” by Jan Commelin, 1676.

The second major step was the realization that elimination of disease organisms that needed to be controlled through disinfestation was feasible in container-grown production while being virtually impossible in soil-grown plants. In the United States a key document was the description of a production system that provided a manual for the use of substrates in conjunction with disease control for production of container-grown plants in outdoor nursery production. Entitled “The U.C. System for Producing Healthy Container-grown Plants through the use of clean Soil, Clean Stock, and Sanitation” (Baker, 1957), it was a breakthrough in container nursery production in the 1950s and 1960s and helped growers to such an extent that it became universally adopted since growers using the system had a dramatic economic advantage over competitors that did not use it. This manual described several growing media mixes consisting of sand and organic matter such as peat, bark, or sawdust in various specific percentages (Matkin and Chandler, 1957). These became known as “UC mixes.” It should be noted that in this manual these mixes were called “soil” or “soil mixes,” largely because prior to that time most container media consisted of a mix of soil and various other materials. That convention is not used in this book; here, we treat the term “soil” as

meaning only a particular combination of sand, silt, clay, and organic matter found in the ground. Thus when we talk about soilless substrates in this book, these may include mineral components (such as sand or clay) that are also found in soil, but no soil directly. The term “compost” was also used as a synonym to “soil mix” for many years, especially in Europe and the United Kingdom (Robinson and Lamb, 1975), but also in the United States (Boodley and Sheldrake, 1973). This term included what is now usually termed substrate or growing medium and, in most cases, suggests the use of mix of different components, with at least one of them being of organic origin. In this book we use the term growing medium and substrate interchangeably, while the term compost is used to define organic matter that underwent a composting process, aiming at stabilizing it and killing plant and zoonotic pathogens. More on this is discussed in [Chapter 8](#), Organic Soilless Media Components.

The abovementioned scientific developments dispelled the notion that growing media can be assembled by haphazardly combining some soil and other materials to create “potting soil.” This notion was supported in the past by the fact that much of the development of ideal growing media was done by trial and error. Today, we have a fairly complete picture of the important physical and chemical characteristics (described in [Chapter 3](#): Physical Characteristics of Soilless Media, and [Chapter 4](#): Chemical Characteristics of Soilless Media, respectively), which are achieved through the combination of specific components (e.g., UC mix) or through industrial manufacture (e.g., stone-wool slabs) to exacting specifications.

Throughout the world there are many local and regional implementations of these concepts. These are generally driven by both horticultural and financial considerations. While the horticultural considerations are covered in this book, the financial considerations are not. Yet this factor is ultimately the major driving force for the formulation of a particular substrate mix that ends up in use in a soilless production setting. The financial factor manifests itself through availability of materials, processing costs, transportation costs, and costs associated with production of plants/crops as well as their transportation and marketing. Disposal of used substrates is, in some cases, another important consideration of both environmental and economical implications. For example, one of the major problems in the horticultural use of mineral wool (stone- and glass-wool) is its safe disposal, as it is not a natural resource that can be returned back to nature. Various methods of stone-wool recycling have been developed but they all put a certain amount of financial burden on the end-user. Additional environmental factor that may limit the use of various materials as growing substrates is the depletion of natural resources. Thus there is a growing trend to minimize mining of Peat, one of the most popular substrates in temperate regions, in most parts of Europe (Alexander et al., 2017).

In countries where peat is readily available, perhaps even harvested locally, growers find this material to be less expensive than in countries where it has to be imported from distant locations. As prices of raw materials fluctuate, growers must evaluate whether to use a “tried and true” component (e.g., peat) or a replacement (e.g., coconut coir) in a recipe that may have proven to work well over the years. In some years, the financial situation may force consideration of a change. Since the properties of all substrates and mixes differ from each other, replacement of one particular component (such as peat) with another component, might result in other costs or lower quality crops (which may be valued less in the market place) especially if the substitution is a material with which the grower has less experience. Thus growers throughout the world face the challenge of assembling mixes that will perform as desired at the lowest possible overall cost.

The result of this is that the substrates used throughout the world differ significantly as to their makeup, while attempting to adhere to a specific set of principles. These principles are quite complex, relating to physical and chemical factors of solids, liquids, and gases in the root zone of the plant.

Today the largest industries in which soilless production dominates are greenhouse production of ornamentals and vegetables, and outdoor container nursery production. In urban horticulture virtually all containerized plants are grown without any field soil.

1.2 HYDROPONICS

Growing plants without soil has also been achieved through water culture without the use of any solid substrates. This type of soilless production is frequently termed “hydroponics.” While this term was coined by [Gericke \(1937\)](#) to mean water culture without employing any substrate, currently the term is used to mean various things to various persons. Many use the term to refer to systems that do include some sort of substrate to anchor or stabilize the plant and to provide an inert matrix to hold water. Strictly speaking, however, hydroponics is the practice of growing plants in nutrient solutions. In addition to systems that use exclusively nutrient solution and air (e.g., Nutrient Film Technique (NFT), Deep Flow Technique (DFT), aerohydroponics), we also include in this concept those substrate-based systems where the substrate contributes no nutrients nor ionic adsorption or exchange. Thus we consider production systems with inert substrates such as stone wool or gravel to be hydroponic. In this book, we will generally use of the term “hydroponics” in conjunction with qualifying terms to clarify the distinction between “liquid-culture hydroponics” and “substrate-based hydroponics”; that latter involves a substantial amount of substrate that is inert and has little ion-exchange capacity, while the former has none (or nearly none) of any substrate.

Initially scientists used hydroponics mainly as a research tool to study particular aspects of plant nutrition and root function. Progress in plastics manufacturing, automation, production of completely soluble fertilizers and especially the development of many types of substrates complemented the scientific achievements and brought soilless cultivation to a viable commercial stage. Today various types of soilless systems exist for growing vegetables and ornamentals in greenhouses. This has resulted with a wide variety of growing systems; the most important of these are described in [Chapter 7](#), Inorganic and Synthetic Organic Components of Soilless Culture and Potting Mixtures, [Chapter 8](#), Organic Soilless Media Components, and [Chapter 12](#), Liquid Culture Hydroponic System Operation. Modern technical equipment used to enable plant production in these systems is discussed in [Chapter 13](#), Technical Equipment in Soilless Production Systems.

1.3 SOILLESS PRODUCTION AGRICULTURE

World agriculture has changed dramatically over the last few decades, and this change continues, since the driving forces for these changes are still in place. These forces consist of the rapid scientific, economic, and technological development of societies throughout the world. The increase in worlds’ population and the improvement in the standard of living in many countries have created a

strong demand for high-value foods and ornamentals and particularly for out-of-season, high-quality produce. The demand for floricultural crops, including cut flowers, pot plants and bedding plants has also grown dramatically with the increase in life standard. The result of these trends was the expanded use of a wide variety of protected cultivation systems, ranging from primitive screen or plastic film covers to completely climate-controlled greenhouses. Initially this production was entirely in the ground where the soil had been modified so as to allow for good drainage. Since the production costs of protected cultivation are higher than that of open field production, growers had to increase their production intensity to stay competitive. This was achieved by several techniques; prominent among these is the rapid increase in soilless production relative to total agricultural crop production.

The major cause for shift away from the use of soil was the proliferation of soil-borne pathogens in intensively cultivated greenhouses. The ban on the use of methyl bromide (which is considered as a reducing agent of the atmosphere ozone layer) accelerated the move from soil to substrates growing systems. This is further described in [Chapter 5](#), Root Zone Microbiology of Soilless Cropping Systems. Soil was replaced by various substrates, such as stone wool, polyurethane, perlite, scoria (tuff), etc., since they are virtually free of pests and diseases due to their manufacturing process. Also in reuse from crop to crop these materials can be disinfested so as to kill any microorganisms. The continuing shift to soilless cultivation is also driven by the fact that in soilless systems it is possible to have better control over several crucial factors, leading to greatly improved plant performance.

Physical and hydraulic characteristics of most substrates are superior to those of soils. A soil-grown plant experiences relatively high water availability immediately after irrigation. At this time, the macropores are filled with water followed by relatively slow drainage which is accompanied by entry of air into the soil macropores. Oxygen, which is consumed by plant roots and soil microflora, is replenished at a rate which may be slower than plant demand. As water is drained or lost to evapotranspiration, the porosity of the soil is such that atmospheric oxygen diffuses into the root zone. At the same time some water is held by gradually increasing soil matric forces so that the plant has to invest a considerable amount of energy to take up enough water to compensate for transpiration losses due to atmospheric demand. Most substrates, on the other hand, allow a simultaneous optimization of both water and oxygen availabilities. The matric forces holding the water in substrates are much weaker than in soil. Consequently, plants grown in porous media at or near container capacity require less energy to extract water. At the same time, a significant fraction of the macropores are filled with air so that oxygen diffusion rate is high enough so that plants do not experience a risk of oxygen deficiency, such as experienced by plants grown in a soil near field capacity. This subject is quantitatively discussed in [Chapter 3](#), Physical Characteristics of Soilless Media, and its practical translation into irrigation control is described in [Chapter 9](#), Irrigation in Soilless Production.

Another factor is that nutrient availability to plant roots can be better manipulated and controlled in soilless cultivation than in most arable soils. The surface charge and chemical characteristics of substrates are the subjects of [Chapter 4](#), Chemical Characteristics of Soilless Media, while plant nutrition requirement and the methods of satisfying these needs are treated in [Chapter 6](#), Nutrition of Substrate-Grown Plants. [Chapter 11](#), Analytical Methods Used With Soilless Substrates, is devoted to the description of the analytical methods, used to select adequate substrate for a specific aim, and other methods, used to control the nutritional status during the cropping

period, so as to provide the growers with recommendations, aimed at optimizing plant performance.

Lack of suitable soils, disease contamination after repeated use, and the desire to apply optimal conditions for plant growth are leading to the worldwide trend of growing plants in soilless media. Most such plants are grown in greenhouses, generally under near-optimal production conditions. An inherent drawback of soilless cultivation over soil-based cultivation is the fact that in the latter the root volume is unrestricted, while in containerized culture the root volume is restricted. This restricted root volume has several important effects, especially a limited supply of water (Poorter et al., 2012; Sinclair et al., 2017; Dambreville et al., 2017) and nutrients (Dubik et al., 1990; Bartal, 1999). The limited root volume also increases root-to-root competition since there are more roots per unit volume of medium. Chapter 2, Root Growth, Physiology, and Potential Impact of Soilless Culture on Their Functioning, discusses the main functions of the root system while Chapter 14, Growing Plants in Soilless Culture: Operational Conclusions, quantitatively analyses the limitations imposed by a restricted root volume. Various substrates of organic origin are described in Chapter 8, Organic Soilless Media Components, while Chapter 7, Inorganic and Synthetic Organic Components of Soilless Culture and Potting Mixtures, describes substrates of inorganic origin and the issue of potting mixes. In both the chapters, subjects such as production, origin, physical and chemical characteristics, sterilization, reuse, and waste disposal are discussed.

Container production systems have advantages over in-ground production systems in terms of pollution prevention since it is possible, using these growing systems, to minimize or eliminate the discharge of nutritional ions and pesticide residues thus conserving freshwater reservoirs. Simultaneously, water- and nutrient-use efficiencies are typically significantly greater in container production, resulting in clear economic benefits. Throughout the developed countries more and more attention is being directed to reducing environmental pollution and in the countries where this type of production represents a large portion of agricultural productivity, regulations are being created to force recirculation so as to minimize or eliminate run-off from the nurseries and greenhouses. The advantages and constraints to closed and semiclosed systems is an area that is currently seeing a lot of research and the state-of-the-art is described in Chapter 10, Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses. The risk of disease proliferation in recirculated production and the methods to avert this risk are described in Chapter 5, Root Zone Microbiology of Soilless Cropping Systems, together with other issues related to substrate's microbiology.

The book concludes with Chapter 14, Growing Plants in Soilless Culture: Operational Conclusions, dealing with operational conclusions. In many cases, practitioners are treating irrigation as separate from fertilization and, in turn, as separate from the design and creation of the substrate in which the plants are grown. This chapter addresses the root zone as a dynamic system and shows how such a system is put together and how it is managed so as to optimize crop production, while at the same time respecting the factors imposed by society (run-off elimination, labor savings, etc.). Another subject which is mentioned in this chapter is the emerging trend of "organic hydroponics" which seems to gain an increasing popularity in some parts of the world.

One of the main future challenges for global horticulture is to produce adequate quantities of affordable food, in the less-developed countries. Simple, low-cost soilless production systems may be part of the solution to the problems created by the lack of fertile soils and know-how. The fact that a relatively small cultivated area can provide food for a large population can stimulate this

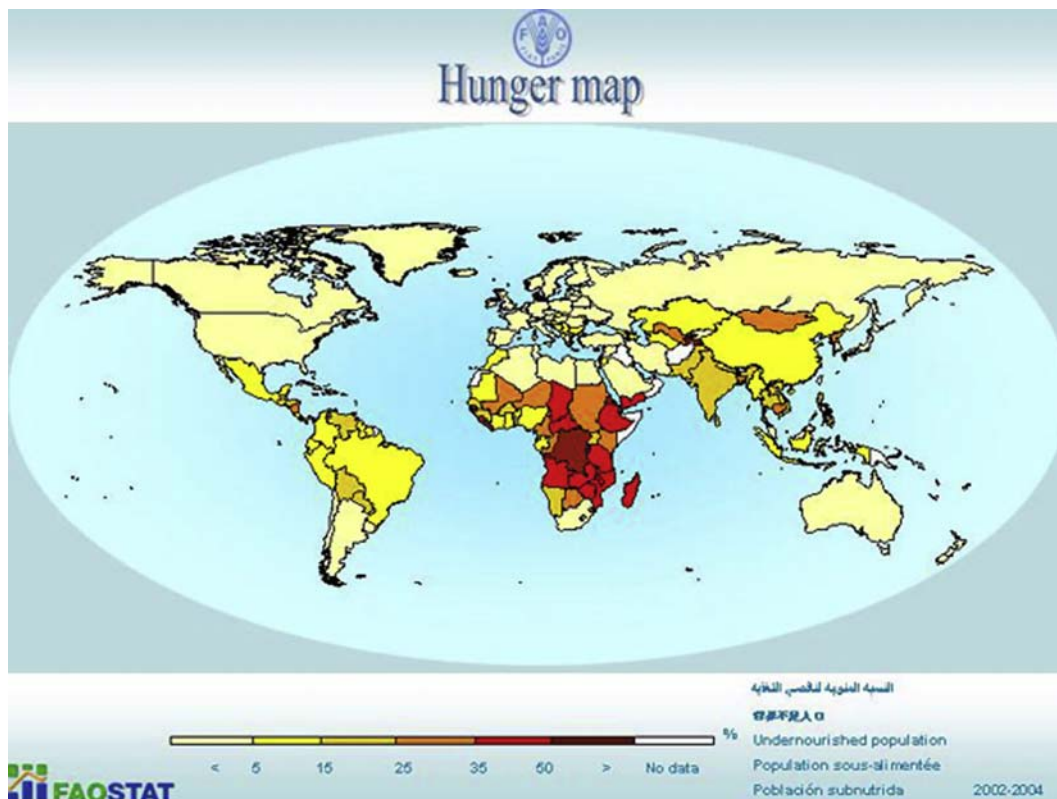
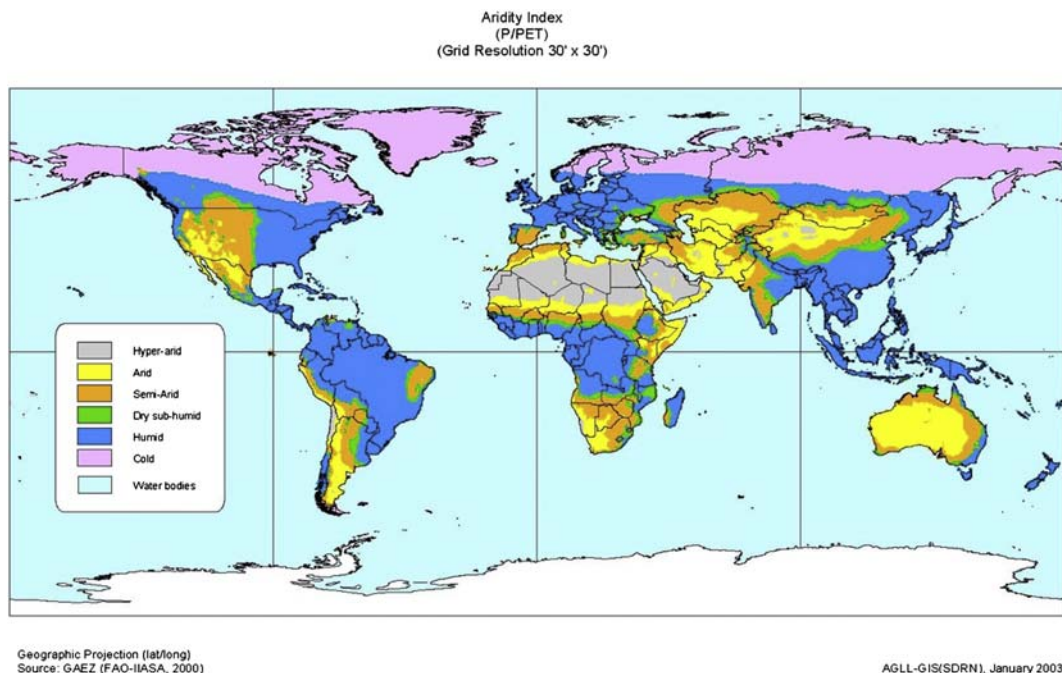


FIGURE 1.4

Percentage of undernourished population around the globe.

With kind permission of the FAO.

development. This, in turn, should stimulate professionals to find alternatives to current expensive and high-tech pieces of equipment and practices, to be suitable and durable for the needs of remote areas. One of the most important advantages of soilless cultivation deserves mentioning in this context: in most of the developing countries water is scarce and is of low quality. By superimposing the FAO's hunger map (Fig. 1.4) on the aridity index map (Fig. 1.5), it is clear that in many regions of the world such as sub-Saharan Africa, Namibia, Mongolia, etc., a large part of the population suffers hunger mainly due to water scarcity. Since water-use efficiency of soilless plant production (and especially in recirculated systems) is higher than that of soil-grown plants, more food can be produced with such systems with less water. Also, plants growing in such systems can cope better with higher salinity levels than soil-grown plants. The reason for this is the connection between ample oxygen supply to the roots and their ability to exclude toxic ions such as Na^+ and to withstand high osmotic pressure (Kriedemann and Sands, 1984; Drew and Dikunwin, 1985; Drew and Lauchli, 1985). It is interesting to note, in this respect, that soilless cultivation is practiced in large

**FIGURE 1.5**

Aridity index around the globe. With kind permission of the FAO.

scale in very arid regions such as most of Australia, parts of South Africa, Saudi Arabia, and the Southern part of Israel. In none of these countries hunger is a problem.

Another facet is an ever-increasing interest in sustainability and circularity. One obvious facet is to somehow use urine and manure from humans or animals to create nutrient solutions for plants, either to generate fertilizer products or even systems directly (Wald, 2017). This is, of course, a throwback to earlier times where farmers took animal waste and spread it in fields to be fertilizer for the plants. In much of the developed world this is no longer done as it pollutes the drinking water. But there are still many areas of the developing world where this is the main fertilizer input to agricultural crops. It is worth exploring how such circularity can be brought back to the current very modern production systems which totally exclude this currently. A key hurdle in this is that such fertilizers are not ionic so that the sensors used today are not effective. There is also a lot of interest in systems where this type of circularity is imposed directly: aquaponics is the combination of hydroponics and aquaculture. The theory is that the waste from the aquaculture can be used by the plants and that unmarketable plant material can be fed to the aquatic organisms as food. The fundamental question is whether the two complete cycles can be run in a way where both plants and aquatic organisms grow well. Unfortunately there are today a lot of hurdles for getting these two cycles working together. While some hurdles are fairly well understood (i.e., management of ammonium) other hurdles still require innovative new solutions.

Another emerging trend in controlled environment agriculture is the use of plant growing systems that do not expose the plants directly to sunlight, but instead use electricity to create photo-synthetically active radiation. With the advent of light-emitting diode lamp technologies, plant growth lighting can convert energy into photons of specific wavelengths so as to effectively create irradiance for the plants which is usable in its entirety by the plants. As of the publication of this book, there have been dramatic changes in this, so that in-door agriculture is today already a viable production method for many plants grown as specialty foods, herbs, or medicinal plants. This trend to lower cost lighting will make more such systems feasible. It is only feasible if all variables are optimized and as such hydroponics is always a part of such systems.

The use of fully automated hydroponic systems in production systems using LED-generated lamps light facilitates another trend in in-door controlled environment agriculture: vertically stacking of production systems. Vertical farms have become very interesting as a way to put modern production systems into cities. While there are numerous hurdles to this, it is clearly a rapidly growing area of agriculture. In such systems there is then also a new type of water which has not been available in field or greenhouse plant production: condensate from the air conditioning system. With full recirculation of all water molecules in such systems, it is possible to achieve water-use efficiencies which cannot even be dreamed of in field agriculture. This then offers an opportunity to grow plants in places where few plants grow today (severe desert and arctic conditions) and all the information in this book will be highly relevant to achieving success.

The science of plant production in soilless systems is still young and although much work has been done, many questions still remain unanswered. One of the purposes of this book is to focus on the main issues of the biological, physical, and chemical environment of the rhizosphere and to identify areas where the future research is needed so as to take further advantage of the available substrates and to propose desirable characteristics for future substrates and growing practices to be developed by the next generation of researchers.

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ROOT GROWTH, PHYSIOLOGY, AND POTENTIAL IMPACT OF SOILLESS CULTURE ON THEIR FUNCTIONING

2

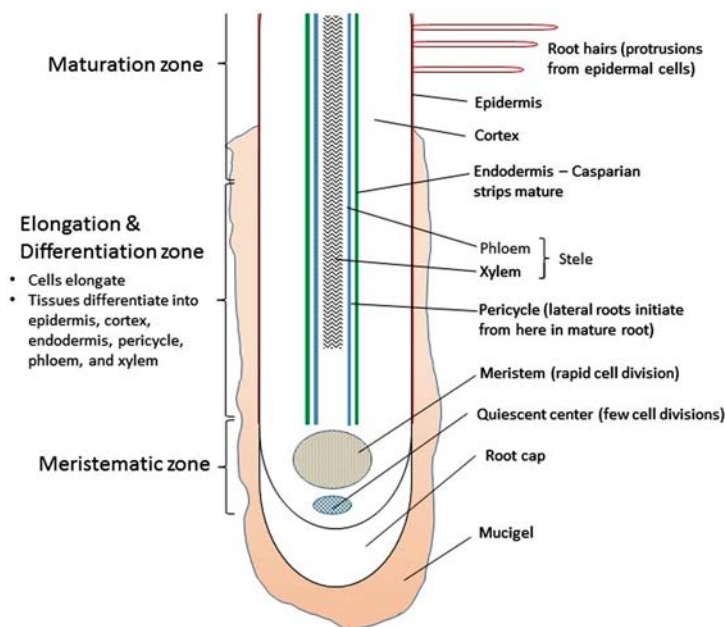
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2.1 BASIC ROOT ANATOMY

Organization of root tissue is optimized for efficient water and nutrient capture, as well as transport to the shoot. Basic organization is similar across most root types, although those roots that undergo secondary growth will have more pronounced vascular tissue. Primary root growth results from cell division in the root apical meristem, which produces a range of different root cells. The apical meristem is protected by multiple layers of root cap cells at the distal side. The root cap also produces mucilage, which facilitates root elongation. Root cap cells die and slough off at the root tip and are replaced by new cells produced by the apical meristem. Other root cells produced by the apical meristem increase in both width and length in the elongation zone just proximal of the apical meristem. Root cells subsequently start differentiating in the differentiation zone, proximal of the elongation zone. From the outside to the inside, cells differentiate into a layer of epidermal cells, some with root hairs, sometimes an exodermis, multiple layers of cortex cells, a layer of endodermis cells where Casparian strips start developing, the pericycle, and vascular ground tissue with phloem and xylem (Fig. 2.1). These tissues mature and become fully functional in the maturation zone (Dolan and Roberts, 1995; Malamy and Benfey, 1997; Peret et al., 2009; Rost, 2011).

Each of these cell types have specific functions and the degree to which they are present and/or modified depends strongly on their immediate environment and to some extent on the carbon flow to the root. The length of each of these zones (cell division, elongation, differentiation, and maturation) also strongly depends on the species, environmental conditions, and the ratios of auxin (which promotes cell division), cytokinin (which promotes cell differentiation), and gibberellin (which inhibits cytokinin signals). Epidermal cells initiate contact with microorganisms (bacteria and fungi) in the rhizosphere, exude acids, and carbon-rich compounds and produce root hairs that increase absorptive root surface area and increase the volume of soil or substrate explored by the root system (Dolan, 1996; Gianinazzi-Pearson et al., 2007; Rost, 2011). In some species a well-developed exodermal area may protect the inner root surface from dehydration (Hose et al., 2001; Lynch et al., 2014), while specialized passage cells allow the continued flow of water (Peterson and Enstone, 1996). The cortex layer forms the main body of the root where metabolic processes and exchange processes with symbiotic bacteria or fungi take place. In this layer, programmed cell

**FIGURE 2.1**

Generic longitudinal section of a root (Taiz and Zeiger, 2010).

death may lead to the formation of aerenchyma or other types of root lacuna, which protects roots from dehydration, increases oxygen diffusion and reduces overall tissue respiratory costs (Postma and Lynch, 2011; Cuneo et al., 2016). The cortex also plays an important role in carbohydrate metabolism and storage. The next layer is the endodermis where the root can control both water and nutrient diffusion and uptake into the stele tissue (Barberon, 2017). Just inside the endodermis is the pericycle, a single layer of cells with the founder cells that initiate lateral root formation. The center of the root is the stele where vascular tissue that moves water, nutrients, and sugars between roots and shoots is located. Roots generally are very active organs where xylem loading takes place, while phloem activity in the root generally is dominated by unloading of compounds translocated from the shoots. Following primary growth, roots of many dicotyledonous plants exhibit secondary growth, which produces more vascular tissue and increases root diameter. The rate of development of secondary xylem is strongly linked to the function of the root within the overall root system framework (Bagniewska-Zadworna et al., 2012, 2014).

Investment in water-transporting tissues is determined by genetics and environment. Even under identical, optimal, growing conditions, roots of six different genotypes of soybean showed considerable variation in root hydraulic conductivity to water—with ~35% of that variation explained by differences in the surface area of the stele (Rincon et al., 2003). Cortex volume and axial

conductance also varied significantly among these genotypes. Lateral root growth is started from primer cells in the pericycle, where the initials that will form a new root meristem are activated. The newly forming root must then push through existing endodermal, cortex, and epidermal tissue to reach the rhizosphere. This process is helped by auxin-induced acidification of cell walls which loosens bonds between cell walls, allowing the new root to push past existing cells.

2.2 DIFFERENCES IN ROOT ANATOMY/PHYSIOLOGY DUE TO SOILLESS ENVIRONMENT

Roots growing in soils experience fluctuations in soil-water environment, temperature, and nutrient availability, while being exposed to shifting soil particles and decomposing or herbivorous soil organisms. Under such conditions, roots tend to develop more protective tissues in response to their external environment than under much more stable hydroponic or aeroponic conditions (Soukup et al., 2004; Meyer et al., 2009; Tylova et al., 2017). Modifications in these protective tissues include increased lignin and suberin content of epidermal cells (McKenzie and Peterson, 1995), reducing the impact of soil strength on the roots, and under these conditions roots tend to have a slightly lower length per mass ratio (i.e., are more dense). Root browning occurs at a faster rate in shallow roots than deeper growing roots (Hendrick and Pregitzer, 1992), indicating that fluctuating conditions and disturbance play a role in facilitating the rate of browning. Once roots have turned brown, their metabolic activity is much reduced (Comas et al., 2000), presumably due to much reduced active nutrient uptake, which can account for as much as 50%–70% of the root respiratory costs for unsuberized roots of fast- and slow-growing herbaceous species (Poorter et al., 1991, 1995).

When roots are permanently immersed in aerated nutrient solution, they maximize root–water–nutrient interactions by increasing the amount of length produced per unit root mass and investing very little in protection from the environment or supportive tissue (Goodman and Ennos, 2001). The latter is not necessary as the nutrient solution supports root mass; however, when these roots are regularly disturbed by bending or flexing they show a marked increase in tissue density and rigidity (Goodman and Ennos, 2001). The use of aeroponic techniques provides the roots with ample oxygen and, when applied properly, will provide ample moisture to prevent dehydration and nutrients to meet plant needs. More supportive tissue is needed to maintain structural integrity when roots are hanging in air, thus relatively more biomass is invested in cortex and epidermal cells, compared to hydroponics. Under soilless conditions of low disturbance, the characteristic browning of roots as epidermal cells accumulate suberin and lignin is absent or delayed and presumably these roots remain much more metabolically active.

2.3 IMPACT OF ENVIRONMENTAL CONDITIONS ON ROOT GROWTH

The supply of carbohydrates to the root has a tremendous impact on root growth (Sloan and Jacobs, 2008). The idea of a functional equilibrium whereby newly assimilated carbon is preferentially allocated to organs that acquire the most limiting factor to plant growth was first posed in the

1960s (Brouwer, 1963, 1983; Lambers, 1983; Bloom et al., 1985) and has been very influential in our general understanding of carbon allocation patterns and tradeoffs ever since (Lambers, 1983; Bloom et al., 1985; Wright et al., 2004; Perez-Ramos et al., 2013; Kramer-Walter et al., 2016). Under conditions of nutrient or water stress, plants invest a larger fraction of their carbohydrates in the root system, presumably as these tissues become stronger carbon sinks when shoot growth is limited by nutrient and water availability. Root tissue nutrient needs are lower than those of shoots, thus when nutrient uptake is limiting, new roots can still be produced while shoot growth ceases, resulting in an increase in root mass fraction (e.g., van Iersel, 1997; Fig. 2.2). The investment of carbohydrates into root growth will help plants acquire more nutrients and may thus help minimize the impact of a limiting nutrient supply.

Under conditions of high CO₂, as well as high light, carbon assimilation is stimulated and biomass production can become limited by nutrient availability. Increased allocation of carbon to the root system, that is, an increase in root mass fraction compared to plants grown under ambient CO₂ or low light conditions, is a general response when nutrients become limited. In soilless conditions, when plants are often provided with a continuous supply of water and high nutrient conditions, allocation patterns strongly shift to investing more mass in shoot growth, leading to root mass fractions as low as 0.05 (Vannoordwijk and Dewilligen, 1987).

Environmental conditions not only affect the balance between root and shoot growth but also affect root anatomy. For example, shading of eight different grass species to 30% of the daylight led to a greater proportion of root stele tissue and larger xylem vessels (Wahl et al., 2001),

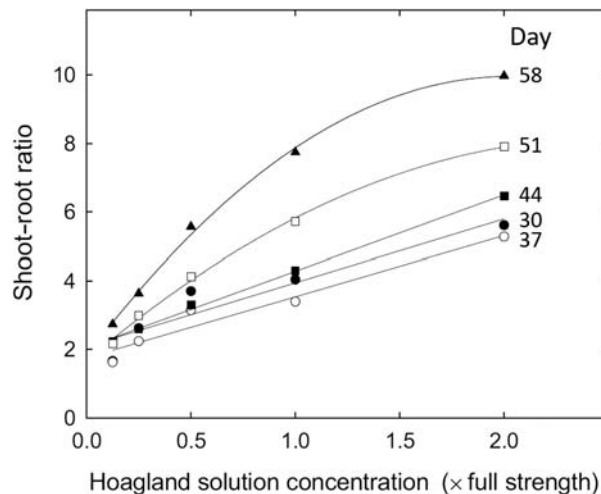


FIGURE 2.2

The effect of nutrient solution concentration on the shoot–root ratio of *Salvia splendens*. Data were collected at 30, 37, 44, 51, and 58 days after transplanting. Note that the shoot–root ratio generally increases with increasing nutrient solution concentration, as well as over time.

Adapted from van Iersel (1997).

reflecting a greater investment in water transport capacity to support a proportionally greater investment in transpiring leaf area.

Impacts of temperature on root growth are not well described. In general, root respiration increases as temperature increases. However, increased metabolism does not necessarily equate with increased root growth or even nutrient uptake rates. Several papers have assessed the impact of root zone temperatures on root growth, depending on the native habitat of the species. For example, alpine plants were found to have a low temperature limit of 0.8°C–1.4°C (Nagelmüller et al., 2017), while polar semidesert plants exhibit some growth at 1°C–3°C but had optimal growth at soil temperatures of 12°C–20°C (Bell and Bliss, 1978). More temperate species such as walnut trees appear to have optimal root growth when soil temperatures are within 20°C–25°C (Contador et al., 2015). Concord grape roots allowed to acclimate to soil temperatures exhibited their highest respiration rate (as a proxy for metabolic activity) at 25°C, after which respiration declined and membrane leakage increased strongly (Huang et al., 2005). In this study there was also a clear link between soil-water content and membrane leakage, where adequately watered roots at 25°C exhibited very little membrane leakage, while roots growing in soils with less than 5% water content exhibited drastic increases in membrane leakage and decreases in respiration. Similarly, *Dichanthelium lanuginosum* showed marked decreases in stomatal conductance and carbon gain when soil temperatures increased from 35°C to 50°C, and negative effects could only be partially offset by increasing soil-water content, suggesting that reduced water uptake was leading to plant water stress (Germino and Wraith, 2003). For more drought resistant species such as citrus (Bryla et al., 1997; Eissenstat et al., 1999), root respiration was affected by temperature, but not by short term low water availability (Bouma et al., 1997). As field studies describing root growth often have confounding effects of seasonal phenology and water availability with soil temperatures, it is difficult to pinpoint exactly what optimal temperatures for root growth are. For field grown plants, the temperature gradients in the soil may affect root distribution, with continued winter root growth in deeper soil layers with moderate temperatures than in colder surface layers.

In a greenhouse study with aeroponically grown pepper (*Capsicum annuum*), a constant root zone temperature of 20°C resulted in more leaves, greater leaf area, and increased dry weight compared to plants with their root zone at ambient temperatures (which fluctuated between 25°C and 40°C) (Dodd et al., 2000). Although the effects of the different root zone temperatures on root growth were not quantified, root zone temperature clearly can have a major impact on shoot growth. Such effects may be much larger in soilless culture than in soil-based crop production. In soil-based production, the large thermal mass of the soil can minimize temperature fluctuations in the root zone. Such a temperature buffering effect is much less pronounced in soilless production. As a matter of fact, in container production, root zone temperatures can fluctuate widely and rapidly, especially if the containers are exposed to direct sunlight. Ingram et al. (2015) reviewed root growth responses to high substrate temperatures in containers and reported that most species exhibit reduced root growth at root zone temperatures of ~38°C–40°C for 6 hours per day, a situation which can easily occur on a sunny day when plants are grown in black plastic pots. Short exposure (~20–30 minutes) to root zone temperatures of 46°C–52°C damages root cell membranes irreversibly. The vast majority of energy that heats up container substrates comes from radiative exposure of the side walls of the containers (Martin and Ingram, 1993; Fig. 2.3), where larger containers have more buffering capacity in the center of the pot than smaller containers. The geometry of the container also has an impact on the ability to buffer temperatures, media in pots with a greater

**FIGURE 2.3**

Photograph showing heat stress damage to the roots of pot grown *Thuja occidentalis* on the south side of the root ball.

Picture courtesy: John Ruter.

surface area exhibited higher media temperatures than media in pots with the same volume but lower surface area (Heller et al., 2015). Irrigation with water with a temperature below that of the root zone temporarily lowers root zone temperature; however, the amount of water applied needs to fully saturate the container to “flush” out the heat energy. Lower amounts of water led to evaporative cooling at the surface. However, the temperature at the center of the container increased when less than saturating amounts were applied (Martin and Ingram, 1991). The opposite effect was obtained when the temperature of the irrigation water was higher than that of the media (Heller et al., 2015).

Even when roots are exposed to nondamaging substrate temperatures, high root zone temperatures affect roots in different ways. Root respiration increases exponentially with increasing temperatures (Atkin et al., 2000), leading to enhanced oxygen use as well as increased media CO₂ concentrations. The ability of roots to acclimate to higher substrate temperatures is strongly species

dependent (Atkin et al., 2000), and temperature and temperature gradients strongly affect root growth. For example, diurnal root elongation rates in hydroponically grown maize were constant when temperature was held constant, and 50% higher at 26°C than at 21°C (Walter et al., 2002). This study also showed two expansion zones within the elongation zone, that exhibit different temperature sensitivity. Their 3-D method of root observation also allowed for the observation of regular (40–70 minutes interval) oscillations in root-tip growth direction, even under stable conditions, potentially giving some explanation as to the cork–screw-like appearance of many roots. Switching temperature from 21°C to 26°C led to new steady state growth rates within 1 hour of switching—showing that root growth can rapidly respond and adjust to changes in thermal environment. The effect of growth media temperature on root growth and development is affected drastically by the nitrogen source, ammonium or nitrate as described in detail in Chapter 6, Nutrition of Substrate-Grown Plants. The temperature gradient itself can also strongly impact plant and root growth; a controlled experiment with barley showed that when roots were exposed to a gradient from 20°C to 10°C, they had strongly accelerated development and after 30 days these plants had ~3.5 times greater shoot and root biomass than plants exposed to a uniform 20°C root zone temperature (Fullner et al., 2012). Such temperature gradients are generally absent in aeroponic and hydroponic systems, while for potted plants a strong horizontal temperature gradient may be present in addition to vertical gradients.

The availability of water in the root zone is essential for root growth. Without adequate water, newly divided cells cannot uptake enough water to exert the pressure needed for cell elongation, and roots exposed to water stress tend to have smaller cells and denser tissue. A confounding factor for soil-grown plants is that soil strength increases when soils dry (Bengough, 1997), increasing the pressure needed for roots to push through the soil. More moist soils reduce friction between soil particles and between particles and the root surface, thus making it easier for the roots to push particles aside and push through the soil. The high porosity of soilless substrates makes it unlikely that substrate strength significantly impairs root elongation.

Maintaining adequate soil-water availability also improves nutrient uptake by the roots as most ions move to the root surface via bulk flow or diffusion. The rate of bulk flow of soil water to the roots depends on transpiration by the shoots. If low soil-water availability limits transpiration, that will in turn reduce mass flow of water and dissolved nutrients to the roots. Diffusion of nutrients depends on the concentration gradient from the bulk soil solution to the roots and the tortuosity of the diffusion pathway. Tortuosity greatly increases at lower soil-water content, as does decreasing diffusion and reducing nutrient movement into the rhizosphere. Interestingly, root growth itself decreases tortuosity in soilless substrates, presumably by connecting pore spaces within the substrate to each other (Kerloch and Michel, 2015).

Roots exhibit a hydrotropism response that is independent from gravitropism—roots sense moisture gradients and across short distances grow toward moister soil parcels. Bao et al. (2014) showed that root branching patterns and growth direction are strongly affected by water availability and its gradient and that hydropatterning in response to water gradients determines the distance between founder cells in the pericycle, and thus lateral root density. Although water deficit generally leads to smaller plants and overall smaller root systems, water deficit does increase relative carbon allocation to the roots, decreasing the shoot–root ratio. Labeled carbon experiments with *Arabidopsis* have shown that sucrose transport in the phloem is increased, as are transporters involved in phloem loading and unloading in response to drought (Durand et al., 2016). This study showed

considerably greater levels of sucrose, glucose and fructose in *Arabidopsis* roots in response to water deficit, allowing for increased osmotic pressure in the roots. This should assist in both water uptake and cell expansion under drought conditions.

At the other extreme, when all pore spaces are filled with water, roots can rapidly run out of the oxygen necessary to maintain cellular respiration. When oxygen is available, roots can carry out oxidative phosphorylation, generating ~60 adenosine triphosphate (ATP) per sucrose. Under low oxygen conditions, oxidative phosphorylation is no longer a viable pathway and maximum ATP yield is ~4 ATP per sucrose through glycolysis only—a 93% reduction in respiratory efficiency. Such conditions immediately feedback on the two processes that take most energy in a growing root system: cell division is strongly reduced and active nutrient uptake, which requires ATP to maintain H⁺ gradients across cell membranes, is severely reduced. It is therefore not surprising that many overwatered plants develop signs of severe nutrient deficiency and the problem with such plants is often misdiagnosed. Low oxygen conditions can also induce the production of ethylene. Tomato plants exposed to low oxygen through overwatering showed leaf symptoms very similar to wilting in response to enhanced transport of 1-aminocyclopropane-1-carboxylic acid, an ethylene precursor, to the shoot where it was converted to ethylene (Bradford and Dilley, 1978; Jackson et al., 1978; Grichko and Glick, 2001). Many plant species, paradoxically, respond to low oxygen conditions by closing their stomates, even though plenty of water is available for transpiration (Sanchezblanco et al., 1994; Kallestad et al., 2007; Else et al., 2009). Interestingly, elevated O₂ conditions in the root zone had little effect on plant growth: only when levels reached 30 mg/L or above (compared to 8.5 mg/L ambient), did tomato plants show reduced growth (Zheng et al., 2007). Such conditions are unlikely on earth but may be important when considering growing plants under controlled conditions in space where plants reduce CO₂ levels and increase oxygen levels.

Because of the importance of oxygen for root respiration, hydroponic solutions need to be properly aerated to prevent hypoxia. When plants are grown in soilless substrates, the high porosity of the substrate facilitates diffusion of oxygen into and throughout the substrate. As a result, substrate water content has little effect on the partial pressure of oxygen in substrate pores and no hypoxia was detected even very wet substrates (van Iersel and Dove, 2013). However, their results do not rule out that hypoxic conditions may occur in the rhizosphere of roots, where oxygen demand is greatest.

Under uniform high nutrient conditions (e.g., hydroponics), plants tend to allocate relatively more biomass to the shoot (van Iersel, 1997). However, work on barley by Drew in the 1970s, has shown strong responses of lateral root production to localized patches of high nutrients such as P and N, but not K (Drew et al., 1973; Drew, 1975; Drew and Saker, 1975, 1978). Such localized nutrient distribution can occur in soilless substrates when controlled release fertilizers are used or when drip irrigation provides nutrient solution to a specific part of the substrate. Localized nutrient availability can override whole plant root allocation patterns. High local lateral root system density may become disadvantageous if resources are highly mobile and lateral roots compete with one another for nutrients (Postma et al., 2014a). Root system plasticity, the ability to rapidly respond the changing water and/or nutrient availability, may be an important attribute for plant survival. More recent work has shown that conditions of low phosphorus reduce the root length to the first lateral root, as well as the distance between lateral roots, while increasing lateral root length (Williamson et al., 2001). Responses to nitrate are similar. Zhang and Forde (2000) showed that

localized high nitrate conditions enhance lateral root branching. This response is modulated by localized high auxin levels in the roots. However, high internal nitrate levels block the signal that induces lateral root formation. In *Arabidopsis*, root hair formation is also very responsive to nutrient availability. Under low P conditions, root hair density increases and root hairs become longer. Bean plants grown under low P conditions had greater rates of soluble sugar unloading and higher sugar concentrations in growing root tips (Ciereszko et al., 1998). Both the pools of sucrose and glucose increased under low P conditions, with the ratio of sucrose:glucose decreasing. The resulting increase in osmotic pressure could facilitate cell elongation and root foraging (Pritchard, 1994). Such root plasticity is likely more important in natural ecosystems and low-input agriculture than in crop production in soilless substrates or hydroponics, where fertilizer application rates and nutrient availability are typically high.

The physical space available in the root environment to support root system expansion has tremendous influence on the size and traits of the root system and subsequent feedback to shoot growth (Fig. 2.4). A recent study expressly designed to test whether wheat responses to elevated CO₂ conditions were consistent across genotypes grown under greenhouse conditions in two different pot sizes showed that container size altered ranking of root traits and CO₂ responsiveness (Bourgault et al., 2017). This finding corroborated earlier CO₂ research that also showed that pot size restricted plant response to elevated CO₂ conditions (Bernstson et al., 1993; McConaughay



FIGURE 2.4

Rootbound *Yucca filamentosa*. Leaving the plant too long in a small container results in excessive root proliferation near the container wall (left). Despite the high root density near the container wall, much of the center part of the substrate contains few roots (right).

Pictures courtesy: Marc van Iersel.

et al., 1993). These findings should be kept in mind with recent interest in high throughput root phenotyping under controlled conditions where most methods involved use some version of limiting the root system for easier observation, either in small pots for magnetic resonance imaging or X-ray imaging (e.g., Leitner et al., 2014; Knipfer et al., 2015; Pflugfelder et al., 2017), on petri dishes or filter paper where plants can only grow for a short duration (e.g., Shi et al., 2013), in space-limited transparent observation systems (e.g., Wright and Wright, 2004; Vegapareddy et al., 2010), or a combination of these (e.g., Le Marie et al., 2014, 2016). Growing plants in pots in general can affect proper root development, not only through limited space but also by causing unnaturally high root zone temperatures and modified temperature gradients, accentuated soil-water extremes, and altered textural and structural properties of the growing media compared to natural soils.

Restriction of root space can lead to structural root deformities in annual and perennial plant species, such as root girdling, and strongly reduces shoot growth (Hsu et al., 1996; van Iersel, 1997), which may be seen as a positive if dwarfing of trees is the goal in containerized production (Hsu et al., 1996; Wall et al., 2000). Root restriction enhanced nitrate uptake kinetics, but reduced net nitrate uptake in grapevines, reducing root and leaf N concentration and leaf photosynthetic rates (Zhu et al., 2006; Yang et al., 2007). This suggests that upregulating uptake rates at the root level, as was also found in tomato (Bar-Tal et al., 1995), in these vines did not compensate for nitrate efflux and/or overall reduced root surface area.

Many container-grown plants are transplanted into larger containers, gardens, or agricultural fields after they are sold. Root restriction during the production of transplants can impact posttransplant performance, although results from research trials have been inconsistent. The severity of posttransplant growth reductions caused by root restriction likely depends on container size, species, and growing conditions (NeSmith and Duval, 1998).

When woody plants have been grown in a too tight container, or potted up in a manner that twists or compresses the roots system, they can form circling roots inside a container (Fig. 2.5). These roots can be kinked, J-rooted, twisted, etc. As they form the future permanent structural roots of the root system when perennials are transplanted in the field, these roots can start girdling the tree trunk which leads to unstable trees and defects that may not become apparent until later in the life of the transplanted tree (Shaw, 1977). Root circling of potted plants may be reduced by using root pruning pots, that use an air layer that restricts root growth (Marler and Musser, 2016), or copper-treated pots that discourage roots from growing near the container edge (Arnold and Struve, 1993), although the efficacy of copper-treated containers depends on container volume (Svenson, 2002). Other types of pots that are intended to minimize circling of structural roots and enhance fine-root growth are air-pots, decomposable pots, or fabric-based pots. Red oak trees produced in root pruning, air-potted or fabric containers showed less water stress under well-watered conditions in the first year after transplanting into the field compared to bare rooted trees, suggesting some benefit of these fine-root stimulating pots. However, these benefits did not persist in subsequent years nor were they present in drier urban conditions (Levinsson et al., 2014).

Within species, the environment has a much greater impact on root traits than genotype. A recent study using 24 genotypes of wheat in 12 different soil environments showed that genotype only accounted for 12% of the variation in root depth, environment explained 48% of the variation and genotype by environment interactions accounted for 40% of the variation in rooting depth (Acuna and Wade, 2012). The large impact of environment on root depth makes sense as soil



FIGURE 2.5

Girdling root on a tree. Although girdling roots are often produced during container production, the negative impacts of such roots, including mortality, may not become evident till years later.

Image courtesy: Heiner Lieth.

penetration resistance varied strongly across the sites. However, traits that led to deeper rooting depth of genotypes in one site did not necessarily predict rooting depth at another site.

2.4 ROOT PHYSIOLOGY AND DEVELOPMENT

Most of the total root system length is made up of fine roots [first and second-order lateral roots, counting from the outside to the inside (Eissenstat and Yanai, 1997)]. Fine roots can grow into smaller soil pores, bringing more root surface area in contact with the soil solution, increasing overall uptake of water (Macfall et al., 1991), and nutrients such as N (Sullivan et al., 2000) and P (Santner et al., 2012). Growth of the primary root is largely indeterminate, while lateral roots and tertiary roots exhibit determinate growth. The gravitropic response of primary roots is much

stronger than that of secondary or tertiary roots, where the response may also be modified by drought conditions, with steeper angles when roots develop under drought stress (Lynch, 2013). Root growth angles and branching patterns also play a major role in effectiveness of N acquisition (Postma et al., 2014b; Dathe et al., 2016). Branching patterns are tightly controlled and branching increases when roots encounter favorable patches in heterogeneous environments (Hodge, 2004). As the most external roots age, the diameter of those roots tends to decrease. Roots of the lowest order tend to have the smallest diameter, although in many species the first 1–3 orders may not differ much in diameter. Root systems in natural soil need to continuously explore new soil volumes and renew as depletion zones are formed. Few people have studied turnover of fine roots in soilless systems, and there are no estimates root lifespan and turnover rates in hydroponic or aquaponic systems.

The inner and outer parts of the cortex form the endodermis and exodermis, respectively. From a physiology point of view the exodermis and endodermis form major control points for water and nutrient uptake. Older parts of the root system and aging determinate roots form suberin layers in the exo- and endodermis where water flow with dissolved nutrients is forced to enter a symplastic pathway. Maturation of endodermal and exodermal tissue occurs in three stages (Ma and Peterson, 2003). Stage I is the formation of Casparian bands between walls of endodermis cells, stage II includes the deposition of suberin lamella between the cell walls and cell membranes of individual endodermis cells, and stage III includes the addition of cellulose and lignin and sometimes suberin into the cell walls, strengthening those walls. Since suberin inhibits apoplastic water flow, the solution is forced to cross the plasmalemma to move through the endo- or exodermis. This provides points of control for the root system, where nutrients can be actively or passively taken up into cells. The exo- and endodermis also form significant barriers to pathogen entry. Development of the exodermis as well as the presence of lignin in the epidermis is likely under environmental control. In addition, as roots encounter more soil resistance, roots thicken and tissue density increases, likely as a result of reduced cell elongation and the presence of more cell wall material per unit root volume. In soilless systems roots generally encounter very little resistance and this may reduce the production of exodermal tissue and the presence of lignin in the epidermis.

Root hair formation generally occurs in the maturation zone of lateral roots. Root hairs generally increase in density under low P conditions (e.g., Gahoonia and Nielsen, 1997; Bates and Lynch, 2001; Bayuelo-Jimenez et al., 2011; Peret et al., 2011) but also have been found to be strongly affected by growing media. For example, when peanut was grown in loamy sand, 4:1 coarse sand:fritted clay mixture or fine sandy loam in the greenhouse, 47%, 17%, or 14% of the roots developed root hairs (Meisner and Karnok, 1991). Root hair formation has been extensively studied in *Arabidopsis* (Dolan, 2001; Ma et al., 2001; Datta et al., 2011), where root hair formation is controlled by auxin gradients.

Nutrient uptake rates per unit of root surface vary by the nutrient studied, external concentration (Kronzucker et al., 1995a, 1995b; Glass et al., 2002), internal concentration (Siddiqi and Glass, 2002; Gorska et al., 2010), species specific root properties (Ristova and Busch, 2014), soil-water content (Eissenstat et al., 1999; Buljovic and Engels, 2001), root age (Volder et al., 2005, 2009), and interactions between a wide range of environmental and other internal factors (e.g., Marschner et al., 2011; Ruffel et al., 2014; Barberon et al., 2016). High- and low-affinity nitrate transporters can quickly up- or down-regulate in response to changing nitrate concentrations in the environments (Glass et al., 2002), although this plasticity changes as roots age (Garnett et al., 2013). Root

hydraulic properties change in response to nitrate presence and this response is species specific and likely linked to internal N demand. For example, in an experiment linking the effect of nitrate on water flow and nitrate uptake rate under high N conditions across seven species varying in relative growth rate, there was a strong positive linear relationship between nitrate uptake rate and effect of nitrate on relative water flow (Gorska et al., 2010).***

2.5 SUMMARY

Roots are highly plastic organs, whose anatomical, morphological, and physiological characteristics depend on both genotype and environment. Although roots evolved to support plant performance in natural soils, their plasticity allows them to adjust to grow and function in soilless substrates, including hydroponics, aeroponics, rockwool, and substrates based on organic materials (e.g., peat, coconut coir, bark). Soilless substrates are characterized by high porosity, low physical strength, and typically readily available water and nutrients. All these factors affect root growth and development. Despite the importance of the root system, roots in soilless substrates have received less research attention than those of soil-grown crops. A better understanding of root systems in soilless substrates may help improve water and nutrient use efficiency.

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PHYSICAL CHARACTERISTICS OF SOILLESS MEDIA

3

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3.1 PHYSICAL PROPERTIES

All media are composed of three phases: solid, aqueous, and gaseous. In the following sections, the physical characteristics of these three phases will be discussed separately and in combination.

3.1.1 BULK DENSITY

Bulk density (BD) of a medium is defined as its dry mass per unit of volume (in a moist state) and is measured in g/cm^3 . Numerous methods for the measurement of BD (as well as other physical parameters) can be found in the literature (e.g., [de Boodt and Verdonck, 1972](#); [Wever, 1995](#); [Raviv and Medina, 1997](#); [Gruda and Schnitzler, 1999](#); [Morel et al., 2000](#)). Some methods are used primarily for research purposes (e.g., the standard ISHS method, as described by [Verdonck and Gabriels 1992](#)). Others are used as industrial standards in certain countries or regions of the world (e.g., BS EN 12580:2001 in the United Kingdom, both the LUFA and DIN methods in Germany and the CEN method in the EU). All of them, however, are based on one principle: wet material is allowed to settle within or compressed using known pressure into a cylinder of a known volume. It is then dried down completely and weighed. For specific details see [Chapter 11](#), Analytical Methods Used With Soilless Substrates.

As many media are composed of more than one ingredient, the characteristics of each ingredient contribute to the total BD of the medium. These are individual and combined particles' arrangements, BD of the ingredients and compaction qualities. In particular, media components that differ significantly in particle size have higher BDs as a mix ([Pokorny et al., 1986](#)). Similarly, they have lower total porosity (TP), water-holding capacity and air-filled porosity (AFP) than media composed of similar particle sizes.

BD affects the choice of media in various ways. For example, outdoor production of tree saplings requires high BD media to prevent container instability under windy conditions. This can be achieved by the inclusion of heavy mineral constituents, such as sand, soil, clay or tuff in the mix. On the other hand, high-intensity greenhouse crops, which are frequently irrigated and may be exposed to oxygen deficiency if hydraulic conductivity and AFP are not high, require media of low BD. Another consideration is that the mixing and transportation of low BD media are easier than those of high BD media.

3.1.2 PARTICLE-SIZE DISTRIBUTION

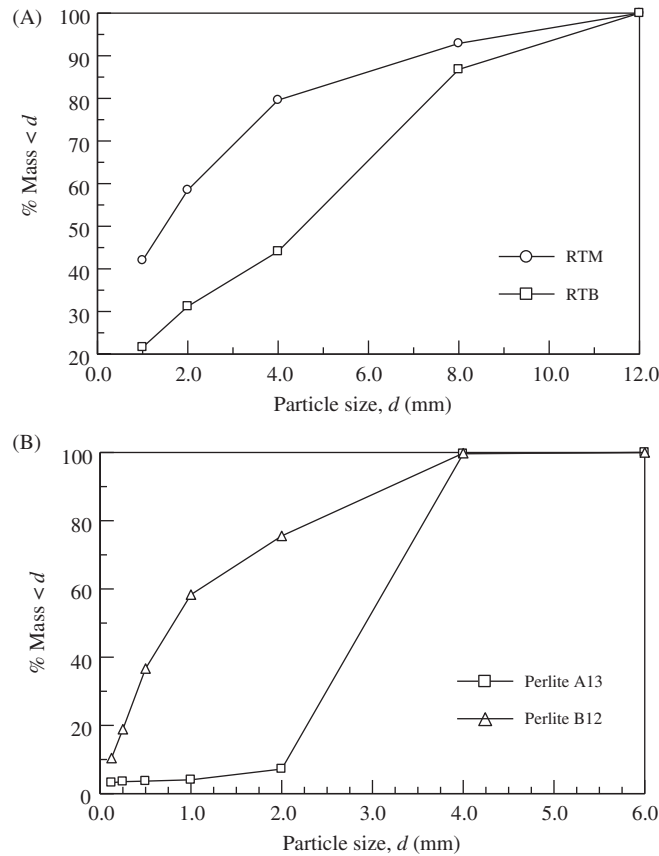
Particle-size distribution (PSD) is the most fundamental physical property of a porous medium and defines its texture. The particle sizes present, and their relative abundance, have a significant influence on most of its physical properties. Particle-size analysis consists of isolating various particle sizes or size increments and then measuring the abundance of each size. The material of which the medium solid phase is composed includes discrete particles of various shapes and sizes, as well as amorphous compounds such as colloidal organic matter. The particle size and mineral composition largely determine the nature and behavior of the medium: its internal geometry and porosity, its interactions with fluids and solutes, as well as its compressibility and strength.

The term medium texture refers to the size range of particles in the medium, that is, whether the particles composing a particular medium are of a wide or relatively narrow range of sizes, and whether they are mainly large, small, or of some intermediate size. As such, the term carries both qualitative and quantitative connotations. Qualitatively, it indicates whether the material is coarse and gritty, or fine and smooth. Quantitatively, it denotes the precisely measured distribution of particle sizes and the proportions of the various size ranges of particles within a given medium. As such, medium texture is an intrinsic attribute of the medium and the one most often used to characterize its physical makeup.

The PSD attempts to divide what in nature is generally a continuous array of particle sizes into discrete fractions. The PSD curves for two types of tuff (scoria) (data from Wallach et al., 1992a) and perlite (data from Orozco and Marfa, 1995) are shown in Fig. 3.1. The shape of PSD is related to the pore-size (radii) distribution which was related, as will be discussed in the following, to the water retention characteristics and the hydraulic conductivity.

The information obtainable from this representation of PSD includes the diameter of the largest grains in the assemblage and the grading pattern, that is, whether the substrate is composed of distinct groups of particles each of uniform size or of a more or less continuous array of particle sizes. Poorly graded media have a dominance of particles of several distinct sizes and are represented by a step-like distribution curve. A medium, with a flattened and smooth distribution curve, is termed well graded. Based on the shape of tuff and perlite PSD (Fig. 3.1), it can be concluded that RTB tuff 0–8 mm (Volcanic ash, scoria, Merom Golan, Israel) and A13 perlite, whose distribution curves are more step-like, are poorly graded relative to RTM tuff (Volcanic ash, scoria, Merom Golan) and B12 perlite, respectively.

PSD is used in soil science for estimating soil hydraulic properties, such as the water retention curve (RC), $\theta(\psi)$ (Section 3.2.3) and saturated as well as unsaturated hydraulic conductivities (Gupta and Larson, 1979; Arya and Paris, 1981; Campbell, 1985; Vereecken et al., 1989). Mathematical relationships between the particle size and hydraulic properties tend to be fairly good for sandy soils, but not as accurate for soils with larger fractions of clay (Cornelis et al., 2001). Significant contributions were made by Arya and Paris (1981) to predict water RCs $\theta(\psi)$, using the PSD. Their physico-empirical approach is based mainly on the similarity between shapes of the cumulative PSD and $\theta(\psi)$ curves. Various authors have developed similar models (Haverkamp and Parlange, 1986; Wu et al., 1990; Smettem and Gregory, 1996; Zhuang et al., 2001). To date, no attempts have been made to find such relationships for soil-less media.

**FIGURE 3.1**

Particle-size distribution for (A) tuff and (B) perlite.

Arya et al. (1999) also derived a model to compute the hydraulic conductivity function, $K(\psi)$ (Section 3.3.2), directly from the PSD. Unlike other models, the need for both measured $\theta(\psi)$ and the saturated hydraulic conductivity, K_s , is eliminated.

3.1.3 POROSITY

TP and its components are expressed as a percentage of the total volume of the medium. The combined volume of the aqueous and the gaseous phases of the medium are defined as its total pore space or TP. TP is related to the shape, size, and arrangement of media particles. The various methods of measuring BD are also relevant to the determination of porosity and its components. Some of these methods are described in Chapter 11, Analytical Methods Used With Soilless Substrates.

In many cases, BD is inversely related to TP (Bugbee and Frink, 1983; Bunt, 1988). However, the medium BD cannot accurately determine TP if components that have closed pores, such as

perlite or pumice, are used (Bunt, 1988; Bures et al., 1997). Not only mineral substrates contain inaccessible pores. Evans et al. (1996) sampled different coir batches. Contrary to their expectations, the samples exhibited lower BD with lower TP. This, of course, can only happen if a significant fraction of the pores was not saturated with water during the saturation stage of the measurement, suggesting that those pores were probably of a strong hydrophobic nature. Unlike the case of closed pores, unsaturation of organic substrates may be regarded as an inherent drawback of the testing method, as can be expected during the practical use of such substrates, their hydrophobicity will be lost and all nonaccessible pores will be frequently filled up with water.

The volumetric amount of water, θ , which saturates a given volume of a substrate, is defined as its effective pore space. The difference between total pore space and effective pore space constitutes the volume of closed pores that are not accessible to water. In horticulture, the AFP is defined as the volumetric percentage of the medium filled with air at the end of free (gravitational) drainage. Most media and mixes have an AFP of 10%–30%. Given that perched water table rests at the bottom of the growing containers after the end of free drainage, optimal AFPs may vary greatly according to the size of the container and the irrigation frequency. A wider discussion of this topic is given in Section 3.2.3. For the rooting of cuttings under intermittent mist, AFPs of >20% are essential. A somewhat lower AFP is required for bedding plants grown in shallow trays or plugs. On the other hand, an AFP as low as 10% may suffice for deep containers with slow growing, infrequently irrigated tree saplings. For all types of containers and media, it is important to consider the tendency of most root systems to grow gravitropically and to form a dense layer of roots at the bottom of the container. This may be the reason for less-than-optimal performance of plants in media that are otherwise considered adequate in terms of AFP (Raviv et al., 2001).

3.1.4 PORE DISTRIBUTION

Porous medium structure consists of a 3D network of pores. Large pores play an important role in allowing roots, gas, and water to penetrate into the medium. The higher the large pore (macropore) density, the more the soil can be exploitable by plant roots (Scott et al., 1988). Similarly, the more continuous the macropores are, the more freely gases can interchange with the atmosphere. Continuous macropores also have a direct effect on water infiltration and solute transport in soil. Although existing pores constrain the penetration of roots, they provide favorable conditions for root growth. Several studies have shown that the presence of continuous macropores significantly benefits root growth (Bennie, 1991). One of the most important factors influencing soil fertility, besides water and nutrient content, is soil aeration (Glinski and Stepniewski, 1985; Hillel, 1998). Large pores are the paths available for gas exchange between the growing medium and the atmosphere. Intuitively, large and continuous pores facilitate water transport. It is now well known that the size and connectivity of soil pores play a major role in the flow characteristics of water and the transport of solutes through soil (Ma and Selim, 1997). Jury and Flühler (1992) stated that fluid transport through well-defined structural voids is not predictable unless the distributions of the voids, aperture sizes and shapes, depths of penetration, and interconnectivity are known.

The capacity of a medium to store water and air, as well its ability to provide them to the plant (via its hydraulic conductivity and rate of gas exchange), is determined by its TP and pore-size distribution, tortuosity, and connectivity (Allaire et al., 1996). Tortuosity is one of the most meaningful 3D parameters of pore structure. It is defined as the ratio of the effective average path in the

porous medium to the shortest distance measured along the direction of the pore (Jury et al., 1991). Tortuosity is a dimensionless factor always greater than one, which expresses the degree of complexity of the sinuous pore path. Tortuosity can easily be related to the conductivity of a porous medium since it provides an indication of increased resistance to flow due to the pore system's greater path length (Dullien, 1992). Connectivity is a measure of the number of independent paths between two points within the pore space (Dullien, 1992). In other words, connectivity is the number of nonredundant loops enclosed by a specific geometrical shape. Each macropore network has a connectivity, which is a positive integer equal to the number of different closed circuits between two points in the network. If there is only one open circuit, the connectivity is equal to 0; the connectivity is 1 if the circuit is closed.

For growing media, classical concepts and values related to physical properties such as AFP, BD, available water, hydraulic conductivity, gas diffusivity, and field capacity need to be adapted to reflect distinct differences in their composition, structure, and stability compared with mineral soils (Caron et al., 2015). An important problem associated with the characterization of soil/soilless-growing media pores is the lack of standard terminology related to their classification into distinct size ranges. Several researchers have identified the need for a standard classification scheme, and suggestions for such a classification have been made. Typical examples are the proposed index for soil pore-size distribution by Cary and Hayden (1973) and the suggested classification of micro-, meso-, and macroporosity by Luxmoore (1981). Although the pore-size distribution of many porous materials could be measured by different techniques, for example, water desorption and mercury intrusion methods (Danielson and Sutherland, 1986), none of them have been applied to soilless container media.

It is recognized by soil physicists that the water RC (Section 3.2.3) is essentially the pore-size distribution curve. The expression relating pore size to the equivalent suction of water in porous media is the capillarity equation (Eq. 3.10). Pore-size distributions have been used to develop expressions for water RCs (Section 3.2.3) by Arya and Paris (1981), Haverkamp and Parlange (1986), Kosugi (1994), Assouline et al. (1998), and Or and Tuller (1999).

3.1.5 WETTABILITY

The theory and experimental results that are described in Sections 3.2–3.6 represent the era when soil–water repellency (SWR) was rarely considered in soil physics theory in general and for soilless substrates in particular. Whereas evidence for the existence of SWR has emerged, particularly in field studies where the associated generation of fingered-shaped flow in the root zone have been observed (e.g., Dekker et al., 2005; Diehl, 2013, and references therein), the effect of nonzero contact angle (CA) between soil particles and a water drop (Fig. 3.2) on soil hydraulic properties and flow and its modeling have lagged behind. This is valid for soilless substrates as well, in spite of the fact that water repellency of organic substrates was within the scope of knowledge and mentioned in the relevant literature. The fact that the research on water repellent soilless media have gained momentum in the last decade or so led us to update the chapter on physical characteristics of soilless media with the theory and current knowledge on wettability in general and that its relation to soilless media in particular. Section 3.1.5 presents the theory on wettability, whereas Section 3.7 describes methods to determine water repellency and its effect on physical properties and flow.

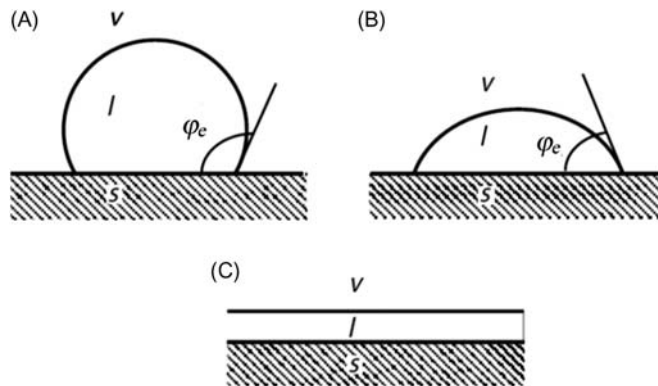


FIGURE 3.2

Different wetting situations of a small droplet in contact with a solid surface: (A) and (B) correspond to partial wetting ($\varphi_e > 0$ degree) and (C) corresponds to complete wetting ($\varphi_e = 0$ degree).

Wetting involves the interaction of a liquid with a solid. It can be the spreading of a liquid over a surface, the penetration of a liquid into a porous medium, or the displacement of one liquid by another. Wetting refers to the macroscopic manifestations of molecular interaction between liquids and solids in direct contact at the interface between them. Such manifestations include the formation of a CA at the solid/liquid/fluid interline, the spreading of a liquid over a solid surface (displacing the fluid initially in contact with that surface), and the wicking of a liquid into (and displacement of a fluid from) a porous medium. The interactions of interest in wetting in the current review exclude the dissolution or swelling of the solid by the liquid or chemical reactions between the phases that change the system composition.

The quantitative measure of the wetting of a solid by a liquid is often taken as the CA φ and defined at a point along the solid/liquid/gas interline as the angle between the normal to the liquid/gas and liquid/solid interfaces at that point. This is often equivalent to the angle observable in profile between the tangents to the intersecting liquid/gas and liquid/solid interfaces, drawn in the liquid (Fig. 3.2). Systems in which the liquid exhibits a CA of 0 degree against the solid are those in which the liquid “wets out” the solid (Fig. 3.2C), partially wets the solid, $\varphi_e > 90$ degrees, and 0 degree $< \varphi_e < 90$ degrees (Fig. 3.2A and B). The latter two cases are noted as “water repellent” and “subcritical repellent” surfaces, respectively.

Measurements of CAs in real systems reveal significant differences, depending on whether the liquid is advancing or receding across the solid surface. When a drop of liquid contacts a surface, one of the two phenomena generally occurs: (1) the liquid wets the surface; a circular three-phase contact line is formed that continues to increase in radius and liquid spreads over the entire surface and (2) the droplet contacts the surface and the three-phase contact line advances to a certain radius and stops, creating a sessile droplet in the shape of a section of a sphere with a CA that depends on the chemical compositions of the liquid and the surface and the topography of the surface. Which of these phenomena occurs and the value of the CA that forms in the second case are described in

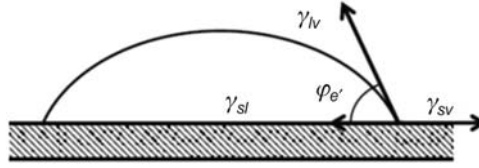


FIGURE 3.3

Young's surface tension force balance giving the equilibrium contact angle.

quantitative terms by Young's equation which relates the CA to the three interfacial free energies (Fig. 3.3):

$$\cos \varphi = \frac{\gamma_{sv} - \gamma_{sl}}{\gamma_{lv}} \quad (3.1)$$

where γ_{sv} , γ_{sl} , and γ_{lv} are the surface-free energy per unit area of solid/vapor, solid/liquid, and liquid/vapor interfaces, respectively. The units of the γ_s in Eq. (3.1) are erg/cm² (energy per unit area).

From Eq. (3.1) it can be seen that the behavior of a liquid on a solid surface is controlled by competition between the interfacial energies δ . Zisman (1964) stated that the surface of a solid is wetted by a liquid when the surface energy of the liquid is smaller than the so-called critical surface energy δ_c of the solid. This critical surface energy δ_c is characterized by a CA of 0 degree and can be determined by using repeatedly liquids of different surface energy.

There is ample experimental evidence that when a sessile drop is first placed on a solid surface, the apparent CA is larger than the equilibrium CA, sometimes by several degrees (Chen et al., 1991). In a similar way, if liquid is removed from the droplet, the apparent CA is smaller than the equilibrium CA (Chibowski, 2007). Although hysteresis in CA has generally been traced to solid surface roughness and/or chemical heterogeneity, one can measure multiple values of CAs, on surfaces that are smooth and homogeneous. This intrinsic CA hysteresis (Schwarz, 1980), that is, hysteresis not associated with the presence of rough or heterogeneous surfaces, can be explained by the presence and thickness of an adsorbed film in the immediate vicinity of the triple-phase contact line (Butt et al., 2007).

Advancing CAs are larger than receding angles and thus correspond to the interline locating itself on the less wettable parts of the surface. Both advancing and receding CAs are also found to depend on the rate of interline movement, as discussed in detail in de Gennes et al. (2004). Among the various possibilities, the most generally useful CA for describing wetting behavior is the static (zero-velocity) advancing angle, and it is this angle that will be implied when simply referring to "the CA."

According to Young's equation, the CA φ_e is a unique function of the interfacial tensions γ_{sv} , γ_{sl} , and γ_{lv} . However, real surfaces can be rough or heterogeneous to some extent, anisotropic, non-inert, and deformable in contact with liquids. In most practical situations, a given liquid usually exhibits a range of CAs on a given solid surface. Each CA gives rise to a mechanically stable liquid meniscus. When the liquid is advanced over a previously unwetted surface, a maximum CA at the three-phase contact line is termed an advancing angle (φ_a). A minimum CA is measured at the contact line when the liquid is retracted over a previously wetted surface (e.g., when

the contact area shrinks). In this case, it is termed a receding angle (φ_r). CA hysteresis is defined as the difference between the (maximum) advancing and (minimum) receding angle.

It is known that the roughness of a solid surface has an important effect on the spreading of a liquid on that surface. It is difficult to determine the intrinsic CA of such materials unless the surfaces are clean and smooth. A simple relation between surface roughness and the value of the CA was derived by [Wenzel \(1936\)](#). He suggested that when a droplet contacts a rough surface and forms a sessile droplet with a smooth liquid–vapor interface, Young’s equation (Eq. 3.1) should be modified by multiplying the numerator of the right side of Eq. (3.1) by a roughness factor, r , that is, the actual contact area divided by the projected area of surface that the droplet contacts:

$$\cos \varphi_{\text{rough}} = r \cdot \cos \varphi_{\text{smooth}} \quad (3.2)$$

However, [Wenzel \(1936\)](#) assumed that the value of the CA was a constant independent of the motion of the contact line (defined as the line where the liquid–air and solid surfaces meet), whereas it is well known that for an advancing contact line, the value of the CA is greater than that for a receding contact line (hysteresis). Following [Wenzel \(1936\)](#) where the water droplets were assumed to be in full contact with the rugged surface (referred as the wetted contact), [Cassie and Baxter \(1944\)](#) addressed the issue of composite surfaces where water droplets are in contact with peaks of the rugged surface as well as the “air pockets” trapped between surface grooves and proposed to predict the CA of a binary composite surface containing two components with CAs of φ_1 and φ_2 and area fractions of f_1 and f_2 , where $f_1 + f_2 = 1$.

It is widely accepted that soils/soilless-growing-medium render subcritical ($0 \text{ degree} < \varphi < 90$ degrees) and critical ($\varphi \geq 90$ degrees) repellent by hydrophobic organic compounds that coat their mineral surfaces ([Bisdorn et al., 1993](#); [Doerr et al., 2000](#)) as well as by organic interstitial material ([Franco et al., 2000](#)). Note that in horticulture, the influence of organic matter is even more important due to the widespread use of organic-based substrates (peat, coir, bark, etc.) that are inherently water repellent when dry. The surface tension of inorganic soil particles generally exceeds the surface tension of water. In this case, a continuous water film on the dry particle surfaces is rapidly formed, which allows the complete wetting of unsaturated soils. However, soils with organic materials often do not wet readily due to the small surface tension of organic compounds ([Zisman, 1964](#)). Numerous investigations to identify chemical compounds causing water repellency have been carried out ([Ma’shum et al., 1988](#); [Horne and McIntosh, 2000](#); [Doerr et al., 2005](#)). Many of these investigations suggest that the structural composition and the arrangement of molecules and functional groups rather than the amount of certain organic substances affect water repellency ([Doerr et al., 2005](#)). The factors, however, that establish the structural composition and arrangement of molecules on the outer side of the organic coatings, and which lead to water repellency, are still poorly understood ([Diehl and Schaumann, 2007](#)). Generally, repellent soils become wettable if they stay in contact with water long enough ([Doerr et al., 2000](#)). Thus, it can be assumed that molecules at organic matter/water interfaces undergo conformational or structural rearrangement. Differences of wetting kinetics between wettable and water repellent soil/soilless medium samples may give information about processes, which break down the repellent character. They may, therefore, also allow conclusions about processes inducing water repellency in soils.

A detailed description on assessment of growing media wettability, and its effect on the hydraulic characteristics and flow, is provided in section infiltration and consecutive flow, and its wetting by using wetting agents appears in [Section 3.7](#).

3.2 WATER CONTENT AND WATER POTENTIAL IN SOILLESS MEDIA

3.2.1 WATER CONTENT

Water content or wetness of a porous medium is the volume or mass of water V_w occupying space within the pores (Gardner, 1986). Water content is typically expressed in one of the three ways: mass basis, volume or depth basis, and percent or degree saturation.

Mass wetness (sometimes referred to as gravimetric water content) is determined by extracting a soil/soilless-growing-medium sample, oven-drying (generally 105°C for 24 hours), and determining the amount of water lost through the drying process. Oven-drying is necessary to remove hygroscopic water adhering to particles that cannot be removed by air drying. This is sometimes a significant amount, depending on small particles of the growing medium and specific surface area; however, not all hygroscopic water can be removed. Mass wetness, ω , is described by

$$\omega = \frac{V_w \rho_w}{V_s \rho_s} \quad (3.3)$$

where V_s and V_w are the volume fractions of the solid and liquid phases, ρ_s is the density of the solid phase, and ρ_w and ρ_b are the wet and dry bulk densities (for analytical processes of bulk densities see Chapter 7: Inorganic and Synthetic Organic Components of Soilless Culture and Potting Mixtures).

Volumetric water content is generally more useful for field and laboratory studies, because it is the form in which soil/growing-medium–water content (as a fractional basis, e.g., 0.34, 0.48, etc.), and is reported in the results from gamma attenuation, neutron probe, and time domain reflectometry (TDR) water content measuring devices. Like mass wetness, it is generally reported as a percentage but is compared to total volume rather than the volume of solids present. Because both the V_w and V_t units of measure are cm^3 , they cancel in the equation, leaving θ dimensionless. Volumetric water content is given by

$$\theta = \frac{V_w}{V_t} \quad (3.4)$$

where V_t is the total soil/growing-medium volume. Volumetric water content may also be obtained by its relation to mass wetness and BD, as follows:

$$\theta = \omega \frac{\rho_b}{\rho_w} \quad (3.5)$$

where ρ_w is the density of water at measured temperature.

Equivalent depth of water is a measure of the ratio of depth of water per unit depth of porous media, described by volumetric water content:

$$d_w = \theta d_t \quad (3.6)$$

where d_w is the equivalent depth of soil/growing-medium–water if it were extracted and ponded over the surface (cm), and d_t is the total depth of soil/growing-medium under consideration (cm). Depth of water is a very important concept in agricultural and irrigation practices.

Degree of saturation, S , expresses the volume of water in relation to the volume of pores. Therefore, S can be as high as 100% in porous medium that remains constantly moist. In most

soil/growing medium, S will not reach 100% because of air entrapped in dead end pores. The degree of saturation is expressed by

$$S = \frac{V_w}{V_w + V_a} \quad (3.7)$$

Effective saturation, S_e , or reduced water content, is defined by

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad (3.8)$$

where θ_s is the volumetric water content at saturation and θ_r is the residual water content. The residual water content is somewhat arbitrarily defined as the water content at which the corresponding hydraulic conductivity is appreciably zero, but very often it is used as an empirical constant when fitting hydraulic functions. When $\theta_r = 0$, S_e approaches S (Eq. 3.8). Note that S_e varies between 0 and 1.

Water content measurement techniques are often classified as either direct or indirect. Direct methods involve some form of removal or separation of water from the soil/soilless-growing-medium matrix with a direct measurement of the amount of water removed. Indirect methods measure some physical or chemical property of a soil/soilless-growing-medium that are related to its water content such as the dielectric constant, electrical conductivity, heat capacity, and magnetic susceptibility. In contrast to the direct methods, the indirect methods are less destructive or nondestructive, that is, the water content of the sample is not necessarily altered during measurement. Thus, these methods allow continued, real-time measurement of water content, which enable to increase water availability to the plant and more efficient irrigation.

The most known and simple direct and destructive method to measure the weight-based moisture content in soil science is the thermogravimetry method. The sample is weighed at its initial wetness and then dried to remove interparticle absorbed water in an oven at 105°C until the soil mass becomes stable; this usually requires 24–48 hours or more, depending on the sample size, wetness, and soil characteristics (texture, aggregation, etc.). The difference between the wet and dry weights is the mass of water held in the original soil sample (Eq. 3.3). The gravimetric method is considered the standard against which many indirect techniques are calibrated. The primary advantages of gravimetry are the direct and relatively inexpensive processing of samples. The shortcomings of this method are its labor- and time-intensive nature, the time delay required for drying (although this may be shortened by use of a microwave rather than a conventional oven, the methodology has not yet been standardized), and the fact that the method is destructive, thereby prohibiting repetitive measurements within the same soil volume. These shortcomings are intensified in soilless-growing media where changes in moisture content are fast compared to field soils and its volume within the container is limited so that intensive sampling may use significant parts of the total medium volume. The gravimetric method is rarely practiced in soilless culture to monitor moisture content variation.

Many indirect methods to measure moisture content are used in soil science. A detailed description can be found in [Gardner \(1986\)](#). The indirect method that has been widely used for soilless culture so far is the TDR method. This method is based on the measurement of reflected electromagnetic signals traveling in transmission cables and waveguides embedded in the growing medium to measure the apparent dielectric constant of the growing medium surrounding the

waveguide. The propagation velocity of an electromagnetic wave along a transmission line (waveguide) of length embedded in the growing medium is determined from the time response of the system to a pulse generated by the TDR cable tester. The medium bulk dielectric constant is governed by the dielectric constant of liquid water ($\varepsilon_w \sim 80$), as the dielectric constants of other medium constituents are much smaller, for example, growing medium minerals ($\varepsilon_s \sim 3-5$) and air ($\varepsilon_a = 1$). This large disparity of the dielectric constants makes the method relatively insensitive to medium composition and texture (other than organic matter and some clays), and thus, a good method for liquid water measurement.

TDR calibration establishes the relationship between the growing medium bulk dielectric constant, ε_b , and θ . The empirical relationship for mineral soils as proposed by [Topp et al. \(1980\)](#)

$$\theta = -5.3 \times 10^{-2} + 2.92 \times 10^{-2} \varepsilon_b - 5.5 \times 10^{-4} \varepsilon_b^2 + 4.3 \times 10^{-6} \varepsilon_b^3 \quad (3.9)$$

provides adequate description for many soils and for the water content range $\theta < 0.5$ (which covers the entire range of interest in most mineral soils), with an estimation error of about 0.013 for θ . However, Eq. (3.9) fails to adequately describe the ε_b – θ relationship for water contents exceeding 0.5, and for organic rich soils, mainly because Topp's calibration was based on experimental results for mineral soils and concentrated in the range of $\theta < 0.5$.

Numerous studies have been conducted during the past 20 years, focusing on TDR and capacitive methods for water content determination in growing media ([Pépin et al., 1991](#); [Paquet et al., 1993](#); [Anisko et al., 1994](#); [Kipp and Kaarsemaker, 1995](#); [Lambiny et al., 1996](#); [da Silva et al., 1998](#); [Morel and Michel, 2004](#)). [da Silva et al. \(1998\)](#) calibrated the TDR for different growing media ([Fig. 3.4](#)) This opened the way to the complete in situ description of water storage and flow characteristics using TDR ([Paquet et al., 1993](#); [Caron et al., 2002](#); [Nemati et al., 2002](#)).

3.2.2 CAPILLARITY, WATER POTENTIAL, AND ITS COMPONENTS

Water that has entered a porous medium but has not drained deep out of the sample bottom will be retained within pores or on the surface of individual medium particles. The particles are typically surrounded by thin water films that are bound to solid surfaces within the medium by the molecular forces of adhesion and cohesion. Because of this, a simple measurement of water content is not sufficient to enumerate the complete status of water in medium. An example of this may be observed in two different media that have been treated in the same manner: though the same amount of water has been applied to each, they will have different water content and varying abilities to contain water due to individual physical and chemical properties. As a result, defining water content alone will not give any indication of the osmotic, or other water potentials, in the medium and their availability to the plant roots.

While the quantity of water present in the growing medium is very important (affecting such processes as diffusion, gas exchange with the atmosphere, soil temperature), the potential or affinity with which water is retained within the medium matrix is perhaps more important. This potential may be defined as the amount of work done or potential energy stored, per unit volume, in moving that mass, m , from the reference state (typically chosen as pure free water). In this manner, one may think of matric potential as potential energy per unit volume, E (J/m^3). Energy is work with units N (Newton) per distance (N m). Consequently, $\text{J}/\text{m}^3 = \text{N}/\text{m}^2$ or N/m^2 , which is also expressed as Pascal (Pa). A Pascal is a force per unit area or pressure, which explains the use of

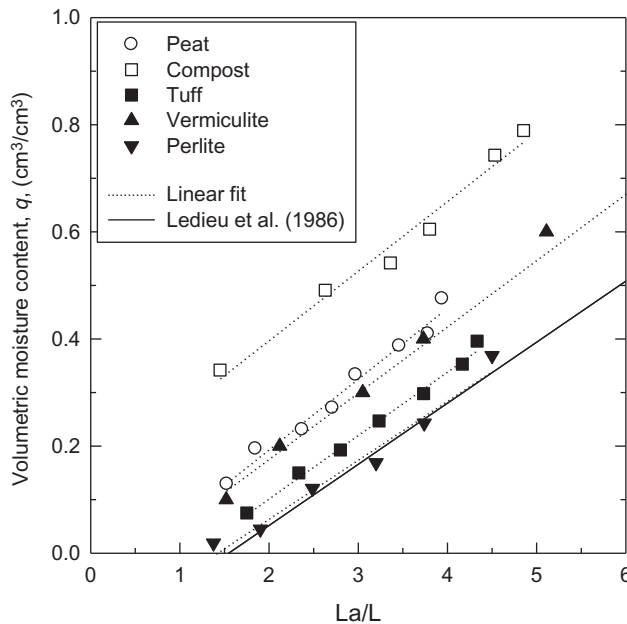


FIGURE 3.4

TDR calibration curves: measured data and corresponding linear regression for the different soil substitutes. The solid line represents Ledieu's equation (Ledieu et al., 1986). TDR, Time domain reflectometry.

Modified from da Silva, F.F., Wallach, R., Polak, A., Chen, Y., 1998. Measuring water content of soil substitutes with time domain reflectometry (TDR). *J. Am. Soc. Hort. Sci.* 123, 734–737.

the term pressure potential and is the reason why soil physicists refer to matric potential as soil pressure or if it is divided by the BD as pressure head.

Capillarity deals with both the macroscopic and statistical behavior of interfaces, rather than their molecular structure. This phenomenon is associated with water retention in porous media. Surface tension, normally expressed by the symbol γ , occurs at the molecular level and involves two types of molecular forces: adhesive forces, which are the attractive forces of molecules of dissimilar substances (the force varies depending on the substance), and cohesive forces, which are the attractions between molecules in similar substances. Cohesive forces decrease rapidly with distance and are strongest in the order solids, liquids, gases. Being work per unit area, the units of γ are dynes/cm, J/m², or N/m.

The relationship between surface tension and capillarity is well illustrated by the water rises in a cylindrical capillary tube (Fig. 3.6). The meniscus at the top of the water column in the tube is spherical in shape and concave upward. By letting r equal the tube radius, the excess pressure above the meniscus compared to the pressure directly below it can be described under various assumptions by $2\gamma/r$. As the pressure on the water surface outside the capillary tube is atmospheric, the pressure in the liquid below the meniscus will be less than the atmospheric pressure above the meniscus by $2\gamma/r$ and has been widely noted as negative pressure. This will force the fluid up the

tube until the hydrostatic pressure of the fluid column within the tube equals the excess pressure of $2\gamma/r$. Because the circumference of the tube is $2\pi r$, the total force (upward) on the fluid is $2\pi r\gamma \cos \theta$. This force supports the weight of the fluid column to the height h_c . The height of capillary rise can be given as

$$h_c = \frac{2\gamma \cos \theta}{\rho g r} \quad (3.10)$$

The radius of curvature, R , of the capillary meniscus in Fig. 3.6 can be determined by $R = r / \cos \theta$. It should also be noted that for a CA of zero, $r = R$.

Eq. (3.10) can be used to calculate the height of water rise in a specific porous medium for which the largest effective pore size is known and also to calculate the diameter of the largest effective pore (assuming one knows h_c). These equations are exact for capillary tubes; however, because soils/growing media do not behave as a single capillary such as a glass tube, these equations are at best approximations of water behavior in these media.

The pressure potential above the concave portion of the meniscus in the tube and at the bottom of the capillary is zero (Fig. 3.5). The pressure potential of the water in the capillary tube, ψ_p , decreases with height to offset the increasing gravitational potential, ψ_g (both potential components are discussed in the following). As a result, the pressure of the water within the capillary tube is

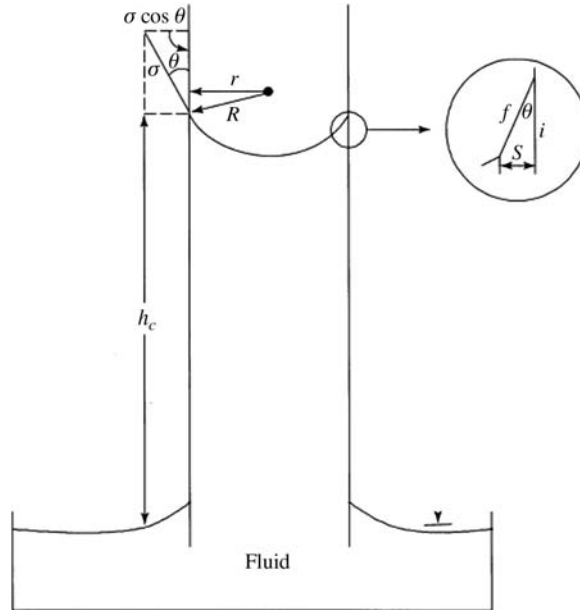


FIGURE 3.5

Capillary rise in a cylindrical tube, where θ is the contact angle, r is the tube radius, R is radius of curve of the gas–liquid interface, and h_c is the height of capillary rise. i is the area of the solid–water interface associated with the triangular volume, and f is the air–water interface.

less than atmospheric pressure, creating a pressure difference on both sides of the meniscus. Because $r = R \cos \theta$, a general relation for the pressure difference across the meniscus interface with radius of curvature R is $\Delta p = 2\sigma/R$. It should be noted here that the highest pressure is on the concave side of the meniscus. This results in the pressure difference across the interface being inversely proportional to R .

Capillary potential relates directly to the air-entry value of porous media. If water is expelled from a capillary by a positive gas pressure, that pressure is termed the bubbling pressure. Because h_c is the pressure head at the air–water interface of the meniscus, this concept can be extended to soil/growing medium pores. For example, if the pressure potential in a porous medium pore is lower than the air-entry value, both cohesive and adhesive forces cannot hold the water any longer. This will result in water draining from the pore until the pressure potential at the air–water interface is equal to that of the air-entry value. Since porous media are of many irregularly shaped capillaries and not a clean glass tube, the concepts discussed here can only be applied qualitatively to capillary water in soils/growing media. Being a threshold of desaturation, air-entry value is generally small in coarse-textured and in well-aggregated soils having large pores but tends to be larger in dense, poorly aggregated, medium-textured, or fine-textured soils. The physical properties of growing media used to grow plants in containers are significantly different to field soils in the sense that their texture is coarse with low air-entry value that enables fast free drainage of most of the water following irrigation. The fast and intensive drainage enables air to replace the draining water and prevent water clogging. Note that the coarseness of the growing media should be adapted to the height of the growing container, namely, it should be lower than the container height.

The total potential of soil/soilless-growing-medium–water has been defined as the amount of work that must be done per unit quantity of pure water in order to transport reversibly and isothermally an infinitesimal quantity of water from a pool of pure water at a specified elevation at atmospheric pressure to the medium–water at the point of consideration. This definition was retained in the Glossary of Terms of the [Soil Science Society of America \(1996\)](#). The transformation of pure water from the reference state to the medium–water state is actually broken into a series of steps. These steps are generally reversible and isothermal. Because of this, the total potential is actually a sum of each sequential step. Soil/growing-medium–water is subjected to a number of possible forces, each of which may cause its potential to differ from that of pure, free water at the reference elevation. Such force fields result from the mutual attraction between the solid matrix and water, from the presence of solutes in the medium solution, as well as from the action of external gas pressured and gravity. Accordingly, the total potential of soil/growing-medium–water can be considered as the sum of the separate contributions of these various factors.

$$\psi_t = \psi_g + \psi_m + \psi_o + \cdots \quad (3.11)$$

where ψ_t is the total potential, ψ_g is the gravitational potential, ψ_m the matric potential, and ψ_o the osmotic potential. Additional terms are theoretically possible in Eq. (3.11).

All objects on the Earth are attracted downward due to gravitational force. This force is equal to the weight of an object, which is a product of mass and gravitational acceleration. Gravitational potential is the energy of water (on a unit volume basis) that is required to move a specific amount of pure, free water from an arbitrary reference point to the medium–water elevation. If the medium–water elevation is above the reference point, z is positive; if the medium–water elevation is below the reference, z is negative, where z is the vertical distance from the reference point to the

point of interest. This means that the gravitational potential is independent of medium properties and is solely dependent on the vertical distance between the arbitrary reference point and the medium—water elevation or elevation in question. Gravitational potential has the value

$$\psi_g = \rho_l g z \quad (3.12)$$

where ψ_g has the units of J/m^3 and is assumed to be positive upward.

The primary effect of salts in the soil/growing-medium—water is that of lowering the vapor pressure. However, this does not directly affect the mass flow of water in the medium, except in the presence of a membrane barrier of some type and in the case of vapor diffusion. Plants make up part of a dynamic medium—water—plant system, and their roots are often affected by the presence of salts in the soil solution. The membranes of plant roots transmit water more readily than salts and are influenced by osmotic forces. Osmotic potential may be thought of more as a suction than a pressure. In the presence of a selective permeable membrane (as found in plant roots), the pressure or energy potential (a suction) on the solution side is less than that of pure, free water, and the water first will pass through the membrane to the solution side. One can prevent flow across the membrane and maintain static equilibrium by exerting gas pressure, P , on the medium solution side in excess of atmospheric pressure. This will result in the total potential at the surface of the solution, ψ_t , being

$$\psi_t = \frac{1}{\rho_w} P + \psi_g + \psi_o \quad (3.13)$$

where P is the air pressure (Pa) in excess of atmospheric. The term $1/\rho_w$ is included due to the small difference between the density of the solution and that of pure water and is expressed in units of J/kg . However, in practice, it is often neglected, and the simple term P is used (in which case the units are expressed in J/m^3). At the surface of the pure water, the total potential is

$$\psi_t = \psi_g + \psi_m + \psi_o = \psi_g \quad (3.14)$$

Because the solution is pure water, we can ignore the matric potential, ψ_m , and osmotic potential, ψ_o . At static equilibrium, $\psi_t = \psi_g$. Consequently, expressed in J/m^3 ,

$$P + \psi_o = 0 \quad (3.15)$$

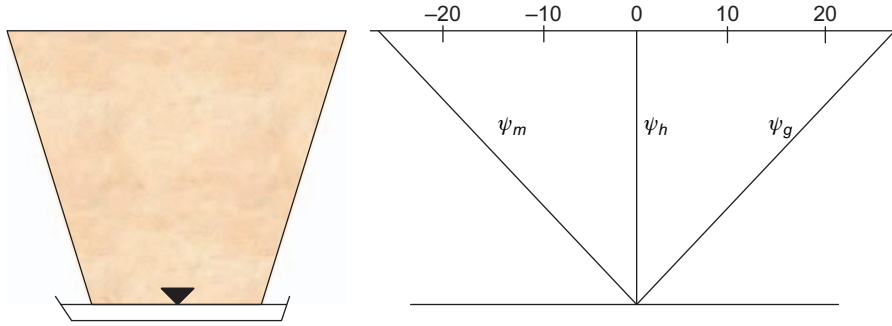
To maintain equilibrium, the excess air pressure needed must be equal to what is commonly referred to as the osmotic pressure, Π , of the solution, resulting in an expression for osmotic potential of

$$\psi_o = -\frac{\Pi}{\rho_w} \quad (3.16)$$

If we assumed the presence of soil/growing-medium and water, Eq. (3.16) (in J/kg) could be expanded to include the total potential of soil/growing-medium—water components and written as

$$\psi_t = \frac{\Pi - \psi_m}{\rho_w} + g z \quad (3.17)$$

The matric potential of a porous medium, ψ_m , results primarily from both the adsorptive and capillary forces due to medium matrix properties and is, thus, a dynamic medium property. The matric potential is often referred to as capillary or pressure potential and is usually expressed with a

**FIGURE 3.6**

The relationship between the components potential in a container, 26 cm high, filled with growing medium after free drainage (container capacity).

negative sign. By convention, pressure is generally expressed in positive terms. Assume a pot, 26 cm high, filled with soil substrate with free water at its bottom as a result of free drainage after saturation or irrigation (Fig. 3.6). The matric potential above the free water would be negative; the potential at the free water would be zero. If the pot was initially saturated with water table at the medium surface, the pressure potential below the water table varies with height and is equal ρgh , where h is the distance below the water table. It should be noted that an unsaturated medium has no pressure potential, but only a matric potential with negative units.

The matric potential of soils and other growing media are measured by tensiometers. A tensiometer consists of a porous cup, usually made of porous ceramic connected to a vacuum gauge through a water-filled tube. The porous cup is placed in close contact with the bulk medium at the depth of measurement. When the measured matric potential is lower (more negative) than the equivalent pressure inside the tensiometer cup, water moves from the tensiometer along a potential energy gradient to the medium through the saturated porous cup, thereby creating suction sensed by the gauge. Water flow into the medium continues until equilibrium is reached and the suction inside the tensiometer equals the soil matric potential. When the medium is wetted, flow may occur in the reverse direction, that is, soil–water enters the tensiometer until a new equilibrium is attained. The tensiometer equation is

$$\psi_m = \psi_{\text{gauge}} + (z_{\text{gauge}} - z_{\text{cup}}) \quad (3.18)$$

The vertical distance from the gauge plane (z_{gauge}) to the cup (z_{cup}) must be added to the matric potential measured by the gauge to obtain the matric potential at the depth of the cup, when potential is expressed per unit of weight (named as suction head). This accounts for the positive head at the depth of the ceramic cup exerted by the overlying tensiometer water column.

Electronic sensors called pressure transducers often replace the mechanical vacuum gauges. The transducers convert mechanical pressure into an electric signal which can be more easily and precisely measured. The combination of electronic sensors with data logging equipment provides continuous measurements of soil matric potential. A problem that one should be aware of is the time required for matric potential sensors to reach equilibrium. There is a significant difference between tension (and moisture content) variation in field soils and limited-volume containers that are filled

with coarse soilless materials to grow plants in greenhouses. The rate of change of these variables in a limited-volume container is much higher than in field soils. As such, tensiometers with ceramic cups of larger pores with lower air-entry values should be used in order to follow in real time the fast changes in matric potential. Owing to the relatively low matric potential values that are usually practiced in soilless culture, air does not invade the pores of the ceramic cup and their fast response to changes in the containerized growing medium is attained.

An example of matric potential variation, during a typical summer day in a growing container, is illustrated in Fig. 3.7. These tensiometric data were taken from an experiment showing the relationship between irrigation frequency and water availability (Wallach and Raviv, 2005). The containers, cylinders of 20 cm diameter and 21 cm height, were filled with RTM tuff 0–8 mm (volcanic ash, scoria, Merom Golan, Israel), a substrate that is frequently used as a growing

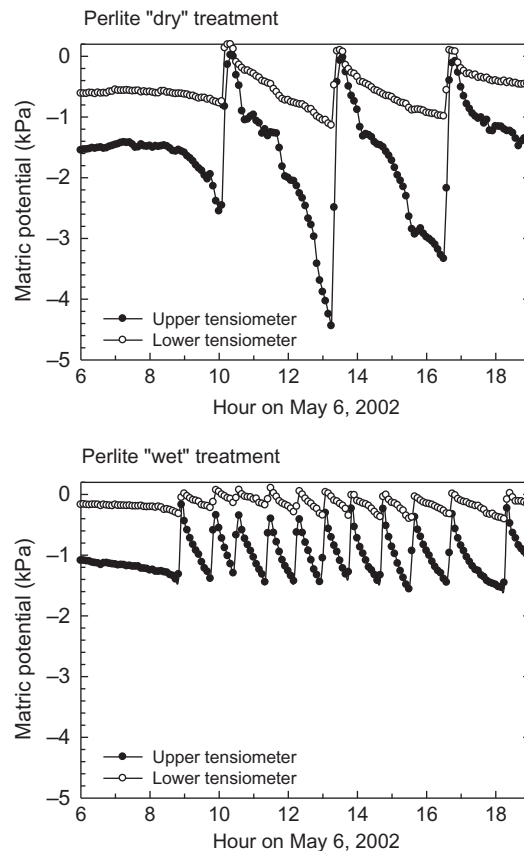


FIGURE 3.7

Matric potential variation during an arbitrary chosen day within an irrigated pot filled with perlite. The tensiometers were located 7 (lower) and 14 (upper) cm above the base of a 21-cm-high pot. The grown rose plants were irrigated at two irrigation frequencies (Wallach and Raviv, unpublished).

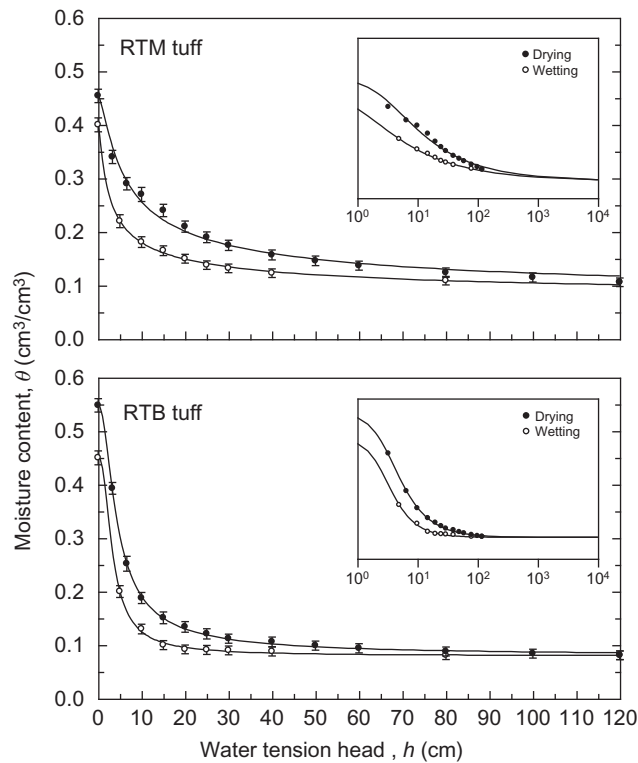
medium for ornamentals and vegetables in greenhouses and to fill trenches in problematic soils. Rose plants were grown in these containers and were drip irrigated with four l/hour drippers per container. The matric potential was measured by two high-flow electro-tensiometers that were installed horizontally in each container at 7 and 14 cm from the bottom. The tensions were measured every 10 seconds and averaged for 5-minute intervals. Two irrigation treatments are shown in Fig. 3.7: (A) wet and (B) dry. The irrigation scheduling in this experiment was automatically controlled according to predefined container-weight thresholds. The dry treatment was obtained by lower weight thresholds at which irrigations were turned on. Matric potential decreases upon the medium wetting by irrigation and increases during periods when water is depleted by root water extraction (Fig. 3.7). A similar threshold high tension was obtained in each of the two treatments owing to the weight threshold that started the irrigation. Although container weight was used here to control irrigation, tensiometers can be an alternative controller for irrigation scheduling (Karlovitch and Fonteno, 1986; Hansen and Pasian, 1999; Lebeau et al., 2003). The rate at which matric potential decreases (tension increases) depends on the water depletion rate or transpiration rate, which themselves depend on the momentary balance between the atmospheric demand and water availability in the medium. This issue will be elaborated later in the chapter.

3.2.3 WATER RETENTION CURVE AND HYSTERESIS

The water characteristic curve of porous media [abbreviated in the soil physics literature by “SWC” (soil–water characteristic)], also known as the moisture retention curve (RC), describes the functional relationship between the water content, θ , and matric potential, ψ_m , under equilibrium conditions. This curve is an important property related to the distribution of pore space (sizes, inter-connectedness), which is strongly affected by texture and structure, as well as related factors including organic matter content. The RC indicates the amount of water in the porous medium at a given matric potential and is frequently used in soilless culture to estimate the water availability to plants and for irrigation management. It is also a primary hydraulic property required for modeling water flow in soils/growing media. RCs are highly nonlinear functions and are relatively difficult to obtain accurately (Figs. 3.8 and 3.9).

When suction is applied incrementally to a saturated porous media, the first pores to be emptied are the relatively large ones that cannot retain water against the suction applied. From the capillary equation (Eq. 3.10), it can be readily predicted that a gradual increase in suction will result in the emptying of progressively smaller pores, until, at high suction values, only the very narrow pores retain water. Similarly, an increase in medium–water suction is associated with decreasing thickness of the hydration envelopes adsorbed to the particle surfaces. Increasing suction is thus associated with decreasing medium wetness (and a decrease in the osmotic potential of the aqueous phase). The amount of water remaining in the medium at equilibrium is a function of the sizes and volumes of the water-filled pores and the amount of water adsorbed to the particles; hence, it is a function of water suction.

The traditional method of determining the water RC involves establishing a series of equilibria between water in the porous medium sample and a body of water at known suctions. The medium–water system is in hydraulic contact with the body of water via a water-wetted porous plate or membrane. At each equilibrium, the volumetric water content, θ , of the medium is determined and paired with a value of the matrix pressure head, ψ , determined from the pressure in the body of

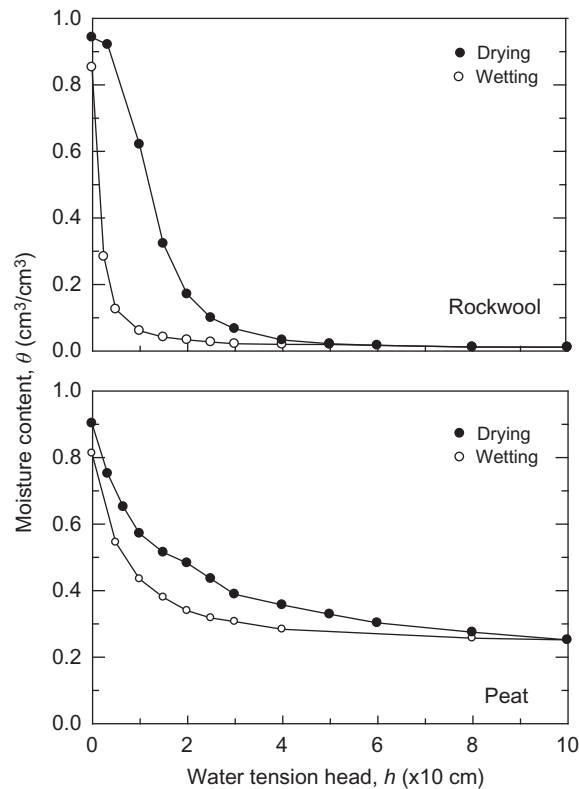
**FIGURE 3.8**

Measured and fitted water retention curves for tuff RTM and RTB.

Based on Wallach, R., da Silva, F.F., Chen, Y., 1992a. Unsaturated hydraulic characteristics of tuff (scoria) from Israel. *J. Am. Soc. Hort. Sci.* 117, 415–421; Wallach, R., da Silva, F.F., Chen, Y., 1992b. Unsaturated hydraulic characteristics of composted agricultural wastes, tuff and mixtures. *Soil Sci.* 153, 434–441.

water and the gas phase pressure in the substrate. The data pair, (θ, ψ) , forms one point on a RC. A summary of methods used in soil science to determine the RC can be found in Klute (1986), and for analyzed procedure mostly used for soilless-growing media research, see Chapter 7, Inorganic and Synthetic Organic Components of Soilless Culture and Potting Mixtures. Owing to its unique texture, the RC of a soilless-growing media is usually determined for very low suctions (up to 300 cm) using a suction funnel (Goh and Mass, 1980), suction table (Ball and Hunter, 1988), or a pressure plate system (Fonteno et al., 1981). A record of measured RCs for different soilless-growing media is given in Table 3.1.

RCs of soilless-growing media have an intrinsically different shape than field soils, a property that is needed to properly manage the air–water relationship within the containers at low suctions range. As an illustration, RCs for RTM and RTB tuff (Wallach et al., 1992a), rockwool (da Silva et al., 1995), and peat (da Silva et al., 1993a) that were measured over a 0–120 cm suction range are presented in Figs. 3.8 and 3.9, respectively. The tension head (cm) shown in this figure is the

**FIGURE 3.9**

Measured and fitted water retention curves for rockwool (A) and peat (B).

(A) Based on da Silva, F.F., Wallach, R., Chen, Y., 1995. Hydraulic properties of rockwool slabs used as substrates in horticulture. *Acta Hortic.* 401, 71–75 and (B) based on da Silva, F.F., Wallach, R., Chen, Y., 1993a. Hydraulic properties of sphagnum peat moss and Tuff (scoria) and their potential effects on water availability. *Plant Soil* 154, 119–126.

matric potential divided by ρg . A notable characteristic of soilless-growing media is their high value of saturated water content, θ_s , and low value of air-entry value, for which the media remains saturated and air does not replace water. Low air-entry values are needed to keep growing containers, usually with height ranging from 5 to 20 cm, aerated soon after their wetting (irrigation). Temporary shortage of oxygen can reduce root and shoot growth, and anaerobic conditions for only a few days will result in the death of some roots. As can be seen in Figs. 3.8 and 3.9, and in other measured RCs for container media (list appears in Table 3.1), the air-entry value for these media can be practically taken as zero. Beyond the air-entry suction, any suction increase (up to 25–40 cm) is associated with a sharp decrease in water content until almost constant moisture content is reached (for this measured suction range). The rate at which moisture content decreases per unit suction increase ($C = d\theta/d\psi$, noted as the specific water capacity) is significantly different for soilless-growing media and soils, being steeper in the former. The sharp decrease in Figs. 3.8 and

Table 3.1 Sources for Retention Curves ($\theta(\psi)$) and Unsaturated Hydraulic Conductivities ($K(\psi)$) of Representative Media

Material	Reference	Comments
Sand	Riviere (1992)	$\theta(\psi)$
Rockwool	da Silva et al. (1995)	$\theta(\psi)$ and $K(\psi)$
Rockwool	Riviere (1992)	$\theta(\psi)$
Perlite	Orozco and Marfa (1995)	$\theta(\psi)$ and $K(\psi)$ for different types available in Spain
Vermiculite	Fonteno and Nelson (1990)	$\theta(\psi)$
Vermiculite	Riviere (1992)	$\theta(\psi)$
Zeolite	Riviere (1992)	$\theta(\psi)$
Tuff-Scoria (RTM, RTB)	Wallach et al. (1992a)	$\theta(\psi)$ and $K(\psi)$
Sphagnum peat moss	da Silva et al. (1993a)	$\theta(\psi)$ and $K(\psi)$, including its mixtures with tuff
Sphagnum peat moss	Heiskanen (1995a, 1995b, 1999)	$\theta(\psi)$, including its mixture with coarse perlite
Sphagnum peat moss	Heiskanen (1999)	$\theta(\psi)$ and $K(\psi)$, including its mixtures with perlite and sand
Canadian sphagnum peat	Fonteno and Nelson (1990)	$\theta(\psi)$
Composted agricultural wastes	Wallach et al. (1992b)	$\theta(\psi)$ and $K(\psi)$, including its mixtures with tuff
Composted cow manure	Raviv (1998)	$\theta(\psi)$
Coir	Raviv et al. (2001)	$\theta(\psi)$ and $K(\psi)$
Pine bark	Fonteno and Nelson (1990)	$\theta(\psi)$
Pine bark mixed with hardwood bark	Bilderback (1985)	$\theta(\psi)$
Sawdust	Goh and Haynes (1977)	$\theta(\psi)$
UC mix	Raviv et al. (2001)	$\theta(\psi)$ and $K(\psi)$
Peat and vermiculite mix (1:1)	Fonteno (1989)	$\theta(\psi)$
Pine bark peat and sand mix (3:1:1)	Fonteno (1989)	$\theta(\psi)$

3.9 is initiated at very low suction, as the air-entry value is close to zero. The constant θ values that follow a sharp decrease shape a unique shape of the characteristic hydraulic conductivity curve for container media, as will be discussed later.

The difference between RC shapes of soilless-growing media and those of regular soils is not solely due to differences in their particle size. This is illustrated in Fig. 3.10, which shows the RCs measured for a 1–2 mm fraction of RTM tuff and similarly textured sand (Wallach et al., 1992a). In spite of the similar particle size, the tuff porosity ($0.58 \text{ cm}^3/\text{cm}^3$) is much higher than the sand porosity ($0.30 \text{ cm}^3/\text{cm}^3$). The porosity disparity is attributed to the inner microporosity of the tuff particles (Chen et al., 1980), which is not present in the sand. Sand has a typical S shape RC, which characterizes soils of different types, while the tuff RC lacks the first vertical part of the S shape

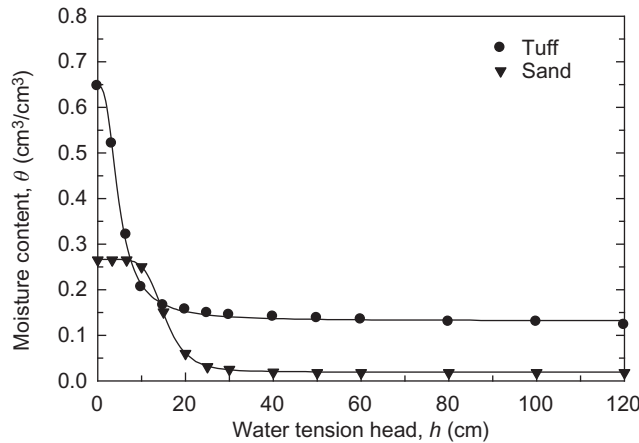


FIGURE 3.10

Measured (symbols) and fitted (lines) drying retention curves for 1–2 mm fractions of red tuff and quartz sand.

Based on Wallach, R., da Silva, F.F., Chen, Y., 1992a. Unsaturated hydraulic characteristics of tuff (scoria) from Israel. *J. Am. Soc. Hort. Sci.* 117, 415–421; Wallach, R., da Silva, F.F., Chen, Y., 1992b. Unsaturated hydraulic characteristics of composted agricultural wastes, tuff and mixtures. *Soil Sci.* 153, 434–441.

and looks more like a decreasing hyperbolic function, similar to the RCs shown in Fig. 3.10. The value of the air-entry suction of the tuff could not be determined from the measured RC. It is probably a few millimeters compared to 10 cm for the sand. The difference in air-entry value can be attributed to the particle shapes, which affect their spatial distribution in the container. Sand particles are regular and smooth compared to tuff particles, which are rough and irregular. The water content approaches 0.125 and 0.02 for tuff and sand, respectively, when suction increases beyond 30 cm. The difference in the residual water content values can be attributed to the inner microporosity of the tuff particles.

Measured values of water content and suction ($\theta-\psi$) are often fragmentary and are usually based on relatively few measurements over the wetness range of interest. For modeling and analysis purposes and for the characterization and comparison of different substrates and scenarios, it is essential to represent the RC in continuous and parametric form. A parametric expression of a RC model should contain as few parameters as possible to simplify its estimation and describe the behavior of the RC at the limits (wet and dry ends) while closely fitting the nonlinear shape of the $\theta-\psi$ data.

An effective and commonly used parametric model for relating water content to the matric potential was proposed by van Genuchten (1980) and is denoted as VG:

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \left[\frac{1}{1 + (\alpha\psi)^n} \right]^m, \quad S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \left[\frac{1}{1 + (\alpha h)^n} \right]^m \quad (3.19)$$

where S_e is the effective saturation (Eq. 3.8), $h = \psi_m/\rho g$ is the matric or tension head [$-L$].

The parameters α , n , and m in Eq. (3.19) determine the shape of the RC and are determined by curve-fitting techniques. The subscripts s and r in Eq. (3.19) refer to the saturated and residual values

of θ , respectively. The latter is defined as the water content at which the gradient $d\theta/d\psi$ becomes zero (excluding the region near θ_s , which may also present a zero gradient). Both θ_s and θ_r can be either measured or estimated along with α , n , and m . As opposed to θ_s that has a clear physical significance the meaning of θ_r and its estimation has not yet been resolved. Stephens and Rehfeldt (1985) reported improved model accuracy using a measured value of θ_r . Fonteno and Nelson (1990) determined θ_r as the water content at $\psi = 300$ cm. However, Ward et al. (1983) and van Genuchten (1978, 1980) suggested that θ_r should be viewed as a fitting-parameter rather than a soil property. van Genuchten and Nielsen (1985) considered not only θ_r but also θ_s to be empirical parameters that should be fitted to the $\theta-\psi$ data. The VG model can be used in conjunction with predictive models for unsaturated hydraulic conductivity, as will be discussed in the next section.

For small m/n ratios, parameter α in Eq. (3.19) approximately equals the inverse of the air-entry value. For large values of m/n , this parameter roughly equals the inverse of the suction at the inflection point of the RC (van Genuchten and Nielsen, 1985). Parameter n is related to the pore-size distribution of the medium and the product $m \cdot n$ determines the slope of the $\theta(\psi)$ curve at large suction values. Therefore, n may be viewed as being mostly affected by the structure of the medium (van Genuchten and Nielsen, 1985).

The VG model (Eq. 3.19) was applied to soilless-growing media by Milks et al. (1989a,b,c) who found that it fits the measured $\theta-\theta_r$ data better than the cubic polynomial. It has also been used by Fonteno (1989), Wallach et al. (1992a,b), da Silva et al. (1993a, 1995), Orozco and Marfa (1995), and Raviv et al. (2001). Examples of the VG model fit to measure RCs of different growing media VG model are shown in Fig. 3.8 for RTB and RTM tuffs and in Fig. 3.9 for rockwool and peat.

There are two initial stages at which RCs are measured: a drainage curve is obtained by establishing a series of equilibria by drainage from zero suction to higher suctions. A wetting curve is obtained by equilibrating samples wetted from a low water content or high suction to zero suction. The RC is hysteretic, that is, the water content at a given suction for a wetting medium is less than that for a draining medium (e.g., Figs. 3.8 and 3.9). The drainage curve that starts on complete saturation of the medium is called the initial drainage curve and the main wetting curve is obtained by wetting the medium from low moisture content. The measured initial drainage and main wetting curves were successfully fitted by the VG model, as shown in Figs. 3.8 and 3.9. As the medium is wetted along the main wetting curve and the suction approaches zero, the water content approaches a value that is less than the TP, ϕ , due to the presence of entrapped air (Figs. 3.8 and 3.9). This value is about $0.8\phi-0.9\phi$ and is called the natural saturation of the satiated water content (Klute, 1986). The drainage curve, starting at natural saturation, is called the main drainage curve. This curve merges asymptotically with the initial drainage curve as the suction increases. The values of θ_s [at saturation and at the end of one cycle of drying and wetting, together with the parameters α , n , and $(m = 1/n)$ for the VG retention model] are shown in Table 3.2, for both drying and wetting.

The main hysteresis mechanism is referred to as the inkbottle effect, due to its similarity with the behavior of bottles of ink with very narrow spouts. Another source of hysteretic behavior is hysteresis in the CA. The CA often exhibits a different value in advancing and receding cases (see Sections 3.1.5 and 3.7.2 for further details). A central question is when and why hysteresis must be included when describing container media processes. Naasz et al. (2008) noted that most growing media can exhibit hysteresis during drying/wetting cycles, which greatly affects their hydraulic properties. In the case of organic substrates, hydrophobicity during desiccation could be considered as one of the main factors leading to hysteretic behavior. The adjustment of the theory used in soil

Table 3.2 Curve Fitting of Drying (d) and Wetting (w) Retention Data of Tested Media Using the VG Retention Model: Fitted Parameter Values, Standard Error Coefficients (in Parenthesis), and Coefficient of Determination (R^2). θ_s is the Water Content at Saturation, and at the End of One Cycle of Drying and Wetting

Medium	θ_s	α		n		θ_f		R^2
Tuff RTM	d 0.454	d 0.346	(0.038)	1.529	(0.030)	0.063	(0.023)	0.990
	w 0.400	w 1.000	(0.086)	1.455	(0.014)			0.999
Tuff RTB	d 0.548	d 0.324	(0.063)	2.186	(0.034)	0.079	(0.020)	1.000
	w 0.450	w 0.387	(0.026)	2.534	(0.080)			0.999
Coarse sand	d 0.265	d 0.067	(0.001)	6.532	(0.060)	0.018	(0.003)	0.985
	w 0.260	w 0.080	(0.000)	6.615	(0.022)			1.000
Rockwool	d 0.935	d 0.083	(0.001)	3.725	(0.059)	0.007	(0.003)	1.000
	w 0.750	w 0.739	(0.069)	2.157	(0.081)			1.000
Sphagnum peat	d 0.901	d 0.264	(0.014)	1.390	(0.098)	0.0	(a)	0.998
	w 0.810	w 0.582	(0.124)	0.345	(0.029)			0.989

physics, and its application to container media, is not straightforward. This is due to the special conditions that exist in the limited container volume where (1) wetting and drying frequently occur while the lowest water content does not reach low values that impose a wetting of a dry porous media for which the “water entry value,” namely, the lowest water content at which the wetting front spontaneously propagates in the porous medium, is higher, (2) special conditions that exist at the container boundaries (the bottom may be open to the atmosphere or in some cases continuously saturated), and (3) the unique textural and structural properties of soilless substrates compared to field soils. Thus, additional research is needed to increase knowledge of the effects of hysteresis on static and dynamic distributions of moisture content versus suction.

The efforts to combine methods and knowledge that have been developed to characterize the physical properties of soils for mineral and organic soilless media follow [Caron et al. \(2015\)](#). They studied how mineral soil concepts can be adapted for horticultural growing media characterization and provided citations regarding the application of these concepts for additional organic substrates.

3.3 WATER MOVEMENT IN SOILLESS MEDIA

3.3.1 FLOW IN SATURATED MEDIA

Following the definition of hydraulic potential (Eq. 2.1.8), the hydraulic head, H , is defined as the potential per water specific weight, ρg . Water will therefore flow from regions of high-to-low hydraulic head. The flow equation in saturated media is Darcy’s law

$$J_W = -K_s \frac{\Delta H}{\Delta s} \quad (3.20)$$

where K_s [$L T^{-1}$] is the saturated hydraulic conductivity, H [L] is the hydraulic head, s [L] is the distance along a stream line in the flow field, $\Delta H/\Delta s$ is the hydraulic-head gradient along the stream line, and J_w is the flux density or flow per unit area opposite to the direction defined by hydraulic-head gradient. If water flow takes place in the horizontal or vertical directions, s becomes x or z , respectively. Water pressure in saturated porous media is greater than zero (although a medium is also saturated at negative pressure that is higher than the water entry value). Darcy's law is empirically based in spite of its similarity to classical relationship (e.g., Poiseuille's law). If the streamlines do not superimpose on the Cartesian coordinates (x,y,z), Darcy's law is written in a more general way

$$\vec{J}_w = -K_s \text{ grad } H \quad (3.21)$$

where "grad" is the vector gradient of H . In Cartesian coordinates, Eq. (3.21) may be expressed by individual components:

$$\begin{aligned} J_x &= -K_s \frac{\partial H}{\partial x} \\ J_y &= -K_s \frac{\partial H}{\partial y} \\ J_z &= -K_s \frac{\partial H}{\partial z} \end{aligned} \quad (3.22)$$

We assume that the growing medium is isotropic, namely, the hydraulic conductivity value does not depend on the flow direction (anisotropy). As opposed to many problems in field soils and groundwater, this assumption holds in most of the cases in soilless culture.

Deviations from Darcy's law can occur at high-flow velocities, when J_w eventually becomes nonlinear with respect to the hydraulic gradient (Fig. 3.11). The departure of the measured curve from the linear line obtained by the Darcy's law (Eq. 3.20) indicates that the latter does not apply at large flow velocities. As flow velocity increases, especially in systems with large pores, the occurrence of turbulent eddies or nonlinear laminar flow results in dissipation of effective energy by the internal mixing of the liquid. As a result, the hydraulic potential gradient becomes less effective in inducing flow. Owing to the coarse texture of soilless container media, high velocities are obtained even at relatively low hydraulic gradients. A criterion for departure from laminar flow is based on the Reynolds number, Re ,

$$Re = \frac{J_w \cdot \rho \cdot d}{\eta} \quad (3.23)$$

where d [L] is an effective pore diameter. The critical Reynolds number is much lower for porous media than for straight tubes, such as those assumed for flow in pipes. Wide ranges have been observed, but generally, for $Re < 1$, laminar conditions are expected and the Darcy law is valid. At a higher Reynolds number, the inertial forces become significant relative to viscous forces and Darcy's law cannot be used. The deviations from Darcy's law in Fig. 3.11 can therefore be related to the development of turbulent flow at $Re > 1$.

The value of the hydraulic conductivity of a saturated porous medium, K_s [$L T^{-1}$], depends on the properties of the medium and the flowing fluid

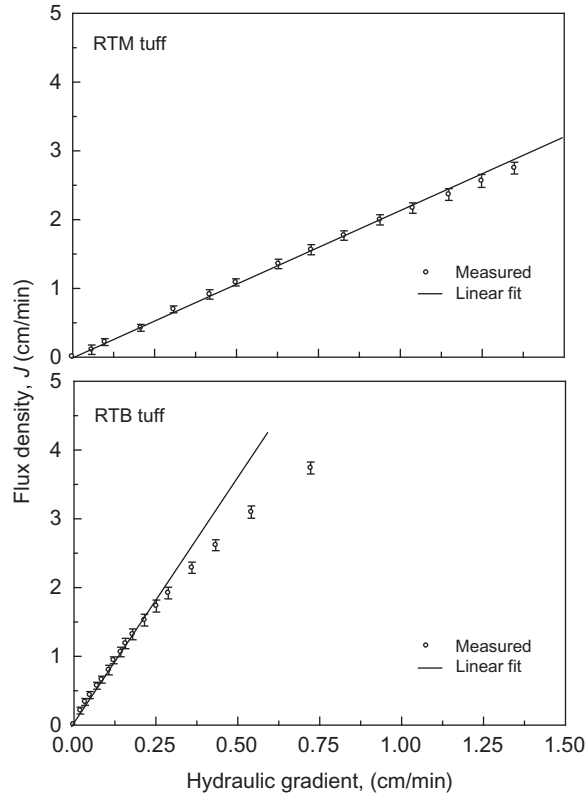


FIGURE 3.11

Deviation from Darcy's law for measured flux density as a function of hydraulic gradient for tuff RTB and RTM.

$$K_s = \frac{k \cdot \rho g}{\eta} \quad (3.24)$$

K_s can be separated into two factors: fluidity (defined as $\eta/\rho g$, where ρ is the fluid density and η is its kinematic viscosity) and intrinsic permeability, k . The intrinsic permeability of a medium is a function of pore structure and geometry. Particles of smaller sized individual grains have a larger specific surface area, increasing the drag on water molecules that flow through the medium. Thus, the result is a reduced intrinsic permeability and K_s .

The measurement of K_s is based on the direct application of the Darcy's law (Eq. 3.20) to a column of uniform cross-sectional area that is filled with the saturated medium. A hydraulic-head difference is imposed on the column and the resulting water flux is measured. A constant or variable head difference is maintained throughout the experiment (Klute and Dirksen, 1986). Note that the applied hydraulic gradient should be limited to the range within which Darcy's law is valid (Fig. 3.11) where measured flux density is a linear function of the hydraulic gradient (Wallach et al., 1992a).

3.3.2 FLOW IN AN UNSATURATED MEDIA

In most cases of irrigated soils and substrates, the pores of the growing media will not be saturated nor filled by both air and water. Assuming that pressure of the air in the pores is atmospheric, the flux density for the water–air system is

$$\vec{J}_w = -K [\text{grad}(h) + \text{grad}(z)] \quad (3.25)$$

where h [L] is the tension head (matric potential divided by ρg) and K is the unsaturated hydraulic conductivity that depends on water status expressed as a function of the tension head h or water content θ .

The value of K decreases rapidly as h increases or θ decreases. Most of the flow occurs in the liquid phase (until the medium gets very dry); consequently, as the water volume decreases, a tortuous path is necessary to move from position to position. Just as the SWC curves of θ versus h can exhibit hysteresis, so can the conductivity function.

Although the saturated hydraulic conductivity, K_s , is usually determined with little difficulty, many technical problems are involved in the measurement of the hydraulic conductivity under unsaturated conditions. It would be desirable to determine $K(\theta)$ or $K(h)$ by direct measurement. However, this is often not possible because the values of the hydraulic conductivity may vary by several orders of magnitude within the water content range of interest. In addition, the measurements are tedious and expensive, and most measurement systems cannot efficiently cover such a wide range of variation.

A method to measure K – h relationship is by the steady-state flux control method (Klute and Dirksen, 1986). According to this method, a constant flux of water, J_w , is established at the upper end of a vertical uniformly packed substrate column, while its lower end is maintained at atmospheric pressure. The flux, J_w , which should be lower than the saturated hydraulic conductivity, K_s , is applied to a previously saturated column, which drains to a condition of steady-state downward flow. Upon reaching this condition, the suction distribution along the column is expected to be relatively constant throughout its upper region. A unit hydraulic gradient should, therefore, be established in that part of the column, and under these conditions K is numerically equal to J_w .

This method was used by Wallach et al. (1992a, 1995) to measure the $K(h)$ of tuff RTM and RTB, and rockwool. The experimental setup included a 50 cm long column (10 cm i.d.), four tensiometers, a pressure transducer, a manual scanning valve system, and a peristaltic pump. Each tensiometer consisted of a high-flow ceramic porous cup, which was mounted in a horizontal position, extending about 6 cm across the column. The vertical distance between tensiometers was 10.5 cm. Starting at saturation, a controlled flow was maintained until the tensiometer readings stabilized and the volumetric outflow rate was constant. Due to the unit-gradient, the constant flux yielded K , while the suction was measured to obtain the related suction. This process was then repeated at a series of decreasing flow rates, and each time J_w and h were recorded. For each flux, measurements continued until a steady-state condition was attained. This usually took from several hours to several days. The setup for $K(h)$ measurement is shown in Figure 2 in Wallach et al. (1992a). The measured $K(h)$ for tuff RTM and rockwool is shown in Fig. 3.12A and B, respectively.

Given the curve-fitting equation for $\theta(h)$, the unsaturated hydraulic conductivity $K(h)$ can be directly calculated by means of a predictive equation based on the fitted $\theta(h)$ curve and a single measurement of the saturated hydraulic conductivity, K_s . The combination of a curve-fitting

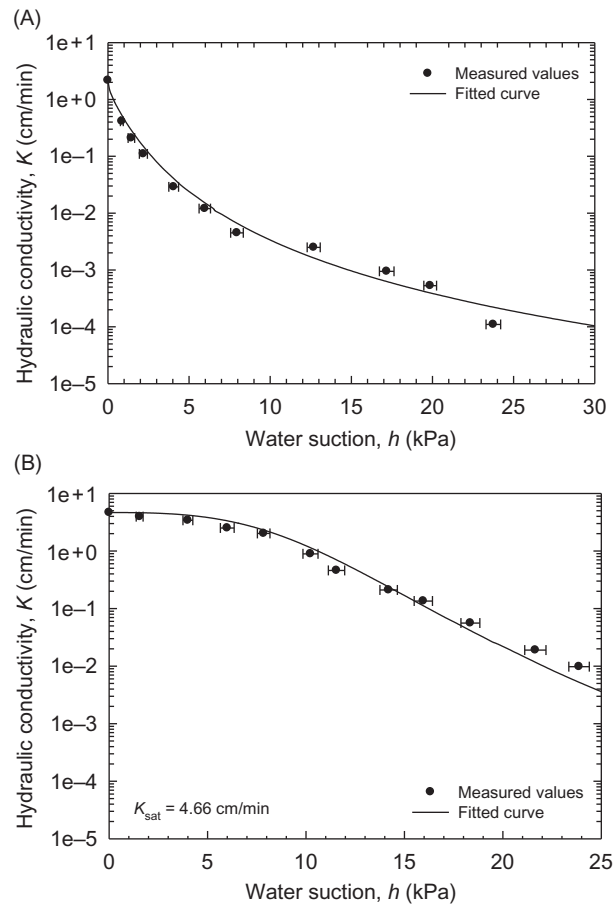


FIGURE 3.12

Measured (symbols) and calculated (lines) hydraulic conductivity, $K(h)$, for (A) RTM tuff and (B) rockwool.

(A) Data for RTM is based on Wallach, R., da Silva, F.F., Chen, Y., 1992a. Unsaturated hydraulic characteristics of tuff (scoria) from Israel. *J. Am. Soc. Hort. Sci.* 117, 415–421 and (B) for rockwool on da Silva, F.F., Wallach, R., Chen, Y., 1995. Hydraulic properties of rockwool slabs used as substrates in horticulture. *Acta Hortic.* 401, 71–75.

equation for $\theta(h)$ with a predictive model for K produces the structure of a combined model for the determination of the soil/substrate's hydraulic properties. To obtain an accurate predictive equation for the unsaturated hydraulic conductivity, an analytical expression that accurately describes the medium–water RC, over the whole relevant suction range, is required. This prerequisite is essential, and any attempt to predict the unsaturated hydraulic conductivity from retention data will fail if the assumed function cannot describe the data over the whole range of interest (van Genuchten and Nielsen, 1985). The accuracy of the predictive equation for unsaturated hydraulic conductivity also depends on the theory and assumptions on which that equation is ultimately based.

Several efforts have been made to relate pore-size distribution to the unsaturated (as well as saturated) hydraulic conductivity. Two of these general relationships are by [Mualem \(1976\)](#) and [Burdine \(1953\)](#). Both of these general relationships are based largely on interconnected capillary bundle. As mentioned previously, the pore-size distribution curve can be defined in terms of the SWC curve by the capillary rise equation. Mualem developed his relationship using an interactive capillary bundle theory with the result.

$$K_r = \frac{K}{K_s} = \frac{S_e^p \left[\int_0^{S_e} \frac{dx}{h(x)} \right]^2}{\left[\int_0^1 \frac{dx}{h(x)} \right]^2} \quad (3.26)$$

where S_e is the effective saturation (Eq. 3.8) and x is an integration variable. Introducing the [van Genuchten \(1980\)](#) SWC relationship (Eq. 3.19) into Eq. (3.26), under the assumption of $m = 1 - 1/n$ (where α , m , n are positive empirical constants in Eq. 3.19 and $0 < m < 1$), results in the relative hydraulic conductivity

$$K_r = S_e^p \left[1 - \left(1 - S_e^{1/m} \right)^m \right]^2 \quad (3.27)$$

The value $P = .5$ in Eq. (3.4) was proposed by [Mualem \(1976\)](#) and is widely used in Eq. (3.5). In terms of h , Eq. (3.5) becomes

$$K_r(h) = \frac{\left[1 - (\alpha h)^{n-1} (1 + (\alpha h)^n)^{-m} \right]^2}{[1 - (\alpha h)^n]^{m/2}} \quad (3.28)$$

$K(h)$ that was calculated by Eq. (3.28) with α , m , and n values obtained by fitting Eq. (3.19) to measured RCs for RTM tuff and rockwool ([Table 3.2](#)) was successfully compared to independently measured $K(h)$ data for the drying branch ([Fig. 3.6](#)). Predicted hydraulic conductivity curves, $K(h)$, by Eq. (3.28) for RTB and RTM tuff and rockwool are shown in [Figs. 3.13A and B](#) and [3.14](#), respectively, for both wetting and drying branches. The parameters of Eq. (3.28) for these $K(h)$ curves were obtained by fitting Eq. (3.19) to the measured wetting and drying RCs ([Table 3.2](#)) that are shown in [Fig. 3.8](#). Predicted hydraulic conductivity curves for other soilless-growing media that were calculated by Eq. (3.28) can be found in [Wallach et al. \(1992b\)](#) for composted agricultural wastes and their mixtures with tuff, in [da Silva et al. \(1993a\)](#) for sphagnum peat moss and its mixture with tuff, in [Orozco and Marfa \(1995\)](#) for perlite, in [Raviv et al. \(1999\)](#) for two types of pumice, and in [Raviv et al. \(2001\)](#) for coir and UC mix. An overall conclusion that can be made for soilless-growing media is that $K(h)$ decreases by several orders of magnitude over a narrow range of suctions. The extreme variation in $K(h)$ at a narrow range of suction that was determined by de Boodt and Verdonck (1972) as easily available water (EAW) has a tremendous effect on water dynamics in the growing container and its availability to the container-grown plants. This topic will be further discussed in the following sections.

3.3.3 RICHARDS EQUATION, BOUNDARY, AND INITIAL CONDITIONS

Darcy's law can be coupled with the conservation of mass principles to derive a continuity equation. The continuity equation is derived by taking a cube of infinitesimal dimensions, dx ,

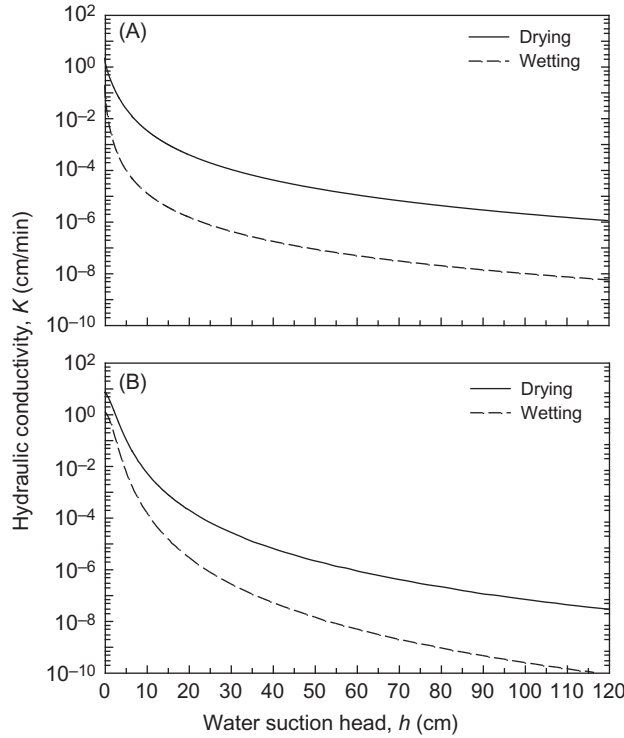


FIGURE 3.13

Predicted $K(h)$ for (A) RTM tuff (A) and (B) RTB tuff.

(A) Based on Wallach, R., da Silva, F.F., Chen, Y., 1992a. Unsaturated hydraulic characteristics of tuff (scoria) from Israel. *J. Am. Soc. Hort. Sci.* 117, 415–421.

dy, dz, and balancing the in- and out-fluxes at each one of the directions. The outcome is (Hillel, 1998)

$$\frac{\partial \theta}{\partial t} = -\nabla \cdot \vec{J}_w \quad (3.29)$$

Substituting Darcy's law (Eq. 3.25) in Eq. (3.29) leads to

$$\frac{\partial \theta}{\partial t} = -\nabla \cdot (K \nabla H); \quad H = h + z \quad (3.30)$$

or

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial x} \left(K \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(K \frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} \left(K \frac{\partial h}{\partial z} \right) + \frac{\partial K}{\partial z} \quad (3.31)$$

The last two equations, along with their various alternative formulations, are known as the Richards equation. Eq. (3.31) has two dependent variables, θ and h . The number of dependent

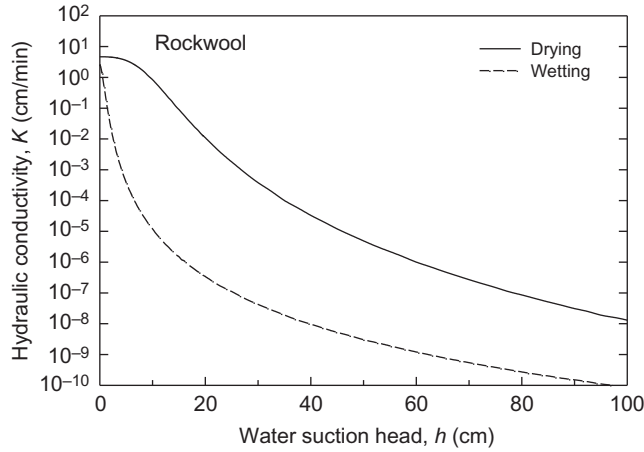


FIGURE 3.14

Predicted drying and wetting $K(h)$ for rockwool

Based on da Silva, F.F., Wallach, R., Chen, Y., 1995. Hydraulic properties of rockwool slabs used as substrates in horticulture. *Acta Hortic.* 401, 71–75.

variables may be reduced from two to one, provided a SWC relationship exists, either as $h = h(\theta)$ or $\theta = \theta(h)$. The θ -based version of Eq. (3.31) is

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial \theta}{\partial x} \right) + \frac{\partial}{\partial y} \left(D \frac{\partial \theta}{\partial y} \right) + \frac{\partial}{\partial z} \left(D \frac{\partial \theta}{\partial z} \right) + \frac{\partial K}{\partial z} \quad (3.32)$$

where $D = K \cdot \partial h / \partial \theta$ is called the hydraulic diffusivity. The h -based version of (3.31) is

$$C \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} \left(K \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(K \frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} \left(K \frac{\partial h}{\partial z} \right) + \frac{\partial K}{\partial z} \quad (3.33)$$

where $C(h) = \partial \theta / \partial h$ is the slope of the water RC and is called the specific moisture capacity. The advantage of the θ -based form is that D does not vary with θ nearly as much as K varies with h .

For a well-posed problem, not only the domain boundaries should be specified but also a description relating the dependent variables along the boundaries. For example, the conditions at the growing media surface (e.g., flooded or flux-controlled irrigation), the lower part of the root zone or the bottom of the growing container (e.g., free- or controlled-drainage), the walls of the growing container or at a symmetry cross section or line within the container, and so on. For time-dependent problems, the initial conditions within the domain should be specified. A formal specifications of boundary conditions help not only to assure completeness in a mathematical sense but also to bring attention to necessary assumption and data limitations.

Some common choices of initial cases include.

1. Constant matric potential (suction head) or water content

$$h(x, y, z, 0) = \text{Constant}$$

$$\theta(x, y, z, 0) = \text{Constant}$$

2. Constant head (static conditions)

$$H(x, y, z, 0) = \text{Constant}$$

$H(z, 0) = -z$ (in 1D problems with z the elevation) that is widely used for container capacity (end of drainage and water redistribution)

3. General matric potential or water content distributions

$$h(x, y, z, 0) = f(x, y, z)$$

$$\theta(x, y, z, 0) = f(x, y, z)$$

3.3.3.1 Boundary conditions

There are two kinds of boundary conditions: “Dirichlet” (first kind), for which the dependent variable is given; “Neumann” (second kind), for which the flux or gradient of the dependent variable is given. Common boundary conditions include

1. h or θ is specified on the boundaries (Dirichlet).
2. Vertical water flux as from irrigation at the growing media surface (Neumann).
3. Zero flux specified at the growing container vertical or inclined surfaces, along symmetry cross sections between plants (Neumann).
4. Evaporation through the soil surface (Neumann).

3.3.4 WETTING AND REDISTRIBUTION OF WATER IN SOILLESS MEDIA—CONTAINER CAPACITY

The small volume of growing containers and pots filled with growing substrates, compared to field soils, enables to analyze the average moisture content changes with time during irrigation, moisture redistribution and free drainage afterward, and water extraction by the plant roots. The container in this type of analysis is an analog to a weighing lysimeter. The typical container-weight variation prior, during, and after an irrigation event is shown in Fig. 3.15. The substrate wetting started at 1240 hours.

A close look at the container-weight variation in Fig. 3.15 indicates that it may be divided into four stages, dominated by different driving forces for moisture distribution within the container. The first stage (noted as stage I in Fig. 3.15) is the time period when water is added to the top of the container. The container weight is then markedly increasing during a peak value. The weight peak is determined by the irrigation duration, if it stops before equilibrium between the input (irrigation flux) and output (the drainage flux through the holes at the container bottom) is reached. If irrigation continues beyond this period, the peak weight sustained until irrigation terminates, as can be seen in Fig. 3.15. The second stage is characterized by a high rate of free drainage flux that leaves the container bottom. This stage starts when irrigation stops and terminates when the container-weight loss becomes small (Fig. 3.15). The drainage flux depends on the momentary distribution of the medium hydraulic conductivity, which itself depends in a nonlinear manner on the moisture content. Thus, the drainage flux is high when irrigation ends, as the moisture content is high, and then diminishes markedly, in parallel with the continuous decrease in moisture content. The duration of this stage and its intensity depend mainly on the hydraulic properties of the medium and the growing container volume and geometry. Given that the time interval between any two data points in Fig. 3.15 is 5 minutes, the duration of this stage for the case shown in this

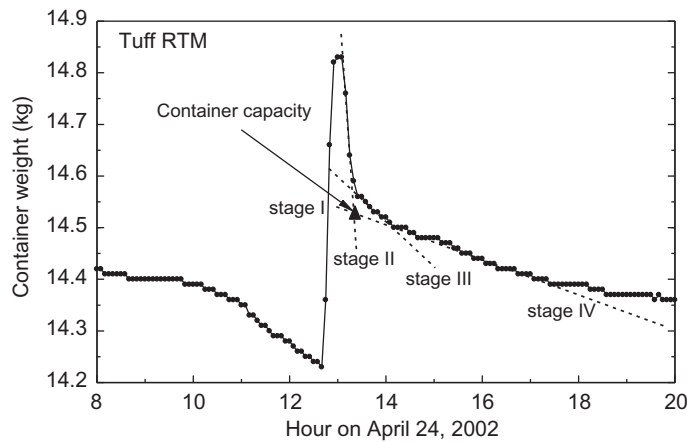
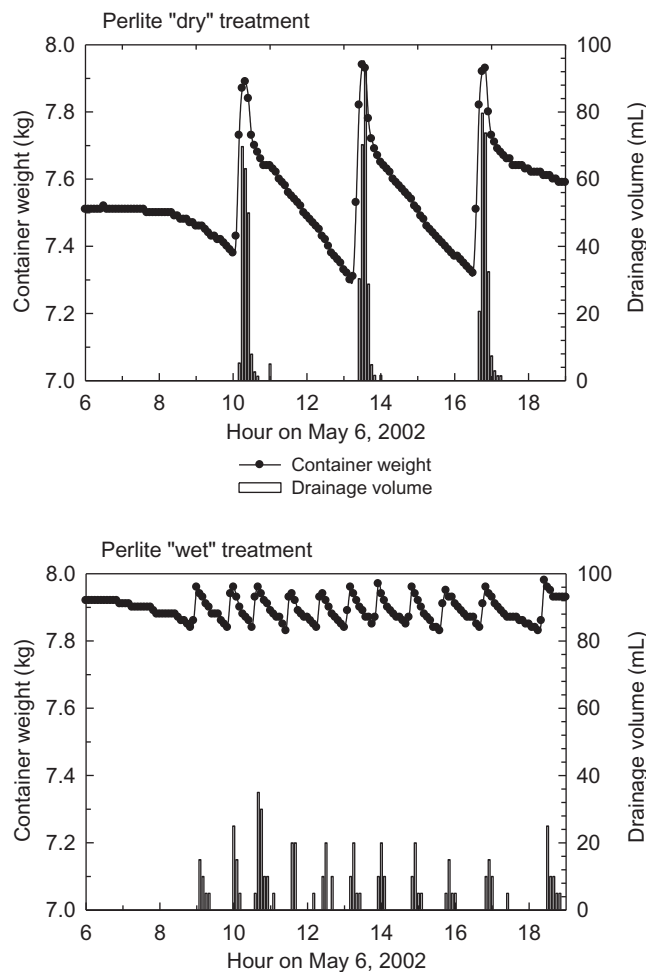


FIGURE 3.15

The four stages of container-weight variation during and after an irrigation event. The data are for a rose plant grown in a container filled with tuff RTM.

figure (tuff substrate and an 8-L container volume) was between 15 and 20 minutes. Following the sharp container-weight decrease, the weight continued to decrease, but at a much lower rate, during a few minutes afterward. This stage, designated as the third stage, is a transition period between the high drainage flux at the second stage, and the low weight decrease rate afterward owing to evapotranspiration, noted as the fourth stage. If the drainage flux from the container bottom is monitored, for example, by tipping buckets (Fig. 3.16), the determination of the initiation and termination of the third stage is obvious. If the container surface is fully covered by the plant canopy, the contribution of evaporation to the weight loss is much smaller compared to transpiration (Urban et al., 1994) and can be practically ignored for this type of an analysis. Thus, the rate of container-weight decrease during the fourth stage may be considered as almost solely related to the transpiration-driven water uptake rate. The calculation of momentary transpiration rate and its distribution along the day hours can be performed by tracking the change of container weight with time ($\Delta W/\Delta t$).

Fig. 3.16 shows the weight variation of two containers filled with perlite and the related drainage that leaves the container through holes at its bottom. These data were taken from an experiment on the relationship between irrigation frequency and water availability, as the tensiometers data shown in Fig. 3.7 (Wallach and Raviv, 2005). The two irrigation regimes, “wet” and “dry,” were predetermined by threshold lower and upper container weights. These figures indicate that the average moisture content (over time) in the “wet” treatment is much higher than in the “dry treatment,” as also indicated by the suction variation in Fig. 3.7, which potentially provide higher water availability. The cumulative drainage in Fig. 3.16 was 650 g (about 220 mL per irrigation) in the “dry” treatment and 585 (about 53 mL per irrigation) in the “wet” treatment. The upper threshold weight that controls the drainage was setup in a way to keep a certain predefined drainage to prevent salt accumulation in the container. The threshold-container-weights method used to control irrigations in Fig. 3.16 provided a higher irrigation frequency during the hours when water demand is high and less frequent during the early morning and afternoon hours, when water demand is lower.

**FIGURE 3.16**

The variation of container weight and drainage leaving the bottom of the pots during an arbitrary chosen day for an irrigated pot filled with perlite. The grown rose plants were irrigated at two irrigation frequencies. Note that these data match the tensiometers data in [Fig. 3.7](#).

The term “container capacity” is widely used in the soilless culture literature to designate the upper moisture content threshold of the available water range. This term is analogous to “field capacity,” a term used by soil physicists and irrigation scientists for field-irrigated crops. Moisture content at field capacity is usually determined as the moisture content in the root zone at the end of the free drainage following an irrigation event. As the upper threshold of available water moisture content range is dictated by the soil/growing medium type and its hydraulic properties, the lower moisture content threshold depends on the decision when to turn the irrigation on. The issue of

water availability will be discussed in detail in the next section. Different methods have been suggested for determining the container capacity. White and Mastalerz (1966) defined container capacity as the amount of water retained in a containerized medium after drainage from saturation has ceased, but before evaporation has started. A combination of mathematical functions for the water characteristic curves and container geometry was later shown to provide a more consistent description of container capacity (Bilderback and Fonteno, 1987; Fonteno, 1989). According to this approach, container capacity is the total volume of water in the container, as given by its water RC, divided by the container volume. The assumption made while determining container capacity is that the pressure distribution along the container height when drainage stops is static (Fig. 3.6).

The basic assumption in the RC-related methods to determine container capacity is that the medium drainage starts from saturation. However, owing to the high porosity of soilless media, saturation rarely occurs during irrigation by conventional on-surface methods. In addition, the RC used to determine container capacity by these methods is usually measured independently for smaller medium volumes, without roots, and independent of the container geometry. Referring to Fig. 3.15, container capacity is reached sometime during the third stage when both free drainage and root uptake take place simultaneously. The question is how can the free drainage be separated from the root uptake in order to find the moment when free drainage ceases. It is suggested herein that container capacity is reached at the intersection between the continuation of the lines fitted to the weight variation during the second and fourth stages (Fig. 3.15). The point of intersection between the two lines is signified by a solid triangle in Fig. 3.15. As container weight changes continuously with plant growth, flower cutting, fruit picking, pruning, trimming, etc., its use for container capacity determination is problematic. However, when tensiometers are used together with load cells, the container weight at container capacity can be related to suction that can be further translated to moisture content by using the independently measured RC of the specific growing medium. An implicit assumption in this procedure to determine container capacity is that moisture content is uniformly distributed along the container height during moisture redistribution (throughout the second stage and early third stage).

3.4 UPTAKE OF WATER BY PLANTS IN SOILLESS MEDIA AND WATER AVAILABILITY

3.4.1 ROOT WATER UPTAKE

Water is transported through the growing medium into the roots and plant xylem toward the plant canopy where it eventually transpires into the atmosphere. The continuous uptake of water is essential for the growth and survival of plants because they lose quite large amounts of water throughout the day, and mainly during periods of high evaporative demand. Unless water loss is replaced immediately, dehydration-related stresses may start to affect the performance of the plant. Thus, water availability to the plant roots and its relation to irrigation scheduling deserve a careful attention in general and in soilless-growing media in particular owing to the limited root volume and root density.

One key function of plant roots is their ability to link the growing substrate, where water and nutrients reside, to the organs and tissues of the plant, where these resources are used. Hence roots

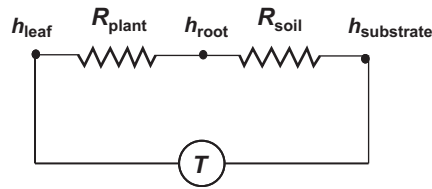
serve to connect the growing substrate environment to the atmosphere by providing a link in the pathway for water fluxes from the medium through the plant to the atmosphere. Fluxes along the growing substrate–plant–atmosphere continuum (SPAC) are regulated by above-ground plant properties, for example, the leaf stomata, which can regulate plant transpiration through interaction with the atmosphere and with other plant organs, and plant root–system properties like depth and spatial distribution, roots permeability, or hydraulic conductance. The root density is often expressed as root length density in centimeters of roots per cubic centimeters of soil. There is a significant difference among plants that are grown in soils and those grown in limited volumes (container, pots, sleeves, etc.) filled with substrates. The limited growing volume affects the spatial root distribution and length density, and other physiological and environmental parameters.

Uptake of water occurs along gradients of decreasing potential from the growing medium to the roots. However, the gradient is produced differently in slowly and rapidly transpiring plants, resulting in two uptake mechanisms. Active or osmotic uptake occurs in slowly transpiring plants where the roots behave as osmometers whereas passive uptake occurs in rapidly transpiring plants where water is pulled in through the roots, which act merely as absorbing surfaces. When the growing medium is warm and moist and transpiration is slow, as at night and on cloudy days, water in the xylem often is under positive pressure (root pressure) as indicated by the occurrence of guttation and exudation of sap from wounds. In moist growing medium, uptake at night and early in the morning is largely by the osmotic mechanism, but as daytime transpiration increases the demand for water in the leaves, uptake increasingly occurs by the passive mechanism and the osmotic mechanism becomes less important (Kramer and Boyer, 1995).

The force bringing about uptake of water by transpiring plants originates in the leaves and is transmitted to the roots or the lower end of cut stems through the sap stream in the xylem. Evaporation of water from leaf cells decreases their water potential, causing water to move from the xylem of the leaf veins. This reduces the potential in the xylem sap, and the reduction is transmitted through the cohesive water columns to the roots where the reduced water potential causes inflow from the growing media. In this situation, water can be regarded as moving through the plant in a continuous, cohesive column, pulled by the matric or capillary forces developed in the evaporating surfaces of stem and leaf cell walls.

When considering root water uptake, we accept the continuum approach as presented by van den Honert (1948), assuming that flow through the growing substrate–plant–atmosphere continuum (SPAC) is at a steady state for an unspecified time period and that water potential across the SPAC is continuous and determined by the cohesion theory. For this condition, an Ohm's law analog between water flow and electrical current is valid and the electric analog model assumes that the water flux ($\text{cm}^3/\text{cm}^2/\text{s}$) through the root zone and the root–stem–leaf–stomata path, namely transpiration rate, T , is proportional to the total head difference Δh_{total} (cm) and inversely proportional to the total resistance R_{total} (expressed in seconds) of the system. A steady-state condition implies that water does not accumulate/deplete in the different sections of the SPAC so that water flow within each section of the SPAC pathway can be determined by the ratio of water potential gradient and flow resistance within each section.

$$T = -\frac{\Delta h_{\text{total}}}{R_{\text{total}}} = -\frac{h_{\text{root}} - h_{\text{substrate}}}{R_{\text{substrate}}} = -\frac{h_{\text{leaf}} - h_{\text{root}}}{R_{\text{plant}}} = \frac{h_{\text{leaf}} - h_{\text{substrate}}}{R_{\text{substrate}} + R_{\text{plant}}} \quad (3.34)$$

**FIGURE 3.17**

Potentials and resistances in the substrate-plant system. T is the transpiration rate, h_{leaf} is the leaf water potential, h_{root} is the root water potential, $h_{\text{substrate}}$ is the substrate-water potential, R_{plant} is the plant resistance, and R_{soil} is the soil resistance.

where T (cm/s) is the transpiration rate, $h_{\text{substrate}}$, h_{root} , and h_{leaf} (cm) are pressure heads in the growing medium (substrate), at the root surface, and in the leaves, respectively, $R_{\text{substrate}}$ and R_{plant} (s) are liquid-flow resistances of the substrate and the plant (Fig. 3.17). Hence R_{plant} does not include stomatal resistance. When the transpiration demand of the atmosphere on the plant system is high or when the substrate is rather dry, $R_{\text{substrate}}$ and R_{plant} influence h_{leaf} in such a way that transpiration is reduced by closure of the stomata. Eq. (3.34) can be applied to the root system as a whole by measuring T , $h_{\text{substrate}}$, and h_{leaf} during two periods, thus obtaining two equations with two unknowns, from which $R_{\text{substrate}}$ and R_{plant} can be computed. The relative magnitude of $R_{\text{substrate}}$ and R_{plant} has been an important object of many studies, mainly in soil-grown plants. Under wet conditions, R_{soil} is small. Generally one can state, except for very dry soil, that $R_{\text{substrate}} > R_{\text{plant}}$. Most of the plant resistance is concentrated in the roots, to a lesser extent in the leaves, and a minor part in the xylem vessels. Eq. (3.34) is applied to the macroscopic flow of water across a complete rooting system (Gardner and Ehlig, 1962) but it was also used to quantify water transport across a single root in a microscopic approach, where T denotes the volumetric uptake rate per unit length of root per unit root surface area (Molz, 1981). A review of the simplifications and implications of Eq. (3.34) was presented by Philip (1966). The constant, time-independent resistances in the electrical analog theory (Eq. 3.34) rarely exist in reality. The plant system is much more complex, resembling more a series-parallel network of flow paths, each characterized by different resistances. There has been considerable discussion concerning the relative importance of soil and root resistance with respect to water uptake. Early investigators, for example, Gardner (1960) and Gardner and Ehlig (1962), concluded that resistance in the soil would exceed resistance in the roots at a soil-water potential of -100 to -200 kPa, and this view was supported by Cowan (1965) and others. Later studies examined the effect of root density soil resistance (Newman, 1969) and concluded that root densities used by previous investigators were much lower than those usually found in nature, and if normal root densities were used, soil resistance in the vicinity of roots would not be limiting until the soil-water content approached the permanent wilting percentage. Regarding the plant resistance in Eq. (3.34), it is likely to vary with transpiration rate (Steudle et al., 1988; Passioura, 1988) and water potential gradients, for example, due to reduced plant conductance by cavitation. The steady-state assumption when using Eq. (3.34) is valid at short time scales but is less likely to apply at time scales longer than a day in field soils and or than couple of hours in limited-volume growing substrates.

The validation of the steady-state resistive flow model for the SPAC (Eq. 3.34) was investigated by Li et al. (2002b) by studying the effects of soil moisture distribution on water uptake of drip-irrigated corn and simultaneously monitoring the diurnal evolution of sap flow rate in stems, of leaf water potential, and of soil moisture, during intervals between successive irrigations. The results invalidate the steady-state resistive flow model. High hydraulic capacitance of wet soil and low hydraulic conductivity of dry soil surrounding the roots depressed significantly diurnal fluctuations of water flow from bulk soil to root surface. On the contrary, sap flow responded directly to the large diurnal variation of leaf water potential. In wet soil, the relation between the diurnal courses of uptake rates and leaf water potential was linear. Water potential at the root surface remained nearly constant and uniformly distributed. The slope of the lines allowed calculating the resistance of the hydraulic path in the plant. Resistance increased in inverse relation to root length density. Soil desiccation induced a diurnal variation of water potential at the root surface, the minimum occurring in the late afternoon. The increase of root surface water potential with depth was directly linked to the soil desiccation profile. The development of a water potential gradient at the root surface implies the presence of a significant axial resistance in the root hydraulic path that explains why the desiccation of the soil upper layer induces an absolute increase of water uptake rates from the deeper wet layers.

3.4.2 MODELING ROOT WATER UPTAKE

Models for water transfer have been developed for the substrate–plant–atmosphere continuum (SPAC) in order to improve the understanding of plant behavior in the environment. The major difficulty in general and in these models in particular is in estimating effective soil–water availability for the plant, which is a fundamental limiting factor of plant transpiration. In this context, models have been developed to simulate the water budget and water movement in the soil on two different scales with appropriate models for each scale: the *microscopic scale* that considers individual roots and the associated volumes of soil, and the *macroscopic scale* that includes part or the entire root system. For overviews of root water uptake models, the reader may refer to Feddes (1981), Molz (1981), and Hopmans and Bristow (2002).

The *microscopic* analysis generally considers the convergent radial flow of water toward and into a representative individual root, taken to be a line or narrow-tube sink uniform along its length, that is, of constant and definable thickness and absorptive properties. The microscopic root system as a whole can then be described as a set of such individual roots, assumed to be regularly spaced in the soil at specified distances that may vary within the root volume in the container/soil profile. The Richards equation (Eq. 3.31), written for the microscopic scale, is solved for the distribution of soil–water pressure heads, water contents, and fluxes from the root outward. As the flow to a single root has a radial symmetry, the flow equations are written in cylindrical coordinates

$$\frac{\partial \theta}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left\{ rK(h) \frac{\partial h}{\partial r} \right\}, \quad \frac{\partial \theta}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left\{ rD(\theta) \frac{\partial \theta}{\partial r} \right\} \quad (3.35)$$

where r is the radial coordinate from the center of the root, $D(\theta) = K(h) \cdot \partial h / \partial \theta$ is the diffusivity (Eq. 3.32). The boundaries for Eq. (3.35) are at $r = r_0$ and $r = r_l$, where r_0 is the radius of the plant root, the internal radius of the equivalent cylindrical shell of soil associated with the plant root, and r_l is the external radius of the equivalent cylindrical shell of soil associated with the plant root. The

radius r_l can be half the distance between adjacent roots, or a distance from the root surface where the effect of water extraction at the root surface on the moisture content or tension head variation is approaching zero. Boundary conditions at the two radii and an initial condition for the dependent variable in the entire domain should be specified in order to obtain a specific solution for Eq. (3.35).

Gardner (1960) was among the first to use Eq. (3.35) to study the extraction of water from soil. He considered, in his analysis, uptake of water by a plant root with radius r_0 surrounded by a cylindrical shell of soil with outer radius r_1 . He have considered an approximate solution after linearizing Eq. (3.35) and assuming that $r_0 \ll r_1$ and a constant flux into a line sink. Gardner (1960) used the line sink solution to calculate water depletion patterns around individual roots. But more importantly, he also used it as a point of departure for the formulation of a simpler model, in which the depletion resulting from uptake by a single root is treated as a series of steady flows in the cylindrical shell of soil surrounding the root, with the soil–root interface at the inner edge and the water coming from the outer edge. This simple model has been used ever since for more sophisticated microscale as well as macroscale models of water uptake. In the single-root model of Gardner (1960), the root is viewed as a cylinder of infinite length having uniform water-absorbing properties. For steady-state conditions, ($\partial\theta/\partial t = 0$), in the soil shell surrounding the root with water flowing from the outer cylindrical surface at $r = r_1$ to the inner cylindrical soil–root interface at $r = r_0$, the solution of Eq. (3.35) under the assumption of constant hydraulic conductivity K gives an expression for the flux q_r at the soil–root interface

$$q_r = \frac{2\pi K}{\ln(r_1/r_0)}(h_l - h_0) \quad (3.36)$$

where q_r (cm³/cm/day) is the rate of water uptake per unit length of root, h_l (cm) is pressure head of the soil at $r = r_1$, h_0 (cm) is the pressure head at the soil–root interface. Eq. (3.36) is an analogous steady-state flux toward a well per unit length of well in groundwater hydrology. Cowan (1965) realized that the assumption of constant K used in Eq. (3.36) can be replaced by an average K , defined by

$$\bar{K} = \frac{\int_{h_0}^{h_l} K(h) dh}{h_l - h_0} \quad (3.37)$$

The integral in Eq. (3.37) can be evaluated from any of the commonly used expressions to represent the $K(h)$ relationship. The sharp decline of $K(h)$ within a small range of h values in containerized substrates (Figs. 3.13 and 3.14) raises doubts about the use of constant or average K in the microscale models.

In spite of the assumptions made in Gardner's studies, they were insightful and simulating and inspired the models that have been developed later. On the other hand, the single-root approach is not practical when a whole rooting system with complex geometries has to be considered. The flow processes in the SPAC can be highly dynamic, thereby requiring transient formulations of root water uptake. Consequently, later studies of water extraction by plants roots have considered the macroscopic approach.

On the microscopic scale, the uptake is represented by a flux across the soil–root interface. That flux is a consequence of the interaction of processes in the soil and in the plant. In the soil, flow of water toward or away from individual plant roots may be described by a nonlinear diffusion

equation, subject to appropriate initial and boundary conditions. The microscale model involves at least two characteristic lengths describing the root–soil geometry; for an individual plant root, one characteristic length is the internal radius of the plant root r_0 and the external radius of the soil associated with the plant root r_1 , and two characteristic times describing, respectively, the capillary flow of water from soil to plant roots and the ratio of supply of water in the soil and uptake by plant roots. Generally, at a certain critical time, uptake will switch from demand-driven to supply-dependent. The resulting microscopic expressions for the evolution of the average water content can be used as a basis for upscaling to the macroscopic scale.

On the *macroscopic* scale, the uptake of water by plant roots is represented by a sink term in the volumetric mass balance Eq. (3.29), representing the rate of water extraction by roots

$$\frac{\partial \theta}{\partial t} = -\frac{\partial J_w}{\partial x_i} - S(x_i, h) \quad (3.38)$$

where S ($\text{cm}^3/\text{cm}^3/\text{day}$) is the sink term, h is the spatially distribute tension head, and x_i is the spatial coordinate ($i = 1, 2, 3$). The volumetric flux J_w is given by Darcy's law (Eq. 3.21). The depth-dependent sink term may be represented by different root water uptake functions ranging from linear to nonlinear (e.g., Raats, 1974; Hoogland et al., 1981; Prasad, 1988) and from 1D to 3D according to the considered dimensionality of the system (Vrugt et al., 2001). Thus, the local root uptake rate, S , depends on location in the root zone, time, the water tension head in the medium, root density, or a combination of these variables. Boundary conditions can be included to allow for specified medium–water potentials and fluxes at the growing-medium surface and the bottom boundary of the growing medium domain, whereas user-specified initial conditions and time-varying source/sink volumetric flow rates should be specified.

The macroscopic models of root water uptake can be further divided into two groups. The first group contains water potential and hydraulic parameters inside plant roots (Nimah and Hanks, 1973; Hillel et al., 1976; Kramer and Boyer, 1995; Molz, 1981), which are difficult to quantify. In the second group, the root water extraction rate is calculated from plant transpiration rate, rooting depth and soil–water potential (Molz and Remson, 1970; Feddes et al., 1974, 1978; Raats, 1976; Gardner, 1983; Prasad, 1988). The parameters in the second group are relatively easy to obtain; therefore, the approach is widely used and has been implemented into commonly used numerical models (SWMS, Simunek et al., 1992; HYDRUS, Simunek et al., 1999). The root water extraction models by Molz and Remson (1970) and Raats (1976) ignore the effect of medium–water content on the distribution of root water uptake. For example, in Molz and Remson's (1970) model, the root system always extracts 40% of the total transpiration from the top quarter of root zone, even if the top layer is desiccated by evapotranspiration.

The sink term, for a 1D vertical mass balance equation (Eq. 3.38)

$$C(h) \frac{\partial h}{\partial t} = \frac{\partial}{\partial z} \left[K(h) \left(\frac{\partial h}{\partial z} + 1 \right) \right] - S(z, h) \quad (3.39)$$

according to Feddes et al. (1978) is

$$S(z, h) = \gamma(h) \cdot S_{\max}(z) \quad (3.40)$$

with $S_{\max}(z)$, the maximal root water uptake as a function of depth [T^{-1}] and $\gamma(h)$ a dimensionless reduction function that simulates the effects of medium–water deficit on root water extraction.

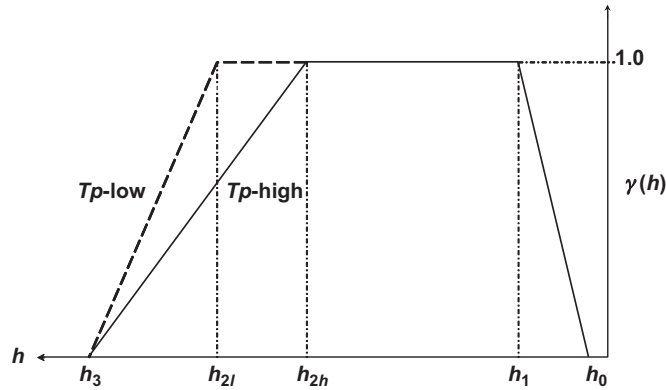


FIGURE 3.18

Reduction function, $\gamma(h)$.

After Feddes, R.A., Kowalik P.J., Zaradny, H. (Eds.), 1978. *Water uptake by plant roots*. In: *Simulation of Field Water Use and Crop Yield*. John Wiley & Sons, Inc., New York, pp. 16–30.

This function is characterized by different tension head values h_0 , h_1 , h_2 (low and high according to the climatic demand) and h_3 , as shown in Fig. 3.18. Above h_0 , $\gamma(h)$ is zero, between h_1 and h_{2l} or h_{2h} is 1, and between h_3 and h_3 the values of the reduction function are given by

$$\gamma(h) = \frac{h(z) - h_3}{h_2 - h_3} \quad (3.41)$$

The actual root water uptake given by Eq. (3.39) is subsequently integrated from the medium surface downward until it reaches either the potential transpiration Tp (no water stress) or the rooting depth (water stress if the integrated root water uptake is smaller than Tp). Specifically, the tension head threshold at which water stress initiates reduction in root water uptake is determined by Tp , with water stress occurring earlier at the less negative values h_{2h} , if Tp is high. This type of functional dependence allows for less favorable water-supplying medium moisture conditions with increasing plant transpiration. Simulation of root water uptake using the above-described model with VG models for the RC and hydraulic conductivity function, with a model to calculate Tp during a relatively short period for which the crop is fully developed and crop growth negligible, requires the specification of about 17 parameters; 6 are medium related and 11 plant related.

The volume and shape of growing containers have a significant effect on the root density distribution (Fig. 3.19). The container's impervious walls and bottom (except holes to allow drainage) increase the root density along these walls. Moreover, the vertical distribution of moisture content in containers, pots, and growing sleeves differs substantially from field soils owing to the existence of a lower boundary. This moisture distribution has an effect on the vertical root density distribution. As such, the use of the 1D model (Eq. 3.39) for simulations of water extraction by roots in pots and containers will thus provide erroneous predictions. The complexity in representing the spatial and temporal root density distribution in growing containers discards the option of using the multidimensional (two/radial and 3D models) mathematical model as well. If one is interested to predict $\theta(x,y,z,t)$ or $h(x,y,z,t)$ within the growing container, its progress depends on the evaluation

**FIGURE 3.19**

Root distribution in limited-volume pots, containers, and sleeves.

of root density distribution as a function of x,y,z,t . However, if the prediction of water availability, under different water demand scenarios, is of interest, alternative methods should be used. As an alternative, root water uptake in containerized media can be indirectly inferred from soil–water balance studies, such as container-weight variation or characterizing the moisture content variation at different spatial and temporal scales by tensiometers and/or TDRs. These methods enable to separate the vertical moisture flow and the water depletion curve by root extraction (Figs. 3.7, 3.15, and 3.16) and to evaluate the overall and local momentary root water extraction.

3.4.3 DETERMINING MOMENTARY AND DAILY WATER UPTAKE RATE

The knowledge of the momentary crop transpiration rate over time may serve to improve irrigation control in soilless culture, since an accurate and dynamic control of the water supply is needed to meet the plants' water requirements in the low water-holding capacity and limited-volume systems. The crop transpiration rate can be evaluated by various direct and indirect methods. One approach is to extrapolate from porometer measurements on single leaves to values for the plant canopy. This method suffers from the major disadvantage that the measurement changes the microclimate at the leaf surface, which can influence the observed transpiration. The heat-pulse method is also used to measure sap flow (Jones et al., 1988; Green et al., 2003; Möler et al., 2004) that can be related to transpiration rate. However, a calibration for each crop is needed and frequent measurements may yield noisy data. Crop transpiration can be modeled using the Penman–Monteith equation (Monteith, 1990) and has been evaluated at all levels—from single leaves to whole canopies—in greenhouses growing both vegetable (Jolliet and Bailey, 1992; Boulard and Jemaa, 1993; Hamer, 1996) and

ornamental crops (Bailey et al., 1993; Baille et al., 1994; Montero et al., 2001). This requires the internal, or stomatal, and the external, or boundary layer, resistances to be known along with the environmental variables within the plant canopy. Most of these studies have been conducted in a specific growth phase of the crop, while there is little information available on the transpiration rate over the plant's ontogeny and its modeling over the entire crop cycle. The relation of the evaluated transpiration rates to water availability in the growing containers has not been widely studied.

A relatively old and simple method to evaluate the daily and hourly transpiration rate is by using frequent measurements of the weight of a growing container (Teare et al., 1973; Yang et al., 2003; Rouphael and Colla, 2005; Medrano et al., 2005). The weight measurement frequency varies from minutes to hours and even a day (e.g., Ray and Sinclair, 1998). It is often argued that lysimeter-grown plants do not accurately represent plants grown in normal soil (Girona et al., 2002). This argument is not valid for container-grown plants, and the method has found wide use when the transpiration rate was related to other environmental and physiological variables. This method includes a single plant or set of plants grown in a container of specific volume and known initial weight that is weighed at each of n predetermined time points, $t_1, t_2, \dots, t_n$. The momentary water uptake rate is then determined by calculation of the numerical derivative of the measured weight

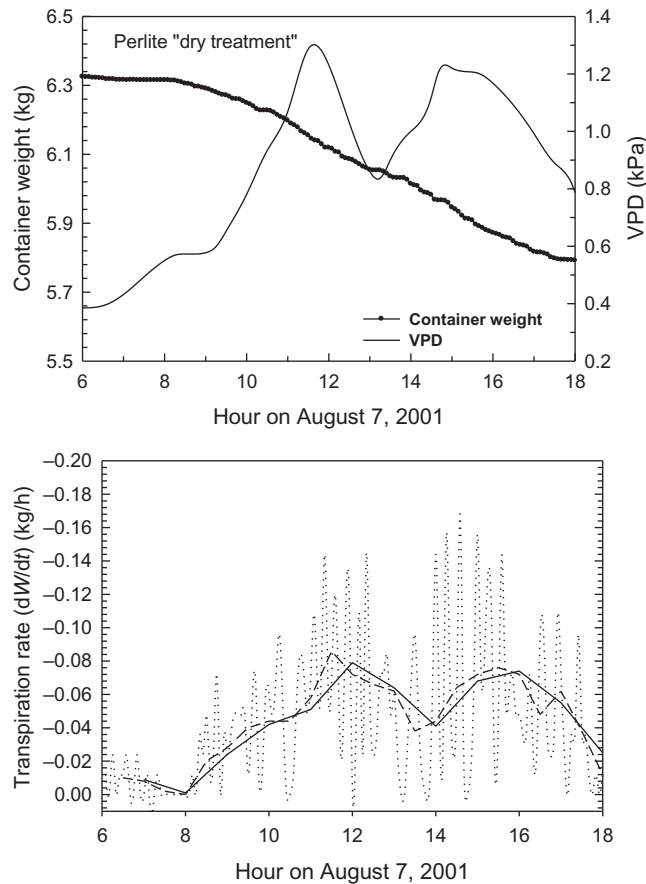
$$T = \frac{dW}{dt} \approx \frac{W_{k+1} - W_k}{t_{k+1} - t_k} \quad (3.42)$$

where W_k is the measured weight of the growth container at time t_k .

Differentiation with time (Eq. 3.42) significantly amplifies the high-frequency noise in the measured time series. Given that measurement errors cannot be avoided, additional attention should be paid when this method is used to calculate the transpiration rate. An additional analysis to the time series of container weight prior to differentiation may improve the accuracy of the calculated momentary water uptake rate.

The calculated momentary transpiration rate depends on the time interval among successive data points, $\Delta t = t_{k+1} - t_k$. Different average rates with different errors are obtained as time intervals among samplings expand. On the one hand, the greater the time interval among samplings is, the smaller the error of calculated average rate is. On the other hand, different average rates for different time intervals carry different (often incomparable) information about the studied system. Therefore, the time intervals at which average transpiration is calculated should correspond to the purpose of the sampling.

The time series of the measured perlite-filled pot weight during August 7, 2001 (Fig. 3.20A) is used herein to demonstrate the role of the sampling time on the calculated rate of water uptake. Irrigation events took place at 1540 hours on the previous day—August 6, and at 1820 hours on August 7 (not shown). Being covered by plant canopy, the evaporation from the substrate surface was neglected and the container-weight decrease (Fig. 3.20A) can be related to transpiration (fourth stage). The momentary transpiration rate of a rose plant grown in this pot was calculated first for $\Delta t = 5$ minutes (shown as dotted lines in Fig. 3.20B), which was the time interval at which data were logged (an average over 5 minutes of weight sampling every 10 seconds). The negative derivative values stand for weight loss. The momentary transpiration rate for $\Delta t = 5$ minutes is very noisy (Fig. 3.20B), which hinders the identification of the actual transpiration pattern. This noisy time derivative was obtained in spite of the relatively smooth measured weight pattern in

**FIGURE 3.20**

(A) Perlite container weight and VPD variation during August 7 and (B) transpiration rate during this day, calculated by $\Delta W/\Delta t$, for different sampling interval Δt .

Fig. 3.20A. This noisy derivative cannot be used for studying the relationship between water uptake rate and other physiological or environmental parameters or as a basis for irrigation management without any further analysis. The effect of larger sampling intervals on the time derivative is demonstrated by producing new time series for $\Delta t = 30$ and 60 minutes by omission of measured data points from the original measured time series (with $\Delta t = 5$ minutes). The transpiration rates calculated by $\Delta W/\Delta t$ for these time series are shown in **Fig. 3.20B**. The conclusion from this figure is that the time series of less-frequent sampled data filter noises associated with numerical derivatives. However, a careful look on the transpiration rate obtained for the increased sampling interval time series reveals that the temporary variation of momentary transpiration rate and its minimum and maximum values, mainly during the midday hours that may be indicative for water availability shortage and plant stress development, are not unique—it depends on the sampling time interval

(Fig. 3.20B). A comparison between the calculated transpiration rate and the vapor-pressure deficit (VPD) variation during the day (Fig. 3.20A) indicates that the deviation between the VPD and transpiration peaks, and the in-between dip timing, increases with Δt . Note that the VPD is commonly used to express the atmospheric water demand. Such meaningful dependence on data-sampling intervals raises questions regarding the method of increasing time interval to filter out derivative noise and what is the adequate sampling interval. The shortcomings of using sparsely measured data to “smooth” noises associated with numerical differentiation while gaining the advantages of the additional information associated with more frequently measured data can be rectified by applying filtering methods to the time series prior to differentiation.

Noises in calculated transpiration rates can be reduced, while preserving the information accompanying frequently measured time series, by smoothing (detrend) the time series prior to differentiation. This smoothing filters out the noise from the measured data while keeping the leading variation patterns. Many methods are available for smoothing noisy data in time-series analysis (Harvey, 1993). These methods can be categorized as nonparametric smoothing (e.g., moving average, Savitzky–Golay, and fast Fourier transform (FFT) filtering) and nonparametric regression (fitting polynomials of various orders, exponential functions, symmetrical and asymmetrical transition functions, etc. to the measured data). Smoothing should be carried out with care since substantially different results may be obtained after differentiation, in spite of the high R^2 values that are usually obtained for the different smoothing methods. Three procedures are used herein to smooth the container-weight time series prior to differentiation (Fig. 3.20A): fitting a linear and a fourth order polynomial, and noise filtering by Savitzky–Golay method. The transpiration rate patterns obtained by time derivative of the smoothed weight time series are shown in Fig. 3.21. A remarkable difference was obtained between these patterns, in spite of the extremely high R^2 values. Note that the transpiration rate pattern calculated by the Savitzky–Golay method is the only pattern that follows the VPD variation (Fig. 3.20A). The smoothed time series provides a smooth transpiration rate that

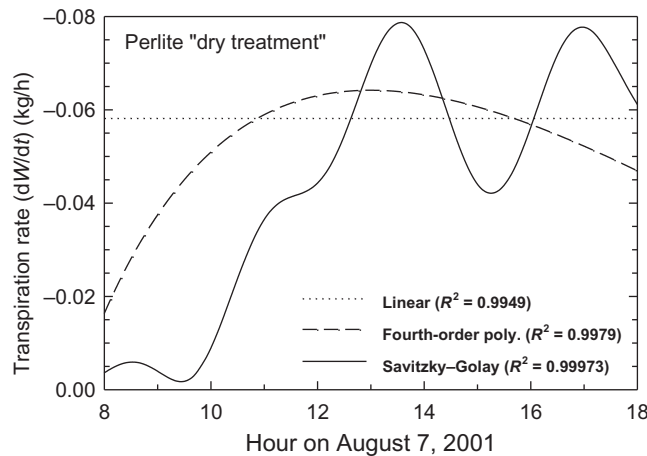


FIGURE 3.21

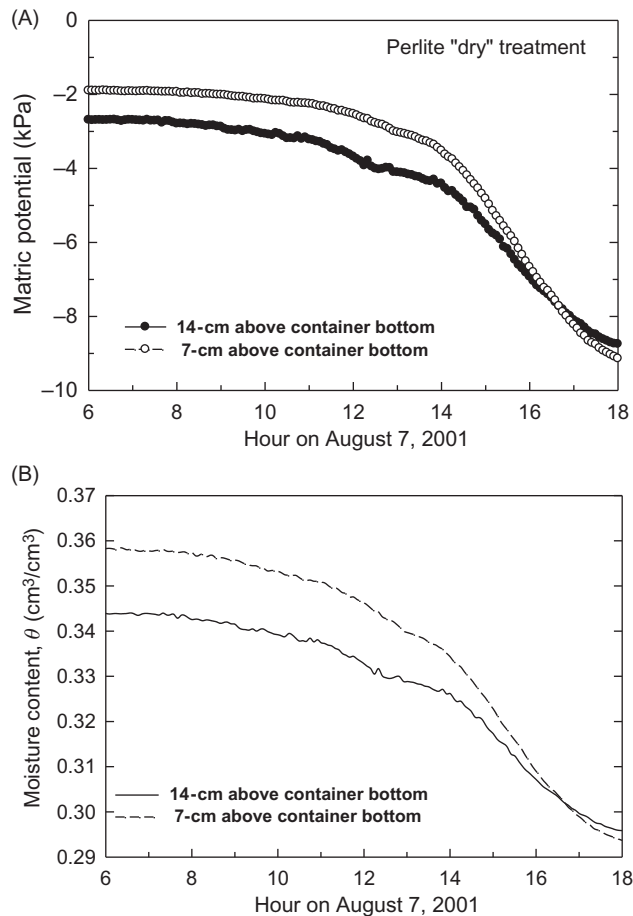
Comparison of three procedures to smooth the container-weight time series prior to differentiation.

enables to figure out momentary values. Many other smoothing methods that were not demonstrated here can also be used to filter noise from the container-weight time series. However, a pertinent question regarding the smoothing procedure is how intensive should the smoothing be in order to preserve the relevant characteristics of the time series. A further discussion on this issue is beyond the scope of this chapter.

3.4.4 ROOTS UPTAKE DISTRIBUTION WITHIN GROWING CONTAINERS

The calculated, or directly measured, whole-plant transpiration rate provides information on the temporal variation of the average moisture content within the root zone (container in our context) but fails to provide information on the spatial variation of moisture content within the root zone/container and an insight on the dynamics of water extraction by roots. Such information may be interpreted to root density variation and temporary water availability at different parts of the container that can be translated to a better irrigation management. Kerloch and Michel (2015) and references therein discussed thoroughly the relationship between root growth and the porous media at the container scale.

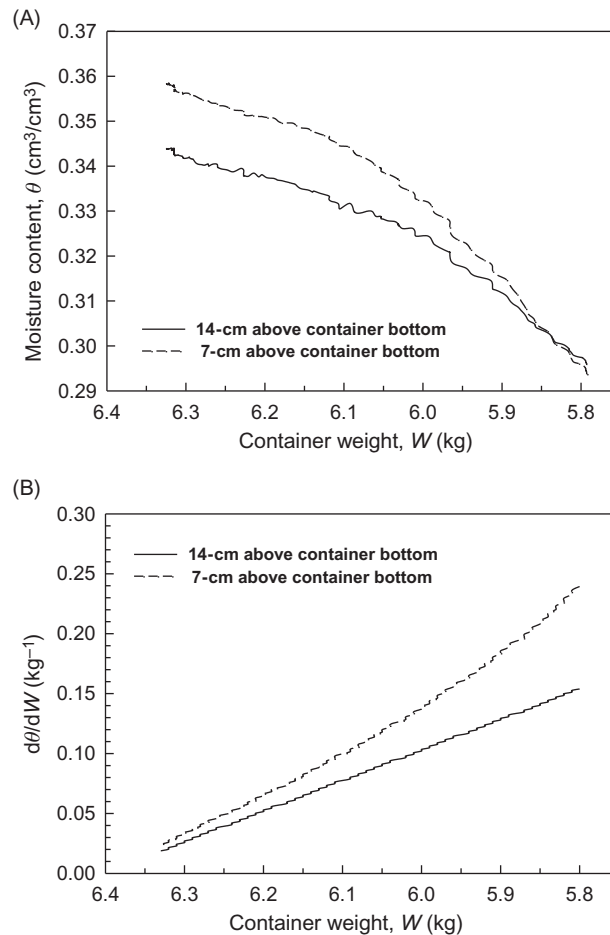
The matric potential at two levels, 7 and 14 cm above the container bottom that was measured by high-flow tensiometers (see experimental setup in Wallach and Raviv, 2005), can be transformed into moisture content by the independently measured RC. The matric potential and the associated transformed moisture content at these two levels within the container are shown in Fig. 3.22. The moisture content is usually higher at the lower part of the container between frequent subsequent irrigation events. However, this tendency may change at longer periods for less-frequent irrigation events, as appears in Fig. 3.22B, for the late afternoon hours. The relative contribution of moisture content at the two levels (7 and 14 cm) above the container bottom to the container weight for the data of August 7 (Fig. 3.22) is shown in Fig. 3.23A. The derivative of the curves in Fig. 3.23A, $d\theta/dW$, was calculated and shown in Fig. 3.23B versus the container weight. This figure highlights the differences in the relative contribution of moisture extraction rate at the two levels to the transpiration rate while the container weight decreases. The curves in Fig. 3.23B indicate that the contribution to the container weight decrease (transpiration rate) was low in the morning when the container weight was high, increased during noon time, and intensified during afternoon hours when the container weight was low (note that the subsequent irrigation started when the container reached the lowest weight value). The comparison between Figs. 3.23B and 3.20A indicates that the rate at which water was extracted, as well as the root density, was higher at the lower part of the container, as shown qualitatively in Fig. 3.19. The initial preference of the plant is to extract water from the lower part of the container where moisture content is high and water is easily available. When moisture content decreases and so its availability, the intensive water extraction moves upward (Fig. 3.23B). Note that local water extraction rate depends on the combination of local root density and water availability (moisture content). The migration of the zone where intensive water extraction rate takes place is also known as “the moving source” and was observed in a field soil by Li et al. (2002a), but at an opposite direction, namely, from the upper layers downward. The actual transpiration rate deviated from the potential one at given ambient conditions when the moisture content at sections where root density is high is low. The sections with low root density cannot extract the appropriate amount of water, even at relatively high moisture contents.

**FIGURE 3.22**

Matric potential (suction head) (A) and moisture content (B) variation during August 7 in the perlite container for the "dry" treatment.

3.4.5 WATER AVAILABILITY VERSUS ATMOSPHERIC DEMAND

The term "available water" is a key parameter in crop irrigation. The amount of water which is "available" for root water uptake is defined in filled soils as the amount of soil–water between field capacity and wilting point. A soil matric pressure of -10 or -33 kPa for field capacity and -1500 kPa for wilting point are commonly used values (FAO, 1979). The widely used determination of water availability in soilless culture was introduced by de Boodt and Verdonck (1972), as shown in Fig. 3.24. They introduced the concept of "EAW," which they defined as the difference between the water content at suctions of 1 and 5 kPa and the "water buffering capacity" as the difference between the water content of the medium at 5 and 10 kPa. These definitions and others, for

**FIGURE 3.23**

Moisture content (A) and moisture content derivative by container weight (B) variation with container weight for the perlite container during August 7.

water availability, have not taken into account the accessibility of the “available” water to meet the evaporation demand or the potential rate of transpiration. Wallach et al. (1992a,b) and da Silva et al. (1993a,b) hypothesized that $K(\theta)$ of the medium bulk indicates the availability (amounts and rates) of medium–water to plant roots and significantly affects the performance of the plant. The differences in potential water availability for peat moss and tuff substrates at container capacity are demonstrated in Fig. 3.25 where the moisture content and $K(\theta)$ distributions along a 25 cm container height are plotted. As tension head (expressed in the figure as height from the container bottom) increases from 0 to 25 cm and water content decreases accordingly, the hydraulic conductivity decreases by approximately three orders of magnitude for peat and by approximately four orders of

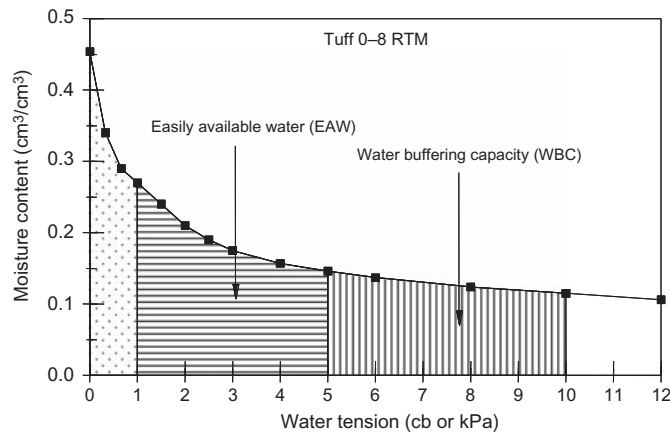


FIGURE 3.24

Definition of water availability following de Boodt and Verdonck (1972).

From de Boodt, M. and Verdonck, O. (1972). *The Physical Properties of the Substrates in Horticulture*. Acta Hortic. 26, 37–44, <https://doi.org/10.17660/ActaHortic.1972.26.5>

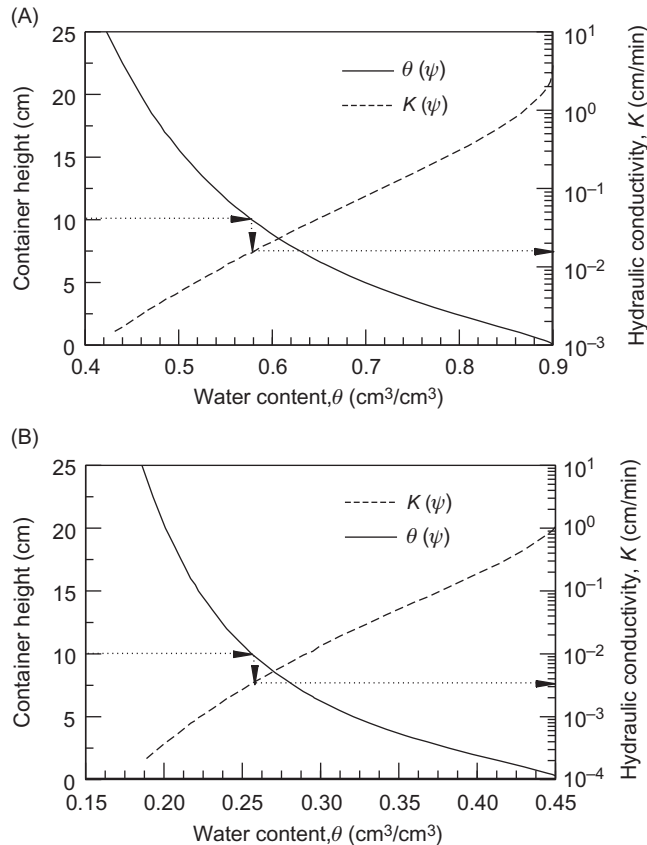


FIGURE 3.25

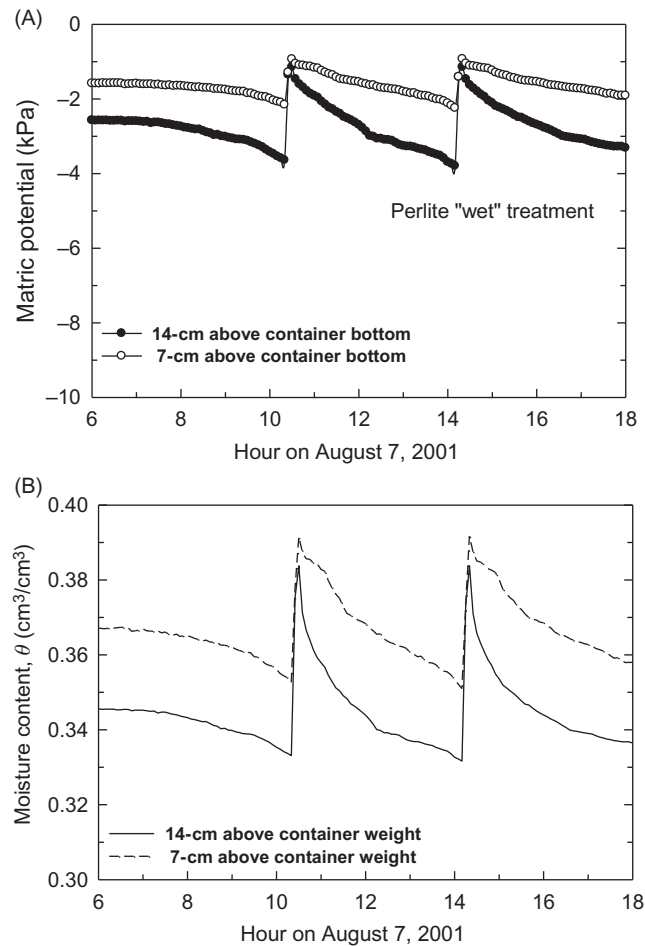
Water content and hydraulic conductivity distributions throughout a container of 25 cm height filled with (A) peat moss and (B) tuff RTM at container capacity

Data taken from da Silva, F.F., Wallach, R., Chen, Y., 1993a. *Hydraulic properties of sphagnum peat moss and Tuff (scoria) and their potential effects on water availability*. Plant Soil 154, 119–126.

magnitude for tuff. As such, the hydraulic conductivity variation in the pot will have a major influence on water availability and should be considered when criteria for water availability are postulated. Denmead and Shaw (1962) verified that transpiration could be restricted and plants could wilt over a wide range of soil–water contents, depending on root density, the soil hydraulic properties, and on the transpiration demand of the atmosphere. They found that a soil–water potential of only -0.1 MPa limited uptake at high rates of transpiration, but -1.0 MPa was not limiting at low rates. The limiting effect of low soil–water potential increases as atmospheric conditions favor high potential rates of transpiration.

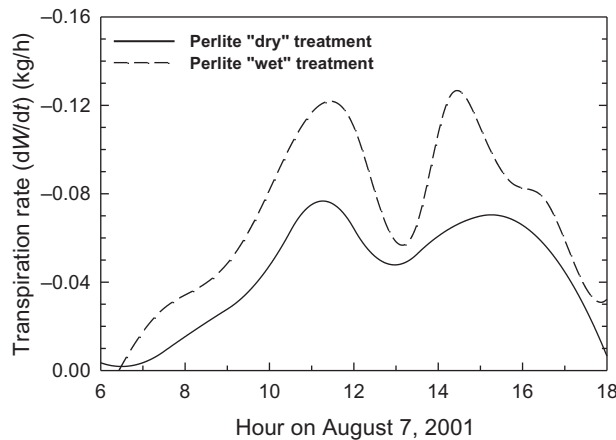
The momentary balance between the actual water flux from the medium volume surrounding the root and the potential water extraction rate (which depends mainly on the momentary atmospheric conditions) determines whether medium moisture is fully available to the plant. By “fully available” we mean that the water extracted from the vicinity of the root is fully replenished at a rate that coincides with the potential transpiration. When the root rhizosphere cannot be immediately replenished, water uptake by the root will be limited by the transport ability of medium–water toward the roots, that is, by the medium’s conductivity near the roots. Thus, when the transpiration rate is plant–atmosphere demand-driven, the water in the container is fully available. However, when water uptake becomes substrate supply–dependent, water is still available, but not fully. This concept of water availability and its dependence on the momentary hydraulic conductivity of the growing substrate was demonstrated in Raviv et al. (1999, 2001). In the first study (Raviv et al., 1999), rose plants were grown in two substrates, tuff and pumice, and were irrigated frequently. The yield differences between the two substrates were greater in the summer than in the cooler period, in spite of the frequent irrigations, large amounts of irrigation water, and average low tension that were maintained in the containers. The transpirational demand in the summer exceeded the maximum water flux that can flow from the pumice media to the roots within the relevant range of hydraulic conductivity, for longer periods than in the winter, accentuating the negative effects of low water availability. In the second study (Raviv et al., 2001) rose plants were grown in UC mix [42% composted fir bark, 33% peat, and 25% sand (by volume)] or in coconut coir. Low specific transpiration rate (transpiration rate per leaf area) values found at tensions between 0 and 1.5 kPa in UC mix suggest that this medium has insufficient free air space for proper root activity within this range. Above 2.3 kPa, unsaturated hydraulic conductivity of UC mix was lower than that of coir, possibly lowering specific transpiration rate values of UC mix-grown plants. As a result of these two factors, specific transpiration rate of plants grown in coir was 20%–30% higher than that of plants grown in UC mix. Specific transpiration rate of coir-grown plants started to decline only at tensions around 4.5 kPa. Yield (number of flowers produced) by coir-grown plants was 19% higher than UC mix-grown plants. This study demonstrated the superiority of hydraulic conductivity over tension to be a measure of water availability.

An example for the relationship between atmosphere-demand and substrate supply–driven transpiration is given in the following by comparing the perlite container with “dry” treatment (Figs. 3.20–3.23) with a perlite container with “wet” treatments. The only difference between the two treatments is the irrigation frequency, namely, the threshold weight at which the irrigation was turned on. Contrary to a single irrigation on August 7, at 1800 hours, in the “dry” treatment (Fig. 3.20A), the “wet” treatment was irrigated three times during that day—1020, 1400, and 2130 hours. The matric potential and the moisture content (determined as for the “dry” treatment from the measured matric potential and the independently determined RC) variation for the “wet”

**FIGURE 3.26**

(A) Matric potential (suction head) and (B) moisture content variation during August 7 in the perlite container for the “wet” treatment.

treatment are shown in Fig. 3.26A and B, respectively. Pronounced differences in matric potential and moisture content were obtained between the two treatments (Figs. 3.22 and 3.26). The lower matric potential and higher moisture content in the “wet” treatment indicate that the momentary hydraulic conductivity of the perlite is higher and so is the water availability. The transpiration rate for the two “wet” and “dry” treatments is shown in Fig. 3.27. Comparison of these patterns with the VPD variation (Fig. 3.20A) reveals that both transpiration rate patterns followed the VPD pattern, with higher transpiration rate for the “wet” treatment. The deviation between the two transpiration rate patterns was high during the peak VPD hours and decreased significantly during the period of lower VPD in-between the two VPD peaks. The deviation pattern between the

**FIGURE 3.27**

Transpiration rate for rose grown in perlite containers for the “dry” and “wet” treatment.

transpiration patterns demonstrates the meaning of water availability as a balance between input and output fluxes. When atmospheric demand is high (peak VPD values), the higher moisture content (and water availability) in the wet substrate (Fig. 3.26B) enables a higher transpiration rate than from the drier substrate. When there is a relief in the atmospheric demand (lower VPD values), water in both treatments were fully available and the two transpiration rates became close, in spite of the significant differences in moisture content between the two treatments. The existing differences in momentary transpiration rate among the two irrigation treatments that still exist during the periods of lower VPD values are probably due to the differences in plant size; the plant in the “wet” treatment was larger than the one in the “dry” treatment.

3.5 SOLUTE TRANSPORT IN SOILLESS MEDIA

3.5.1 TRANSPORT MECHANISMS—DIFFUSION, DISPERSION, CONVECTION

Soil scientists and agricultural engineers have traditionally been interested in the behavior and effectiveness of agricultural chemicals (fertilizers, pesticides) applied to soils for enhancing crop growth, as well as in the effect of salts and other dissolved substances in the soil profile on plant growth. The movement and fate of solutes in soil is affected by a large number of physical, chemical, and microbiological processes requiring a broad array of mathematical and physical sciences to study and describe solute transport. A vast amount of work on solute transport can be found in the soil science literature. Soil scientists have paid much attention to water movement and mainly chemical transport in the absence of roots. However, much less attention has been devoted to solute movement and transport mechanisms that include nutrient uptake by roots in containerized media of limited volume with the unique structure of soilless media, frequent irrigations, and chemicals that are continuously added with the irrigation water (fertigation). Consequently, both crop

simulation and water flow models tend to treat the solute transport and root uptake mostly by empirical means, thereby limiting their general applicability.

The scope of this chapter permits only an introductory treatment of the subject. First, the standard transport mechanisms relevant to the fundamental convection–dispersion equation (CDE) will be introduced. This equation is most often used to model solute transport in porous media. The movement of a solute that undergoes adsorption by the solid particles requires modifications of the CDE, particularly if several solute species are present that may participate in a number of different reactions. The nutrient uptake mechanisms will be discussed as well, both qualitatively and quantitatively.

The mass balance equation for a solute species at the macroscopic level in a three-phase substrate–air–water system that is subject to arbitrary reactions is given as

$$\frac{\partial \theta C}{\partial t} = - \frac{\partial J_s}{\partial x_i} - \theta R_s(x_i) \quad (3.43)$$

where θ is the volumetric water content [$L^3 L^{-3}$], C is the solute concentration expressed as solute mass-per-water volume [$M L^{-3}$], t is time [T], x_i is the spatial coordinate ($i = 1, 2, 3$) [L], J_s is the solute flux expressed in solute mass-per-cross-sectional area of soil-per-unit time [$M L^2 T^{-1}$], and R_s denotes arbitrary solute sinks (<0) or sources (>0) [$M L^{-3} T^{-1}$]. The 1D mass balance equation is obtained by replacing x_i by x in Eq. (3.43). The solute flux is usually distinguished in an advective and a dispersive component according to

$$J_s = J_w C + J_D \quad (3.44)$$

where J_w is a vector quantifying the water flux [$L T^{-1}$], namely, the Darcian velocity (Eq. 3.25) and J_D is a solute flux to quantify transport caused by a gradient in the solute concentration [$M L^{-2} T^{-1}$], also per unit area of soil.

Molecular diffusion is an important mechanism for solute transport in soils/substrates in directions where there is little or no water flow. A net transfer of molecules of a solute species usually occurs from regions with higher to lower concentrations as the result of diffusion as described by Fick's first law. The 1D mass flux, J_{diff} [$M L^2 T^{-1}$], due to molecular diffusion given by

$$J_{diff} = - D_0 \frac{\partial C}{\partial x_i} \quad (3.45)$$

where D_0 is the coefficient of molecular diffusion for a free or bulk solution [$L^2 T^{-1}$]. Many publications exist that provide data on D_0 (Kemper, 1986; Lide, 1995). To characterize diffusion in soil/substrate, the diffusion in a free solution is typically adjusted with terms accounting for a reduced solution phase (a smaller cross-sectional area available for diffusion), and an increased path length. A general treatment of diffusion in soils can be found in Olsen and Kemper (1968) and Nye (1979). The macroscopic diffusive flux per unit area of soil is

$$J_{diff} = - \theta D_{diff} \frac{\partial C}{\partial x_i} \quad (3.46)$$

where the molecular diffusion within the soil–liquid D_{diff} can be expressed in terms of the diffusion in free water D_0 by

$$D_{diff} = \xi \theta D_0 \quad (3.47)$$

where ξ is the tortuosity factor. One form of ξ is (Milington and Quirk, 1961; Jury et al., 1991)

$$\xi = \frac{\theta^{7/3}}{\phi^2} \quad (3.48)$$

In some publications, the moisture content, θ , in Eq. (3.47) is combined in the tortuosity factor, providing $\xi = \theta^{10/3}/\phi^2$.

Mechanical dispersion is formed by local variations in water flow in a porous medium. Several mechanisms contribute to mechanical dispersion: (1) the development of a velocity profile within an individual pore such that the highest velocity occurs in the center of the pore, and presumably little or no flow at the pore walls; (2) different mean flow velocities in pores of different sizes; (3) the mean water flow direction in the porous medium being different from the actual streamlines within individual pores, which differ in shape, size, and orientation; and (4) solute particles converging to or diverging from the same pore. All of these processes contribute to the increased spreading, in which initially steep concentration fronts become smoother during movement along the main flow direction.

The effects of dispersion can be measured in a laboratory experiment in which water and a dissolved tracer are applied to an initially tracer-free, or vice versa, uniformly packed soil/substrate column of length L . The column is subjected to steady-state water flow with uniform water content. As more of the tracer is added, the initially very sharp concentration front near the soil surface becomes spread out because of dispersion. Eventually, a smooth and sigmoidally shaped effluent curve can be monitored at the column exit as shown in Fig. 3.28 for a 30-cm-long column filled with 1-mm saturated sand. In the absence of dispersion, the front of a perfectly inert tracer will travel as a square wave through the column (a process often called “piston flow”) to reach the bottom of the column at time $t = L/v$, where v is the average pore water or interstitial velocity (Fig. 3.28). This velocity is the ratio of the Darcian water flux density (J_w) and the volumetric

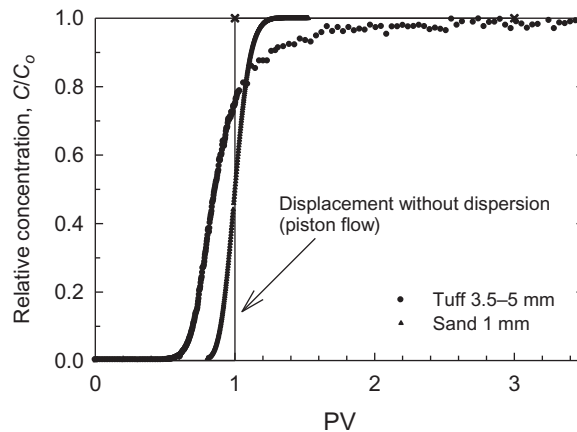


FIGURE 3.28

Measured BTCs for saturated sand (1 mm) and tuff (3.5–5.0 mm) column. Water discharge of 10 mL/min was displacing by 6 g/L NaCl from bottom to top of a 30-cm-long column. BTCs, Breakthrough curves.

water content (θ). Note that v is given per unit area of fluid whereas J_w is defined per unit area of (bulk) soil. For piston flow, the tracer reaches the column exit exactly after one pore volume of tracer solution has been injected (or collected at the column exit). Pore volume is defined as the amount of water stored in that column.

The 1D solute flux, due to mechanical dispersion in a uniform isotropic soil/substrate, may be approximated in a similar way as Fick's law:

$$J_{dis} = -\theta D_{dis} \frac{\partial C}{\partial x} \quad (3.49)$$

where J_{dis} is the dispersive solute flux [$M L^{-2} T^{-1}$]. It is commonly assumed that D_{dis} is a linear function of the flow velocity

$$D_{dis} = \alpha v \quad (3.50)$$

where α [L] is the dispersivity. When the above 1D geometry is too simplistic for a given problem, 2D or 3D dispersion is quantified with a dispersion tensor with α_L and α_T as the longitudinal and transverse dispersivity.

Hydrodynamic dispersion is a nonsteady, irreversible process (i.e., the initial tracer distribution cannot be obtained by reversing the flow) in which the tracer mass mixes with the nonlabeled portion of the solution. This is due to the presence of flow through a complicated system of pathways within the soil/substrates matrix. Hydrodynamic dispersion consists of mechanical dispersion and molecular diffusion. These two contributions are usually artificially separated but in actuality are totally inseparable in form because both occur together; however, the dependence of each on different parameters can vary due to changes in physical and chemical conditions. As an example, mechanical dispersions are more prominent at high water content and greater flow velocities because it is here that nutrient particles mix more freely with water within the soil pores, as that water meanders around individual particles. Molecular diffusion predominates at low water content and low-flow velocity since at this stage, chemical phenomena associated with a tracer continue, even though mechanical dispersions due to water movement have ceased. Molecular diffusion alone takes place at the molecular level in the absence of motion while dispersion occurs at the pore level. Hydrodynamic dispersion is also generally associated with early breakthrough of the tracer.

The structure of the substrate affects dispersion and the resultant shape of the breakthrough curve (BTC) in numerous ways. As particle size increases or the medium is aggregated, the graphic representation of the BTC has more tailing due to diffusion into stagnant regions, but the effluent appears at the end of the column sooner due to larger pore size. However, to get $C/C_0 = 1$, a larger volume of effluent would be required, due to diffusion into both the stagnant areas and into the tortuous path of large aggregates. Smaller particle sizes yield a more even distribution in pore size, which means there is less stagnant water, so the effluent appears at the later time and requires less volume for $C/C_0 = 1$ than for a soil with large aggregates.

The macroscopic similarity between diffusion and mechanical dispersion has led to the practice of describing both processes with one coefficient of hydrodynamic dispersion ($D = D_{dis} + D_{diff}$). This practice is consistent with results from experiments which do not permit a distinction between mechanical dispersion and molecular diffusion. The hydrodynamic dispersive flux (J_D) (Eq. 3.44) consists of contributions from molecular diffusion (Eq. 3.45) and mechanical dispersion (Eq. 3.49) as

$$J_D = J_{diff} + J_{dis} \quad (3.51)$$

Hydrodynamic dispersion is often simply referred to as “dispersion,” as will be done in the remainder of this chapter.

Dispersion coefficients may be determined by fitting a mathematical solution to observed concentrations (Toride et al., 1995). Additional procedures to determine dispersion coefficients are given by Fried and Combarnous (1971) and van Genuchten and Wierenga (1986). Values of the longitudinal dispersivity (δ_L) for laboratory experiments typically vary between 0.1 and 10 cm with 6–20 times smaller values for δ_T (Klotz et al., 1980).

3.5.2 CONVECTION—DISPERSION EQUATION

The expressions for the convective and dispersive solute fluxes can be substituted in the mass balance (Eq. 3.43). The 1D CDE for solute transport in a homogeneous soil/soilless-substrate becomes

$$\frac{\partial(\theta C)}{\partial t} = -\frac{\partial}{\partial x} \left(J_w C - \theta D \frac{\partial C}{\partial x} \right) + \theta R_s \quad (3.52)$$

In the case where the water content is invariant with time and space, the CDE may be simplified to ($v = J_w/\theta$)

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - v \frac{\partial C}{\partial x} + R_s \quad (3.53)$$

This is a second-order linear partial differential equation, which like the diffusion equation is classified as a parabolic differential equation. To complete the mathematical formulation of the transport problem, several concentration types, mathematical conditions as well as analytical and numerical solutions can be found in different textbooks and papers (e.g., van Genuchten and Alves, 1982; Warrick, 2003).

A variety of solute source or sink terms may be substituted for R_s . The most common source/sink term is due to adsorption/desorption and ion exchange stemming from chemical and physical interactions between the solute and the soil solid phase. Many other processes such as radioactive decay, aerobic and anaerobic transformations, volatilization, photolysis, precipitation/dissolution, reduction/oxidation, and complexation may also affect the solute concentration. A further refinement of the transport model is necessary in the case of nonuniform interactions between the solute and the soil, or if there is adsorption on moving particles and colloids. In the following, only interactions at the solid–liquid interface will be considered.

3.5.3 ADSORPTION—LINEAR AND NONLINEAR

Dissolved substances in the liquid phase can interact with solid particles and inorganic or organic colloids. Adsorption of solute by the soil/substrate is an important phenomenon affecting the fate and movement of solutes. The CDE for 1D transport of an adsorbed solute may be written as

$$\frac{\partial C}{\partial t} + \frac{\rho_b}{\theta} \frac{\partial S}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - v \frac{\partial C}{\partial x} \quad (3.54)$$

where S is the adsorbed concentration, defined as mass of solute per mass of dry soil [M M^{-1}]. The above equation can be expressed in terms of one dependent variable by assuming a

suitable relationship between the adsorbed and liquid concentrations. This is typically done with a simple adsorption isotherm to quantify the adsorbed concentration as a function of the liquid concentration at a constant temperature. In addition to temperature, the adsorption isotherm is generally also affected by the solution composition, total concentration, the pH of the bulk solution, and sometimes the method used for measuring the isotherm. A mathematically pertinent distinction is often made between linear and nonlinear adsorption. Although most adsorption isotherms are nonlinear, the adsorption process may often be assumed linear for low solute concentrations or narrow concentration ranges.

There are three common types of adsorption isotherms for which the CDE has been written: linear,

$$S = K_d C \quad (3.55)$$

Freundlich,

$$S = k_1 C^n \quad (3.56)$$

and Langmuir,

$$S = \frac{k_2 C}{1 + k_3 C} \quad (3.57)$$

where K_d is the partition coefficient, often referred to as the distribution coefficient, expressed in volume of solvent per mass of soil [$L^3 M^{-1}$], k_1 , k_2 , k_3 , and n are empirical constants. Using the derivative chain rule $\partial S / \partial t = \partial S / \partial C \cdot \partial C / \partial t$, the CDEs for the linear, Freundlich, and Langmuir isotherms are

$$\left[1 + \frac{K_d}{\theta} \right] \frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - v \frac{\partial C}{\partial x} \quad (3.58)$$

$$\left[1 + \frac{nk_1}{\theta} C^{n-1} \right] \frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - v \frac{\partial C}{\partial x} \quad (3.59)$$

$$\left[1 + \frac{k_2}{\theta(1+k_3 C)^2} \right] \frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - v \frac{\partial C}{\partial x} \quad (3.60)$$

respectively. The term that multiplies the derivative $\partial C / \partial t$ in Eqs. (3.57)–(3.59) is noted as the retardation factor, R . The convective and dispersive fluxes are reduced by a factor R as a result of adsorption. The movement of the solute is said to be retarded with respect to the average solvent movement. If there is no interaction between the solute and the soil, the value for R is equal to unity. The value for R can be readily calculated from the parameters values as obtained from chemical analyses of the solution and adsorbed phases. Alternatively, R can be estimated from solute displacement studies on laboratory soil columns.

3.5.4 NONEQUILIBRIUM TRANSPORT—PHYSICAL AND CHEMICAL NONEQUILIBRIA

Solute BTCs for aggregated substrates exhibit asymmetrical distributions or nonsigmoidal concentration fronts, as shown in Fig. 3.28 for the 3–3.5 mm tuff substrate. The concept behind physical nonequilibrium models is that differences between regions of the liquid phase lead to mostly lateral gradients in the solute concentration, resulting in a diffusive type of solute transfer process.

Depending upon the exact pore structure of the medium, asymmetry is sometimes enhanced by desaturation when the relative fraction of water residing in the marginally continuous immobile region increases.

Since most of the sorption sites are only accessible after diffusion through the immobile region of the liquid phase, a corresponding delay in adsorption will occur. The delayed adsorption can also be explained with a kinetic description of the adsorption process. Both cases may be described with chemical nonequilibrium models, which distinguish between sites with equilibrium and kinetic sorption. The two-site chemical nonequilibrium model was applied successfully to describe solute BTCs by [van Genuchten \(1981\)](#), and [Nkedi-Kizza et al. \(1983\)](#), among others. Bicontinuum or dual-porosity nonequilibrium models are the most widely used. Only two concentrations need to be considered and the equilibrium CDE (Eq. 3.54) can be readily modified for this purpose. The same dimensionless mathematical formulation can be used for physical and chemical nonequilibrium models. If necessary, the CDE can be modified to incorporate additional nonequilibrium processes and continua.

The *physical nonequilibrium* approach is based on a partitioning of the liquid phase into a mobile or flowing region and an immobile or stagnant region. Solute movement in the mobile region occurs by both convection and dispersion, whereas solute exchange between the two regions occurs by first-order diffusion ([Coats and Smith, 1964](#)). Following [van Genuchten and Wierenga \(1976\)](#), the governing equations for the two region model are

$$(\theta_m + f\rho_b K_d) \frac{\partial C_m}{\partial t} = \theta_m D_m \frac{\partial^2 C_m}{\partial x^2} - \theta_m v_m \frac{\partial C_m}{\partial x} - \alpha(C_m - C_{im}) \quad (3.61)$$

$$[\theta_m + (1-f)\rho_b K_d] \frac{\partial C_{im}}{\partial t} = \alpha(C_m - C_{im}) \quad (3.62)$$

where f represents the fraction of sorption sites in equilibrium with the fluid of the mobile region, α is a first-order mass transfer coefficient [T^{-1}], and the subscripts m and im , respectively, refer to the mobile and immobile liquid regions (with $\theta = \theta_m + \theta_{im}$), while ρ_b and K_d are the soil BD and distribution coefficient for linear sorption. Transport equation, Eq. (3.61) follows directly from the addition of a source/sink term (R_s) to Eq. (3.54).

Growing media are coarser materials compared to regular soil. Pockets of air, to some extent, are generally present within the pores regardless of water content. The mobile and immobile regions within soils of similar structure have been widely used in direct relation to this phenomenon. However, the use of solute transport models to growing substrates is exceptional. [Fig. 3.28](#) introduces a BTC that was measured for a saturated tuff growing medium packed in a 30-cm column. Comparing the asymmetrical tuff with the symmetrical sand, BTCs ([Fig. 3.28](#)) highlight the role of the immobile water in the tuff substrate on solute transport. The following parameters for Eqs. (3.61) and (3.62) were determined by using the CXTFIT software ([Toride et al., 1995](#)) for the measured tuff BTC: $v = 0.012$ cm/s and $n = 0.61$ (both directly measured), $D_m = 2.9 \cdot 10^{-3}$ cm/s, $\theta_m = 87\%$, $\theta_{im} = 13\%$, and $\alpha = 7.1 \cdot 10^{-5}$ s $^{-1}$. Note that θ_m does not exist in the sand.

3.5.5 MODELING ROOT NUTRIENT UPTAKE—SINGLE-ROOT AND ROOT SYSTEM

Uptake of chemicals by the root system depends mainly on two factors. These are the characteristics of the root system (e.g., morphology, growth rate, nutrient absorption rate) and the interaction

of the chemical species with the soil and the soil–water supply (e.g., soil adsorption, mass transport, diffusion). In recent years, several mathematical models have been developed to simulate root growth and its interaction with both the soil–water and nutrient supply.

Modeling root nutrient uptake started with simulation of mass flow and diffusion of nutrients to a single root, approximated by a cylinder of constant radius, solving a simple form of the CDE for a circular geometry (Passioura and Frère, 1967) considering water but no nutrient uptake. Nye and Marriott (1969) included nutrient uptake, using a Michaelis–Menten-type boundary condition at the soil–root interface, and explored the sensitivity of the solution to changes in some of the equation parameters (i.e., diffusion coefficient, soil buffer capacity, root absorbing power, and nutrient flux at the root surface), whereas Barley (1970) simulated uptake in a system of irregular, parallel roots. Later models have included root competition (Baldwin et al., 1973) and root growth (Claassen and Barber, 1976; Barber and Cushman, 1981), but the root system was modeled only in terms of root length per unit depth, and no attention was paid to the interaction between 3D distribution of the roots, and water and solute transport patterns. Advances have also been made in the modeling of root growth, including the analytical approach of Hackett and Rose (1972a,b), the numerical simulations of Lungley (1973), the 3D simulation models of Diggle (1988) and Pages et al. (1989), and the stochastic approach of Jourdan et al. (1995).

Coupling of such efforts as presented above has led to models capable of simulating various aspects of root–soil dynamics. Models have been developed focusing on the transport and uptake of highly immobile nutrients transported mainly by diffusion (Silberbush and Barber, 1984) or mobile nutrients such as nitrate moving through the soil mainly by mass flow. Jones et al. (1991) simulated 1D root growth of a user-selected crop as influenced by soil physical properties, soil–water and nutrient content, and temperature. Clausnitzer and Hopmans (1994) simulated 3D root architecture and the interaction between root growth and soil–water movement, emphasizing the effects of soil strength on root growth, whereas Benjamin et al. (1996) modeled the influence of rooting patterns on root water uptake and leaching in two dimensions. Somma et al. (1998) included the interaction between plant growth and nutrient concentration in their model, thus providing a tool for studying the dynamic relationships between changing soil–water and nutrient status and root activity.

The general transport model for root uptake that follows the development of the mass balance equation in Section 3.5.3 to solve the 3D form of the CDE for solute concentration C [M L^{-3}], as fully described in Simunek et al. (1999), is

$$(\theta + \rho K_d) \frac{\partial C}{\partial t} = \frac{\partial}{\partial x_i} \left(\theta D \frac{\partial C}{\partial x_i} \right) - J_{w,i} \frac{\partial C}{\partial x_i} - S' \quad (3.63)$$

where K_d is the linear adsorption coefficient, $J_{w,i}$ is the water flux component in the i th direction, and S' [T^{-1}] is the sink term to account for root nutrient uptake. Many more rate constants can be added to the CDE, for example, to allow for reactions of the solute in the dissolved or adsorbed phase such as microbial degradation, volatilization, and precipitation. The solution of Eq. (3.63) yields the spatial and temporal distribution of nutrient concentration and fluxes at the same time resolution as the solution of Eq. (3.39). Note that the solution of Eqs. (3.39) and (3.63) yields macroscopic quantities, that is, values of matric potential, concentration, fluxes, in resolution which is usually much larger than root diameter and root spacing.

The Nye–Tinker–Barber model for nutrient uptake by a single cylindrical root in an infinite extent of the soil was first developed in the 1970s (Nye and Tinker, 1977; Barber, 1984; Tinker and Nye, 2000). In that model, the nutrient uptake by the root is considered to occur from the liquid (water) phase of the soil only. The soil is assumed to be homogeneous and isotropic, and the changes in the moisture conditions are assumed to be negligible since the soil is assumed to be fully saturated. The classical Nye–Tinker–Barber model also neglects the effect of root exudates, microbial activity, mycorrhizae, etc. on the plant nutrient uptake and nutrient movement in the soil. The movement of nutrient to the root surface takes place by convection due to water uptake by the plant and by diffusion of nutrient ions in the soil pore water.

The relation between the net influx of nutrients into the root and the concentration of nutrients at the root surface is described in the Nye–Tinker–Barber model by an experimentally measured heuristic Michaelis–Menten-type uptake law. Classically, the Michaelis–Menten kinetics is derived from the most basic enzyme catalyzed reaction. This is compatible with the nutrient “carrier” theory in which the nutrients are considered to be carried through the root surface cell membranes by the so-called nutrient carrier proteins. Thus, the Michaelis–Menten uptake law appears to include the characteristics of symplasmic pathway of nutrients into the root. Alternatively, the Michaelis–Menten-type uptake law could also be derived similarly to that of Langmuir adsorption isotherm. This describes the fraction of free and bound nutrient ion-binding sites on a membrane or on a solid surface. This results in what looks like a Michaelis–Menten-type equilibrium condition for the number of binding sites that are bound at any given time. However, as mentioned, the Nye–Tinker–Barber model does not consider the exact mechanism of nutrient uptake by the plant root, instead it uses an experimentally measured heuristic Michaelis–Menten nutrient uptake law.

For the specific case of a cylindrical root, the problem can be written in terms of polar radius r , and moreover the Darcy flux (satisfying $\nabla \cdot u = 0$) is given as $u = -aV/r$, where V is the water flux to the root and a is the root radius. The CDE for a single root, written in radial coordinates, is

$$(\theta + Kd) \frac{\partial C}{\partial t} = \frac{\theta D}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C}{\partial r} \right) \quad (3.64)$$

The boundary condition at the root surface ($r = a$) is an uptake flux of Michaelis–Menten type.

$$\theta D \frac{\partial C}{\partial r} + VC = \frac{F_m C}{K_m + C} \quad \text{on } r = a \quad (3.65)$$

where F_m and K_m are properties of the root surface. The boundary condition far from the root surface is the initial concentration, C_0

$$C \rightarrow C_0 \quad \text{as } r \rightarrow \infty \quad (3.66)$$

The initial soil concentration is prescribed and equal to the far-field concentration away from the root:

$$C = C_0 \quad \text{at } t = 0 \quad (3.67)$$

A combined model for water and nutrient uptake has not been applied so far to containerized growing media with its unique features of frequent fertigation management and limited root volume. Such a model could quantitatively highlight the effect of irrigation frequency in such systems on the availability of different nutrients to the roots, the relative effect of diffusion and convection on nutrients transport to the soil–root interface at different stages during and between subsequent irrigation events, etc.

3.6 GAS TRANSPORT IN SOILLESS MEDIA

3.6.1 GENERAL CONCEPTS

The understanding of gas transport in growing media is important for evaluation of soil aeration or movement of O_2 from the atmosphere to the media. Media aeration is critical for plant root growth because roots generally cannot get enough O_2 from leaves. Gas in unsaturated substrates is generally moist air but has higher CO_2 concentrations than atmospheric air because of plant root respiration and microbial degradation of organic compounds.

Kerloch and Michel (2015) found that root development decreases the pore tortuosity and the gas diffusivity that is equal to the ratio between AFP and pore tortuosity of the growing medium. Successive drying–wetting cycles are generating physical consolidation that decreases the TP and air porosity. However, gas diffusion increased with time, despite the decrease in TP and AFP, owing to better pore connections by root effects.

The volumetric gas content, θ_g , is defined as

$$\theta_g = \frac{V_g}{V_T} \quad (3.68)$$

where V_g [L^3] is the volume of the gas and V_T [L^3] is the total volume of the sample. This definition is similar to that used for volumetric water content. In many cases the volumetric gas content is referred to as “the gas porosity.” The saturation with respect to the gas phase S_g is

$$S_g = \frac{V_g}{V_v} \quad (3.69)$$

where V_v [L^3] is the volume of pores.

In unsaturated systems, water wets the solids and is in direct contact with them, whereas gas is generally separated from the solid phase by the water phase. Water fills the smaller pores whereas gas is restricted to the larger pores. Gas and water differ greatly. The gas density is much lower than liquid density; the density of air ranges generally from 1 to 1.5 kg/m^3 , depending upon its composition, whereas the density of water is close to 1000 kg/m^3 . When water flow is considered, it is assumed that its velocity at the solid surfaces is zero (no slip flow), whereas gas velocities are generally nonzero (slip flow). As the dynamic viscosity of air is ~ 50 times lower than that of water, significant air flow can occur at much smaller pressure gradients. The density and viscosity differences between gas and water induce a lower hydraulic conductivity (an order of magnitude) for gas than for water in the same material. Because gas molecular diffusion coefficients are about four orders of magnitude greater than those of water, gas diffusive fluxes are generally much greater than those of water.

3.6.2 MECHANISMS OF GAS TRANSPORT

As in liquid transport (see details in Section 3.5), transport in the gas phase may also be described by convection (advection) and dispersion. Although some studies have found mechanical dispersion or velocity-dependent dispersion to be important for chemical transport in the gas phase (Rolston et al., 1969), in most cases, mechanical dispersion is ignored because gas velocities are generally too small and the effects of diffusion are generally much greater than dispersion in the gas phase.

The 1D diffusive flux of a given gas can be described by Fick's law, similar to Eq. (3.45) for solute transport,

$$J_g = -D_{gs} \frac{dC_g}{dx} \quad (3.70)$$

where J_g is the flux of gases, D_{gs} is the effective gas diffusion coefficient, and C_g is the gas concentration [$L^3 L^{-3}$] in the air phase. Soil gas diffusivity and its dependency on θ_g and growing media type (texture, structure) control gas transport and fate where diffusive gas transport is normally dominant compared with convective gas transport. Accurate predictive models for D_{gs} are needed to evaluate, for example, soil aeration (Buckingham, 1904; Taylor, 1949), the diffusion and emission of fumigants at soil fumigation sites (Call, 1957; Jin and Jury, 1995), the diffusion and volatilization of organic chemicals from polluted soil sites (Petersen et al., 1996), and the diffusion and biodegradation of greenhouse gases such as methane (Kruse et al., 1996). Numerous predictive—descriptive models for D_{gs} as a function of ε are available and were divided by Moldrup et al. (2004) into six groups:

1. The first group consists of predictive $D_{gs}(\theta_g)$ models based only on θ_g . The first θ_g -based models were introduced a century ago. Buckingham suggested that the relative oxygen diffusion coefficient in soil was proportional to θ_g^2 (Buckingham, 1904). Other classical $D_{gs}(\theta_g)$ models in the first group are the linear $D_{gs}(\theta_g)$ models by Penman (1940),

$$\frac{D_{gs}}{D_{g0}} = 0.66\theta_g \quad (3.71)$$

van Bavel (1952), and Call (1957), and the nonlinear models by Marshall (1959) and Millington (1959). The latter two can be considered mechanistically based (cutting and randomly rejoining pores) models (Ball et al., 1988; Collin and Rasmuson, 1988).

2. The second group consists of simple, empirically, or mechanistically based, nonlinear $D_{gs}(\theta_g)$ models that take into account both θ_g and soil TP, ϕ . These predictive models introduce a minor soil type effect through ϕ that is dependent on, for example, soil texture and management. Among the numerous models within this group are the Millington and Quirk (1960) model, as reintroduced by Jin and Jury (1996) and the Millington and Quirk (1961) model

$$\frac{D_{gs}}{D_{g0}} = \frac{\theta_g^{10/3}}{\phi^2} \quad (3.72)$$

that is almost universally accepted and applied in unsaturated soil transport and fate models to describe both gas and solute diffusivity (Eq. 3.48). The frequent use of the Millington and Quirk (1961) model is noteworthy since the model has never been validated against gas diffusivity data for undisturbed soils representing a broad interval of soil types and porosities.

3. The models in the third group use the SWC curve as an additional input to take into account soil type effects on gas diffusivity. Moldrup et al. (1996) introduced the Campbell (1974) SWC parameter b as the third model parameter, together with θ_g and ϕ , in $D_{gs}(\theta_g)$ models. Moldrup et al. (1999) combined the Campbell b -dependent $D_{gs}(\theta_g)$ model with the Buckingham (1904) expression for gas diffusivity in dry soil (void of water) to develop the so-called Buckingham—Burdine—Campbell model. Moldrup et al. (2000) further introduced the AFP at 100 cm H_2O of soil—water matric potential to describe soil structure effects on gas diffusivity.

4. The fourth group of models consists of generalized power law models that introduce additional, empirical model parameters and thereby can provide a good fit to $D_{gs}(\theta_g)$ data within the θ_g interval where measurements are available. The most frequently used within this group is the Troeh et al. (1982) model where two additional fitting parameters are introduced. The Troeh et al. (1982) model was successfully used in several studies to fit and subsequently represent measured $D_{gs}(\theta_g)$ data in gas transport and fate models (Petersen et al., 1994, 1996). Although one of the Troeh et al. (1982) model parameters can be interpreted as the AFP where gas diffusion ceases due to interconnected water films (creating blocked pore space), no relationships between the Troeh et al. (1982) model parameters and soil physical properties have been identified. The model at present is descriptive rather than predictive (Moldrup et al., 2003).
5. The fifth group consists of two- or three-region $D_{gs}(\theta_g)$ models that partition the pore space into, for example, easily accessible, difficult accessible, and nonaccessible pore space (Arah and Ball, 1994). The models have typically been developed for highly structured or highly aggregated artificial porous media or repacked soil aggregates and include the classical model by Millington and Shearer (1971). The models contain additional pore shape and pore region parameters and are mainly descriptive models that can be used to fit detailed $D_{gs}(\theta_g)$ data. Moldrup et al. (2004) introduced a three-porosity model that requires only one point (at -100 cm H₂O) on the SWC and predicts the gas diffusion coefficient as well as those who use the entire SWC and better than typically used soil type independent models.
6. The sixth group of $D_{gs}(\theta_g)$ models consists of macroscopic pore-size distribution models based on equivalent pore radius capillary tube, jointed tubes of different radii, or multidimensional capillary tube networks (Ball, 1981; Nielson et al., 1984; Steele and Nieber, 1994). Further, Freijer (1994) established links between this type of model and the multiparameter Mualem–van Genuchten SWC model (van Genuchten, 1980). The $D_{gs}(\theta_g)$ models in this group at present have several empirical constants that must be fitted to actual $D_{gs}(\theta_g)$ data for the soil. Caron and Nkongolo (2004) have recently introduced a rapid method for the estimation of gas diffusivity in situ in peat substrates that is based on water storage and flow measurements.

3.6.3 MODELING GAS TRANSPORT IN SOILLESS MEDIA

Under the assumption that gas transport in soils and growing media is basically diffusion-dependent, both vertical (long distance) and horizontal (short distance) transport is driven mainly by concentration gradients. In horizontal transport, the geometry of aggregates and the liquid phase are the major components of resistance for diffusion.

Eq. (3.70) describes the flux of gas (oxygen, for example) but it does not account for its consumption. Thus, to account for more realistic cases, the mass balance equation with a sink term should be used in order to account for local concentration changes and gas component depletion. Eq. (3.53) and its radial and cylindrical forms can be used to model different cases of gas transport. Note that the solute concentration and diffusion coefficient should be replaced by the proper terms for the relevant gases.

3.7 WETTABILITY EVALUATION AND ITS EFFECT ON HYDRAULIC PROPERTIES AND FLOW

3.7.1 DIRECT MEASUREMENT OF SOILLESS MEDIA WETTABILITY

There are several methods to measure the resistance of porous media (soil, soilless media, powders, etc.) to wetting in general and by water in particular, but two general concepts form the basis of most methods: (1) indirect quantification of the affinity of the soil to the liquids by imbibition/infiltration and (2) measuring the liquid–solid CA. The two approaches are detailed in the following sections.

3.7.1.1 *Water drop penetration time and molarity of ethanol droplet tests*

SWR is characterized by its persistence and intensity (severity). The common method of quantifying water repellency persistence is the water drop penetration time (WDPT) test (Doerr, 1998). The WDPT test consists of placing couple of droplets (three and higher are suggested to enable statistics calculations) of distilled water on the soil/substrate surface and recording the time taken for it to penetrate completely into the soil/substrate. The water repellency persistence indirectly expresses the dynamic change in CA, the direct measurement of which is more complicated. The average time for the droplets is reported as the WDPT. Five classes are usually used to classify the soil/substrate repellency degree: class I, wettable (infiltration will start within 5 seconds); class II, slightly water repellent ($5 < \text{WDPT} \leq 60$ seconds); class III, strongly water repellent ($60 < \text{WDPT} \leq 600$ seconds); class IV, severely water repellent ($600 < \text{WDPT} \leq 3600$ seconds); class V, extremely water repellent ($\text{WDPT} \geq 3600$ seconds) (Bisdorn et al., 1993). The water drops that penetrate the soil usually maintain a well-defined, spherical shape and can be easily removed with a spatula.

The molarity of ethanol droplet (MED) test is another popular rapid method for assessing SWR under field and laboratory conditions. The difference between the WDPT and MED tests is that the molarity of water and ethanol mixtures is varied repeatedly, until droplets of the mixture infiltrate into the dry soil in a pre-given length of time (Watson and Letey, 1970; King, 1981; de Jonge et al., 1999). A penetration time of 5 seconds for a droplet defines a CA of 90 degrees. From the volume fraction of the ethanol in the mixture, the corresponding surface energy of the liquid is determined and defined as the 90 degrees surface energy (Letey et al., 2000). Roy and McGill (2002) suggested a standardized procedure for MED test to improve its reproducibility.

Both WDPT and MED tests are useful because of the easy application, especially in the field/greenhouse. However, both methods have disadvantages. WDPT test possibly alters the solid surface during the measurement. Therefore the result is likely affected by both the impact of the initial CA and the persistence of the repellency (Roy and McGill, 2002). Moreover, dissolution of soil organic and inorganic compounds during the measurement may change the liquid surface energy either to smaller (Anderson et al., 1995) or larger values. Graber et al. (2007) studied whether the reduction of δ_{lv} can result from dissolution of soil-borne surface-active organic compounds into the water drop by measuring the drop penetration time of water extracts from repellent soils obtained after varying extraction times and at different soil:water ratios. It was indeed found that soil extracts had lower surface tensions (δ_{lv} , approximately 51–54 mN/m) than distilled water. However, the drop penetration time of the soil extracts in

water repellent soils was generally the same or greater than that of water. In contrast, drop penetration time of ethanol solutions prepared with the same δ_{lv} , electrical conductivity, and monovalent/divalent cation ratio as the soil extracts was much faster. Graber et al. (2007) then concluded that the surface-active, soil-derived organic substances in aqueous soil extracts do not contribute to wetting dynamics, and as such, this mechanism for explaining kinetics of water penetration into water repellent soils is rejected. It is also concluded that the rapid penetration of ethanol solutions, which are usually used as an agent to reduce δ_{lv} and, as such, to reduce drop penetration time, is must due not only to changes in δ_{lv} , but to also to changes in either or both δ_{lv} , δ_{sl} and the solid–vapor interfacial energy (δ_{sv}). Moreover, Bachmann et al. (2003) noted that MED test is somewhat unsatisfactory, because of its empirical nature and the lack of a rational foundation

3.7.1.2 Sessile drop method

The methods discussed so far assess the CA and surface energy of porous materials in an indirect way. Other methods aim at measuring CAs directly.

Bachmann et al. (2000) suggested a method for direct measuring the CA of a soil while eliminating the effect of liquid penetration through its surface. The method includes a small glass plate (“slide”) that is covered with a double-sided adhesive tape and pressing onto it a homogeneous one-grain layer of soil particles of similar diameter. Droplets of water are then placed on the surface with a syringe and the CA of each droplet is measured using the tangent of a special goniometer fitted microscope.

Unfortunately, because of capillary forces, a thin layer wicking effect on wettable samples can occur, and therefore this so-called sessile drop method (SDM) is restricted to hydrophobic particles. However, experiments applying pressed powder pellets (Valat et al., 1991; Chibowski et al., 2002) showed that SDM can provide reasonable results.

As the initial CA is a measure of the initial interfacial energies at the solid surface (γ_{sl} or γ_{sv} for soil–liquid and soil–vapor, respectively) prior to their change by rearrangement of organic molecules coating soil–particle surfaces, it can be used as an indirect measure for water repellency intensity. The SDM can be further used to study the decrease in CA during the contact time of the liquid with the soil particles (Bugici and Wallach, 2016). Graber et al. (2007) found that the reduction in γ_{sl} can result from conformational changes, hydration, or rearrangement of organic molecules coating soil–particle surfaces as a result of contact with water. The initial and decrease of CA with contact time between a water drop and a single-grain layer of two hydrophobic soils are depicted in Fig. 3.29. The water repellent soils were taken from the surface of two adjacent olive orchards of different ages (RM1 orchard is younger than RM2) grown in sandy soil (87% sand). The soil rendered water repellent mainly by the waxes of trees litter and roots (Mao et al., 2014). The initial CAs measured for the single-particle layer were below 90 degrees, despite the fact that soils from these two plots showed a certain degree of water repellency (WDPT > 60 seconds). Note that the initial CA measured at the surface of the bulk soils was >90 degrees. The initial CA for the RM2 soil was higher than for the RM1 soil, indicating a higher intensity of water repellency for the former. The respective rate terms for the RM1 soil were four time larger than those for the RM2 soil and could be used as a quantitative measure of water repellency persistence.

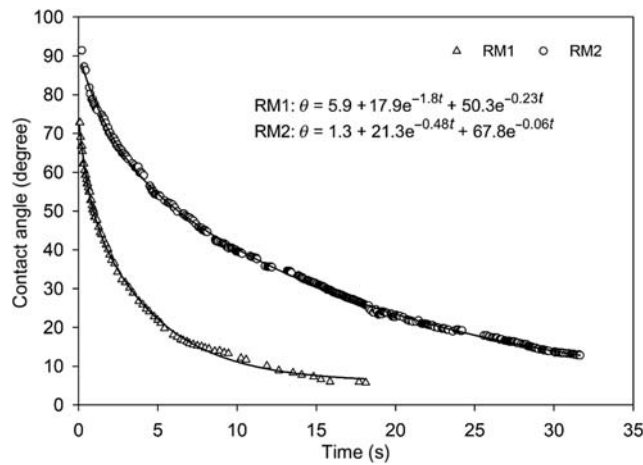


FIGURE 3.29

Measured and fitted change of contact angle with time as measured for a sessile drop on a single-grain layer for soils that rendered hydrophobic olive litter. RM1 notes a young olive orchard whereas RM2 a mature olive orchard.

Data taken from Bugici, T., Wallach, R., 2016. Formation of soil–water repellency in olive orchards and its influence on infiltration pattern. *Geoderma* 262, 1–11. doi:10.1016/j.geoderma.2015.08.002.

3.7.1.3 Capillary rise method

The capillary rise method is an alternative procedure for determining CAs by using a completely different physical principle. It examines the imbibition rate of porous materials by a specific liquid (Siebold et al., 1997, 2000) and its final height and has become an accepted technique for measuring advancing CAs of powders and for deriving other parameters of solid surfaces. A major disadvantage of the capillary rise method is that the testing fluid does not rise into the sample, if the liquid/solid CA is larger than 90 degrees. This excludes hydrophobic soil material, when water is used as the testing liquid, as the CA decreases with contact time between the particles with water (note that this change in CA with time is the basis for the WDPT test). Yet, similar to the use of ethanol/water mixtures to replace the WDPT, methanol–water mixtures of varying composition (and varying surface energies) can replace the pure water as wetting agents for hydrophobic soil/substrates with initial CA of 90 degrees (Bachmann et al., 2003).

The air-dry porous medium to be studied is packed in a glass tube, which is closed at the lower end with a filter. The base of the upright tube is brought into contact with water or other test fluid in a reservoir (Fig. 3.5), and the mass gain of the tube recorded with an electronic balance has been usually used to estimate the wetting from height in the tube. In the early stages of the liquid penetration, the hydrostatic pressure can be neglected and according to Washburn equation

$$h^2 = \frac{R\gamma_{lv}\cos\varphi}{2\eta} \cdot t \quad (3.73)$$

where R is the mean radius of pores in the tube, assuming a bundle of parallel capillaries, γ_{lv} is the surface energy of the liquid–air interface, φ is the advancing CA, and η the viscosity of the liquid.

The maximum height h_m that is reached by the wetting front above the base of the tube can be expressed, similar to Eq. (3.10), as

$$h_m = \frac{2\gamma_w \cos \varphi}{\rho g R} \quad (3.74)$$

where φ is the static CA, ρ the density of the fluid, g the acceleration due to gravity. Eq. (3.74) is known also as Jurin equation. Attention should be paid that Eqs. (3.73) and (3.74) cannot be used to determine the CA directly because h_m is a function of two unknown parameters, R and φ . An additional capillary rise experiment is then needed, using a fully wetting fluid (e.g., dodecane, octane) for which $\cos \varphi \approx 1$, for determining R . This calculated R can then be introduced to Eq. (3.73) to calculate the contact angle φ when water is used (Wallach et al., 2013). Additional liquids of low surface tension that can be used for capillary rise experiments appear in Table 1 in Siebold et al. (2000).

Michel et al. (2001) and Kerloch and Michel (2015) used the capillary rise method with Washburn equation (Eq. 3.73) to measure the CA for peat, coir, pine bark, and wood fiber. In spite of the fact that these substrates are known as water repellent when they are air-dry, all measured CAs were <90 degrees (Table 1 in Kerloch and Michel, 2015) which categorize these substrates as subcritically water repellent. The reason for the relatively low CA could be related to the experimental conditions where the substrates were equilibrated at -10 kPa water potential prior to packing in the tube. The dependence of CA on water content was studied by King (1981), de Jonge et al. (1999), and Regalado and Ritter (2005) who found that water repellency varies nonlinearly with the water content. In general soils are wettable close to saturation, becoming water repellent up to a maximum as moisture content decreases. After this, maximum water repellency diminishes monotonically with decreasing water content or rises again to a second local maximum (de Jonge et al., 1999). Repellency is usually measured at fixed water content, without considering soil moisture dependency. Therefore, it would be useful to derive parameters from the repellency versus water content curve and integrate the water dependence behavior of soil repellency, which must be considered as a dynamic property (Regalado and Ritter, 2005). To the best of our knowledge, the dependence of substrates' CA on water content has not been measured so far.

3.7.2 WATER REPELLENCY EFFECT ON CAPILLARY PRESSURE—SATURATION RELATIONS (RETENTION CURVES)

The dependence of capillary pressure—saturation relations of porous media on wettability has been rarely studied for soils, and particularly for soilless media. Morrow (1975, 1976) studied this dependence by preparing smooth plates and rough capillaries using PTFE (polytetrafluoroethylene) as a test material, and for porous medium. By preparing smooth plates, rough capillaries, and a porous medium of the same material, he was able to demonstrate the systematic dependence of the advancing and receding CA on the intrinsic CA φ . The change in CA was imposed by liquids with different surface energies. Morrow showed that drainage and imbibition capillary pressure—saturation relations of the porous PTFE medium were proportional to the cosine of receding and advancing CAs observed at rough surfaces. Diamantopoulos et al. (2013) showed that the water retention functions and the unsaturated hydraulic conductivity functions

estimated from the dynamic experiments are strongly dependent on the degree of water repellency and the wetting/drying process.

Naasz et al.'s (2008) study aimed to estimate the influence of changes in wettability on the CA and water RCs of peat and pine bark during a drying/wetting cycle. Peat showed considerable variation in wettability related to water potential, and considerable hysteresis in CA measurements (20%) and in the water RC (21%). The wettability of pine bark also varied with water potential, but very limited hysteresis was observed in CA measurements (<5%) and the response was much less pronounced in the RC (10%). The hysteresis in the RCs measured by Naasz et al. (2008), with higher water contents at similar matric potentials for the wetting than drying curves, contradicts the common findings for mineral soils for which the water contents along the drying curve are higher than for a wetting curve at similar matric potentials. Note that the measured hysteresis for peat and rockwool that is presented in Fig. 3.9 agrees with the common hysteresis pattern for mineral soils. The general explanation for the lower water content along the wetting curve is the higher advancing CA that yields a higher water entry value. The relationship between the wetting and drying RC in Naasz et al. (2008) was in spite of the measured higher advancing CA upon wetting. The discrepancy between the typical wetting/drying curves for mineral soils and the measured organic substrates could result from shrinkage/swelling of the latter that was not measured in Naasz et al.'s (2008) study.

3.7.3 WATER REPELLENCY AND FLOW IN SOILLESS SUBSTRATES

The effect of water repellency on flow pattern in water repellent soils, and the formation of fingered-shaped flow that is caused by wetting front instability in particular, has been widely studied for mineral soils in the field and laboratory alike (Hendriks et al., 1993; Ritsema and Dekker, 1996; Ritsema et al., 1998; Bauters et al., 1998; Wang et al., 2000; Wallach and Jortzick, 2008; Xiong et al., 2012; Bugici and Wallach, 2016; Rahav et al., 2017). Moreover, Wallach et al. (2013) found, using a flow chamber study, that fingered flow develops for subcritically water repellent porous media ($\varphi < 90$ degrees). Furthermore, Lamparter et al. (2006) studied the effect of subcritical repellency in a sandy soil on water infiltration and preferential flow formation by using an infiltrometer. They found that water infiltration rates were reduced due to subcritical repellency by a factor of 3–170 compared with the rate of infiltration of ethanol (φ assumed to be 0 degree). Although preferential flow was not observed on their experimental scale (infiltrometer), such a flow regime was observed in a different tracer experiment at the same site, where a 10 m transect was examined under natural rainfall conditions (Deurer et al., 2001).

Despite the fact that resistance to wetting of dry organic substrates and the formation of preferential flow in pots have been known to growers and researchers, studies similar to the abovementioned ones for mineral soils have not been carried out, to the best of our knowledge, for organic water repellent substrates. Combining the findings that the CA for peat, coir, pine bark, and wood fiber categorized them as subcritically water repellent (Michel et al., 2001; Naasz et al., 2008; Table 1 in Kerloch and Michel, 2015) with the abovementioned findings on finger flow development in subcritically water repellent mineral soils reassures the applicability of the theory that has been developed for water repellent mineral soils to mineral soilless substrates.

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CHEMICAL CHARACTERISTICS OF
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ABBREVIATIONS

AEC	anion exchange capacity
CEC	cation exchange capacity
DCPD	$\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$
HAP	$[\text{Ca}_5(\text{PO}_4)_3\text{OH}]$
OCP	$(\text{Ca}_4\text{H}(\text{PO}_4)_3 \cdot 2.5(\text{H}_2\text{O}))$
P_w	solution P concentrations
PZSE	point of zero salt effect
SAI	specifically adsorbed ion
βTCP	$(\text{Ca}_3(\text{PO}_4)_2)$
ZPT	suspension pH prior to the addition of protons or hydroxyls

Typical soilless media are commonly defined by root constraint and limited volume of substrate and accordingly, low buffer capacity for solution composition and limited supply of nutrients. It is generally accepted in the horticultural industry that fertilization of soilless-grown plants is more intensive from that of soil-grown plants (Sonneveld and Voogt, 2009) and, therefore, optimal plant production necessitates some specific and different attention to the chemical characteristics of the substrates. The surfaces of particles and other solids in horticultural substrates bear permanent and/or variable electrical charges. The surface charge properties and the dissolution characteristics of the substrate solids are important since they affect the ionic composition of nutrient solutions. Ordinarily, the chemical properties of substrates are established prior to their use for growing plants; therefore undisrupted, homogeneous solid phases are addressed. However, throughout the growth period, organic compounds excreted from plant roots or resulting from decomposition processes accumulate in the substrate (Tate and Theng, 1980; Huang and Violante, 1986; Tan, 1986; Silber and Raviv, 1996). Thus the surfaces of the newly formed solids become heterogeneous and the chemical properties of the mixture may be significantly different from those of the single, well-defined component that was characterized before use (Silber and Raviv, 1996). Plants growing in soilless culture typically have smaller root-system volume than plants growing in soil. Therefore the root density of soilless-grown plants is higher so that such plants have a greater impact on the rhizosphere than soil-grown plants. Despite extensive research related to irrigation and fertilization

management or plant growth in soilless culture, the literature regarding the chemical properties of soilless media is scarce. Although peat, perlite, stone wool, and mixtures of organic with inorganic materials are abundant in the horticulture industry, the literature regarding the chemical properties of these media is insufficient. Tuff is a volcanic material used as a substrate for horticultural crops in Italy, Spain, France, Turkey, and Israel, which has been the subject of considerable research with respect to chemical properties as a horticultural substrate. Thus tuff is used in this chapter as a model for illustration and elucidation of the chemical processes taking place in soilless media systems.

4.1 CHARGE CHARACTERISTICS

The surfaces of solids in horticultural substrates of volcanic or organic origin carry permanent and/or variable positive and negative electrical charges. Permanent negative charge results from isomorphous substitutions (Gast, 1977; McBride, 1989, 2000; Sumner and Miller, 1996; Sposito, 1989, 2000), that is, substitution of structural cation by lower valency cation that has the same coordination number and size in layer silicates (common examples: Si^{4+} by Al^{3+} or Al^{3+} by Mg^{2+}). The extent of cation adsorption to the surfaces is referred to as the *cation exchange capacity* (CEC; cmol/kg) and is used to characterize the cation exchange properties of the medium (Gast, 1977; Sposito, 1989).

The CEC of several inorganic and organic horticultural substrates is presented in Table 4.1. These values represent CEC measurement of intact materials prior to use, and it is important to note that plant growth may affect the chemical properties of the substrate solids. CEC values of soil components are usually referred by weight (i.e., cmol/kg). However, soilless production

Table 4.1 Cation Exchange Capacity (CEC) of Several Horticultural Substrates

Substrate	CEC		Reference
	cmol/kg	cmol/L	
Inorganic substrates			
Perlite	25–35	2–4	Doğan and Alkan (2004)
Stone wool	34	5	Argo and Biernbaum (1997)
Tuff	10–60 ^a	10–60	Silber et al. (1994)
Clinoptilolite (zeolite)	200–400	400–800	Mumpton (1999)
Organic substrates			
Coconut coir	39–60	2–4	Evans et al. (1996)
Peat	90–140	7–11	Puustjarvi and Robertson (1975)
Pine bark	98 ^b	10	Daniels and Wright (1988)
Compost	160–180	15–20	Inbar (1989)

^aCEC in pH 7, tuff has variable charge surfaces as shown in Fig. 4.2.

^bCEC in pH 7, pine bark has variable charge surfaces as shown in Fig. 4.4B.

systems involve a relative small fixed root zone volume and, therefore, it is more practical to relate substrates CEC to volume unit (cmol/L). By using volume-based comparison, it is possible to properly compare both “heavy” and “light” media.

The variable charge (both negative and positive, depending on the solution-pH) is generated mainly from the adsorption of H^+ and OH^- on solid surfaces such as metal oxides, hydroxides, microcrystalline silicates (allophane and imogolite), or on functional organic groups (Stevenson, 1994). The effects of solution-pH on the magnitude of the surface charge may be determined experimentally by measuring the CEC and the *anion exchange capacity* (AEC) over a range of pH values. The pH at which the AEC is equal to the CEC is referred to as the *point of zero net charge* (Parker et al., 1979; Sposito, 1981, 1984, 2000) and is frequently used to characterize the charge properties of the medium. Note that in all cases, pH is referred to the solutions. An alternative experimental method is used to conduct a pH titration curve using an indifferent electrolyte (Gast, 1977; McBride, 1989, 2000; Sparks et al., 1996). A typical potentiometric titration for woody peat is presented in Fig. 4.1 (Bloom, 1979).

The pH value, at which a series of titration curves for various electrolyte concentrations intersect, is referred to as the *point of zero salt effect* (PZSE; Parker et al., 1979; Sposito, 1981, 1984, 2000). However, common substrates (compost, peat moss, tuff, stone wool) rarely consist of a single, well-defined component. They typically contain a mixture that presents both permanent and

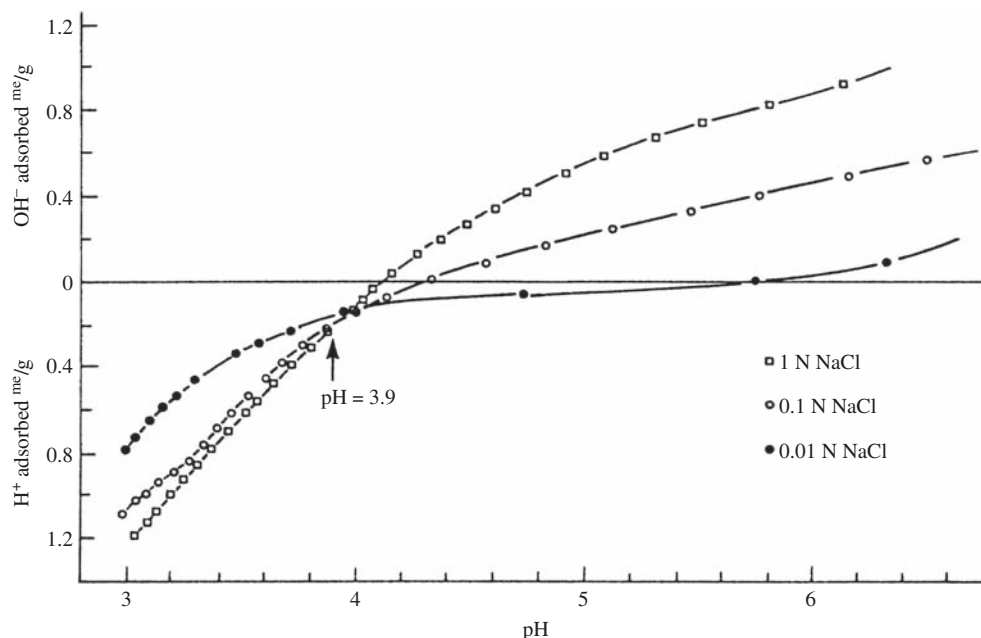


FIGURE 4.1

Acid–base potentiometric titration curves of woody peat.

Reprinted from Bloom, P.R., 1979. Titration behavior of aluminum organic matter. *Soil Sci. Soc. Am. J.* 43, 815–817, with kind permission from the Soil Science Society of America Journal.

variable surface charges. The interpretation of the experimental CEC, AEC, or pH titration data obtained from these mixtures is, therefore, complicated. Moreover, the existence of additional sources or sinks for H^+/OH^- derived from the dissolution/precipitation of minerals during the CEC, AEC, or pH titration analyses makes direct interpretation of analytical results even more ambiguous.

The difficulties involved in the determination of the actual surface charge of a heterogeneous material that contains both pH-dependent and permanently charged surfaces, together with primary and secondary minerals, can be seen in the case of tuff (Figs. 4.2 and 4.3). Yellow, red, and black tuff, commonly used in greenhouse production, are characterized by the degree of weathering, with yellow tuff being most weathered and black tuff being the least weathered. The CECs of yellow, red, and black tuff were found to be pH-dependent, with the CEC increasing by 6.6, 4.2, and 1.9 cmol/kg, respectively, for every unit increase of pH (Silber et al., 1994).

The differing values of CEC slope among the three types of tuff reflect differences in their contents of amorphous materials with variable charged surfaces. Silber et al. (1994) found however that it was not possible to determine the AEC of these tuffs because phosphorus released from indigenous Ca–P minerals during their analysis was readsorbed on the tuff surfaces and thus interfered with the AEC determination. The CEC of tuff at pH 3.5, especially that of the weathered yellow tuff, was relatively high (35 cmol/kg) and may indicate the presence of components having a permanent charge. The pH titration curves of the tuff types differed significantly from those of homogenous variable charged materials. In the cases of the less-weathered tuffs (red tuff and black tuff), the pH titration lines obtained with three different NaCl concentrations overlapped (data not presented). Distinction of an ionic strength effect was possible only for the yellow tuff at a pH above 7 (Fig. 4.3A).

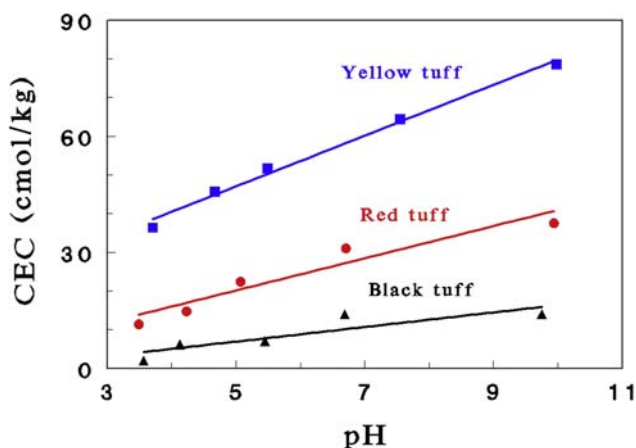


FIGURE 4.2

Cation exchange capacity versus pH for black, red, and yellow tuffs from northern Israel.

Reprinted from Silber, A., Bar-Yosef, B., Singer, A., Chen, Y., 1994. Mineralogical and chemical composition of three tuffs from northern Israel. *Geoderma* 63, 123–144, with kind permission from Elsevier Science.

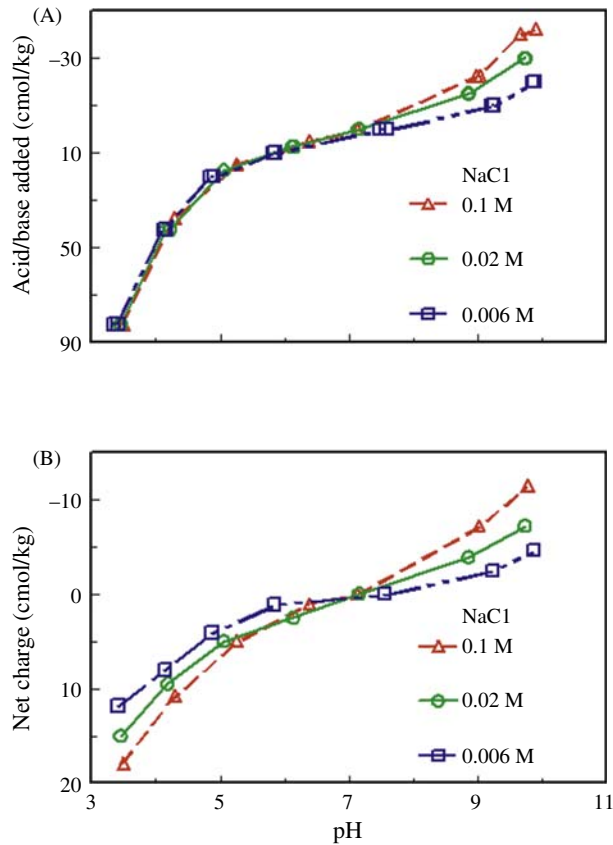


FIGURE 4.3

Potentiometric titration curves of yellow tuff: (A) acid/base addition (0.1 N HCl and NaOH, respectively) as functions of pH and NaCl concentrations, tuff/solution ratio of 1/40, equilibrium period of 1 week. ZPT is the suspension pH prior to the addition of acid or base and (B) net surface charge calculated with Eq. (4.1).

Based on Silber, A., 1991. *Chemical Properties and Surface Reactions of Pyroclastic Materials From Mt. Peres, the Golan Heights* (Ph.D. thesis). Hebrew University of Jerusalem (in Hebrew with English abstract), pp. 171.

The overlap of potentiometric titration curves at low pHs in volcanic material from Chile was attributed to the exchange of added H^+ ions with cations associated with a permanent negative charge (Espinoza et al., 1975; Gast, 1977). Wann and Uehara (1978) reported that specific adsorption of P caused overlapping of potentiometric titration curves of soil; therefore readsorption of P may be partially responsible for the overlap of the pH titration curves in the case of tuff. Nevertheless, on the assumption that H^+/OH^- consumption occurs in cation/anion exchange, the net charge at each pH value can be evaluated by subtracting the net quantity of cations/anions

accumulated in the liquid solution of the substrate (Δcharge) from the quantity of acid/base added at the pertinent pH. Hence, Δcharge (cmol/kg) is defined as

$$\Delta\text{charge} = (\sum \text{cat} - \sum \text{anion})\text{pH} - (\sum \text{cat} - \sum \text{anion})\text{ZPT} \quad (4.1)$$

where ZPT is the suspension pH prior to the addition of protons or hydroxyls (in Fig. 4.3A: 7.1, 7.2, and 7.6 in 0.1, 0.02, and 0.006 M, respectively) and $\sum \text{cat}$ and $\sum \text{anion}$ include all the cationic and anionic species in the solution, according to speciation calculations. The differences between the quantities of acid/base added to tuff suspensions and the net surface charge calculated with Eq. (4.1) (Fig. 4.3A and B, respectively) demonstrate that the major quantities of acid/base added to tuff solutions were exchanged with adsorbed cations/anions or consumed in the dissolution of indigenous Ca–P minerals and of very fine amorphous particles (Silber et al., 1999).

The CEC of common organic substrates, such as peat, pine bark, or composts, is generally high (80–160 cmol/kg) and is pH-dependent (Brown and Pokorny, 1975; Puustjarvi, 1977; Ogden et al., 1987; Daniels and Wright, 1988). The charge is derived mainly from ionization of COOH groups and, to a lesser extent, from phenolic OH (Stevenson, 1994). The pH effect on CEC of organic material was found to be more pronounced than that of soil inorganic materials (Hallsworth and Wilkinson, 1958; Helling et al., 1964; Ogden et al., 1987; Daniels and Wright, 1988; Stevenson, 1994). The contribution of a unit pH increase to the CEC of soil organic material was found to be 51 cmol/kg of organic C (Fig. 4.4A).

Assuming that organic C constituted 58% of the organic matter (Stevenson, 1994), then each unit increase in pH raised the CEC of soil organic matter by 29.6 cmol/kg. Cation exchange properties of pine bark growing media increased linearly between pH 4 and 7 and the contribution of a unit pH increase to the CEC was 23.1 cmol/kg for particles of size smaller than 0.05 mm (Fig. 4.4B). Note that the increase in measured CEC slope of the fine pine bark was close to the calculated value of soil organic matter and that these were an order of magnitude higher than the aforementioned slopes of 1.9, 4.2, and 6.5 cmol/kg per pH unit measured for black, red, and yellow tuff, respectively (Fig. 4.2).

The charge characteristics of composts were found to be dependent on composting time (Harada and Inoko, 1980; Inbar, 1989; Inbar et al., 1989, 1991; Iglesias-Jimenez and Perez-Garcia, 1992; Saharinen, 1996; Jokova et al., 1997), mainly because of transformations of organic constituents such as C/N ratio, humic material, and lignin contents during the composting process (Inbar et al., 1989, 1991; Saharinen, 1996). The C/N ratio and the lignin concentration correlated with the composting time (Raviv et al., 1987; Tarre et al., 1987; Inbar, 1989; Inbar et al., 1989, 1991) and could be used in estimating the CEC (Fig. 4.5) and maturity indices of compost quality.

4.1.1 ADSORPTION OF NUTRITIONAL ELEMENTS TO EXCHANGE SITES

The affinity of cations for negatively charged surfaces under equal concentration in the aqueous portion of the substrate is affected by ion characteristics such as valence, size, and hydration status. The affinity of divalent cations is higher than that of monovalent cations because of their greater charge and usually, the relative adsorption strength follows the order of hydration, that is, $K \approx \text{NH}_4 > \text{Na}$ and $\text{Ca} > \text{Mg}$ (Barber, 1995). However, concentrations of each of the cations in a typical irrigation solution for greenhouses crops are not equivalent, with the K concentration

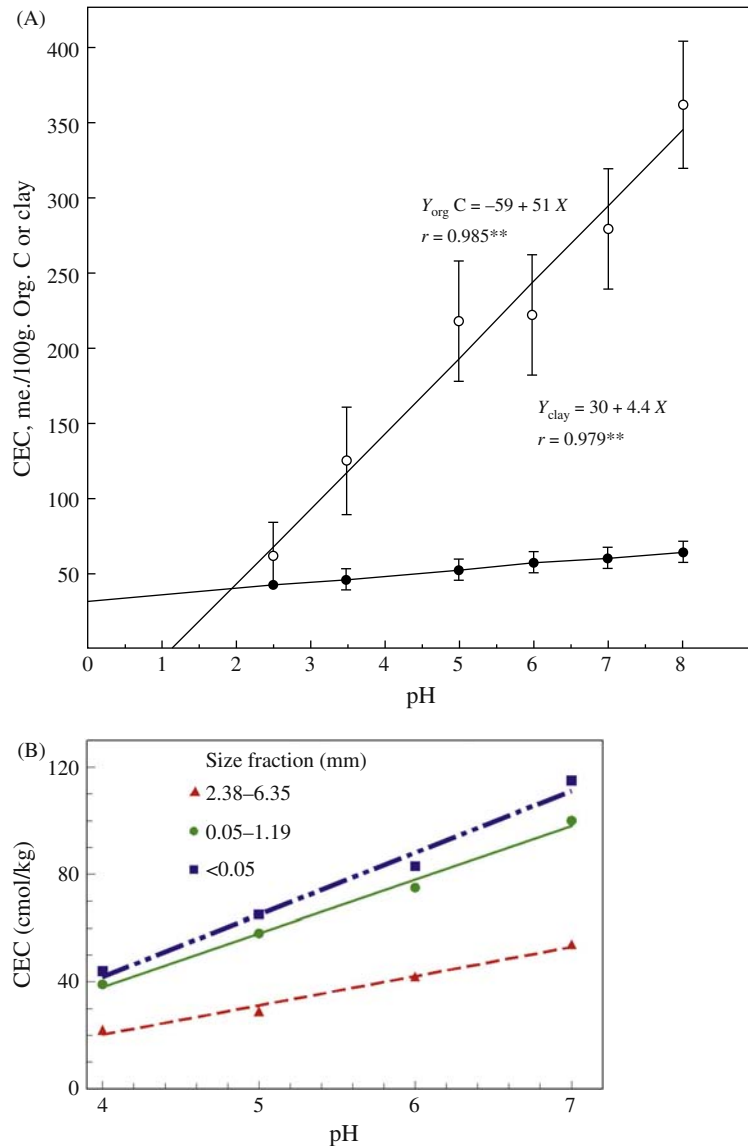
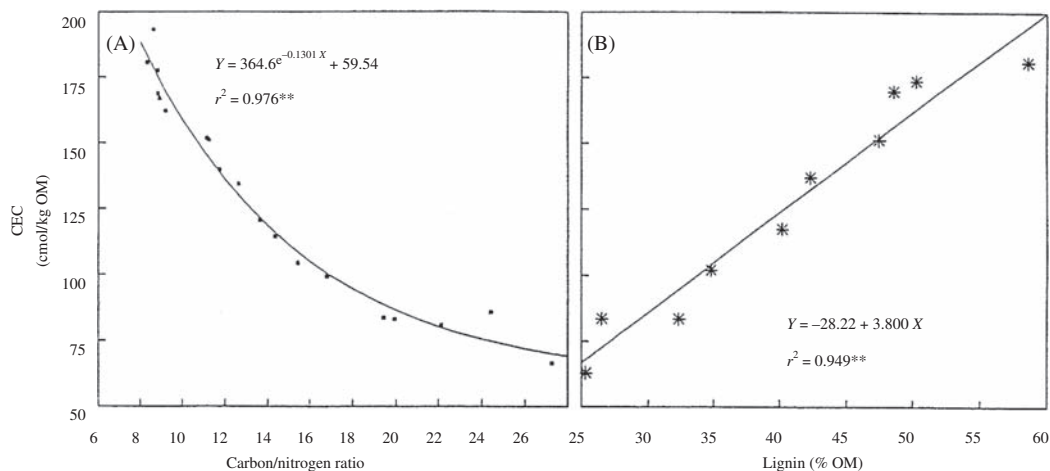


FIGURE 4.4

Effect of pH on the CEC of (A) soil organic C ($Y_{\text{org C}}$) and clay (Y_{clay}). Vertical lines show standard errors of the individual values and (B) several particle size fraction of pine bark growing media. AEC, Anion exchange capacity.

(A) From Helling, C., Chesters, G., Corey, R.B., 1964. Contribution of organic matter and clay to soil cation-exchange capacity as affected by the pH of the saturating solution. *Soil Sci. Soc. Am. J.* 28, 517–520, with kind permission from the Soil Science Society of America Journal. (B) Based on Daniels, W.L., Wright, R.D., 1988. Cation exchange properties of pine bark growing media as influenced by pH, particle size, and cation species. *J. Am. Soc. Hortic. Sci.* 113, 557–560.

**FIGURE 4.5**

Cation exchange capacity of compost during the course of the composting process as functions of (A) carbon/nitrogen ratio, and (B) lignin content.

Based on Inbar, Y., 1989. *Formation of Humic Substances During the Composting of Agricultural Wastes and Characterization of Their Physico-Chemical Properties* (Ph.D. thesis). Hebrew University of Jerusalem.

commonly exceeding the concentration of Ca, NH_4 , or Mg (Adams, 2002; Sonneveld, 2002). As an example, recommendations for K, NH_4 , Ca, and Mg concentrations in the nutrient solution for soilless production of ornamental and vegetable crops in the Netherlands (Naaldwijk) consist of 5.0, 1.0, 4.5, and 2.5 mmol/L, respectively (Sonneveld, 2002). Thus it is expected that exchange reaction, that is, displacement of cations from the substrate solids by K^+ , will reduce its concentration and subsequently will increase the concentration of other cations in the liquids within the substrate. The exchange reaction may be horticulturally beneficial in cases where (1) concentration of a toxic or detrimental cation such as heavy metals or Na^+ in the liquid solution of the growing medium declines as a result of adsorption or (2) concentration of a beneficial cation in that liquid increases as a result desorption. Alternatively, the opposite could occur, with expected detrimental effects, of cation exchange including (1) increased concentrations of a toxic or detrimental cation in the solution followings desorption or (2) decreased concentrations to a deficiency level of beneficial cation followings its adsorption to substrate solids.

4.2 SPECIFIC ADSORPTION AND INTERACTIONS BETWEEN CATIONS/ANIONS AND SUBSTRATE SOLIDS

The solubility and the interactions of nutritional elements with the substrate solids are associated with ion characteristics such as valence, size, and hydration status as detailed above. The solubility of NO_3^- , the major N source for plants grown in soilless culture (Chapter 5: Root Zone

Microbiology of Soilless Cropping Systems), is very high, its affinity to positive charged surfaces is very low compared with that of H_2PO_4^- or SO_4^{2-} , and therefore, the concentrations of NO_3^- in the substrate solution are nearly unaffected by precipitation and/or adsorption reactions. The effects of microbes on NO_3^- availability in the rhizosphere are detailed elsewhere (Chapter 5: Root Zone Microbiology of Soilless Cropping Systems). NH_4^+ is also a N source for plants in soilless production and, similar to NO_3^- , generally does not precipitate. Unlike NO_3^- , the high affinity of NH_4^+ to negatively charged surfaces affects its availability to plant. Nitrification/denitrification reaction and the effects of NH_4^+ nutrition on rhizosphere pH and the availability of other nutritional elements are discussed in Chapter 5, Root Zone Microbiology of Soilless Cropping Systems. Potassium is an important nutritional element and is also not involved in precipitation reactions to an appreciable extent. However, as a charged cation, K^+ may be adsorbed by negatively charged surfaces, and this may reduce its availability to plants. Note that the partitioning of K^+ as well as other cations between the aqueous and the solid parts of the substrate depend on its relative concentration in the solution and composition of the exchange sites. The affinity of the divalent cations Ca^{2+} and Mg^{2+} to negatively charged surfaces is very high (Barber, 1995) and they are highly involved in precipitation reactions, especially with H_2PO_4^- and SO_4^{2-} . Actually, it is well accepted that P availability in the rhizosphere is governed by Ca^{2+} and Mg^{2+} activities (Lindsay, 1979). Metal micronutrients such as Cu, Fe, Mn, and Zn are considered as specifically adsorbed ions (SAIs); their affinity to charged surfaces is very high and considerably higher than other cations. These metals are also highly involved in precipitation reactions so that their availability to the plant is very low. Fertiligation managements used in the horticulture industry for guaranteeing the availability of nutritional elements for plants, including that of micronutrients, are discussed in Chapter 5, Root Zone Microbiology of Soilless Cropping Systems.

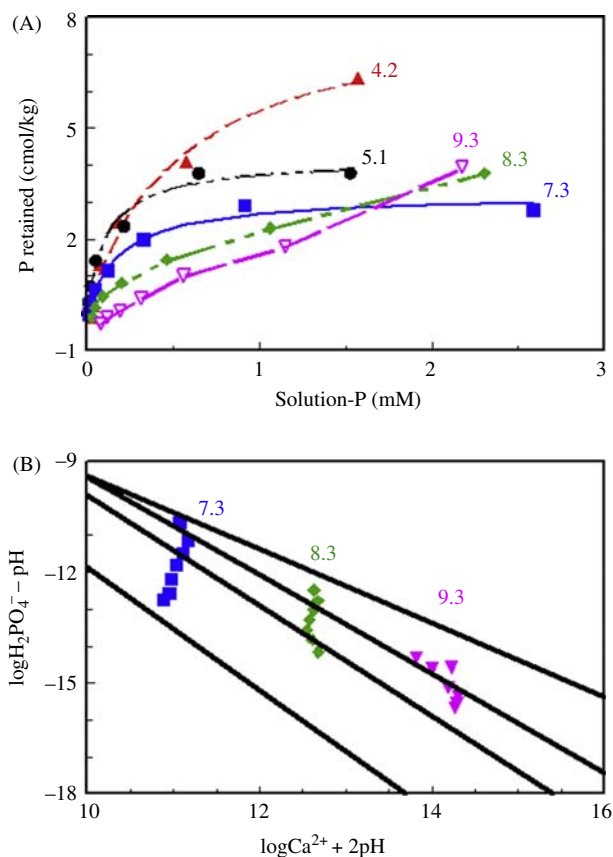
Phosphorus and zinc were selected below for detailed discussion to illustrate the interactions of cations and anions with the solid constituents of substrate because (1) both are important nutritional elements for plants; (2) under common agricultural practices, low availability in the vicinity of the roots of either of these elements often limits crop growth and productivity, even if P and Zn concentrations in the irrigation solutions are adequate (Marschner, 1995); (3) ions of both elements are classified as SAIs because of their high affinity to charged surfaces; and (4) both elements easily form new solid compounds as a result of precipitation reactions.

4.2.1 PHOSPHORUS

Continuous decline of solution P concentration (P_w) following fertilizer application is a widespread phenomenon, and temporary P deficiency often restricts crop productivity. P_w decreases through two mechanisms: (1) very fast (on a scale of seconds or minutes) electrostatic reactions of adsorption onto the substrate, as a result of the high affinity of P ions to charged surfaces, and (2) slow (on a scale of hours or days) formation of new solid metal-P compounds (with Al, Fe, and Mn under acidic pHs and with Ca and Mg under neutral or basic pHs; Lindsay, 1979).

4.2.1.1 Effect of pH on P partitioning between substrate solids and liquids in tuff suspensions

Solution-pH plays the dominant role on P partitioning between substrate solids and liquids as presented in Fig. 4.6A for yellow tuff. Negative values for P retention indicate that the quantities of

**FIGURE 4.6**

(A) pH-dependent adsorption of phosphate onto yellow tuff (pH values are shown above each isotherm), constant ionic strength (0.01 M NaCl), tuff/solution ratio of 1/40, 24 h of reaction. Arrows indicate experimental points above the K_{sp} of OCP. (B) Experimental activities in tuff extracts at pH 7.3, 8.3, and 9.3 in relation to the theoretical solubility lines for calcium-phosphate minerals (HAP: $\text{Ca}_5(\text{PO}_4)_3\text{OH}$, βTCP : $\text{Ca}_3(\text{PO}_4)_2$, OCP: $\text{Ca}_4\text{H}(\text{PO}_4)_3 \cdot 2.5(\text{H}_2\text{O})$, and DCPD: $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$). OCP, ($\text{Ca}_4\text{H}(\text{PO}_4)_3 \cdot 2.5(\text{H}_2\text{O})$); βTCP , ($\text{Ca}_3(\text{PO}_4)_2$).

Based on Silber, A., 1991. *Chemical Properties and Surface Reactions of Pyroclastic Materials From Mt. Peres, the Golan Heights* (Ph.D. thesis). Hebrew University of Jerusalem (in Hebrew with English abstract), p. 171.

soluble P in tuff suspensions were greater than the added quantities, as a result of dissolution of indigenous hydroxyapatite (HA) during the analysis procedure (Silber et al., 1994, 1999). Retention of P by yellow tuff at acidic pHs could be fitted to a pH-specific Langmuir isotherm (Fig. 4.6A). The activity products of Al^{3+} , Fe^{3+} , and PO_4^{3-} in acidic tuff suspensions (pH 4–7) were below the K_{sp} of variscite ($\text{AlPO}_4 \cdot 2\text{H}_2\text{O}$) and strengite ($\text{FePO}_4 \cdot 2\text{H}_2\text{O}$), respectively; therefore it is logical to deduce that adsorption reactions governed P solubility in acidic tuff suspensions. This conclusion is in accord with the high content of amorphous materials in the yellow tuff (51.5 g per 100 g of

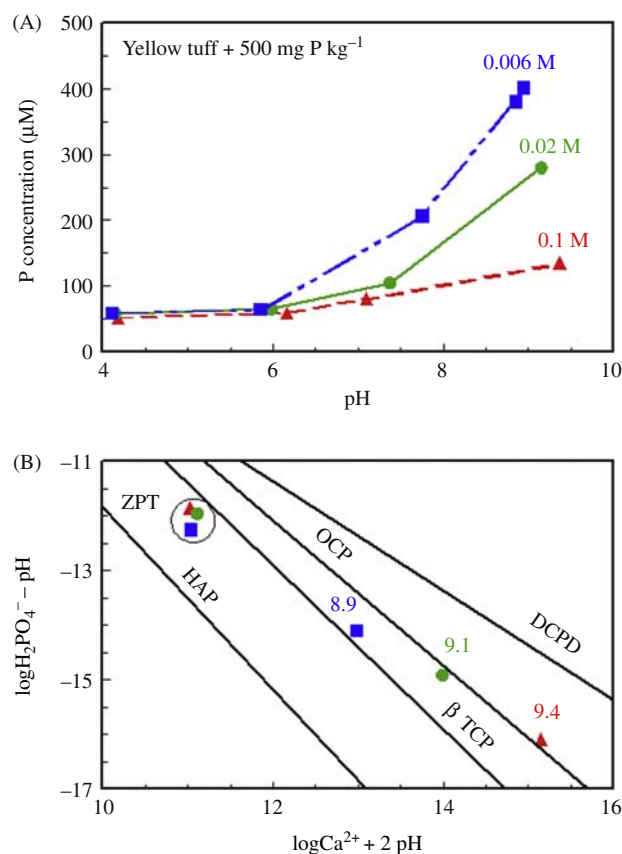
halloysite-like allophane) having variable charged surfaces with high P-binding capacity (Silber et al., 1994). However, the formation of an X-ray-amorphous analogue of variscite or AlOHNaPO_4 (Veith and Sposito, 1977) is also possible. At higher pH values, the Langmuir rule of declining slope with increased P_W was not maintained: at pH 8.3 and 9.3 and at P_W above 0.46 and 0.31 mM, respectively, an increase in the slope of P-sorption isotherm was observed (see arrows in Fig. 4.6A). The increased slope indicates that a change in the phase-controlling P partitioning took place: increasing P addition rates enhanced the feasibility of Ca–P mineral precipitation because above pH 7, all the activity product of Ca^{2+} and PO_4^{3-} in tuff solutions exceeded the K_{sp} of HA ($\text{Ca}_5(\text{PO}_4)_3\text{OH}$) and shifted toward the solubility line of βTCP ($\text{Ca}_3(\text{PO}_4)_2$) or OCP ($\text{Ca}_4\text{H}(\text{PO}_4)_3 \cdot 2.5 \text{H}_2\text{O}$) (Fig. 4.6B). HA is the most thermodynamically stable Ca–P mineral, but more soluble compounds such as βTCP or even OCP temporarily controlled P in tuff solutions. This conclusion is supported by the data of Imas et al. (1996) and Salinger et al. (1993) who found that on a time scale of a few days, P solubility in the Ca– CO_3 – PO_4 system was controlled by a metastable Ca phosphocarbonate phase. Adopting the conclusion of Sanyal and De Datta (1991) that βTCP governs P dissolution in neutral and basic soil solutions, adsorption reactions may govern P solubility in tuff suspensions at pH 7.3, as long as P_W does not exceed 0.9 mM at the pertinent Ca concentration (0.4 mM). At pH 8.3, the P_W value was below the βTCP line, at 0.09 mM whereas at pH 9.3, P_W was above βTCP solubility even at zero P addition (Fig. 4.6B). If the solution concentration of Ca and Mg is assumed to be 2 mM, which is typically required for plant growth, then the calculated P_W concentration at pH 7.3, according to βTCP solubility, will not exceed 2.5 μM ; its actual value will depend on the CO_2 and SO_4 concentrations as well. It is, therefore, possible to assume that adsorption reactions govern P_W in tuff substrate at basic pHs for a short time (minutes or hours), but as time proceeds, P_W is expected to decline until the activity product of Ca^{2+} , Mg^{+2} , and PO_4^{3-} attains the K_{sp} of a thermodynamically stable compound. Thus the reaction kinetics must be taken into account when evaluating the relative importance of adsorption and precipitation reactions in tuff solutions.

4.2.1.2 Effect of solution ionic strength on P solubility

The solution ionic strength (I) may affect P solubility via three mechanisms:

1. Alteration of the ion activity coefficient: increase of I decreases the ion activity coefficient, in turn increasing the possible concentration of P in the substrate solution.
2. Variation of charge density: the solution ionic strength affects the charge density in the diffuse double layer; therefore the effect of I on P adsorption will be pH-dependent. Similarly to the effect of I on the surface charge characteristics presented in Figs. 4.1 and 4.3, below the PZSE of the sorbent, increase of I causes a reduction of anion adsorption whereas above the PZSE, the opposite trend is expected and increase of I causes increased anion adsorption (Sposito, 1984).
3. Variation in anion/cation concentrations in the substrate solution: increases of I may affect the release of anions and cations from the solid phases, on which they might have been adsorbed or of which they might have been an indigenous part.

The effect of I on P_W in yellow tuff solutions can be used to illustrate these mechanisms (Fig. 4.7). The activity products of Ca^{2+} and PO_4^{3-} with no acid/base addition (ZPT) were above the HA solubility line but below that of βTCP (Fig. 4.7B). P_W in yellow tuff suspensions at the

**FIGURE 4.7**

(A) pH-dependent ionic strength (NaCl) effect on solution-P concentration in tuff extracts after addition of P at 500 mg/kg, tuff/solution ratio of 1/40, 24 h of reaction, ZPT is the suspension pH prior to the addition of acid or base. (B) Experimental activities in tuff extracts at pH 8.9, 9.1, and 9.4 in relation to the theoretical solubility lines for calcium-phosphate minerals.

Based on Silber, A., 1991. *Chemical Properties and Surface Reactions of Pyroclastic Materials From Mt. Peres, the Golan Heights* (Ph.D. thesis). Hebrew University of Jerusalem (in Hebrew with English abstract), p. 171.

ZPT decreased as a result of increases in I (Fig. 4.7A). In the light of the conclusions of Sanyal and De Datta (1991) that β TCP controls P_w dissolution/precipitation reactions, it may be deduced that adsorption governed P_w in this pH range. The PZSE of the yellow tuff was estimated to be 6.2 (Silber, 1991); therefore the increase of P_w with decreasing I was consistent with mechanism 2, described above.

At higher pH, the electrostatic potential of the solid surfaces became less negative (Figs. 4.1 and 4.3) and P_w increased significantly with decreasing values of I (Fig. 4.7A). An increase in I affected metal concentrations as well (mechanism 3, above), and the Ca and Mg concentrations in

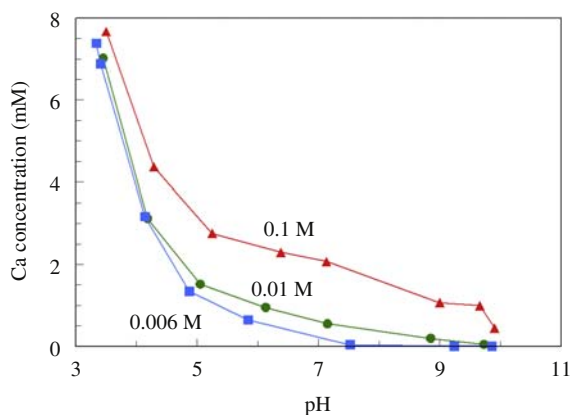


FIGURE 4.8

pH-dependent ionic strength (NaCl) effect on solution-Ca concentration in tuff, tuff/solution ratio of 1/40, 24 h of reaction.

tuff suspensions increased (only data for Ca concentrations are presented in Fig. 4.8). The activity products of Ca^{2+} and PO_4^{3-} in tuff solutions were between the solubility lines of βTCP and OCP (Fig. 4.7B). Adsorption reactions probably controlled P_W at high pH but it would be expected that at equilibrium, the P_W in 0.1 and 0.02 M solutions would decline further, until the activity products of Ca^{2+} and PO_4^{3-} attained the K_{sp} of βTCP . In acidic tuff solutions, the effect of I on the P_W was insignificant (Fig. 4.7A), probably because of P release from indigenous Ca–P minerals (Silber, 1991; Silber et al., 1994, 1999) during the analyses.

4.2.1.3 Horticultural implications

Phosphorus retention by substrates has an important effect on P fertilization management. For example, a typical substrate quantity used for vegetables or flowers consist of 50–60 kg of tuff per m^2 . The daily evapotranspiration rate during the growth period of roses (*Rosa × hybrida*) in a semiarid climate is usually 5–10 L/m^2 ; therefore the daily irrigation supply is 0.1–0.2 L/kg of tuff. The recommended P concentration in the irrigation water for cut rose culture is usually 30–45 mg/L (Sonneveld and de Kreij, 1987; Jones, 1997; Sonneveld, 2002), therefore the daily input of P is 3–9 mg/kg. Under this daily P rate and ignoring leaching, the possible P_W at pH 7.3 according to the P retention curve will be lower than 0.05 mmol/L (Fig. 4.6A). In light of the above data and on the assumption that adsorption reactions control P availability in tuff suspensions, it is expected that the decline of P solubility in the solution will be very fast so that shortly after irrigation, the P_W will be deficient and insufficient for rose production. Johanson (1978) found that the yield of cut roses fell to 40% following low P application (0.16 mmol/L), compared with that under an adequate P management (1.3 mmol/L). Thus low P concentrations in tuff substrates may restrict plant production. If a P concentration of 0.5 mmol/L is taken as the minimum for adequate growth of cut roses, the amounts of P adsorbed by yellow tuff at pH 6 and 7.3 should be 840 and 600 mg/kg, respectively (Silber, 1991). If the quantity of P added daily via fertilization is taken to be 4.5 mg/kg (0.15 L/kg at a P

concentration of 30 mg/L) then, with solution-pH of 6 or 7.3, at least 187 or 133 days would be required, respectively, to fill all the available sites for P adsorption on the yellow tuff surface and to attain the desired P_w . During this period, cut rose development would certainly be impaired. With a solution-pH above 7 and Ca and Mg concentrations assumed to be 1.5–3 mmol/L, as required for optimal plant growth, P deficiency will restrict plant growth because the activity products of Ca^{2+} , Mg^{2+} , and PO_4^{3-} are far above the K_{sp} values of known Ca–P and Mg–P minerals (Lindsay, 1979).

The direct outcome of the adsorption and the precipitation reactions detailed above is that, under equilibrium conditions, P_w may be lower than plant requirements throughout the pH range of plant growth in tuff substrates. It is important to note that conditions of P deficiency are not exclusive to tuff media. Precipitation of P with Ca and Mg at neutral and high pH is expected to be the general phenomenon in all substrates, as long as nutrient concentrations in the irrigation water are kept at adequate level for plant growth. Considering the effect of pH on adsorption and precipitation reactions, it is reasonable to expect that optimal P availability should be obtained at pH 6.0–6.5.

P adsorption by organic compounds is well known. High P retention by pine bark media has been reported (Cotter, 1979; Yeager and Wright, 1982; Ogden et al., 1987), and low P availability inhibited plant growth (Cotter, 1979). Shortening the time between consecutive P additions, that is, increasing the frequency of fertigation and inducing nonequilibrium conditions between successive fertigation events, may be a preferable management approach to enhance P availability in the rhizosphere and to improve P uptake by plants (Silber et al., 2003, 2005; Xu et al., 2004). The beneficial effects of high irrigation frequency on P availability to plant roots are detailed in Chapter 5, Root Zone Microbiology of Soilless Cropping Systems. Alternatively, addition of organic components, such as mature compost, which contain humic and fulvic substances to tuff substrates might significantly reduce P retention by the substrate (Gottesman, 1989).

4.2.2 ZINC

Zn ions are SAIs that have affinity to charged surfaces. Zn adsorption increases as pH increases, reflecting typical cation behavior. Zn solubility may be controlled by hopeite $[Zn_3(PO_4)_2 \cdot 4H_2O]$ solubility (Lindsay, 1979). Similar to P, Zn concentration in the rhizosphere may be lower than plant requirements throughout the pH range of plant growth in many substrates. The effect of solution-pH on Zn adsorption by tuff and perlite: two common substrates differing by their CEC (Table 6.1) and surface properties are presented in Fig. 4.9A and B, respectively.

Zn adsorption by substrates is expected to follow the CEC values presented in Table 6.1, and therefore some substrates commonly used in soilless culture (tuff, zeolite, compost) exhibit high adsorption capacity for Zn (as presented in Fig. 4.9A for yellow tuff), as well as other micronutrients such as Fe, Mn, and Cu. Zn adsorption to less active substrates, such as perlite and stone wool, is expected to be lower on volume basis (as presented in Fig. 4.9B for perlite). However, plant growth may alter the surface properties of their solid phases and subsequently, Zn adsorption may be significant even in “inert” substrate (Silber et al., 2010, 2012), as presented for Zn adsorption onto used perlite in Fig. 4.9C. The used perlite had served at the Besor Experimental Station, Israel, as a growing medium for sweet peppers during 6 months and had been exposed to the standard irrigation and fertilization regime recommended by the Israeli Extension Service. The

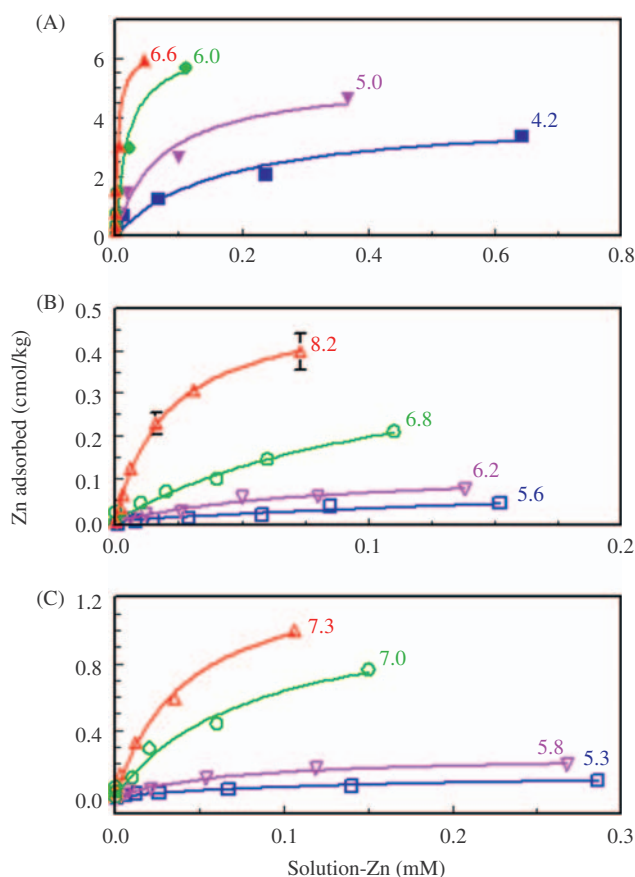


FIGURE 4.9

pH-dependent adsorption of zinc (pH values are displayed above each isotherm) onto yellow tuff (A), intact new perlite (B), and used perlite (C). Constant ionic strength (0.01 M NaClO₄), 25°C, solid/solution ratio of 1/40, 24 h of reaction for the yellow tuff, and 3 h for the new and used perlite. Symbols are experimental data $\pm$ SE (SE not shown when smaller than the symbol).

Based on Silber, A., 1991. *Chemical Properties and Surface Reactions of Pyroclastic Materials from Mt. Peres, the Golan Heights* (Ph.D. thesis). Hebrew University of Jerusalem (in Hebrew with English abstract). pp. 171; and Silber, A., Bar-Yosef, B., Suryano, S., Levkovitch, I. 2012. Zinc adsorption by perlite: effects of pH, ionic strength, temperature, and pre-use as growth substrate.

Geoderma 170, 159–167.

difference between Zn adsorption by used and by new perlite could be attributed to Zn adsorption by the accumulated organic matter that originates from root excretion, degradation, and rhizosphere biota, which contribute 6 g of organic matter per kilogram of perlite.

Thus under the solution-pH conditions commonly used in soilless production (pH of 5.5–7.5), adsorption reactions will reduce the concentrations of metal micro-nutritional elements to

deficiency levels. Bearing in mind the effect of pH on adsorption and precipitation reactions, it is reasonable to expect that optimal Zn availability should be increased at acidic pHs. Increasing the solution-metal concentration is useless for the same reasons as detailed above for phosphorus. Chelating agents such as ethylenediaminetetraacetic acid, diethylenetriaminepentaacetic acid), or ethylenediamine di(*o*-hydroxyphenylacetic acid) are commonly used to maintain adequate availability of micronutrients (Parker et al, 1995). In addition, as in the case of P, increasing the fertigation frequency was found to be an effective means of enhancing metal uptake by plants (Silber et al., 2003, 2005).

The pH-dependent dissolution of Ca–P minerals from indigenous sources in the yellow tuff and the perlite released phosphorus to the solution phase. Addition of Zn to yellow tuff suspensions decreased the P_W at all the pHs examined (Fig. 4.10A) and similar phenomenon was observed for the used perlite. Each unit of Zn adsorbed onto the used perlite suspension caused the retention of an additional 0.36 units of P on the surfaces (Fig. 4.10B), probably because of elevation of the

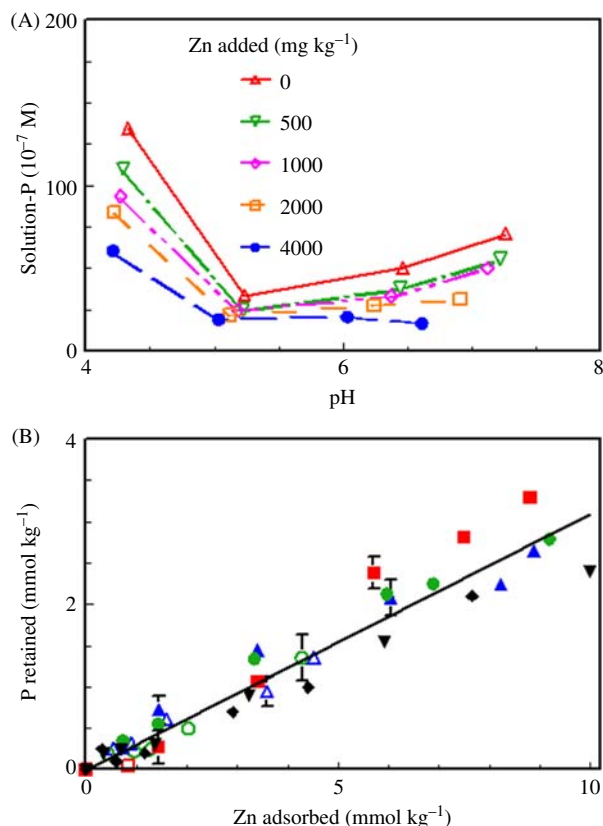


FIGURE 4.10

(A) Solution-P concentrations as a function of zinc addition and pH in tuff extracts. Constant ionic strength (0.01 M NaClO_4), tuff/solution ratio of 1/40, 24 h of reaction. (B) P retention as function of Zn adsorption onto perlite. Symbols are experimental data $\pm$ SE (SE not shown when smaller than the symbol) during 3 h of equilibration. The plotted line was calculated according to the equation (SE in parentheses): $P = 0.31(0.008) \times \text{Zn adsorbed}$.

(A) Based on Silber, A., 1991. *Chemical Properties and Surface Reactions of Pyroclastic Materials From Mt. Peres, the Golan Heights* (Ph.D. thesis). Hebrew University of Jerusalem (in Hebrew with English abstract), p. 171.

surface positive charge. For all examined P concentrations in both tuff and perlite suspensions, the calculated activity products of Zn^{2+} and H_2PO_4^- were below the solubility product constants (K_{sp}) of Zn-phosphate or Zn-oxide minerals, which indicate that the aforementioned Zn–P interaction was elicited by surface adsorption reactions and not by precipitation.

However, under common agricultural practices, P_{w} is higher by at least three orders of magnitude than the Zn concentration (Sonneveld, 2002); therefore the opposite trend is to be expected, and usually Zn availability is restricted through hopeite solubility (Lindsay, 1979) and/or the increase of negative charge that follow P adsorption on the solid phase surfaces (Diaz-Barrientos et al., 1990; Pardo, 1999).

4.2.3 EFFECTS OF P AND Zn ADDITION ON SOLUTION-SI CONCENTRATION

Addition of SAI anion or cation, respectively, enhances the negative or positive surface charge (Sposito, 1984). Silicon is an important constituent of substrate solids and Si ions have pH-dependent adsorption properties (Hingston and Raupach, 1967; Goldberg and Glaubig, 1988; Hansen et al., 1994; Nartey et al., 2000). Therefore the effects of the addition of SAI anion or cation on solution-Si concentration are of interest; Si concentrations increased with decreasing pH in all tuff suspensions (Fig. 4.11) because of the dissolution of volcanic glass and halloysite-like allophane (Silber et al., 1999).

Addition of P to tuff solutions induced an increase of Si concentrations, in accordance with the findings of Rajan (1975), Rajan and Perrott (1975), Rajan and Fox (1975), Veith and Sposito (1977), and Pardo and Guadalix (1990) that P retention was followed by displacement of SiO_2 from aluminosilicates, either because of displacement from the surface or because of partial disruption of the structure of the substrate solids. Contrary to the effect of anion addition, Zn addition to

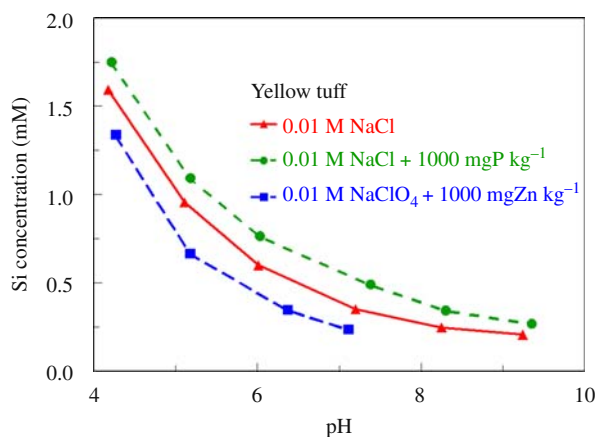


FIGURE 4.11

Solution-Si concentrations as a function of phosphorus and zinc addition to tuff suspensions. Ionic strength and electrolytes used are indicated inside the figure, tuff/solution ratio of 1/40, 24 h of reaction.

Based on Silber, A., 1991. *Chemical Properties and Surface Reactions of Pyroclastic Materials from Mt. Peres, the Golan Heights* (Ph.D. thesis). Hebrew University of Jerusalem (in Hebrew with English abstract), p. 171.

yellow tuff suspensions decreased Si concentrations at all the pHs examined (Fig. 4.11). Therefore the effects of P and Zn addition on Si concentrations may be of interest.

4.3 PLANT-INDUCED CHANGES IN THE RHIZOSPHERE

4.3.1 EFFECTS ON CHEMICAL PROPERTIES OF SURFACES OF SUBSTRATE SOLIDS

Plant growth may affect the chemical properties of the substrate solids through three main mechanisms:

1. Changes of the surface charge and chemical properties of the solids through the addition of SAIs such as orthophosphates in the fertigation nutrients.
2. Addition of new solid materials, that is, organic compounds, mainly because of root growth.
3. Accumulation of root exudates and decomposition products, especially humic and fulvic acids that may cause decomposition of substrate solids (Tan, 1986) and/or modification of the surface charge (Tate and Theng, 1980; Huang and Violante, 1986).

The effect of SAI addition on the chemical properties of substrate solids (mechanism 1) is well understood (Sposito, 1984) and as long as the solution-pH is not extreme (below pH 4) and does not cause substantial dissolution of the solids, this effect is expected to be reversible. The direct effect of plant growth on the chemical properties of the substrate, that is, alteration of the chemical characteristics of the media resulted from formation of new organic material in the form of roots, is less understood. Generally, the CEC of root dry matter has the same order of magnitude as soil constituents (10–100 cmol/kg). The root CEC was found to be pH-dependent, and adsorption of protons and copper to root cell walls could be evaluated by the same surface complexation model (single capacitance model) used for adsorption to soil (Allan and Jarrell, 1989), suggesting that the same mechanism of adsorption is similar to that of soil. It is, however, doubtful that intact roots contribute to the charge characteristics of a medium, because the high CEC values reported in the literature were related to root components such as cell walls. Therefore although the high CEC values may reflect the “potential” CEC, the “actual” CEC of intact plant root *in vivo* is probably lower because of the low accessibility of ions from the external solution to the internal apoplast (Marschner, 1995).

The effects of root excretions and root-induced pH changes or modifications of nutrient availability in the rhizosphere were reviewed by Marschner and Römheld (1996), Bar-Yosef (1996), Jones (1998), and Hinsinger et al. (2003). The effects of plant growth on the chemical characteristics of the substrate have been studied by Silber and Raviv (1996). Surface characteristics of red tuff (T0) were evaluated prior to using it rose culture as well as after it had been exposed to fertigation solution in the absence and presence of plants (Tf and Tp, respectively). Extractions were carried out at various pH levels with solutions differing in ionic strength. The ionic strength and pH were selected to reflect the relevant conditions at the rhizosphere (Silber and Raviv, 1996). The pH titration lines of intact unused red tuff (T0) overlapped, and distinction of an ionic strength effect was not possible (Fig. 4.12A).

Red tuff consists mainly of primary minerals and unweathered volcanic glass. It contains only 19% of amorphous material that may contribute to the pH-dependent charge (Silber et al., 1994).

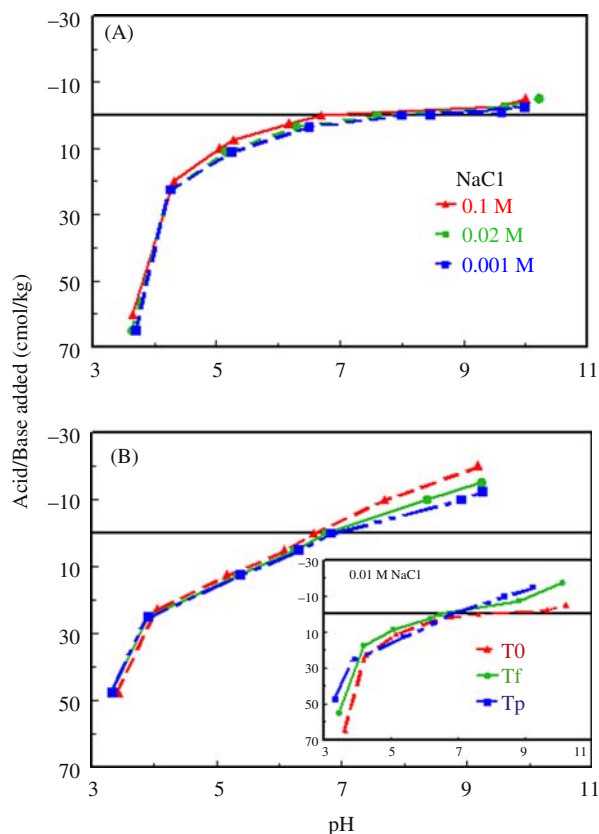


FIGURE 4.12

Acid/base addition (0.1 N HCl and NaOH, respectively) as functions of pH and NaCl concentrations, tuff/solution ratio of 1/40, equilibrium period of 1 week. (A) intact unused red tuff and (B) red tuff after 2 years of rose growing. The inset curves show a comparison of acid–base addition at constant ionic strength of 0.01 M NaCl to unused tuff (T0) and tuff irrigated with fertilization solution in the absence and presence of plants (Tf and Tp, respectively).

Based on Silber, A., Raviv, M., 1996. Effects on chemical surface properties of tuff by growing rose plants. *Plant Soil* 186, 353–360.

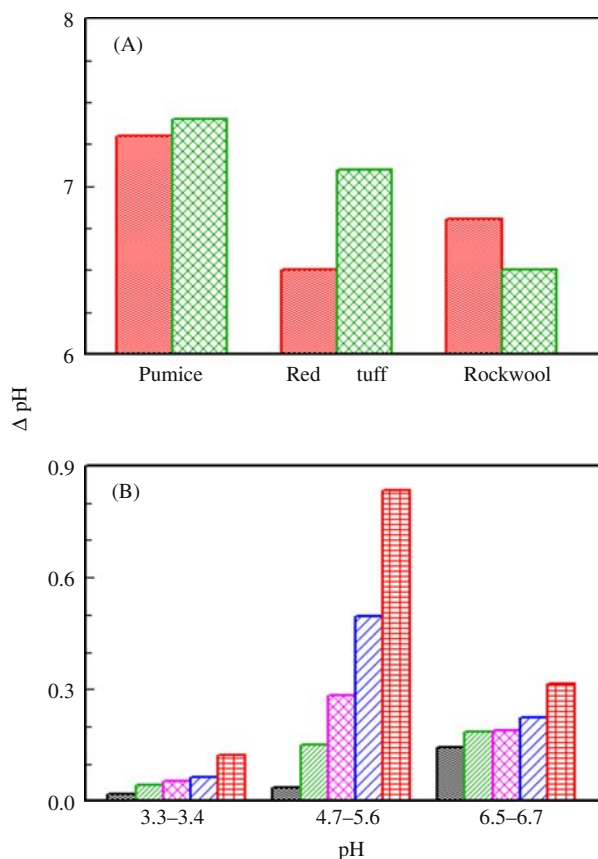
Therefore major proportions of added acids/bases were probably consumed either through exchange with cations/anions adsorbed on permanently charged sites, through the dissolution of indigenous Ca–P minerals and very fine amorphous particles, or both, as suggested by Silber et al. (1999). As a consequence, these processes masked the effect of solution ionic strength on variable charged surfaces (Fig. 4.12A).

Plant growth induces an increase in the negatively charged surfaces (Fig. 4.12B) analogous to the trend observed after the addition of citric acid to goethite and clay minerals (Lackovic et al., 2003). In addition, the effect of solution ionic strength at pH values above 7 was apparent

(Fig. 4.12B) and was similar to that observed for yellow tuff (Fig. 4.3). The increase of negatively charged surfaces of red tuff in the presence of plants (Tp) was probably due to the specific adsorption of root-excreted organic compounds. Two years of irrigating tuff substrate with fertilizer solution in the absence of plants (Tf) also induced an increase of positive charge on red tuff particle surfaces, but this effect was less significant than under plant growth (inset in Fig. 4.12B). Specific adsorption of orthophosphate ions added via fertigation, and the accumulation of soluble low-molecular-weight compounds exuded from roots or released during organic matter decomposition in the tuff substrate, induced pH decrease in Tf and Tp solutions, respectively (Silber and Raviv, 1996). In almost all cases, addition of equal quantities of acid or base lowered the pH in tuff solutions, with intact unused tuff being most affected, and tuff exposed to fertigation solutions in the presence of plants being the least (inset in Fig. 4.12). Similar trends, that is, increased negatively charged surfaces and enhanced water-soluble P, Ca, and Mg concentrations were also reported for perlite (Silber et al., 2010). It is important to note that all the root-mediated changes in the chemical properties of substrate solids are affected by the solid characteristics, the plant type and growth stage, and the prevailing agricultural management (rate of fertilization, proportions of fertilizer components, such as the $\text{NH}_4\text{:NO}_3$ ratio, fertigation frequency, and the volume of medium available for root growth). Morel and Hinsinger (1999) discussed the effect of root-induced modifications of P partitioning between the substrate solid and the substrate solution and concluded that it was notably influenced by root density. They also found that growth of crops with low root density had little or no effect on soil properties over a long period of 80 years (Morel et al., 1994) while 1 week of intensive plant growth with 100-fold increase in root density significantly affected soil properties (Morel and Hinsinger, 1999). Soilless culture is commonly characterized by root-system constraints that differ from those found in conventional soil culture and that lead to higher root density of plants in soilless production. The conclusion by Morel and Hinsinger (1999) suggest that in soilless production, plants have a significantly greater impact on medium properties than plants growing in soil.

The effects of root-mediated pH changes in the rhizosphere are well documented (for review see Nye, 1986 and Hinsinger et al., 2003); they are caused by excretion of ions (mainly H^+ , OH^- , and organic bases or acids) which balance the electric charge, following cation or anion uptake by roots, root respiration, carbon exudation, or redox processes. All these processes are strongly dependent on plant type and nutritional status (Marschner et al., 1986; Nye, 1986; Neumann and Römhild, 1999; Abadía et al., 2002; Hinsinger et al., 2003). The effect of plants on the rhizosphere pH is largely related to the chemical properties of the solid phases. Accumulation of soluble organic acid excreted from roots in the presence of an inert medium may induce a pH decrease because the adsorption reactions are negligible and the ligand acts as weak acid, whereas, in contrast, the quantities of OH^- released following ligand adsorption onto a substrate with a high surface charge may be significant and much higher than that of H^+ added via ligand dissolution. The effects of tomato (*Lycopersicon esculentum* Mill.) plants on the pH in leachate from pots filled with various substrates are presented in Fig. 4.13A.

Citric acid, which is an important organic acid excreted from the roots, was added to tuff suspensions to simulate the effect of roots on the pH of the medium. The substrate was equilibrated at three pH levels (3.3, 4.8, and 6.5) which approximately fit the $\log K^0$ of citric acid (3.1, 4.8, and 6.3) (Martell and Smith, 1976–1989; Smith and Martell, 1989). At the three pH values examined, the pH rose as a result of increasing citric acid addition, but the size of the change in pH resulting

**FIGURE 4.13**

(A) pH in leachates from pots filled with pumice, red tuff, and rockwool in the presence and absence of plants. The solution $\text{NH}_4\text{--N}/\text{NO}_3\text{--N}$ ratio applied was 1:3 and the solution-pH is indicated by the horizontal line. (B) The effect of citric acid addition to tuff suspensions equilibrated at three pHs (3.3, 4.8, and 6.5), Δ pH values represent the difference between actual pH and the initial pH, that is, before citric acid addition.

(A) Based on Bar-Yosef, B., Silber, A., Markovitch, T., Keinan, M., Levkovitch, I., Soriano, S., 1997. The effect of substrate volume and type and fertilizer level on greenhouse tomato growth. In: Report Submitted to the Chief Scientist of the Israeli Ministry of Agriculture and Rural Development (in Hebrew), p. 33 (Bar-Yosef et al., 1997). (B) Based on Silber, A., Ganmore-Neumann, R., Ben-Jaacov, J. 1998. Effects of nutrient addition on growth and rhizosphere pH of *Leucadendron* 'Safari Sunset'. *Plant Soil* 199, 205–211 (Silber et al., 1998).

from the citric acid addition (Δ pH) varied (Fig. 4.13B). The predominant citrate species at pH 3.3 were H_3Cit^0 and H_2Cit^- (38% and 60%, respectively, of the total citrate ions in the substrate solution) whereas the monovalent H_2PO_4^- was the main P species (Lindsay, 1979). Thus the competing effect of citrate ions was relatively small and, therefore, only small quantities of OH^- were released from tuff surfaces after citric acid addition. In addition, dissolution of indigenous Ca–P

minerals during the pH equilibration process and the partial dissolution of tuff surfaces (Silber et al., 1999) probably added competing anions. Evaluation of the impact of citric acid on pH changes at each pH should be based on the quantities of OH^- released. The largest quantity of citric acid added to tuff (8 g/kg) at initial solution-pH values of 3.3, 4.8, and 6.5 induced increases of 3.1×10^{-10} , 1.4×10^{-7} , and 1.6×10^{-6} mol, respectively, of OH^- per kilogram of tuff (calculated from solution-pH changes in Fig. 4.13B). In general, in a multifaceted system consisting of substrate, solution, and plant, numerous factors may have significant impacts on the direction and extent of ΔpH . Such factors include organic ions excreted from roots or inorganic specifically adsorbed anions (mainly P) added via the irrigation solution, biological processes in the rhizosphere (nitrification/denitrification), surface charge characteristics of the solid phase, root density, and, especially, the nature of the plant and its growth conditions.

4.3.2 EFFECTS ON NUTRIENTS AVAILABILITY

The effects of plant growth on the solubility of nutrient elements in the rhizosphere are specific and primarily related to the formation constants between the inorganic ions and root-excreted ligands. For example, the formation constants ($\log K^0$) of citric and oxalic acids with K are 1.2 and 1.1, respectively, as compared with 4.8 and 3.8, 4.7 and 3.4, and 9.9 and 13.0, for the same two acids with Ca, Mg, and Al, respectively (Martell and Smith, 1976–1989). Therefore in the presence of plants and with the consequent build-up of root-excreted organic acids in the rhizosphere, the soluble complexes of Ca-, Mg-, and Al-organic ligands will be predominant while the proportions of K-ligands will be minor. These differences may account for the inconsistency between the effects of plants on solution-K and on solution-Ca in tuff suspensions (Fig. 4.14): solution-K concentrations decreased while those of solution-Ca and solution-Mg (data for Mg are not shown because of their similarity with those for Ca) increased in the presence of plants compared with their concentrations in the same tuff leached with fertilizer solution (Fig. 4.14A and B, respectively).

The solution concentrations of Ca and Mg increased as a result of two consecutive processes: increase in adsorption of Ca and Mg by tuff surfaces during the growth period and subsequent displacement by electrolyte ions during the titration procedure. The Ca and Mg adsorption increased because of two independent, yet simultaneously acting mechanisms: (1) augmentation of the negatively charged surface area (Fig. 4.12) which resulted in increased adsorption of Ca and Mg to inorganic particles and (2) augmentation of negatively charged soluble complexes (e.g., Ca-citrate) that adsorbed onto positively charged surfaces.

Enhancement of P availability may be the most important outcome of plant growth, because solution-P concentrations are governed by adsorption/desorption and precipitation/dissolution reactions; they frequently fall to deficiency levels, as detailed above. Thus any modification of the surface properties that enhances P availability may be valuable. Numerous studies have found significant enhancement of P mobilization toward roots as a result of rhizosphere modification by organic acids (mainly citric) exuded by plant roots (Gardner et al., 1983; Dinkelaker et al., 1989; Dinkelaker and Marschner, 1992; Hoffland, 1992; Gerke et al., 1994; Bar-Yosef, 1996; Jones, 1998; Jones and Brassington, 1998; Geelhoed et al., 1998, 1999; Neumann and Römhild, 1999, 2001; Bertin et al., 2003; Lambers and Poot, 2003). The high root density of plants in soilless production probably accounts for a greater effect on P availability than that of soil-grown plants. Rose

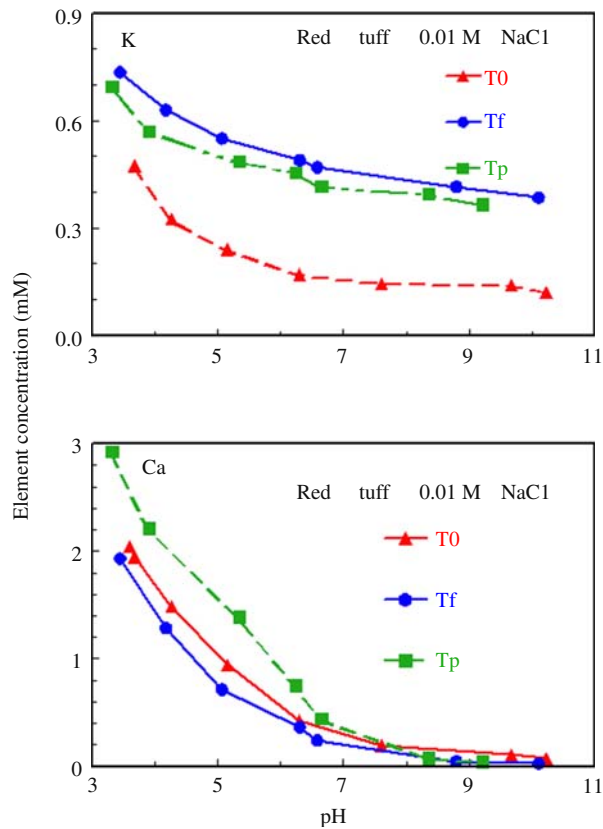


FIGURE 4.14

Potassium and Ca concentrations in three tuff extracts unused tuff (T0), and tuff irrigated with fertilization solution in the absence and presence of plants (Tf and Tp, respectively) as a functions of pH at constant ionic strength (0.01 M NaCl).

Based on Silber, A., Raviv, M., 1996. Effects on chemical surface properties of tuff by growing rose plants. *Plant Soil* 186, 353–360.

growth in tuff substrate induced a considerable increase in solution-P concentrations compared with that in tuff without plants, irrigated with identical fertilizer solution (Fig. 4.15). Surface reactions probably governed P solubility in the absence of plants, since the activity products of Al^{3+} , Fe^{3+} , and PO_4^{3-} at pH 4–7 were below the K_{sp} of variscite and strengite, respectively, and that of Ca^{2+} and PO_4^{3-} at higher pHs were between the solubility lines of βTCP and OCP (only data of Ca–P solubility lines are presented, inset in Fig. 4.14A). The P partitioning between the solid and the aqueous phases was typical to specifically anions, that is, the concentration increased with increasing pH, and there was a clear effect of ionic strength (I). Below the PZSE, an increase of I increased the positive surface charge of the sorbent, whereas above the PZSE, it decreased the positive surface charge, as illustrated in Fig. 4.1. Hence, if P solubility was governed solely by surface

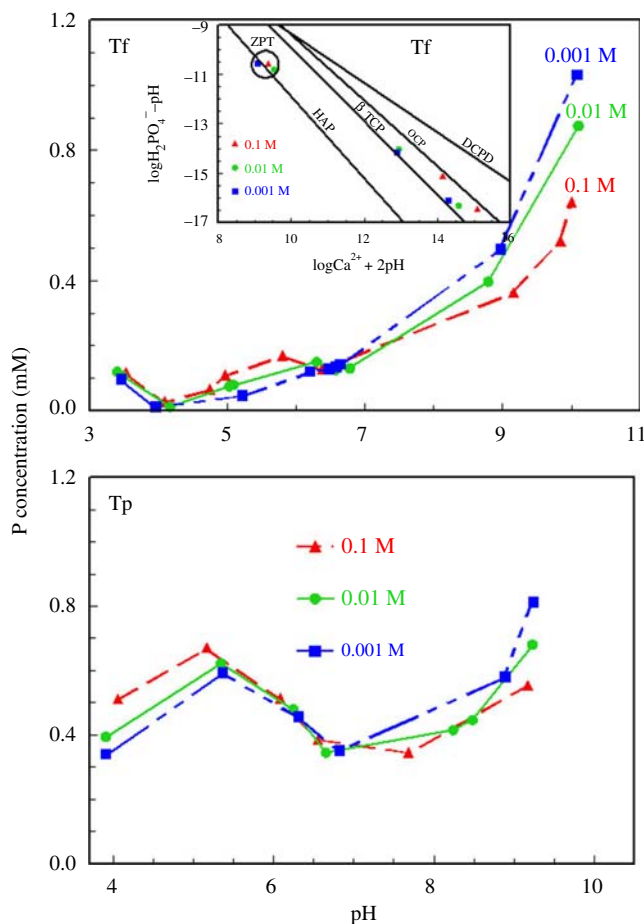


FIGURE 4.15

Phosphorus concentration as a function of pH at three initial ionic strength (NaCl): (A) tuff irrigated with fertilized solution in the absence of plants (Tf), the inset present the experimental activities in tuff extracts in relation to the theoretical solubility lines for calcium-phosphate minerals, ZPT is the suspension pH prior to the addition of acid or base; (B) tuff irrigated with fertilized solution in the presence of plants (Tp).

Based on Silber, A., Raviv, M., 1996. Effects on chemical surface properties of tuff by growing rose plants. *Plant Soil* 186, 353–360.

reactions, then P adsorption would be expected to increase or decrease with increasing I below or above the PZSE, respectively, as indeed presented in Fig. 4.15A. The overlap around pH 4 probably resulted from fast dissolution of tuff surfaces. The clear intersection around pH 6.5 (Fig. 4.15A) is consistent with previous results that indicated pH 6.5 to be the PZSE of the red tuff (Silber, 1991).

In contrast to intuitive thought, plant growth significantly improved P availability in the rhizosphere. Despite consumption by plants, P_w was higher in Tp than in Tf extracts at all ionic

strengths, especially at acidic pH levels (Fig. 4.15A and B). The higher Ca and Mg concentrations in Tp extracts than in Tf ones (Fig. 4.14) did not prevent the P_w from being higher as well, probably because the Ca^{2+} and Mg^{2+} activities were low because of organic complexing agents excreted by the roots (Inskeep and Silvertooth, 1988; Grossl and Inskeep, 1991). The effect of I on P_w in Tp extracts was almost negligible compared with that in Tf extracts, and no clear intersection could be discerned (Fig. 4.15B). The increase of negative charged surface induced by plant growth (Fig. 4.12), which were probably caused by specific adsorption of carboxylic acids excreted from roots, also reduced P adsorption on tuff surfaces, in accord with known mechanisms reported in the literature (Nagarajah et al., 1970; Earl et al., 1979; Lopez-Hernandes et al., 1979; Traina et al., 1986a,b, 1987; Kafkafi et al., 1988; Geelhoed et al., 1998, 1999; Jones, 1998; Jones and Brassington, 1998; Morel and Hinsinger, 1999; Lackovic et al., 2003).

4.3.3 ASSESSING THE IMPACT OF PLANTS: THE EFFECT OF CITRIC ACID ADDITION ON P AVAILABILITY

Adding citric acid to tuff suspensions allows assessment of the role of low-molecular-weight organic acids excreted from roots in P partitioning between the aqueous and solid parts of the substrate. This also simulates the beneficial effect of plant growth on P availability that was detailed above as shown by Silber and Raviv (1996) (Fig. 4.15). They also found that the isotherms of citric acid adsorption by yellow tuff fell as pH increased from 3.5 to 4.9 and to 6.4 (Fig. 4.16) and that these curves were pH-specific Langmuir curves. This was consistent with previous findings of specific adsorption of citrate by variable charged surfaces (Bowden et al., 1980; Bar-Yosef, 1996; Jones and Brassington, 1998; Geelhoed et al., 1999; Lackovic et al., 2003). On the other hand, it is inconsistent with citric acid isotherms that the quantities of P released are greatest for pH 3.5, intermediate for pH 6.4, and lowest for pH 4.9 (Fig. 4.17).

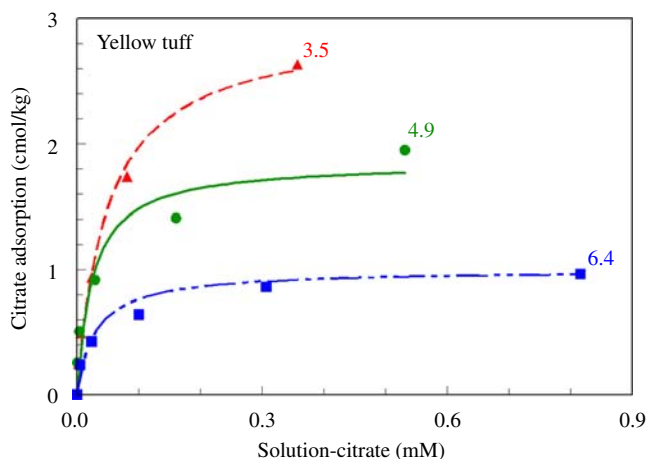
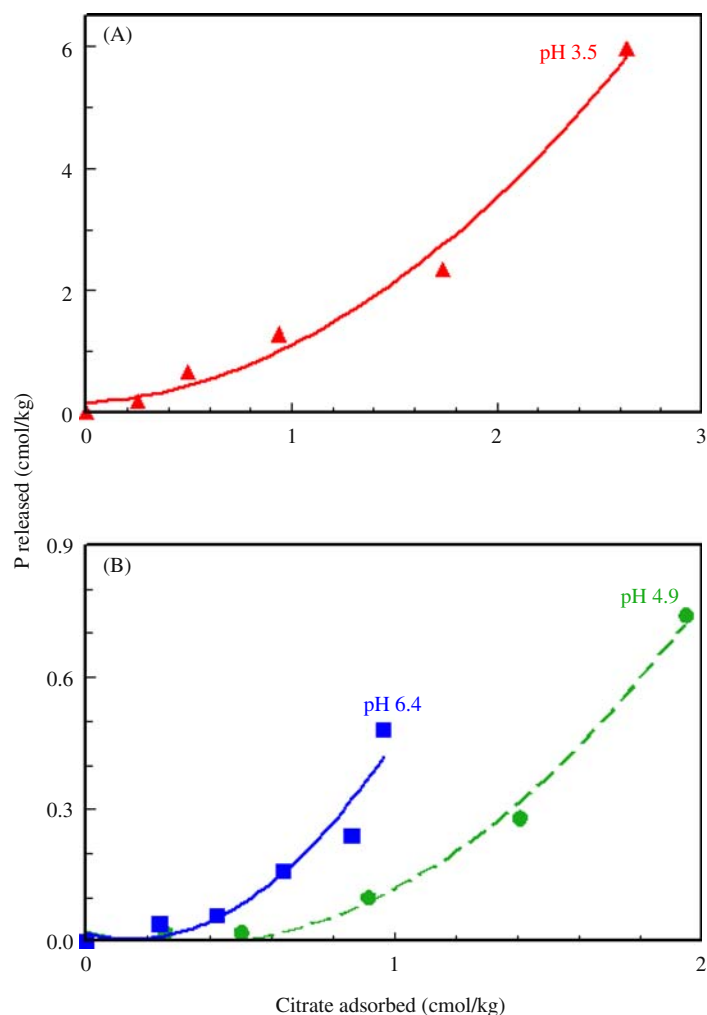


FIGURE 4.16

pH-dependent adsorption of citric acid onto yellow tuff (pH values are shown above each isotherm), constant ionic strength (0.01 M $NaClO_4$), tuff/solution ratio of 1/40, 24 h of reaction.

**FIGURE 4.17**

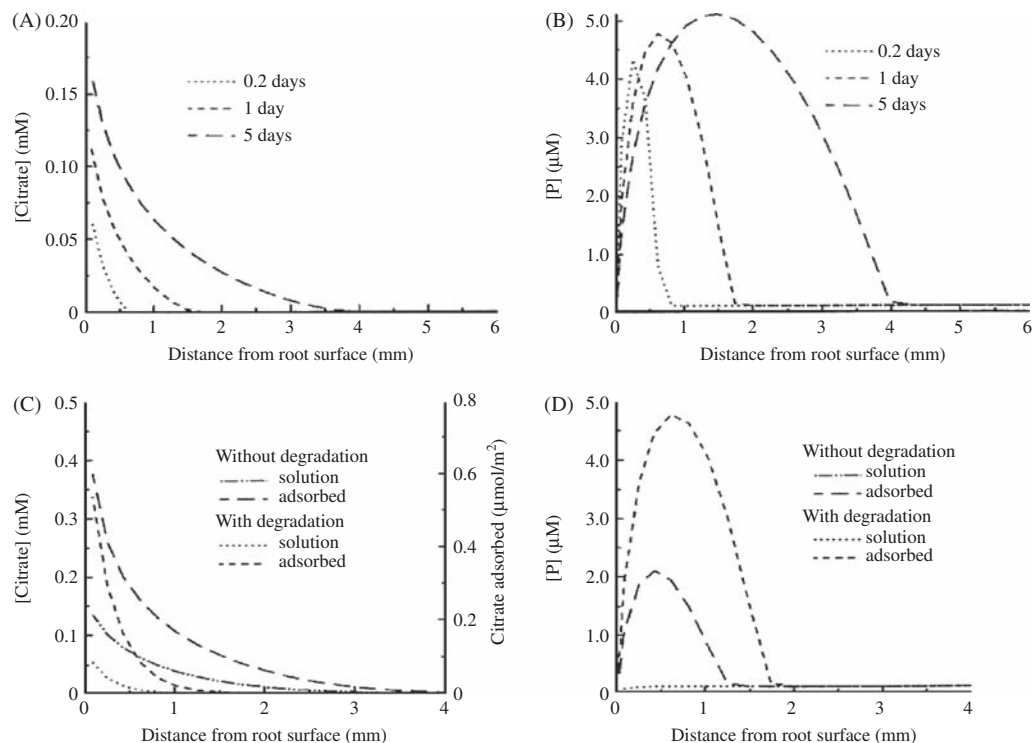
Quantities of P released as a result of citric acid addition to yellow tuff, constant ionic strength (0.01 M NaClO₄), tuff/solution ratio of 1/40, 24 h of reaction. (A) initial pH 3.4 and (B) initial pH 4.9 and 6.4.

Citrate adsorption onto an iron oxide (goethite) surface has been suggested to be as an inner-sphere complex (no molecule of the bathing solvent is interposed between the surface functional group and the molecule unit it bounds; Zelancy et al., 1996) at pH < 7 (Geelhoed et al., 1998, 1999; Harsh, 2000; Lackovic et al., 2003), or as outer-sphere adsorption (nonspecific adsorption by electrostatic attraction) at pH 8.8 (Lackovic et al., 2003). However, the inconsistencies in the order and slope of the curves of P release versus citric acid adsorbed (Fig. 4.17) suggest that an additional process may be involved in the release of P into the substrate solution. It is plausible that, in

addition to the mechanism of citrate-P competition for the adsorption sites, citrate addition accelerated the dissolution of indigenous P through the formation of soluble metal-citrate complexes. Assessment of the effects of low-molecular-weight organic acids, such as citric acid excreted from roots, on rhizosphere-P solubility, should take into account miscellaneous factors such as rate of excretion by plant roots, the involvement of microflora that may secrete or consume organic acids, fertilization management, content in the system of metal-P compounds that might be prone to dissolution, and properties of the substrate solids. The relation between plant-P status and the rate of root excretion is well established and usually solution-P concentrations and the amounts of organic acids excreted exhibited opposing trends. For example, root excretion significantly decreases as solution-P increases (Gardner et al., 1982, 1983; Lipton et al., 1987; Gerke et al., 1994; Johnson et al., 1994; 1996a,b; Imas et al., 1997; Keerthisinghe et al., 1998). Thus with the solution-P concentrations normally used in soilless culture (0.5–1.5 mM), the enhancement of P availability by organic acids is expected to be low. In addition, rapid utilization and degradation by microorganisms may diminish the content of organic acid in the rhizosphere (Jones, 1998). Therefore even if adsorption by solids reduced the rate of degradation (Jones and Edwards, 1998), the actual contribution of organic acids to nutrient acquisition by plants grown under proper fertilization regime would be minor (Jones et al., 2003).

The diffusion coefficients of organic acids in the rhizosphere are very low (Darrah, 1991) and therefore the concentration of organic acid is estimated to be heterogeneous in the root zone (Jones et al., 2003). Models can provide insight into the complex processes occurring in the rhizosphere, specifically with regard to examination of the effects of root excretions on nutrient availability under diverse growth conditions as illustrated by Geelhoed et al. (1999) in Fig. 4.18A–D. In particular, solution-citrate concentrations in the substrate solution increase with citric acids excretion from roots and decrease through degradation. According to model simulations, the profiles of citrate concentration are expected to decrease sharply with increasing distance from the root surface over short time (few hours) so that the citrate concentration will fall almost to zero at a distance of few millimeters from the root (Fig. 4.18A). This phenomenon is valid as long as (1) electrostatic adsorption reactions are instantaneous, (2) no degradation processes are taking place, (3) only competition between citrate and P on the solid phase surface sites is involved (i.e., no involvement of chelating effect of citrate on indigenous metal-P compounds). However, over time and in the absence of degradation, citrate is diffused to few millimeters from the root surfaces. As citrate accumulates in the boundary layer around the roots, adsorbed phosphate is released from the surfaces of substrate solids into the substrate solution and P is mobilized (Fig. 4.18B). The presence of a microflora population and its utilization of citrate significantly decrease citrate quantities in both the solid and aqueous parts of the substrate (Fig. 4.18C), thus reducing the effect of root-excreted citrate on P partitioning (Fig. 4.18D).

The patterns resulting from this model simulation (Fig. 4.18) are valid only under the assumptions of the model and for the parameters used. The results obtained may be affected significantly by the presence of different amounts of P and citrate in the system or different values for the model's parameters such as P and citrate adsorption rates and the function relating citrate degradation rates to pH. The model result may also differ with alternate submodels of related processes such as the rate dependence that pH has on the activities of the microflora population. Also, other factors used in the model of Geelhoed et al. (1999) such as root density, diffusion coefficients, tortuosity factor, or volumetric water content may significantly affect the model predictions.

**FIGURE 4.18**

(A) Solution-citrate concentration as a function of distance from root surface in the absence of degradation processes after 0.2, 1, and 5 days of excretion. Calculations for $95 \text{ m}^{-2}/\text{kg}$ goethite, $1.3 \text{ } \mu\text{mol P}/\text{m}^2$, and excretion of $0.5 \text{ } \mu\text{mol citrate}/\text{m}/\text{day}$; (B) Solution-P concentration as a function of distance from root surface in the absence of degradation processes after 0.2, 1, and 5 days of excretion, calculations as for Fig. 4.17A; (C) predicted concentration profiles of citrate in solution and adsorbed onto goethite with and without degradation. Calculations for $95 \text{ m}^{-2}/\text{kg}$ goethite, $1.9 \text{ } \mu\text{mol P}/\text{m}^2$, and excretion of $0.5 \text{ } \mu\text{mol citrate}/\text{m}/\text{day}$; (D) effect of citrate exudation, with and without degradation, on the predicted P concentration in solution, calculations as for Fig. 4.17C.

Reprinted from Geelhoed, J.S., Van Riemsdijk, W.H., Findenegg, G.R., 1999. Simulation of the effect of citrate exudation from roots on the plant availability of phosphate adsorbed on goethite. *Eur. J. Soil Sci.* 50, 379–390, with kind permission from Blackwell Science, Ltd.

4.4 NUTRIENT RELEASE FROM INORGANIC AND ORGANIC SUBSTRATES

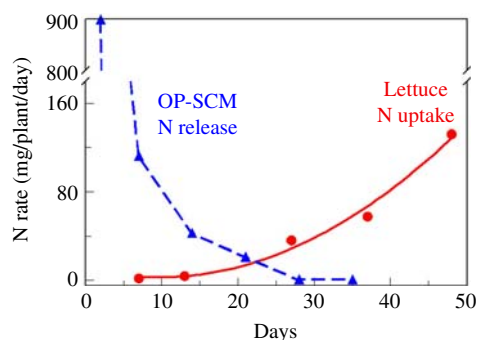
Reducing the pollution hazards posed to the environment by excess fertilization in soilless production has led to efforts to regulate nutrient availability in the rhizosphere and to develop substrates that might contribute nutrients during the growth period of horticultural crops. The goal of reducing excess fertilization may be achievable through two approaches: (1) precharging the substrate with nutrients that are subsequently released during the growth period, that is, the substrate functions as

slow-release fertilizer, and (2) release of indigenous nutrients from the substrate as a result of chemical/biological processes such as N mineralization in organic media or dissolution of indigenous nutrients from inorganic components.

Precharging of substrates with nutrients requires that the material have high CEC or AEC values. The low values of these characteristics of most inorganic substrates, such as stone wool, perlite, and pumice, suggest that these widely used horticultural materials may not be suitable for this. Zeolites, however, possess extremely high CEC values of 200–400 cmol/kg (Mumpton, 1999). In fact, zeolites are used as nutrient sources in infertile soils and substrates because of their high affinity for K or NH_4 ions (Hershey et al., 1980; Chen and Gabelman, 1990; Notario del Pino et al., 1994; Williams and Nelson, 1997; Dwairi, 1998; Mohammad et al., 2004). Although the use of zeolites as a single-component growing substrate is not recommended because of the high bulk density (1.9–2.3 Mg/m^3 ; Ming and Mumpton, 1989), mixed substrates which may include organic or inorganic materials as well as zeolites are used worldwide in the production of flowers and vegetables. Theoretically, if all the exchange sites of a zeolite substrate were precharged with K ions, the total potential pool for K would be 80–160 g/kg. Since the total K acquisition by tomato or pepper (*Capsicum annum* L.) crops under intensive conditions is 15–20 g per plant (Bar-Yosef, 1999), approximately 200 g of zeolite is required to satisfy the total plant demand for K throughout a growing season. Thus from the quantitative point of view, the precharging approach can be regarded as potentially successful; however, the release rate may present problems. Basically, the optimal rates of nutrient release must match the plant demand, which is obviously low at the beginning of the growing period and increases gradually during the various growth stages. However, the precharged nutrients are released from the surface sites by exchange reactions with competing ions (in the case of K^+ these are NH_4^+ , Na^+ , Ca^{2+} , and Mg^{2+}). Thus factors such as concentrations of competing ions in the substrate solution, irrigation doses, and plant activity all affect the release rate; therefore nutrient availability is not actually regulated and may be different from the plant's demands.

A combination of approaches (1) and (2) to reducing excess nutrient runoff was implemented by mixing phosphate rock with zeolite precharged with NH_4^+ ions (Allen et al., 1995; Williams and Nelson, 1997; Pickering et al., 2002). While the NH_4^+ ions served as an N source for plants during the growth period, the main purpose of this process was to induce phosphate rock dissolution, either through acidification or through exchange with Ca ions (Allen et al., 1993; Notario del Pino et al., 1994) so as to slowly release phosphate over time from the zeolite. Zeolites can also be used as slow-release sources of micronutrients and K (Opena and Williams, 2003; Puschenreiter and Horak, 2003).

The contribution of indigenous nutrients released from inorganic substrates is usually very small and, except for a few cases of utilization of phosphate rock, the release of indigenous nutrients is practically negligible. Organic media, on the other hand, and especially composts, contain large quantities of N and, therefore, the production of composts that may contribute N during the growth period [approach (2)], especially for organically grown crops, may be of interest. Generally, part of the initial N content of organic constituents is lost during the composting process. In fact, high-quality mature composts are characterized by their low mineralization rate and low N availability (Hadas and Portnoy, 1994, 1997). It is, however, possible to reduce the N loss during the composting process by using a high C/N ratio, lowering the pH and manipulating the air flow and the composting temperature (Raviv, 2005). Raviv et al. (2005) found that composts enriched with wheat

**FIGURE 4.19**

Comparison between the rate of N release from compost OP-SCM and the N requirements of lettuce. The N release data were computed from Fig. 4.2 of [Raviv et al. \(2005\)](#), assuming application of 3 L of compost per plant. The lettuce N uptake data were taken from [Silber et al. \(2003\)](#). OP-SCM, Containing orange-peel amendment to separated cattle manure.

straw or orange peels (imparting a high initial C/N ratio and a low pH, respectively) satisfied the N requirement of tomato plants for several months under conditions of minimal leaching fraction but they were, nevertheless, not able to match the rate of N release to the plant requirement. The rate of N release from a successful N-conservation compost consisting of orange peels and separated cattle manure ([Raviv et al., 2005](#)) was very high during the first days. The rate far exceeded plant demand for a crop such as, for example, lettuce ([Fig. 4.19](#)). However, the release rate declined very rapidly so that after 22 days, it was lower than the N requirement of the lettuce plants, and the discrepancy between the two rates increased with time. Nitrogen release from composts is usually described by first-order kinetic equations ([Hadas and Portnoy, 1994, 1997](#)) and is influenced by temperature, water content, and other environmental factors that affect biological and chemical processes. If the amount of N required daily by mature plants grown under intensive growth condition in greenhouses is taken to be 120–170 mg ([Bar-Yosef, 1999](#)), it is not likely that organic composts could contribute sufficient N, even under less favorable conditions for N degradation as presented in [Fig. 4.19](#). Nonetheless, inclusion of high-N composts in substrates may allow for reduced N application rates during the initial growth period. This is particularly important under organic agriculture management and in situations where reduction of pollution hazards is important. This finding may be particularly applicable in short-term crops such as organic transplant production which usually takes up to 35 days ([Raviv et al., 1998](#)).

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ROOT ZONE MICROBIOLOGY OF SOILLESS CROPPING SYSTEMS

5

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5.1 INTRODUCTION

The root zone of soilless cropping systems is a complex world of microorganisms which play important roles as both beneficial or deleterious entities. Designing soilless cropping systems has traditionally involved little consideration for this except to disinfect any constructed root zone or its components. There are possibilities for improving the overall approach to dealing with microbes in the root zone to harness them for greater benefits. To this end, it is important to know: What life forms are present there, where do they come from, what do they do, and how do they interact with the components of the root zone environment? To understand the whereabouts and functionality of the microbiota associated with soilless systems, these systems must be viewed from the perspective of a microbe or a consortium of microorganisms and in terms of habitat stratification, such as the rhizosphere and its different layers. Due to the interactions between microbes and their environment, the quantity, structure, and function of the microbial community may vary widely over time. Standard growth factors such as light, temperature, humidity, available water, and nutrition are obvious components of good plant management practice. However, the impact of the microbiota associated with the root environment (except pathogens) is generally neglected in practice. Although factors related to microbes tend to be similar to those that are crucial for good plant management, the factors which influence root permeability are particularly important for microorganisms in root-related contexts, for example, plant root health, phytotoxic effects, plant nutrition, pH, and stability of the growing medium. These include anaerobic conditions, temperature (Pettersson and Bååth, 2003), and light intensity (Larcher, 2003; Neumann and Römheld, 2001; Rovira, 1969), but also microbial competition. These positive or negative aspects have to be considered in evaluation and development of growing media and/or complex soilless cropping systems.

5.2 MICROBIAL PROPERTIES OF ROOTS, GROWING MEDIA, AND NUTRIENT SOLUTIONS

From a microbial perspective, soilless cropping systems entail four decisive components:

- The root and the zone immediately surrounding and influenced by the root (rhizosphere) (Hiltner, 1904; Hartmann et al., 2008)
- The growing medium, which consists of solid substrate
- The nutrient solution, which is primarily a liquid phase
- The gasses occupying the pore space

These are closely connected and have shared boundary layers. They differ in their physical state and nutritional (*trophic*) conditions. Water and nutrient availability, pH, atmosphere, temperature, and salt content determine microbial survival, proliferation, and lifestyle. Thus, different microbial colonization patterns and dynamics occur in the different compartments of soilless culture systems. The majority of microbial organisms inhabiting the root as well as rhizosphere, growing medium, and nutrient solution employ a heterotrophic lifestyle, thus depend on the supply of organic compounds from primary organic carbon sources. These organic compounds provide nutrients to microorganisms (microbial nutrients). Due to their lifestyle and capacity for rapid proliferation, bacteria are usually more abundant than fungi in all of these subentities. Under acidic conditions, however, as for example in untreated peat, growth of fungi is favored over growth of bacteria.

5.2.1 THE ROOT AND THE RHIZOSPHERE

The root is the main driver of microbial colonization in all soilless culture regardless of presence of absence of growing medium. Both dead and living roots contribute to the pool of organic nutrients, but the processes behind their release differ. Dead roots are subject to microbial decomposition and represent a finite reservoir, while a living root replenishes its habitat with organic nutrients (root exudates) that are, at least partly, readily available to microorganisms. Root exudates may include water-soluble, low molecular weight compounds such as sugars, amino acids, and carboxylic acids which are released from the roots into the rhizosphere. The rate of deposition may be influenced by colonizing microbes (Phillips et al., 2004). Active release of, for example, high molecular weight compounds, is believed to take place through anion channels and exocytosis (Neumann and Römheld, 2001). As the root is covered with a water film of varying thickness, this allows microbes to move into the root zone passively with the water film or actively to attractive sites (*chemotaxis*) (De Weert et al., 2002).

The root zone (rhizosphere) is characterized by complex trophic interactions which involve these compartments: the interior of the root, known as the *endorhizosphere*, the root surface (also called “*rhizoplane*”), and the exterior of the root known as the *ectorhizosphere*. For many variables, such as exudate concentration and ion concentrations, there is a gradient from the root interior to increasing distance from the root surface. This gradient differs with the amount of substrate used in the soilless culture system, especially between substrate-based systems and liquid culture (Fig. 5.1), and is a function of irrigation frequency and irrigation volume. In liquid culture, continuous dilution of plant-derived organic nutrients can be expected, leading to a more far-reaching impact of

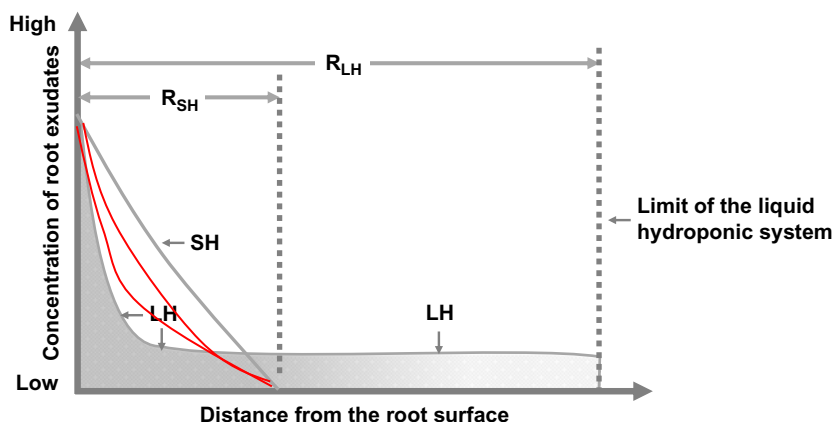


FIGURE 5.1

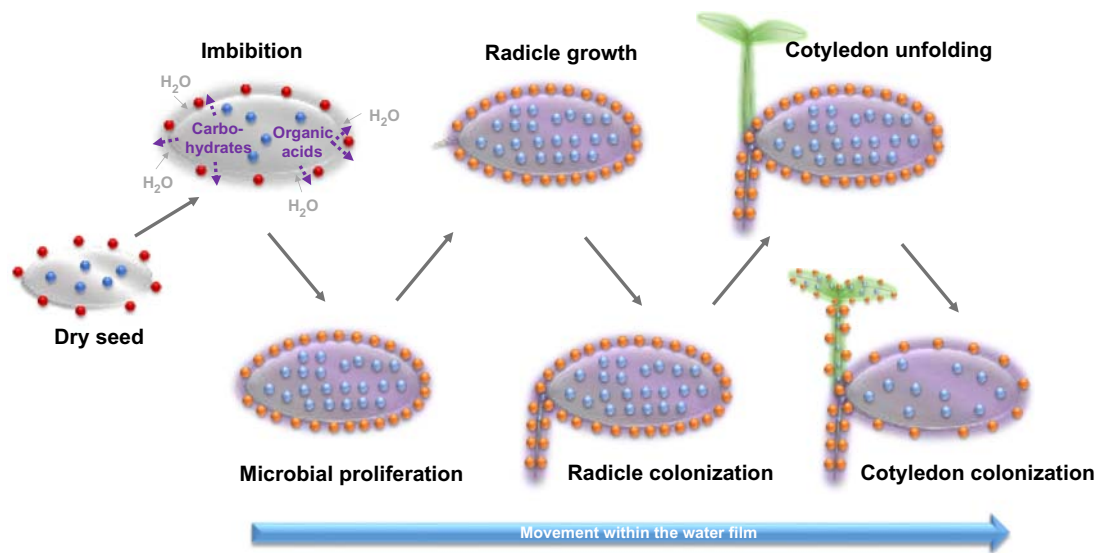
Comparison of the extent and intensity of the root exudate effect in a substrate-based SH versus liquid culture (LH) with little or no growing medium. R_{SH} = extent of the root exudate effect in the rhizosphere. R_{LH} = extent of the rhizosphere effect in liquid hydroponics. LH, Liquid culture; SH, Soilless culture system.

Modified from Hockenhull J. and Funck-Jensen D. (1983). Is damping-off, caused by Pythium, less of a problem in hydroponics than in traditional growing systems? Acta Hort. 133, 137–146. DOI: 10.17660/ActaHortic.1983.133.14.

the root on its environment than in substrate culture with infrequent irrigation. With pulse irrigation, the dilution effect on exudates in the growing media could be comparable to liquid culture. Microbial interactions in aeroponics have not received the same attention as other soilless culture systems, and no detailed figures are at hand. Most likely the situation in aeroponics is similar to the one in liquid culture, that is, nutrient film technique (NFT). Although the compartments in aeroponics are physically separated, the misted nutrient solution can act as a vehicle for microbes to move between the compartments as experiences with nutrient mist bioreactors indicate (Sharaf-Eldin and Weather, 2006).

The above compartments are each colonized by a wide variety of microorganisms. Their community structure is influenced by the richness and quality of available organic nutrients and the biological properties of the resident microbiota. Microbial growth rate and motility are important attributes in exploiting this nutrient-rich habitat. Microbes that are in direct contact with the root may profit directly from plant exudates released actively or passively by the root, while microorganisms living at increasing distance from the root will exploit those compounds which are not used, broken down, or transformed by the microbes that are in physical contact with the root. Competition for nutrients, and thereby community structure, is also affected by plant and microbial defense metabolites, such as quorum-sensing inhibitors, siderophores, and antibiotic compounds (Brimecombe et al., 2007; Lynch, 1990).

Plant species and plant age are the most important drivers of microbial biomass and community structure in the root zone. Microbial colonization of roots varies over time as the plant develops. It starts already during seed germination (Fig. 5.2). Shifts in source–sink relationships during plant growth and development, and thus carbon partitioning, lead to shifts in the quantity and quality of

**FIGURE 5.2**

Microbial colonization during germination and emergence. The dry seed is colonized with dormant microbial cells. While soaking (imbibition) the dry seeds, carbohydrates and organic acids are exuded through the seed coat. The moist and nutrient-rich conditions allow the dormant microbial cells to switch to an active lifestyle, proliferate, and passively (through the water film) or actively prospect for new colonization sites on the emerging seedling. Dark blue/red dots indicate dormant microbial cells, while light blue/red dots denote living cells. The gray and purple shadow around the seed represents the zone affected by the seed (spermosphere).

(Illustration: B. Alsanius)

organic carbon deposition in the rhizosphere and thus affect the community structure. Using community-level physiological profiling (CLPP), [Rosberg et al. \(2014\)](#) demonstrated that the substrate utilization pattern of rhizosphere microorganisms differs between decisive growth stages of hydroponically grown tomato (*Lycopersicum esculentum*), thus indirectly indicating shifts in available organic nutrients. These shifts are corroborated by community structure alterations detected by DNA-based methods ([Rosberg et al., 2014](#)).

Apart from direct plant effects, the growing medium ([Khalil and Alsanius, 2001](#); [Khalil et al., 2009](#)) and its moisture content ([Khalil and Alsanius, 2011](#); [Van Der Gaag and Wever, 2005](#)), electrical conductivity ([Khalil, 2011](#)), and temperature ([Panova et al., 2004](#)) have been shown to be decisive for root colonization. Moreover, secondary metabolite production and/or disease development depends on these environmental factors ([Van Rij et al., 2004](#)). Growing medium additives can also impact the relative abundances of microbial taxa in the rhizosphere, most probably through physiochemical changes, as demonstrated for hydroponically grown strawberries and lettuce ([De Tender et al., 2016](#); [Debode et al., 2016](#)). Several studies have concluded that a microbially diverse community structure in the rhizosphere or growing medium is more resilient to root disease attack ([Rosberg et al., 2018](#); [Postma et al., 2000a, 2005a](#)) (see [Sections 5.3.2 and 5.3.3](#)).

5.2.2 GROWING MEDIUM

Organic carbon turnover in a growing medium is a critical factor for the microbial community in that medium (Alsanjus and Wohanka, 2009). However, as the organic matter content and the ratio between organic carbon and nitrogen (C/N ratio) differ fundamentally between organic and inorganic growing media, microbially managed processes also deviate substantially between these two types of media. A narrow C/N ratio favors organic matter breakdown and concomitant loss of physical structure, whereas a wide C/N ratio immobilizes nitrogen and thus decreases its availability (Barrett and Burke, 2000; Nicolardot et al., 2001). The fate of the organic carbon pools in substrate-based and liquid culture is described in Fig. 5.3. From the carbon pool perspective, five phases can be discriminated, namely, events linked to the (1) production site and conditioning of the growing medium, (2) storage, (3) first use, (4) treatment prior to reuse as well as (5) reuse. The last two phases might be repeated if the growing medium is reused during additional growing cycles which would lead to changes in the amount of organic matter prone to breakdown. Different phases are applicable with respect to the choice of soilless system (solid, liquid) and the nature of the growing medium (inorganic, organic), as described below.

Substrate-based soilless culture systems employing inorganic growing media, for example, mineral wool, perlite, or pumice, are usually free from organic carbon prior to first use. Exposure to high temperature during manufacturing sterilizes these growing media of microbial contamination, and therefore, they are often described as being “sterile” and having a “microbial vacuum” (Paulitz and Bélanger, 2001). Once plants are introduced into inorganic growing media, however, organic carbon is continuously fed into the system through the plant roots. Although living and dead roots, living and dead microbiota, and macroorganisms accumulate and enrich the organic carbon pool during first use, the organic carbon gradient between the rhizoplane and ectorrhizosphere is more pronounced than in organic growing media. Thermal treatment is a usual measure to decontaminate inorganic growing medium prior to reuse (Gudehus, 2005; Runia, 2000; Diara et al., 2012). Although heat treatment favors organic matter mineralization and concomitantly boosts microbial growth, the organic carbon pool in reused inorganic growing medium is still richer (Raviv et al., 1998), and thus, gradual organic carbon enrichment occurs over multiple growing cycles. The organic carbon pool during subsequent growth cycles on inorganic growing media is thus replenished by readily available organic carbon and organic matter accumulated during previous growth cycles, dead roots, and dead microbiota.

In contrast, organic growing media are influenced by the history of their organic constituents. Conditioning of organic growing media, such as pH adjustment, also improves the conditions for microbial activity. Moreover, the microbial dynamics in organic growing media are influenced by the source of nutrient adjustment (inorganic or organic) and vary between different stages of processing and storage (Storey et al., 2015). Storage conditions, such as temperature and moisture, are thus essential to maintain desired physical and biological properties of organic growing media. The total and readily available organic carbon content is initially higher in organic growing media than in inorganic. Breakdown of organic fibers, replenishment through root exudation, and decay of plant material and microbes result in shifts in the microbial community structure in organic growing growth media, which in turn affect microbial activity and biomass and cause secondary shifts

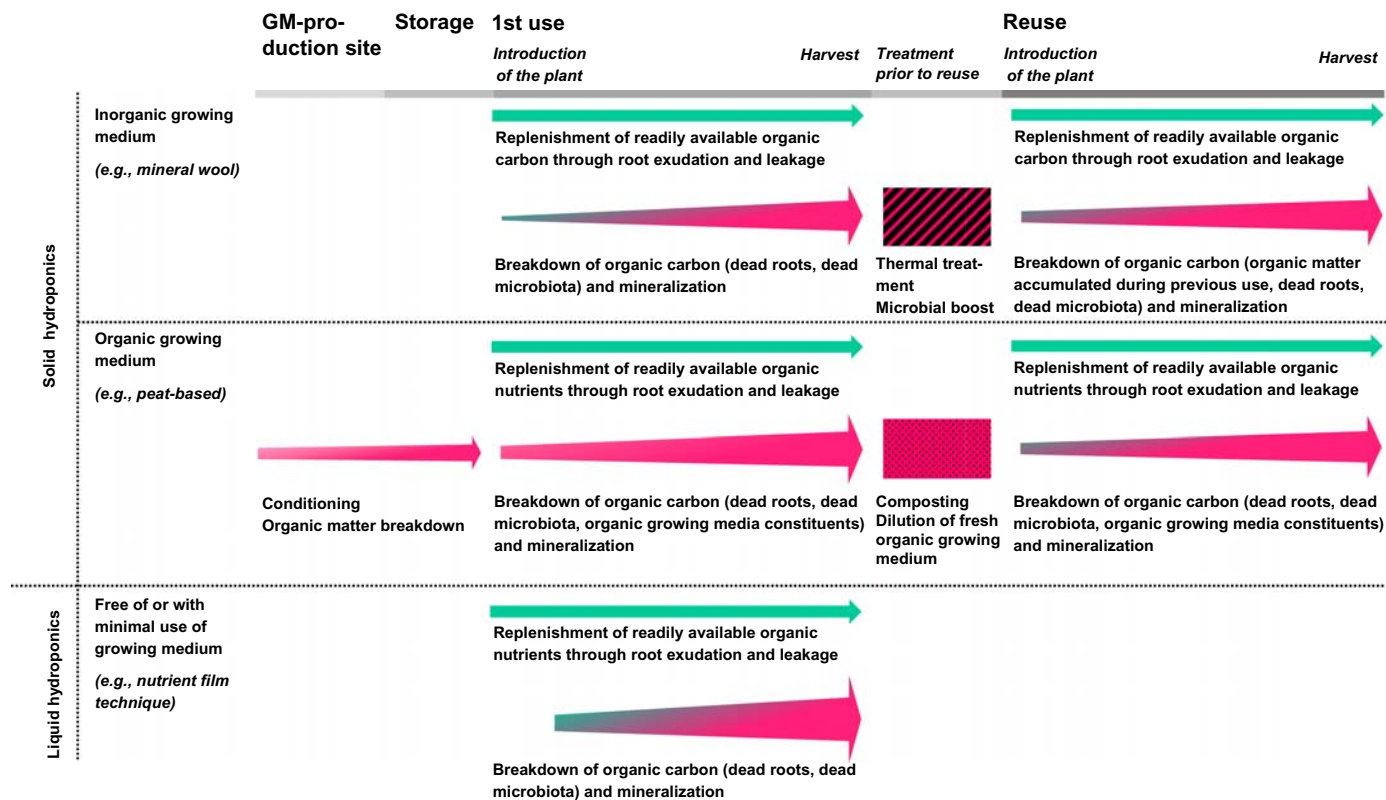


FIGURE 5.3

Fate of the organic carbon pool in solid hydroponics (using inorganic or organic growing medium) and liquid hydroponics (free from or with minimal use of growing medium; e.g., the nutrient film technique). The green arrows denote the continuous organic carbon flow deriving from the plant roots through exudation and leakage. The red arrows indicate breakdown and mineralization of organic carbon. The size of the arrows indicates the dynamics in carbon flow. The nature of the treatment prior to reuse is decisive for mineralization of organic matter.

(Illustration: B. Alsanius)

Growing Medium	Phase			Bacteria	Fungi	Reference
	Pre-pl	Post-pl	Reuse			
Rockwool		● ^a		Log 6.2 CFU/g moist rockwool	Log 1–2 CFU/g moist rockwool	Postma et al. (2000b)
		● ^b		Log 7.8 CFU/g moist rockwool	Log 3.9 CFU/g moist rockwool	Postma et al. (2000b)
				Log 7 CFU/g moist rockwool	Log 3.5 CFU/g moist rockwool	Postma et al. (2000b)
				Log 7.6 CFU/g moist rockwool	Log 4.5 CFU/g moist rockwool	Postma et al. (2000b)
Peat-based growth medium	●			2×10^4 CFU/mL	3.5×10^5 CFU/mL	Martínez et al. (2013)
Coir fiber	●	●		2.96×10^7 CFU/mL	4.7×10^5 CFU/mL	Martínez et al. (2013)
				5.01×10^6 CFU/mL	9×10^4 CFU/mL	Martínez et al. (2013)
		●		4.5×10^6 CFU/mL	1×10^4 CFU/mL	Martínez et al. (2013)

The studied phases were classified depending on incidence. Note that the units vary between different studies. ^a7 days after sowing; ^b35 days after sowing; ^{c,d}reused, non autoclaved; ^c7 days after sowing; ^d35 days after sowing. post-pl, Postplantation; pre-pl, preplantation.

in the microbial community structure. If reused, organic growing media residues are composted and may be reintroduced into fresh organic growing medium.

Initially, the situation in liquid culture systems resembles that in inorganic growing media, since before transplanting there is nothing organic in the water. Under the specific conditions provided, a considerable root mat develops. However, absence of a supporting structure leads to gradual compaction and oxygen depletion within this root mat, which affects the microbial community structure and metabolism. The continuous flow of nutrient solution across the root dilutes the concentration of organic carbon exuded by the plant, leading to a gradient of decreasing organic carbon content with increasing distance from the root surface (Fig. 5.3).

Table 5.1 shows typical microbial densities in different growing media.

5.2.3 NUTRIENT SOLUTION

Although the nutrient solution in hydroponic systems is much lower in organic carbon content than the root rhizosphere and growing medium, it must be considered eutrophic (Alsanius et al., 2014). It is by definition rich in inorganic nutrients for both plants and microbes. The content of organic nutrients is an outcome of the organic carbon content of the raw water used for preparation of the nutrient solution, leakage from system devices, and particles and dissolved carbon leached from the rhizosphere and growing medium or originating from the organic growing medium, decaying roots, or dead microbiota. Thus, the nutrient solution is less densely colonized [bacteria: 10^5 – $10^{7.1}$

colony-forming units (CFU)/mL; fungi: $<1-10^{3.4}$ CFU/mL] (Khalil and Alsanius, 2001; Rosberg et al., 2014; Alsanius and Jung, 2004) than the root zone (bacteria: $10^{9.2}-10^{10}$ CFU/g; fungi: $10^{4.2}-10^{5.5}$ CFU/g (Koohakan et al., 2004; Rosberg et al., 2014) and the growing medium (see Table 5.1). Nevertheless, it is in a state of dynamic change. However, it is worth noting that these comparisons are hampered by the fact that they are expressed on a different basis, that is, per unit mass and volume. The most dramatic shift in microbial density of the nutrient solution occurs after transplantation, resulting in a 1000-fold increase in bacteria within 20 hours. However, shifts also occur during the growing season (Renault et al., 2017).

Proteobacteria and Parcubacteria have been found to be the dominant microbial taxa associated with the nutrient solution in soilless cropping systems (Renault et al., 2017). It may be tempting to draw conclusions on their importance from these studies. However, to actively exploit microbes and their actions in hydroponic systems, knowledge about their function is needed, rather than on their taxonomy or content of certain genes. Therefore, more attention should be paid to functional biodiversity in relation to phenological mapping of the cropping system.

5.3 POSITIVE AND NEGATIVE EFFECTS OF ROOT ZONE MICROBIOTA

In principle, the microbiota in soilless growing systems will interact with crop plants in a similar way as microbiota interacts with plants growing in soil (e.g., Alabouvette et al., 2004; Rheinberg and Shaw, 1976). Since microbe–plant interactions are known to affect plant health, nutrition, and growing media decomposition in soil-based system, it is interesting to identify the degree to which this is the case in soilless culture. Furthermore, the growing medium may be colonized by interfering microbial contaminants (e.g., plant or human pathogens, molds, phytotoxin-forming microorganisms). These positive or negative effects have to be considered in evaluation and development of growing media and/or complex soilless cropping systems.

5.3.1 PLANT HEALTH AND ROOT ZONE MICROBIOTA

Growing media used in soilless growing systems ought to be free from plant pathogens. This is true for growing media with manufacturing process that include high temperatures, for example, mineral wool, perlite, or vermiculite. Plant pathogens may occur if the constituents with inferior hygienic properties are used. However, such pathogens may be introduced and dispersed due to (1) poor hygienic conditions during production or handling of the growing medium, or during reconditioning before reuse; (2) introduction of infected seeds and plant material; and (3) soilless cropping systems without adequate sanitation (see Section 5.5.3). Beneficial microorganisms in the root zone of a soilless cropping system may contribute to mitigate the risk of plant diseases. For cultivation in soil, such disease suppressive effects have been recognized for more than 100 years and have been well studied (e.g., Alabouvette et al., 1979; Chandrashekara et al., 2012; Cook and Baker, 1983; Huber and Schneider, 1982). Disease suppressive effects have also been documented for organic growing media and growing medium components, for example, compost (see Chapter 8: Organic Soilless Media Components) and peat and can also be induced by biocontrol agents (see Section 5.5.2).

5.3.2 DISEASE SUPPRESSIVE EFFECTS OF PEAT

In contrast to compost, untreated peat is usually poorly colonized by microorganisms before first use. Therefore, various authors considered peat as a “disease conducive control” in their studies about diseases suppression (Boehm et al., 1997; Borrero et al., 2004; Van Der Gaag et al., 2004; Hoitink and Fahy, 1986; Noble and Coventry, 2005). However, after adjusting the pH to 6.0–6.5 and the water content to about 80% of maximum water holding capacity of peat substrate, a significant increase in viable counts (bacteria) has been observed (Wohanka et al., 2010). Various publications have shown that peat can provide some disease suppressiveness, compared with sterilized peat in particular (Hunter et al., 2006; Inbar et al., 1991; Tahvonen, 1982; Wohanka et al., 2010; Wolffhechel, 1988). There are some indications that this effect might be attributable to microbial community structure, metabolism, and activity, and to the decomposition level of the peat. Experiments with Canadian sphagnum peat of different decomposition levels have shown that root rot of poinsettia (*Euphorbia pulcherrima*) and development of *Pythium ultimum* are greatest in dark peat (H4 on the von Post decomposition scale), intermediate in slightly decomposed peat (H3), and suppressed in light peat (H2) (Boehm and Hoitink, 1992) (Fig. 5.4). The highest microbial activity [fluorescein diacetate (FDA) activity] in that study was detected in light peat and the lowest in dark peat. However, compared with compost, peat mixes tend to have lower disease suppressive effects (see Chapter 8: Organic Soilless Media Components). Disease suppressive effects generally decline during decomposition of peat, while the pattern of change in composts is more complex. The crucial role of the microbiota is highlighted by the fact that heat sterilization eliminates disease suppressiveness (Boehm and Hoitink, 1992; Wohanka et al., 2010). However, there are no clear factors determining the suppressiveness of a specific peat, although the level of decomposition, the active microbial biomass, and the composition of the microbial community are reported to be important (Boehm et al., 1993; Chen et al., 1988a,b; Hoitink and Boehm, 1999).

5.3.3 DISEASE SUPPRESSIVE EFFECTS IN HYDROPONIC SYSTEMS

In contrast to organic growing media, inorganic growing media lack a source of readily available carbon for microbial growth. Therefore, while microbial colonization is initially limited, as soon as plants are introduced a significant increase in microbe numbers is observed (Postma et al., 2000a; Berkelmann et al., 1994). Various studies show that the microbiota have the potential to suppress plant diseases in soilless culture (McPherson, 1998; Postma et al., 2000a, 2001; Tu et al., 1999). In closed tomato cropping systems where the nutrient solution is recirculated, suppressiveness against *Phytophthora cryptogea* (McPherson, 1998) and *Pythium* sp. has been reported (Tu et al., 1999). In another study, nutrient solution from a 9-week-old tomato crop very strongly inhibited mycelial growth of *Fusarium oxysporum* f.sp. *lycopersici* in an in vitro test. Heat sterilization or membrane filtration (<0.2 µm) eliminated this antagonistic effect completely, suggesting the cause to be biological (Berkelmann, 1992). Furthermore, reuse of nonsanitized rockwool slabs resulted in the build-up of a suppressive microbiota against *F. oxysporum* in tomato (Srinivasan et al., 2009) and against *Pythium aphanidermatum* in cucumber (*Cucumis sativus*) (Postma et al., 2001, 2005a). Suppressiveness was also reestablished in sterilized rockwool slabs in those studies when the initial microbiota extracted from a healthy crop was reintroduced. In experiments with *P. aphanidermatum* in cucumber, the suppressiveness was correlated with bacterial diversity and the number of filamentous actinomycetes (Van Os et al., 2004b).

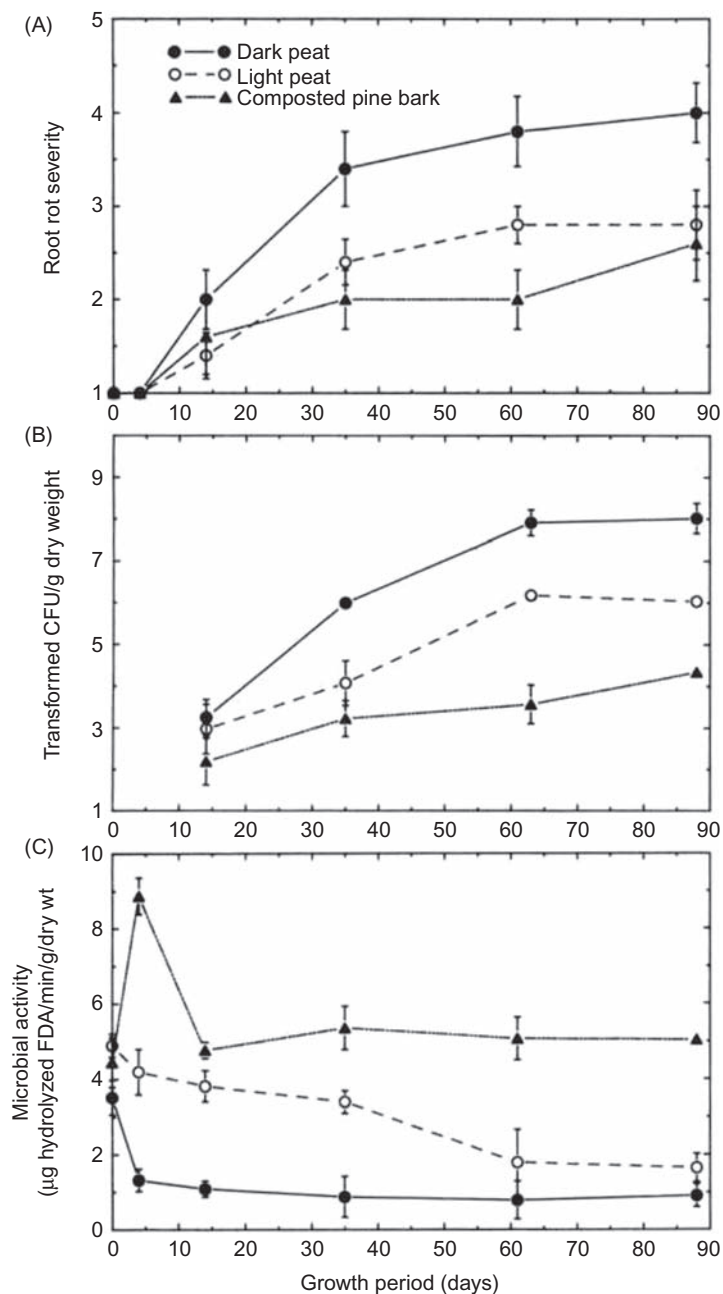


FIGURE 5.4

Comparison of *Pythium* root rot severity, population density of *Pythium ultimum*, and microbial activity in a light (H3) and a dark (H4) peat mix, and a composted pine bark-amended potting mix. (A) Root rot severity; 1 = symptomless, 2 = mild root rot, 3 = severe root rot, 4 = severe root rot and crown infection, and 5 = dead plant. Vertical bars represent standard error ($n = 5$). (B) Population density of *P. ultimum*. Population density data were transformed using Taylor's power law ($b = 0.76$). Vertical bars represent standard error ($n = 3$). (C) Microbial activity based on the rate of hydrolysis of fluorescein diacetate. Vertical bars represent standard error ($n = 4$).

Reproduced, by permission, from Boehm, M. J., and Hoitink, H. A. J. 1992. Sustenance of microbial activity in potting mixes and its impact on severity of *Pythium* root rot of poinsettia. *Phytopathology* 82, 259–264. Copyright The American Phytopathological Society.

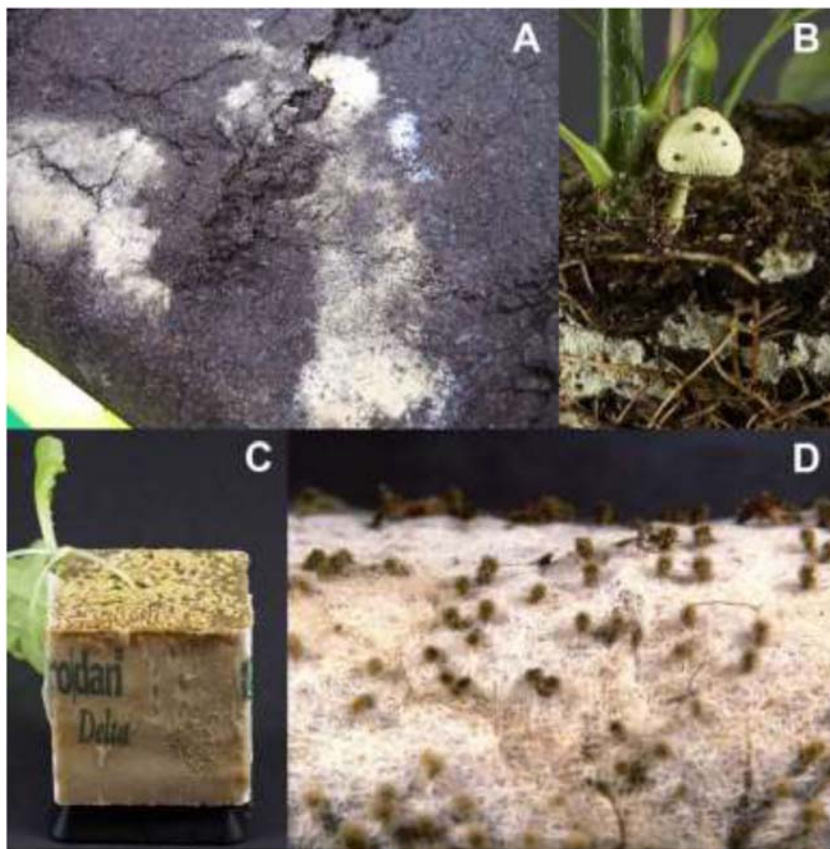
The mechanisms of disease suppression in soilless culture systems are as yet poorly understood. The release of antibiotics and other metabolites (Hultberg et al., 2000; Jung, 2003; Ongena et al., 1999), the production of biosurfactants (Dáes et al., 2010; Hultberg et al., 2010; Stanghellini and Miller, 1997) or siderophores (Benizri et al., 1995; Van Peer et al., 1988), the competition for nutrients (Zheng et al., 2000), and microbially induced resistance in the plant (Ongena et al., 1999; Paulitz, 1997) have been suggested as probable mechanisms for the disease suppressive effects.

Unfortunately, disease suppressive effects are not always strong enough to prevent serious plant damage and are not clearly predictable. Therefore, introduction of biological control agents (BCA) and/or microbe habitat management favoring ambient microbiota capable of antagonizing plant root pathogens should be considered (Section 5.5.2).

5.3.4 INTERFERING COLONIZATION BY NONPHYTOPATHOGENIC MICROBES

Development of visible fungal structures, such as mold layers of *Peziza* spp. (most often *Peziza ostracoderma*), in peat-based growing media have been reported in many cases (Hennebert and Korf, 1975; Schlechte, 1997; Meinken et al., 2016). Such fungi can also develop on surfaces of wet rockwool and various organic growing media (Fig. 5.5). Small seedlings may be overgrown by fast developing fungi. Hydrophobic mycelium, often originating from *Leucocoprinus birnbaumii*, has been observed in various growing media (peat, coco fiber), decreasing the wettability. A positive correlation between fungal biomass and hydrolyzable organic carbon (and C/N ratio) has been found for a variety of organic growing media when subjected to artificial inoculation with *P. ostracoderma* (Meinken et al., 2016). White mycelia and other microsclerotia of *Athelia turficola* often occur on peat-based growing media stored in plastic bags, contributing to nitrogen immobilization (Schlechte and Hoffmann, 2000; Schlechte and Schmielewski, 2010). Allergic reaction of consumers to spores of higher fungi (e.g., *Aspergillus* spp., *Penicillium* spp.) developing on the substrate surface of pot plants can pose a potential health risk (Schlechte and Schmielewski, 2010; Staib et al., 1978). Even if fungal layers or fruiting bodies do not have negative effects on plant growth, their presence may still be a quality problem for the grower or the consumer. A comprehensive overview of many fungi colonizing organic growing media is available in a compendium (Schlechte and Schmielewski, 2010).

Some microorganisms present in soilless systems may interfere with human health. Sanitary hazards can originate from the growing medium or from raw water used for preparation of the nutrient solution or for dissolving or diluting chemical protection agents, or can be introduced by plant material (seeds, small plants). Growing medium amended with improperly composted animal wastes can be an important port of entry for human pathogens transmitted through the fecal–oral route. European legislation lists appropriate animal wastes or bioproducts that may be introduced into compost and sets sanitary quality standards (European Commission, 2011; European Council, 2002). *Salmonella* spp., *Escherichia coli*, and *Enterococcaceae* are indicator organisms. Composted products should be free of *Salmonella* in any sample. The average number of viable counts for *E. coli* or *Enterococcaceae* isolated from five representative samples may not exceed 1000 CFU/g and only one of the five samples may contain 1000–5000 CFU/g. However, any presence of the extremely pathogenic enterohemorrhagic *E. coli*, infective at very low doses (<100 cells), is not acceptable in compost products. To ensure the sanitary quality of composts, any raw compost feedstock based on animal waste material should be checked with respect to its contaminant load.

**FIGURE 5.5**

Examples of interfering colonization of growth media with nonphytopathogenic microbes. (A) *Athelia turficola* (Schlechte and Hoffmann, 2000); (B) *Leucocoprinus birnbaumii* (Corda, 1839; Singer 1961); fruiting body and sclerotia; (C) *Peziza ostracoderma* (Korf, 1961); (D) *Chaetomium globosum* (Kunze ex Fr., 1829).

Photo: (A), (C), and (D): W. Wohanka; (B) W. Schönbach.

The temperatures obtained during the thermophilic phase of composting inactivate the majority of enteric bacteria, such as *Enterobacteriaceae* (Franke-Whittle and Insam, 2013). However, the feedstock needs to be thoroughly blended, heat needs to be able to distribute evenly throughout the entire feedstock during composting, and the final compost needs to be stored adequately to avoid recontamination of the mature product.

Studies from various countries report the presence of *Legionella* species in green waste composts and organic growing media (Casati et al., 2009, 2010; Cramp et al., 2009; De Jong and Zucs, 2010; Koide et al., 1999; Lindsay et al., 2012; McCabe et al., 2011; Pravinkumar et al., 2010; Steele et al., 1990; Velonakis et al., 2010). Moreover, cases of Legionnaire's disease and of Pontiac fever have been linked to use or contact with growing media before onset of the disease

(Cramp et al., 2009; Koide et al., 1999; Wallis and Robinson, 2005; Hughes and Steele, 1994; Den Boer et al., 2007; Kingston and Padwell, 1994). The *Legionella* organism has been isolated from composts and growing media containing wooden debris, bark, green wastes (including fresh green plant residues), and peat (Casati et al., 2009, 2010; Koide et al., 1999; Velonakis et al., 2010). According to tests using different determination methods, it is present in very high levels in some of these substrates (viable count: 10^3 – 10^5 CFU/g; qPCR: 10^4 – 10^6 GU/g) (Casati et al., 2009). *Legionella longbeachae* serogroup 1 and *Legionella pneumophila* are reported to persist in garden soil or organic growing media for several months (Van Heijnsbergen et al., 2016). Aerosol formation and hence aspiration, but also ingestion, have been cited as routes of transmission. Apart from the substrate per se, other factors can also add to growing medium-associated *Legionella* infections, such as smoking, proximity to drips from hanging baskets, and eating and drinking while gardening without washing the hands (O'Connor et al., 2007). Health-risk perceptions regarding *Legionella* and growing media vary widely, leading to recommendations to label consumer bags with a warning (Cramp et al., 2009), use protective clothing (face masks) (Cramp et al., 2009; Wallis and Robinson, 2005), and issue information to users to increase health awareness and hand hygiene (O'Connor et al., 2007).

Microbes, heterotrophic bacteria in particular, can utilize organic matter present in soilless systems as an energy source and proliferate, which can cause operational problems with the irrigation system and lead to uneven water supply. Beside physical and chemical factors, microbially formed exopolysaccharides are a major source of clogging in drip irrigation systems (Bucks et al., 1979; Dosoretz et al., 2011). Various chemicals can be applied continuously or intermittently to mitigate the risk of biofouling, such as chlorine (Carrillo et al., 1996; Liu and McAvoy, 2012), peracetic acid, hydrogen peroxide, or other oxidizing agent (Benoit and Ceustermans, 2004; Nederhoff, 2000a; Van Der Burg, 1996). Flushing the tubing with acid to remove inorganic particles may also reduce microbial growth. Furthermore, clogging may be reduced through choice of appropriate nozzles, as recently reviewed by Dorais et al. (2016).

5.3.5 MICROBIAL PRODUCTION AND DEGRADATION OF PHYTOTOXINS

Under certain conditions, microbes can produce phytotoxic compounds in the organic growing media or nutrient solutions of hydroponic systems. Under anaerobic conditions, various bacteria (e.g., *Micrococcus denitrificans* or *Thiobacillus denitrificans*) can produce nitrate reductase, resulting in enrichment of phytotoxic nitrite (Payne and Grant, 1981), which can cause serious root damage (Fischer and Meinken, 1988). However, nitrite accumulation may also occur under aerobic conditions, due to the initial lack of nitrifying bacteria (e.g., *Nitrobacter* spp.) in new synthetic substrates (Bar-Yousef, 2008). Another inorganic phytotoxin produced under anoxic conditions is hydrogen sulfide. It has been detected in toxic concentrations in nutrient solution in deep ornamental containers filled with expanded clay (H.-D. Molitor, pers. comm. 2018).

Inhibited plant growth in hydroponic systems has also been ascribed to various organic molecules. For example, Yu and Matsui (1993, 1994), Jung et al. (2002), and Kitazawa et al. (2005) detected organic acids, predominantly phenolic acids, in the nutrient solution of hydroponic systems with tomato, cucumber, and strawberries (*Fragaria* sp.). A similar finding has been reported for organic growing media (Politycka et al., 1984). However, the claimed growth depressions caused by phenolic acids have not always been proven under ambient conditions (Caspersen et al., 1999;

Jung et al., 2002, 2004). Below concentrations of 100–200 μM of individual phenolic acids (i.e., benzoic acid, caffeic acid, chlorogenic acid, p-hydroxy-benzoic acid, salicylic acid, vanillic acid), growth is commonly not significantly affected. However, root growth, root branching, root hair formation, and root and shoot biomass have been reported to be impaired at higher concentrations of individual phenolic compounds. Apart from the conditions in the hydroponic system, high irradiance lowers the phytotoxic effect, as shown for ferulic acid and lettuce (*Lactuca sativa*) under gnotobiotic conditions, that is, when grown from sterilized seeds and under aseptic conditions except for experimental colonization by a set of known or defined bacterial species (Caspersen et al., 1999). Furthermore, under aerobic conditions phenolic compounds in the nutrient solution of hydroponic systems are efficiently degraded by the associated microbiota (Alsanius and Jung, 2004). Coinoculation of nutrient solution with ferulic acid (200 μM) and a multiple-strain cocktail of *Pseudomonas* spp., or amendment with biodegraded olive mill wastewater (*bio-margine*), which is rich in phenolics, may even increase plant growth and biomass production (Caspersen et al., 2000; Kistner et al., 2004).

Although hydroponics seems to be a self-regulating system with respect to phenolic acids by means of the indigenous microbiota, this does not apply to all organic molecules that may occur, especially in the nutrient solution of closed hydroponics, for example, pesticide residues. Different studies indicate the persistence of active fungicidal and insecticidal compounds (fenhexamid, pyrimethanil, imidacloprid, etridiazole, cyromazine) in the nutrient solution of hydroponic systems (Alsanius and Bergstrand, 2014; Alsanius et al., 2013; Crum et al., 1985; Dekker et al., 1995). However, the residues detected differ between studies using different active compounds, which might point to a compound-dependent effect on persistence (Dunsing et al., 1988; Patakioutas et al., 2007).

A special problem is the phytotoxicity that occurs with self-heated peat. Microbially induced self-heating (50°C–70°C) can occur in many types of organic material, such as hay, compost, or peat, under unfavorable storage conditions. Self-heated peat is usually characterized by darker color, specific smell [similar to the aroma of lovage (*Levisticum officinale*)], and occasionally increased visible fungal growth following transport and storage (Belke, 1997; Wever and Hertogh-Pon, 1993). During self-heating, a succession from mesophilic to thermophilic organisms is assumed (Göttlich, 1990; Kuester, 1972; Ranneklev and Bååth, 2001; Schlechte, 2003). Common fungi found at relatively high densities in self-heated peat include *Penicillium* spp., *Mortierella* spp., *Trichoderma* spp., and others (Schlechte, 2003; Tahvonen and Kemppainen, 2008). On the basis of phospholipid fatty acid (PLFA) analysis, Ranneklev and Bååth (2003) showed that the total amount of PLFAs was significantly higher in self-heated peat than in its nonheated counterpart. Furthermore, they found that two PLFAs (T1, T2) related to bacteria belonging to the genus *Alicyclobacillus* indicated self-heating events. Although there is a clear change in the chemical composition of self-heated peat, the reduction in plant growth seems to be the result of phytotoxic compounds of microbial origin (Wever and Hertogh-Pon, 1993; Wever and Kipp, 1997). Experiments with Chinese cabbage (*Brassica rapa* var. *chinensis*), lettuce, and cucumber grown on self-heated peat have demonstrated reduced seed emergence, growth inhibition, and occurrence of chlorotic leaf spot symptoms, with lettuce being particularly sensitive (Tahvonen and Kemppainen, 2008). From these experiments, it can be concluded that compounds harmful to plants can develop in stored peat material during the end phase of the self-heating process. The nature of these harmful compounds has not been identified, but it can be assumed that they are a result of excessive

microbiological activity (Tahvonen and Kemppainen, 2008). After storage of self-heated peat, the plant growth reduction effect tends to disappear (Wever and Hertogh-Pon, 1993). Growth tests to monitor the phytotoxicity of compost are described in detail in Section 11.5.3.

5.3.6 MICROBIAL INTERACTION WITH PLANT NUTRITION AND STABILITY OF GROWING MEDIA

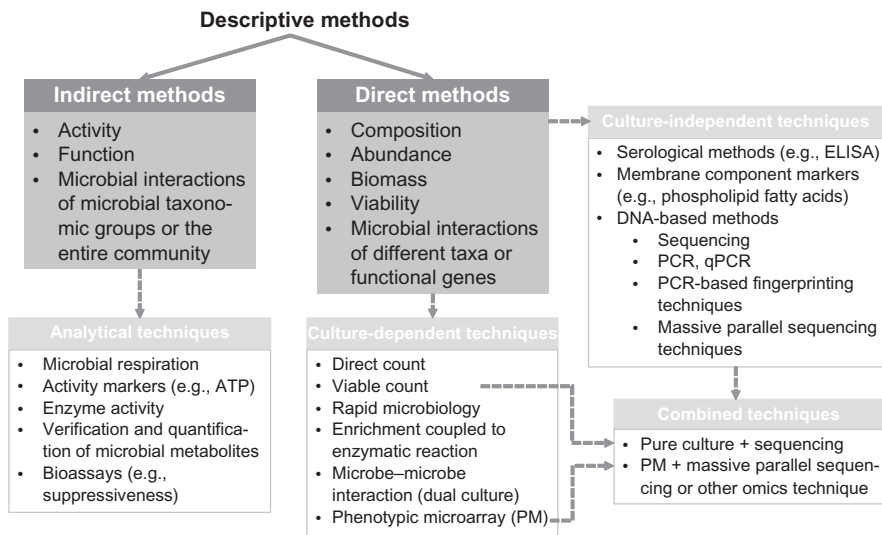
Impact of the root zone microbiota on plant nutrition and the stability of growing media are described in detail in Chapter 6, Nutrition of Substrate-Grown Plants, and Chapter 11, Analytical Methods Used With Soilless Substrates. Many microbes are involved in the availability of nutrients, nitrogen in particular. Depending on oxygen availability, specialist bacteria transform ammonia into nitrite and finally nitrate. Under anaerobic conditions, the phytotoxic nitrite can accumulate. In organic and inorganic growing media, microbes are responsible for the mobilization and immobilization of plant nutrients. Silber et al. (2008) found that Mn^{2+} is oxidized by microorganisms to Mn^{3+} which reacts with phosphate to produce insoluble salts that are unavailable to plants. A high degree of nitrogen immobilization has been observed particularly with nonimpregnated wood fiber (Gruda et al., 2000).

5.4 METHODS FOR ANALYSIS OF THE MICROBIOTA ASSOCIATED WITH ROOTS, GROWING MEDIA, AND NUTRIENT SOLUTION

Microbial analysis of microorganisms in the root zone, growing medium, and nutrient solution of hydroponic systems has undergone major advances with respect to commercial horticulture realities and scientific possibilities. Plant health and substrate stability were the primary drivers of interest in early studies, but the potential of the associated microbiota has received increasing attention as more advanced and culture-independent microbiological techniques have entered the scene. Basically, the available methods can be grouped into indirect and direct methods, where indirect methods mainly focus on various microbial activities, without differentiating a specific organism or strain, and direct methods focus on the presence, abundance, viability, or function of different taxa (Fig. 5.6). Direct methods can be divided into culture-dependent and culture-independent methods. A precondition for culture-dependent methods is that the organism or group of microbial organisms can be grown in artificial microbiological medium. Culture-independent methods are based on the identification of microbial taxa through DNA-based methods or biochemical methods. Both culture-dependent and culture-independent methods can provide information on microbial function or expression of certain properties in a specific context.

Thanks to advances in culture-independent methods, interest in microbial interactions in the root zone, growing medium, and nutrient solution of soilless systems has increased exponentially. However, to exploit the full potential of microbes and their functions in these contexts, paradigm shifts are needed.

First, there is a need to abandon the existing man-made division of the microbiota in the root zone, growing medium, and nutrient solution of hydroponic systems into beneficial, neutral, deleterious, or pathogenic organisms. We postulate that

**FIGURE 5.6**

Methods for characterizing and quantifying the composition and function of the microbiome associated with growth medium, plant roots, and nutrient solution in hydroponic systems.

- Microbes are unsuited for this type of grouping, in fact they do not align to the human perspective in their life forms. The microbial lifestyle is about making a living, and thus survival and proliferation, through successful exploration and attachment to new nutrient-rich habitats that suit the preferences and level of activity of the particular microbe
- Plant health, including plant health-related effects such as suppressiveness, conduciveness, and pathogenicity, is a function of habitat, plant, and resident microbiota.

To exploit the full potential of the microbiota associated with the root zone, growing medium, and nutrient solution in hydroponics, mechanisms of interest need to be identified and their mode of action and preconditions defined. Such processes may take place in microloci and may not be affected by general (coarse, nonspecific) manipulation of the habitat. Thus, entry into a new area of microbiology in soilless culture will require multidisciplinary approaches, a microbial perspective, a good understanding of ecological processes, and dedicated analysis and modeling, rather than a technical perspective and modifying one environmental/biotic factor or another. The methods for description of microorganisms, microbial communities, and microbial processes associated with the root zone, medium, and nutrient solution in hydroponics can therefore be divided into:

- Quantitative methods revealing microbial biomass and microbe numbers
- Functional methods describing microbial activity
- Methods for monitoring microbial diversity and community structure
- Methods describing plant-microbe interactions.

The purposes of these groups of methods are summarized in [Table 5.2](#).

Table 5.2 Methods Used for Microbial Characterization With Respect to Disease Suppressiveness and Contamination of Growing Media, Roots, and Nutrient Solution, and Use and Reuse of Growing Media in Hydroponic Systems

	Suppressiveness	Contamination	Use and reuse
Bioassays	Mandatory	Limited	Useful
Quantitative analysis	Microbial biomass = basic information	Specific groups or single organisms	Dynamics of relevant microorganisms “monitoring”
Functional analysis	Specific groups		
	Enzyme activities relevant for disease suppression	Enzyme activities	Enzyme activities include nitrogen metabolism
Microbial diversity and community structure		Toxic compounds	Toxic compounds
	Specific groups (viable counts, DNA-based methods)	Specific groups (viable counts, phenotypic microarray, DNA-based methods)	Specific groups (viable count, DNA-based methods)
	Physiological and chemical profiles		Physiological and chemical profiles

Modified from Alsanius, B.W., Wohanka, W. 2009. Prospects for biological characterization and evaluation of growing media. Acta Hortic. 819, 99–110.

In contrast to other branches within microbiology, for example, soil, food, and water microbiology and plant pathology, microbiologists working with issues related to the biology of hydroponic systems have not agreed on a gold standard for microbial analysis. Therefore, various methods, sometimes inappropriate in a given context, are used. The range of methods employed leads to difficulties in comparing results and drawing general conclusions. There is an urgent need to scrutinize microbial analysis methods relating to hydroponics and to agree on a basic set of suitable and scientifically sound methods, like the set identified for characterization of chemical and physical properties of growth media.

5.4.1 QUANTITATIVE METHODS

Quantitative methods are based on microscopy or biochemical or chemical approaches. The term “quantitative” per se may create unrealistic expectations for this group of methods, since they produce estimates rather than accurate quantifications. As for other procedures, pretreatment and/or extraction of the microbiota is important for the results obtained. The main types are direct cell count, plate count, enrichment-coupled enzymatic reactions, molecular markers, and biochemical and physiological techniques.

5.4.1.1 Direct cell count

Direct cell count, using count chambers and light microscopy, is assumed to be a reliable tool for estimation of number and volume of microbial cells (Alef, 1991). Microbial groups and structures and living and dead cells can be differentiated by staining, as previously described by Alsanius and Wohanka (2009).

5.4.1.2 Plate count

Plate counts are based on the use of general or semiselective media, formulated with respect to the demands of the specific microbial groups or organisms. Plate count results reflect the number of colonies that can emerge under the given physical and chemical conditions (atmosphere, temperature, pH, available nutrients, presence of growth inhibitory compounds). Colonies are aggregates of living microbial cells, and hence, the results cannot be compared with those from direct counts. Plate counts underestimate the presence of microorganisms, as quiescent, viable but not culturable, and nonculturable microorganisms are omitted from the count. Several procedures have been used in microbial analysis of growing medium, roots, and nutrient solution from hydroponic systems, for example, the plate pour method, plate spread method, and split plate method (Madigan et al., 2017; Thomas et al., 2015). The nutritional formula may be more or less specific for microbial groups or taxa. Chromogenic media may allow differentiation of colonies based on colony size, shape, and color, but this needs to be confirmed biochemically, serologically, or by DNA-based methods. Apart from the nutritional formula, incubation temperature and duration of incubation are essential to obtain comparable results.

5.4.1.3 Enrichment-coupled enzymatic reactions

The enzymatic cleavage of specific substrates by specific taxa may result in fluorogenic or chromogenic reactions, and thereby allow the identification and quantification of specific taxa. In the case of chromogenic reactions, the specific substrate is conjugated with a chromophore, which is released and displays a specific and characteristic color upon cleavage. Likewise, enrichment coupled to fluorogenic reaction makes use of a fluorophore resulting from enzymatic cleavage of a specific substrate. For example, *E. coli* metabolizes 4-methylumbelliferyl-beta-D-glucuronide through formation of beta-D-glucuronidase. This reaction yields the 4-methylumbelliferyl moiety, a compound which fluoresces in the presence of ultra-violet (UV) light. Test methods based on similar fluorogenic and chromogenic reactions have been developed for important microbial taxa, such as *Enterobacteriaceae* and enterococci, for important culturable human pathogens (e.g., *Salmonella*, *Listeria*, *Yersinia*, *Vibrio*, *Legionella*), and for indicator organisms of contamination (e.g., total coliform bacteria). Identifications performed on the basis of the chromogenic or fluorogenic colony response need to be confirmed by biochemical and/or serological methods or by DNA-based technique (sequencing).

5.4.1.4 Molecular markers

PLFAs are constituents of all cell membranes. They do not have any storage function. Under natural conditions, they are a relatively consistent fraction of cell mass and the composition of PLFAs is characteristic for different life forms. The total amount of PLFAs can be used for quantification, as an indicator of viable microbial biomass, as phospholipids are rapidly degraded upon cell death. Certain single fatty acids are specific signature biomarkers for specific microbial groups. PLFA analysis has been successfully employed to describe the microbial community structure in various subentities of hydroponic systems. The analytical procedure, consisting of lipid extraction followed by fractionation, methanolysis, and gas chromatographic analysis, and statistical analysis of the data have been demonstrated in various publications (Debode et al., 2016; Khalil and Alsanius, 2001;

Khalil et al., 2001a). As PLFAs reflect the biomass of living microbiota, they are often used in combination with massive parallel sequencing (MPS) techniques for biomass quantification.

Sterols account for approximately 1% of fungal biomass, with ergosterol as the dominant compound. Ergosterol is considered a sensitive measure of fungal biomass in soils, and this method has also been transferred to growing media studies (Meinken et al., 2016; Lohr et al., 2017).

5.4.1.5 Biochemical and physiological techniques

The fact that microorganisms respond quickly to environmental changes has been used as a basis for estimation of microbial biomass. Various laboratory methods have been established for quantification of microbial activity, such as

1. *Substrate-induced respiration*, which reflects the evolution of CO₂ in response to addition of glucose. This technique can be combined with fumigation to differentiate between microbial groups (Alef, 1991; Alsanusi and Wohanka, 2009; Amha et al., 2015; Amha et al., 2011; Kjöllér and Struwe, 1994).
2. *Chloroform extraction*, translating the impact of cell lysis due to fumigation and extraction into increases in various cell components (Brookes and Joergensen, 2006).
3. *ATP extraction*, since ATP is present in living cells and ceases rapidly after cell death, it is a measure of microbial activity and living microbial biomass. A variety of ATP extraction methods are available from soil studies. In the context of growing media and hydroponic systems, suitable ATP extraction methods need to be selected carefully with regard to organic matter content, clay content, and content of potentially inhibitory compounds, in order to obtain reliable results. A method suitable for composts has been described by Turner and Carlile (1984).

5.4.2 FUNCTIONAL METHODS

5.4.2.1 Microbial respiration

Microbial respiration, involving oxygen uptake and carbon dioxide evolution by bacteria, fungi, algae, and protozoa, is a measure of microbial activity and covers gas exchange in both aerobic and anaerobic metabolism (Schinner et al., 1996). It is linked to the degradation of organic matter and is hence a fundamental process in all subentities of hydroponic cropping systems, with carbon mineralization as a function of the composition and availability of the carbon-containing material and the prevailing environmental conditions (see Sections 5.2.1 and 5.2.2). There are various field and laboratory methods available for measurement of respiration without additional substrate, such as glucose. The microbial metabolic quotient (respiration-to-biomass ratio), or $q\text{CO}_2$, is used to determine the qualitative influence on biomass. It is an indicator of the energetic efficiency of a microbial community; the lower the $q\text{CO}_2$, the more efficient the microbial turnover.

5.4.2.2 Nitrogen mineralization

The conversion of organic nitrogen into the inorganic state (nitrogen mineralization) is mediated by microorganisms. This process is divided into two steps, namely, (1) ammonification, that is, the conversion of organic nitrogen-containing substances into ammonium, and (2) nitrification, consisting of the microbial conversion of ammonium into nitrite and of nitrite into nitrate (see also

Chapter 4: Chemical Characteristics of Soilless Media). Nitrogen mineralization can be studied in the field and laboratory and involves determining the activity of arginine deaminase and urease, or ammonium oxidation. It may also be monitored in laboratory studies. Existing test methods designed for assessment of nitrogen mineralization under open-field conditions can be relevant for hydroponic systems, but incubation conditions need to be standardized, especially with respect to temperature, incubation duration, and moisture.

5.4.2.3 Enzyme activity

Measurements of enzyme activity are an expression of the biologically generated turnover of organic matter and fluxes of compounds (carbon, nitrogen, etc.). Methods for enzyme activity analysis have been developed for soil studies. Enzyme activities indicating general and specific microbial activities have been utilized in studies on hydroponic systems or growing media. These include

1. *Dehydrogenase activity*, indicating biological redox systems, which can be interpreted as a measure of microbial metabolic intensity. Dehydrogenase activity has been used with regard to growing media and composted material (Carlile and Wilson, 1991).
2. *FDA hydrolysis*, a measure of global hydrolytic capacity and used as a broad-spectrum indicator for biological activity (Shaw and Burns, 2006).
3. *Cellulase and xylanase activity*, which are of interest both with respect to the stability of organic growing media (with cellulose being a crucial constituent) and to the control of certain fungus-like and fungal organisms, as cellulose and xylan are essential cell wall components. At least three different synergistic enzymes (endo- β -1,4-glucanases, exo- β -1,4-glucanases, and β -glucosidases) are involved in microbial degradation of cellulose (Eriksson and Wood, 1985; Sinsabaugh and Linkins, 1989). The carbon polymer xylan is a substantial constituent in wood and grasses, and is thus abundant in organic growing media. Xylans are more susceptible to degradation than cellulose. Many fungi and bacteria produce endoxylanases and exoxylanases to degrade xylan to oligosaccharides and monomers, respectively.
4. *Chitinase and β -1,3-glucanase activity*, which are of interest for predicting disease suppressiveness. Chitin is a common structural substance and an important constituent of fungal cell walls. Hydrolysis of chitin yields chitobiose and chitotriose. Chitin is utilized by a broad range of bacteria (*Streptomyces*, *Nocardia*, *Pseudomonas*, *Flavobacterium*, *Bacillus*, *Cytophaga*) and fungi (*Mortierella*, *Aspergillus*, *Penicillium*) (Schinner et al., 1996). In hydroponic systems, chitinolytic and β -1,3-glucanase activities have primarily been analyzed with respect to induction of disease suppressiveness in growing media, as BCA, and in slow filters for purification of the circulating nutrient solution (Alsanius et al., 2007; Brand and Alsanius, 2003, 2004a,b,c; Brand et al., 2001; Furtner et al., 2007a,b; Khalil, 2011; Khalil and Alsanius, 2011; Khalil et al., 2009).

5.4.2.4 Phytotoxic compounds

Phytotoxic compounds are a very diverse group of substances which may derive from the growing medium, the plants present, metabolites released by the associated microbiota, raw water, additives (e.g., plant protection products and fertilizers, especially organic fertilizers with variable input), or the equipment. They may be organic or inorganic. The chemical nature of the substance governs the analytical procedure for quantification. Indirect procedures are used for monitoring the impact

of phytotoxic compounds on plant performance (see Sections 10.4.4.1 and 10.4.4.2). Direct quantification approaches have been developed for phenolic acids, involving sampling method and analysis (Jung et al., 2002, 2004; Kistner et al., 2004).

5.4.3 METHODS TO STUDY MICROBIAL DIVERSITY AND COMMUNITY STRUCTURE

5.4.3.1 Viable counts and specific groups

Calculation of viable counts of specific groups considers the culturable aerobic heterotrophic microbiota. The basic concept and procedure for viable counts are described in Section 5.4.1.2. Through the use of semiselective media, specific taxonomic groups are favored. However, as the nutritional conditions differ between media, the results obtained from different media or formulations of different brands are not always comparable. For interpretation, information on the choice of medium, incubation conditions, and plating technique is important. The conditions in and on agar plates provide an ecological habitat with competition for nutrients and space between different species. In general terms, viable counts underestimate the presence of microorganisms, but are an important complement to modern DNA-based techniques, such as MPS for description of microbial diversity and community structure.

5.4.3.2 Phospholipid fatty acid patterns

PLFAs (see Section 10.4.1.4) can also be used for characterization of microbial community structure and quantification of living microbial taxa. This method does not allow detection of individual species, but PLFA patterns have been successfully used to describe changes in microbial community structure in nutrient solution, growing medium, and the root zone of hydroponic systems and in hydroponically grown crops under various conditions (Debode et al., 2016; Khalil and Alsanius, 2001; Khalil et al., 2001a, 2001b; Renault et al., 2017; Waechter-Kristensen et al., 1996).

5.4.3.3 Community-level physiological profiling

The idea of using microbial substrate utilization for description of functional variations in microbial communities inhabiting surface water was first tested several decades ago (Garland and Mills, 1991). As with viable counts, it reflects the heterotrophic culturable microbiota metabolizing specific substrates. Originally a commercial platform for identification of gram-positive (GP panel) or gram-negative (GN panel, GN2 panel) bacterial strains was used. This platform contains 95 sole carbon sources and one control substrate, and the method is often referred to as sole carbon source utilization. Substrate utilization is measured spectrophotometrically for each compound in response to the color change in a redox dye. Alterations in utilization are expressed with respect to the average well color development of all 95 substrates, individual substrates, or groups of substrates, and the number of compounds utilized. These indicate functional shifts or stability rather than changes in taxonomic community composition or taxonomy, although these might be caused by the changes in community structure. EcoPlates™ are based on the same approach. The limited number of substrates (three sets of 31 sole carbon sources plus one control substrate) offers repetitions of the sample sets within the same platform. Standard procedures for extraction, dilution, inoculation, and incubation are described by Rutgers et al. (2006). Sole carbon utilization using GN panels has been employed in various studies related to hydroponic systems (Alsanius and Jung, 2003, 2004;

Alsanius et al., 1998; Khalil and Alsanius, 2001; Khalil and Alsanius, 2009; Khalil et al., 2001b; Renault et al., 2017; Waechter-Kristensen et al., 1998). CLPP has been refined on the basis of the automated commercial Omnilog platform, displaying a set of almost 2000 substrates and conditions and allowing continuous reading. This technology has been used for microbial dynamics studies of the rhizosphere and in vertical flow filters in hydroponics (Rosberg et al., 2014; Gruyer, 2012). However, the system is not accompanied by a sophisticated software package that explores the full potential of the big data sets generated in these assays. Statistical packages like those for single-strain approaches (Vaas et al., 2012a,b) would be a very welcome future development.

5.4.3.4 DNA-based methods

During recent years, culture-independent methods have increasingly been used to characterize microbial community structures related to hydroponic systems, mostly focusing on taxonomy using metagenomic analysis, and have revealed relative abundances of associated microbial taxa (Rosberg et al., 2018; Grunert et al., 2016; Storey et al., 2015; De Tender et al., 2016; Deboode et al., 2016). However, the very rapid technological development within this domain makes comparison, generalization, and conclusions difficult.

DNA-based techniques reflect the taxonomic diversity and richness of microbial community structures. During recent years, these techniques have revolutionized the potential to actively use microbial tools in hydroponics. However, there are some limitations to DNA-based methods, for example, (1) they show relative abundances (often restricted to 1% or 0.1% depending on the analytical depth), and not amounts of taxon; and (2) taxonomic grouping does not give information on function. They should therefore not be considered a stand-alone method but should instead be integrated with other methods for characterization of the biological, chemical, and physical properties in the respective habitat. In this context, the value and the variability over time of “general” parameters needs to be considered and their significance determined for the “niches” studied. Thus, substantial attention should be given to the experimental design, to generate robust tools for microbial modification and stabilization in soilless systems. Different fingerprinting and MPS methods have been used to characterize the microbiota associated with hydroponic systems.

Fingerprinting techniques make it possible to demonstrate microbial community composition and its dynamics. In contrast to MPS methods, they need to be complemented by cloning, sequencing, and alignment, in order to determine the identity of different bands or peaks. Terminal restriction fragment length polymorphism, denaturing gradient gel electrophoresis (DGGE), and single-strand conformation polymorphism have been used to analyze the microbial community structure associated with soils and growing media (Postma et al., 2000b, 2005b, Smalla et al., 2007) and the impact of biological or chemical additives on the microbiota inhabiting the nutrient solution in closed cropping systems (Alsanius et al., 2013; Renault et al., 2017). Early studies using DGGE by Postma et al. indicated a close interaction between actinomycetes and suppressiveness of *P. aphani-dermatum* in hydroponically grown cucumber.

Examples of *MPS methods* are Sanger sequencing, 454 pyrosequencing, and Illumina/Solexa using the amplification of either the bacterial 16S rRNA gene or the nuclear ribosomal internal transcribed spacer region as a universal fungal barcode marker. These methods differ in price, run time, and number of bases per run, and also read length. Technological development within this area is intense, leading to rapid changes in available technologies. For example, 454 sequencing was phased out in 2015. Emerging techniques have been increasingly used in studies on hydroponic

systems in recent years. Using 454 pyrosequencing, [Rosberg et al. \(2018\)](#) described the dynamics of the rhizosphere microbiota of hydroponically grown tomato as related to plant age and health status. [Debode et al. \(2016\)](#), using Illumina, demonstrated that the relative abundance of bacterial genera belonging to taxa with well-documented antagonistic properties increases substantially upon addition of chitin to the growing medium. In another study using the Illumina technique, increased biodiversity and altered susceptibility to *Botrytis cinerea* have been found on supplementing growing medium with biochar ([De Tender et al., 2016](#)). MPS methods enable the characterization of relative abundances of microbial taxa, microbial biodiversity, the core microbiome, and specific community members. Changes in the core microbiome may occur as a result of biological or environmental disturbances. Depending on the health status of the crop and cropping systems, beneficial or deleterious organisms may be identified among the environment-specific and/or plant-specific taxa.

5.4.4 BIOASSAYS

Microbes associated with the growing medium, plant root, and nutrient solution interact with their environment in various ways, and these complex interactions often cannot be detected by monitoring a single distinct parameter. This problem can be overcome using bioassays performed on different levels of complexity.

5.4.4.1 Incubation tests

Incubation tests are laboratory tests to assess the interactions (e.g., biocontrol capacity, inhibition) between different distinct microorganisms (dual culture tests) and distinct compounds or extracts and single microbial species or culturable microbial communities (agar diffusion tests).

5.4.4.2 Plant bioassays

Plant bioassays are mostly conducted to assess disease suppressiveness or phytotoxicity. Standardized test systems for growing media assaying suppressiveness are not yet available. However, systems for estimating the disease suppressiveness of soils and composts have been presented in various studies ([Alabouvette et al., 2006](#); [Fuchs and Larbi, 2004](#); [Termorshuizen et al., 2006](#)). Standardized assays for soilless systems, especially to evaluate growing media, should consider the most common plant-pathogen complexes in hydroponics (e.g., *Pythium*, *Phytophthora*, *Fusarium*, and *Rhizoctonia*). Such assays also need to be highly reproducible. A potential approach has been described by [Wohanka et al. \(2008\)](#) and [Alsanius and Wohanka \(2009\)](#).

In contrast, standardization of bioassays for assessment of phytotoxicity has received more attention from certification organizations ([Wever, 2004](#)), which has resulted in the release of a standardized low-cost test kit based on image analysis ([Blok et al., 2008](#)).

5.5 DISEASE MANAGEMENT IN SOILLESS CROPPING SYSTEMS

From a plant protection point of view, the main advantage of soilless cropping is avoiding soil as a permanent source of infection. However, as described in [Section 5.3.1](#), plant pathogens may be

introduced and dispersed through poor hygiene during growing medium production or reconditioning, infected seeds and plant material, and inadequate sanitation in soilless cropping. Soilless systems are usually combined with complete or partial reuse of irrigation water or nutrient solution. Such closed or semiclosed systems generally pose a certain risk of spreading root-infecting pathogens through the reuse of irrigation water or nutrient solution. A comprehensive overview by Stewart-Wade (2011) showed that all types of root-infecting pathogens, oomycetes (e.g., *Pythium* spp., *Phytophthora* spp.) in particular, are more or less easily dispersed throughout the whole irrigation system. To mitigate the disease risk in soilless systems, the main task is to prevent the dispersal of pathogens. However, ensuring a disease suppressive root environment and using disease-resistant varieties, BCA or pesticides, and good sanitization practices can also contribute to a healthy crop. Detection and identification of the relevant plant pathogens are essential to guide the choice of approach.

5.5.1 DISEASE SUPPRESSIVENESS IN SOILLESS CROPPING SYSTEMS

As part of integrated plant disease management, disease suppressiveness makes a fundamental contribution to crop health. Disease suppressiveness in soilless cropping systems is discussed in detail for peat and hydroponics in Sections 5.3.2 and 5.3.3, respectively, and for compost in Chapter 8, Organic Soilless Media Components. The disease suppressing effect can be supported by introducing BCA (see Section 5.5.2).

5.5.2 BIOLOGICAL CONTROL

BCA (also called biocontrol agents) can play an important role in suppressing root pathogens in soilless systems. BCA are those products that control plant pathogens or pests, or reduce their amount or their effect, and can comprise one or more organisms (i.e., viruses, bacteria, fungi, insects). An overview of biological control principles and many examples is given by among others Campbell (1989), Alabouvette et al. (2006), and Heydari and Pessarakli (2010).

Compared with field soil, where it is difficult to introduce BCA in sufficient concentrations to the lower parts of the root system, the limited volume of the matrix around the roots in soilless systems facilitates introduction of the antagonist into the root environment. In general, soilless systems allow good interaction between host, pathogen, and antagonist. Establishment of antagonists is also easier in soilless systems, with a new substrate with an unbalanced microflora, than in a microbiologically buffered system such as soil, with tremendous competition with the microorganisms already present (Paulitz and Bélanger, 2001).

Many antagonists (i.e., microorganisms that inhibit other organisms) of root pathogens are known and have been tested in greenhouse cropping systems (Khalil and Alsanious, 2010; Lee and Lee, 2015; Paulitz and Bélanger, 2001; Whipps and Postma, 2008; Whipps, 2001). However, limited numbers of antagonists are available as commercial products. Commercially available BCA to control root-borne (fungal) pathogens are based on *Trichoderma asperellum* (syn.: *Trichoderma harzianum*), *Trichoderma gamsii* (syn.: *Trichoderma viride*), *Gliocladium virens*, *Gliocladium catenulatum*, nonpathogenic *F. oxysporum*, *Coniothyrium minitans*, *Streptomyces griseoviridis*, *Bacillus amyloliquefaciens*, and *Bacillus subtilis*. Much research has also been done on *Pseudomonas* species, which are good antagonists, but a product for soilless systems is not yet available.

One problem with several *Pseudomonas* species is poor survival of the bacterial cells after drying and storage (so-called shelf life), since they do not produce spores.

Various groups of microorganisms are able to produce so-called biosurfactants (Mulligan, 2009). A pioneering study demonstrated the potential for using biosurfactant-producing microorganisms to control specific plant pathogens (e.g., *Pythium* spp. and *Phytophthora* spp.) through zoospore lysis (Stanghellini and Miller, 1997). These findings have been confirmed in more recent research (De Jonghe et al., 2005; De Souza et al., 2003; Hultberg et al., 2010; Nielsen et al., 2006). Examples are *Pseudomonas aeruginosa* controlling *Phytophthora* root rot (Stanghellini and Miller, 1997), *Pseudomonas fluorescens* controlling root rot in hyacinth (*Hyacinthus orientalis*) bulbs caused by *Pythium intermedium* (De Souza et al., 2003), *Pseudomonas koreensis* suppressing *P. ultimum* in a hydroponic tomato crop (Hultberg et al., 2010), and *Lysobacter enzymogenes* controlling cucumber root and crown rot caused by *P. aphanidermatum* (Folman et al., 2003).

Unfortunately, biocontrol of root diseases often shows inconsistent results (Postma et al., 2001). This can be due to poor survival or activity of the biocontrol agents, or to insufficient colonization of the infection sites by the biocontrol agents. In particular, if the antagonist was originally isolated from soil or from a crop different from that to which it is applied, it may not be fully adapted to soilless systems. For effective biocontrol, the activity of the antagonist and that of the pathogen should be synchronized in time and in space. The influence of location of the antagonist in relation to location of the pathogen is illustrated by the failure of suppression of *Fusarium* wilt in carnation (*Dianthus caryophyllus* L.) in a recirculation system when the antagonist was added on top of the stone wool blocks and the pathogen was introduced via the nutrient solution (Rattink and Postma, 1996). When introduced via the nutrient solution where the pathogen was located, the antagonist proved to be extremely effective in that study.

In contrast to the above-mentioned BCA, beneficial microorganisms living on or in the plant (endophytes) have the advantage of being protected from environmental stresses and might be able to protect the plant in a soilless system from the propagation stage and throughout the crop cycle. A recent review showed high potential of endophytes to control plant diseases, vascular wilts in particular (Kaouthar et al., 2016). Nonpathogenic *Fusarium* species present on plants can, for example, protect carnation against wilt disease caused by *F. oxysporum* (Postma and Luttikholt, 1996), or tomato against the plant-parasitic nematode *Meloidogyne incognita* (Hallmann and Sikora, 1994). An example of a bacterial endophyte is the endophytic colonization of tomato tissue with a *Pseudomonas* strain that inhibits wilt caused by *Verticillium dahliae* (Sharma and Nowak, 1998).

Another group of microorganisms are those that protect the plant through induced systemic resistance (ISR) (e.g., Bakker et al., 2007; Chen et al., 1999). Experiments to test whether a biocontrol agent acts through ISR usually involve either using a split root system or challenging the roots with the control agent, followed by evaluating the symptoms of a leaf pathogen. For various *Pseudomonas* strains, such effects have been demonstrated for *Pythium* root rot in cucumber (Zhou and Paulitz, 1994) and *Fusarium* wilt and root rot in tomato (Duijff et al., 1998; Pastor et al., 2016). However, little is known about the importance of this mechanism for an entire crop, since most research on ISR of root pathogens has been performed with individual young plants using a split root system.

Biological control has the potential to solve problems with root diseases in soilless systems. However, there is a need for better products and more knowledge on the optimal inoculation

procedures and the environmental conditions under which the biocontrol agent is active. There is also a need for organisms that are better adapted to soilless systems and synchronized to the pathogens occurring in such systems. For commercial application of biocontrol agents, the legal requirement for registration of these products is an economic barrier. To get a biocontrol agent registered, safety for the user, the consumer, and the environment and biocontrol efficacy need to be proven. This registration procedure is relatively expensive for products which can only be applied within a relatively small market.

5.5.3 DISINFECTION OF NUTRIENT SOLUTION

Once introduced in closed cultivation systems, root-infecting pathogens can be dispersed rapidly by the recycled irrigation water. Many examples have been described (Hong et al., 2014, Part II, pp. 55–121). Depending on the irrigation system, the disease risk may vary widely. Optimization of the irrigation design, use of best management practices, disease suppressiveness, introduction of BCA, and (restricted) use of pesticides are strongly recommended to reduce the disease risk in the recirculated irrigation water. Various disinfection techniques are available, such as filtration, pasteurization, irradiation, or chemical treatment. In general, the target of treatment is a 99.9% reduction in the original pathogen density (also called a log 3 reduction).

5.5.3.1 Filtration

Filtration can be used to remove undissolved material from the nutrient solution (see Table 5.3). Various types of filters are available to match the range of particle sizes. Rapid sand filters are often used to remove large particles from drainage water before adding, measuring, and control of electrical conductivity and pH, and addition of new fertilizer. A fine synthetic filter (50,000–80,000 nm) is often built into the water flow system after the fertilizer unit, to remove undissolved fertilizer salts or precipitates and thus prevent clogging of the irrigation nozzles. A synthetic filter may also be used as a pretreatment for water disinfection. Rapid sand and standard synthetic filters can be cleaned manually or automatically, but they do not remove plant pathogens to a sufficient extent.

Table 5.3 Overview of Filtration Processes Used for Disinfection of the Nutrient Solution in Hydroponic Systems

Principle	Pore Size (nm)	Removal of
Rapid sand filter	>100,000	Algae, roots, leaves
Mechanical synthetic filter	10,000–100,000	Small particles, sand grains
Slow filter	1,000–10,000	Roots, pathogens
Membrane filter		
Microfiltration	100–1000	Colloidal and suspended particles, bacteria, fungi
Ultrafiltration	10–100	Proteins, bacteria, virus
Nanofiltration	1–10	Sugars, divalent ions, larger monovalent ions
Reverse osmosis	<1	Small ions

Flow is slowed with declining pore size, so removal of very small particles requires a combination of a large array of fine filters or higher pressures and frequent cleaning of the filter(s) to prevent clogging. Removal of pathogens requires relatively small pore size (see Table 5.3).

Various membrane filtration technologies are available, where water is pushed through a membrane under high pressure. The water is separated into the desired clean water (filtrate) and residual water with concentrated salts (the so-called brine). This brine has to be discarded and, depending on the country and situation, there may be strict rules concerning its disposal, as the amount of brine can represent as much as 40% of the water being treated. Theoretically, membrane filtration eliminates all kind of plant pathogens, including viruses (Werres and Wohanka, 2014). In practical use, the membrane technologies experienced some problems, such as clogging and leaking (Van Os, 1999) and the investment cost is rather high. While today's membranes are more reliable, the disinfection effect of this technique is only regarded as a good side-effect in the horticulture industry (e.g., using reverse osmosis to obtain high-quality irrigation water).

Slow sand filtration (Fig. 5.7) is considered to be a reliable, low-cost solution to eliminate soil-borne pathogens from recirculated irrigation water (Ehret et al., 2001; Lee and Oki, 2013; Van Os, 2009; Van Os et al., 1997). Since its introduction in the late 1980s, slow sand filtration has been widely utilized in closed cultivation systems in the horticulture industry. The principle is based on

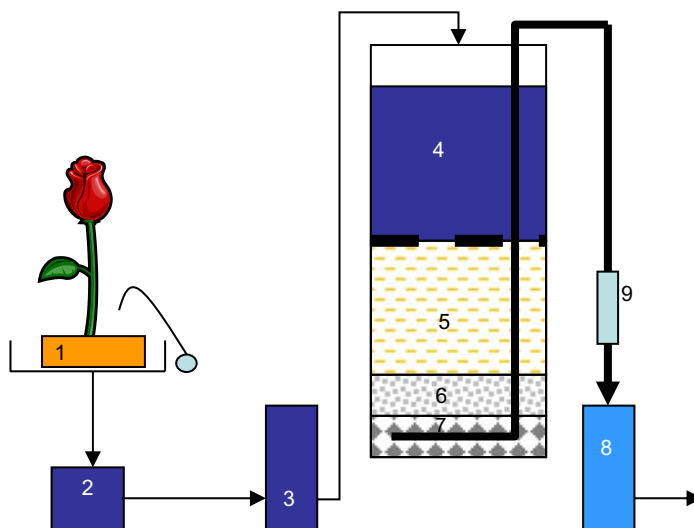


FIGURE 5.7

Flowchart of slow sand filtration. Nutrient solution drains from substrate (1) to the collection tank (2). From there it is pumped to the day storage tank (3) and into the top of a large container or metal silo (4), from which it drips into a sand layer of about 1 m thickness (5). The top layer between (4) and (5) is called the *Schmutzdecke* or filter skin. (6) and (7) are a 10 cm fine gravel layer and a 15 cm coarse gravel layer, respectively. The filtrate is pumped out of the gravel layer to container (8). For initial filling of the filter, clean water is reverse-pumped from (8) into the gravel layers (7) and (6) and to above the sand layer. Flow meter (9) controls the filtration rate. From container (8), the filtrate is mixed with fresh water to create a new nutrient solution for the plants.

a supernatant water layer, which percolates slowly (100–300 L/m²/h) through a bed of fine filter sand or other filter material (e.g., mineral wool). During an initial ripening phase, a so-called *Schmutzdecke* (i.e., “dirt layer”) forms on the surface of the filter bed (Ellis, 1985). Beside the mechanical sieving effect of the sand, this layer of organic and inorganic material, which has a high level of biological activity, contributes substantially to the effectiveness of a slow sand filter (Brand and Wohanka, 2000). *Phytophthora* and *Pythium* can be eliminated completely by this method, but fungi with small spores (e.g., *Fusarium* spp.), bacteria, viruses, and nematodes, are only partly removed (90%–99.9%). An overview of the many efficiency studies reported in the literature is given by Werres and Wohanka (2014).

5.5.3.2 Heat treatment

Heating the drainage water to lethal temperatures is one of the most reliable methods for disinfection of recycling water and is highly effective against all types of root-infecting plant pathogens (Van Os, 2009). The function of this pasteurization technique is shown in Fig. 5.8. Excess nutrient solution returns from the plants and is collected in a tank. From there, it is pumped into a pasteurization unit, where it enters a first heat exchanger to be preheated to a temperature of about 80°C by heat recovery from the hot water exiting the unit. In the next part of the unit, the solution is heated to a disinfection temperature of between 90°C and 97°C, using an external heat source (boiler, 7), which has an incoming temperature of T3 (95°C–105°C) and an outgoing temperature T4 (85°C–95°C). The disinfected solution flows back to the first heat exchanger to be cooled down and is then stored in a clean water tank. Mainly based on results in an early study (Runia et al., 1988), a top temperature of 95°C for 30 seconds is recommended, which involves relatively high energy costs. Later studies

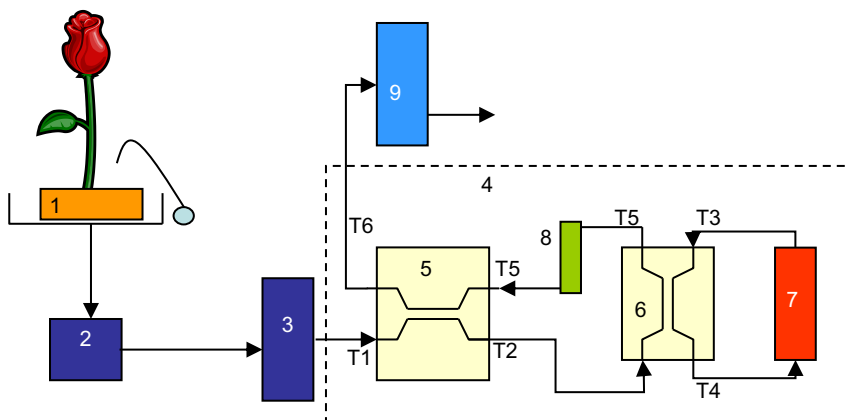


FIGURE 5.8

Flowchart of heat treatment. Nutrient solution drains from substrate (1) to the collection tank (2). From there it is pumped to the day storage tank (3) and into heat treatment unit (4). The solution is pumped with temperature T1 (about 17°C–25°C) into heat exchanger (5) and preheated to T2 (70°C–80°C). In the heat exchanger (6), it is further heated to T5 (85°C–97°C) by an external heat source (boiler, 7), which has an incoming temperature of T3 (95°C–105°C) and an outgoing temperature T4 (85°C–95°C). The water is kept at temperature T5 in unit (8) for an exposure time of about 30 seconds and then cooled to temperature T6 (22°C–30°C) and stored in container (9) to be mixed with fresh water and nutrients for watering the plants.

have revealed that using extended exposure times and lower temperatures can produce similar efficacy of disinfection (Runia and Amsing, 2001a, 2001b). The use of lower temperatures in combination with longer exposure times has the advantage that no external heat source is needed to deliver the high temperatures in the second heat exchanger (Fig. 5.8, temperatures T3 and T7). The amount of fuel needed to heat the water is about 1 m³ natural gas per 1 m³ water for a disinfection temperature of 95°C, but this declines to 0.6 m³ gas per 1 m³ water for a temperature of 60°C.

The disinfected water leaving the first heat exchanger (T6 in Fig. 5.8) is at most 5°C warmer than the incoming drain water. By refreshing it with clean water, the temperature can be reduced to an adequate level. However, if all disinfected water is returned to the plants without much refreshing (e.g., in NFT systems), the temperature of the nutrient solution will rise to unacceptable levels.

Another aspect of achieving high temperatures is precipitation of scale (CaCO₃) on the plates of the heat exchanger at temperatures around 100°C. Practical experience shows that decreasing the pH of the solution to about 4.5 prevents this problem. The plates of the heat exchangers are usually made of stainless steel, but titanium-coated heat exchanger plates are recommended if the chloride concentration in the solution is higher than 4 mmol/L.

5.5.3.3 Ultra-violet irradiation

UV irradiation is electromagnetic radiation at a wavelength between 200 and 400 nm. Wavelengths between 200 and 280 nm (UV-C), with an optimum at 254 nm, have a strong killing effect on microorganisms, because they minimize the multiplication of DNA chains. The energy dose of the radiation is expressed in mJ/cm² (=mW s/cm²), where mW is the quantity of UV-C energy, s is the duration of the exposure, and cm² is the area to which the energy is applied. From experiments, it is known that different levels of radiation are needed for different organisms to achieve a certain level of efficacy (Wohanka, 1992). For example, 100 mJ/cm² is recommended for eliminating bacteria and fungi and up to 250 mJ/cm² for eliminating viruses (Runia, 1988, 1994b). However, Ehret et al. (2001) report some successful results with lower dosages.

Several types of UV-lamps available on the market create the required dose. The main types are low- and high-pressure lamps, both of which are effective if the required dose can be achieved. Low-pressure lamps have a low capacity (110–300 W), produce relatively less heat, and are less expensive, but more lamps are needed to disinfect a certain amount of water. High-pressure lamps are mainly high-powered lamps (3–8 kW) for disinfecting large quantities of water. The temperature of the disinfected water is elevated compared with the incoming solution. However, at a suitable flow rate, the temperature difference is less than 1°C for low-pressure lamps and less than 2°C for high-pressure lamps. Depending on the disinfection capacity required, one or more lamps may be included. The nutrient solution flows in a thin layer (<1.5 cm) in a turbulent manner around the UV-lamp in a specified time to give it good exposure to the UV radiation.

Good UV disinfection units measure the transmittance of water with a light cell. The composition of the nutrient solution and organic load, iron chelates in particular, may considerably reduce UV-C transmission. Furthermore, pathogen propagules in particles of peat or plant debris are completely protected. Hence, prefiltration with at least a standard rapid sand filter is usually recommended (Nederhoff, 2004). At low transmission rates, the nutrient solution may be blended with rain water, and the water layer in the UV apparatus should be as thin as possible (preferably <10 mm). UV intensity is also affected by precipitation of minerals on the surface of the quartz glass tube. Therefore, the UV-C intensity should be checked using a UV sensor. If necessary, the

glass tube should be cleaned (automatic cleaning is possible) or the radiators renewed. Under certain conditions, the interaction of UV radiation with iron chelates can substantially reduce the available iron and even cause deficiency (Schwartzkopf et al., 1987; Stanghellini et al., 1984; Yoganathan et al., 1980), which has to be taken into account during fertilizer application. The most UV-stable iron chelate is Fe-EDDHA, followed by FE-DTPA and Fe-Na-EDTA (Acher et al., 1997). Another option is to increase the dose of UV-C, which can be done automatically in more sophisticated installations with high-pressure lamps that measure transmittance automatically and adapt the dose accordingly.

5.5.3.4 Chemical treatment

Most chemical water disinfection methods use chemical oxidation as the mechanism for killing microorganisms. An oxidizing agent reacts with the organic matter present in the solution or in the pipes through which the solution is passing. As such, it reacts with pathogens and nonthreatening organisms (such as algae), as well as other organic matter. To reduce the reaction with nonpathogenic organic matter, prefiltration is essential in such systems.

5.5.3.4.1 Hydrogen peroxide

Hydrogen peroxide (H_2O_2) is a strong, unstable oxidizing agent that reacts to form H_2O and an O-radical. The latter reacts with any type of organic material, including pathogens. Commercial “activators” (mostly formic acid or acetic acid) are added to stabilize the original solution and to increase the efficacy. Various dosages are recommended (Runia and Paternotte, 1993): 0.005% against *Pythium*, 0.01% against fungi such as *Fusarium*, and 0.05% against viruses. As a weak acid is added to the solution, the pH decreases by 1–3 U. Before reuse of the nutrient solution, the pH has to be adjusted. The 0.05% dose is also harmful for plant roots. Currently, H_2O_2 is considered especially helpful for cleaning the watering system, while it is regarded as not sufficiently efficient for disinfection at nonphytotoxic concentrations (Nederhoff, 2000b).

5.5.3.4.2 Ozone

Ozone (O_3) is a powerful oxidizing agent which reacts with all organic matter including pathogens. It also reacts with iron chelates. Ozone is produced from dry air and electricity using an ozone generator (converting $3\text{O}_2 \rightarrow 2\text{O}_3$). The ozone-enriched air is injected into the water to be sanitized and it is stored for a period of 1 hour. Various studies report the effectivity of ozone against plant pathogenic bacteria, fungi, oomycetes, and viruses (Laplante et al., 2005; Runia, 1996, 1994a,b; Vanachter et al., 1988). Based on several trials, Runia (1996, 1994a,b) concluded that an ozone supply of 10 g/m^3 drainage water with an exposure time of 1 hour is sufficient to eliminate all pathogens, including viruses. Supply water can be treated with 5 g/h/m^3 , because the organic load is much lower. Human exposure to the ozone that vents from the system or the storage tanks must be avoided, since even short-term exposure to a concentration of 0.1 mg/L of ozone may cause irritation of the mucous membranes. A disadvantage of ozonation is the inability to process large quantities of water at a time. Another drawback is that ozone reacts with iron chelate, precipitating it from solution, depositing the iron in the irrigation system, breaking down the chelate, and rendering the iron unavailable for uptake by the plant. Consequently, higher dosages of iron are needed and measures need to be taken to deal with iron deposits in the system.

5.5.3.4.3 Chlorine

Common sources of chlorine for disinfection of irrigation water are chlorine gas, sodium hypochlorite, calcium chlorite, and potassium hypochlorite. On-site systems for electrolytical generation of sodium or potassium hypochlorite solutions (electrolyzed oxidized water) are also in practical use (Bandte et al., 2016). When any of these chlorine species is added to water, molecular Cl_2 (at pH <5 only), hypochlorous acid (HOCl), and hypochlorite ions (OCl^-) are formed (Fisher et al., 2014). These three forms of chlorine are also referred to as “free chlorine.” Hypochlorous acid has the highest germicidal effect and this form dominates between pH 5 and 6.5 (Bernard, 1991). Chlorine is a strong oxidizer, reacting with inorganic and organic compounds in the nutrient solution. Hence, the efficiency may be reduced and toxic residues may be formed (Bernard, 1991). In particular, carcinogenic organic-chlorine compounds, including trihalomethanes, are formed through the reaction with humic substances (Ellis, 1991). Accumulation of chlorate (ClO_3^-) in produce from soilless systems with chlorination also has to be considered (Dannehl et al., 2016).

Many studies have shown the efficacy of chlorination against phytopathogens (e.g., Fisher et al., 2014). Maintaining the chlorine concentration continuously at 2.5–5 ppm in the nutrient solution of NFT systems is reported to give good control of *Pseudomonas solanacearum* and *Pythium* spp. on tomato and *Pythium* spp. on cucumber, without phytotoxicity (Jenkins, 1981). In another study, *Phytophthora cinnamomi* was eradicated at 2 ppm free chlorine within 5 minutes (Armitage, 1993). To control fungi such as *F. oxysporum* and *Rhizoctonia solani*, much higher doses (14 and 12 ppm, respectively) are necessary (Cayanan et al., 2009). However, the threshold chlorine concentration at which plant growth is not adversely affected is less than 2.5 ppm (Cayanan et al., 2008; Cook and Devine, 1979). Runia (1988) found no effect of 1–5 ppm free chlorine against cucumber green mottle mosaic virus. As shown by Bandte et al. (2016), Pepino mosaic virus seems to be more sensitive, with complete inactivation of this virus by 0.2 ppm free chlorine applied as electrolytically generated potassium hypochlorite solution.

Based on the many studies on irrigation water treatment with chlorine, a dose of about 2 ppm free chlorine seems to be most appropriate for soilless cropping system. However, as the chlorine effect depends on concentration, time of exposure, and many other factors (e.g., temperature, pH, organic load, ammonium content), no general recommendations can be given. To minimize the chlorine demand by organic particles, suitable prefiltration is essential. In addition, any fertilizer containing ammonium will rapidly react with chlorine to form chloramines (Fisher et al., 2014). While chloramines have a certain disinfecting capacity, their efficacy is only about 4% of that of hypochlorous acid.

Despite a series of drawbacks (e.g., pH-dependence, reactions with fertilizers and organics, potentially toxic by-products), chlorination can be a useful low-cost water treatment technique when applied properly.

5.5.3.4.4 Chlorine dioxide

Chlorine dioxide (ClO_2) is a greenish-yellow gas which, in contrast to chlorine, does not hydrolyze in water. It has biocidal properties and its composition is unaltered in water with pH 4–8.4 (Armitage, 1993). Laboratory tests have shown that *P. cinnamomi* is effectively killed by 2 ppm ClO_2 (Armitage, 1993). Moreover, ClO_2 at a concentration of at least 3 ppm for 2 minutes controls *P. cinnamomi*, *F. oxysporum*, and other fungi, as demonstrated by Mebalds et al. (1995). Recently,

Scarlett et al. (2016) demonstrated the high efficiency of chlorine dioxide in the range 1–5 ppm against eight plant pathogens and their different propagules.

Water of lower quality, typical of that used after recycling, will require higher concentrations of ClO_2 (Mebalds et al., 1995; Scarlett et al., 2016). Hence, prefiltration is recommended and sensors should be placed in the irrigation system to account for ClO_2 depletion by impurities in the water. In regions where the horticulture industry has to struggle with high pH levels in the irrigation water, ClO_2 seems to be a suitable disinfectant for recycling water. In contrast to chlorine, no chlorinated toxic by-products (e.g., trihalomethane) are formed. However, under certain conditions chlorite and chlorate may be formed.

In general, up to 2 ppm chlorine dioxide is recommended for treatment of irrigation water. So far, no phytotoxic reactions have been reported at that or lower concentrations.

5.5.3.4.5 Ionization

Electrolysis of water by copper or silver electrodes releases positively charged free copper (Cu^+) or silver (Ag^+) ions, which can attach to the negatively charged cells of microorganisms. There, they destroy cell wall permeability, denature proteins, and interfere with electron transport in respiration systems (Pommerville, 2010). For irrigation water treatment, copper electrodes are usually used.

The efficiency of Cu^{2+} against various phytopathogenic bacteria and fungi was evaluated by Wohanka (2014) in experiments where standard fertilizer solution was contaminated with the respective pathogen and treated by an Aqua-Hort-ionizer (Aqua-Hort, DK-8220 Brabrand) to achieve concentrations of 1, 2, and 4 ppm Cu^{2+} . The most sensitive pathogens proved to be the bacteria *Ralstonia solanacearum* and *Erwinia carotovora* pv. *carotovora*, followed by *Xanthomonas hortorum* pv. *pelargonii*, *Clavibacter michiganensis* var. *michiganensis*, and *Agrobacterium tumefaciens*. *F. oxysporum* and *P. aphanidermatum* were not completely eliminated, even at the longest contact time of 24 hours and the highest Cu^{2+} concentration of 4 ppm. Runia (1988) did not obtain a log 3 reduction for tomato mosaic virus or *F. oxysporum* after a treatment time of up to 4 days. Regrettably, there have been few field studies on copper or silver ionization in horticulture. Jamart (1998) cites an azalea (*Rhododendron* sp.) nursery using silver/copper, where the mean number of *Phytophthora* propagules was reduced from 4.5 CFU/mL before to 2.7 CFU/mL after treatment (44% reduction rate) over the whole growing season. Experiments with gerbera (*Gerbera jamesonii*) pot plants grown in ebb-and-flow systems showed that disease incidence was significantly reduced (by 40%–70%) at a concentration of 0.28 ppm Cu^{2+} compared with 0.07 ppm in nutrient solution infected with *P. cryptogea* zoospores (Toppe and Thinggaard, 1998). Similar disease-inhibiting effects have been demonstrated for *P. cinnamomi* in ivy (*Hedera* sp.) grown in ebb-and-flow systems (Toppe and Thinggaard, 2000).

For most crops, the critical toxicity level of copper is considered to be >20–30 ppm in soil solution. In soil and organic growth media, copper (Cu^{2+}) is strongly bound to humic and fulvic acids, while in soil water or nutrient solution, free copper ions are rapidly complexed to copper organic compounds (Marschner, 1995). Copper ionization of irrigation water is mainly applied to crops grown in soil or in organic growth media (e.g., peat mixes). No phytotoxic effects have been reported from such applications. However, when using copper ionization for hydroponically grown crops, the situation might be different. Various experiments have determined critical levels of Cu^{2+}

in the nutrient solutions to be approximately within the range 0.2–0.5 ppm (Hill et al., 2000; Zheng et al., 2004; Zheng et al., 2005).

5.5.3.5 Combined strategies

For proper crop management, all environmental factors should be taken into account. Crop growth depends on a suitable substrate, climate conditions, and cultural and sanitation measures. For effective disease control, several practices should be combined, for example, using strong plants, reducing the numbers of pathogens, and improving the disease suppressiveness of the system. Chemical pesticides can be applied if they are available and permitted for use. Previous sections of this chapter described various methods for reducing the pathogen incidence in the irrigation water or improving the disease suppressiveness of a particular system, and the technical possibilities for monitoring pathogens. The following examples are combinations of these strategies.

5.5.3.5.1 Combining biological control agents and disinfection

BCA should be added as early as possible, so as to achieve a stable microbial community with a maximum of beneficial organisms before pathogens are present. If BCA have to be added again at later stages, adding them to a disinfected solution has the advantage that the number of microorganisms present in the solution is lower and nutrients derived from dead cells are present.

An example of combining BCA with a disinfection treatment is described by Garibaldi et al. (2003), who found that biocontrol strains of *Trichoderma* and *Fusarium* in combination with slow sand filtration showed good efficacy in reducing *P. cryptogea* root rot on gerbera.

5.5.3.5.2 Nonpathogenic microbiota after disinfection

Immediately after disinfection treatment, the microflora in the nutrient solution can be expected to change dramatically. Disinfection by UV radiation eliminates 90%–99.9% of microorganisms, with only a few bacterial species surviving (Van Os et al., 2004b). After slow sand filtration, over 90% of the total aerobic bacterial population may still be present, but it shows a shift in composition, with the numbers of fluorescent pseudomonads, filamentous actinomycetes, and fungi decreasing more drastically than the total number of bacteria (Van Os et al., 2004a). The potential of the microbiota to utilize various carbon sources and the genetic profile of the bacterial population are also altered by disinfection (Van Os and Postma, 2000; Van Os et al., 2004a). Therefore, it has been suggested that slow sand filtration might result in higher disease suppressiveness in the growing system compared with almost complete sterilization using for example, UV (Van Os et al., 2004b). However, this is not the case in practice, as a decrease in disease suppressiveness due to disinfection treatment has not been detected. Thus, the plant-driven microbiota is probably not disturbed. Measurements show that the microbiota in the nutrient solution around the roots changes only little, or not at all. This is explained by the fact that only part of the solution, and thus only part of the microbiota, is removed from the growing medium around the roots.

5.5.3.5.3 Addition of beneficial microbes to slow sand filters

Slow sand filters always need some time before they become highly efficient. The addition of suitable microorganisms, which can be obtained from existing effective filters, can enhance this process. In one study, the biological activation of a filter unit was very significantly enhanced by the addition of three *Pseudomonas putida* and two *Bacillus cereus* strains (Déniel et al., 2004).

The bacteria-amended filter became effective for eliminating *F. oxysporum* after 1 month, whereas the noninoculated filter needed six months to become effective.

5.6 CONCLUDING REMARKS

- The very complex domain of microbial interactions in the root zone, growth medium, and nutrient solution of soilless cropping systems is emerging from the shadow of the conventional descriptive discourse on physical and chemical properties.
- Determining and understanding microbial interactions in soilless cropping systems demand a shift in perspective, “microbial spectacles” that allow this sphere to be viewed from the perspective of microbial lifestyles.
- The rapid development of novel DNA-based methods is substantially helping to explore and explain decisive phenomena occurring in the root zone, growing medium, and nutrient solution of soilless systems and thus to disentangle the processes occurring and make them available for exploitation in commercial cropping systems.
- Successful characterization of microbial interactions in soilless systems demands gold standard methods such as those developed for the characterization of physical and chemical parameters in growing media.
- Selecting suitable tools for characterization of growing media and for implementing microbial properties in new applications (such as GENIII growing media which entail microbially based and predictable plant beneficial functions) using the vast amount of information that can be generated from microbial community studies calls for greater awareness and embedding of approaches in ecological theory. This should preferably be done before the start of experimental work.

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PART

SPECIFIC
COMPONENTS OF
A SOILLESS PLANT
PRODUCTION
SYSTEMS

II

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NUTRITION OF SUBSTRATE-GROWN PLANTS

6

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6.1 GENERAL

The basic principles of mineral nutrition of crops have been reviewed by many authors (Marschner, 1995; Mengel and Kirkby, 2001; Epstein and Bloom, 2005). While the theory of plant nutrition for soilless-grown plants is not different from that of soil-grown plants, some aspects, however, are different. One of the objectives of this chapter is to highlight these differences.

Universal problems, such as effects of deficiency or of toxic conditions on plant development, and the relevant diagnostic tools have been explored and widely reviewed in the literature (Marschner, 1995; Loneragan, 1997; Mengel and Kirkby, 2001; Epstein and Bloom, 2005); therefore, these issues are not discussed in the present chapter. The main factor that distinguishes between fertilization management of soil-grown from that of soilless-grown plants is the limited volume of substrate in the latter case; this means lower buffer capacity for solution composition and limited supply of nutrients (capacity factor). Consequently, soilless culture methods offer unique benefits such as capabilities to control water availability, pH, and nutrient concentrations in the root zone. At the same time, there are higher risks because of the smaller root system and low buffering capacity for water, nutrients and because of the increased risk of exposure to extreme ambient temperatures. These subjects are further discussed in Chapter 13, Technical Equipment in Soilless Production Systems. The general objective of this chapter is to exemplify nutritional issues of substrate-grown plants and to discuss differences between fertilization/nutrition of soil-grown plants and of soilless-grown ones. The specific objectives are (1) to present the nutritional requirements of soilless-grown crops; (2) to illustrate the impacts of the NH_4/NO_3 ratio in the irrigation water on rhizosphere pH and on the development of soilless-grown plants under various temperatures and aeration conditions; (3) to discuss the integrated effect of irrigation frequency and nutrient availability on crop growth; (4) to present the crop responses to changes in the electrical conductivity (EC) of the root-zone solution and to the ionic composition of the water, in soilless culture; (5) to describe the mode of fertilization and the potential nutrient sources; and (6) to discuss possible interactions among various fertilizers, and between the roots, the solution, and the substrate, including the effects of the medium, temperature, and oxygen concentration. Special attention will be devoted to pH effects on plant root functions and on availability of specific nutrients.

6.2 NUTRIENT REQUIREMENTS OF SUBSTRATE-GROWN PLANTS

6.2.1 GENERAL

Plants absorb many elements through their roots: >50 elements have been found in various plants. However, not all are considered to be essential elements. An essential element is defined as one that is required for the normal life cycle of a plant and whose role cannot be assumed by another element. Twenty elements are thought to be essential to the growth of most plants, and these are classified as macronutrients and micronutrients. The former are those which are required in relatively large amounts: carbon, hydrogen, oxygen, nitrogen, phosphorus, calcium, sulfur, potassium, and magnesium. The micronutrients are those required in small amounts, such as chlorine, iron, manganese, boron, zinc, copper, molybdenum, sodium, and selenium (Marschner, 1995; Mengel and Kirkby, 2001; Epstein and Bloom, 2005). Silicon is usually not considered as essential—except in case of some algae and horsetails (*Equisetaceae*)—but it plays a significant role in plant resistance to stress conditions and to some diseases and is, therefore, designated as “quasi-essential” (Epstein, 1994, 1999; Epstein and Bloom, 2005). Nickel was found to be an essential element for legumes (Loneragan, 1997). The elements required in the largest quantities are the main structural elements, C (carbon), H (hydrogen), O (oxygen), N (nitrogen), P (phosphorus), K (potassium), and S (sulfur), whereas elements required in very small amounts, such as Ni (nickel) and Mo (molybdenum), are needed for only one or very few enzymes (Loneragan, 1997). The essential macroelements were first recognized in the 19th century by De Saussure and Boussingault, who showed by chemical analysis that plants contain C, H, O, and N. Soilless (hydroponic) culture has served as the major tool in the investigation of plant nutrition from the early studies of Sachs and Knop during the 19th century in Germany; they identified N, P, S, K, Ca, Mg, and Fe as essential elements (Marschner, 1995). Most of the essential microelements were recognized in the 20th century, from 1930 and onward, thanks to the development of more highly purified chemicals and more sensitive methods for analyzing trace concentrations (Loneragan, 1997). The range of concentrations of the essential elements in plant tissues and the required annual amounts for maximum yields are given in Table 6.1.

Nutrients can enter the plant through two organs: roots and leaves, with the former involving nutrients dissolved in water, while the latter generally involves nutrients in gasses which constitute the air. Carbon and oxygen are absorbed predominantly by the leaves from the air as carbon dioxide or oxygen. Other essential nutrients are taken up by the roots in forms dissolved in water. Carbon is also taken up by the roots in the form of HCO_3^- , and oxygen additionally enters the roots through both gaseous O_2 and as part of water molecules. Hydrogen is also taken up as the ion H^+ . In this chapter, following the practice of commercial horticulture, we will define as nutrients those elements that are taken up predominantly by the roots (Table 6.1). Nevertheless, it is important to note that some nutrients, especially micronutrients such as iron, zinc, copper, and manganese, are often applied via foliar spray. Most of the nutrients are taken up as cations or anions, except for boron, which is absorbed as boric acid or as the borate ion, depending on the pH. Thus, nutrient solutions are composed of mineral salts, acids, and bases. The effects of the composition of the nutrient solution and the ratios of nutrient uptakes by plants on the pH of the solution and of the rhizosphere will be discussed later, in the section Impact of watering-N source. Nitrogen is a unique

Table 6.1 Ranges of the Essential Element Concentrations in Nutrient Solutions and Plant Tissues, and the Required Annual Amounts for Maximum Yields

Element	Chemical Symbol	Form Available to Plants	Nutrient Solution	Plant Tissues	Annual Consumption
Macroelements			mg/L	g/kg	kg/ha/year
Calcium	Ca	Ca ⁺²	40–200	2.0–9.4	10–200
Magnesium	Mg	Mg ⁺²	10–50	1.0–2.1	4–50
Nitrogen	N	NO ₃ [−] , NH ₄ ⁺	50–200	10–56	50–300
Phosphorus	P	HPO ₄ ^{−2} , H ₂ PO ₄ [−]	5–50	1.2–5.0	5–50
Potassium	K	K ⁺	50–200	14–64	40–250
Sulfur	S	SO ₄ ^{−2}	5–50	2.8–9.3	6–50
Micronutrients			mg/L	μg/g	g/ha/year
Boron	B	H ₃ BO ₃ , HBO ₃ [−]	0.1–0.3	1.0–35	50–250
Copper	Cu	Cu ⁺ , Cu ⁺²	0.001–0.01	2.3–7.0	33–230
Iron	Fe	Fe ⁺³ , Fe ⁺²	0.5–3	53–550	100–4000
Manganese	Mn	Mn ⁺²	0.1–1.0	50–250	100–2000
Molybdenum	Mo	MoO ₄ ^{−2}	0.01–0.1	1.0–2.0	15–30
Zinc	Zn	Zn ⁺²	0.01–0.1	10–100	50–500

nutrient that can be absorbed either as cation or anion, NH₄⁺ and NO₃[−], respectively. This characteristic of nitrogen influences plant nutrition in general and it has a strong impact on fertilization in soilless culture.

Based on Liebig's law of the diminishing returns, E.A. Mitscherlich, in 1909, developed an equation that related growth to the supply of plant nutrients. He observed that when plants were supplied with adequate amounts of all but one nutrient, their growth was proportional to the amount of this one limiting element, which was supplied to the soil. Plant growth increased as more of this element was added, but not in direct proportion to the amount added. Mitscherlich expressed this mathematically as

$$\frac{dy}{dx} = (A - y)C \quad (6.1)$$

where y is the yield, x is the nutrient amount, A is the potential yield that would be obtained by supplying all growth factors in their optimum amounts, and C is a proportionality constant that depends on the individual growth factor. By integration, one obtains

$$y = A(1 - e^{-Cx}) \quad (6.2)$$

The value of C was found by Mitscherlich to be 0.122 for N, 0.60 for P, and 0.4 for K. Numerous workers since Mitscherlich have found that C is not, in fact, constant but varies rather widely with crops grown under various climatic conditions. Therefore, the value of C has to be

determined for the element of interest under the specific climatic conditions for each specific crop. An extension of this concept has been developed for the case in which two or more factors limit growth:

$$y = A(1 - e^{-C_1 \times 1})(1 - e^{-C_2 \times 2})(1 - e^{-C_3 \times 3}) \dots (1 - e^{-C_n \times n}) \quad (6.3)$$

Note that in this concept, the rate of application is not considered, nor are the growth rate and nutrient consumption, as functions of time.

The main issue that distinguishes between fertilization management of substrate-grown and of soil-grown plants is the limited volume of substrates, which means lower buffer capacity for solution composition and limited supply of nutrients (capacity factor). These two factors can be defined as the quantity and intensity factors. The quantity factor is the demand by plant as described in the next section. The intensity factor is the solution concentration. Root and nutrient concentrations have complementary effects, since uptake rate is the integral of the [flux $\times$ root surface area (or length)] in a given soil subvolume, over the total number of subvolumes in the soil profile. The flux F [mol/(cm root)/h] is determined by the nutrient concentration in the soil solution at the root surface, C_r (mol/L), as shown by the Michaelis-Menten equation:

$$F = F_{\max} \frac{C_r}{K_m + C_r} \quad (6.4)$$

$F_{\max}$ mol/(cm root)/h and K_m (mol/L) are plant coefficients obtained in flowing or well-stirred nutrient solution experiments. The integrated uptake approach and the presented equation are the basis for most models, which simulate nutrient uptake and plant growth (Barber, 1995; Bar-tal et al., 1994b; Claassen and Steingrobe, 1999; Hopmans and Bristow, 2002; Silberbush, 2002; Massa et al., 2008, 2009). Such models can be developed as fertigation decision tools, by comparing the computed and the required uptake rates at each time step during the simulation process. If the estimated uptake and required uptake rates do not match, a corrective action must be taken, subject to an economic optimization procedure.

Unlike cultivation in soils, in soilless culture, there is a need to supply most of the essential elements, including the micronutrients, continuously, because of the limited buffer capacity of the medium and its limited supply of nutrients (Savvas, 2001). Since the development of all-purpose nutrient solutions by Hoagland and Arnon (1950), many authors and organizations have published recommended tables of solution composition for different crops grown in soilless culture (Joiner et al., 1983; Steiner, 1984; Wright and Niemiera, 1987; Sonneveld and Straver, 1994; Schwartz, 1995; Nelson, 1996, 2003; Hanan, 1998; Resh, 1998; Adams, 2002; Sonneveld, 2002; Jones, 2005; Sonneveld and Voogt, 2009). However, the exact composition of the nutrient solution varies according to crop, stage of development, environmental conditions, and irrigation regime. Therefore, in the present chapter, we will concentrate on the main principles of nutrition in soilless culture rather than presenting recipes for growing specific plant species in soilless culture.

6.2.2 CONSUMPTION CURVES OF CROPS

6.2.2.1 Dry weight accumulation

The above description of crop responses to fertilization does not address the time factor. However, the demand for nutrients varies widely and dramatically during crop growth. There are considerable

differences in uptake rate and in the time at which maximum consumption rate occurs, among crops and among varieties of the same species. In many cases, the consumption function is not monotonic but exhibits sharp changes at critical physiological stages. Basically, the rate of nutrient requirement at each growth phase is associated with two predominant processes: (1) formation of new vegetative plant tissues and (2) formation of reproductive organs (flowers, fruits, seeds, etc.). The nutrient requirements for dry weight (DW) increase are primarily related to the photosynthesis rate, which is affected by various meteorological factors such as photosynthetically active radiation (PAR), air temperature and humidity, wind speed, and solar azimuth position (Thornley and Johnson, 1990; Goudriaan and van Laar, 1994). Theoretically, the net CO_2 assimilation rates by leaves, at 20°C under high light and CO_2 levels, are 1200 and 2000 $\mu\text{g}/\text{m}^2/\text{s}$ for C_3 and C_4 plants, respectively (Goudriaan and van Laar, 1994). In light of these data, and assuming that the average C content in the dry matter of vegetative crops to be 45% and the C content of CO_2 to be 27.3%, the calculated DW production of C_4 plants during 10 hours of daylight, at PAR values above a threshold of $200 \text{ W}/\text{m}^2$ (Thornley and Johnson, 1990), is $43.6 \text{ g}/\text{m}^2/\text{day}$. However, in addition to climatic aspects, other growth factors may limit biomass production; thus, for example, the actual DW production rate in the Netherlands during the summer has been reported to be only $15\text{--}25 \text{ g}/\text{m}^2/\text{day}$ (Goudriaan and van Laar, 1994). The initial growth phase is commonly characterized by an exponential rate, as each new leaf contributes to the light interception and to the biomass production, whereas during further growth phases, mutual shading leads to a gradually decreasing growth rate. At advanced growth phases, translocation of carbohydrates to nonphotosynthetic organs such as root, flower, fruit, or seed further diminishes the growth rate, and nonvegetative phases are characterized by a linear rate followed by a senescence rate. Representative rates of DW accumulation are illustrated in Fig. 6.1A–D. Note that lettuce (*Lactuca sativa* L.) production, which is characterized by fast vegetative growth, is characterized by an exponential rate (Fig. 6.1A), whereas vegetable fruits such as tomato (*Lycopersicon esculentum* Mill) and bell pepper (*Capsicum annuum* L.) are characterized by initial exponential growth rates that continue until 92 and 102 DAT (days after transplanting), respectively, followed by linear growth (Fig. 6.1C and D), and even senescence in the case of bell pepper. The decrease in DW accumulation rate of bell pepper plants between 170 and 145 DAT resulted from too low temperature in the winter season (Tzipilevitz et al., 1996). In the spring season, the climate became more favorable for vegetative growth (indicated by arrow, Fig. 6.1D) and the rate of DW accumulation increased to its former value. Note that the rate of DW accumulation (slopes of the curves in Fig. 6.1A–D) of lettuce in the course of the vegetative phase is significantly higher than those of tomato and bell pepper during the reproductive phase (18.5, 8, and $10.5 \text{ g}/\text{m}^2/\text{day}$, respectively).

6.2.2.2 Nutrient accumulation

Maynard and Lorenz (1979) reviewed the advances made in vegetable mineral nutrition and fertilization during the first 75 years of the 20th century, and Hochmuth (2003) has described the more recent progress in the field. The main technical change in this period was the application of drip irrigation with liquid fertilizers, which offers the benefits of nutrient application to the crops in amounts and at times when it is most required by the plant, and at the location where it is most likely to be absorbed by the roots. The principles of fertigation and the solute dynamics in the root zone have been reviewed by Bar-Yosef (1999), Mmolawa and Or (2000), and Bar-Tal et al. (2015). In the recent two decades, a major issue in fertilizer management has been to generate the greatest

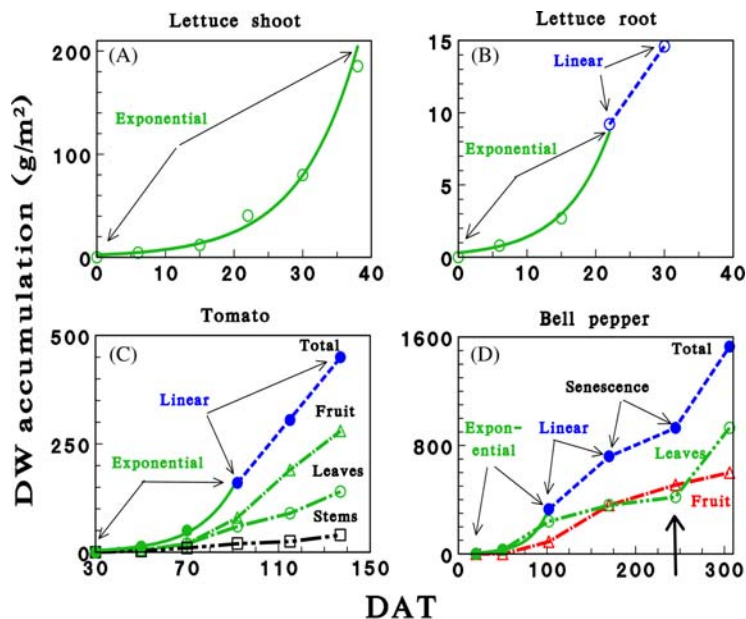


FIGURE 6.1

DW accumulation as a function of time (DAT) of several crop organs: (A) lettuce shoot; (B) lettuce root; (C) tomato fruits, leaves, stems, and total plant; and (D) bell pepper fruits, leaves + stems, and total plant. *Solid lines* denote the initial intensive growth phases (exponential phase), and *broken lines* denote the subsequent growth phases (linear and senescence phases). *Arrow* indicates the beginning of spring season as climate became more favorable for vegetative growth. DW, Dry weight; DAT, days after transplanting.

(A) and (B) Based on Xu, G., Levkovitch, I., Soriano, S., Wallach, R., Silber, A., 2004. Integrated effect of irrigation frequency and phosphorus level on lettuce: yield, P uptake and root growth. *Plant Soil* 263, 297–309; (C) based on Tanaka, A., Fujita, K., Kikuchi, K., 1974. Nutrio-physiological studies on the tomato plant. I. Outline of growth and nutrient absorption. *Soil Sci. Plant Nutr.* 20, 57–68; (D) based on Tzipilevitz, A., Silverman, D., Omer, S., Reznik, D., 1996. Consumption curves of high quality greenhouse bell pepper (*Capsicum annuum* cv. Mazurka). *Gan Sade VeMeshek* 2, 54–62. (in Hebrew).

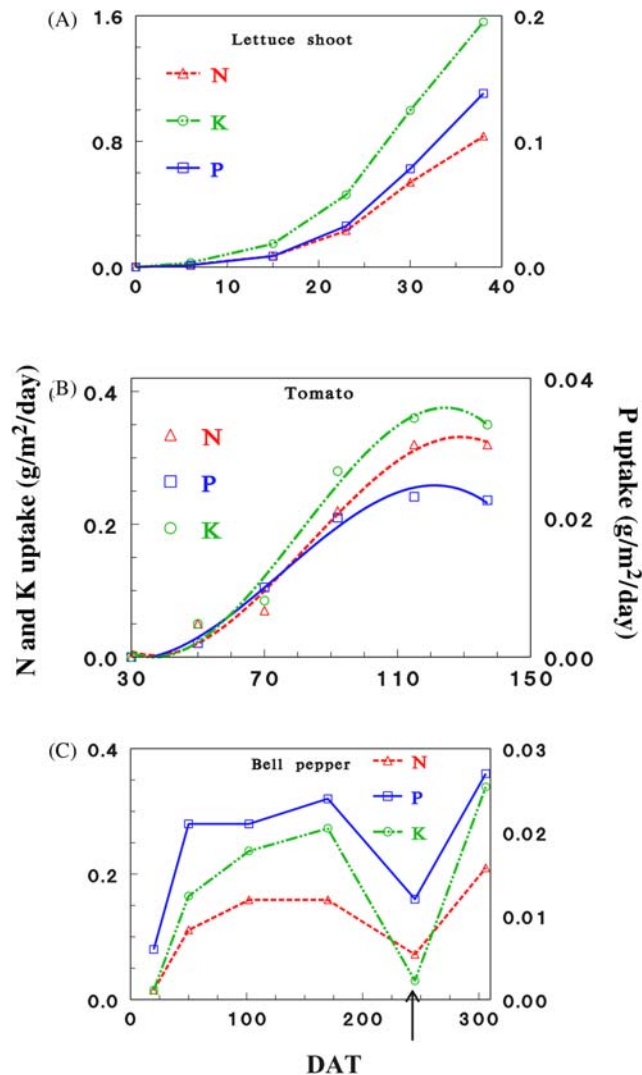
yield with the least negative impact on the environment. A schedule for nutrient application is the key to meeting the environmental requirements and satisfying the crop demand. Consumption curves provide the basic information required for optimal fertilization management, and fertigation is the tool with which to apply optimal management. Kläring (2001) reviewed the strategies for controlling water and nutrient supplies to greenhouse crops. The aim of the present section is to describe the specific features of nutrition of substrate-grown plants.

There are considerable differences among the shapes of consumption curves of crops and among those of varieties of the same species. In many cases, the consumption curve of a nutrient exhibits sharp changes that are related to the physiological stages of plant development (Bar-Yosef, 1999). The major differences between consumption curves of different crops stem from the development stages that the crops pass through, and the duration of each stage. The general nutrient consumption

curve described by Hochmuth (1992) is principally analogous to that of DW accumulation illustrated in Fig. 6.1A–D, that is, an exponential rate during the initial vegetative growth, followed by a linear and a senescence rate as reproductive organs develop. This consumption curve closely fits published measurements of nutrient uptake by determinant crops such as maize (*Zea mays* L.), fruit vegetables such as topped tomatoes (Tanaka et al., 1974), and melon (*Cucumis melo* L.) grown under environmental conditions that retard continuous fruit production. When nonterminating fruit tomatoes were grown continuously under well-controlled climatic conditions, the nutrient uptake rate grew steadily until production of the first fruit truss and then became monotonic (Tanaka et al., 1974). Similar results were obtained with greenhouse tomato (Bar-Tal et al., 1994a; Bar-Yosef, 1999) and pepper (Bar-Tal et al., 2001b). The nutrients consumption curve depends on the dry matter production curve, but there are differences between the two curves, which vary with developmental stage and between specific nutrients.

The major differences between the consumption curves of various fruit vegetables stem from differences between their patterns of production, that is, concentrated or continuous production. Daily nutrient uptake rates can be derived from the consumption curves, and those that result in optimum yield and product quality are crop specific and depend on climatic conditions but are independent of soil characteristics and of irrigation techniques. Lack of attention to the changes in uptake rate with time may lead to periods of overfertilization and underfertilization. Overfertilization may increase soil salinity and environmental contamination, whereas underfertilization may result in nutrient deficiency and yield reduction (Bar-Yosef, 1999). The rates of nutrient uptake by a typical vegetative crop (lettuce) and two widespread fruit crops (tomato and bell pepper) at various growth phases are illustrated in Fig. 6.2A–C.

Note that the rate of nutrient uptake by the leafy vegetable (lettuce) is characterized by a purely exponential curve [what I see looks very linear from day 23 on; since the time before day 23 is not presented in the other Fig. 6.2B and C, it is hard to draw the conclusion that you have here] and increases sharply as time proceeds (Fig. 6.2A), whereas those of tomato and bell pepper (Fig. 6.2B and C) became linear and even decreased in the course of fruit development. The sharp decrease in nutrient uptake rate of bell pepper plants (Fig. 6.2C) resulted from nonoptimal climatic conditions (mainly too low temperature, Tzipilevitz et al., 1996) in the winter season, which delayed fruit ripening and decreased the rate of fruit DW accumulation (indicated by arrow, Figs. 6.1D and 6.2C). Nutrient consumption curves of nontopped and topped tomato plants were studied by Tanaka et al. (1974), who found that the rate of accumulation was slow in the early stages but accelerated as the rate of dry matter production increased (only curves for nontopped plant are presented in Fig. 6.2B). This active accumulation continued until the end of the experiment in the nontopped plants but slowed down in the topped plants. The uptake of nutrients was continuous throughout growth, especially in nontopped plants, indicating that the physiological status of the plant did not appear to change with flowering; in other words, vegetative growth continued throughout growth without any marked effect upon fruit development. There was, however, some redistribution of N and P from vegetative organs at lower strata to the developing fruits during rapid fruit development. The amounts of N, K, and Ca absorbed were greater than those absorbed by cereal crops under ordinary cultural conditions (Tanaka et al., 1974). Similar to other greenhouse crops, tomato tissues are specific in that they have a high mineral content but, nevertheless, the mineral content of the fruits is lower than that of the vegetative organs (Gary et al., 1998); a larger and more continuous supply of these elements is needed by tomato than by cereal crops. In fruits, there is a continuous

**FIGURE 6.2**

Rate of NPK uptake by: (A) lettuce plant; (B) terminated (eight trusses) tomato plant; and (C) bell pepper plant. The *arrow* indicates nonoptimal climatic conditions (mainly too low temperature) in the winter season that delayed fruit ripening and decreased the rate of fruit DW accumulation. DW, Dry weight.

(A) Based on Xu, G., Levkovitch, I., Soriano, S., Wallach, R., Silber, A., 2004. Integrated effect of irrigation frequency and phosphorus level on lettuce: yield, P uptake and root growth. *Plant Soil* 263, 297–309; (B) based on Tanaka, A., Fujita, K., Kikuchi, K., 1974. Nutrio-physiological studies on the tomato plant. I. Outline of growth and nutrient absorption. *Soil Sci. Plant Nutr.* 20, 57–68; and (C) based on Tzipilevitz, A., Silverman, D., Omer, S., Reznik, D., 1996. Consumption curves of high quality greenhouse bell pepper (*Capsicum annuum* cv. Mazurka). *Gan Sade VeMeshek* 2, 54–62. (in Hebrew).

accumulation of carbohydrates in the vegetative organs during rapid growth, whereas in cereals, there is an apparent retranslocation of stored carbohydrates from the vegetative organs to the grains during ripening; development proceeds from the lower to the upper leaves and truss units. Each unit grows by using the currently absorbed elements and the currently produced photosynthates. The fruits grow by using the current photosynthates and the elements absorbed from the root during fruit growth.

Several examples of recommended nutrient solution compositions, adjusted for the stage of development and season in Israel, are presented in Table 6.2.

However, extrapolation of known NPK uptake data to environmental conditions different from those specified should be done carefully and treated only as a first approximation (Bar-Yosef, 1999). For example, Xu et al. (2001) reported that the total consumption of N by pepper plants in the summer was about 2.2–2.8 times higher than that in the winter, although the nutrient solution contained the same concentration of N.

Not only can the total demand fluctuate but the specific demand for individual nutrients can vary independently of fluctuations in total demand. The total uptake of nutrients is more or less determined by the growth and transpiration rates, but the uptake of individual nutrients depends more on the stage of growth (Voogt, 2003).

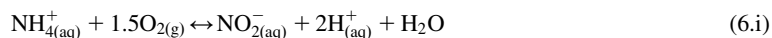
Table 6.2 Recommended Nutrient Solution Compositions Matched to the Growth Phase in Soilless Culture in Israel

Growth Phase	N	P	K	Ca	Mg
	mg/L				
Strawberry in greenhouse					
Transplanting	55–60	20–25	45–60	60–70	35–40
Anthesis and first fruit wave	70–85	20–25	70–90	100	45
Second fruit wave	80–85	25–30	80–90	100	45
Third fruit wave	80–85	25–30	80–90	100	45
Fourth fruit wave	55–60	20–25	55–60	80	35
Summer sweet pepper in greenhouse and net-house					
Transplanting to blooming	50–60	50–60	75–80		
Anthesis to fruit growth	80–100	80–100	100–120		
Fruit ripening and harvesting	100–120	100–120	140–160		
Fruit harvesting	130–150	130–150	180–200		
Fall-winter tomato					
Transplanting	80–90	30–40	120–140	180–220	40–50
Blooming and anthesis	120–150	30–40	180–220	230–250	40–50
Fruit ripening and harvesting	180–200	30–40	230–250	180–220	40–50
Fruit harvesting	120–150	30–40	180–220	180–220	40–50
From the Israeli Extension Service Recommendations.					

6.3 IMPACT OF N SOURCE

The nitrogen source may significantly affect the rhizosphere pH (Nye, 1981; Marschner, 1995; Marschner and Römheld, 1996; Bar-Yosef, 1999; Bloom et al., 2003) especially that of soilless-grown plants due to lower volume and buffer capacity (Bar-Yosef, 1999). The common N fertilizer sources in soil systems are NH_4^+ , NO_3^- , and urea. Rapid adsorption on the surface sites of the solid phases and nitrification reactions diminish the NH_4^+ content in soils and media, and hence, rhizosphere NH_4^+ concentration is commonly low even under irrigation with a high $\text{NH}_4\text{-N}/\text{NO}_3\text{-N}$ ratio (R_N). However, the NH_4^+ concentrations in soilless media that have low buffer capacity and are subjected to frequent fertigation events may be high. Consequently, in soilless culture, the impact of varying irrigation solution NH_4^+ concentration and R_N on rhizosphere pH and on physiological processes becomes crucial and controlling the R_N is of utmost importance. In addition, it is well accepted that the nitrifying bacteria exist as isolated colonies rather than being homogeneously distributed through the substrate volume (Darrah et al., 1987a; Strong et al., 1997). Consequently, the heterogeneity within low-buffered media in pH and in other chemical features driven by nitrification processes may be large.

Transformation of NH_4^+ to NO_3^- is a two-step process, in which each step is mediated by a highly specific group of bacteria. These are autotrophic bacteria that use inorganic material as energy sources (ammonia and nitrite as donors of electrons) and carbon dioxide as the carbon source for cell construction. In the first step, NH_4^+ is oxidized to NO_2^- by ammonium-oxidizing bacteria (of which the most investigated are the *Nitrosomonas* group) [Eq. (6.i)] and in the second step NO_2^- is oxidized to NO_3^- by nitrite-oxidizing bacteria including the *Nitrobacter* group [Eq. (6.ii)].



High levels of NH_4^+ can inhibit the nitrification processes (Focht and Verstraete, 1977; Niemiera and Wright, 1987b; Lang and Elliot, 1991; Russell et al., 2002) and it has been shown that nitrite (NO_2^-) oxidizing bacteria are more sensitive to high NH_4^+ levels than the NH_4^+ oxidizers (Nakos and Wolcott, 1972). In addition to the specific effect of NH_4^+ , high solute concentrations, in themselves, might inhibit nitrification, and the degree of inhibition was found to be related to both osmotic pressure and the specific anion that accompanied the NH_4^+ ions (Darrah et al., 1985, 1987b). The osmotic pressure threshold for inhibition was found to be 3.3 atm; however, chloride inhibited nitrification to a greater extent than its contribution to the osmotic pressure would indicate (Darrah et al., 1987b). The specific inhibitory effect of chloride must be considered carefully, because progressive chloride accumulation in the root zone as a result of high uptake of water and low frequency of irrigation is common under semiarid conditions.

Nitrification increases with increasing temperature, within moderate limits (5°C – 35°C), and the relationships between temperature and nitrification rates are usually described with the Arrhenius equation (Focht and Verstraete, 1977; Rodrigo et al., 1997; Russell et al., 2002) as presented in Fig. 6.3A. Nitrite-oxidizing bacteria were found to be more sensitive to low temperature than the NH_4^+ oxidizers; therefore, NO_2^- , the intermediate product of the nitrification process, may accumulate at low temperatures (Russell et al., 2002). Nitrite is well known to be toxic to plants; therefore,

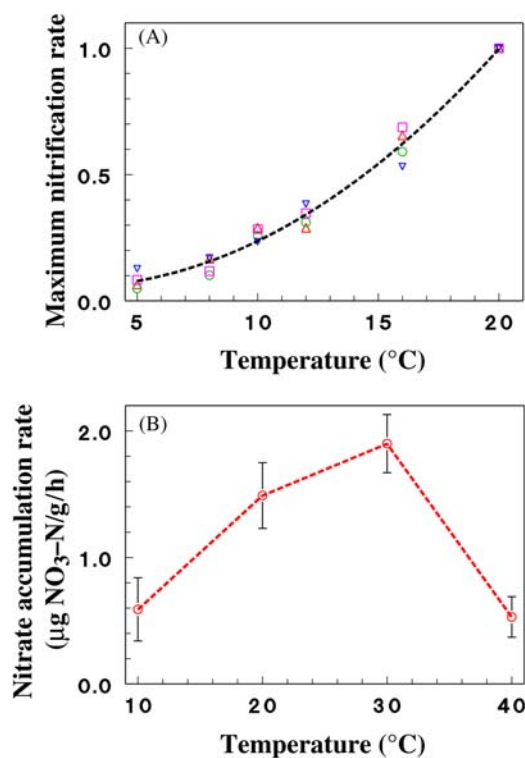


FIGURE 6.3

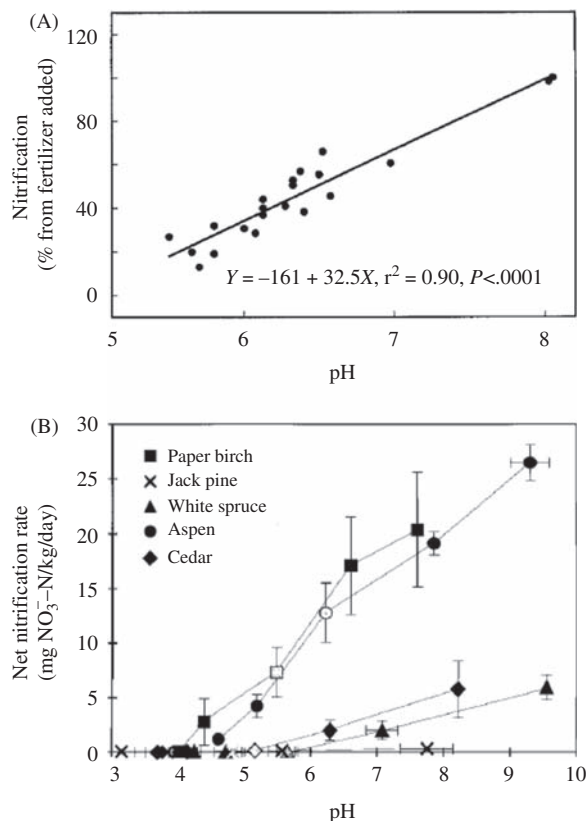
(A) Effect of temperature on the relative maximum nitrification rates in sandy soil amended with four N sources. (B) Influence of temperature on nitrate accumulation rate ($\mu\text{g NO}_3^- \text{N/g/h}$) in a pine bark medium.

(A) Based on Russell, C.A., Fillery, I.R.P., Bootsma, N., McInns, K.J., 2002. Effect of temperature and nitrogen source on nitrification in a sandy soil. *Commun. Soil Sci. Plant Anal.* 33, 1975–1989; (B) based on Niemiera, A., Wright, R.D., 1987a. Influence of temperature on nitrification in a pine bark medium. *HortScience* 22, 615–616.

NH_4^+ fertilization in a cold climate should be treated with caution. However, under hot conditions, the general trend of increasing nitrification rates with elevating temperature (Fig. 6.3A) changes direction, and above 30°C, nitrification rates drop (Fig. 6.3B).

The pH of the medium has been reported to be an important influencing factor in nitrification processes, and the tendency for nitrification rates to decrease under acid conditions, as depicted in Fig. 6.4A and B, is generally recognized (Niemiera and Wright, 1987b; Lang and Elliot, 1991; Olness, 1999; Ste-Marie and Paré, 1999; Russell et al., 2002; Kyveryga et al., 2004). However, the shape and magnitude of the effect of pH depend on medium properties, water content, and N sources (Stevens et al., 1998; Ste-Marie and Paré, 1999; Russell et al., 2002).

Similar to the effect of temperature, NO_2 oxidizing bacteria were found to be more sensitive to acidic conditions than the NH_4^+ oxidizers. Accumulation of toxic levels of NO_2 under low pH has been reported (Russell et al., 2002). The pH has a direct impact on the oxidation process of NH_4^+

**FIGURE 6.4**

Effects of pH on nitrification. (A) Relationship between soil pH and percentage nitrification of fertilizer N; (B) pH effect on net nitrification rates in the nitrifying forest floor of different stand types.

(A) From Kyveryga, P.M., Blackmer, A.M., Ellsworth, J.H., Isla, R., 2004. Soil pH effects on nitrification of fall-applied anhydrous ammonia. *Soil Sci. Soc. Am. J.* 68, 545–551, with kind permission from the Soil Science Society of America Journal; (B) from Ste-Marie, C., Paré, D., 1999, Soil pH and N availability effects on net nitrification in the forest floors of a range of boreal forest stands. *Soil Biol. Biochem.* 31, 1579–1589, with kind permission from Elsevier Science.

to NO_2^- , as H^+ is a product of this reaction [Eq. (6.i)], and consequently, accumulation of free protons in the solution inhibits further oxidation of NH_4^+ . The direct effects of pH on microbial activity, or the size or composition of populations, are not clear, because changes in pH frequently induce changes in additional vital features, for example, membrane permeability, nutrient availability, and element toxicity. Suzuki et al. (1974) showed that the change in the rate of oxidation in the pH range of 6.5–9.1 was relatively small when expressed on the basis of free ammonia rather than the ammonium ion. The relation between NH_4^+ , H^+ , and $\text{NH}_3(\text{aq})$ under equilibrium is given by the following equation:



It is well accepted that ammonia is the real substrate for ammonia oxidizers (Suzuki et al., 1974). The ammonia molecule diffuses easily through the cytoplasm into the cells (Kleiner, 1981). When the pH is low, the bacteria have to ingest ammonium actively through the cytoplasm into the cells and to maintain a higher pH in order to obtain higher concentrations of free ammonia (De Boer and Kowalchuk, 2001). An acidic pH causes low concentrations of carbonates (Lindsay, 1979), and a lack of the carbon sources required for microbial growth could be the direct cause of the reduction of nitrification rates (Kinbursky and Saltzman, 1990; Stevens et al., 1998).

Addition of NH_4^+ to common soilless growth medium (perlite) under four initial pH profiles (4.5, 5.5, 6.2, and 7.4) induced fluctuations in solution pH, NH_4^+ , and NO_3^- concentrations (Cytryn et al., 2012). Nitrification rate (calculated from ammonium oxidation and corresponding nitrate formation) was pH and time dependent (only results under initial pH of 7.4 are presented in Fig. 9.4). Under initial pH value of 7.4, the nitrification rate was highest on day 3 (pH value of 6.2), sustained stable between days 6 and 9 (pH values of 5.5 and 4.9, respectively), and dropped as pH further decreased (Fig. 9.4). During this period, nitrite did not accumulate in the vessels (except for day 3 of the initial pH 7.4 profile where a concentration of 0.04 mM was measured), indicating substantial activity of nitrite oxidizing bacteria. Cytryn et al. (2012) concluded that intensive ammonia-based fertilization in soilless culture media may result in a significant difference in both nitrification dynamics and ammonia oxidizing prokaryotes composition, relative to traditional soil-based agricultural systems. Under neutral and slightly acidic pH levels, the “r-strategists”-like *Nitrosomonas* strains were the dominant ammonia oxidizing bacteria due to their low substrate affinity and high maximum activity. However, under acidic pH levels, the “k-strategists”-like *Nitrospira* strains, characteristically dominant in soils, become more abundant. The main effects of nitrogen source on soilless-grown plants, that is, modification of the rhizosphere pH, availability of further nutritional elements, NH_4^+ toxicity, and incidence of physiological disorders such as chlorosis and blossom-end rot (BER), will be detailed and discussed below.

Urea is the cheapest N source (per N unit) and is the most concentrated N fertilizer (46%); it is highly soluble, moves easily with the irrigation water, and is, therefore, widely used in agriculture. However, urea is not taken up directly by plants and, therefore, only the derivative of urea hydrolysis— NH_4^+ ions—and the product of ammonium oxidation— NO_3^- ions—are absorbed by plant roots. The hydrolysis reactions are time-dependent processes (Lad and Jackson, 1982); therefore, urea is not commonly used in soilless culture, unless recycling systems are used (see Chapter 9: Irrigation in Soilless Production) so that NH_4^+ and NO_3^- ions are the main N source (Barker and Mills, 1979).

6.3.1 MODIFICATION OF THE RHIZOSPHERE pH AND IMPROVEMENT OF NUTRIENT AVAILABILITY

The nitrogen source may affect the rhizosphere pH via three mechanisms (Marschner, 1995; Marschner and Römhild, 1996; Bar-Yosef, 1999): (1) displacement of H^+/OH^- adsorbed on the solid phase, (2) nitrification/denitrification reactions, and (3) release or uptake of H^+ by roots in response to NH_4 or NO_3 uptake, respectively. Mechanisms (1) and (2) are not associated with any plant activity and affect the whole volume of the fertigated substrate, whereas mechanism (3) is directly related to the uptake of nutritional elements and may be very effective because it affects a limited volume in the immediate vicinity of the roots (Moorby et al., 1984; Gahoonia and Nielsen,

1992; Gahoonia et al., 1992; Marschner and Römheld, 1996; Taylor and Bloom, 1998; Bloom et al., 2003) as presented in Fig. 6.5. The extent of the pH alterations caused by the three mechanisms described above depends on medium properties, medium volume, plant activity, and all the environmental factors that affect nitrification rate, as detailed above.

Reduction of the medium pH, driven by NH_4^+ nitrification and root excretion of protons, is commonly used for overcoming growth disorders induced by micronutrient deficiencies, such as chlorosis and “little leaf” or “rosette” (Tagliavini et al., 1995; Bar-Yosef et al., 2000; Savvas et al., 2003; Silber et al., 1998, 2000a,b, 2004). The effects of N level and source throughout the growth cycle on the pH of the leachate from soilless-grown rice flower (*Ozothamnus diosmifolius*, Asteraceae) are illustrated in Fig. 6.6A and B, respectively.

Increasing the NH_4^+ supply, either by increasing the total N concentration (fixed R_N) or by increasing the R_N (fixed N concentration), resulted in a sharp decrease in the pH (Silber et al., 2004). The pH during the growth period was significantly affected by the amount of irrigation solutions applied, as well as by the N concentration and source. The higher volumes of irrigation solutions that were applied during the summer (up to 120 DAT) in response to the increased transpiration rate enhanced the supply of NH_4^+ , and the leachate pH fell to 4.1 at high NH_4^+ application. This trend changed during the winter (from 180 DAT) as the transpiration and irrigation rates and, consequently, the NH_4^+ doses decreased, and the leachate pH rose to close to that of the irrigation solution (Fig. 6.6). The growth of rice flower plants exposed to low-N fertilization or low R_N was poor, and the plants exhibited growth disorders such as tip burn, and severe chlorosis and necrosis (Silber et al., 2004). The pH in the root environment was probably the most important factor affecting rice flower growth and yield. A linear regression was obtained between the total dry mass production and the average pH in leachates within the pH range of 4.5–8.0 (Fig. 6.7A). It is reasonable to consider that the pH in itself did not affect rice flower growth but did so indirectly, through its effect on nutrient availability. The only nutrient that was significantly correlated with pH and yield parameters (total fresh and dry mass production, or the number of flowering stems)

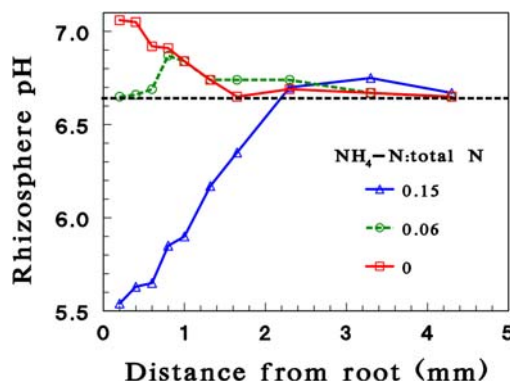


FIGURE 6.5

pH as a function of NH_4^+ -N fraction (of the total N application) and the distance from the roots of rape (*Brassica napus*) plant. Solid line indicates the pH of the unplanted soil.

Based on Gahoonia, T.S., Nielsen, N.E., 1992. Control of pH at the soil-root interface. *Plant Soil* 140, 49–54.

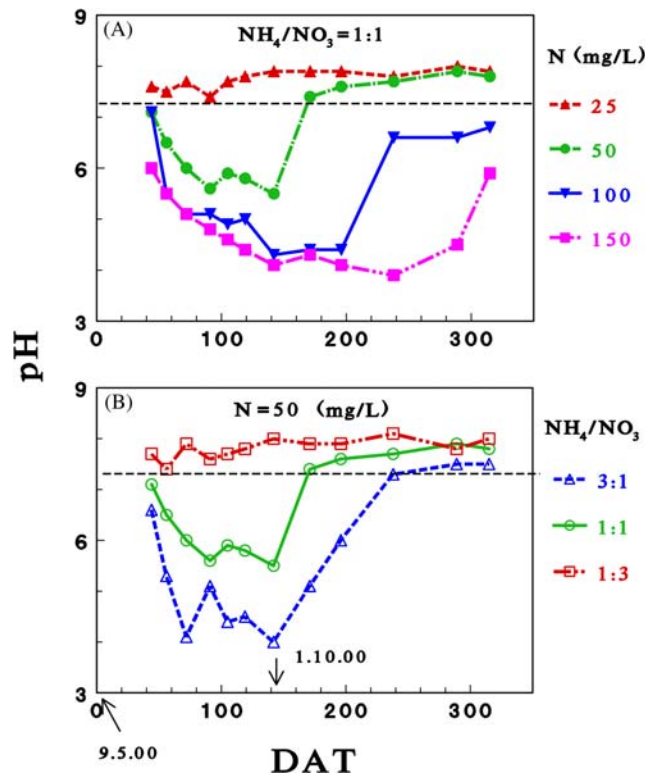


FIGURE 6.6

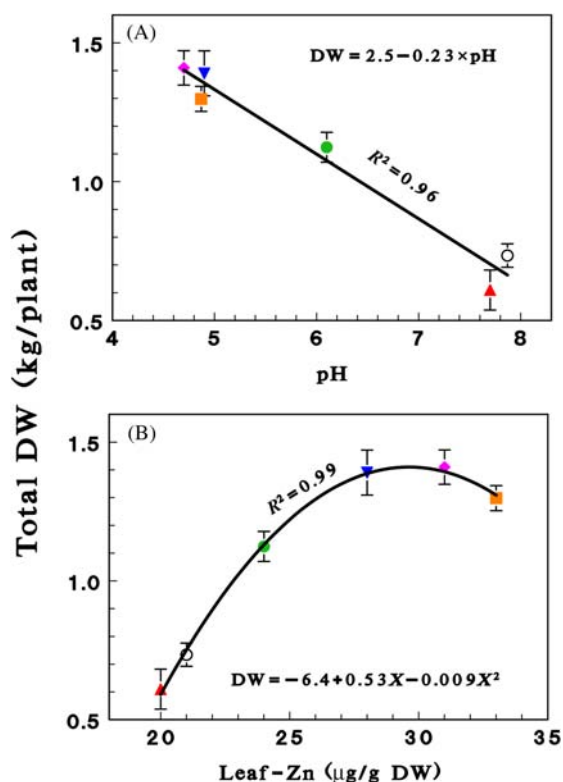
Effects of N level and source throughout the growth cycle on the pH of the leachate from soilless-grown rice flower. (A) Different N concentrations at constant $\text{NH}_4^+/\text{NO}_3^-$ ratio; and (B) Different $\text{NH}_4^+/\text{NO}_3^-$ ratio at constant N concentration.

(B) Based on Silber, A., Ben Yones, L., Dori, I., 2004. Rhizosphere pH as a result of nitrogen level and $\text{NH}_4^+/\text{NO}_3^-$ ratio and its effect on Zn availability and on growth of rice flower (*Ozothamnus diosmifolius*). *Plant Soil* 262, 205–213.

was Zn. Zinc solubility is strongly pH dependent (Barrow, 1993; Lindsay, 1979; Marschner, 1993), therefore, Zn-deficient conditions are induced by a high rhizosphere pH. A significant quadratic regression was obtained between the yield and leaf-Zn concentrations (Fig. 6.7B).

Similar effects of R_N on the drainage solution pH and on diverse growth disorders induced by high pH have been observed for other soilless-grown plants such as chrysanthemum (*Dendranthema × grandiflorum* Kitam), impatiens (*Impatiens walleriana* Hook. F.), ‘Safari Sunset’ [*Leucadendron salignum* Bergius × *Leucadendron laureolum* (Lam.) Fourc.], wax flower (*Chamelaucium uncinatum* L.), and gerbera (*Gerbera jamesonii*) (Siraj-Ali et al., 1987; Argo and Biernbaum, 1997a,b; Silber et al., 1998, 2000a,b; Bar-Yosef et al., 2000; Savvas et al., 2003).

It is important to note that decreasing the NO_3^- nutrition, by decreasing either the N level or R_N , may reduce metal mobilization toward the roots, as a result of diminished organic acid exudation

**FIGURE 6.7**

Relationships between total dry weight production of rice flower and: (A) averaged pH in leachates during the main growth season and (B) leaf-Zn concentrations. Vertical bars indicate the standard error.

(B) Based on Silber, A., Ben Yones, L., Dori, I., 2004. Rhizosphere pH as a result of nitrogen level and NH_4/NO_3 ratio and its effect on Zn availability and on growth of rice flower (*Ozothamnus diosmifolius*). *Plant Soil* 262, 205–213.

(Fig. 6.8). Formation of carboxylic anions is increased as a result of NO_3^- assimilation in plants (Fig. 6.9) and some of the carboxylates are exuded from the roots to the rhizosphere (Kirkby and Mengel, 1967; Kirkby and Knight, 1977; Touraine et al., 1990; Marschner, 1995).

Organic acids play a key role in mobilization of metal ions toward the roots (Gardner et al., 1982a,b, 1983; Bar-Yosef, 1996); therefore, the data presented in Figs. 6.8 and 6.9 clearly indicate that R_N augmentation may play a dual role in affecting micronutrients availability. On the one hand, micronutrients availability is enhanced through rhizosphere acidification (Tagliavini et al., 1995; Silber et al., 1998, 2000a,b, 2004; Bar-Yosef et al., 2000; Savvas et al., 2003) but, on the other hand, it is diminished because of the decrease of organic acid exudation (Figs. 6.8 and 6.9).

In addition to the indirect effect of R_N on nutrient availability in the rhizosphere, through pH modification as detailed above, R_N affects the apoplastic pH and, consequently, Fe^{III} reduction and mobilization in plants (Mengel et al., 1994; Kosegarten et al., 1998, 1999a,b, 2004; Zou et al., 2001).

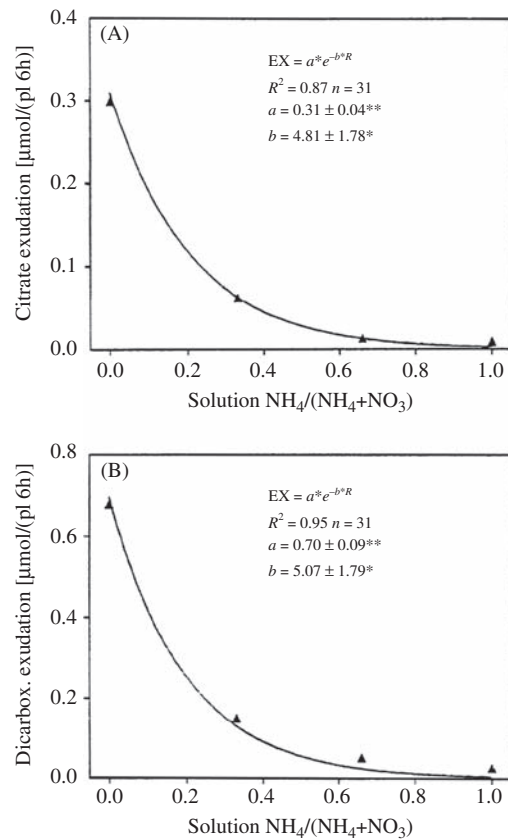


FIGURE 6.8

Relationships between R_N and carboxylic acids exuded from tomato roots. (A) Citric acid and (B) dicarboxylic acids.

From Imas, P., Bar-Yosef, B., Kafkafi, U., Ganmore-Neumann, R., 1997. Release of carboxylic anions and protons by tomato roots in response to ammonium nitrate and pH in nutrient solution. *Plant Soil* 191, 27–34, with kind permission from Elsevier Science.

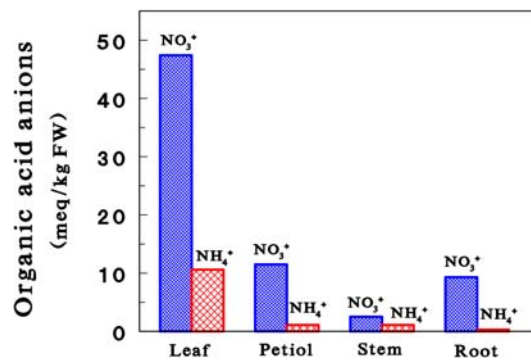


FIGURE 6.9

Influence of N source on citric acid concentration in tomato tissues.

Based on Kirkby E., Mengel, K., 1967. Ionic balance in different tissues of the tomato plant in relation to nitrate, urea, or ammonium nutrition. *Plant Physiol.* 42, 6–14.

Mengel (1994) found that inhibition of iron uptake from the root apoplast into the cytosol of root cells, rather than low iron availability in the rhizosphere, was the main cause of chlorosis caused by iron deficiency. Kosegarten et al. (1999b) found that the pH in the apoplast was different from that of the outer solution because of the strong H^+ buffer capacity of the apoplast, but that NO_3^- nutrition raised the apoplastic pH and depressed the ferric-chelate reductase activity. Therefore, high NO_3^- or $CaCO_3$ in the rhizosphere induced chlorosis incidence even when the concentration of iron in plants was normal and not different from that in nonchlorotic plants fed with high R_N in the irrigation solution.

6.3.2 CATION–ANION BALANCE IN PLANT AND GROWTH DISORDERS INDUCED BY NH_4^+ TOXICITY

The nitrogen source may significantly affect numerous physiological processes in plants (Marschner, 1995; Forde and Clarkson, 1999; Mengel and Kirkby, 2001; Epstein and Bloom, 2005). The cation–anion balance in plant tissues is maintained by diffusible and nondiffusible organic and inorganic ions and has been found to be notably affected by the sources of N nutrition (Marschner, 1995; Mengel and Kirkby, 2001; Epstein and Bloom, 2005). It has been widely reported that NH_4^+ nutrition depressed the uptake of cations, especially in leaves and petioles, and that NO_3^- nutrition depressed that of anions (Kirkby and Mengel, 1967; Kirkby and Knight, 1977) (Table 6.3).

Excess of cations and anions in plants is balanced by organic acids; therefore, the decrease of cation uptake followings NH_4^+ nutrition induced lower organic acid content in plants (Kirkby and Mengel, 1967; Touraine et al., 1990) and excretion from their roots (Imas et al., 1997) as depicted in Fig. 6.9.

Table 6.3 Influence of N Source on Cation and Anion Concentrations in Tomato Tissues

Tissue	N Source	K	Ca	Mg	SO_4^{2-}	$H_2PO_4^-$
(meq 100 g ⁻¹ DW)						
Leaf	NO_3^-	58	161	30	22	13
	NH_4^+	29	62	25	35	15
Petiole	NO_3^-	176	126	38	12	15
	NH_4^+	90	61	17	29	20
Stem	NO_3^-	162	86	35	8	13
	NH_4^+	54	50	18	14	15
Root	NO_3^-	93	44	40	10	27
	NH_4^+	43	38	11	15	21

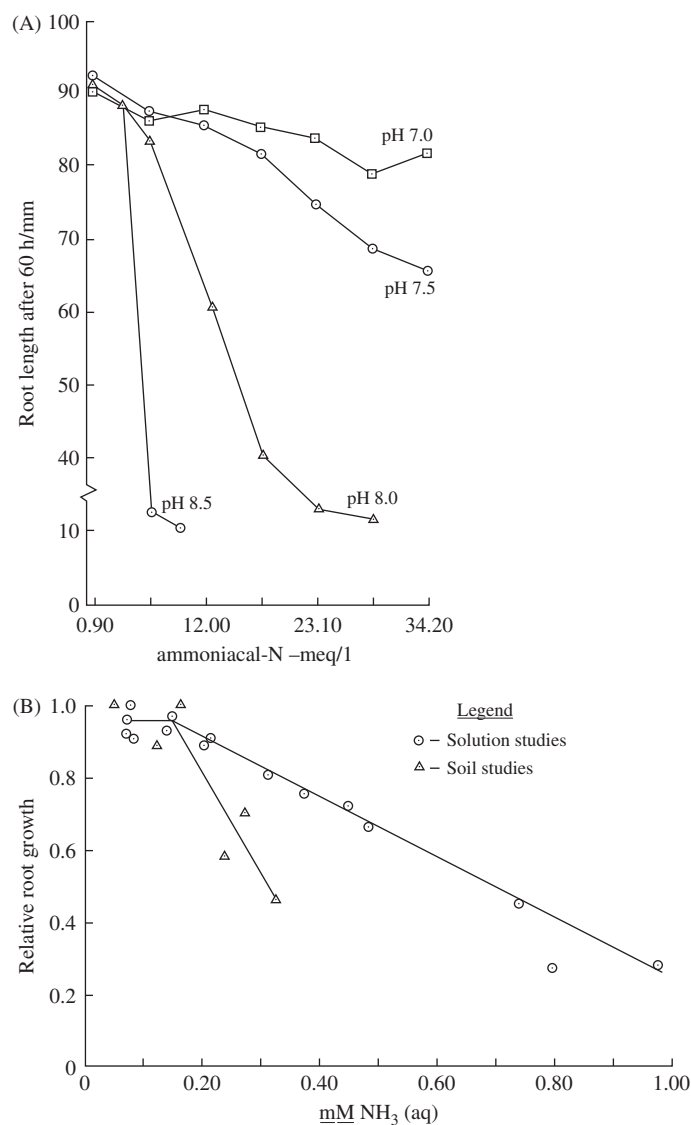
DW, Dry weight.

Based on Kirkby E., Mengel, K., 1967. Ionic balance in different tissues of the tomato plant in relation to nitrate, urea, or ammonium nutrition. *Plant Physiol.* 42, 6–14.

Although N is an essential nutrient for plant growth, high NH_4^+ concentrations are toxic to most plants, especially at high root temperatures and under high salinity (Kafkafi, 1990; Forde and Clarkson, 1999; Adams, 2002; Sonneveld, 2002; Britto and Kronzucker, 2002). Excluding some exceptional plants such as *Lisianthus* (*Eustoma grandiflorum*, Raf.) that positively responded to high NH_4^+ supply (Mendoza-Villarreal et al., 2015), it is well accepted that high NH_4^+ concentrations are harmful. The detrimental effect of NH_4^+ on vegetative growth, fruit production, and fruit quality (BER and flat fruit) of pepper and tomato is well documented (Ganmore-Neumann and Kafkafi, 1980a,b; Bar-Tal et al., 2001a; Claussen, 2002; Aki et al., 2003; Silber et al., 2005a). Yet, the direct cause of NH_4^+ toxicity remains obscure and, in spite of numerous studies, the causative mechanism has not been identified (Britto and Kronzucker, 2002).

NH_4^+ toxicity has been hypothesized to be associated with the solution concentration of $\text{NH}_3(\text{aq})$ (Bennett and Adams, 1970a,b; Bennett, 1974). $\text{NH}_3(\text{aq})$ can enter easily through cell membranes, and even at a very low concentration (0.17 mM in the nutrient medium), and might inhibit vital processes such as photosynthesis, respiration, and enzyme activity (Bennett and Adams, 1970b; Bennett, 1974). Inhibition of root elongation as a result of NH_4^+ nutrition has been found to be pH dependent and was directly related to the calculated solution concentrations of $\text{NH}_3(\text{aq})$ (Bennett and Adams, 1970b), as depicted in Fig. 6.10A and B. However, the interpretation of Bennett and Adams (1970b) is not straightforward, because the severe inhibition of root growth at pH above 8 (Fig. 6.10A) could be related to deficiency of Ca (Bennett and Adams, 1970a) or of other nutrients.

It has been reported (Britto et al., 2001a,b) that disruption of the influx regulation and consequent accumulation of NH_4^+ in the cytosol caused useless cycling of N across the plasma membrane of root cells. This futile cycling induced an abrupt increase of respiration, with high energy cost and subsequent decline in growth (Britto et al., 2001b). Almost all the NH_4^+ ions taken up from the irrigation solution are rapidly assimilated in the roots (Raven, 1985, 1986; Marschner, 1995); therefore, the demands for carbohydrates and oxygen in the roots are higher under high- $\text{NH}_4\text{-N}$ than under high- $\text{NO}_3\text{-N}$ treatment (Ganmore-Neumann and Kafkafi, 1983; Kafkafi, 1990). High temperature directly reduces solution- O_2 concentrations; it also indirectly reduces carbohydrate availability and solution- O_2 concentrations, through increases in plant respiration (Gur et al., 1972). Thus, growth impairment and yield reduction of plants that are susceptible to ammonium toxicity are expected to be more important under hot than under cold or temperate climates (Ganmore-Neumann and Kafkafi, 1980a, 1983; Kafkafi, 1990, 2001; Marschner, 1995; Mengel and Kirkby, 2001). The uptake rate of most nutrients is reduced under nonoptimal temperatures (Kafkafi, 1990; Rachmilevitch et al., 2006b). NH_4 absorption by roots is more sensitive to low temperature than NH_4 because of the lower energy required for NH_4 absorption (Kafkafi, 1990); however, an increase in NO_3 uptake in suboptimal temperature (10°C) has been reported by Calatayud et al. (2008). Root temperature also has strong impact on nutrient transport and assimilation in plants (Gorbe and Calatayud, 2010). Nutrient transport from the root to above-ground organs decreases under suboptimal temperature (Kafkafi, 1990; and Chapter 2: Root Growth, Physiology, and Potential Impact of Soilless Culture on Their Functioning). Under root low temperature, nitrate reductase activity in leaves (Calatayud et al., 2008) and roots (Rachmilevitch et al., 2006a) increased, while under high root temperature, nitrate assimilation rate decreased (Rachmilevitch et al., 2006b). It should be noted that the detrimental combination of hot temperature and high R_N occurs more frequently in soilless than in soil culture, because of the lower buffer and heat capacities of the former that is dependent of the volume and shape of the growing containers

**FIGURE 6.10**

(A) The effect of solution concentration of NH_4^+ at different pH values, on root growth of cotton in nutrient solution during 60 h and (B) the relationship between growth inhibition of cotton roots and the calculated concentration of $\text{NH}_{3(\text{aq})}$ in nutrient and in soil solution.

From Bennett, A.C., Adams, F., 1970b. Concentration of $\text{NH}_{3(\text{aq})}$ required for incipient NH_3 toxicity to seedlings. *Soil Sci. Soc. Am. Proc.* 34, 259–263, with kind permission from the Soil Science Society of America Journal.

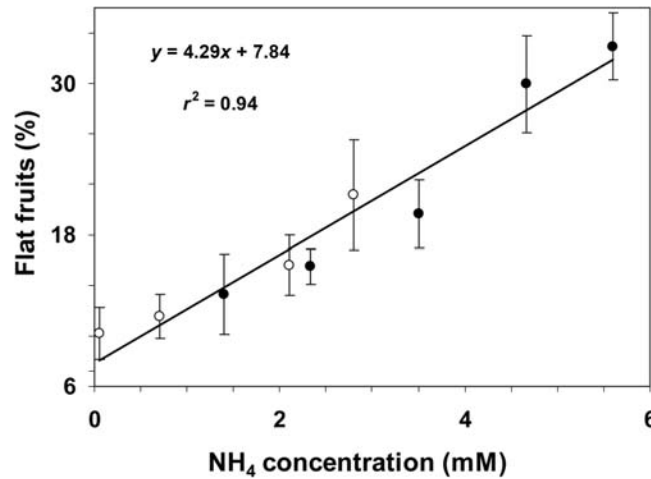


FIGURE 6.11

Relationship between incidence of flat fruit (sweet pepper) and solution concentration of NH_4^+ . The empty symbols are for Expt 1 (N varied from 0.25 to 14.0 mmol/L, constant R_N of 0.25), and the full symbols are for Expt 2 (R_N varied from 0.25 to 2.0, constant N concentration of 7.0 mmol/L), the line is the best-fit linear regression. Vertical bars indicate the standard error.

From Bar-Tal, A., Aloni, B., Karni, L., Oserovitz, J., Hazan, A., Itach, M., et al., 2001a. Nitrogen nutrition of greenhouse pepper. I. Effects of nitrogen concentration and $\text{NO}_3^-/\text{NH}_4^+$ ratio on yield, fruit shape, and the incidence of blossom-end rot in relation to plant mineral composition. *HortScience* 36, 1244–1251, with kind permission from the American Society of Horticultural Science.

(Heller et al., 2015). On the other hand, high temperatures (up to 30°C–40°C, see Niemiera and Wright, 1987a; Walden and Wright, 1995) enhance the nitrification rate and therefore curtail exposure of the roots to the toxic NH_4^+ -N concentrations in the rhizosphere. In soilless culture, the high air-filled porosity of most media (Raviv et al., 2002) ensures the availability of oxygen for nitrification processes that rapidly reduce the NH_4^+ -N concentration in the rhizosphere.

The NH_4^+ concentration is affected by both the total N concentration and the R_N , therefore, it is more practical to express NH_4^+ activity in terms of the solution NH_4^+ -N concentration. Bar-Tal et al. (2001a) found that the incidence of BER and flat pepper fruit increased as the concentration of NH_4^+ in the nutrient solution increased, because of increases in the concentration either of total N or specifically of NH_4^+ (only the relationship between incidence of flat fruits and NH_4^+ concentration is depicted in Fig. 6.11).

However, the threshold values of NH_4^+ -N concentration in the irrigation solution, as determined in field experiments, are not definitive. Nitrification is a time-dependent process and therefore, the transient NH_4^+ concentration in the rhizosphere is significantly affected by the interval between successive irrigations. Thus, under irrigation schedule of one to four daily irrigation events, roots are exposed to apply NH_4^+ only during short periods immediately after the irrigation events, whereas during most of the time between successive irrigations, NO_3^- rather than NH_4^+ is the main N source.

However, under frequent fertigation, with 10–30 daily irrigation events, the transient NH_4^+ concentration in the vicinity of the roots may be high (Silber et al., 2005a). It is well documented that high NH_4^+ concentration and high R_N enhance the incidence of BER in vegetable fruits (Wilcox et al., 1973; Morley et al., 1993; Bar-Tal et al., 2001a; Silber et al., 2005a). It is generally accepted that the effect of the NH_4^+ concentration on the incidence of BER is through its effect on Ca uptake (Ho et al., 1993; Adams, 2002; Sonneveld, 2002). Since NH_4^+ has been found to depress the leaf Ca content (Table 6.3), it was suggested that in susceptible crops such as tomato and sweet pepper, $\text{NH}_4\text{-N}$ should be restricted to 15% or even 10% of the total N supplied via irrigation (Adams, 2002; Sonneveld, 2002). High solution- NH_4^+ concentration restricted the vegetative and fruit production of solanaceous crop (eggplant, *Solanum melongena*) (Savvas et al., 2010). However, the harmful effect was probably associated with the specific period of NH_4^+ application. Supplying NH_4^+ before bloom promoted the vegetative growth of eggplants and fruit occurred earlier than on similar plants supplied with nitrate (Claussen and Lenz, 1995). Furthermore, the number of flowers and the fruit yield were increased.

6.4 INTEGRATED EFFECT OF IRRIGATION FREQUENCY AND NUTRIENTS LEVEL

The mechanisms of water and nutrients transport in the rhizosphere, and the mechanisms of ion uptake by root have been widely reviewed by Barber (1995), Tinker and Nye (2000), Hopmans and Bristow (2002), Jungk (2002), and Silberbush (2002). A quantitative approach is presented in Chapter 3, Physical Characteristics of Soilless Media, of this book. Therefore, these issues are not discussed in the present chapter, and attention is focused on the interrelationships between irrigation management, especially under nonstress conditions, nutrient availability in soilless media, and several nutritional aspects of plant growth.

The beneficial effects of high-frequency irrigation were recognized some decades ago, and it is considered a useful tool for optimizing the root environment. Rawlins and Raats (1975) stated that “...we should not overlook the possibilities for conserving and making most effective use of our water, land and fertilizers resources by using high-frequency irrigation,” and Clothier and Green (1994) provided further support. Although these conclusions were based on studies of soil and of soil-grown plants, their basic approach is valid for soilless media as well. It is important to note that in the 1970s and until the 1990s, high-frequency irrigation was defined as having intervals of <7 days (Martin et al., 1990), but nowadays the time between successive irrigation events of soilless-grown plants has diminished to hours, or even less. Recently, microdrip irrigation systems have been developed that provide emitter discharges of <0.5 L/h. These systems are mostly based on low-pressure gravity flow and have been studied mainly with soilless-grown crops. Preliminary results have indicated promising potential. The discussion of the effects of irrigation frequency on root-zone matric potential and water and nutrients uptake by plant roots in the present chapter is mainly focused on nonstress conditions, that is, the irrigation frequency does not lead to water-stress conditions, and the amount of available water in the root volume between two successive irrigation events is sufficient and does not limit uptake by the root.

6.4.1 NUTRIENT AVAILABILITY AND UPTAKE BY PLANTS

Adsorption on the solid phases and precipitation of insoluble compounds decrease the concentrations of nutrients in the root area (Chapter 4: Chemical Characteristics of Soilless Media). Thus, the nutrient concentrations in the rhizosphere may be high or even excessive immediately after irrigation and may subsequently fall to deficit levels as illustrated in Fig. 6.12A. These processes are time dependent; therefore, reducing the time interval between successive irrigations to maintain constant, optimal water content in the root zone may also reduce the variations in nutrient

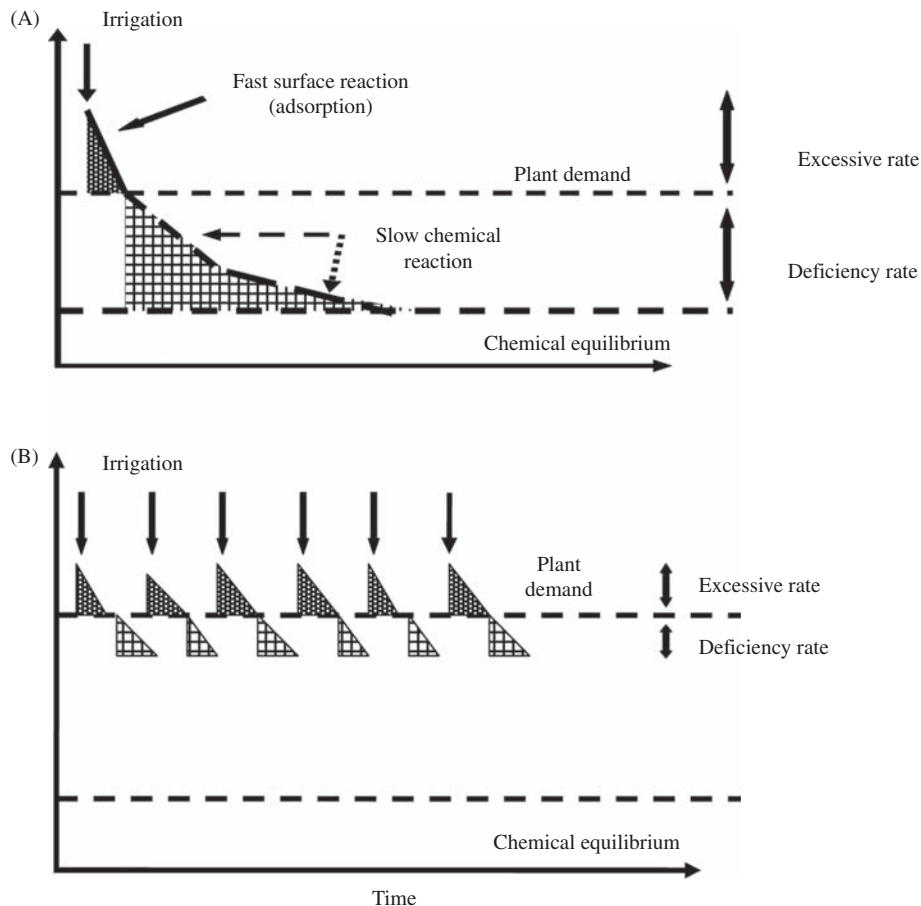


FIGURE 6.12

Schematic presentation of the time variation of nutrient concentration in the rhizosphere under conventional and frequent irrigation (A and B, respectively). Excess and deficiency rates correspond to nutrient concentration above or below plant demand, respectively; chemical equilibrium corresponds to nutrient concentration governed by equilibrium processes.

concentration (Fig. 6.12B), thereby increasing their availability to plants and reducing their leaching out of the root zone.

Water and nutrients acquisition by plants, and the formation of a depletion zone in the immediate vicinity of the roots, drive solute movement toward the roots. Nutrient transport to the root surface takes place by two simultaneous processes: convection in the water flow (mass flow), and diffusion along the concentration gradient (Barber, 1995; Tinker and Nye, 2000; Jungk, 2002). Medium properties, crop characteristics, and growing conditions affect the relative importance of each mechanism, but the general situation is that the mobile NO_3^- ion supply is taken up mainly through mass flow, whereas for less mobile elements such as P and K, diffusion is the governing mechanism (Barber, 1995; Claassen and Steingrobe, 1999; Jungk, 2002; Mmolawa and Or, 2000; Tinker and Nye, 2000). The diffusion takes place through the aquatic phase and the diffusion coefficient (D_e) is given by the following equation (Nye, 1966):

$$D_e = D_L \theta f \left(\frac{dC_l}{dC_s} \right) \quad (6.5)$$

in which D_L is the diffusion coefficient of the ion in water, θ is the volumetric water content, f is the tortuosity or the medium-impedance factor, and dC_l/dC_s is the reciprocal of the medium buffer power for the ion in the solution, C_l is the ion concentration in solution and C_s is the total diffusible amount of the ion. Nutrient diffusion takes place in water-filled pores and, therefore, θ has an important effect on D_e (Kovar and Claassen, 2005). The tortuosity factor (f) is strongly affected by θ ; therefore, the effect of θ on D_e is usually described by a quadratic equation (Claassen and Steingrobe, 1999). Increasing θ from 0.1 to 0.4 resulted in increases of D_e for K (Kuchenbuch et al., 1986a) and P (Bhadoria et al., 1991) by factors of about 10. In addition, θ affects the shape of the depletion zone (water- or nutrient-depleted area surrounding plant roots) so that reducing θ increased the width of the depletion zone and, consequently, reduced the soil volume available for root acquisition and limited the K uptake per unit root area (Kuchenbuch et al., 1986b), as well as that of other nutrients. However, simulations demonstrated that the root was able to partially compensate for the decrease of diffusion by increasing the concentration gradient (Claassen and Steingrobe, 1999). Simulation of NO_3^- uptake showed that it was less sensitive than K uptake to changes in θ (Claassen and Steingrobe, 1999). The nitrate ion is almost not bound to the solid phases; therefore, the soil buffer power (dC_l/dC_s) is significantly lower for NO_3^- than for K. In light of the simulations by Claassen and Steingrobe (1999) and taking into account the soil buffer power for the nutritional elements, it is to be expected that P uptake will be strongly affected by θ variations. Bar-Tal et al. (1994b) used a simulation to predict that increasing the θ of the medium would compensate for low P concentration in the irrigating solution. In addition, soil-P concentration is governed by adsorption and precipitation reactions (Chapter 6: Nutrition of Substrate-Grown Plants); therefore, the mechanism by which increasing the nutrient concentration in the vicinity of the root compensates for the above-described decrease in diffusion is not valid for P. Indeed, altering θ significantly affected the depletion of inorganic P in the rhizosphere (Gahoonia et al., 1994). In addition to the effect of θ on D_e , frequent watering enhanced the mixing of solute between pores containing mobile and immobile water, respectively, and thereby improved P and K uptake by plants (Kargbo et al., 1991).

Claassen and Steingrobe (1999) commented that under adequate supply conditions, even NO_3^- is transported mainly by diffusion, and they concluded that diffusion was a much more efficient

mechanism for nutrient transport than mass flow. However, these conclusions were based mainly on experiments and model calculations in which nutrients were supplied independently of, and much less frequently than, irrigation water. Furthermore, in most of those experiments, nutrients were supplied only before planting seedlings or sowing; therefore, on a time scale of days or weeks, nutrient transport from the soil solution to the root interface took place mainly by diffusion. Nevertheless, the simultaneous application of irrigation water and nutrients, together with the high values of saturated hydraulic conductivity for container media, ensure that the root surface and its vicinity are frequently supplied with fresh nutrient solution during the irrigation events and the subsequent redistributions. These frequent replenishments diminish the depletion zone formed at the root surface by uptake of nutrients during the period between successive irrigation events, decrease the concentration gradient between the medium solution and the root interface, and diminish the role of diffusion in nutrient transport toward the roots (Bar-Tal et al., 1994b). Consequently, the contribution of convection to the nutrient uptake of soilless-grown plants is much more significant than that of diffusion.

Nitrate, the main N source for soilless-grown plants (Sonneveld, 2002), is hardly ever involved in adsorption or precipitation reactions; therefore, the concentration of NO_3^- in the irrigation water and its actual concentration in the vicinity of the roots are quite similar. In contrast, P availability to plant roots is time dependent, as a result of adsorption and precipitation reactions (Chapter 6: Nutrition of Substrate-Grown Plants). Potassium ions are hardly ever involved in precipitation reactions but may be adsorbed on negatively charged surfaces. Therefore, the difference between the K concentrations in the irrigation solution and the rhizosphere lies between those between the respective NO_3^- and P concentrations. Consequently, it can be expected that the impact of fertigation frequency on the uptake of nutritional elements by plants will be related to both their mobility and their availability, as indeed has been reported (Mbagwu and Osuigwe, 1985; Kargbo et al., 1991; Silber et al., 2003). Although the effects of irrigation frequency on nutrient concentration in soilless-grown lettuce leaves presented in Fig. 6.13 followed the expected order of $\text{P} > \text{K} > \text{N}$, the magnitudes of the nutrient increases in the plant were found to be closely related to the fertilization level.

The increases in the leaf P and K concentrations were attributed to both direct and indirect effects of irrigation frequency on the P and K concentrations at the root surface. The direct effect is the frequent elimination of the depletion zone at the root surface by the supply of fresh nutrient solution during and soon after the irrigation events. This supply was fully available to the roots soon after the irrigation events, at which times its uptake rate behaved purely in accordance with the Michaelis–Menten equation [Eq. (6.4)]. Moreover, a higher irrigation frequency maintains higher dissolved P and K concentrations in the substrate solution, by shortening the period during which precipitation may take place. The indirect effect of irrigation frequency on nutrient availability is manifested through the higher convective and diffusive fluxes of dissolved nutrient from the substrate solution to the root surface, which increase with increasing irrigation frequency. The findings that increasing the fertilizer rate improved nutrient uptake and plant yield, and that increased irrigation frequency resulted in systematic diminution of the nutrients uptake enhancement, may indicate that the main effect of increased fertigation frequency was related to an improvement in nutritional status, mainly with regard to P (Silber et al., 2003, 2005b; Xu et al., 2004). Thus, increasing the irrigation frequency may compensate for certain nutrient deficiencies and in the above examples, the lower yields of plants fertigated at low frequency might have been a result of nutrient shortage rather than water shortage.

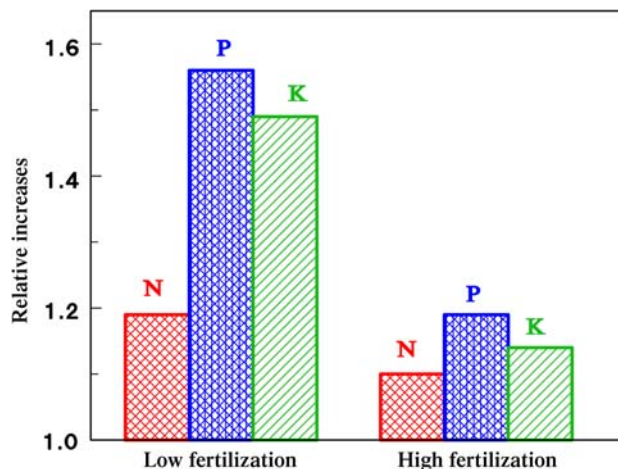


FIGURE 6.13

Relationships between fertilization level and irrigation frequency. Relative nutrient uptake was defined as the increases of N, P, and K in lettuce plants under high-frequency irrigation (18 daily irrigation events) relative to those under low frequency irrigation (2 daily irrigation events).

6.4.2 DIRECT AND INDIRECT OUTCOMES OF IRRIGATION FREQUENCY ON PLANT GROWTH

6.4.2.1 Root growth and root/shoot ratio

The adaptive changes in root growth and morphology in response to water stress and nutrient deficiency in the rhizosphere were thoroughly described in the scientific literature (Waisel et al., 2002). Alterations of growth conditions generally lead to modifications of the root system; therefore, irrigation frequency may have an effect on the root system through two main mechanisms: (1) the direct effect on the wetting patterns and water distribution in the soil volume, which modulate root distribution and growth (Phene et al., 1991; Clothier and Green, 1994; Coelho and Or, 1996, 1999) and (2) indirect effect on nutrient availability (Lorenzo and Forde, 2001), especially that of P, which significantly modify root system efficiency (Glass, 2002; Lynch and Ho, 2005), including their effects on root hair density (Bates and Lynch, 1996; Ma et al, 2001) and root system architecture (Liao et al., 2001; Williamson et al., 2001).

Separation between the direct effect of irrigation frequency, that is, through the increases of θ in the root area, and the indirect effect of increased P availability is not straightforward. The effect of irrigation frequency on the root/shoot ratio has been reported to be smaller than that of P concentration (Xu et al., 2004; Silber et al., 2005b) and to be very sensitive to plant age (Xu et al., 2004). The reasons for the age-linked diminution of the effect of irrigation frequency on root/shoot ratio (Fig. 6.14A) could be (1) during early growth, the roots were mainly located at the top of the pots and were more sensitive to the drying and rewetting processes than later on; (2) adsorption of the

added P by the sand substrate induced stronger deficiency conditions in the early growth period than later; and (3) the young roots in the early stages were mostly active roots, whereas in the later stages part of the roots became inactive and probably masked the changes (Xu et al., 2004).

Similarly to the findings with lettuce, observations of bell pepper plants showed high sensitivity of the root/shoot ratio to variations in irrigation frequency under low P application and a diminished response under high P application (Fig. 6.14B). Note that the root/shoot ratio under low

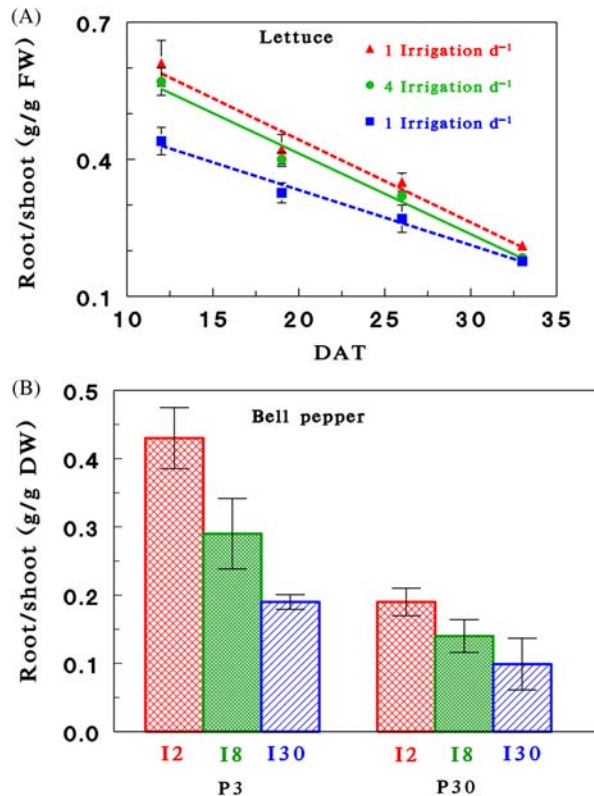


FIGURE 6.14

Root:shoot ratio of (A) lettuce plants under three irrigation frequencies (same daily amount of water)—1, 4, and 10 irrigation events per day—during lettuce growth (DAT) and (B) bell pepper under two water-P levels: P1 and P3-3 and 30 mg/L, respectively, and three irrigation frequencies (same daily amount of water): I1 (two irrigation events per day), I2 (four irrigation events per day), and I3 (1.5 min every 30 min throughout the day). Vertical bars indicate the standard error (not shown when smaller than the symbol). DAT, Days after transplanting.

(A) From Xu, G., Levkovitch, I., Soriano, S., Wallach, R., Silber, A., 2004. Integrated effect of irrigation frequency and phosphorus level on lettuce: yield, P uptake and root growth. *Plant Soil* 263, 297–309, with kind permission from Springer Science and Business Media; (B) based on Silber, A., Bruner, M., Kenig, E., Reshef, G., Zohar, H., Posalski, I., et al., 2005b. High fertigation frequency and phosphorus level: effects on summer-grown bell pepper growth and blossom-end rot incidence. *Plant Soil* 270, 135–146.

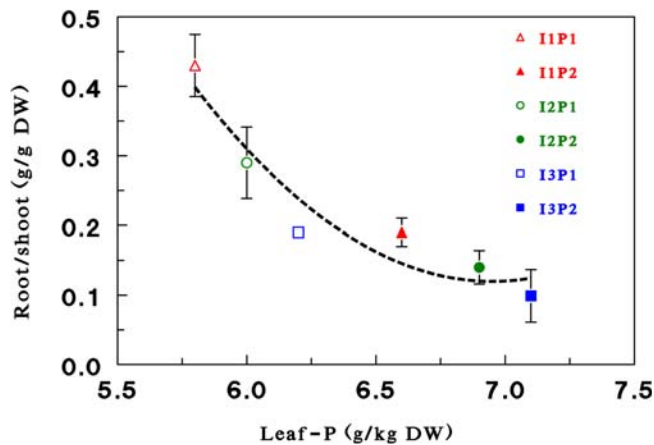


FIGURE 6.15

Relationships between root:shoot ratio and leaf-P concentration in bell pepper plants. The treatments included three irrigation frequencies (same daily amount of water): I1 (two irrigation events per day), I2 (four irrigation events per day) and I3 (1.5 min every 30 min throughout the day); and two water-P levels—P1 and P2—3 and 30 mg/L, respectively. Vertical bars indicate the standard error (not shown when smaller than the symbol).

Based on Silber, A., Bruner, M., Kenig, E., Reshef, G., Zohar, H., Posalski, I., et al., 2005b. High fertigation frequency and phosphorus level: effects on summer-grown bell pepper growth and blossom-end rot incidence. Plant Soil 270, 135–146.

P and high-frequency irrigation was very similar to that under high P and low-frequency irrigation, which may indicate that both treatments affect the same mechanism. Therefore, if a single mechanism is assumed and if the main impact of irrigation frequency is actually arises from the increase of P availability and the consequently higher P uptake by the plant, then the root/shoot ratio should be correlated with a single plant indicator, common to both factors. Actually, irrespective of the experimental causes for leaf-P variations (P level or irrigation frequency), the values of root/shoot ratio were correlated negatively with leaf-P concentrations (Fig. 6.15).

6.4.2.2 Yield and physiological aspects

It has been shown that yield gained under high irrigation frequency can be primarily related to increased availability of nutrients, especially P (Mbagwu and Osuigwe, 1985; Silber et al., 2003, 2005b). The relationships between DW production of several crops and leaf-P concentrations induced by irrigation frequency are depicted in Fig. 6.16. Multiple stepwise regressions relating nutrient concentrations in the plant to yield revealed a significant correlation between the DW production and P concentration in leaves (Silber et al., 2003, 2005a, Xu et al., 2004), indicating that the main effect of fertigation frequency was related to improvements in P mobilization and uptake.

In addition to its effects on soil–water tension and uptake, P availability, root system geometry and performance, and DW production, detailed above, irrigation frequency directly and indirectly influenced other processes in plants. Indirect effects of irrigation frequency on the concentrations of starch in the leaves and of sucrose and reducing sugars in the fruits have been reported (Silber et al., 2005b). Increased starch and reduced sucrose and hexose concentrations have previously

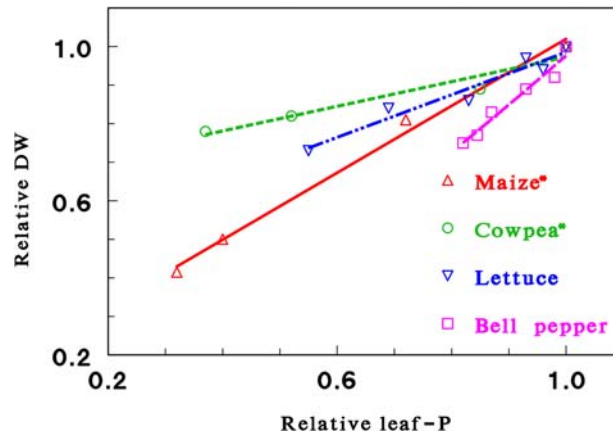


FIGURE 6.16

Relationships between DW production and leaf-P concentrations. The values are relative to the DW and leaf-P concentrations obtained in the best treatment. DW, Dry weight.

Data for maize and cowpea were taken from Mbagwu, J.S.C., Osuigwe, J.O., 1985. Effects of varying levels and frequencies of irrigation on growth, yield nutrient uptake and water use efficiency of maize and cowpeas on a sandy loam ultisol. *Plant Soil* 84, 181–192, for lettuce from Silber, A., Xu, G., Levkovitch, I., Soriano, S., Bilu, A., Wallach, R., 2003. High fertigation frequency: the effects on uptake of nutrients, water and plant growth. *Plant Soil* 253, 467–477 and for bell pepper from Silber, A., Bruner, M., Kenig, E., Reshef, G., Zohar, H., Posalski, I., et al., 2005b. High fertigation frequency and phosphorus level: effects on summer-grown bell pepper growth and blossom-end rot incidence. *Plant Soil* 270, 135–146.

been found in phosphorus-deficient plants (Craft-Brandner, 1992; Qiu and Israel, 1992; Paul and Stitt, 1993; Plaxton and Carswell, 1999); therefore, the differences in the concentrations of starch in the leaves and of sucrose and reducing sugars in the fruits were attributed to the variations in leaf P (Silber et al., 2005b).

A considerable and important effect of irrigation frequency on BER incidence has been reported (Silber et al., 2005b; Fig. 6.17A). The cause(s) of high BER incidence under low-frequency fertigation is/are unclear, but it is generally accepted that BER incidence may be associated with water stress, for example, soil water deficit, high osmotic pressure, or high salinity (Adams, 2002; Ho et al., 1995; Saure, 2001). BER has also been related to Ca deficiency and, especially, to low Ca transport to the fruits, particularly to the distal fruit tissue (Ho et al., 1993, 1995; Marcelis and Ho, 1999; Ho and White, 2005; Barickman et al., 2014). However, unlike BER incidence, the fruit Ca concentrations were almost unaffected by the fertigation frequency (Silber et al., 2005b). The discrepancy between the Ca concentrations in the fruits and BER incidence was consistent with the general remark of Saure (2001, 2014) that the role of Ca in BER (in tomato fruit) should be reassessed.

Several studies in the last decade reported that features of physiological disorders such as BER included the production of oxygen free radicals, and decreases in antioxidative compounds, enzymatic activities in the fruit tissue (Aktas et al., 2005; Mestre et al., 2012; Turhan et al., 2006a), a massive lipid peroxidation event that is caused by the breakdown of cellular glutathione

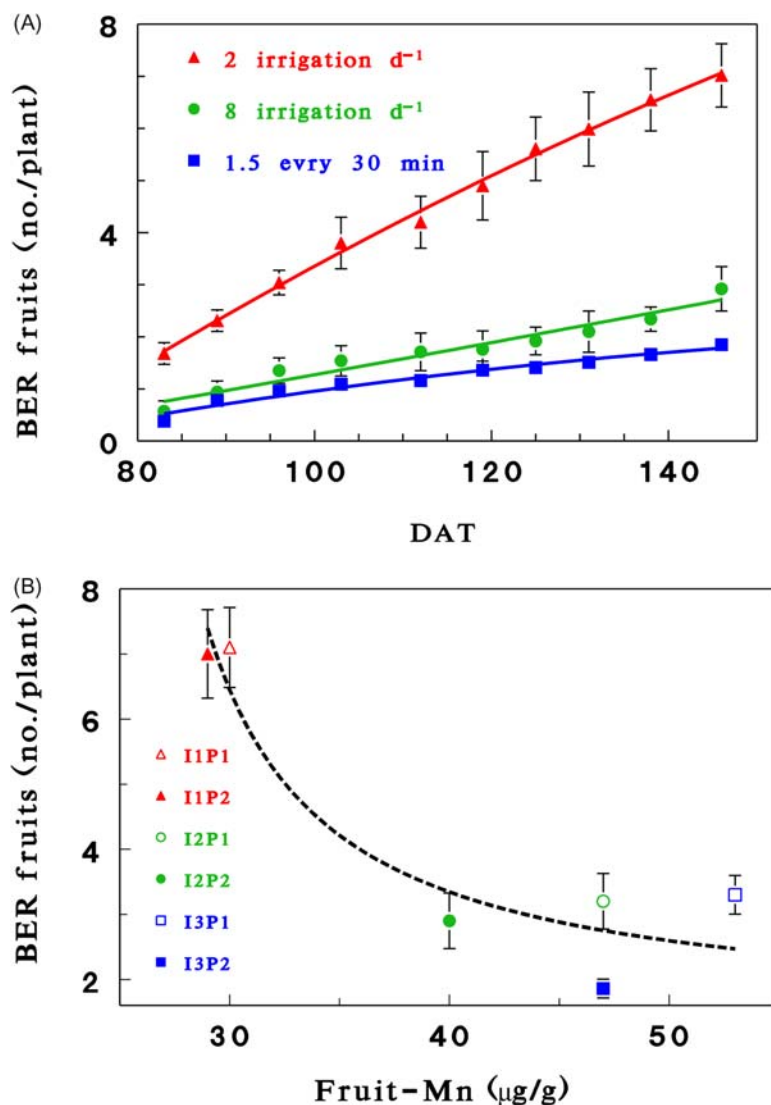


FIGURE 6.17

(A) Effect of irrigation frequency on the cumulative number BER-affected fruits per plant. (B) Relationships between cumulative number of BER-affected fruits and fruit-Mn concentration. Vertical lines indicate the standard error (not shown when smaller than the symbol). BER, Blossom-end rot.

From Silber, A., Bruner, M., Kenig, E., Reshef, G., Zohar, H., Posalski, I., et al., 2005b. High fertigation frequency and phosphorus level: effects on summer-grown bell pepper growth and blossom-end rot incidence. *Plant Soil* 270, 135–146, with kind permission from Springer Science and Business Media.

homeostasis (Mestre et al., 2012), and it is associated with impairment of sugar uptake and metabolism in the fruits (Turhan et al., 2006b). Manganese has been found to play a crucial role in enzyme activities and in detoxification of oxygen free radicals; therefore, it was suggested that BER is related to Mn deficiency (Aktas et al., 2005; Silber et al., 2005b). Interestingly, Silber et al. (2005b) found a negative correlation between the number of fruits with BER and the fruit's Mn concentrations (Fig. 6.17B).

However, further studies did not support this hypothesis (Silber et al., 2009). Instead, it was found that increasing Mn supply enhanced antioxidant activity and reduced oxidative damage to the pepper fruit skin (Silber et al., 2009; Yasuor et al., 2015). In the last decade, several studies showed that Ca deficiency in the fruit tissue is the main cause to BER development (Barickman et al., 2014; De Freitas et al., 2011, 2012, 2014; Mestre et al., 2012) and it was shown that abscisic acid application increased fruit calcium uptake and prevented BER development in tomato fruit (Barickman et al., 2014; De Freitas et al., 2011, 2014), whereas gibberellins consistently reduced fruit Ca uptake and water-soluble apoplastic Ca concentration, leading to leakier plasma membranes and an increase in BER development in fruit tissue of tomato (De Freitas et al., 2012).

6.5 SALINITY EFFECT ON CROP PRODUCTION

6.5.1 GENERAL

High salt concentrations usually harm crops through two main mechanisms (Munns and Tester, 2008). The first is increase of the osmotic potential, both outside the roots where it depresses the external water potential, and in plant organs as a result of salt accumulation (Munns and Tester, 2008). The second mechanism comprises specific ion effects, for example, ion toxicity and/or disturbed mineral nutrition (Läuchli and Epstein, 1990). Note that an increase of osmotic potential in plant organs is not necessarily deleterious and is frequently used to improve fruit quality. Growth reduction caused by salinity is a time-dependent process, and Munns (1993, 2002) proposed a two-phase model to depict the detrimental effects of salinity on plant growth. The first phase includes a rapid (minutes, hours) apparent impairment caused by the increase of external osmotic potential (the first of the above mechanisms) and is basically identical to water stress (Munns, 2002). The second phase is much slower, taking days or weeks to develop, and leads to internal injuries caused by excessive salt accumulation in transpiring leaves.

The majority of crops grown in intensive agriculture are defined as glycophyte plants, which are sensitive to salinity. The yields of such plants diminish as the salinity increases above a threshold value, which is specific for each crop (Maas and Hoffman, 1977; Maas, 1990; Shanon and Grieve, 1999; Sonneveld et al., 1999). The model of Maas and Hoffman (1977) is often used to describe the response of soil-grown crops to salinity; however, pertinent growth conditions such as temperature, humidity, irradiance, the atmospheric CO₂ concentration, and irrigation frequency affect both threshold values and yield reduction gradients (Hoffman and Rawlins, 1971; Feigin et al., 1988; Salim, 1989; Zeroni and Gale, 1989; Pasternak and De Malach, 1995; Sonneveld et al., 2004; Rouphael et al., 2008; Rouphael and Colla, 2008). Nutritional elements added to soil-grown plants via fertilization normally contribute only a minor part of the total ion concentration, whereas in soilless culture, this contribution is relatively more significant: it has been reported that in Dutch

greenhouses, N and K formed one-third of the total ion concentration (Sonneveld et al., 1990). Sonneveld et al. (2004) argued that salinity and nutrition effects would be mixed in the linear model of Maas and Hoffman (1977) and used polynomial and exponential models to describe the effect of the solution EC on substrate-grown crops. The various osmotic effects were induced by addition of varied amounts of nutritional elements (mainly NO_3 and K); therefore, the yield reductions at low EC values reported by Sonneveld et al. (2004) probably resulted from simple nutrient deficiencies rather than real osmotic effects.

The EC in the soil solution is usually higher than that in the irrigating water (Maas and Hoffman, 1977; Sonneveld, 1988), as a result of slow and inefficient leaching processes caused by narrow and tortuous soil pores. In contrast, most of the soilless media contain wide pores that enable fast and efficient leaching (Raviv et al., 2002; Chapter 3: Physical Characteristics of Soilless Media) so that the EC in the medium solution can be controlled. Consequently, the actual EC in the root zone of soilless-grown plants may be lower than that of soil-grown plants under similar irrigation conditions. Salinity may build up in a recirculating system; this will be presented and discussed in Chapter 10, Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses. The salinity of the water is determined by the ion contents in the raw water and in the salts added as fertilizers. As the goal in soilless culture is to maintain the optimal growth conditions for the crops, it is necessary to minimize the contribution of fertilizers to the salinity of the irrigation water and the substrate solution.

Within the relevant range, there is a linear relationship between the salinity and the EC of the solution; therefore, the EC is widely used as a measure of the salinity of the solution. A 1-cmol/L solution has an EC of about 1 dS/m and an osmotic pressure of ~ -0.036 MPa (at 25°C). The general recommendation for source water is that the EC should be below 1.0 dS/m (Schröder and Lieth, 2002), but higher values are common in semiarid and arid zones.

6.5.2 SALINITY—NUTRIENTS RELATIONSHIPS

Salinity may interfere with mineral nutrition acquisition by plants in two ways (Grattan and Grieve, 1992): (1) the total ionic strength of the soil solution, regardless of its composition, can reduce nutrient uptake and translocation and (2) uptake competition with specific ions such as sodium and chloride can reduce nutrient uptake. These interactions may lead to Na-induced Ca and/or K deficiencies (Volkmar et al., 1998) and Cl^- -induced inhibition of NO_3^- uptake (Xu et al., 2000; Massa et al., 2009). Navarro et al. (2002) found that the negative effects of salinity on pepper production and fruit damage by BER were more severe under exposure to chlorides than to sulfates. Application of a large amount of K might increase the plant's capacity for osmotic adjustment in saline habitats (Cerdeira et al., 1995), because K is the most abundant cation in the cytoplasm of glycophytes (Marschner, 1995). Chow et al. (1990) found that spinach (*Spinacia oleracea* L.) required more K for shoot growth under high- than under low-salinity conditions, and Kafkafi (1987) suggested that the salinity tolerance of crops could be improved by the appropriate use of nutrients. There is abundant evidence that Na and the Na/Ca ratio can affect K uptake and accumulation within plant cells and organs (Lahaye and Epstein, 1971; Kent and Läuchli, 1985; Lynch and Läuchli, 1985; Cramer et al., 1985, 1986; Ben-Hayyim et al., 1985, 1987; Chow et al., 1990; Massa et al., 2009), and salt tolerance seems to be correlated with selectivity for K uptake over Na (Läuchli and Stelter, 1982; Wrona and Epstein, 1985), although salinity had no effect on the leaf K

content of two pepper cultivars (Chartzoulakis and Klapaki, 2000). Ben-Hayyim et al. (1987) found that growth of cultured citrus cells in solutions containing various concentrations and proportions of NaCl and CaCl₂ was a function of internal K concentration, irrespective of the NaCl concentration. It has been hypothesized that K application could reduce the deleterious effects of salinity on plant development (Khalil et al., 1967; Kafkafi, 1984; Ben-Hayyim et al., 1987). However, contradictory results were obtained regarding the effects of K fertilization under saline conditions on whole plant and field crops. These results included a reduction in salinity damage to various crops when high concentrations of K were present in the media (Lagerwerff and Eagle, 1962; Helal and Mengel, 1979; Jeschke and Nassery, 1981; Chow et al., 1990; Ben Asher and Pacardo, 1997), no response to K under salinity (Jeschke and Wolf, 1988; Feigin et al., 1991), and even a negative effect of K nutrition on salt tolerance (Lunin and Gallatin, 1965; Lauter et al., 1988; Bar-Tal et al., 1991).

It has been reported that the detrimental effect of salinity was closely related to the oxygen regime in the root zone (Drew and Dikunwin, 1985; Drew and Läuchli, 1985; Drew et al., 1988). The combination of high NaCl level and anoxic conditions accelerated Na⁺ transport to maize shoots while depressing K⁺ transport and, consequently, the shoot Na/K ratio increased significantly. Breakdown of the Na⁺ exclusion mechanism under oxygen deficiency may be responsible for the salinity damage of salt-sensitive plants (Drew and Läuchli, 1985).

Antagonism between Cl⁻ and NO₃⁻ uptake by plants was demonstrated in numerous publications and was reviewed by Xu et al. (2000). This antagonism was found in various plants, including substrate-grown crops such as melon and lettuce (Feigin, 1985), strawberry (*Fragaria × ananassa* Duchesne) (Wang et al., 1989), tomato (Kafkafi et al., 1982; Kafkafi, 1984; Feigin et al., 1987; Heuer and Feigin, 1993), rose (Massa et al., 2009). However, no positive effects were obtained in tomato by adding high levels of NO₃⁻ at high salinities (Heuer and Feigin, 1993). The competition of Cl⁻ with NO₃⁻ was found to be stronger in salt-sensitive than in salt-tolerant plants (Xu et al., 2000) so that Bar et al. (1997) found that when the irrigation water contained NO₃⁻ at about 8–16 mM, even sensitive plants such as avocado (*Persea americana* Mill.) could survive at chloride concentrations of 8–16 mM, and Cordovilla et al. (1995) found that supplying NO₃⁻ considerably relieved the suppressive effects of Cl salinity on faba bean (*Vicia faba* L.). In the light of the above findings, Xu et al. (2000) recommended increasing the NO₃⁻ supply to sensitive crops irrigated with water containing high chloride concentrations.

Much information is available on the effect of saline water irrigation on rose (*Rosa × hybrida*) production (Yaron et al., 1969; Hughes and Hanan, 1978; Fernandez Falcon et al., 1986; De Kreij and Van Den Berg, 1990; Urban et al., 1994; Baas et al., 1995; Raviv and Blom, 2001). The range of sodium and chloride concentrations that allow optimal production of roses is well defined (Bernstein, 1964; Yaron et al., 1969; Hughes and Hanan, 1978; Fernandez Falcon et al., 1986; De Kreij and van den Berg, 1990), and damage to the plants is known to occur under exposure to Cl or Na at as little as 4 mmol/L (Hughes and Hanan, 1978). Irrigation with saline water increased the concentration of chloride in rose tissue (Yaron et al., 1969; Bernstein et al., 1972; Hughes and Hanan, 1978; Baas and van den Berg, 1999; Cabrera and Perdomo, 2003; Bernstein et al., 2006). The reported threshold leaf chloride concentration for rose plant damage is 4.5 g/kg (Bernstein et al., 1972; Feigin et al., 1988). Several studies found that the sodium content in rose leaves was unaffected by salinity (Sadasivaiah and Holley, 1973; Cabrera and Perdomo, 2003; Bernstein et al., 2006). However, Na exclusion is not general to all rootstocks; other reports described the

stimulation of Na accumulation under salinity (Fernandez Falcon et al., 1986; Baas and van den Berg, 1999; Massa et al., 2008).

Ben Asher and Pacardo (1997) proposed a conceptual model in which enhanced fertilization with KNO_3 under saline conditions might reduce the toxic effect of salinity, even if KNO_3 increased the soil osmotic potential. At two soil salinity levels (2 and 8 dS/m) application of KNO_3 was associated with a significant increase in the DW of the shoot of groundnut (*Arachis hypogaea* L.) by up to 150–300 kg KNO_3 /ha. Both the growth and the development of tomato plants irrigated with a saline solution of EC 5.5 dS/m and containing 50 mM NaCl were improved by adding 2 mM KNO_3 to the saline solution, and to a lesser extent by adding 20 mM $\text{Ca}(\text{NO}_3)_2$, at EC of 7.5 dS/m (Satti et al., 1994). However, an external K supply was not required for root growth of castor bean (*Ricinus communis* L.) under saline conditions (Jeschke and Wolf, 1988). Tip-burn symptoms in Chinese cabbage (*Brassica rapa*), induced by NaCl plus CaCl_2 salinity, were not alleviated by the addition of KNO_3 (Feigin et al., 1991).

The uptake interaction between chloride and phosphorus appears to be complex; there is no direct competition, because of the great differences in their physical and physiological properties (Xu et al., 2000). However, in several studies, phosphate uptake was stimulated when chloride concentration was low and suppressed when it was high, whereas in other studies, no effect was obtained (Xu et al., 2000). Only a few studies found that addition of adequate P could also help to alleviate salt stress (Champagnol, 1979; Awad et al., 1990) and that it increased the yields of fox-tail millet (*Setaria italica* L.) and clover (*Trifolium alexandrinum* Fahli) grown in a saline soil (Ravikovich and Yoles, 1971).

Calcium plays an important role in the integrity, selectivity, and permeability of plant membranes (Hanson, 1984; Grattan and Grieve, 1999; Hirschi, 2004). It has been shown that high salinity and high sodium concentration displace calcium from membrane-binding sites and so cause dysfunctioning and alteration of the membrane permeability and selectivity (Cramer et al., 1985). Several studies with various plant species demonstrated that calcium could ameliorate the adverse effects of sodium on plant membranes, roots, and whole plants (Lahaye and Epstein, 1971; Kent and Läuchli, 1985; Cramer et al., 1986). Carvajal et al. (2000) reported that high-NaCl solution (50 mM) decreased the passage of water through detached roots of melon by reducing the activity of water channels (aquaporins), but pretreatment with addition of 10 mM CaCl_2 solution prevented the decrease in root hydraulic conductance. A similar effect was observed when the flux of Ca^{2+} into the xylem and the Ca^{2+} concentration in the plasma membrane of the root cells were determined. The ameliorative effect of Ca^{2+} on NaCl stress could be related to the functioning of the water-channel. The negative effects of salinity (50 mM NaCl) on hydroponically grown pepper plants with respect to tissue calcium concentration and water relations were mitigated if there was a supply of Ca^{2+} , and this effect increased when the root-zone temperature was increased (Cabanero et al., 2004). Supplementary Ca partially ameliorated the negative effects of moderate salinity (35 mM NaCl) on plant growth and fruit yield of strawberry (Kaya et al., 2002), and a similar effect was obtained with tomato (Navarro et al., 2000).

Several reviewers concluded that the results reported in the literature on the interaction between salinity and nutrients were contradictory (Kafkafi, 1984; Feigin, 1985). However, part of this conflict can be removed by using the definitions of Bernstein et al. (1974), who described three different types of idealized salinity/nutrition interactions: (1) increased salt tolerance at suboptimal nutritional levels; (2) independent effects of salinity and nutrition at optimal and suboptimal nutritional

levels; and (3) decreased salt tolerance at suboptimal nutritional levels. These definitions (Bernstein et al., 1974) were used for defining the interaction on the basis of plant performance at optimal fertilization relative to that at suboptimal fertilization (Maas, 1990; Grattan and Grieve, 1992) and depicted in Fig. 6.18. This approach can be used to analyze data from a variety of publications and to find out what type of interaction(s) exist(s). Some of the conclusions on salinity/nutrition interaction published in the early literature were based on a misleading interpretation. A schematic illustration of the interactions between salinity and fertility levels is presented in Fig. 6.19. Under low salinity, nutrient-deficient control plant growth was accompanied by an increased salt tolerance function (left part of Fig. 6.19). A misleading interpretation in such case is that fertilization relieved the salinity stress. As the salinity level increased, plant stress imposed by salinity equaled that imposed by nutrient deficiency (unaffected salt tolerance function), but as the salinity level increased further, salinity was the main yield-limiting factor, and a decreasing salt-tolerance function was obtained (right part of Fig. 6.19).

Shalhevet (1994) concluded in a review on the use of water of marginal quality for crop production that fertilizer application does not materially change the salinity response function of crops. In some cases, a better response to fertilizer was found under nonsaline than under saline conditions, but only rarely could a relative advantage to fertilizer application under saline conditions be found. Grattan and Grieve (1999) concluded that “Despite a large number of studies that demonstrate that salinity reduces nutrient uptake and accumulation or affects nutrient partitioning within the plant, little evidence exists that adding nutrients at levels above what is considered optimal in nonsaline environments, improves crop yield.”

6.5.3 YIELD QUALITY INDUCED BY SALINITY

The effects of salinity on crop quality depend mainly on the plant part that is marketed. Fruit quality is generally more tolerant to high salinity than fresh leaf yield (Xu et al., 2000). Salinity improves both the taste and visual quality of tomato and melon fruits (Mizrahi, 1982; Mizrahi and Pasternak, 1985; Sonneveld and Welles, 1988; Adams, 1991; Li et al., 2000). The increase in quality under saline conditions was attributed to the significantly higher contents of total soluble solids, sugars, and aromatic constituents (Ehret and Ho, 1986; Adams, 1991; Sonneveld and van den Burg, 1991; Li et al., 2000; Mulholland et al., 2002). Mulholland et al. (2002) found, in a split root experiment, that tomato fruit quality could be enhanced by high EC in half of the root system. In contrast, salinity reduced product quality and yield of leafy vegetables such as lettuce and cabbage (*Brassica oleracea* L.), because of tip burn (Feigin et al., 1991); under high salinity the leaves accumulated chloride, whereas both soluble sugars and vitamin C contents were significantly lower than under nonsaline conditions (Xu et al., 2000).

The negative effects of salinity are often related to Ca deficiency. Salinity increases the occurrence of BER in tomato (Adams, 1991; Adams and Ho, 1992, 1993; Ho and White, 2005) and pepper (Sonneveld, 1988; Tadesse et al., 1999; Navarro et al., 2002), internal rot in eggplant (Savvas and Lenz, 1996), tipburn in lettuce and cabbage, and blackheart in celery (*Apium graveolens* L.) (Feigin et al., 1991; Sonneveld, 1988). Adams and Ho (1993) found that increasing the EC of nutrient film technique solution from 3 to 17 dS/m by adding NaCl had no effect on the occurrence of BER in tomato fruits whereas achieving the same EC increase by adding the major nutrients resulted in almost 100% BER. Flower production is generally more sensitive to salinity than

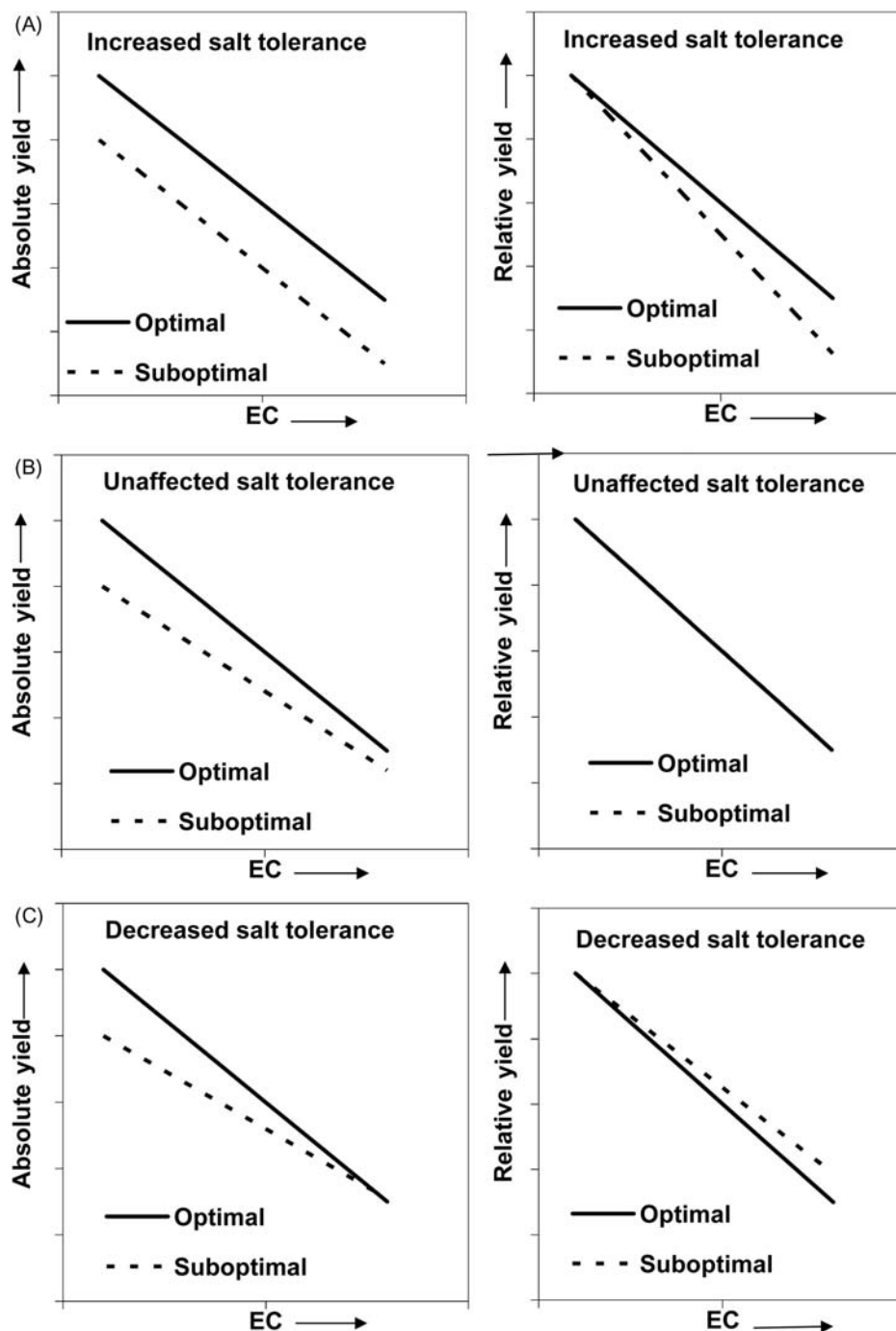


FIGURE 6.18

Types of idealized interactions between salinity and nutrients level, in their effects on absolute and relative yields. (A) increased salt tolerance under deficient nutritional levels; (B) independent effects of salinity and nutrition at optimal and deficient nutritional levels; and (C) decreased salt tolerance under deficient nutritional levels.

Based on Bernstein, L., Francois, L.E., Clark, R.A., 1974. Interactive effects of salinity and fertility on yields of grains and vegetables. *Agron. J.* 66, 412–421.

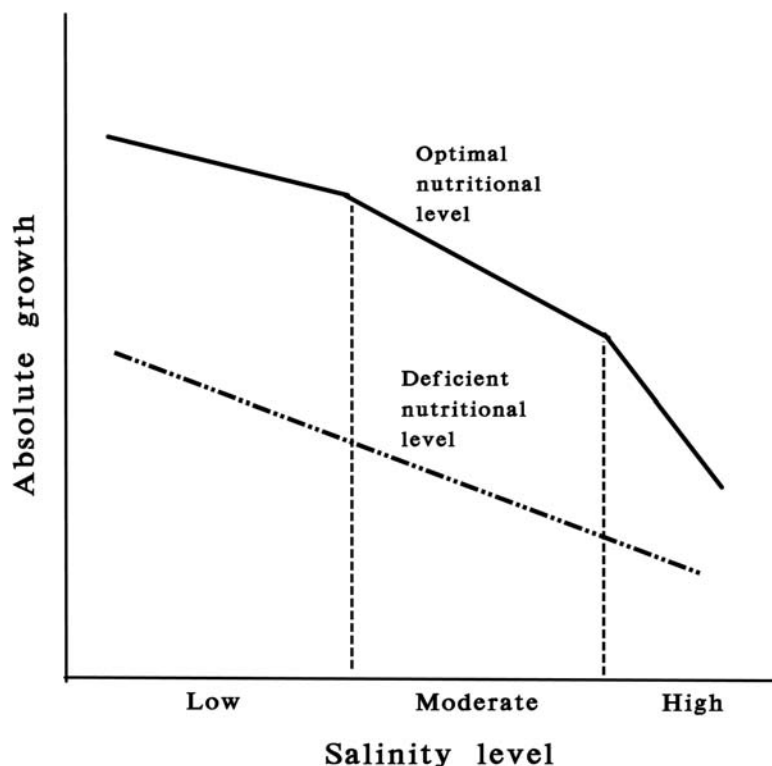


FIGURE 6.19

Effect of salinity levels on absolute yield under optimal (*solid line*) and deficient (*broken line*) nutrition levels.

Based on Bernstein, L., Francois, L.E., Clark, R.A., 1974. Interactive effects of salinity and fertility on yields of grains and vegetables. *Agron. J.* 66, 412–421; Grattan, S.R., Grieve, C.M., 1992. Mineral element acquisition and growth response of plants grown in saline environments. *Agric. Ecosyst. Environ.* 38, 275–300.

vegetable yield. For example, the recommended optimal EC values for anthurium (*Anthurium*) are 0.7 and 1.0 dS/m in the applied solution and the drainage water, respectively (Sonneveld and Voogt, 1993). The yield of various cut flower crops decreases as salinity increases above a specific threshold (Sonneveld, 1988; Sonneveld et al., 1999; Adams, 2002). Baas et al. (1995) found that salinity reduced the yield and quality of carnation (*Dianthus caryophyllus* L.) and gerbera flowers. They reported that a similar yield decrease occurred in a high-EC treatment which used increased concentrations of major elements and concluded that the yield decrease was not related to specific ion-toxic effects of sodium and chloride. Pre- and postharvest application of calcium improved the quality and vase-life of cut flowers such as gerbera (Gerasopoulos and Chebli, 1999) and rose (Halevy and Mayak, 1981; Michalczuk et al., 1989; Starkey and Pedersen, 1997; Torre et al., 1999, 2001; Bar-Tal et al., 2001c). Maintaining low Ca/K and Ca/Na ratios in the solution can have considerable negative impacts on fruit and flower quality in soilless culture under saline conditions.

6.6 COMPOSITION OF NUTRIENT SOLUTION

The source of fertilizer to be used in fertigation has to be chosen very carefully. Solubility and compatibility with the specific substrate and irrigation water characteristics must be considered. Loss of nutrients such as P, K, or NH_4^+ , or of micronutrients, through adsorption on strongly adsorbing substrates, and/or through precipitation processes detailed elsewhere (Chapter 4: Chemical Characteristics of Soilless Media), may lead to deficiency levels in the rhizosphere. Thus, nutrient deficiency is a common occurrence even if the nutrient concentration in the irrigation water is adequate. The quality of the irrigation water plays an important part in determining which kind of fertilizers should be used. An improper choice may cause difficulties by clogging the irrigation system or by corrosion, or it may lead to losses of nutrient elements. Generally, the fertilizers used should have complete solubility, quick dissolution, and high nutrient content and should be free of any toxic constituents. Methods for preparing nutrient solutions for various crops, based on target pH, EC, and nutrient concentration, ratios have been methodically explored and widely reviewed in the literature (Steiner, 1961, 1984; Graves, 1983; Sonneveld and Straver, 1994; Schwartz, 1995; Resh, 1998; De Kreij et al., 1999; Savvas and Adamidis, 1999; Savvas, 2001; Adams, 2002; Sonneveld, 2002; Jones, 2005) and, therefore, are discussed only very briefly in the present chapter.

6.6.1 pH MANIPULATION

The rhizosphere pH plays a key role in the determination of nutrient availability to the plant; therefore, pH management, which can lead to optimization of the nutritional regime, attracts a lot of interest in both soil-based and soilless culture research. However, manipulation of rhizosphere pH is more feasible in soilless media, as these are characterized by low buffer capacity and subjected to frequent applications of nutrient solutions. The optimal pH values recommended in soilless culture are usually within the range of 5.0–6.0 (Adams, 2002; Sonneveld, 2002) but, nevertheless, pH values between 6.0 and 7.0 are adequate for the majority of crops (Graves, 1983).

High pH values (> 7.0) in the irrigation water are undesirable, because Ca and Mg carbonates and orthophosphates may precipitate in the irrigation lines and drippers. In addition, at pH above 7.2, the proportion of H_2PO_4^- species decreases and that of HPO_4^{2-} increases (Lindsay, 1979). Since the H_2PO_4^- ion is much more available to plants than HPO_4^{2-} (Hendrix, 1967), high pH may induce P-deficient conditions, even if there is adequate total P in the solution, and high substrate pH may reduce micronutrients availability to plants, because of precipitation reactions (Lindsay, 1979). However, it is possible to ensure an acceptable rhizosphere pH range by fertilizer management, with higher concentrations of micronutrients compensating for lower solubility at high pH (Smith et al., 2004a,b). To avoid precipitation at $\text{pH} > 5$, and to facilitate sufficient transport toward roots in soil, micronutrients are added in solutions as chelates of organic ligands. Chelating agents such as ethylenediamine tetraacetic acid, diethylenetriamine pentaacetic acid, or ethylenediamine-di-(*o*-hydroxyphenylacetic acid) may be used to maintain micronutrient availability for plants. The synthetic chelates lose their capacity to retain microelements at $\text{pH} < 3$ and, therefore, it is advisable to store chelated microelements separately from acid fertilizers, at $\text{pH} > 5$, and to inject them

into the water after the acid stock solution. In general, it is not recommended to use any component, such as ammonia, that might raise the solution pH when injected into the irrigation water. Compounds that may reduce the irrigation water pH are nitric (HNO_3) and orthophosphoric (H_3PO_4) acids. Depending on their price and the raw water composition, these acids may be used to lower the irrigation water pH. The acid dose required to attain a target pH depends primarily on the HCO_3^- concentration in the tap water. However, at HCO_3^- concentrations above 2–3 mM, it is not possible to use only H_3PO_4 to adjust the pH to 5.0–5.5 without adding excessive amounts of phosphate to the solution. For instance, if the raw water contains 3 mM HCO_3^- , about 2–2.5 mmol of H_3PO_4 should be added per liter of nutrient solution, in order to neutralize as much HCO_3^- as is necessary to achieve the target pH. However, the H_2PO_4^- concentration in the nutrient solution should normally not exceed values above 1.0 mM. The use of nitric acid is not limited by such considerations, since the target NO_3^- concentration in the nutrient solutions is normally >10 mM, and the HCO_3^- concentrations never reach this level. It is important to note that the response of the medium pH to any chemical treatment should be expected to vary according to the components of the medium, the chemical properties of the tap water and plant interactions (Bishko et al., 2003). Acidic pHs (below 5.0) may be detrimental to root membranes and may increase the Al and Mn concentrations in the substrate solution to toxic levels. Soilless media based on un-amended sphagnum peat and pine bark are characterized by low pH, and liming materials, that is, $\text{Ca}(\text{OH})_2$ and $\text{Mg}(\text{OH})_2$, or carbonates, that is, CaCO_3 and MgCO_3 , must be added to raise the pH (Argo and Biernbaum, 1996, 1997a,b; Elliott, 1996; Bishko et al., 2002).

The approach presented by Steiner (1961, 1984), Sonneveld and Straver (1994), De Krijg et al. (1999), and Savvas and Adamidis (1999), who focused basically on attempting to attain an optimal pH in the irrigation water (target pH), is generally accepted (Graves, 1983; Schwartz, 1995; Resh, 1998; Savvas, 2001; Adams, 2002; Sonneveld, 2002; Jones, 2005). However, there is no doubt that the target pH of the irrigation water is only one part of the many aspects of nutrient availability in real circumstances, that is, in the presence of a microbial population and plant activity. Even in the absence of plants, the H^+ contributed from the nitrification process should not be ignored. Two moles of H^+ is released during nitrification of 1 mol of NH_4^+ [Eqs. (6.i) and (6.ii)]. The quantity of H^+ released to the solution via nitrification per liter of substrate is 2×10^{-4} mol, under the following conditions: irrigation- NH_4^+ concentration, 1 mmol/L (10% of the NH_4^+ that corresponds to a total N concentration of 10 mmol/L); 10 L of substrate per plant; and irrigation dose of 1 L/day per plant. Thus, according to this simple calculation, there could be more H^+ added via nitrification than via acidification, and this should be considered. Furthermore, the chemical properties of the root environment are notably different from those of the bulk medium, as a result of changes induced by the plants, as detailed elsewhere in this book (Chapter 4: Chemical Characteristics of Soilless Media, section dealing with plant-induced changes of the chemical properties of solid surfaces). Roots strongly affect their environment to a range of 1–2 mm, as depicted in Figs. 6.5 and 6.17, by secreting H^+ and citric acid. Published data on root length (Bar-Yosef, 1999; Xu et al., 2004; Silber et al., 2005b) indicate that the volume of substrate that is strongly affected by root activity amounts to only 10%–15% of the total volume. Thus, the concentrations of H^+ , OH^- , and organic components excreted from roots into this limited volume may exceed those in the bulk substrate by an order of magnitude. The straightforward conclusion to be drawn from these remarks is that the real challenge lies in managing nutrient availability and pH in the vicinity of the roots (rhizosphere) rather than in the irrigation solution.

6.6.2 SALINITY CONTROL

According to uptake data presented in the previous section, and assuming a daily irrigation of 5 mm (50 m³/ha), N and K concentrations in the irrigation water at the time of maximum consumption rate may reach 15–20 mmol/L, which correspond to an EC of 1.5–2.0 dS/m. Under such conditions, especially when the water EC is above 1.0 dS/m, care should be taken to minimize the amount of accompanying ions added with the applied nutrients. For example, KCl, which is a cheap source of K, should be replaced with KNO₃ and K₂HPO₄, and NH₄NO₃ and urea should be preferred over (NH₄)₂SO₄. Under high chloride concentrations, K₂SO₄ should be the preferred source of K, because of its lower salt index and absence of Cl (Zehler et al., 1981; von Braunschweig, 1986). Sodium-based fertilizers (e.g., NaNO₃ or NaH₂PO₄) are not advisable sources of N and P, because of the adverse effects of Na on plant functioning.

In most cases, the source of N will be NH₄NO₃, (NH₄)₂SO₄ and multinutrient fertilizers such as KNO₃, Ca(NO₃)₂, and (NH₄)₃PO₄. The multinutrient fertilizers should be preferred when the salt content of the water is high. Potassium application in irrigation water is relatively problem free, thanks to the high solubility of most K salts; KCl is more soluble than K₂SO₄, and in high-Ca water there is a possibility of CaSO₄ precipitation. On the other hand, when the Cl concentration is high, KCl is not to be recommended for sensitive crops, but KNO₃ is a very efficient source of both K and N.

6.7 EFFECTS OF SILICON AMENDMENT ON SOILLESS-GROWN PLANTS

The most debatable element in the plant physiological literature over decades is probably silicon (Si) (Epstein, 2005; Jana and Jeong, 2014). Although Si content in soil-grown plants is high above that of P and in the same order of magnitude of other macronutrients such as N and K, Si is still not considered as an essential element. Almost all the plants can grow without added Si, and the apparent lack of interaction of Si with plant metabolism reinforces the debated hypothesis that Si is not an essential element (Fauteaux et al., 2006). Actually, “minus-Si plants do not exist in nature or agriculture” (Epstein and Bloom, 2005) and therefore, Si should be added to non- or low-Si media through the fertilization system in order to avoid from experimental artifacts as suggested by Epstein (2005).

Si availability for proper development of soils-grown plants is probably unlimited, especially in abundant clay minerals soils. Nevertheless, in soilless-grown plants, especially in limited-volume media, Si deficiency may restrict plant development. Although many substrates such as stonewool contain considerable amount of Si (47% of SiO₂), the limited root volume and small quantities of solid phases available for plant may limit Si uptake by plants and, consequently, restrict plant development (Voogt and Sonneveld, 2001; Muir et al., 2011). Voogt and Sonneveld (2001) reported that Si concentration in young leaves of several horticultural plants grown in soilless media was significantly lower in comparison with soil-grown plants and recommended for further researches on Si–plant interactions. The range of Si concentrations in the leaves of soilless-grown plants varied between 0.1% and 5% on a DW basis (Voogt and Sonneveld, 2001). Jana and Jeong (2014) stated that silicon is the most underappreciated element in horticultural crops. They reviewed many horticultural research results which focused on the beneficial effects of Si supplementation,

especially to soilless-grown plants. Mattson and Leatherwood (2010) reported that the development of several soilless-grown floricultural species was improved by addition of exogenous Si application.

Si in soil is present in the uncharged monosilicic acid (H_4SiO_4) which is the main form adsorbed by plants (Iler, 1979; Epstein, 1994; Guntzer et al., 2012). Following Jones and Handreck (1967), the passive uptake of Si by roots through water mass flow was well accepted in the past. Positive correlations between transpiration and Si uptake which may support the passive uptake approach were reported by Mayland et al. (1991) for wheat (*Triticum aestivum*), Voogt and Sonneveld (2001) for cucumber (*Cucumis sativus*) and Kamenidou et al. (2011) for sunflower (*Helianthus annuus* L.). However, Barber and Shone (1966) firstly stated that “the entry of silica into plants appears to require the expenditure of metabolic energy since the process is sensitive to both metabolic inhibitors and variations in temperature.” Later on Raven (2001) reported that Si adsorption by rice (*Oryza sativa*) required a metabolic control and Henriët et al. (2006) described dual mechanisms for Si uptake by banana (*Musa* spp.) plants. Si uptake by hydroponically grown banana at low solution-Si concentrations (<0.8 mM) were above the calculated uptake by mass flow, indicating an active Si transport. However, at higher solution-Si concentrations (1.66 mM), Si uptake was driven by passive transport of water mass flow and the distribution of Si in plant organs indicated the importance of Si transportation through the transpiration channel (Henriët et al., 2006). Thus, active and passive mechanisms may coexist (Liang et al., 2007; Guntzer et al., 2012). Rice and maize are Si accumulator plants and worldwide important crops, and therefore, mechanisms of Si uptake by roots have been studied on these plants. The first Si transporters were isolated in rice by Ma et al. (2004, 2006, 2007) and in maize by Mitani et al. (2009). Later on Si transporters were isolated in pumpkin (*Cucurbita pepo*) by Mitani-Ueno et al. (2011), and soybean (*Glycine max*) by Deshmukh et al. (2013).

The scientific community generally accept the statement of Epstein (2005) that “The significance of silicon is most clearly apparent in plants that are under some environmental stress, whether biotic or abiotic.” Although the exact mechanisms associated with the beneficial effects of Si application are still poor understood, Savvas and Ntatsi (2015) proposed four possible mechanisms: (1) Si accumulation in plant tissues enhances the mechanical strength improves nutrient and water movement; (2) modifications of oxidative stress and enhances activity of antioxidant; (3) complexation or coprecipitation of toxic metals with Si; and (4) modulation of gene expression and signaling through phytohormones. Among the abiotic stress, salinity has, in the last few decades, become a worldwide important problem following the reduction in both, quantity and quality of water resources. Si application through the fertilization system alleviated salinity damage in various soilless-grown plants such as tomato, zucchini (*C. pepo* L.), cucumber, lettuce, rose, and gerbera. Yet, the recovery of the salinity-stress plants by Si application was only partial and the yield remained lower in comparison with no salinity-stress plants. The beneficial effects of exogenous Si application to salinity-stress plants were generally associated with two main mechanisms: (1) reduced translocation of Na and Cl from the roots to the vegetative organs (Stamatakis et al., 2003; Savvas et al., 2007, 2009; Li et al., 2015); and (2) modifications of oxidative stress and enhances activity of antioxidant (Zhu et al., 2004; Savvas et al., 2009; Milne et al., 2012; Li et al., 2015). In addition, water status in leaves, stomatal conductance, and photosynthesis and shoot growth were improved following the decreases of Na and Cl accumulation. Extreme temperatures are an additional abundant abiotic stress, and it has been reported that Si may be used to enhance temperature

tolerance (Sivanesan et al., 2014). Generally, Si application through the fertilization system was found to be more effective than foliar application (Savvas and Ntatsi, 2015), and high foliar sprays (150 mg Si/L of NaSiO₃) even caused stem shortening and flower deformation of gerbera flower (Kamenidou et al., 2010). Exogenous Si application affected indirectly leaf-macro and -micro concentration, the effects were inconsistent and varied with the Si source and plant species (Kamenidou et al., 2010, p. 11).

In addition to the abiotic stress described above, Si application may play an important role in disease suppression (Guntzer et al., 2012; Jana and Jeong, 2014). Si supplied through the fertilization system reduced the incidence of powdery mildew (*Podosphaera xanthii*) in zucchini (Savvas et al., 2009), powdery mildew (*P. xanthii*, formerly *Sphaerotheca fuliginea* Schlech ex Fr. Poll. and *Golovinomyces cichoracearum*, formerly *Erysiphe cichoracearum* DC ex Merat) in melon (Buttaro et al., 2009), powdery mildew (*Oidium neolycopersici*) in tomato (Garibaldi et al., 2011), downy mildew (*Bremia lactucae*) in lettuce (Garibaldi et al., 2012), *Alternaria japonica* in rocket (*Eruca vesicaria*) (Cogliati et al., 2013), Fusarium wilt (*Fusarium oxysporum* f. sp. *lactucae*) in lettuce (Chitarra et al., 2013), Fusarium wilt (*F. oxysporum* f. sp. *Lycopersici*) in tomato (Chitarra et al., 2014). Although the beneficial effects of Si supplementation on plant pathogens, especially fungus is well documented, the precise mechanisms involved are poorly understood (Guntzer et al., 2012). In the past, the beneficial effect of Si was attributed to mechanical barrier induced by amorphous Si precipitation (Jones and Handreck, 1967). Later on it was suggested that Si enhances the expression of natural defense mechanisms, leading to induced resistance against pathogen fungi (Fauteux et al., 2005). Complete transcriptome analysis of control and powdery mildew-stressed *Arabidopsis* plants, with or without Si application indicated that only 2 out of ~40,000 transcripts analyzed were affected by Si addition to control plants (Fauteux et al., 2006). Subsequently, it was concluded that the beneficial effect of Si addition on plant development is probably indirect and mostly associated with stress alleviation (Fauteux et al., 2006).

6.8 OXYGEN AVAILABILITY IN THE RHIZOSPHERE OF SOILLESS-GROWN PLANTS

Oxygen (O₂) depletion to suboptimal concentration (hypoxia) may inhibit root respiration or even injure root tissues in case of complete depletion of dissolved O₂ (anoxia). Soil-grown plants are prone to O₂ as a result of waterlogging and/or poor aeration conditions in the root environment (Morard and Silvestre, 1996). Likewise, the restricted root volume of soilless-grown plants, as well as the low buffer capacity, high root respiration, and high microbial activity, may stimulate O₂ depletion (Marfà et al., 2005). Dissolved O₂ concentration (DOC) in pure water is significantly affected by temperature fluctuation and may be calculated from the partial O₂ concentration in the gaseous phase by using Henry's law, that is, DOC is proportional to the partial pressure of oxygen in the air as depicted in Fig. 6.20. Temperature increases in the solution phase accelerated O₂ consumption resulting from higher respiration rate of root and microbial population (Morard and Silvestre, 1996; Atkin et al., 2000; Yuste et al., 2007). Thus, hot periods (summer) are more susceptible for hypoxia or even anoxia events. Following the simulation analyses of Naasz et al. (2008a), the fraction of O₂ consumed by microbial respiration may be more than half from the total

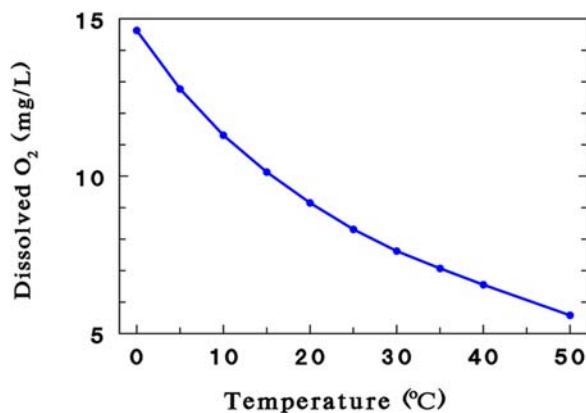


FIGURE 6.20

The relationship between DOC and the temperature. *DOC*, Dissolved O₂ concentration.

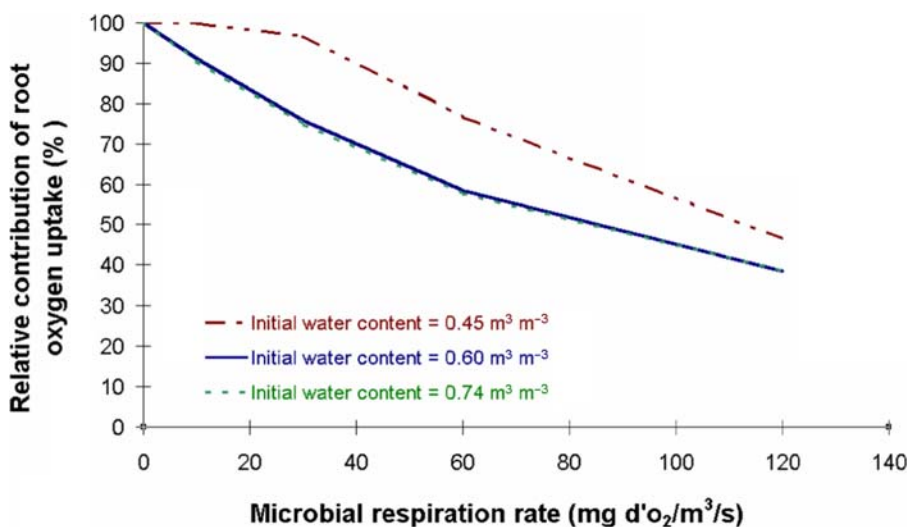


FIGURE 6.21

Evolution of oxygen availability and oxygen consumption in a peat substrate (θ initial = 0.60 m³/m³) during 1 day of simulation.

From Naasz, R., Michel, J.-C. and Charpentier, S. (2008). Microbial Respiration and Its Consequences on Oxygen Availability In Peat Substrate. *Acta Hortic.* 779, 91–96, <https://doi.org/10.17660/ActaHortic.2008.779.9>.

O₂ content in a peat substrate (Fig. 6.21) and might induces conditions of O₂ stress (Fig. 6.22). By definition, both high water and O₂ content is the ultimate demand from any substrate used in the horticulture industry, but unfortunately, this wishful demand cannot coexist and it is not possible to simultaneity maintain high water and oxygen availability (Chapter 3: Physical Characteristics of

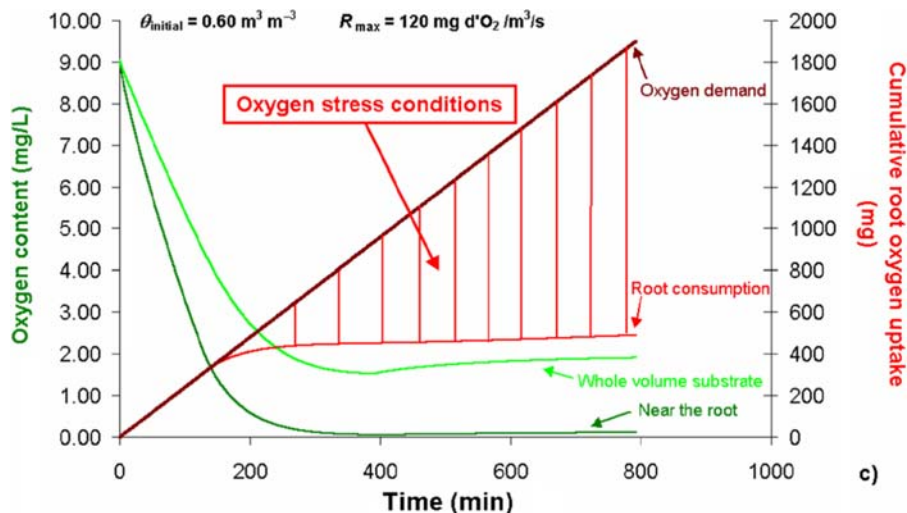


FIGURE 6.22

Evolution of oxygen content and oxygen consumption in peat (θ initial = $0.60 \text{ m}^3/\text{m}^3$) during 1 day of simulation and microbial respiration rate of $120 \text{ mg of O}_2/\text{m}^3/\text{s}$.

From Naasz, R., Michel, J.-C. and Charpentier, S. (2008). Microbial Respiration and Its Consequences on Oxygen Availability in Peat Substrate. *Acta Hortic.* 779, 91–96, <https://doi.org/10.17660/ActaHortic.2008.779.9>.

Soilless Media). Daily-water and $-\text{O}_2$ content in porous substrate notably fluctuate in accord with the meteorological conditions, irrigation events, and water and/or O_2 uptake by the plant. Consequently, low unsaturated hydraulic conductivity followed by low water content or, alternatively, low DOC resulted from low diffusivity rate of O_2 may induced water- and O_2 -stress during different part of a single day (Fig. 6.23). It is essential therefore to establish a compromise management to ensure adequate water content and reasonable unsaturated hydraulic conductivity on one hand and sufficient DOC on the other hand (Naasz et al., 2008a,b).

Overcoming O_2 deficiency under conditions of hot temperature and high rate of root-respiration is possible by using the oxyfertilization method based on supplying dissolved oxygen in the irrigation water at oversaturated concentrations using fertigation equipment (Bonachela et al., 2005; Carazo et al., 2005; Marfà et al., 2005). Oxyfertilization treatments significantly increased O_2 concentrations in the gaseous phase (inside the substrate pore) and slightly elevated DOC (Bonachela et al., 2005; Marfà et al., 2005). Yet the effect on DOC was not significant and the DOC levels in the control were higher than the threshold level of 3 mg/L designed by Morard (1995). This proposed threshold is in accord with different reports that DOC levels above $2.5\text{--}3 \text{ mg/L}$ were adequate for proper root activity and accordingly, supported proper yield (Bonachela et al., 2005, 2010; Acuña et al., 2008; Ehret et al., 2010; Bar-Yosef and Lieth, 2013). In contrast, O_2 enrichment above this threshold induced higher yield of pepper and melon grown in perlite bags (Marfà et al., 2005; Urrestarazu and Mazuela, 2005), and consequently, additional parameters such as a specific sensitivity of plant species to DOC level and/or growth conditions may affect the optimal DOC threshold.

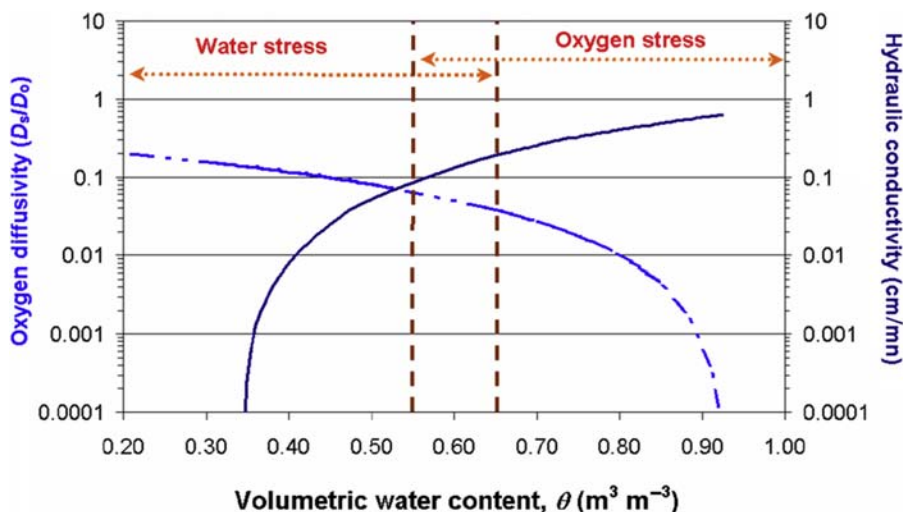


FIGURE 6.23

The effect of water content on O_2 diffusivity and water stress in a peat substrate during one day of simulation.

From Naasz, R., Michel, J.-C. and Charpentier, S. (2008). *Microbial Respiration and Its Consequences on Oxygen Availability in Peat Substrate*. *Acta Hortic.* 779, 91–96, <https://doi.org/10.17660/ActaHortic.2008.779.9>.

Several studies clearly indicated that the uptake of nutrients and water by plant roots exposed to low DOC levels significantly decreased, mainly because of low energy pool in the root cell consequent of low generation of Adenosine triphosphate (Drew, 1988; Morard and Silvestre, 1996; Morard et al., 2000; Armstrong and Drew, 2002). Morard and Silvestre (1996) stated that the depressive effects of O_2 depletion followed the order: $K > N > P > H_2O > Mg-Ca$. Yet, this schematic order is probably affected by several experimental factors such as plant species, rate of O_2 depletion, time length of O_2 starvation, and thus, different experimental conditions may yield different order. Tomato plants grown in nutrient solution without aeration rapidly induced a sharp decrease of the DOC level (Morard et al., 2000). Six hours after starting the experiment, the DOC was $<2\%$ (hypoxia condition), and after additional six hours, further O_2 consumption totally depleted O_2 and induced anoxia condition. Uptake of K by tomato plant in this experiment almost stopped after 10 hours as DOC level was below 1%, while that of Ca and Mg stopped after 12 hours (anoxia condition), and that of P and SO_4 after 16 hours. Surprisingly, O_2 depletion almost did not affect NO_3^- uptake, while NO_2^- in the nutrient solution was detected 12 hours after starting the experiment and accumulated until the end of this experiment (Morard et al., 2000). In accord with the above, hypoxia condition (DOC of 1.5–2 mg/L) did not affect leaf-N concentration of rose flower grown in nutrient solution during winter, spring, and summer (Bar-Yosef and Lieth, 2013). But conflicting with the data of Morard et al. (2000), hypoxia did not significantly affect leaf-K and -S concentrations during 4 months of experiment, even during the summer period as DOC dropped below 1.5 mg/L. Leaf-P, -Ca, -Zn, and -Mn in plant exposed to hypoxia at the end of the experiment significantly reduced by 25, 8, 36, 37, and 68%, respectively, from the control plants (Bar-Yosef and Lieth, 2013). Diurnal oxyfertiligation of rose plant grown in perlite container

increased leaf-K and -Ca concentrations but not that of -N, -P, -Mg and -Fe (Carazo et al., 2005). Over-saturation up to 30 mg/L O₂ in the irrigation water almost not affected the development of tomato grown in rockwool cubes while 40 mg/L impaired root growth (Zheng et al., 2007). Thus, DOC level above the ambient level of O₂ saturation is probably not effective.

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INORGANIC AND SYNTHETIC ORGANIC COMPONENTS OF SOILLESS CULTURE AND POTTING MIXTURES

7

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7.1 INTRODUCTION

Though the same physical principles apply to both soilless substrates and field soils, their physical and hydraulic properties are vastly different. In addition, there are fundamental differences with regard to dynamic water, air, and nutrient distribution processes and root growth and development between spatially confined soilless production systems and unconfined field soils. While water flow and nutrient transport in growth containers is restricted, water drains and redistributes to much deeper layers in agricultural soils unless natural impediments exist. This leads to vastly different infiltration and redistribution dynamics requiring more intensive management of soilless systems to provide a stress-free rhizosphere environment for optimum plant growth. Besides physical aspects, substrate chemistry and plant nutritional considerations are equally important for intensive soilless production systems. While the basic principles of mineral plant nutrition that have been extensively reviewed by numerous authors (e.g., Marschner, 1995; Mengel and Kirkby, 2001; Epstein and Bloom, 2005), in general do not differ from soil-based production, certain aspects, mainly related to volume restrictions, require special consideration. Proper fertilization management needs to account for reduced buffer capacity for supplied nutrient solutions and associated limited nutrient supply due to volume restrictions (capacity factor).

The components of horticultural soilless substrates and potting mixtures are primarily selected based on their physical and chemical characteristics and, in particular, their superior ability to simultaneously supply roots with sufficient levels of oxygen and water. The substrates should be porous and retain water with weak matric forces to ease extraction by plant roots. To assure sufficient aeration and to also ease handling and transportation, the bulk density (BD) of substrates is

commonly much lower than the BD of agricultural soils, which also leads to higher total and air-filled porosities.

Various commercial soilless substrates such as stone wool, polyurethane, perlite, and vermiculite are virtually pest-free and can easily be disinfested between growth cycles in case of disease contamination. They also enable relatively easy control of the pH and the availability of nutrients in the root zone. An inherent limitation of soilless cultivation is the fact that root volume is restricted and confined by growth containers. The limited root volume increases root-to-root competition for water, oxygen, and nutrients since there are more roots per unit medium volume. This topic is discussed in [Chapter 14, Growing Plants in Soilless Culture: Operational Conclusions](#). Description of substrates used in horticulture, as well as soilless culture management, has been thoroughly reviewed by [Bunt \(1988\)](#), [Schwartz \(1995\)](#), [Hannan \(1998\)](#), [Resh \(1998\)](#), [Kipp et al. \(2000\)](#), [Raviv et al. \(2002\)](#), [Nelson \(2003\)](#), and [Jones \(2005\)](#).

It is impossible to cover all substrate types available on the market due to the wide variety in origin, physical and chemical characteristics, and frequent introduction of new products. Thus only the most representative and most widely applied media are describe here. In this chapter, substrates are grouped into two categories of materials: (1) “inorganic substrates” are classified as natural unmodified materials (sand, tuff, pumice) and (2) “synthetic substrates” consist of processed materials (such as stone wool, perlite, vermiculite, expanded clay, zeolite, and foamed glass). Synthetic organic materials include substrates such as phenolic resin or polyurethane.

Important properties of the substrates include their chemical activity and surface charge. Therefore, substrates are characterized as active (e.g., peat, tuff) or semiinert (e.g., stone wool and sand) materials. The difference between these two groups is quantitative rather than of qualitative nature. For a more thorough discussion of this issue, see [Chapter 4, Chemical Characteristics of Soilless Media](#). The description of each substrate includes information on production and origin, and general information about their applicability as growing medium. The physical characteristics that include BD, water retention, and hydraulic conductivity are provided, as they are essential for proper irrigation management. The chemical characteristics such as composition, stability as affected by pH, cation exchange capacity (CEC), pH, and salinity discussed, as they are required for proper fertilization and nutrient management. Information on substrate sterilization is provided, as disease control is a major factor for the reusability of substrates. Information on waste treatment is also presented, because today the potential for environmental contamination is a central issue in intensive soilless cultivation.

7.2 COMMONLY USED INORGANIC SUBSTRATES IN SOILLESS CULTURE

7.2.1 NATURAL UNMODIFIED MATERIALS

7.2.1.1 Sand

Production, Origin, and General Information—Sand is the coarse soil mineral fraction that is defined by the International Society of Soil Science as particles with diameters between 0.02 and 2 mm, and it is further separated into fine sand (0.02–0.2 mm) and coarse sand (0.2–2.0) mm. Pure sand is widely used in deserts and coastal plains because it is an inexpensive, local, natural resource. It is often used in growth beds situated on the ground on a polyethylene film that

separates it from the underlying soil. As a natural deposit, the particle sizes and distributions are often variable. The required depth of the sand layer depends on the range of particle diameters. The finer the sand, the deeper the required layer to avoid water logging and poor aeration. Sand is also used as a component of various substrate mixtures, commonly forming the heaviest constituent, serving as ballast for top-heavy plants and in outdoor nurseries against wind.

Physical Characteristics—The BD of sand is high relative to other growth media components; 1.48 and 1.80 g/cm³ for fine and coarse sand, respectively. The total porosity is relatively low, 45% and 30% for fine and coarse sand, respectively, and the volumetric water content at saturation is somewhat low at 39 and 27 vol.%, respectively. Sand has a narrow pore-size distribution; therefore the small pore fraction retains almost a constant water volume over a matric potential range from 0 to −10 cm water for coarse sand and from 0 to −20 cm water for fine sand. A further decrease in matric potential results in a steep decline in water content (Fig. 7.1). The hysteresis phenomenon is negligible. Its retention curve (da Silva, 1991; Wever et al., 1997) indicates that aeration problems are expected when using fine sand in pots or beds. Bunt (1991) showed that the mean oxygen diffusion rate in the profile of a fine sand bed was 10–100 times lower than that of peat, perlite, redwood bark, and various mixtures thereof. The saturated hydraulic conductivity of coarse and medium sand is relatively high at 5.1 and 7.1 cm/min, respectively (da Silva, 1991). However, the unsaturated hydraulic conductivity of coarse and medium sand sharply declines for matric potentials below −10 and −20 cm, respectively (da Silva, 1991).

Chemical Characteristics—Quartz (SiO₂) is the most common constituent of the sand fraction in soils. It is the second most abundant mineral in the earth's crust and is highly resistant to

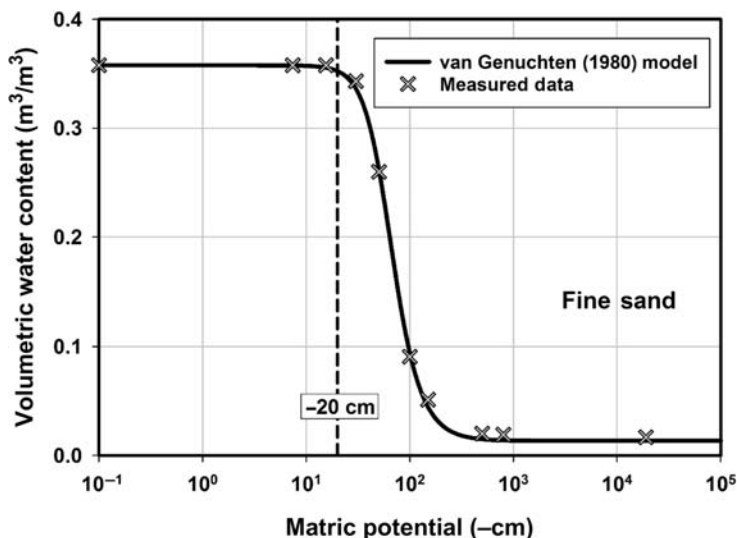


FIGURE 7.1

WRC for fine sand. The dashed vertical line marks the matric potential at −20 H₂O. Note that while the matric potential is a negative (subatmospheric) pressure, its absolute value is commonly plotted for display of the WRC. WRC, Water retention curve.

weathering (Drees et al., 1989). Quartz particle density is high, 2.6–2.65 g/cm³, with a relatively low specific surface, 1–10 m²/g (Drees et al., 1989). Quartz is a stable mineral with a low solubility of 3–7 mg Si/L, independent of pH in the pH range of 2.5–9.0 (Drees et al., 1989). It is one of the purest minerals known with a very low substitution of Si by Al, Fe, and other trace elements. Thus, the charge deficiency that plays a major role in the physicochemical activity of other soil minerals is very low in Quartz (Drees et al., 1989). Therefore, sand is a chemically inactive substrate that can serve as diluent to more reactive components in potting media. It must be emphasized that beach sand and road deicing sand contain high amounts of salts and therefore should not be utilized for horticultural purposes.

Sterilization, Reuse, and Waste Disposal—Sand can be steam-sterilized. With thin layers, the pores may rapidly fill with water due to capillary condensation, which disturbs the steaming process (Kipp et al., 2000). Sand is very durable, because it is neither chemically nor biologically altered in course of its use as a growth medium. Sand waste can be used for construction; thus, there are no concerns with regard to environmental pollution.

7.2.1.2 Tuff

Production, Origin, and General Information—Tuff (volcanic scoria, a.k.a. volcanic cinder, lapillus, tezontle, pozzolana, and picon) is a common name for pyroclastic (Greek pyro “fire,” and klastos “fragment”) volcanic material, characterized by high porosity and surface area. Volcanic rocks are classified according to their silica content as follows (silica percentage in the solid phase): rhyolitic (more than 65), andesitic (50–65), and basaltic (less than 50). Rhyolitic lavas are formed during eruption at relatively low temperatures (800°C–1000°C), and therefore, they contain predominantly light elements, such as Si and Al, whereas their Fe, Mn, Ca, and Mg contents are low. The viscosity of rhyolitic rocks is high and they have a light color. Basaltic lavas are formed at high temperatures (above 1000°C), and therefore have a higher Fe, Mn, Ca, and Mg content, which induces low viscosity and dark color. Andesitic lavas are formed at intermediate temperatures and have intermediate color and a chemical composition between that of rhyolitic and basaltic lavas. Rapid cooling of magma during eruption prevents the formation of primary minerals and, therefore, pyroclastic materials contain mainly vesicular, volcanic glass. The physical and chemical properties of tuff are determined mainly by its mineralogical composition and weathering stages (Silber et al., 1994, 1999). There are different form of tuffs (black, red, and yellow), all of them erupted from the same volcano and having almost the same chemical composition, but a differing primary mineral composition and weathering stage, they had different surface charge characteristics, *P* adsorption capacity and dissolution kinetics (Silber et al., 1994, 1999). In addition, grinding and sieving processes may affect tuff’s physical and chemical properties.

Physical Characteristics—Tuff has a BD of 0.8–1.5 g/cm³, and a total porosity of 60%–80%, depending on its origin and the sieving/grinding process. Retention curve and hydraulic characteristics of tuff are discussed in detail in Chapter 3, *Physical Characteristics of Soilless Media*.

Chemical Characteristics—Tuffs have permanent and variable charge surfaces, resulting mainly from amorphous materials. The chemical composition and the surface characteristics of three tuffs from Northern Israel differing in their weathering stage are shown in Table 7.1 (after Silber et al., 1994). Tuff exhibits a significant buffering capacity and may adsorb or release nutrients, especially *P*, during the growth season (Silber et al., 1999; Silber and Raviv, 1996). The chemical stability of tuff depends on the mineralogical composition. Volcanic glass dissolution is very rapid while that

Table 7.1 Chemical Composition, Expressed as Oxides (%) and Surface Characteristics of Three Tuff Types From Northern Israel

Oxide	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	Na ₂ O	K ₂ O
Black	45.8	16.3	12.1	8.7	7.3	3.9	1.6
Red	43.3	13.3	12.2	10.9	4.3	4.2	1.7
Yellow	39.6	20.4	15.5	4.9	5.3	0.2	0.2
Tuff Type			Black		Red		Yellow
Specific surface area (m ² /g) ^a			7		28		174
pH in H ₂ O ^b			9.3		8.1		7.9
pH in 1 M-KCl ^b			7.8		6.6		6.6
CEC in pH 7 (cmol/kg)			10.7		28.5		60.1
CEC, Cation exchange capacity.							
^a N ₂ adsorption.							
^b 90 min shaking 0.5 g/25 mL.							

of secondary minerals, such as kaolinite and halloysite, is slower. Hence, nonweathered materials containing a high concentration of volcanic glass, like black tuff, are unstable and dissolve easily in solution below pH 6, while red and yellow tuffs are more stable (Silber et al., 1999). Introducing plants to black or even red tuff, after equilibration with an acidic electrolyte (pH below 5), may be risky due to Al and Mn toxicity. More specific chemical characteristics of tuff are discussed in detail in [Chapter 6, Nutrition of Substrate-Grown Plants](#).

Sterilization, Reuse, and Waste Disposal—Tuff is a stable material, which can last for many years. Growing plants may even improve the chemical properties of tuff due to the accumulation of organic matter (OM) and low molecular weight fulvic acid (Silber and Raviv, 1996). Steaming, solarization, or chemical treatment can be used for disinfestation. Disinfestation treatments of tuff after rose (*Rosa × hybrida*) cut flower production resulted in yields that were superior to those of unused tuff (Raviv et al., 1998). The disinfestation process as such did not affect the solid phase or stability of tuff, but significantly affected the solubilization of OM. The soluble organic compounds released during disinfestation were adsorbed later by the tuff surfaces, thereby changing its surface charge (Silber and Raviv, 2001). Thus, beyond their beneficial effects on pathogen populations, the disinfestation treatments enhanced nutrient availability.

7.2.1.3 Pumice

Production, Origin, and General Information—Pumice, like tuff, is a product of volcanic activity and usually forms from silicic lavas developed in rhyolitic composition, rich in gases and volatiles (Challinor, 1996). Rapid release of pressure during volcanic eruptions leads to gas expansion and the formation of low-density materials composed of highly vesicular volcanic glass. Pumice is common in areas rich in volcanic activity, such as the Portuguese Azores, Greek islands, Iceland, Japan, New Zealand, Russia, Sicily, Turkey, and the United States. The raw material is mined from quarries and ground and sieved to specification. Its physical and chemical properties are affected by its aggregate size. Pumice has been used since Roman times as lightweight aggregate in

construction, for stonewashing in the clothing industry, for polishing and cleaning metal, wood and glass, and as filler in the paper and plastic industries. The physical properties of pumice are very similar to those of tuff, while the chemical properties, especially surface activity and charge, are vastly different. While tuffs have permanent and variable charge surfaces and possess a buffering capacity for nutrients, pumice is an inert aluminosilicate material.

Physical Characteristics—Pumice is a lightweight aggregate with a low BD of 0.4–0.8 g/cm³, and a total porosity of 70–85 vol.% (Boertje, 1994; Challinor, 1996; Raviv et al., 1999), depending on its origin and the sieving/grinding process. Pumice possesses large pores and consequently its volumetric water content decreases sharply with decreasing (more negative) matric potential (Boertje, 1994; Raviv et al., 1999). The water holding capacity of pumice is relatively low when compared to stone wool, perlite, or organic substrates and may limit water and nutrient uptake by plants, especially in arid climates (Raviv et al., 1999). The volumetric total porosity, air-filled porosity, and easily available water (defined at a matric potential of: 0, 0.0 to –1.0 and –1.0 to –5.0 kPa, respectively) of three pumices are listed in Table 7.2. The unsaturated hydraulic conductivity of Sicilian and Greek pumice decreased by almost 4 and 6 orders of magnitude, respectively, as matric potential decreased from 0 to –10 kPa (Raviv et al., 1999). Therefore, very frequent irrigation is required for plants grown in pumice.

Chemical Characteristics—Pumice is an inert aluminosilicate material composed primarily of silica and Al₂O₃ but may also contain metal oxides, calcite, or salts. The chemical composition of pumice is shown in Table 7.3. Pumice has no buffering capacity and possesses a very low surface charge, derived mainly from carbonate and metal impurities (Silber, unpublished data). The material is stable even at a pH of 2.5 (Silber, unpublished data). However, caution is required when applying fresh pumice material because high concentrations of Na may be initially leached.

Sterilization, Reuse, and Waste Disposal—Pumice is biologically inert and contains no pathogens or weeds (Challinor, 1996). It is stable and can be reused practically indefinitely. Being a natural product, it can be disposed of without causing environmental pollution.

7.2.2 PROCESSED MATERIALS

7.2.2.1 Perlite

Production, Origin, and General Information—Perlite is a glassy volcanic rock with a rhyolitic composition and 2%–5% of combined water (Alkan and Dögan, 1998; Dögan and Alkan, 2004; Kaufhold et al., 2014). The major known global perlite resources (about 70%) are located along the

Table 7.2 Physical Properties of Pumice from Several Sources

Pumice Source	Bulk Density (g/cm ³)	Total Porosity (%)	Air-Filled Porosity (%)	Easily Available Water (%)
Iceland ^a	0.4	85	40	5
Sicily ^b	0.7	70	27	4
Greece ^b	0.6	75	41	2

^aFrom Boertje (1994).

^bFrom Raviv et al. (1999).

Table 7.3 Chemical Composition of Pumice, Expressed as Oxides (%)

Oxide	SiO ₂	Al ₂ O ₃	CaO	MgO	Fe ₂ O ₃	Na ₂ O	K ₂ O
Percent	70–75	12–14	1–3	0.1–0.6	0.8–2.0	3–6	4–5

Aegean coast in Turkey. Commercial perlite is produced by heating the ground and sieved material to 760°C–1100°C. Life cycle analysis (LCA) conducted to assess environmental impacts of cherry tomato production using perlite as a substrate suggests that the amount of energy consumed while producing perlite contributes a significant environmental burden, but it minimizes the uncertainty involves in field production of this crop (Romero-Gamez et al., 2017). The combined water in the perlite is converted to gas at high oven temperatures and subsequently the volume expands to 4–20 times of its original volume, resulting in a lightweight, high-porosity material. Perlite is frequently used in potting soil mixtures and as a standalone growth medium (Grillas et al., 2001; Gül et al., 2005). It is produced in various grades, the most common being 0–2.0 and 1.5–3.0 mm in diameter. The various grades differ in their physical characteristics. Perlite is a relatively expensive substrate, and its replacement by bark proved economical (Rodriguez et al., 2006). Similarly, volcanic gravel was a reasonable substitute for perlite (Neocleous and Polycarpou, 2010). However, in other cases, the higher yields obtained in perlite, as compared to pine bark and stone wool compensated for its higher cost (Hanna, 2009).

Physical Characteristics—Due to enclosed air pockets, expanded perlite is very lightweight with a particle and BD of 0.7 and 0.1 g/cm³, respectively. It is very porous and can hold 3–4 times its weight in water. Due to this very low BD, perlite is probably the most used substrate for green roofs (Vijayaraghavan and Raja, 2014); however, one has to consider the high weight of the moist substrate. Perlite is probably the medium ingredient having the highest air-filled porosity (Schindler et al., 2017). Burés et al. (1997a) reported that water retained at –10 kPa is much higher for the coarse fraction (0.5–1.0 mm diameter) than for the fine fraction (0.25–0.50 mm diameter) of expanded perlite. This difference in water holding capacity between the coarse and fine fractions indicates that most of the water is held by the coarse particles in internal pores. However, it is not explained by the volume of internal porosity alone (Burés et al., 1997a). The decrease in water content with decreasing matric potential is moderate relative to sand and stone wool (Burés et al., 1997a,b). The available and nonavailable water in commercial perlite of 0–4 mm diameter was 13.6 and 36.5 vol.%, respectively (Burés et al., 1997b). The water retention curve (WRC) of perlite shows moderate hysteresis (Burés et al., 1997b; Wever et al., 1997). Wever et al. (1997) reported that the saturation of perlite was very rapid, independent of its initial moisture content. The saturated hydraulic conductivity depends on particle diameter (Burés et al., 1997a). For commercial perlite of 0–4 mm diameter with 50% of the particles smaller than 0.5 mm, saturated hydraulic conductivity was 0.3 cm/min (Burés et al., 1997b). A reduction of two orders of magnitude in the hydraulic conductivity was obtained as the matric potential decreased from 0 to –30 cm H₂O (Burés et al., 1997a). This change is moderate when compared to sand. Due to perlite's excellent physical characteristics, it is frequently used as a component in mixtures based on organic substrates such as compost (Raviv, 2013). By this, the positive biological features of such media (see Chapter 8: Organic Soilless Media Components, for details) can be maintained for a relatively long period, while preventing compaction.

Chemical Characteristics—Perlite is neutral with a pH of 7.0–7.5, but it has no buffering capacity and contains no mineral nutrients. The inert nature and neutrality of perlite make it a favorable medium for assessing the usefulness of various forms and levels of nutrients (Arquero et al., 2006; Wik et al., 2006). When the pH is low, there is a risk of toxic Al release into the solution. The chemical composition of the material, as analyzed by Olympios (1992), is provided in Table 7.4.

Sterilization, Reuse, and Waste Disposal—Perlite is a sterile product as it is produced at very high temperatures. Chemically, perlite is a stable material, which can last for several years (Acuña et al., 2013); its stability is not greatly affected by acids or microorganisms. Being an inert material, recycling perlite poses no environmental problems. Reusing perlite without processing to grow successive crops is potentially risky because of media compaction, salt buildup, and pest contamination (Hanna and Smith, 2002). Replacing used perlite with new media to raise successive crops is costly, and recovering the expense by selling the product at a higher price may not work well in a competitive market. Steam sterilization of used perlite before planting a new crop has been recommended to safeguard against pathogen contamination (Wilson, 1988). However, this treatment requires expensive steam generators and may not be adequate to restore perlite's loose structure and to reduce accumulated salt content (Hanna, 2005). In a study aimed to determine if growing tomatoes (*Lycopersicon esculentum* Mill.) in cleaned and disinfected used perlite is more economical than in new perlite and without any negative effect on yield, Hanna (2005) recycled perlite twice a year for commercial tomato production in bag (18.9 L) culture. After the separation of roots from the previous crops, used perlite was treated with hot water (13.25 L water/18.9 L perlite) at temperatures reaching 93.3°C to leach excess salts and disinfect the medium. This treatment raised media temperatures above limits necessary to kill several fungi and nematodes and significantly reduced salt contents (Electrical Conductivity (EC), NO₃–N, and K were reduced by 43%, 50%, and 47%, respectively) with no noticeable change in physical properties [i.e., particle size distribution (PSD)] over 8 years. This contrasts the common notion that because of its plastic nature, perlite is prone to some mechanical compression and disintegration, as was found by Giuffrida et al. (2008) and Giuffrida and Consoli (2016). Cleaning and disinfecting used perlite for recycling saved 56% of the cost to replace the media and gave greater marketable yield and heavier tomato fruit than with new perlite. The observed yield benefit of perlite recycling was attributed to the collective effect of salt reduction, media disinfection, and the presence of an optimum level of nutrients; it usually takes time to build up nutrients to an optimum level in new perlite. Thus, it has been concluded that used perlite can be cleaned and disinfected as needed and recycled for many years because it is not organic in nature and physically and chemically stable (Hanna, 2010). Marfa et al. (1993) also found that perlite retains its physical properties for successive crops. A very important finding is the fact that recycled perlite and peat–perlite mix were more suppressive against *Fusarium oxysporum* f. sp. *radicis lycopersici* than the new, unused media (Clematis et al., 2009). Another efficient manner of disinfecting perlite is by solarization of the growing bags within the greenhouse (Perez-Hernandez et al., 2017).

Table 7.4 Chemical Composition of Perlite, Expressed as Oxides (%)

Oxide	SiO ₂	Al ₂ O ₃	CaO	MgO	Fe ₂ O ₃	Na ₂ O	K ₂ O
Percent	73.1	15.3	0.8	0.05	1.05	3.65	4.5

Table 7.5 Chemical Composition of Vermiculite, Expressed as Oxides (%)

Oxide	SiO ₂	Al ₂ O ₃	MgO	Fe ₂ O ₃
Percent	20–25	5–10	35–140	32–35

7.2.2.2 Vermiculite

Production, Origin, and General Information—The raw material for vermiculite is a natural clay mineral that has a layered structure with water in between the layers. Expanded vermiculite is produced in a similar fashion as perlite by heating the grinded and sieved material to 1000°C. The water is converted to vapor at high temperatures in the oven, which increases interlayer spacing. This process is termed “exfoliation.” As a result, expanded vermiculite consists of accordion-shaped granules that are lightweight and exhibit high porosity (Kipp et al., 2000). Vermiculite is used as a propagation medium and as a component of potting soil mixtures (Pill and Lambeth, 1975). Fine grades are mainly used as mulch in transplant production, while coarse grades are frequently used as rooting media (Wright, 1989).

Physical Characteristics—Expanded vermiculite is produced in various grades and sizes, the most common being 0–2, 2–4, and 4–8 mm in diameter. All grades are very lightweight with a particle density of 0.9 g/cm³ and a BD range from 0.07 to 0.1 g/cm³, for the coarse and fine grades, respectively. It is very porous, easily absorbs water, and can hold 3–4 times its weight in water.

Chemical Characteristics—Vermiculite is a neutral clay, with a pH of 7.0–7.5 and low EC. This clay type contains two tetrahedral sheets for every octahedral sheet. Like the raw material, it has a permanent negative charge. Consequently, it has a CEC value of 15–21 cmol/kg and a buffering capacity for pH and cations. It also adsorbs ions like phosphate due to its high surface area and some positive charge sites located at the edges of the clay tactoids. This might be the reason why in comparative studies performance of vermiculite was better than that of perlite (Gelsomino et al., 2010). The chemical composition of the material, expressed as oxides (%), is given in Table 7.5. When the pH is low, there is a risk of toxic Al release into solution. Mg is an important component of the mineral structure and is the dominant adsorbed cation.

Sterilization, Reuse, and Waste Disposal—Vermiculite is a sterile product as it is produced at very high temperatures. However, it cannot be steam-sterilized as it disintegrates during heating. Vermiculite expanded structure collapses easily. Therefore, it is not suitable for long-term application. It is sensitive to mechanical compression, which may pulverize particles. Disposal of vermiculite is not hazardous to the environment. In the past, asbestos was found in some vermiculite mines. All those mines were closed and now vermiculite is considered a safe growing substrate.

7.2.2.3 Expanded clay granules

Production, Origin, and General Information—Expanded clay is a granular product with a crystalline structure with surface and internal porosity. It is produced by forming the clay into pellets that are baked in rotary kilns at 1200°C; at this temperature, gas is released, which leads to expansion of the clay (Oekotau Easy Green GmbH, 2006; ESCSI, 2006). It is essential that the raw material has a low soluble salt content, so that substances such as lime are not added during the process. Otherwise, salt may be leached during cultivation. The grade size used in horticulture is 4–8 or

4–16 mm diameter (Oekotau Easy Green GmbH, 2006). Expanded clay granules have been used in horticulture since 1936. They are used in hydroculture and in containers for various crops.

Physical Characteristics—Expanded clay granules are lightweight with a BD of 0.28–0.63 g/cm³ and water content of 11–24 vol.% at a matric potential of –10 cm H₂O (de Kreij et al., 1995). Wever and van Leeuwen (1995) reported high BD, 1.2 g/cm³, and 53% porosity. They found that the water content was reduced from 48% to 44%, as the matric potential decreased from –3 to –50 cm H₂O; thus, the amount of easily available water is very small, just 4%. In contrast, the material contains a large amount of enclosed air. It has been shown that there were no changes in physical characteristics after 5 years of intensive cropping. Hydration of the substrate from dry state is relatively slow, but if initiated at matric potential of –100 cm it is rapid (Wever et al., 1997). The WRC does not exhibit insignificant hysteresis. There is no available information on the hydraulic conductivity of expanded clay granules, but they are relatively resistant to compression (Wever and van Leeuwen, 1995).

Chemical Characteristics—Expanded clay granules are neutral, with a pH of about 7.0. The EC is low if low salt clay was used, and no salts were added during the baking process; otherwise, washing is required. It has been classified as an inert material with no cation exchange or buffering capacity. However, Meinken (1997) and Meinken and Fischer (1994) demonstrated that nutrients may accumulate due to diffusion into the granules and they can be released back to the solution.

Sterilization, Reuse, and Waste Disposal—Expanded clay granules are a sterile product as they produced at a very high temperature. After use, they can be washed and sterilized without deleterious effects. Expanded clay granules are very stable and can last for many years. Waste material can be used in the construction industry.

7.2.2.4 Zeolite

Production, Origin, and General Information—Zeolites are crystalline hydrated aluminosilicates of alkali and alkaline cations that exhibit infinite, three-dimensional crystal structures (Ming and Mumpton, 1989; Mumpton, 1999). Zeolites are usually formed by metamorphism of volcanic rocks but may also origin from nonvolcanic materials in marine deposits or aqueous environments (Ming and Mumpton, 1989). Due to their ion exchange, adsorption, hydration/dehydration, and catalysis properties, zeolites are widely used in agriculture and in numerous industries for the removal of pollutants from waste and drinking water (Ming and Mumpton, 1989; Mumpton, 1999). Amelioration of the chemical and physical properties of natural zeolite is achieved by producing synthetic minerals. Sherman (1999) stated that 508 different types of synthetic zeolites with improved physicochemical properties are produced worldwide. Zeolites (mainly clinoptilolite) are used in soil remediation to adsorb nuclear waste or heavy metals (Kapetanios and Loizidou, 1992; Rosen, 1996; Chlopecka and Adriano, 1997; Paasikallio, 1998; Krutilina et al., 1999, 2000). They are used in agriculture as soil amendments for a source of P, K, and NH₄ nutrients in infertile soils and substrates (Hershey et al., 1980; Chen and Gabelman, 1990; Notario del Pino et al., 1994; Allen et al., 1995; Williams and Nelson, 1997; Dwairi, 1998), for reducing N losses and nitrate contamination (Weber et al., 1983; Ferguson and Pepper, 1987; Huang and Petrovic, 1994; Ando et al., 1996; Kithome et al., 1999), and for improving water availability (Huang and Petrovic, 1995; Yasuda et al., 1995). A sophisticated example for the use of zeolite as growth medium is by pre-charging it with aluminum in order to produce blue hydrangea (Opena and Williams, 2003). Additional agricultural uses of zeolites were described in Pond and Mumpton (1984). Currently,

zeolites (mainly clinoptilolite) are widely used as components of growing media, especially in Mediterranean countries such as Greece and Turkey (Gül et al., 2005; Samartzidis et al., 2005).

Zeolites possess extremely high CEC values (220–460 cmol/kg) as well as a relatively high BD (1.9–2.3 g/cm³) (Ming and Mumpton, 1989), and therefore, the use of zeolite as a single component growth substrate is not recommended. However, in mixed substrates, which include organic (peat and compost) or inorganic materials (sand and perlite), zeolites are widely used for flower and vegetable production all over the world (Bulgaria, China, Cuba, Italy, Japan, Jersey, Korea, Russia, United States, and Yugoslavia). The experimental use of zeolite as a single growth substrate has been reported for several crops, such as carnation (*Dianthus caryophyllus* L.) (Challinor et al., 1995), sweet pepper (*Capsicum annum* L.) (Harland et al., 1999), tomato (Rivero-Gonzales and Rodriguez-Fuentes, 1988), gerbera (*Gerbera jamesonii* Bolus ex Hooker f.) (Papadopoulos et al., 1995), and lettuce (*Lactuca sativa* L.) (Gül et al., 2005).

Clinoptilolite is the principal zeolite mineral used in agriculture, and therefore, only its physical and chemical properties are discussed below.

Physical Characteristics—Clinoptilolite has a particle density of 2–2.1 g/cm³ (Ming and Mumpton, 1989) and a saturation water content of 34 vol.% (Mumpton, 1999). However, grinding and sieving affect the chemical and physical properties. The physical properties of Turkish zeolites (clinoptilolite) of varying aggregate sizes were almost identical to that of basaltic and rhyolitic tuff with similar particle sizes (Unver et al., 1989). The volumetric total porosity, air-filled porosity, and easily available water of these zeolites are listed in Table 7.6 (from Unver et al., 1989).

The fact that the BD of zeolite is not as high as soil allows its use in roof gardens—a type of use that became increasingly popular in the last decade (Nektarios et al., 2015). Its relatively high water holding capacity enables efficient storm-water management (Fassman and Simcock, 2012), which is important in urban environments.

Chemical Characteristics—Clinoptilolite is very stable, but dissolves at pH 2 or lower (Ming and Mumpton, 1989). Clinoptilolite has a unit cell formula of [(Na₃K₃)(A₁₆Si₃₀O₇₂)]24H₂O and CEC of 220 cmol/kg (Ming and Mumpton, 1989). For additional discussion on potential nutritional aspects of zeolites, see Chapter 6, Nutrition of Substrate-Grown Plants.

Sterilization, Reuse, and Waste Disposal—Zeolites are not sensitive to mechanical compression and so their physical properties do not change due to handling; this offers possibilities for zeolite reuse. Harland et al. (1999) compared zeolite (clinoptilolite) as a new and recycled substrate under closed (recirculating) and open (nonrecirculating) fertigation systems for growing sweet pepper over a period of 4 years (three recycling cycles). In the recycling system, at the end of each cropping period, the plants were removed and roots were sieved from the substrate and then it was steam-sterilized at 90°C for 3 hours, before being rebagged and used as substrate for the succeeding

Table 7.6 Physical Properties of Zeolite Fractions

Fraction Size (mm)	Bulk Density (g/cm ³)	Total Porosity (%)	Air-Filled Porosity (%)	Easily Available Water (%)
2–6	0.69	50	28	4
6–10	0.65	40	20	10
10–14	0.64	35	10	1

crop. Over the period of the trial, there was no loss of yield or quality attributable to the reuse of zeolite substrate (Harland et al., 1999).

7.2.2.5 Foamed glass

Production, Origin, and General Information—Foamed glass aggregates are made from recycled glass. The production process starts with crushing and grinding recycled glass into a very fine powder that is mixed with calcium carbonate (2% on mass basis) and other additives that act as foaming agents. The mixture is poured into molds and heated at about 850°C in a foaming furnace. Due to the evaporation of the foaming agents, the molten mixture expands, thereby creating a network of fine pores. After cooling to room temperature, the foamed glass slabs are crushed, tumbled, and sieved to various aggregate sizes for applications in horticulture and aquaculture. Foamed glass is also applied as fire-retardant insulation material in the building industry, and for road construction as frost-heave insulation and lightweight bulk fill material. For horticultural applications, foamed glass aggregates are marketed as Growstone (Growstone, LLC, Albuquerque, NM, USA) and are available in different size ranges and as mixtures with coconut coir. Preliminary greenhouse trials at the University of Arizona Controlled Environment Agriculture Center (Chen Lopez, 2011; Costa, unpublished data) revealed that tomatoes grown in Growstone aggregate growth bags showed similar yields and plant development as tomatoes grown in stone wool. Because of better aeration properties of the Growstone aggregates, the sugar content of the fruits was significantly higher in Growstone than in stone wool. A similar trial at Wageningen University and Research Centre (Costa, unpublished data) tomato yield from Growstone aggregates outperformed stone wool by 4% due to higher occurrence of blossom end rot in stone wool.

Physical Characteristics—Foamed glass aggregates are lightweight and dependent on aggregate size distribution, exhibit a BD of about 0.15–0.25 g/cm³ and an accessible specific surface area of about 15 m²/g. They exhibit a typical bimodal pore size distribution with a fraction of small intraaggregated pores and a fraction of larger interaggregate pores. For foamed glass aggregates with 0.85 cm mean diameter, the total pore volume is 95% with the interaggregate porosity accounting for 70% (Chamindu Deepagoda et al., 2013a). The WRC shows a bimodal shape in accordance with the two distinct pore fractions (Fig. 7.2). After saturation, the interaggregate pores rapidly drain over a very narrow matric potential range, thereby contributing to the excellent aeration properties of foamed glass aggregates which lead to a lower risk of *Pythium*, as compared to stonewool (Blok et al., 2011). The intraaggregate pores release water at a much lower rate acting as storage compartment for root accessible water. It is interesting to note that the aeration/water holding properties can be modified and tailored to a specific crop by simply reducing or increasing the aggregate sizes. The saturated hydraulic conductivity for foamed glass aggregates with 0.85 cm mean diameter was determined as 124 cm/min (Chamindu Deepagoda et al., 2013a). Physical and hydraulic properties for a 1:2 (v:v) foamed glass aggregate/coconut coir mixture are reported in Chamindu Deepagoda et al. (2013a).

Chemical Characteristics—There is little published information about the chemical characteristics of foamed glass aggregates, but it is recommended to rinse the aggregates prior to transplant to reduce substrate pH. An interesting chemical aspect is the slow release of mono silicic acid from the foamed glass matrix, which accumulates in plant leaves and shoots. Several studies have shown beneficial effects of elevated silicon concentrations in plant leaves that include resistance to Mg

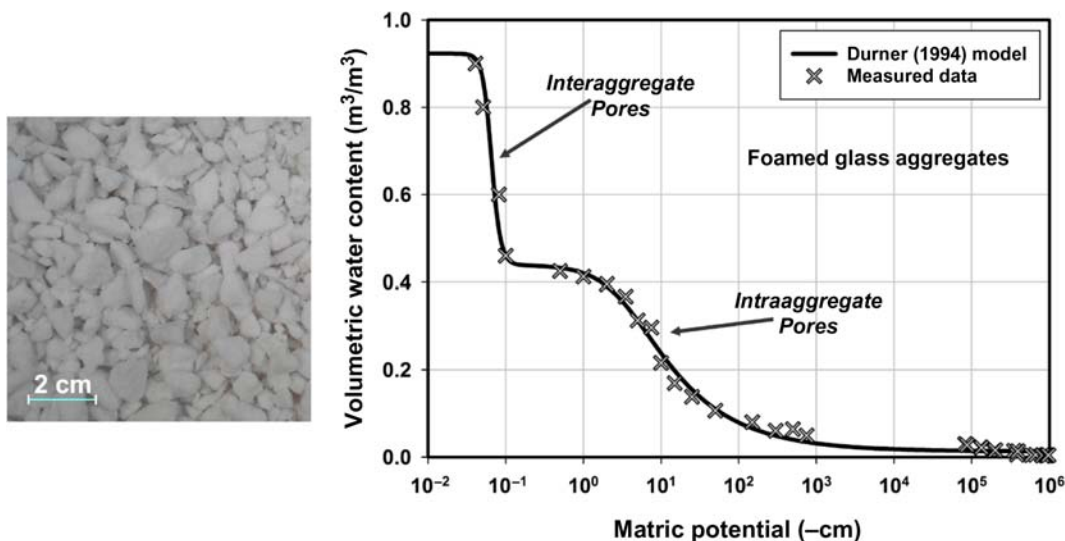


FIGURE 7.2

WRC for foamed glass aggregates. The bimodal shape reflects the water retained within the inter- and intraaggregate pore fractions. The solid line represents the fitted Durner (1994) dual-porosity water retention model. WRC, Water retention curve.

toxicity, fungal diseases such as *Pythium*, and to perforating insects such as leaf miners (Hogendorp and Cloyd, 2008; Kamenidou et al., 2010; Lu et al., 2008).

Sterilization, Reuse, and Waste Disposal—Being a sterile product, new foamed glass aggregates provide a root environment that is initially less susceptible to pathogens. Steam sterilization of foamed glass aggregates does not negatively impact the material structure, suggesting that the aggregates can be reused without losing physical strength.

7.2.3 MINERAL WOOL

7.2.3.1 Stone and glass wool

Production, Origin, and General Information—Mineral wool is a lightweight, artificial material, originally produced for thermal and acoustic insulation in the construction industry. Stone wool products have been initially produced for a variety of purposes by the Danish company I/S H.J. Henriksen and V. Kähler in 1909 (see <http://www.rockwool.com>). The application of modified forms of stone wool as a substrate for horticulture started in Denmark in 1969 (Smith, 1987). Through various corporate transitions the GRODAN Group emerged and by the 1980s its flagship product “rockwool” became widely synonymous with stone wool. The stone wool that is used in horticulture is mainly in form of slabs or blocks of bonded fibers but is also available in granulated form as a component of potting mixtures. Stone wool is manufactured by heating a mixture of three natural raw materials: 60% diabase (a form of basalt rock, dolerite), 20% limestone, and 20% coke.

These materials are fused at high temperature, with the coke acting as fuel in a blast furnace through which air is forced to raise the temperature to 1600°C. The molten mixture is then spun at a high speed into thin fibers of about 0.5 mm diameter, which is cooled by an air stream. The fibers are heated with additives (i.e., phenolic resin and wetting agents) to bind them together and lower the natural hydrophobicity of the material. It is then pressed into blocks or slabs of various sizes (Smith, 1987). Manufacturing of stone wool accounts for a substantial fraction of the total energetic requirements of greenhouse production of tomato and other crops (Anton et al., 2012). Another major drawback of stone wool is its presumed negative health effect (Kudo et al., 2009).

Glass wool is made by melting quartz sand in electric furnaces at 1200°C. The process of production of glass wool fibers and slabs is similar to that of stone wool. In general, both stone wool and glass wool (mineral wool) are sterile, easy to manage, and consistent in performance. Today stone wool is one of the major greenhouse substrates in The Netherlands, Belgium, Germany, Denmark, and other Western European countries, and other regions (Smith, 1987) with current uses in virtually all areas where greenhouse vegetables are produced (Canada, United States, Netherlands, and other countries). It is an effective growth medium for horticultural crops, in which growers can easily manipulate the ratio between water and air and the nutrients in the root zone (Papadopoulos et al., 1999). On the other hand, it is “unforgiving” to management errors, because it lacks a buffering capacity for nutrients, pH, and water, and due to the low volume of the common slabs.

Physical Characteristics—Stone wool is a lightweight substrate with a low BD of about 0.05–0.2 g/cm³ and a pore volume of 92%–98%, depending on the producer (Smith, 1987, 1998; Kipp et al., 2000). It shows high water retention at matric potentials above –1 kPa and the water content sharply declines as the matric potential decreases, so that virtually no water is retained at potentials lower than –5 kPa (Fig. 7.3) (da Silva, 1991; da Silva et al., 1995). As a result, the water buffer capacity is low and a steep gradient in water content occurs from the top to the bottom of the slab following irrigation and free drainage.

The air volume is low at the bottom, just 4% at a height of 1 cm above the base, while the upper layers are dry. The recommended height of a slab for optimal water to air ratio is 7.5–10 cm (Smith, 1987). However, Van der Gaag and Wever (2005) recommended using thicker slabs when growing cucumbers in order to reduce stone wool conduciveness to *Pythium aphanidermatum*, the causal agent for *Pythium* root and crown rot, as a result of stone wool’s high water content in shallow slabs. Pronounced hysteresis has been observed during the initial wetting/drying cycle, but it diminished in subsequent cycles (da Silva et al., 1995). The hydraulic conductivity at saturation is very high (5–20 cm/min) but sharply decreases with decreasing matric potential (da Silva et al., 1995; Chamindu Deepagoda et al., 2013a). Such reductions in hydraulic conductivity may lead to poor uptake of water and nutrients and to rapid development of water stress if irrigation is delayed. Stone wool is a soft and elastic substance that is compressed under pressure and retains its original height following relaxation, but after use it becomes softer and less elastic (Wever and van Leeuwen, 1995). Glass wool is very light and can contain a lot of water and air. In contrast to stone wool, the fiber diameter of glass wool can vary, which affects the water-holding capacity. By having finer fibers in the upper part of the slab than in the lower part, it is possible to obtain better water content distribution over the height of the slab.

Chemical Characteristics—The chemical composition of stone wool, expressed as oxides (%), is given (Verwer, 1976) in Table 7.7. The main chemical characteristic of stone wool is that it is

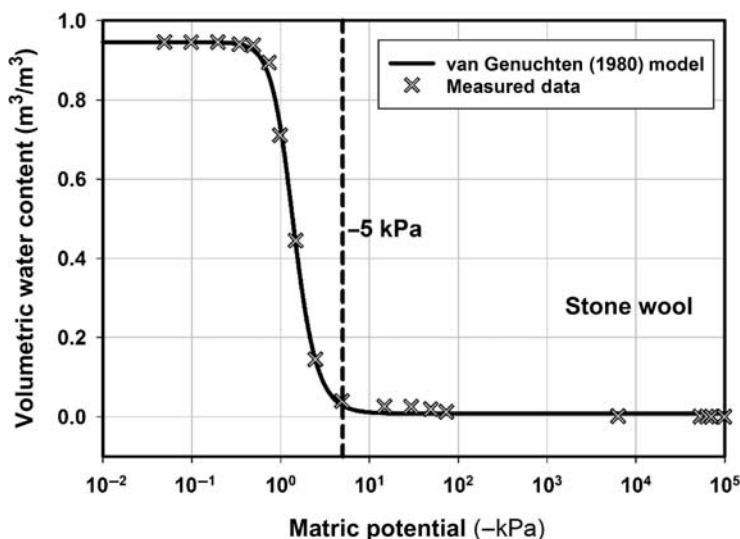


FIGURE 7.3

Water retention curve for stone wool ($BD = 0.15 \text{ g/cm}^3$). To the right of the dashed vertical line virtually no water is retained in stone wool. *BD*, Bulk density.

Table 7.7 Chemical Composition of Stone Wool, Expressed as Oxides (%)

Oxide	SiO ₂	Al ₂ O ₃	CaO	MgO	Fe ₂ O ₃	Na ₂ O	K ₂ O	MnO	TiO ₂
Percent	47	14	16	10	8	1	1	1	1

relatively inert except for some minor effects on pH. In the past, the initial pH of commercial stone wool has been rather high (7.0–8.0) and values of up to 9.5 have been reported (Smith, 1987). However, today values of 6–6.5 are the industry standard. Below pH 5.0, stone wool starts to dissolve; hence low pH values should be avoided. Stone wool itself has no effect on crop nutrition and all nutrients must be supplied with the irrigation water. However, it was demonstrated that a considerable amount of Fe can be taken up from stone wool by rose rootstocks (Rupp and Dudley, 1989) and some vegetables (Sonneveld and Voogt, 1985). Moreover, Rupp and Dudley (1989) reported that considerable concentration of Fe, Mn, Cu, and Zn could be extracted with DTPA from various stone wool products.

Sterilization, Reuse, and Waste Disposal—Being a sterile product, new stone wool provides a root environment, which is initially less susceptible to pathogens, especially in closed-loop fertigation systems (Tu et al., 1999). Stone wool can be steamed before reuse. After one or two crop cycles, stone wool is usually discarded, producing a high volume of waste (approximately, $125 \text{ m}^3/\text{ha}$ of plant production). However, in some cases (e.g., Acuña et al., 2013), stonewool was successfully used for three crop cycles. Because stone wool does not naturally decompose, the excessively

high waste volume is a major environmental concern. Commonly practiced disposal of used horticultural stone wool in landfills is no longer sustainable in many countries (Bussell and McKennie, 2004).

In the last two decades, various methods of stone wool recycling have been developed. Research on the reuse of stone wool has been and continues to be conducted by stone wool manufacturers, particularly GRODAN A/S (Denmark) and GRODAN BNF (The Netherlands). The GRODAN studies summarized by Bussell and McKennie (2004) included several suggestions of reusing horticultural stone wool such as recycling by a stone wool factory, soil amendments (to improve poor physical properties of clay soils), mixing with other substrates (e.g., peat and compost), brick production, landfill cover, mining cover/recultivation, and mushroom casing. The direct reuse of sterilized stone wool is, however, the preferred mode of use. Unfortunately, sterilized inoculated stone wool enhanced disease infections (Minuto et al., 2008). On the other hand, composting used and inoculated stone wool with yard and leaf waste eliminated the *P. aphanidermatum* zoospores' infection capability (Zheng et al., 2009). Reuse of stone and glass wool as a raw material for both horticultural and insulation applications and in brick production are now widely practiced in the United Kingdom and The Netherlands, where nearly all of the used stone wool from crop production is collected and processed at large facilities (Neefjes, 2001). In areas where this type of recycling is not feasible, stone wool is commonly disposed in landfills. Still, an analysis done on various substrates for green roofs suggested that stone wool should be avoided in order to produce environmentally responsible roof systems (Chenani et al., 2015).

One of the potential uses of used stone wool slabs is their incorporation (as chopped small particles) into peat- or compost-based media for horticultural production; this provides an opportunity for used stone wool recycling and for reducing the consumption of a nonrenewable resource such as peat (Dubsky and Sramek, 2008).

In New Zealand, a pilot study explored the possibility of utilizing the used stone wool slabs (after removal of plastic sleeves) as mulch around avocado trees; the mulch successfully suppressed weed growth for 2 years without affecting the growth of avocado trees (Bussell and McKennie, 2004). The bottlenecks to the direct reuse of stone wool slabs for growing succeeding crop(s) include their susceptibility of being broken when moved, the possibility of incomplete steam sterilization between crops and the occurrence of variable amounts of root matter in the used slabs making water content, and EC control much more difficult than in new slabs (Bussell and McKennie, 2004).

7.3 COMMONLY USED SYNTHETIC ORGANIC MEDIA IN SOILLESS CULTURE

7.3.1 POLYURETHANE

Production, Origin, and General Information—Mineral oil is the raw material for the production of various foams, slabs, and granules used for the furniture and construction industries. Mineral oil products, like di-isocyanate, may be mixed with glycol to produce polyurethane. The product is a polymer that contains excess di-isocyanate groups, which react with water to release CO₂ and induce foaming of the polymer. The resulting foams are used in the furniture industry and the

cutting residues provide the raw material for growth substrates. They are ground to granules, which are fused under pressure with additives to form a slab. Steam at 140°C is blown in during the process. Foam is used as a growth medium for vegetables and flowers in Belgium and The Netherlands. The granules are often used as a component in mixtures. Polyurethane is considered as a substitute for perlite in potting mixes (Cole and Dunn, 2002). Similarly, phenolic foam can serve as a substitute to both stonewool and perlite (Han et al., 2014).

Physical Characteristics—Polyurethane is a very lightweight material, with a low particle density and a very low BD, 1.19 and 0.078 g/cm³, respectively. Consequently, the porosity is very high at 95%. Because most of the pores are relatively large, polyurethane holds a lot of air and very little water at low matric potentials (Kipp et al., 2000) similar to stone wool. The available water within the matric potential range from −10 to −100 cm H₂O amounts only to 2% of its volume. There are no available data on its hydraulic conductivity characteristics.

Chemical Characteristics—Polyurethane is chemically inert with a low EC and a pH of 6. It does not contain or release any important nutrients, except Fe, Zn, and B. In the past, the polyurethane foam contained harmful organic substances, but this problem has now been eliminated.

Sterilization, Reuse, and Waste Disposal—Polyurethane is resistant to acids and is not decomposed by microbial enzymes, hence cannot be composted after use. It is a sterile product that can be steamed between crops for several growing cycles. Waste may be reused for the production of polyurethane slabs for construction or may be incinerated. Polyurethane can be reused for as long as 10–15 years and is easy to manage (Benoit and Ceustermans, 1995). The superficial roots can be mechanically removed, and the slabs/mats can be pressed dry, which facilitates the passage of steam, so that sterilization is more efficient. Benoit and Ceustermans (1995) showed that the roots left in polyurethane mats over 4 years of cultivation increased water retention at −1 kPa from 8% to 16%, whereas air volume decreased from 85% to 74%, suggesting a shift of the pore size distribution to smaller pores. In a 10-year trial, Benoit and Ceustermans (1994) observed that in recycled polyurethane substrate, water retention at −1.0, −1.7, and −2.0 kPa increased and air volume decreased over time; after 10 years with eight steam treatments, one each between two successive years, the air/water retention properties of polyurethane substrate were comparable to that of new stone wool. However, owing to its low density, the substrate was susceptible to compression without being able to recover to its original volume; this might be a disadvantage when the substrate is reused and repeatedly steamed. In tomato production, earliness and total production were not affected by the reuse of polyurethane substrate over 10 years (Benoit and Ceustermans, 1994). In cucumber (*Cucumis sativum* L.) production, repeated use of the same polyurethane slab for two consecutive years provided marketable yields comparable to yields obtained with new perlite or organic substrate each year (best yielding media), but polyurethane required higher irrigation water amounts resulting in higher leaching percentage (Böhme, 1995). Hardgrave (1995) reported that when cucumbers were grown on previously used steam-sterilized polyurethane foam slabs, the productivity, and fruit quality were equal or better than for new polyurethane foam slabs or stone wool; such beneficial effects on crop performance were attributed to the improved water-holding capacity due to the presence of root materials within the recycled slabs. Tests also showed that a polyurethane foam slab was stronger, more flexible, and hence suffered much less physical damage than a stone wool slab during the steaming process between two successive crops (Hardgrave, 1995). Because of labor saving and its long-lasting nature with repeated use, polyurethane is considered a beneficial substrate economically, agronomically, and environmentally. Benoit and

Ceustermans (1995) suggested that the higher initial purchase price of polyurethane could be amortized over a minimum of 10 years, and the price for steam sterilization was only half the price as for recycling stone wool slabs.

7.3.2 POLYSTYRENE

Production, Origin, and General Information—Flakes or beads of expanded polystyrene foam are byproducts of polystyrene processing. These were sometimes added to substrate mixes to improve aeration and drainage. Polystyrene is one of the best-known synthetic polymers (others include polyethylene, polypropylene, and polyester), a very commonly found, mass-produced plastic. The individual building block of polystyrene is styrene, a liquid hydrocarbon, derived in the late nineteenth century from Storax balsam, which comes from a tree in Asia Minor called the oriental sweet gum (*Liquidambar orientalis* Mill.). Polystyrene was invented in 1950 primarily to provide thermal insulation. At present, polystyrene includes any grade, class, or type of thermoplastic polymer, copolymer, interpolymer, alloy, or blend composed primarily of polymerized styrene. Polystyrene is made in a process known as suspension polymerization. Styrene (C_8H_8) is first derived from petroleum or natural gas through a two-step reaction between ethylene (C_2H_4) and benzene (C_6H_6). First, ethylene reacts with benzene (C_6H_6) in the presence of a catalyst such as aluminum chloride to form ethylbenzene (C_8H_{10}), which is then dehydrogenated (hydrogen is removed) at $600^\circ C$ – $650^\circ C$ to form styrene. Styrene is subsequently polymerized either by heat or by an initiator such as benzoyl peroxide. Degree of polymerization is regulated by ceasing the polymerization process at some desired point with polymerization inhibitors (also called terminating agents) such as oxygen, sulfur, or quinol. To form low-density, loosely attached cells, polystyrene is then suspended in water to form droplets. A suspension agent, such as specially precipitated barium-sulfate or copolymers of acrylic- and methacrylic-acid and their esters (organic product formed by the reaction between an acid and an alcohol), is then added to the water. The suspension agents are viscous and serve to hold up the droplets, preventing them from sticking together. The beads of polystyrene produced by suspension polymerization are tiny and hard. To expand them, special blowing agents, including propane, pentane, methylene chloride, and the chlorofluorocarbons (CFCs) are used. The resulting polystyrene beads are then cleaned, and beads with anomalous size are removed.

During the 1980s, many producers incorporated polystyrene in growth media. However, it has been shown that polystyrene did not provide any beneficial properties that could not have been contributed more efficiently by other materials. Because the air within polystyrene beads is insulated from the root zone air phase and because the beads do not exhibit distinct water-holding or ion-exchange capacities, the only contribution lies in a reduction of the effective root zone volume that is amenable to root development. The biggest problem for virtually all growers was the environmental contamination around the nurseries due to polystyrene beads, which had blown or washed out of containers and were clogging drains. Virtually all growers have since abandoned the use of polystyrene.

Physical Characteristics—Polystyrene beads have many of the physical characteristics of perlite. They exhibit a particle density of 1.03 g/cm^3 . Polystyrene beads contain approximately 98–99% entrapped air. They are extremely lightweight with a BD of around 0.024 g/cm^3 . When added to growth mixtures, polystyrene increases aeration and drainage, and decreases the BD; it has almost

no water-holding capacity. In a study comparing the addition of 0.38 m³ of polystyrene beads or perlite to a growing mix (0.38 m³ sphagnum peat moss + 2.27 kg, calcitic lime + 1.36 kg of dolomitic lime), Matkin (2005) measured a maximum volumetric water retention of 27% for the polystyrene mix as compared to 42% for the perlite mix.

Chemical Characteristics—Chemically, polystyrene beads are composed of 94%–96% polystyrene, 4%–6% blowing agent, <0.4% styrene monomer, and <0.5% water. They are chemically inert; they supply no nutrients and are devoid of CEC or anion exchange capacity. The pH is neutral.

Sterilization, Reuse, and Waste Disposal—Polystyrene beads are highly resistant to microbial decomposition because of their large particle size and hydrophobic nature (Anonymous, 1994). They may be broken down by high temperature and by certain chemical disinfectants. They are also susceptible to gasoline and similar petroleum hydrocarbon liquids. They should not be steam heated, which could cause the beads to migrate to the top of the growth medium, which because of their light weight promotes wind and water erosion. They can be safely incinerated and will yield only carbon dioxide and water. Handling polystyrene should be done with special care from environmental point of view, because it contains CFCs; CFCs are inert, and harmless to humans and the environment upon their release. However, long after the first use of CFCs, scientists realized that they contribute to the depletion of the ozone layer of the atmosphere. Therefore, and since recycling of polystyrene mixed with other components is not possible, its use is not recommended anymore.

7.3.3 POLYESTER FLEECE

Production, Origin, and General Information—Polyester fleece is a soft, fuzzy fabric generally used for clothing or blankets, and in any other applications where a warm, wool-like material is needed. It is a two-sided pile material, which means that both the front and the back surface of the fabric consist of a layer of cut fibers, similar to corduroy or velvet. Polyester fleece is made from pure polyester fibers, which are mixed and thermally fused. The raw material for polyester fleece, polyester, is produced by heating terephthalic acid (a petroleum derivative) with ethylene glycol (another petroleum derivative, commonly known as antifreeze) at 150°C–210°C. This first reaction produces dihydroxydiethyl terephthalate, which is then pumped into an autoclave, where it is heated under pressure to about 280°C. At this temperature, the chemical transforms into a polymer called polyethylene terephthalate. As the polymer cools, it becomes a thick viscous liquid. This liquid is then extruded through a showerhead-like nozzle (tiny holes in a metal disk called a spinneret). On contact with air, the streams of liquid polymer dry and harden. In England, this polymer was called terylene; Du Pont secured exclusive US rights to the polymer in 1946, calling it polyester. If the polymer, polyethylene terephthalate, is not extruded into fibers, it can be formed into the plastic commonly used for soda bottles. Interest in recycling plastics in the 1980s led to the development of polyester fiber made from used soda bottles. Many polyester fleece garments on the market today are made from a combination of recycled and virgin polyester. The slabs of polyester fleece covered with plastic foils are used as horticultural medium. Polyester fleece is an economically attractive substrate; the amount of substrate required can be as low as 1.7–3.6 t/ha for polyester fleece as compared to 8.8–12.5 t/ha for stone wool.

Physical Characteristics—The water–air–solid ratio of polyester fleece at a matric potential of -0.5 kPa has been reported as 59:39:2 (Schroeder and Forester, 2000), which could be considered as optimum for growing horticultural crops. The pore volume of a polyester fleece slab is 98%, very close to that of a stone wool slab. It is an extremely lightweight medium. Depending on the quality of the fleece and structure of the fibers (compactness), and the height of the slabs, the BD may vary considerably. Schroeder and Forester (2000) used polyester slabs with three different heights (2, 4, and 8 cm) at the same length (100 cm) and width (20 cm), which had BDs of 0.045, 0.045, and 0.026 g/cm^3 , respectively. The water retention of polyester fleece slabs within a matric potential range from -1 to -3 kPa is significantly lower than that of stone wool slabs. As reported by Schroeder and Forester (2000), for a decrease in matric potential from 0 to -1.0 kPa, the water-holding capacity of polyester fleece sharply decreased from 98% to less than 10%, while water retention in stone wool was still remained at around 70%.

Chemical Characteristics—The polyester fleece fibers are pH neutral and pH-stable from pH 2 to 11; hence nutrient levels in the rhizosphere are easy to control. The CEC is close to 0 cmol/kg , and it is an absolutely chemically inert medium.

Sterilization, Reuse, and Waste Disposal—After use, fleece can be steam heated for sterilization, and it is reusable for several subsequent crops; it is resistant to temperatures as high as 160°C . Due to paucity of research results, the performance of used fleece slabs remains unknown. Fleece can be disposed in landfills and used for manufacturing new slabs. It might have a potential for recycling as a soil amendment (to improve physical properties of clay soils) or it could be used in substrate mixes to improve the balance of air/water retention. A key advantage of fleece products is that dried fleece waste may be incinerated.

7.4 SUBSTRATE MIXTURES—THEORY AND PRACTICE

Individual substrates with desirable and complementing properties for plant development and production (e.g., water holding capacity and aeration efficiency) are commonly mixed at varying ratios. Fine textured field soil and organic components, such as peat or compost, often lack coarse particles necessary for adequate aeration, and they hold moisture relatively tight in small pores and on particle surfaces due to capillary and adsorptive forces (Bilderback and Jones, 2001; Or and Tuller, 1999; Tuller et al., 1999). To optimize aeration and water holding properties, they are commonly mixed with coarser materials such as milled bark, sand, or aggregated media to create larger pores that rapidly drain after irrigation, thereby creating optimal rhizosphere conditions that can be tailored for a specific crop. Such mixtures often exhibit bimodal pore size distributions and water retention characteristics, where the fraction of smaller pores mainly retains water and the larger pore fraction allows for optimal aeration. During aging, decomposition of the organic components often results in a shift of particle- and associated pore-size distributions, which in turn results in a modification of the hydraulic and physicochemical mixture properties that need to be taken into account for irrigation and fertigation management. Zeolite possesses unique adsorption capacity with a high cation exchange and dehydration–rehydration ability (Mumpton, 1999; Blonquist et al., 2006). Perlite, on the other hand, is rather inert (low buffering and CEC) with the majority of its water being held superficially and released at relatively low tension, providing excellent drainage of the medium and aeration of the

rhizosphere (Marfa et al., 1993). Diatomite is a chalk-like, soft, friable, earthy, very fine-grained, porous, light-colored siliceous sedimentary rock with low density. Diatomite is essentially inert. A mixture of chemically active substrate, such as zeolite (which has poor water retention capacity) with chemically inert perlite or fine diatomite (with good air/water retention properties), is expected to yield a beneficial substrate, because of the complementary properties of the individual components. Thus, perlite or diatomite, when mixed with zeolite, improves the hydraulic properties of zeolite (Grillas et al., 2001). On the other hand, the high ion-exchange capacity of zeolite imparts the buffering properties of the media mixes, thus restricting the fluctuations of pH and nutrient concentration in the root environment. Zeolite, perlite, and diatomite, all have a relatively stable structure and can, therefore, be reused for several successive crops.

Spomer (1974) stated, “The optimum amount of amendment creates physical conditions that closely match plant requirements.” The “optimum” is commonly selected through trial and error by growing plants in a series of mixtures. Replacing this trial and error approach with physicochemical relationships for prediction of mixture behavior from well characterized constituent properties would significantly advance soilless culture production and eliminate costly mistrials. Based on this premise, the prevailing physical and chemical properties of substrate mixtures are reviewed and discussed below.

7.4.1 SUBSTRATE MIXTURES—PHYSICAL PROPERTIES

Bulk density—BD is an important physical property, which affects water holding and aeration properties and governs the root penetrability of substrate mixtures. When two components are mixed and each component contributes weight in proportion to its volume fraction, the BD of the mixture can be calculated as (Pokorny et al., 1986)

$$D_m = \frac{(V_1 C_1) D_1 + (V_1 C_2) D_2}{V_t} \quad (7.1)$$

where D_m is the predicted BD (g/cm^3) of the volume mixture; V_t is the total sample volume (cm^3); C_1 is the volume fraction of component 1; C_2 is the volume fraction of component 2; D_1 is the BD (g/cm^3) of component 1; D_2 is the BD (g/cm^3) of component 2. The linear increase in BD associated with an increase of the sand fraction in a mixture was used to predict the volume fraction of sand in unknown mixtures of sand and bark of the same particle size (Pokorny and Henny, 1984a).

When two or more substrates with different PSDs are mixed, the resultant mixture volume is smaller than the sum of the individual component volumes (Spomer, 1974) (Fig. 7.4). For example, when 0.0283 m^3 coarse pine bark is mixed with 0.0283 m^3 fine sand the resultant mixture volume amounts to $0.0425\text{--}0.0495 \text{ m}^3$ rather than to 0.0566 m^3 , the sum of the individual component volumes. Because finer particles occupy pore spaces between the particles of the coarser component, the BD of the mixture is commonly higher than the BDs of the individual components and the total and air-filled porosities lower, resulting in higher container capacity and available water (Bilderback et al., 2005; Brown and Pokorny, 1975).

In addition, the moisture content of the individual components prior to mixing is a crucial determinant for resultant mixture properties. If components are mixed in a dry state, particles tend to pack more tightly, resulting in a higher mixture BD when compared to mixing moist components (Pokorny and Henny, 1984b).

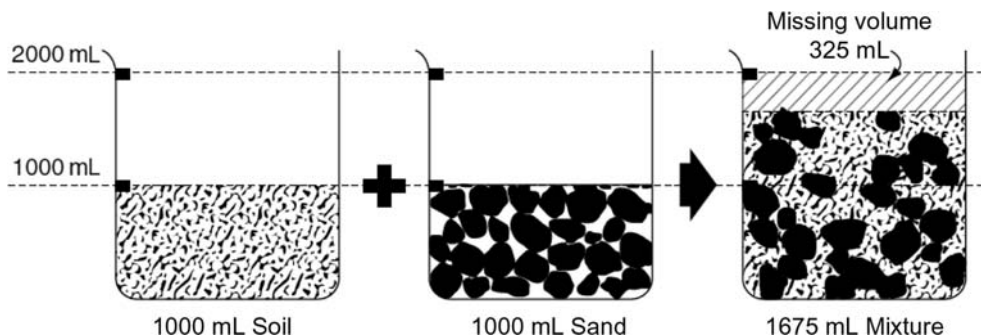


FIGURE 7.4

Sketch illustrating the relationship between individual component and total mixture volumes. While the sum of the individual component volumes prior to mixing is 2000 mL, the total mixing volume amounts to about 1675 mL.

Spomer, L.A., 1974. Hort.Science 14, 75–77, with permission from the American Society for Horticultural Science.

Nash and Pokorny (1990) determined particle size-dependent volume shrinkage for two-component mixtures (Fig. 7.5). The maximum volume reduction relative to the summed individual component volumes has been observed when coarse bark particles were mixed with fine sand particles; no volume reduction has been observed when medium bark particles were mixed with medium sand particles.

Pokorny et al. (1986), Burés and Pokorny (1991), Martinez et al. (1991), Burés and Pokorny (1991), and Burés et al. (1993a,b,c,d) were among the first to demonstrate that effective physical properties of binary substrate mixtures can be predicted from the properties of their base components. Their theoretical calculations matched experimentally determined mixture properties (e.g., particle-size distribution, BD, particle density, total porosity, and water-release patterns) reasonably well. This allows for informed selection of mixing ratios to create substrate mixes with desired rhizosphere properties for specific crops. Burés et al. (1993b,c) applied Monte Carlo simulations to predict the BD and porosity of mixtures of spherical particles of various diameters. Simulations for a mixture of pine bark and sand were in good agreement with scanning electron microscopy observations (Burés et al., 1993a). Another approach was developed by Pokorny et al. (1986), who modified Eq. (7.1) to consider volume shrinkage:

$$D_m = \frac{(V_t C_1) D_1 + (V_t C_2) D_2}{V_t - S} \quad (7.2)$$

with variables defined above for Eq. (7.1) and S is the volumetric shrinkage fraction. Pokorny et al. (1986) successfully applied Eq. (7.2) in combination with measured shrinkage values to predict the BD of sand/pine bark mixtures. Pokorny et al. (1986) modified Eq. (7.2) to predict the BD of mixtures composed of three or more components:

$$D_m = \sum_{i=1,n} \frac{(V_t C_i) D_i}{V_t - S} \quad (7.3)$$

where i is the index of summation and n the number of components. Eqs. (7.2) and (7.3) enable prediction of the BD of mixtures but they include the unknown variable S . Shrinkage curves have

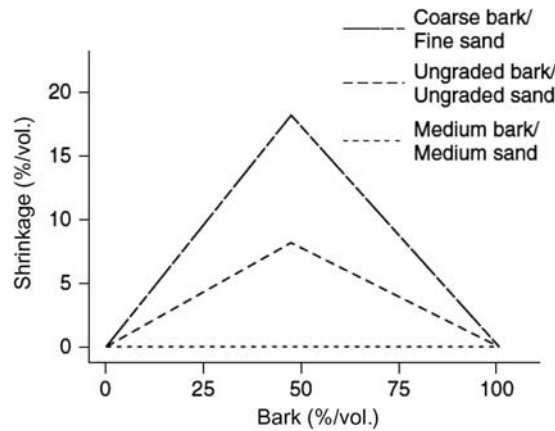


FIGURE 7.5

Shrinkage curves for potting media composed of 3 grades of pine bark (B) and sand (S). Coarse bark-fine sand: 0%–50% bark: $Y = 0.53 + 0.34B$, $df = 28$, $r = 0.98$; 50%–100% bark: $Y = 0.73 + 0.33S$, $df = 28$, $r = 0.96$. Medium bark-medium sand: 0%–50% bark: $Y = 0.36 + 0.00B$, $df = 28$, $r = 0$; 50%–100% bark: $Y = 0.36 + 0.00S$, $df = 28$, $r = 0$. Ungraded bark-ungraded sand: 0%–50% bark: $Y = 0.80 + 0.14B$, $df = 28$, $r = 0.93$; 50%–100% bark: $Y = 0.42 + 0.15S$, $df = 28$, $r = 0.94$.

Nash, M.A., Pokorny, F.A., 1990. Shrinkage of selected two-component container media. *Hort.Science* 25(8), 930–931, with permission from the American Society for Horticultural Science).

been graphically characterized for several binary mixes (Pokorny et al., 1986; Nash and Pokorny, 1990) and a mathematical expression for a binary system has been proposed by Burés and Pokorny (1991):

$$S = 2q_s S_{\max} \quad (7.4)$$

where q_s is the component with volume $< 50\%$ and $S_{\max}$ is the maximum shrinkage. It has been shown for several binary systems that $S_{\max}$ coincides with the 1:1 volumetric mixing ratio (Pokorny et al., 1986; Nash and Pokorny, 1990; Burés and Pokorny, 1991) (Fig. 7.2). Eq. (7.4) has been successfully validated for binary mixtures of peat moss/perlite, pine bark/sand, and peat moss/sand (Burés and Pokorny, 1991). Thus, the BD of any binary mixture may be calculated from the BDs of individual components and measured $S_{\max}$. Pokorny et al. (1986) obtained an excellent agreement between actual and predicted bulk densities of binary mixtures of pine bark and sand. Burés et al. (1993d) extended Eq. (7.4) for ternary systems:

$$S = 2(x_2 S_{\max 12} + x_3 S_{\max 13}) \text{ for } x_1 > 0.5 \quad (7.5)$$

$$S = 2(x_1 S_{\max 12} + x_3 S_{\max 23}) \text{ for } x_2 > 0.5$$

$$S = 2(x_1 S_{\max 13} + x_2 S_{\max 23}) \text{ for } x_3 > 0.5$$

$$S = (1 - 2x_1)S_{\max 23} + (1 - 2x_2)S_{\max 13} + (1 - 2x_3)S_{\max 12} \text{ for } x_1, x_2, x_3 < 0.5$$

Application of Eq. (7.5) has been successfully validated for ternary mixtures of pine bark, sand, and calcined clay for a variety of volumetric mixing ratios. Jones and Or (1998) combined the

fractional packing model proposed by Koltermann and Gorelick (1995) that requires measurement of minimum mixture porosity with an empirical relation of Furnas (1931) for predicting the porosity of a binary mixture of fine and coarse glass beads of different diameter ratios as a function of the coarse component fraction. Measured porosities for coarse component mixtures of 0.33, 0.5, and 0.67 were slightly overestimated by the model ($d/D = 0.5$) (Fig. 7.6). Jones and Or (1998) found that the geometric mean yielded better estimates for porosity than the harmonic mean for a New Jersey soil and for a mixture of coarse and fine sand particles (Fig. 7.3).

Water Retention—The WRC is a crucially important substrate characteristic, as it determines water availability and aeration for growing plants. The WRC is intimately related to the substrate pore size distribution and BD as discussed in Chapter 3, Physical Characteristics of Soilless Media. For mixtures of two or more base materials, the WRC is governed by the distribution and sizes of pores within each individual component and the new pore spaces established due to mixing (Bunt, 1974; Burés et al. 1993a,b,c,d; Spomer, 1974). A sketch illustrating the water retention characteristics and air-filled porosities of field soil, sand, and a mixture thereof (Spomer, 1974) is depicted in Fig. 7.7.

Sahin et al. (2002) found that the water retention properties and pore size distributions of 1:1 volumetric binary mixtures of perlite, pumice, and creek sand at matric potentials > -330 cm H_2O were quite predictable from the properties of the individual base substrates. For matric potentials below -330 cm H_2O (drier conditions), the water retention capacity of the perlite/pumice mixture was higher than for the perlite/sand and pumice/sand mixtures. The ratio of macropores (>100 μm) that determine air-filled porosity and meso- (30–100 μm) and micropores (3–30 μm) that

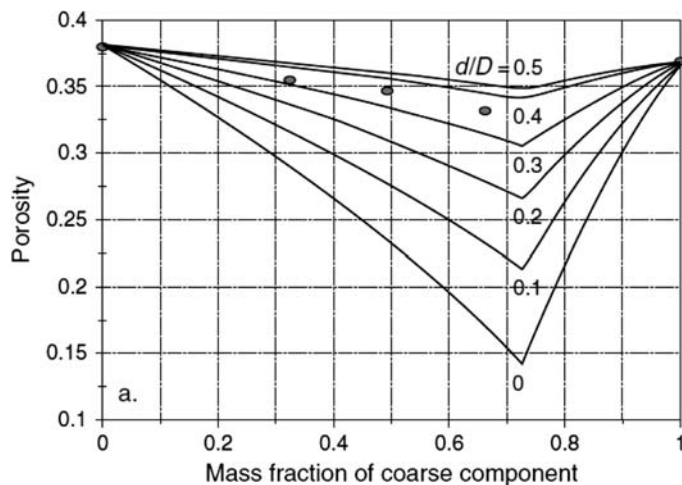


FIGURE 7.6

Estimated porosity as a function of the mass fraction of the coarse component of binary mixtures whose diameter ratios, d/D , vary from 0 to 0.5. Data points are measured porosities for varying mixing ratios of 100 and 200 μm diameter glass beads.

Jones, S.N., Or, B., 1998. Design of porous media for optimal gas and liquid fluxes to plant roots. *Soil Sci. Soc. Am. J.* 62, 563–573, with permission from the American Society of Soil Science.

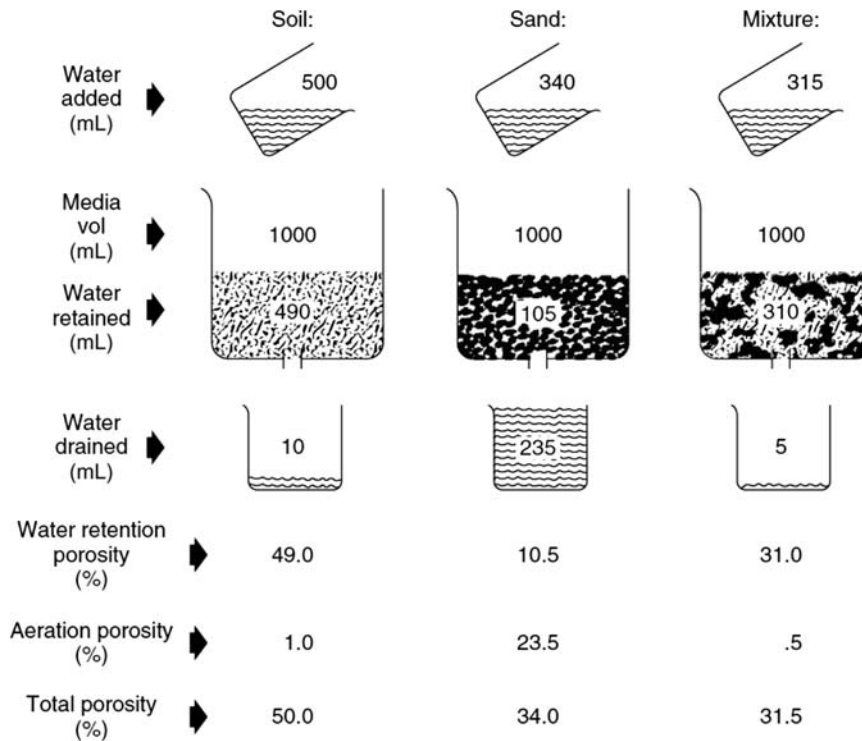


FIGURE 7.7

Sketch illustrating the water retention characteristics and air-filled porosities of field soil, sand, and a mixture thereof.

Spomer, L.A. 1979. Three simple demonstrations of the physical effects of soil amendment. *HortScience* 14, 75–77.

determine water-holding capacity was comparable for the perlite/sand and perlite/pumice mixtures, but higher than for the pumice/sand mixture. The perlite/pumice mixture contained a higher fraction of pores $<3\ \mu\text{m}$ than both other mixtures. Nash and Pokorny (1992) derived a water retention model for mixtures that considers volume shrinkage:

$$\text{WR}(x) = \frac{\sum_{i=1}^n \text{WR}(x)_i C_i}{(V_t - S)} \quad (7.6)$$

where $\text{WR}(x)$ is the water retained within the mixture at matric potential x , i is the index of summation, n is the number of base substrates, $\text{WR}(x)_i$ is the retained water in the base substrates at matrix potential x , C_i is the volume fraction (%) of the base substrates, V_t is the total mixture volume, and S is the volume shrinkage fraction (%). While water retention estimates based on Eq. (7.6) were in reasonable agreement with measurements for a wide range of pine bark/sand mixtures (0%–100% pine bark) at matric potentials of 0, -10 , -50 , and $-100\ \text{cm H}_2\text{O}$, this weighting approach was not independently validated for other substrate mixtures; hence should be applied with caution.

The physico-empirical model proposed by Arya and Paris (1981) for prediction of the WRC from the PSD was adopted for mixtures consisting of two components by Jones and Or (1998). Estimates based on the optimized Arya and Paris (1981) model matched measured retention curves for sand reasonably well (Fig. 7.8A), but estimates for glass beads show a significant deviation from measured data mainly at the dry and wet ends of the SWC (Fig. 7.8B). The application of this approach for substrate mixtures is lacking, but the WRCs for a binary mixtures (at varying ratios) of fine (100 μm) and coarse (200 μm) glass beads fall between the WRCs of the individual glass bead media for a matric potential range from -0.4 to -0.8 m (Fig. 7.9).

Weiss et al. (1998) found that the values of the empirical parameters α and ν of Van Genuchten (1980) model for various peat soils are dependent on the BD and the percentage of botanical components Sphagnum and Carex. They proposed an alternative semiempirical exponential model for the retention curve that contains a single free parameter, k , which is related to BD and the percentage of the botanical component Sphagnum and the lignin content.

Hydraulic Conductivity—The saturated, K_s , and unsaturated, $K(h)$, hydraulic conductivities determine water flow and nutrient transport through the growth substrates. The decrease of $K(h)$ in soilless substrates with decreasing water content (θ) is much steeper than in field soils (Raviv and Blom, 2001). A minute decrease in θ (e.g., of several percent) may lead to a $K(h)$ decrease by an order of magnitude or more; hence, greatly affecting water availability to plant roots (Wallach et al., 1992a,b; Raviv et al. 1999, 2001). It has been shown that in substrates, the main limiting factor to water availability and thus to water uptake by the roots is $K(h)$ and not the matric potential (da Silva et al., 1993; Raviv et al., 2001). For example, $K(h)$ of stone wool is especially low under less than saturated conditions (de Swaef et al., 2012). This concept of water availability and its dependence on hydraulic conductivity of mixed growing substrates is presented and discussed in

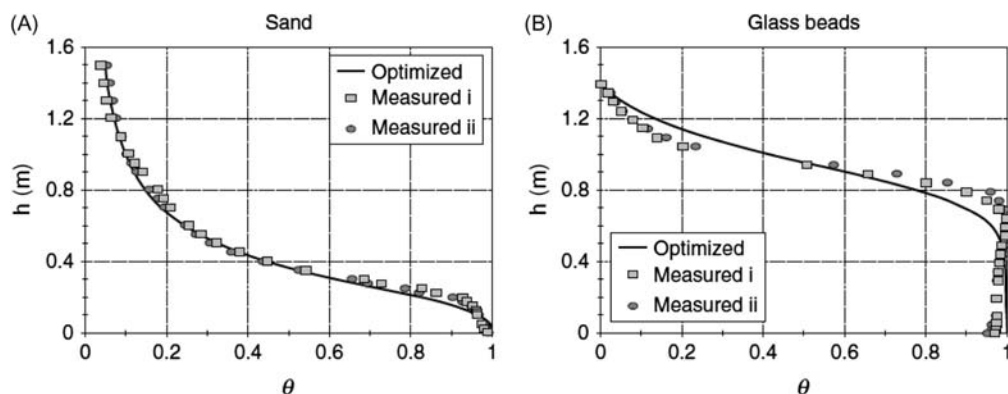


FIGURE 7.8

Comparison of optimized and measured water retention curves for (A) sand and (B) glass beads, where h is the absolute value of the matric potential and θ is the relative water saturation.

Jones, S.N., Or, B., 1998. Design of porous media for optimal gas and liquid fluxes to plant roots. *Soil Sci. Soc. Am. J.* 62, 563–573, with permission from the American Society of Soil Science.

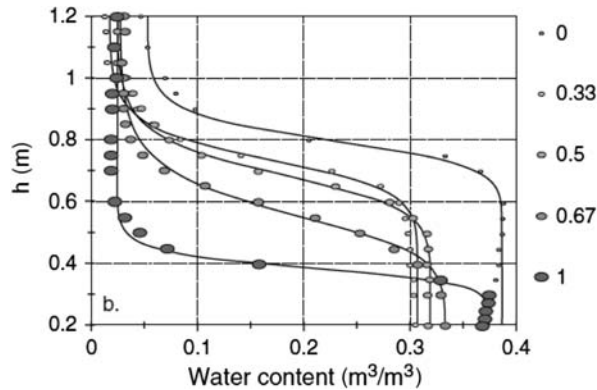


FIGURE 7.9

Water retention curves for mixtures of glass bead sizes of 100 and 200 μm of different mass fractional contents of the coarse (200 μm) component (0%–100%).

Jones, S.N., Or, B., 1998. Design of porous media for optimal gas and liquid fluxes to plant roots. *Soil Sci. Soc. Am. J.* 62, 563–573, with permission from the American Society of Soil Science.

Chapter 3, Physical Characteristics of Soilless Media (Section 3.4.5) and was recently studied by Schindler et al. (2016).

The saturated hydraulic conductivity, K_s , of a mixture can be estimated with the Kozeny–Carman equation (Bear, 1972):

$$K_s = \left(\frac{\rho_w g}{\mu} \right) \frac{d_m^2 \eta^3}{180(1-\eta)^2} \quad (7.7)$$

where ρ_w is the liquid density (Mg/m^3), g is the acceleration due to gravity (m/s^2), μ is the dynamic viscosity (Pa s), η is the total porosity (m^3/m^3), and d_m is the mean grain diameter (m), which is calculated from either the geometric or harmonic mean as described in Koltermann and Gorelick (1995). This method yields K_s estimates for soils that are generally within one order of magnitude of measured values (Koltermann and Gorelick, 1995). Application of this model with the geometric mean grain diameters by Jones and Or (1998) yielded a K_s of $8.6 \times 10^{-5} \text{ m/s}$ for a mixture of sand particles as compared to a measured K_s of $4.6 \times 10^{-4} \text{ m/s}$.

If the K_s is known, $K(h)$ curves for mixtures can be calculated with Eq. [3.3.9] based on the retention curve as shown for composted agricultural wastes and their mixtures with tuff (Wallach et al., 1992b), sphagnum peat moss mixtures with tuff (da Silva et al., 1993), sphagnum peat moss mixtures with coarse perlite (Heiskanen, 1995a,b, 1999), and sphagnum peat moss mixtures with perlite and sand (Heiskanen, 1999).

We started this section with a statement of Spomer dating back four decades about the need to eliminate the inefficient and inaccurate procedure of trial and error in developing mixtures with favorable rhizosphere properties. Since then there has been considerable progress in understanding of the physical properties of substrates and our ability to estimate the BD, porosity and water retention of mixtures based on the properties of their base components and measured maximal shrinkage [Eqs. (7.2)–(7.6)]. Actual measurements of these properties are still advisable, because most

models assume ideal spheres, which are not representative for the intrinsic irregular shape of natural and artificial substrates and mixtures thereof (Bunt, 1988). However, with accepting the concept of water availability and its dependence on the momentary hydraulic conductivity of the growth substrate, there is a need for accurate prediction of K_s and $K(h)$. Though there has been some progress in roughly estimating K_s (Koltermann and Gorelick, 1995; Jones and Or, 1998) as discussed above, an empirical approach is still required to determine mixture K_s . Nevertheless, the knowledge gained about the relationship between K_s , the WRC and $K(h)$ is significant, as it eliminates the need for extremely laborious and challenging measurement of mixture $K(h)$ for matric potential ranges relevant for soilless culture.

7.4.2 SUBSTRATE MIXTURES—CHEMICAL PROPERTIES

Some of the substrates discussed in the preceding sections, such as stone wool, perlite, pumice, polyurethane foam, and others are chemically inert, while others, such as organic materials, tuff and zeolite, exhibit a considerable ion-exchange capacity and buffering properties for fluctuation of pH and nutrient availability. While the hydraulic and physicochemical properties of distinct substrates are well defined, chemical properties (e.g., CEC, pH, buffering capacity, etc.) of mixtures result from complex interactions of the base mixture components.

When a mixture is composed of two substrates that do not react with each other, the chemical properties of the substrate are expected to be in proportion to the values of the individual component fractions. Thus, the CEC of a milled pine bark/sand mixture decreased with increasing sand fraction percentage (Brown and Pokorny, 1975). CEC was most accurately determined on a volume basis because of the large difference in BD between milled pine bark and sand (Brown and Pokorny, 1975). The CEC of two-component (pine bark and sand) mixtures was predicted from laboratory analysis of each component with a model similar to Eq. (7.2) for the BD (Nash and Pokorny, 1990):

$$CEC_m = \frac{C_1 CEC_1 + C_2 CEC_2}{C_1 V_t + C_2 V_t - SV_t} \quad (7.8)$$

Increasing the percentage of sand in a potting substrate mixture of milled pine bark and sand from 0% to 100% resulted in a linear increase of the pH from 4.1 to 5.4 (Brown and Pokorny, 1975). However, there are no published models about the effects of chemical reactions between components on the combined chemical properties. Research to date is rather empirical providing recipes for modifying chemical substrate properties with amendments. For example, fly ash is applied to raise pH and alter chemical properties of substrate mixtures (Bilderback et al., 2005). The produced organic acids and microbial enzymes involved in the decomposition of organic components may also interact both physically (through mere physical adsorption) and chemically (through sorption, complex formation, and dissolution) with inorganic components such as expanded clay granules, vermiculite, pumice, and tuff. Clays are also known to abiotically modify several reactions of OM decomposition through catalytic action (Huang, 2000). Thus, clay can affect the chemical properties of substrate mixtures. The interactions of organic compounds with inorganic minerals, especially clays, are extensively discussed in the soil literature and briefly presented in Chapter 4, Chemical Characteristics of Soilless Media (chemical characteristics). Organic compounds, especially the active and soluble fractions (humic and fulvic acids), may cause

decomposition of the solid phase (Tan, 1986) and/or modification of the surface charge (Tate and Theng, 1980; Huang and Violante, 1986). The effects of organic exudation and root residues on chemical properties of mineral substrates are discussed in Chapter 4, Chemical Characteristics of Soilless Media. The main effect lies in an increase of negative surface charge sites of tuff or other mineral substrates due to organic acids. Chapter 4, Chemical Characteristics of Soilless Media, presents the contrasting effects of low weight, soluble organic acids exudation by roots on the pH of intact substrate and a substrate with a high surface charge; similar effects have been reported for the reaction of soluble organic acids with various inert minerals and those with charged surfaces. In mixtures of organic and inorganic substrates, both mechanisms of decomposition of the solid phase and modification of the surface charge sites of the mineral components may occur simultaneously. Moreover, the potential effect of the inorganic component may affect the structure of the organic component and/or its decomposition products; therefore, it is challenging to predict the chemical properties of mixtures. The principles of creating substrate mixtures to regulate nutrient availability in the rhizosphere and to develop substrates that might contribute nutrients during the growth period of horticultural crops are described in Chapter 6, Nutrition of Substrate-Grown Plants.

7.4.3 SUBSTRATE MIXTURES—PRACTICE

Despite the importance of the hydraulic conductivity for management of soilless growth systems there is little research on this parameter for substrate mixtures. Spomer (1974) suggested that drainage of organic substrates may be improved by either increasing the bed depth or by mixing with coarse inorganic substrate amendments. Mixing sand with organic components is a common practice, resulting in lower infiltration rates of irrigation water, particularly in fresh or less decomposed organic components (Bilderback and Jones, 2001). The percolation rate declined with increasing sand percentage in a milled pine bark/sand mixture (Brown and Pokorny, 1975). While lower infiltration rates in sand-amended substrate mixtures promote more thorough wetting of substrate particles due to longer contact times with the infiltrating water, in purely organic substrates irrigation water rapidly percolates to the bottom (Bilderback and Jones, 2001). da Silva (1991) found that the WRC of a 1:1 (v:v) mixture of tuff and peat was slightly closer to the curve of the peat than to that of the tuff while the hydraulic conductivity of the mixture was very close to that of peat for the studied container height (25 cm).

Another important issue for the management of soilless substrates is the modification of hydraulic and physical substrate properties throughout the growth season due to sequential increase of root mass. There is clear experimental evidence that plant roots successively displace macropore space. Fig. 7.10A illustrates how root-induced dynamic displacement of macropore space modifies the WRC as dwarf rice roots grow within Profile, a substrate consisting of stabilized baked ceramic aggregates, over a 6-week period (Jones et al., 2012). In this experiment the root zone was purposely reduced to a 1000 cm³ volume, creating a confined space for root growth under a high-intensity light source. The water characteristic evolved with time moving from 71% saturation to 60% after 42 days, a pore space reduction of 11%. The two extreme water characteristic curves associated with Days 1 and 42 are shown in Fig. 7.10B. The large difference between the two curves is evidence that root intrusion can significantly alter the porous medium hydraulic properties. Besides plant roots, hydraulic substrate properties may be significantly altered as a result of microbial growth (Martin, 2010; Vandevivere and Baveye, 1992; Seki et al., 1998;

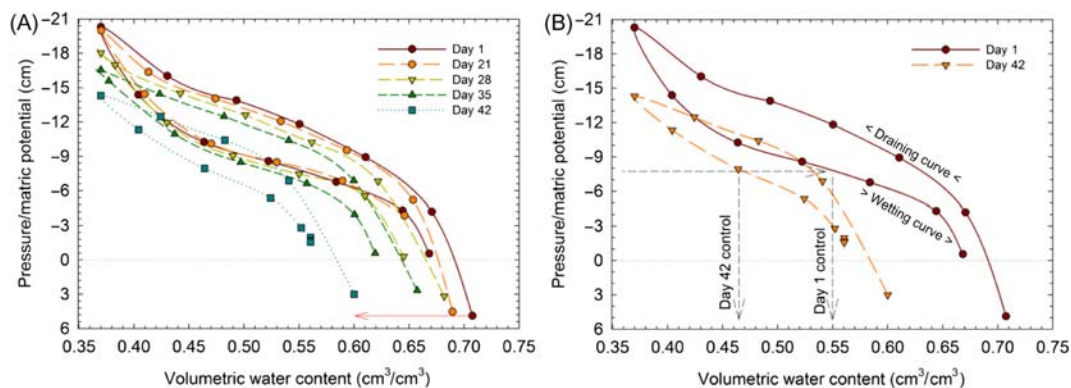


FIGURE 7.10

Substrate water retention curves for Profile (0.25–1 mm) plant growth medium showing (A) interaggregate pore space displacement by rice roots and (B) shifting water contents related to matric potential–based set points (Jones et al., 2012).

Rockhold et al., 2002). Root and microbial growth rates are important factors for selection of container geometry and size.

Peat moss-based substrates are popular due to their high nutrient buffering capacity, good aeration, and suitability for various crops (Caron et al., 2005; Papadopoulos and Tan, 1991; Steidman, 1988; Sheldrake, 1989). When lightweight organic materials such as peat are used in substrate mixtures, a fine sand or clay are often added to increase the BD for improved mechanical stability. Mixtures with very low BD are not suitable for outdoor nurseries due to stability concerns. Sand and clay also increase the wettability of the mixtures (Guttormsen, 1974). However, undesirable effects of adding sand or clay are a reduction in total and air-filled porosities (Bunt, 1974); it is well documented that water and nutrient management is difficult when the air-filled porosity is low. Bunt (1983) determined the principal physical properties of a mixture of peat (100, 75, 50, 25, and 0 vol.%) and fine- and coarse-grained mineral substrates (0, 25, 50, 75, and 100 vol.%). The investigated mineral substrates included perlite, vermiculite, calcined clay, and sand. In general, the total porosity was inversely related to BD, and there was no interaction between the PSD of the mineral substrates and total porosity. Except the fine-grained perlite/peat mixtures exhibited lower air-filled porosity values, which remained fairly constant for all mixing ratios, than observed for the individual components. In contrast, addition of coarse-grained mineral substrates progressively increased the air-filled porosity of the mixture with increasing mineral substrate content; this behavior was observed irrespective of mineral substrate type. Easily available water was not significantly affected by the addition of fine-grained mineral material. However, the coarse-grained mineral material reduced the easily available water, especially when it was present in the mixture at more than 25 vol.%. Addition of fine- and coarse-grained vermiculite or perlite resulted in much higher easily available water than observed for other minerals (Bunt, 1983). Bunt (1991) observed a remarkable increase in BD and decrease in total and air-filled porosities as a consequence of mixing fine (25–50 vol.%) or coarse (25 vol.%) sand with peat. However, fine perlite (25 vol.%) in peat-based substrates caused only a slight increase in BD without affecting total and air-filled

porosities. Brückner (1997), on the other hand, did not observe any noticeable change in the physical and hydraulic properties of substrate mixtures that were prepared by adding various proportions of clay (10–35 vol.%), perlite (15–30 vol.%), or sand (20 and 50 vol.%) to peat in the presence or absence of other organic constituents (wood chips, pine bark, and/or rice husks). Substrate mixes prepared by combining peat moss with the inorganic components perlite, pumice, or creek sand at a 1:1 volume ratio exhibited BD, water retention characteristics, and pore-size distributions that fell in between the values observed for the individual components; similar results were also obtained when peat or sawdust was mixed with perlite, pumice or creek sand at a 1:1 volume ratio (Sahin et al., 2002). Verhagen (2004) observed that when coarse- and fine-grained clay substrates (5–25 vol.%) were mixed with peat, the clay component significantly affected water uptake and retention characteristics of the mixtures, with the fine-grained clays leading to a more distinct increase in BD and decrease in total porosity and easily available water than the coarse-grained clays. The fine-grained clays induced a large increase in the unsaturated hydraulic conductivity, even when the volume fraction added to peat was relatively low (Verhagen, 2004). It was suggested that the effectiveness of clay amendments to substrate mixtures could not always be attributed to specific chemical characteristics of the clays; physical effects and interaction with cultivation systems (irrigation frequency and fertilization) also play an important role (Verhagen, 2004). Riga et al. (2003) evaluated the physical properties of 4 mixtures of peat and used blended stone wool (from 0 to 60 vol.%) as a growth medium for geranium (*Pelargonium peltatum* L.) production and applied two matric potentials (–15 and –10 kPa) as initial set points for fertigation control. Obtained results indicated that addition of used blended stone wool to peat resulted in an increase of air space and a decrease of volumetric water content. When the matric potential decreased from 0 to –10 kPa, the volumetric water content in the blended stone wool decreased by 85%, while the water content of peat only decreased by 62% relative to the saturated water content. However, the easily available volumetric water content was only minimally affected when used stone wool was incorporated with the peat substrate. Used stone wool exhibited a higher soluble minerals content than peat. Nevertheless, plants grown in the mixture with the highest volumetric stone wool content (i.e., 60 vol.%) had inadequate market quality. When fertilizing plants at –10 kPa, all parameters remained similar to the control, except for the plants that have been grown in the mixture with 60 vol.% stone wool, where a lower biomass production and similar production of inflorescences per plant and market quality were observed. It has been concluded that blended recycled stone wool can be used successfully as a component of peat-based media for geranium production as long as the fertigation schedule is modified according to the physical properties of the applied medium. Several previous studies also indicated the viability of substrate mixtures containing previously unused stone wool, wood chips, peat moss, coconut coir and perlite (Kim et al., 2000), or municipal compost (Bilderback and Fonteno, 1993); all these studies reported that the addition of stone wool to soilless substrates improved the physical and hydraulic characteristics, resulting in a favorable rhizosphere environment for crop growth and production.

Stone wool additions to compost, peat or other substrates generally results in improved aeration efficiency and an increase of water-holding capacity of the resultant mixtures (Fonteno and Nelson, 1990; Choi et al., 1999, 2000). Research conducted in Korean (Jeong, 2000; Kim et al., 2000; Lim and Jeong, 2000; Shin and Jeong, 2000; Jeong and Hwang, 2001; Kim and Jeong, 2003) emphasized the importance of (1) grinding the used stone wool and other mixture components to a suitable size distribution; (2) soaking and weathering specific mixture constituents, particularly

bark and wood chips, to remove chemicals harmful to plant growth; and (3) steam-sterilizing used ground stone wool before its incorporation in the mixture.

Granulated or powdered clays are recommended as constituents in substrate mixtures to provide buffering capabilities against pH, EC, and nutrient concentration fluctuations. [Martinez et al. \(1997\)](#) added a granulated (2–8 mm) and a powdered commercial clay material to peat/coconut coir mixtures and evaluated modifications of physicochemical properties as a result of the interaction of the mixture components. The clay amendment resulted in significant increases in BD, easily available water, available water, and pH, and significant decreases in total and air-filled porosities, water retention at -1 kPa, OM content and EC. No significant effect on water buffering capacity and water retention at -2 kPa was observed. The effects of granulated and powdered clay on air/water parameters and EC were similar, but pH was significantly higher when powdered clay was added ([Martinez et al., 1997](#)). When the proportion of coconut coir was increased (from 3:1 to 1:3 vol.% peat:coir), significant decreases of BD and OM content, and increases in total and air-filled porosities and pH were observed relative to the properties of the peat and coir base materials. An increased of the coir proportion led to significant increases in easily available water and available water. Overall, the aeration and water-holding properties of the mixtures exhibited a nonlinear bulking effect of coir when mixed with peat that seemed to be saturated when coir was one-half of the mixture volume. This effect was attributed to the interaction between the rigid coir fibers and the flexible peat fibers; coir fibers can establish a relatively rigid framework that promotes an open rearrangement of peat fibers. Mixtures with organic materials such as peat, pine bark, and coir can be difficult to hydrate initially, and also rewetting after drying might be challenging. Clay and mineral substrates in general can be amended to enhance wettability (i.e., the ability of a dry medium to rapidly absorb water) of peat-based substrates ([Bunt, 1988](#)). [Airhart et al. \(1978\)](#) reported that an air-dry peat/vermiculite mixture required 5 days to reach 70–78 vol.% moisture saturation, while milled pine bark required 48 days to achieve 58–78 vol.% saturation.

[Issa et al. \(2001\)](#) investigated seasonal and diurnal photosynthetic responses and flower yield of two gerbera cultivars to perlite, zeolite, and a 1:1 volume mixture thereof. They concluded that gerbera plants grown in the 1:1 mixture exhibited the best physiological responses. [Samartzidis et al. \(2005\)](#) studied productivity and physiological responses of rose to zeolite (z) and perlite (p) substrates at various volumetric ratios (25z:75p, 50z:50p, 75z:25p, and 100z:0p). Increasing the fraction of perlite in the mixture reduced water uptake and net photosynthesis of rose at various stages. This observation was explained by the increase of easily available water as the fraction of zeolite increased. Both zeolite and perlite used in this study were coarse. During the mixing and production phase, an alteration (fragmentation) of perlite usually occurs, whereas zeolite is not affected because of its rigid structure ([Orozco and Marfa, 1995](#)). It has been hypothesized that perlite particles smaller than 3 mm in diameter decreased the percentage of macropores ($>60\text{ }\mu\text{m}$) and increased the percentage of pores with a diameters between 30 and $60\text{ }\mu\text{m}$ ([Samartzidis et al., 2005](#)). Consequently, an increase of the total available water content of the growth medium with high zeolite fraction occurred, improving water status and nutrient uptake as well as alleviating midday water stress. The studies discussed above ([Issa et al., 2001](#); [Samartzidis et al., 2005](#)) demonstrate that physical and chemical characteristics of substrate mixtures play an important role for plant growth and crop production. However, conclusions from such comparisons have to be drawn with caution, because the optimal irrigation and fertilization strategies are different for each substrate as has been shown in [Riga et al. \(2003\)](#). Irrigation management as well as growth container

heights are crucial determinants for development of optimal mixtures (Jones and Or, 1998; Caron et al., 2005). A high fraction of sphagnum peat mixed with coarse pine bark and sand yielded the best capillary rise and highest growth for nursery trees grown in pots with a subirrigation system (Caron et al., 2005).

7.5 CONCLUDING REMARKS

The selection of an appropriate growth medium for a specific plant species, together with appropriate water and nutrient management, have large impacts on both the yield and quality of soilless culture plant production. As reported by Olympios (1992), inert media do not react with nutrient ions, and they do not possess an adsorption complex. This allows precise adjustment of nutrient levels to match actual plant requirements (Sonneveld and Spaans, 1990). Besides chemical properties, even more important are the physical properties that govern plant available water and aeration efficiency and provide mechanical support for the plant root system. While chemical properties of the rhizosphere can be adjusted during the growth season, little can be done to modify physical and hydraulic substrate properties after planting; hence, the initial plant-specific substrate selection is of crucial importance. Because the hydraulic properties are dynamically changing with increasing root volume and due to microbial activity, they need to be carefully monitored and the irrigation strategy adjusted for optimum yield and high-quality produce. Because soilless substrates are commonly sterile and they can be easily steam-sterilized or fumigated for multiple reuse (van Os et al., 1991), they are advantageous with regard to plant health when compared to field soils.

Despite a worldwide trend of growing plants in various soilless production systems rather than in field soils, the selection of the appropriate soilless substrate for a specific crop remains a largely empirical endeavor, often relying on locally available raw materials and economic considerations rather than on the physical principles that govern water- and air-flow and nutrient transport in porous media (Jones and Or, 1998). Recurrent problems with plant hypoxia and water stress have prompted the need for refined selection criteria for optimal soilless growth substrates. It is also interesting to note that even after five decades of research, an in-depth understanding of the relationships between physicochemical substrate properties and plant response and performance is still lacking (Raviv et al., 2004). The optimal control of water availability and aeration is an essential prerequisite to successfully operate soilless systems. To date there are no accepted universal standards for physical properties of horticultural substrates. Yeager et al. (1997) provided property ranges for “easy-to-manage” (i.e., irrigation and nutrients) substrate mixtures applied for commercial horticultural crop production: total porosity (50%–85%), air-filled porosity (10%–30%), container capacity (45%–65%), available water (25%–35%), unavailable water (25%–35%), and BD (0.19–0.7 g/cm³). Other researchers dispute the usefulness of such property ranges, as the optimal target properties are dependent on the plant variety, growth conditions, and the economic considerations of the growers. The approach proposed by Jones and Or (1998) seems promising. They applied nonlinear least squares multiobjective optimization to maximize liquid and gas phase distributions and fluxes to plant roots, while matching media hydraulic properties with a control system design. However, this approach has only been tested for mixtures of sand particles and glass beads. More recently, Chamindu Deepagoda et al. (2013a) proposed a simple gas diffusivity mixing model and defined a critical window of diffusivity to assist

with the selection of mixing ratios of base substrates to create mixtures with favorable aeration and water retention properties. They also defined integral parameters for water, energy, and aeration properties for a variety of soilless substrates that can potentially aid with the selection of optimum growth media and provide set points for irrigation management (Chamindu Deepagoda et al., 2013b). These new concepts are awaiting validation in greenhouse trials. Finally, another avenue that can potentially advance substrate selection prior to greenhouse trials and provide critically needed tools for control and management of containerized soilless production systems is numerical 3-D modeling of water, gas, and nutrient fluxes. Modeling results can provide a more scientifically sound basis for the design of commercial greenhouse production trials and selection of adequate plant-specific substrates, thereby alleviating the risk of costly mistrials.

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ORGANIC SOILLESS MEDIA COMPONENTS

8

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8.1 INTRODUCTION

Soilless culture of plants in containers involves a restricted root system and a reduced root zone volume when compared with soil-based production. While details related to this are covered in [Chapter 2](#), Soilless Production Agriculture, and [Chapter 14](#), Growing Plants in Soilless Culture: Operational Conclusions, it is important here to note that under these conditions the physical and chemical properties of the growing medium must be such as to provide adequate storage of water and nutrients for the plant whilst maintaining good aeration.

Historically, materials such as locally available soils were used to grow plants in pots and containers. Often, organic materials such as manures and composts were used to supplement the inorganic base materials, improving water retention and airspace as well as offering nutritional benefits. Many formulations of combinations of soils with organic materials, notably peat and manures, are reported in the 19th century articles in magazines and elsewhere ([Anon, 1874](#)), and even a few reports exist of growing media containing exclusively organic materials ([Anon, 1830](#)).

The scientific studies of [Lawrence and Newell \(1939\)](#) in England on development of standardized growing media of consistent pH, absence of pathogens, pest and weeds, and without nutritional imbalances (the so-called John Innes composts) recommended the inclusion of 25% organic matter in the form of moss (sphagnum) peat. However, their media required the acquisition and sterilization of good-quality loam soil, and difficulties in obtaining such loam and cost of its sterilization led to its popularity declining among professional growers post-1950. In the United States during the 1950s, growing media principally based on peat were developed at the University of California ([Baker, 1957](#)). Further studies in Germany ([Penningsfeld and Kurzmann, 1966](#)) and Ireland ([Woods and Kenny, 1968](#)) showed that peat alone could form the basis of growing media. Peat use in horticultural growing media then expanded rapidly. In the early part of the 21st century, growing media used commercially in many parts of the world are prepared mainly from peat along with other organic materials: the major exceptions being mineral wool substrates used for glasshouse crops such as tomatoes, cucumbers, and pepper.

Data from [Schmilewski \(2017; Tables 8.1 and 8.2\)](#) illustrate the dominance of peat within organic growing media manufactured in Europe. Peat production and use has decreased in some countries, for example, in the United Kingdom peat use in horticulture declined from 2.5 M m³ in 1997 to <2.0 M m³ in 2015: Indeed, in the United Kingdom, a progressive phase-out of peat in growing media for retail markets by 2020 and professional markets by 2030 is now government strategy ([Carlile and Waller, 2013](#)). In contrast, peat production for growing media has expanded in some of the Baltic states, for example, in Latvia from 0.65 M m³ in 1997 to 2.85 M m³ in 2009 ([Paappanen and Leinonen, 2010](#)). Elsewhere, Canada, with over 1 M km² of peat reserves, produced about 17 M m³ of peat for use in horticulture in 2015 ([CSPMA, 2017](#)), and Russia may produce similar or slightly greater volumes for horticultural purposes ([Anon, 2015](#)).

Peat is a carbon store, a semifinite resource, regenerating only slowly after extraction, and since the late 1980s, environmental lobbies have urged peat harvest reduction and replacement of peat by other organic materials regarded as renewable ([Carlile, 1997; Carlile and Waller, 2013](#)). Dominant

Table 8.1 Amounts of Peat (in m³ in 000's According to [European Committee for Standardisation, 1999: EN 12580](#)) Used for the Production of Growing Media in 16 EU Countries

Country	Bog (<i>Sphagnum</i>) Peat				Fen Peat	
	H1–H5 ^a		H6–H10 ^a		H1–H10 ^a	
	Pro.	Hob.	Pro.	Hob.	Pro.	Hob.
Austria	40	100	5	40	0	0
Belgium	225	135	275	175	0	0
Denmark	63	60	27	140	0	0
Estonia	677	305	60	96	0	0
Finland	340	200	100	160	0	0
France	475	254	474	253	27	58
Germany	2000	1600	1900	1300	0	0
Ireland	119	20	130	500	0	0
Italy	1100	507	650	185	0	1
Latvia	1366	455	98	33	0	0
Lithuania	1074	366	380	142	0	0
The Netherlands	1294	233	1118	288	38	21
Poland	600	410	220	290	50	250
Portugal	2	2	0	0	0	0
Sweden	180	600	60	150	10	60
United Kingdom	143	549	315	302	0	115
Total	9698	5796	5812	4054	125	505

Pro., Professional market; Hob., hobby market.

^aDegree of peat humification.

Data from [Schmilewski, G., 2017. Growing media constituents used in the EU in 2013. Acta Hortic. 1168, 85–92 and reproduced by kind permission of the International Society for Horticultural Science.](#)

Table 8.2 Amounts of Organic Materials Other Than Peat (in m³ in 000's According to European Committee for Standardisation CEN, 1999: EN 12580) Used for the Production of Growing Media in 16 EU Countries

Country	Bark (Nontreated)		Coir (Pith, Fibers, Chips)		Wood Fibers		Wood (Chips, etc.)		Rice Hulls		Heather Soil ^a	
	Pro.	Hob.	Pro.	Hob.	Pro.	Hob.	Pro.	Hob.	Pro.	Hob.	Pro.	Hob.
Austria	0	0	1	1	5	20	0	0	0	0	0	0
Belgium	15	5	25	5	5	10	0	0	0	0	0	0
Denmark	0	0	1	0	1	0	0	0	0	0	0	0
Estonia	0	0	3	0	0	0	0	0	0	0	0	0
Finland	0	15	0.5	0	0	0	0	4	0	0	0	0
France ^b	170	268	69	22	114	153	0	0	0	0	14	60
Germany ^b	41	48	60	27	139	205	5	1	0	0	0	0
Ireland	0	0	0	77	0	0	0	0	0	0	0	0
Italy	10	50	300	140	0	0	0	0	0	0	0	0
Latvia	0	0	100	50	1	0	0	0	0	0	0	0
Lithuania	0	2	0	0	0	0	0	0	0	0	0	0
The Netherlands ^b	185	20	316	51	14	0	40	14	12	0	0	0
Poland	0	0	0	0	0	0	0	0	0	0	0	0
Portugal	25	25	3.5	0	0	0	0	0	0	0	0	0
Sweden	1	3	3	0	0	0	0	0	0	0	0	0
United Kingdom ^c	32	0	29	30	30	400	40	195	0	0	0	0
Total	479	436	911	403	309	788	85	214	12	0	14	60

Pro, Professional market; Hob, hobby market.

^aHeather soil is defined as surface soil which accumulated in heather covered areas or pine forests.

^bIn addition, France reported 2000 m³, Germany 1000 m³, and the Netherlands 5000 m³ of imported peat moss (*Sphagnum* spp.) for the professional market.

^cThe United Kingdom reported additional but unspecified 8000 m³ for the professional and 14,000 m³ for the hobby market.

Data from Schmilewski, G., 2017. Growing media constituents used in the EU in 2013. *Acta Hort.* 1168, 85–92 and reproduced by kind permission of the International Society for Horticultural Science.

among the latter is coir, the use of which has increased greatly worldwide from the early years of this century (Carlile et al., 2015). Other, often locally available, organic materials are incorporated into growing media, including bark (aged or composted), wood fibers, rice hulls, and composted materials. Inorganic materials such as perlite, vermiculite, expanded clay, and pumice are frequently incorporated into organic growing media to enhance physical and in some cases physico-chemical properties. Inorganic media components are described in Chapter 7, Inorganic and Synthetic Organic Components of Soilless Culture and Potting Mixtures.

Public pressure to recycle industrial by-products is increasing. Alternative organic substrates based on recycled materials that are well characterized and corrected by suitable blending with inorganic components make it possible to produce high-quality horticultural plants (Buck and Evans, 2010; Lamont and Elliott, 2016) and at the same time contribute to the reduction of overexploitation of natural peatlands. As such the trend toward using peat-reduced media is likely to continue (Raviv, 2013; Carlile et al., 2015), and in this respect, as with peat, new organic

growing media components have to be studied on a scientific and technical basis while evaluating the environmental and social impact linked to their large-scale use in horticultural media mixes. As the availability of resources such as peat moss becomes more limited, the importance of reusing substrates, if needed after disinfection, may become a high priority. If this does not prove to be a viable option, responsible methods of disposal of used media should be defined (Raviv, 2016). This subject is further discussed later in this chapter.

Properties of the most important organic materials used alone and in combination form the basis of this chapter. Physical and structural characteristics, physicochemical properties, nutrient status, and, in some cases, biological/microbiological attributes are described. Their stability, sustainability, and end-use are also considered.

8.2 PEAT

Peat is formed by the slow decomposition of mosses, reeds, and sedges in wet ecosystems where biomass accumulates in conditions of low pH and low oxygen content that are not conducive to microbial activity. Peatlands occur over a wide range of climates, but those of principal interest to horticulturists occur in the higher middle latitudes (50–60N) of the Northern Hemisphere, particularly in Canada, Russia, and countries bordering the Baltic Sea. These peat deposits developed in the postglacial period after recession of the ice sheets 14,000–11,000 years ago.

Peat bogs can be classified as either ombrogenous (raised and blanket bogs), where growth of the peat layer is controlled by rainfall, or topographic (basin bogs), where deposition of new peat is controlled by topography and the groundwater table (Hammond, 1975). The initial formation of peat bogs occurred in postglacial lakes or locally wet depressions. Thus peat bogs are formed initially from incomplete decomposition of aquatic plants, and subsequently of sedges and reeds (*Carex* and *Phragmites* spp.). The accumulation of plant debris and the lack of decomposition of senesced plant materials cause the bog to slowly expand which allows the growing plants to encroach into the lake with the upward displacement of the water. This gives rise to higher water tables and allows the development of peat-forming plants beyond the original lake margins. As peat formation continues, the groundwater effect diminishes, and there is an increasing dependence on rainfall to supply nutrients. If this is sufficient, then there is a transition to a raised bog and species of moss such as *Sphagnum* and *Hypnum* become dominant.

The International Peatland Society has proposed a simplified classification system for peat based on its botanical composition, degree of decomposition, and nutrient status (Kivinen, 1980) as follows:

1. Botanical composition
 - a. Moss peat (predominantly sphagnum and other mosses)
 - b. Sedge peat (sedges, grasses, herbs)
 - c. Wood peat (remains of trees and woody shrubs)
2. Degree of decomposition/humification (H)
 - a. Weakly decomposed (H1–H3)
 - b. Medium decomposed (H4–H6)
 - c. Strongly decomposed (H7–H10)

3. Trophic status
 - a. Oligotrophic (low in nutrients)
 - b. Mesotrophic
 - c. Eutrophic (high in nutrients)

The properties of peat depend on the nature of the plant remains and their degree of decomposition. Peats vary in age and are commonly classified by the simple but very practical von Post scale developed in the 1920s (von Post, 1922) where three categories of peat, namely, younger, undecomposed of low humification (H1–H3), partly decomposed (H4–H6) and older, highly decomposed (H7–H10) are identified. The test involves squeezing a small quantity of peat in the palm of a hand by drawing the fingers together to form a fist. The color of water expressed and extent of peat extrusion between the fingers are used to define the degree of decomposition. For horticultural purposes, peats denoted H1 to H4 are often referred to subjectively as “white peat,” more decomposed (H4–H6) as “dark peat” and highly decomposed peats (H7 and above) as “black peat” (Bunt, 1988). Indeed, attempts have been made to define peats more clearly based on measurements of color (Prasad and Maher, 2013), and a strong correlation between color and humic acid content demonstrated (Prasad et al., 2017), opening up the opportunity of using a simple quantitative test to better predict properties of peat. Dark peats have higher lignin/humic contents compared to white peats and are thus less prone to biodegradation. Black peats often undergo freezing in winter and this improves their air and water-holding capacity.

Peatlands are drained prior to extraction. Purpose-designed equipment has been developed for peat harvesting. Milling machinery is now widely employed to remove the surface layer of peat from drained peatlands, breaking up the semi-dried peat as it does so. This layer, often only 1–2 cm, may be further allowed to dry before aggregation into large windrows or piles, the latter then often covered with sheeting to prevent rewetting and left in the field until required. However, much peat is removed from drained peatlands by direct vacuum harvesting (Fig. 8.1). Traditional cutting of peat into large (typically 25 cm × 10 cm × 10 cm) sods (Fig. 8.2) which are then air-dried is still extant, since sods may be a better source of large particle size fractions than milled peat.

Peat became the major component of growing media throughout Europe and North America because of its ready availability, consistent nature, and its highly suitable physical and chemical characteristics. Moss peat has high porosity, provides good aeration, and has high water-holding capacity. Its low pH and nutrient content are readily modified to desired levels by addition of lime and fertilizers. Peat is practically free of weeds as well as pests and plant pathogens; although in many cases, it is conducive to rapid development of the latter (Chen et al., 1988; Borrero et al., 2004). It is light and therefore inexpensive to transport. Finally, peat is relatively stable when used as a growing medium.

8.2.1 PHYSICAL PROPERTIES

The physical properties of peats vary with species composition of the originating bogs as this determines the particle size distribution and degree of decomposition (Puustjarvi and Robertson, 1975). Peats, notably decomposed sphagnum types, are resistant to decomposition (Cattivello, 2009; Prasad and Maher, 2004). *Sphagnum* mosses do not possess lignin of the type found in vascular

**FIGURE 8.1**

Vacuum harvesting of peat.

Image by kind permission of Costantino Cattivello.

**FIGURE 8.2**

Peat sods.

Image by kind permission of Colman Hynes.

plants (Rydin and Jeglum, 2013); but lignin-like polymers form a major part of their cell walls. However, resistance to decomposition may be linked also to unique pectin-like substances (sphagnum) of cell walls in sphagnum mosses (Hajek et al., 2011). *Sphagnum* peats also contain phenolic antimicrobial substances (Thormann, 2010).

Properties of some peats are shown in Table 8.3. In general, the bulk density is low and the total porosity is high, but the bulk density increases with degree of humification. The lightweight nature

Table 8.3 Physical Properties of Sphagnum-Based Peats and Peat Fractions Used as Components of Growing Media

Origin, von Post Indexes, and Particle Size/Fraction Where Noted	Dry Bulk Density (kg/m ³)	Total Pore Space (% by Volume)	Air Content (% by Volume)	Available Water: 1–10 kPa Suction (% by Volume) ^a	Source
Canada (undefined)	84	94.2	41.2	26.9	Abad et al. (2005)*
Latvia H3 0–5 mm	41	96.4	50.0	21.0	Amha et al. (2010)
Finland H3 0–5 mm	51	95.7	26.0	43.3	Amha et al. (2010)
Estonia H5 0–5 mm	98	93.7	13.9	39.6	Amha et al. (2010)
Russia H2/H3 0–16 mm	89	93.9	10.1	47	Bures (1997)*
Belarus H3 0–20 mm	105	93.4	6.8	45.3	Šrámek and Dubský (2009)
Ireland H4 0–10 mm	98.8	93.7	20.4	35.8	From Prasad and Maher (1993)**
Ireland H4 0–3 mm	118	94.4	13.7	42.3	
Ireland H4 6–12 mm	118	92.0	36.3	21.7	
Ireland H4 10–25 mm	129	91.5	40.9	16.9	
Ireland H7 3–6 mm	263	83.8	42.6	8.1	
Ireland H7 6–12 mm	255	83.8	49.3	6.1	

Determinations of total pore space, air content, available water, and water buffering capacity were carried out using suction funnels with a porous plate as described by De Boodt et al. (1974), with sand boxes as described by Leijn van Dijk and de Bes (1987)** or EU standard EN 13041(2000) with, in all cases, air content measured at 1 kPa suction.*

^aAvailable water as defined by Handreck and Black (2010) as easily available water plus water buffering capacity.

of peat allows economically viable transportation by ship and road to both manufacturers of media, and nurseries. High porosity is the single most important physical attribute since growing media based on peat can be well aerated and yet contain an adequate reservoir of water for plants.

Porosity of growing media is influenced by the physical nature of the constituents as well as the manner of extraction, processing, and preparation of these. *Sphagnum* peat moss surfaces contain pores that allow ingress and movement of water. Increasingly detailed knowledge on the porous structure of sphagnum peat in situ is now available, including quantitative data on size, geometry, connectivity, surface roughness, and anisotropy of the pore spaces (Quinton et al., 2009; Rezanezhad et al., 2009, 2016), and these data may have relevance to harvested peat used as a component of growing media. The total porosity of peat includes the relatively large, interparticle pores that can actively transmit water, as well as relatively small, closed, and dead-end pores formed by the remains of plant cells (Hayward and Clymo, 1982; Kremer et al., 2004). Scanning electron microscopy (SEM) reveals both relatively large interparticle macropores as well as smaller dead-end pores. Rezanezhad et al. (2009) found pore distribution in a near-surface sample of sphagnum peat, an area typical of low decomposition, was dominated by a single, large highly connected and complex pore space accounting for 94%–99% of the total interparticle pore volume, whereas in deeper peat layers, likely to be more decomposed, pores were significantly smaller with fewer interconnections.

Pore sizes in undecomposed peat may exceed 5 mm, but significant shrinkage occurs during dewatering, compression, and decomposition (Rezanezhad et al., 2016). Pore structure and size is

linked to the type and degree of decomposition of peat, but the superhydrophilic nature of peat is maintained even after peat extraction and processing, and peat media are able to absorb 10–20 times their own weight of water (Koch and Barthlott, 2009). Undecomposed sphagnum peat with high fiber content and a large active porosity yields as much as 80% of its saturated water content to drainage in its natural state; highly decomposed *situ* peat samples release <10% of their water to drainage (Radforth and Brawner, 1977; Rezaeezhad et al., 2016). This relationship is maintained after peat extraction and processing: the younger less decomposed peats (H2–H4) have a much higher water-holding capacity when compared to older, more decomposed peats (H5–H7) (Michel, 2010). Conversely, more decomposed peats may maintain higher air volumes than younger peats, and the structural stability of the former allows air capacity to be maintained during use of decomposed peats for long-term growth of large plants (shrubs, trees) in containers (Niemininen and Reinikainen, 2011).

Pore structure characteristics, notably the biochemical structure and configuration of molecules there, are not well understood. *Sphagnum* peat moss becomes highly hydrophobic on drying to moisture contents below 40%–50% (depending on the degree of humification), and this may be due to changes in pore structure (Michel, 2015). Evidence suggests the involvement of humic substances and other lignin-type compounds in this process that contribute to the water-retentive abilities of peats (Rydin and Jeglum, 2013). More humified (H5–H7) sphagnum peats generally acquire a more pronounced hydrophobic character than less humified peats if they dry out naturally (Michel et al., 2001). In practical terms, the difficulties experienced in rewetting dried growing media based on peat may be overcome by the addition of wetting agents, such as nonionic ethylene oxide derivatives or by mixing peat with hydrophilic materials such as coir.

Laboratory determinations of bulk density, porosity, air-filled content and water retention characteristics of peat with differing humification values for horticultural use have been widely reported (Table 8.3). Overall, Michel (2010) considered relatively undecomposed sphagnum peats (H2–H3) with an air volume of >20% (v/v) and possessing high water availability (>25%) along with high water buffering capacity to form almost ideal growing media for many horticultural species.

Particle size is an important consideration when designing growing media, and in the case of peat, it is affected by degree of decomposition, extraction, and processing (Michel, 2010). Peat produced by milling or cutting (to form sods) may be mechanically broken down and sieved to produce fractions of different particle size, offering the possibility of providing peat growing media with a wide range of physical properties (Prasad and Maher, 1993). Peat fractions can be produced either by milling peat on the bog, or producing sod peat and breaking these sods into the desired fractions. Two types of sods exist: bin sods, and sod moss. Bin sods are produced by extrusion of peat of a high degree of decomposition (H6–H7), and sod moss refers to sods cut directly from bogs of moderately decomposed peat (H5–H6). Both types of sods are dried on the bog before use. Predictably, the bulk density of fractions from bin sods is higher than fractions from sod moss. Fractions produced from sods are more uniform and stable compared to fractions from milled peat (Prasad et al., 2017). Bin sods have been found to be a good alternative to Irish sod peat because of similar water characteristics (Zevenhoven and Verhagen, 1997). The moisture content of the sods is critical for fraction formation, since a moisture content of 60% + can lead to high percentage of fine particles (<5 mm) during processing (Prasad et al., 2017).

Fractions from milled peat have been reported to contain a higher proportion of fine particles (not as “clean”) and be more susceptible to disintegration than fractions from sod moss: the latter

are reported to have fewer fines (“cleaner”), be more uniform and considered harder than milled peat fractions (Verhagen, 2015 personal communication). In addition, when examined for water holding and release characteristics bin sods had similar characteristics to sod moss (Zevenhoven and Verhagen, 1997).

Retention of water and ease of filling of, for example, plug trays in media designed for plant propagation, as well as enhancement of air content within media in large containers for nursery stock production can be achieved by use of fractioned peats. Peat fractions are especially valuable for plants requiring higher aeration, particularly those raised in large containers for months, or even years. In such cases, consistent aeration during the growing period is important to ensure optimal growth. Thus Prasad and Maher (1993) report increases of air-filled porosity from 17.9% to 40.6% in 0–3 and 5–10 mm fractions respectively of an H5 peat, and from 16.9% to 33.4% in identical fractions of an H2 peat. Conversely, the easily available water [(EAW) at 1 kPa] decreased from 31.9% and 32.3%, respectively, to 12.6% and 16.1%. Scharpf (1997) used mixtures of sieved peat fractions to create substrates with air-filled porosity varying from 6% to 35%.

Several studies have focused on the proportion of fine particles in peat as a broad indicator of air-filled porosity. Verhagen (1997) mixed milled peats from a number of sources and found a strong relationship between the amount of particles <1 mm in the raw peat components and the air-filled porosity of the mixes. Moreover, the relationship was the same for all the types of peat included in the research. The proportion of fine particles in the peats was a better predictor of the air-filled porosity of the peat mixtures than the air-filled porosity values of the individual peats. Prasad and Ní Chualáin (2004) similarly found a good relationship between particle size <1 mm and the air-filled porosity of various peat mixes. In practical terms, this work could be used as a tool to compose mixtures of any given air-filled porosity.

The relevance of air-filled porosity values for peat substrates has been questioned in studies by Caron et al. (Nkongolo and Caron, 2006; Caron et al., 2010), who studied gas diffusivity of peat and bark-based substrates in relation to plant growth. Surprisingly, in the light of development of peat fractions, studies on growth of two pot species (*Poinsettia* and *New Guinea impatiens*) in a range of sphagnum peat (1:1 mixtures of H2/H3 and H4/H5) media of particle size from 1–2 mm up to 10–20 mm by Caron et al. (2010) found growth parameters to be superior in a 2–4 mm fraction of sphagnum peat: this fraction had the highest gas relative diffusivity, and gas diffusivity values correlated much better with plant growth than air-filled porosity values. The authors attributed this feature of good growth in peat media of small particle size to the microporous nature of peat-based substrates and conclude that determinations of gas exchange are a better predictor of plant performance than air-filled porosity measurements. Furthermore, air capacity provides little information on the actual gases, principally oxygen, carbon dioxide, and nitrogen, within the root zone of media. The concentrations of oxygen and CO₂ are respectively much lower and higher in growing media than in the atmosphere due to both root respiration and that of microorganisms there. Among the few studies in this area, Schmitz et al. (2013) found that oxygen diffusion rates differed little in substrates with the same air content but that as might be expected water tension and increasing density of substrate adversely affect oxygen diffusion in media. However, determination of gas exchange in media is more complicated than the procedures used to determine air-filled porosity, and thus the latter is likely to be preferred in routine measurements.

8.2.2 pH AND CHEMICAL PROPERTIES

Plants raised in growing media must be able to assimilate the same range of nutrients as those raised in soil. For this purpose, the pH of peat-based media for most plants other than calcifuges needs to be in the range 5.5–6.5 (Lucas and Davis, 1961). In general, the pH values for raw sphagnum peats (untreated, as harvested from a bog) range from 3.5 to 4.1, and desirable pH levels for most growing media are easily achieved with peat through addition of lime, commonly as dolomitic limestone (which also provides calcium and magnesium) at 2–3 kg/m³ for less decomposed (H2/H3) and 3–7 kg/m³ for more decomposed (H4–H6) peats (Maher and Prasad, 2004). The acidic nature of sphagnum peat makes it an almost indispensable component of media for growth of calcifuges such as *Azalea*, *Camellia*, and *Rhododendron* spp., where, typically, peat-based media receive a dressing of 1–1.5 kg m³ of lime (Maher et al., 2000). Fen, sedge, and reed-derived peats may have a higher pH than sphagnum peat, up to 5.0 (in some cases high lime reed peats may be as high as 7.5) and are used as components of growing media in a few locations (Robbins and Evans, 2006).

Although the cation exchange capacity (CEC) of growing media constituents is frequently reported on a weight basis, since plants grow in a limited volume of media, and in the case of peat-based media, low bulk density, it is more appropriate (as with nutrient concentrations) to express CEC on a volume basis. If the bulk density of materials is known, then conversion from cmol (+)/kg (meq/100 g) to meq/L is straightforward (Handreck and Black, 2010). The CEC of peat has been recorded at 130 cmol (+)/kg for undecomposed sphagnum peat, 80 cmol (+)/kg for sedge peat, and 150–250 cmol (+)/kg for decomposed H4–H6 peats (Puustjarvi and Robertson, 1975). These values approximate to 150–250 meq/L (Landis, 1990), and endow many peats, particularly H4–H6, with good buffering capacity, being resistant to pH changes brought about by alkaline water supplies and minimizing checks to plant growth from fertigation. Further, the high CEC values mean that peat has a good capacity to retain cationic nutrients. However, the anion exchange capacity is very low (Bunt, 1988). This means, for instance, that, unlike in most soils, phosphate can easily be leached from peat-based growing media as will N present in nitrate form.

The low pH and very low basic level of fertility of sphagnum peat requires the addition of nutrients as well as lime to support good plant growth. Of course, the amount of nutrients needed depends on the species being grown and the stage of development. Seeding and propagation require only a low level of nutrients, greater amounts are required for rapidly growing plants. Cost and composition of proprietary fertilizers may determine the nature of these added to peat-based media, but commonly base fertilization of media for seedling modules/plugs is low at 50–100 g/L of a low-N fertilizer, but the actual rate used varies with individual species. Media for pot plants have a higher concentration of base nutrients, usually 1–1.5 kg/m³ of, for example, an 18:6:12 N:P:K fertilizer but again with variations according to species (Bunt, 1988). Growing media for large container plants may contain a similar level of base fertilizer, but this may be further supplemented with an appropriate controlled release fertilizer. Plants in growing media may be additionally or alternatively fertigated with an appropriate liquid feed.

Micronutrients must also be added to peat-based growing media. These can be added as individual inorganic salts, as components of compound fertilizers or in slow-release form, for example, as fritted trace elements (FTE). For example, in Gallagher (1975), the trace elements for a peat-based medium are listed as B—1.0 g/m³ as Borax, Cu—3.5 g/m³ as copper sulfate, Mn—3.5 g/m³ as manganese sulfate, Zn—3.2 g/m³, Mo—1.2 g/m³, Fe—7.0 g/m³ as ferrous sulfate, and 2.5 g/m³ as

iron chelate (EDTA). Alternatively, these could be substituted by FTE (253 A) applied at 400 g/m³. The advantages of the latter-fritted form are greater convenience, fewer problems in ensuring good uniformity during mixing, and potentially greater persistence. The disadvantage is that of greater expense.

8.2.3 BIOLOGICAL CHARACTERISTICS AND STABILITY

Contamination of peat may come from weed seeds arising from plants growing on peat bogs or on bog margins (Keijzer and van Schie, 1997). The principal species involved are rushes (*Juncus* spp.), and sheep's sorrel (*Rumex acetosella* L.). Control of these is usually carried out on site by manual inspection and removal, with a particular focus of attention being the drainage ditches around peat extraction areas. The acidic waterlogged conditions under which peat develops do not favor extensive microbial growth. Harvested peat and indeed peat-based media at the point of manufacture have a low microbial count: total viable counts (TVCs) are often 10⁴ cfu g⁻¹ dry weight or less (Kavanagh and Herlihy, 1975; Dickinson and Carlile, 1995). However, the low population of microorganisms may exert little of the antagonistic effects seen in other media, particularly composted materials, and once pathogens such as *Pythium* or *Phytophthora* spp. arrive in peat media, they can spread very rapidly.

Despite a low inherent microbial population, self-heating is a major problem in stored peat, whether on the bog or in transit, and is considered to be microbially mediated (Tahvonen and Kemppainen, 2008) notwithstanding the low pH of peat. Both thermophilic fungi (Wever and Hertogh-Pon, 1993) and bacteria (including some rare species such as the genus *Alicyclobacillus*—Rannekleiv and Bååth, 2003) have been associated with self-heating. During self-heating, phytotoxic substances may be released (Wever and Hertogh-Pon, 1993); in addition, alterations in the physical, chemical, and microbiological properties may occur in self-heated peats (Cattivello, 2009), and germination and growth of plants adversely affected in media derived from such peats. Indeed, in most cases, peats that have undergone self-heating are not suitable for production of growing media.

Preservation of physical properties, especially air space, within media during plant growth requires structural stability. All organic-based growing media may undergo microbial degradation, and this combined with shrinkage and swelling (Michel, 2010) may have detrimental effects on structural stability. Microbial degradation of peat media is slow compared to media prepared from other constituents (Table 8.4). Biological stability as measured by oxygen uptake rate (OUR) indicates O₂ consumption rates of as little as 1.9 mmol/kg/h (Blok et al., 2017). Grunert et al. (2016) measured respiration as CO₂ release from various growing media components and found Irish peat (10–30 mm) showed the lowest rate of respiration with 16.1 ± 3.1 mg CO₂-C/kg/day, followed by sod peat (10–30 mm) at 25.2 ± 1.5 mg CO₂-C/kg/day and coconut fiber at 82.6 ± 4.0 mg CO₂-C/kg/day. This low rate of decomposition has been linked to the amount of lignin-like substances present in different peats (Freeman et al., 2001; Prasad and Maher, 2004). In addition, sphagnum peats contain phenolic antimicrobial substances as well as humic substances that are highly resistant to microbial degradation (Rydin and Jeglum, 2013). Thus moss-derived peat, even when pH is raised and media fertilized, has a low rate of shrinkage during use,

Table 8.4 CO₂ Respiration Rate of Different Fresh Growing Media Constituents and a Mixture (GB) of Constituents (*n* = 3)

Growing Media Constituents	CO ₂ Respiration Rate (mg CO ₂ -C/kg/day) Mean ± SD	C/N Ratio
Mineral growing media	-2.33 ± 9.6 ^a	3
Compost (green waste)	138.83 ± 20.07 ^c	16
Coconut fiber	82.55 ± 3.96 ^d	103
Sod peat	25.18 ± 1.46 ^b	61
Irish peat	16.05 ± 3.13 ^a	49
Mixture: 20% coconut fiber; 40% sod peat; 40% Irish peat	45.86 ± 5.58 ^c	ND

Figures bearing the same letter do not differ significantly (p 0.05)
Adapted from Grunert, O., Reheul, D., Van Labeke, M-C., Perneel, M., Hernandez-Sanabria, E., et al., 2016. Growing media constituents determine the microbial nitrogen conversions in organic growing media for horticulture. *Microb. Biotechnol.* 9, 389–399. <file:///C:/book%20reviews/Grunert%20et%20al%202016%20co2%20release%20from%20substrates.html>.

and, when compared to other substrates, humified sphagnum peat is often preferred for growth of nursery stock plants in large pots or containers over months or even years (Aendekerk, 1997, 2001; Prasad and Maher, 2004).

8.3 COIR

Coir is the material that forms the middle layers or mesocarp of coconut fruits (*Cocos nucifera* L.). These layers are composed of fibers embedded in the so-called coir pith, also referred to in its dry state as coir dust. Coir is one of the most abundant plant-derived organic waste materials in many tropical and subtropical countries. Long fibers are extracted from the mesocarp after soaking and used for production of matting, brushes, and insulating materials. The remaining material is composed of short fibers and coir pith. Traditionally this material was dumped and accumulated as a waste product, and the material is still considered as a problem in some areas (Indian Coir Board, 2016). However, these coir “dumps” have now mostly gone from major areas of production, and coir pith from freshly harvested coconuts is now predominantly used for substrate production. After separation of fiber, fresh coir is normally aged or “composted” for several months prior to preparation as a substrate for growing media (Indian Coir Board, 2016).

Although considered as a potential growing medium in the 1860s (Anon, 1862) with interest reawakened in the 1940s (Hume, 1949), coir was not extensively used as such until the 1980s, largely due to problems of quality and consistency. This century has seen improvements in quality control and consequent increases in production of coir for use in growing media. Coir pith may contain high salt levels (predominantly Na, Cl, and K) and is washed with water and/or solutions, for example, Ca(NO₃)₂ (the so-called buffered coir), prior to preparation for horticultural use (Poulter, 2014). After washing or leaching the dried coir pith is compressed (ranging from 5:1 to 8:1 v/v, Ross et al., 2012) into blocks (c. 5 kg) or briquettes (c. 0.6 kg) for ease of transport. On

receipt by growing media producers, blocks are then reconstituted with the addition of water, with an expansion ratio of 1:12 (wt:vol) or greater sought by manufacturers (Hynes, personal communication, 2017). Coir chips (5–15 mm) prepared from the hard outer husk of the coconut are also used in growing media, principally to enhance aeration, particularly with media for long-term nursery crops.

Major areas of production include the Tamil Nadu and Kerala areas of India, and Sri Lanka: these areas primarily supply coir to Europe and Australia. Mexico exports coir to the United States. Exports of coir pith from Sri Lanka increased from 97,000 tonnes (equivalent to 1.16 M m³ if reconstituted on a 1:12 wt:vol basis) in 2007 to 120,000 tonnes (1.44 M m³ if reconstituted) in 2010 (Anon, 2011). Exports of coir pith, primarily for horticultural use, from India, have increased markedly, from 6501 tonnes in 2000–01 to 408,897 tonnes in 2015–16 (Indian Coir Board, 2016): if reconstituted on a 1:12 (wt:vol) basis this represents volume increases from 78,000 m³ to 4.8 M m³ (Fig. 8.3). The Netherlands alone imported 73,735 tonnes of coir from India in 2015–16, equivalent to almost 885,000 m³ on 1:12 (wt:vol) reconstitution. Other areas of the world developing coir for horticultural use include the Philippines, Indonesia, Ivory Coast, Costa Rica, and Thailand (Abad et al., 2005). Coir is produced in large quantities in Brazil, but much is applied as a soil amendment (Brasileiro et al., 2013).

Increasing reliability and consistency in quality control and supply has led to coir's widespread adoption as an organic substrate. Coir offers high porosity, good aeration, with very high water-holding capacity. Coir pith, like peat, is practically free of weeds as well as pests and plant pathogens, with most major exporters engaging with bodies such as the Netherlands-based RHP (Regeling Handels Potgronden) to certify products free of unwanted organisms. Finally, coir is relatively stable when used as a growing medium, and indeed may be reused in some horticultural applications, for example, with strawberries in growing modules. Of course, for growing media manufacturers in Europe and North America, transport costs are higher than for peat.

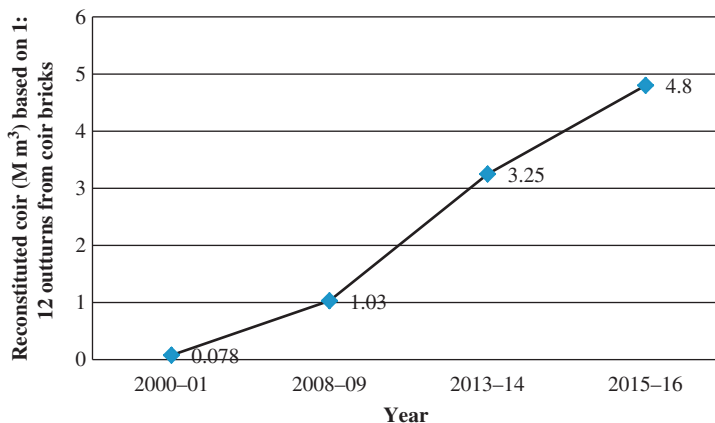


FIGURE 8.3

Quantities of coir pith in Mm³ exported from India 2000–15/16. Calculation based on transforming export tonnes (Indian Coir Board, 2016) to cubic meters based on 1:12 reconstitution from coir bricks.

Sources of coir may differ greatly in physicochemical (pH, electrical conductivity - EC) as well as chemical and physical characteristics, due to source, age, and degree of processing; it is essential that authors state the latter when reporting analyses of coir, but this is not always done.

8.3.1 COMPOSITION AND PHYSICAL PROPERTIES

Composition of macromolecules and their configuration within coir pith may serve to explain its favorable properties as a substrate for plant growth. The dominant constituents vary according to the age and treatment (particularly composting) of coir.

The principal macromolecular constituents of coir pith include lignin: in coir fiber principally a *p*-hydroxyphenyl-guaiacyl-syringyl lignin (Rencoret et al., 2013). The lignin content of coir pith is quoted as low as 30% (Indian Coir Board, 2016) and as high as 50% by weight (Israel et al., 2011; Muthurayar and Dhanarajan, 2013), and the lignin structure may degenerate with age, especially in stored/composting piles (Priya et al., 2016). Cellulose is also a prominent component of coir pith, again varying among authors from 26% (Indian Coir Board) to 35% by weight (Israel et al., 2011; Muthurayar and Dhanarajan, 2013), with some authors quoting values as high as 50% (Paramanandham and Ross, 2013). Hemicelluloses probably account for most of the remaining polymers in coir pith at 15%–35% (Sjostrom, 1993).

The configuration of these macromolecules in coir pith is not well understood but must be associated with the properties of pith that make it such an attractive option as a substrate for plant growth. Resistance to degradation of coir pith may be associated with ligno-cellulose complexes, formed of cellulose, hemicellulose, and lignin linked through covalent and noncovalent bonds, and which are highly resistant to degradation (Carr, 2012). Coir may also contain phenolic substances (Ma and Nichols, 2004).

As with moist peat, coir pith is superhydrophilic (Koch and Barthlott, 2009). Coir from several sources has been shown to hold 5–10 times its weight of water (Arenas et al., 2002; Abad et al., 2005). Both peat and coir are renowned for their ability to absorb and retain not only water but also air, due primarily to their microporous nature (Tsuneda et al., 2001; Fornes et al., 2003) allowing internal retention of both water and air. Few ultrastructural studies of pore characteristics have been undertaken with coir pith. Fornes et al. (2003) using SEM found that coir dusts from Mexico and Sri Lanka did not differ in microstructural parameters, whilst differing markedly in particle size. Fornes et al. (2003) revealed external circular pores in coir pith ranging in diameter from 30 to 80 μm , leading to a relative surface porosity of around 49%: much greater than that of sphagnum peat at 12%. Pores were also observed by Paramanandham and Ross (2015) who, in a SEM study with coir pith from Tamil Nadu, India, observed an open-mouth matrix with a honeycomb-like appearance containing a large number of pores. The frequency and (larger) size of pores on the surface of coir particles compared to those on the surface of sphagnum peats may explain the swifter uptake of water, as well as equally rapid drainage and consequent higher air content than most peats (Fornes et al., 2003). Unlike sphagnum peat, coir is rarely hydrophobic: whereas water-absorbing polymers are frequently added to peat-based media to aid absorption after drying, these wetting agents are unnecessary in media that principally contain coir.

Variation exists in the overall physical properties of air and water capacity of coir pith (Table 8.5): the actual husks from which coir pith is derived, processing of these, and especially

Table 8.5 Physical Properties of Coir

Material, Origin, Particle Size (Where Noted)	Dry Bulk Density (kg/m ³)	Total Pore Space (% By Volume)	Air Content (% By Volume)	Available Water: 1–10 kPa Suction [†] (% by Volume)	Source
Coir pith, Philippines	40–80	85.8–89.5	9.5–12.5		Evans et al. (1996)
Coir pith, Tamil Nadu, India 85% 0.25–2 mm	56	96.3	41.2	28.1	Abad et al. (2005)*
Coir pith, Sri Lanka	74–81	94–96	13.5–29.4	23.8–37.8	Prasad (1997a)**
Coir pith, Mexico, particle size 50% <1 mm	61–72	95.2–95.9	43.7–57.8	19.9–25.8	Abad et al. (2005)*
Coir pith, Sri Lanka	72–89	94.1–95.1	31.7–42.0	26.4–27.8	Abad et al. (2005)*
Coir pith, Ivory Coast	71	95.2	66.7	9.5	Abad et al. (2005)*
Coir chips	72–82	95	55	3	Aendekerck et al. (2000)*

For European samples, determinations of total pore space, air content, available water were carried out using suction funnels with a porous plate as described by De Boodt et al. (1974)*, with sand boxes as described by Leijn van Dijk and de Bes (1987)** or EU standard EN 13041(2000) with, in all cases, air content measured at 1 kPa suction.

the degree of separation of pith and coir fiber, as well as the age of the pith all contribute to this variation (Abad et al., 2005). The particle size of coir may have an important role in determining physical properties. Evans et al. (1996) described samples of coir from the Philippines having pith particles predominantly between 0–5 and 2 mm in diameter, with one source having a high concentration of very fine particles of 0.25–0.5 mm diameter, and correspondingly the highest bulk density, water-holding capacity and total pore space, but the lowest air porosity of the five samples studied. In a comprehensive study of coir from several countries, Abad et al. (2005) found particle sizes in most samples to be between 0.5 and 2 mm but with some significant outliers. Abad et al. (2005) investigated only one sample of coir from India, but Ross et al. (2010) also found five samples of coir pith from Tamil Nadu to contain predominantly particles in size from 0–3.2 mm. The wide divergence of airspace and easily available water in coir (Table 8.5) may be linked to particle size which, especially with particles of <1 mm, is strongly correlated to aeration and water availability (Noguera et al., 2003; Prasad and Ní Chualáin, 2004). Prasad and Ní Chualáin (2004) report the relationship between airspace and particle size of coir to be highly significant ($R^2 = 0.94$; $n = 8$): however, the regression line was different from that of peat, probably due to the fine particles in coir having a higher airspace than in peat. The general protocols of many coir exporters describe a particle size of principally 0.5–1 mm but where individual specifications are required for particular growing systems, the particle size of coir can be adjusted by sieving at the point of production.

8.3.2 pH AND CHEMICAL PROPERTIES

In general coir is less acidic than peat, with considerable variation depending on source as shown in Table 8.6. If coir is the major component of a medium then lime may be unnecessary, but calcium must be provided with gypsum often recommended for this purpose (Handreck and Black, 2010).

Table 8.6 pH and EC Levels in Coir			
pH	EC (dS/m)	Source	Reference
From commercial sources			
5.9–6.6	0.8–2.5 ^a	Philippines	Evans et al. (1996)
5.6–6.3	0.3–2.9 ^a	Sri Lanka	Evans et al. (1996)
5.8–6.9	0.9–2.5 ^a	Indonesia	Evans et al. (1996)
4.8–6.8	0.32–0.97	Sri Lanka	Meerow (1994)
5.5–5.7	0.80–1.90	Sri Lanka	Handreck (1993a)
5.0–5.7	0.12–1.51	India	Prasad (1997a)
5.3	1.96	Malaysia	Asiah et al. (2004)
4.9–6.6	0.32–0.41		Smith (1995)
6.0–6.7	0.2–0.4		Kipp et al. (2000) (coir dust)
5.9–6.1	0.2–0.9		Kipp et al. (2000) (coir chips)
Fresh samples ^b		(Average particle size)	
6.1	4.41	300 µm	all from Ross et al. (2012)
6.5	4.26	750 µm	
6.8	3.67	1.5 mm	
6.6	3.32	2 mm sample all from Tamil Nadu, India	
pH values of Noguera, Evans, and Meerow are based on saturated media extract, Smith by 1:5 water extract and the others by 1:1.5 water extract. EC values have been converted to 1:1.5 water extract (see below).			
^a SME needing conversion			
^b 1:1.5 ratio needing conversion.			

The EC of coir can vary enormously depending on age, processing, whether, and how often it was washed and/or buffered (Table 8.6). The variation in procedures used around the world to measure EC of growing media and their constituents leads to difficulties in comparing published EC data. However, McLachlan et al. (2004) have found a strong correlation between saturated extract values and various ratios of water extract, 1:1, 1:2, 1:5, etc. Thus it is possible to cross-reference EC values with different extraction methods and this has been done in Table 8.6.

Samples of raw coir from the East and West coasts of Tamil Nadu in India had EC values as high as 11.25 dS/cm and 8.08 dS/m (1:1.5 extracts), respectively (Ross et al., 2010). Sequential washing of coir reduces EC values markedly—for example, after five successive washes from 6.5 to 0.14 dS/m (1:1.5 extracts) in a sample of coir from India of 0.2–0.5 mm particle size: however, after these successive washes, the pH rose from 5.9 to 7.8 (Paramanandham and Ross, 2013). Elevated levels of EC are caused mainly by naturally occurring high levels of Na, Cl, and K in water-soluble form, and it is the leaching of these elements that leads to the reduction in EC.

Coir has a lower CEC compared to peat with values ranging from 35 to 95 cmol (+)/kg (approximately 70–150 meq/L) reported (Evans et al., 1996; Abad et al., 2002; Handreck and Black, 2010), and some authors report greater leaching of nutrients from coir than peat (Bragg, 1995; Hongpakdee and Ruamrungsri, 2015). In terms of macronutrients, coir sources are universally low in N, in most cases both NH_4N and NO_3N being recorded at <5 mg/L (Evans et al., 1996;

Table 8.7 Chemical Properties of Coir (Water-Soluble Cations and Cl) mg/L⁻¹					
K	Ca	Mg	Cl	Na	Reference
<i>From commercial sources</i>					
59–926	10–51	8–28	6–753	18–180	Noguera et al. (2003)
84–149	8–9	8–9	–	34–54	Meerow (1994)
4–468			5–246	49–105	Prasad (1997a)
60–102	<5.5	<2	91–169	31–81	Smith (1995)
63–112	–	–	110–276	17–47	Konduru et al. (1999)
59–922	9–51	8–25	11–76	18–123	Abad et al. (2002)
32–62	0–4	0–2	25.74	18–32	Kipp et al. (2000) (coir dust)
23–90	0–4	0–144	0–56	21–116	Kipp et al. (2000) (coir chips)
101	–	–	35	–	Van Doren (2001)
<i>Fresh samples (Ross et al., 2012)</i>					
1402	4.8		720	285	Particle size
1443	4.0		640	316	300 µm
1350	3.8		490	272	750 µm
1242	3.3		405	210	1.5 mm
					2 mm
<i>Values of N, P, Cl, SO₄ of Noguera, Evans, Meerow, and Abad have been converted from saturated media extract to 1:1.5 water extract using the following equation: $y = 0.378x - 4.8$, $R^2 = 0.908$, $n = 96$; for P, $y = 0.377x - 0.1$, $R^2 = 0.913$, $n = 147$ and for K, Na, Ca, Mg $y = 0.446 + 7.3$, $R^2 = 0.935$, $n = 96$ (Prasad, unpublished).</i>					

Table 8.8 Chemical Properties of Coir (CaCl₂/DTPA Extractable Micronutrients) mg/L⁻¹					
Fe	Mn	B	Zn	Cu	Reference
	1.1–1.5	0.12–0.18	0.7–1.3	0.17–22	Handreck (1993a)
0.079–0.157	0.814–154	0.066–0.077	0.429–0.527	0–0.0064	Kipp et al. (2000) coir dust
0.045–0.14	0.484–0.561	0.066–0.154	0.364–0.552	0.0250–0.0448	Kipp et al. (2000) coir chips
4.1–7.7	0.9–5.0	0.2–0.4	0.5–1.1	0.1–0.3	Prasad and Maher, unpublished data

Noguera et al., 2003). A few sources of coir have moderate concentrations of P, up to 30 mg/L, but most have levels much lower than this (Evans et al., 1996; Noguera et al., 2003). Almost all samples of raw coir are rich in K, Na, and Cl (Ross et al., 2012; Table 8.7), and as noted, these cations must be washed out prior to preparation as a horticultural substrate. When the washing water contains Ca(NO₃)₂, it removes exchangeable Na and K and the exchange complex is saturated mainly with Ca. Even after washing especially with water, moderate-to-low residual concentrations of these ions may remain in coir substrates (Table 8.7), and indeed most importers of coir pith for horticultural purposes will specify cation concentrations and EC limits to exporters. Coir, again depending on source, may also contain significant amounts of micronutrients, but their concentrations can vary a great deal (Table 8.8).

8.3.3 BIOLOGICAL CHARACTERISTICS AND STABILITY

Processed and exported coir pith does not generally contain weeds or pests, although some exceptions have been reported (James et al., 2011). However, the microbial content of raw coir is higher than that of raw peat by two orders of magnitude or more (Prasad, 1997a) with TVCs of approximately 10^5 (cfu) bacteria, 10^4 actinomycetes, and 10^3 fungi per gram of raw coir (Paramanandham and Ross, 2016). Slightly higher microbial activity has been recorded in coir-based media than in peat media (Prasad, 1997a) but coir, as peat, is regarded as a stable substrate component. For example, Prasad (1997a) reported volume reduction as a result of biodegradation at 24% over 24 months. Nelson et al. (2004) showed shrinkage of coir during growth of *Impatiens* to be lower (at 19%) in coir than in a sphagnum peat (28%–37%), although the source and H value of the latter were not stated. Notwithstanding these indications of stability, some authors report the need for supplementary N with coir (Handreck, 1993a; Arenas et al., 2002).

The higher microbial counts in coir rather than peat have led to studies on coir's potential for biocontrol of plant pathogens. Coir extracts have been shown to inhibit growth of some wilt fungi (Hyder et al., 2009), although this work was conducted solely in vitro. Successful control of disease in coir-containing growing media following addition of biocontrol agents has been demonstrated (Castaño et al., 2013).

8.3.4 PLANT GROWTH

Coir is used widely as a plant substrate across the world: for many crops, it is incorporated as one component of growing media: for example, along with bark in Southern Hemisphere countries (Handreck and Black, 2010) or as a peat diluent in Europe (Blok and Verhagen, 2009, Carlile and Waller, 2013). Coir is particularly valued as a rooting medium, but its water-holding ability and wettability make it an attractive proposition for bedding and pot plants, where growth in the latter has often been shown to be superior to that in peat (Smith, 1995; de Kreij and van Leeuwen, 2001). In the 2000s, coir has been increasingly adopted as a single substrate for soft fruit crops such as strawberries, raspberries, and blueberries as well as roses, lilies, and some other ornamentals. Movement from soil-based systems where pest and especially disease problems have become more difficult to control following restrictions on pesticide use has led to systems such as the so-called table-top culture of strawberries raised in bags or preformed modules containing coir. The higher cost is mitigated by the ability to reuse coir three or four times in growing modules for soft fruit crops.

8.4 BARK

Bark is a by-product of sawmills, often categorized as “residue,” and bark from both softwood, principally conifer, and hardwood (particularly in North America) species is a major component of growing media, especially in areas where peat is scarce or expensive. Bark was formerly discarded as a by-product of the lumber industry in the United States, but from the 1960s onward its value as a decorative mulch and component of growing media was recognized and use of pine bark in particular steadily increased. In the United States, Australia, and New Zealand nursery stock species

raised in large containers over several months, or even years, are widely grown in media in which bark is the principal constituent (Yeager et al., 2007; Handreck and Black, 2010). In Europe, bark is commonly mixed with peat for growth of species in large containers (Edmonds, 1980; Rivière et al., 2008). Early reviews of bark as a growing medium include those of Aaron (1976) and Raviv et al. (1986).

Although bark may be used fresh from sawmills (Lemaire et al., 1980; Prasad, 1980; Cobb and Keever, 1984), aged bark is preferred as a basic component of soilless substrates used for container nursery production (Pokorny, 1979; Buamscha et al., 2007; Handreck and Black, 2010). Aged bark refers to material that may be left in piles for several months or even years. During this time, potentially phytotoxic materials such as phenols and terpenes may dissipate, and low molecular weight organic materials absorbed by microorganisms leading to biological stability (Solbraa, 1979; Buamscha et al., 2007). Aging is particularly appropriate with bark that contains >10% so-called white wood (xylem): if used fresh, such bark, with high cellulose/hemicellulose content, may immobilize nitrogen very rapidly (Fields et al., 2012).

Composted bark, used particularly in Europe and Australia, involves windrow preparation, application of nitrogen, turning and maturation to produce a biologically stable product (Prasad and Carlile, 2009; Anon, 2016). Aged and especially composted bark are much less hydrophobic than fresh bark. However, if dried to 35% (w/w) or less, bark, irrespective of treatment, will become hydrophobic (Airhart et al., 1978). Details of bark composting have been described by Solbraa (1979). When bark is composted, supplemental N is provided to overcome N immobilization and reduce toxicity. Other elements are typically not added during composting (Prasad and Carlile, 2009), although sometimes phosphorus is also recommended (Solbraa, 1979). Other vital parameters during composting are moisture, gaseous oxygen, and temperature. A rate of urea addition of 3% of the initial bark dry weight was considered to be optimum. As a result of composting, there is a decrease in easily decomposable organic compounds (Solbraa, 1979).

In Europe, bark use in growing media remained at c. 1.9 M m³ from 2005 to 2013 (Schmilewski, 2017). The principal user of bark in Europe is France, with 1.05 M m³ in 2013 (down from 1.2 M m³ in 2005), here derived principally from the maritime pine, *Pinus pinaster* Aiton (Schmilewski, 2009, 2017). Bark from the Monterey pine, *Pinus radiata* D. Don, is the principal organic component of growing media in New Zealand, and as well as *P. pinaster*, the slash pine *Pinus elliotii* Engelm is used in Australia, along with hardwood bark from *Eucalyptus diversicolor* F. Mueli and *E. (Corymbia) calophylla* Lindley (Handreck and Black, 2010). In the United States, the slash pine, *P. elliotii* and loblolly pine, *Pinus taeda* L. in Eastern states, and the Douglas fir, *Pseudotsuga menziesii* (Mirb.) Franco in the Pacific Northwest are the principal sources of bark for use in growing media (Bilderback et al., 2013). Other coniferous species such as the red pine, *Pinus resinosa* Sol. Ex Aiton and the Jack pine, *Pinus banksiana* Lamb. are used in the northern United States and Canada. In some parts of the United States, hardwood bark from redwood (*Sequoia* spp.) and other species also form components of growing media (Bilderback et al., 2013).

Bark supplies for growing media, especially in the Southeast United States, declined due to the slump in house building and thus sawmill activity after 2008, and an increase in use of wood, including sawmill residues, for pellets used as biomass (Jackson and Fonteno, 2013; IEE, 2016). However, in the second decade of the millennium, bark remained the most popular substrate component of nursery stock media in the United States. Bark supplies have also declined in France, due to debarking within the forest rather than at sawmills, leading to increases in the cost of bark

recovery (Rivière et al., 2008). Problems of bark supply for growing media have also been raised in Australia (Handreck, 2014).

8.4.1 PHYSICAL PROPERTIES

Bark, depending on its origin, is very variable in terms of morphology and constitution (Chang, 1954). Light and electron microscopic examination of bark from *Pinus* spp. reveals a porous structure, linked to the spaces within dead cells: principally phloem, axial, and radial parenchyma (Pokorny and Wetzstein, 1984; Nunes et al., 1996). Pokorny and Wetzstein (1984) showed the internal porosity of *P. taeda* bark to account for 42%–44% of its volume, and that plant roots could invade these pores and absorb water there. Indeed, water held internally within bark may provide a valuable reserve since bark substrates are generally free-draining, with a lower water-holding capacity than either peat or coir. Bark mixes generally have a high level of air-filled porosity and low water-holding capacity, but the latter may be improved by mixing with peat (Beardsell et al., 1979).

Bark is hammer-milled or ground and then screened to produce a range of particle sizes, allowing defined mixtures of air and water capacity to be manufactured (Yap et al., 2014; Fields et al., 2015; Table 8.9). In the case of nursery stock in the United States, pine bark is commonly screened between 0.5 and 10 mm (Pokorny, 1975; Bilderback, 1982). Spruce bark in Europe may be screened to 10–20 mm for nursery stock, and for general propagation and/or pot plants between 2 and 10 mm (Maher et al., 2000). Aged bark, whilst having lower air content than fresh bark, has

Table 8.9 Physical Properties of Individual Sieve Size Fractions of Aged Pine Bark (from *Pinus taeda*) at 67% Moisture, and in Mixture (4 mm Sieved Fraction) With Coir and Sphagnum Peat Moss

Source	Container Capacity (%) ^a	Air Space (%)	Total Porosity (%)	Dry Bulk Density (g · cm ³)
Unscreened bark (< 15.9 mm)	51.2	33.1	84.3	0.16
Screened bark				
6.3–4.0 mm	34.8	52.4	87.2	0.13
4.0–2.0 mm	38.5	51.4	89.9	0.14
2.0–1.0 mm	48.2	39.1	87.3	0.16
1.0 mm to pan	61.5	17.7	79.3	0.23
Amended 4–2 mm screened bark				
Amended with 40% (v/v) peat	56.9	29.9	86.8	0.13
Amended with 40% (v/v) coir	58.5	31.9	90.4	0.13

^aContainer capacity is percentage of the sample volume occupied by water after allowing sample to drain for 1 h. Adapted with permission from Fields, J.S., Owen Jr., J.S., Scoggins, H.L., 2015. Exploring the influence of particle size on plant water availability in pine bark substrates. *Proc. South. Nurs. Assoc. Res. Conf.* 60, 19–27.

Table 8.10 Physical Properties of Bark, and Bark Mixtures With Peat

Material, Origin, Particle Size (Where Noted)	Dry Bulk Density (kg/m ³)	Total Pore Space (% By Volume)	Air Content (% By Volume)	Available Water: 1–10 kPa Suction [†] (% By Volume)	Source
Fresh <i>Pinus radiata</i> bark, Australia	250	70	30	9	Handreck and Black (2010)
Aged <i>P. radiata</i> bark, Australia	270	71	20	16	Handreck and Black (2010)
Fresh <i>Pinus pinaster</i> bark (>10 mm) France	173	89	54	9	Lemaire et al. (1980)
Fresh pine bark, United States	170	88.3	39.3	9.8	Harrelson et al. (2004)
Aged pine bark, United States	190	87.3	25.2	26.9	Harrelson et al. (2004)
Fresh fine (>9.5 mm) <i>Pseudotsuga menziesii</i> bark	188	nd	39.0	nd	Buamscha et al. (2007)
Aged fine (>9.5 mm) <i>P. menziesii</i> bark	166	nd	35.8	nd	Buamscha et al. (2007)
Fresh coarse (>22 mm) <i>P. menziesii</i> bark	184	nd	49.2	24.8	Buamscha et al. (2007)
Aged coarse (>22 mm) <i>P. menziesii</i> bark	146	nd	43.2	37.8	Buamscha et al. (2007)
Mixtures with peat (rate of bark as %)					
0	153	90.1	23.8	19.2	All from Maher et al. (2000)
12.5	172	98.5	24.7	16.8	
25	185	88.6	24.4	16.4	
50	210	87.6	26.7	13.8	
100	263	85.5	31.5	7.2	

For European samples, determinations of total pore space, air content, and available water were carried out using suction funnels with a porous plate as described by De Boodt et al. (1974). Handreck and Black (2010) outline the procedures used in Australia for physical determinations, with air content measured at 6 cm (0.6 kPa) suction. Harrelson et al. (2004) and Buamscha et al. (2007) used an in-house porometry system developed at the Horticultural Substrates Lab at North Carolina State University with a mean suction of 3.8 cm (0.38 kPa).

substantially increased water-holding capacity (Table 8.9; Harrelson et al., 2004). Bark from most sources is heavier on a dry bulk density basis than sphagnum peat or coir (Table 8.10).

8.4.2 pH, CHEMICAL COMPOSITION, AND PROPERTIES

The chemical composition of bark varies according to species, growth conditions, age of the trees, time of harvest (Solbraa, 1979) as well as methods of harvesting, particularly the efficiency of separation of bark from underlying wood (Eberhardt, 2012). As with peat and coir, lignin is one of the principal polymer constituents. Lignin analyses reported for bark in many cases include suberin and

Table 8.11 Composition of Various Conifer Barks

Source	Lignin-Like Polymers (%)	Polysaccharide/Monosaccharide* (%)	Solvent-Extractable (%)	Author
<i>Picea abies</i>	20–35	45%–55% cellulose and hemicellulose	nd	Solbraa (1979)
<i>Pinus sylvestris</i>	45	25% cellulose: 15% hemicellulose	nd	Valentín et al. (2010)
<i>Pinus pinaster</i>	43.7	41.7%	11.2	Nunes et al. (1996)
<i>Pinus taeda</i> One of two samples	42	Ara* 1.4. Gal* 2.3. Rha* 0.2:0.4. Glu* 18.3. Xyl* 5.2. Mann* 3.5.	10	Eberhardt (2012)
<i>Pseudotsuga menziesii</i> (0.25–0.45 mm)	27	Ara* 0.8. Gal* 1.0Glu* 18.4. Xyl* 2.1Mann* 3.4	18.4	Ferreira et al. (2015)

other lignin-like compounds (Naasz et al., 2009; Krogell et al., 2012). Thus lignin and associated polymers in *Pinus* spp. are often reported as high as 40%–50% w/w (Table 8.11). The other dominant polymers in bark are polysaccharides, and monosaccharide analyses of these for Norway spruce *Picea abies* (L.) H. Karst, *P. taeda*, and *P. menziesii* spp. (Table 8.11) indicate the presence of hemicelluloses and in some cases pectins (Eberhardt, 2012; Kylliäinen and Holmbom, 2004) as well as cellulose. Bark also contains relatively high concentrations of mono- and disaccharides (Kylliäinen and Holmbom, 2004; Naasz et al., 2009) and uptake of these perhaps along with degradation of cellulose by microorganisms may contribute to nitrogen immobilization (Handreck, 1992). Information on decomposition of hemicelluloses in bark is sparse, but these compounds are readily hydrolyzed in other systems (Dekker, 1985) and their degradation following microbial colonization of growing media containing bark may also contribute to nitrogen immobilization. Barks also contain phenols, condensed tannins, and other compounds, usually defined as soluble and/or solvent-extractable materials: some of these are extracted and used commercially, but in fresh bark at least may exert considerable phytotoxic effects affecting both germination and growth of plants (Nunes et al., 1996). Aging and/or composting will decrease or eliminate such effects (Solbraa, 1979).

Barks contain a range of inorganic compounds, the presence and concentrations of which depend on species (Lambert, 1981). Barks from several sources contain macroelements (P and K; Table 8.12), reflected in EC values. Concentrations of K are such that a reduction in base fertilization with this element is possible. In addition, barks may provide sufficient micronutrients to satisfy the requirements of many plants (Niemeira, 1992; Buamscha et al., 2007), although enhanced growth has been observed at pH > 5 where micronutrients have been added to pine bark media (Wright et al., 1999). Indeed some micronutrients may be present at concentrations that may inhibit plant growth: concentrations of manganese in spruce (*P. abies*) bark have been shown to inhibit plant growth (Solbraa and Selmer-Olsen, 1981) and in such cases, liming is advised in order to precipitate manganese.

The pH of most barks, at 4.0–5.0 for pine bark (Wright et al., 1999) and 5.5 for spruce barks (Naasz et al., 2009) is higher than that of peat, and thus lime may be added at lower rates than to peat: in some cases, plant growth in pine bark media at pHs < 5 did not respond to liming, and

Table 8.12 pH, EC, and Inorganic Constituents of Bark											
			P	K	Ca	Mg	Fe	Mn	Zn	Cu	
Total inorganic elements in mg kg dry weight											
<i>Pinus radiata</i>	Range over five sites		50–100	1230–1490	2190–5920	230–880	70–150	40–135	5–15	nd	Lambert (1981)
<i>Pinus sylvestris</i> (mean: $n = 21$)											
<i>Picea abies</i> (mean: $n = 25$)											
			482	2150	7940	nd	102	216	32.2	3.2	Saarela et al. (2005)
			853	5010	19,600	nd	73	387	148	5.5	Saarela et al. (2005)
Available micronutrients determined by DTPA or water* extraction in mg L ⁻¹											
<i>Pinus taeda</i>	pH	EC (dS/m)					22.7	9.7	3.9	0.10	Niemeira (1992)
<i>Pinus banksiana</i> *	6.01	0.5	nd	194	41	nd	0.29	1.7	0.52	0.13	Naasz et al. (2009)
<i>Abies balsamea</i> *	5.54	0.75	0.5	446	151	nd	0.14	18.7	2.00	0.18	Naasz et al. (2009)
<i>Picea glauca</i> *	5.69	0.90	2.9	240	79	nd	0.18	17.4	3.1	0.15	Naasz et al. (2009)
<i>Pseudotsuga menziesii</i> fresh	4.0–5.0		*8.8–19.2	*77.2–110.0	*17.3–26.7	*7.3–11.0	21.7–30.6	8.1–12.1	2.2–3.6	0.34–0.47	Buamscha et al. (2007)
<i>P. menziesii</i> aged	3.7–4.2		*10.3–28.0	*78.8–162.0	*21.5–71.2	*14.6–44.4	61.4–92.4	7.9–13.0	2.3–3.2	0.30–0.47	Buamscha et al. (2007)

indeed in some species, growth was reduced (Wright et al., 1999). Aged bark generally has a slightly lower pH than fresh bark (Table 8.12), whereas the pH of composted bark may be much higher at 5–6 (Hoitink, 1980; Dickinson and Carlile, 1995). Barks in general have a lower CEC than sphagnum peat: among those researchers reporting CEC on a volumetric basis, Nash and Pokorny (1990) state a value of 96.6 meq L for a milled bark sample, and Altland et al. (2014) report values of 29.9–74.4 meq L for samples of pine bark originating from the Southern United States.

8.4.3 BIOLOGICAL CHARACTERISTICS AND STABILITY

Nitrogen immobilization has been widely reported in bark-based media (Dickinson and Carlile, 1995; Jackson et al., 2009a; Boyer et al., 2012) and is associated with colonization of bark by microorganisms that may assimilate free mono and disaccharides as well as degrade polysaccharide components such as cellulose and hemicelluloses, then assimilating the monosaccharides hydrolyzed from these polymers. Growth of microorganisms also involves acquisition of nitrogen, and other elements such as phosphorus, potassium, and trace elements. Phosphorus depletion may occur in bark media (Handreck and Black, 2010), but nitrogen immobilization is more common, particularly in substrates prepared from fresh bark and may also occur even after composting for example, with spruce bark. (Fig. 8.4 and Dickinson and Carlile, 1995). When composted (usually for 4–8

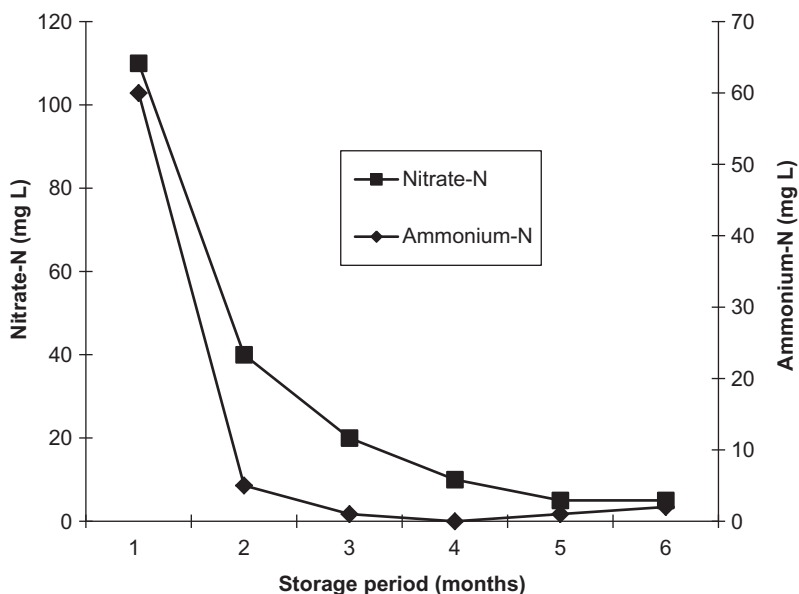


FIGURE 8.4

Rapid immobilization of nitrogen in a (composted) spruce bark-based growing medium.

From Dickinson, K., 1995. *Plant Growth, Nutrient Status and Microbial Activity in Peat Free Growing Media During Storage* (Ph.D. thesis). Nottingham Trent University, UK, p. 175.

weeks), bark is normally supplemented with urea at 2–4 kg m³ depending on species (Prasad and Carlile, 2009; Anon, 2016). Even after composting, supplementary N may be added to media. For example, the nitrogen drawdown index devised by Handreck (1993b) suggests that composted pine bark requires addition of 0.2 g/L Ca(NH₄)₂NO₃ and 0.5 g/L of slow-release N before planting and that aged pine bark needs 0.4 and 0.8 g/L, respectively. Nitrogen immobilization varies with species, for example, spruce bark being more prone to depletion than pine bark (Bilderback, 1982; Prasad and Carlile, 2009), perhaps linked to the much higher concentration of free sugars in bark derived from spruce (Naasz et al., 2009). However, growth reductions in bark-amended substrates may not be entirely due to nitrogen deficiencies and in some instances may be linked to poor root aeration (Naasz et al., 2009), also perhaps to overwatering (Harrelson et al., 2004).

The propensity of barks to inhibit the growth of plant pathogens within growing media has been extensively studied and is attributed to the antagonistic activities of microorganisms present in, especially, composted media. Both hardwood and pine barks have been shown to exert antagonistic effects (Hoitink, 1980), and offer suppression of root diseases including *Pythium* and *Phytophthora* damping off and root rots, as well as *Fusarium* wilts (Hoitink and Zuniga de Ramos, 2004). In addition to direct disease suppression and promotion of root development fostered by high air content, good drainage also serves to reduce sporulation and spread of waterborne root-attacking Oomycete pathogens in media (Hoitink and Zuniga de Ramos, 2004).

8.4.4 PLANT GROWTH

As noted, bark forms the major constituent of nursery stock media in the United States, Australia and is also incorporated into container media in Europe. Spruce bark is a major component of peat-free media destined for retail/hobby markets in the United Kingdom and Ireland (Prasad and Carlile, 2009), with *P. pinaster* bark widely used for both hobby and professional markets in France (Schmilewski, 2017). Bark is also suitable as a propagation medium both under high-humidity conditions and under intermittent mist (Lorenzo et al., 1982; Pokorny and Austin, 1982).

Bark is often mixed with other organic as well as inorganic components to produce nursery stock media: in the United States, sand (0.5–2 mm) is widely used, helping to stabilize containers and retain water (Brown and Pokorny, 1975; Bilderback, 2000) as well as peat (Fields et al., 2015) and other materials such as cotton waste and whole tree substrates (Jackson et al., 2005; Jackson et al., 2009b). In Europe, bark is often mixed with peat and grit (2–5 mm) in container media for shrubs and trees (Edmonds, 1980; Bunt, 1988).

Specialized uses of bark include that of bark chips (20–40 mm) for epiphytes such as orchids and bromeliads. Chips for orchids are usually sterilized by steam (Wang and Gregg, 1992). Bark chips derived from Douglas fir (*P. menziesii*) are used widely in North America for orchids: in Europe, the principal species is *P. pinaster*. Typically, Douglas fir bark is graded to 6–7 mm for growth of genera such as *Cymbidium*; to 12–15 mm for *Cattleya*, *Phalaenopsis*, and *Dendrobium*; and 20–25 mm for *Vandas* (Cullina, 2004). Growing media based on bark chips provide the high air content needed for growth of orchids, and the chips break down only slowly, probably due to the presence of antimicrobial substances (Cullina, 2004): bark chips for orchid media are not composted. Orchids raised in bark chips of high particle size require frequent irrigation, and attempts have been made to introduce coir and other water-retaining substrates into media for orchid

cultivation, with the aim of alleviating water loss during transport of these high value ornamentals (Hwang and Jeong, 2007; Lichty et al., 2015). Materials such as undecomposed sphagnum moss and wood fiber in mixture with pine bark have allowed growth of orchids such as *Phalaenopsis* comparable or even superior (in terms of root growth) to conventional mixtures of pine bark and sphagnum peat (70:30 v/v) (Amberger-Ochsenbauer, 2010).

8.5 WOOD-DERIVED MATERIALS: SAWDUST, WOOD FIBER, AND WHOLE TREE SUBSTRATES

Sawdust is a by-product of the lumber industry whilst wood fiber is manufactured from clean, chipped wood. Whole tree substrates are typically produced by shredding and grinding of complete trees as well as leaves, twigs, branches, and needles removed during forest thinning.

Sawdust is available from lumber operations worldwide and has been used as a component of media in many countries, particularly where peat is scarce or expensive, including Australia (Handreck and Black, 2010; Anon, 2016), New Zealand (Prasad, 1979; Thomas et al., 1980), the United States (Bilderback, 1982), South Africa (Maree, 1994), and Argentina (Sawan and Eissa, 1996). As might be expected, sawdust is produced from many species including conifers such as *P. radiata* in Australia, New Zealand and South Africa, *P. menziesii* and *Tsuga heterophylla* (Raf.) Sarg. in the United States and *Pinus sylvestris* in Europe (Marinou et al., 2013). Sawdust from hardwood species has also been used as a component of growing media (Handreck and Black, 2010; Anon, 2016). Sawdust may be weathered outdoors for a year or more, or composted, before incorporation into growing media (Ehret et al., 1998).

Wood fiber is produced by mechanical defibrillation or more commonly steam-assisted thermal extrusion of clean wood chips through a thermo-screw press (Gumy, 2001). Due to the friction involved in the processing, the product is heated to 80°C–90°C and is therefore free from pathogens and pests, and the process may also drive off volatile toxins (Lemaire et al., 1989). In addition, the system also has the capability to impregnate wood fiber with additives such as nitrogen fertilizers. Furthermore, wood fiber can be produced as coarse or fine grades. As with coir, wood fiber can be compressed for ease of transport (Jackson, 2016). Initial wood fiber products included Hortifibre® manufactured in France and Toresa® produced in Germany. Another wood fiber, Fibralur®, was developed in Spain during the 2000s (Domeno et al., 2010). A major upsurge in the manufacture and use of wood fiber took place during 2010–17 with new manufacturing plants established in the United States, Ireland, Germany and Latvia. In the United Kingdom, volumes of wood fiber used in growing media rose from 62,000 m³ in 2007 (DEFRA, 2009) to 0.5 M m³ in 2013 (DEFRA, 2014).

Interest in the use of whole tree substrates, notably pine tree substrates and associated clean-chip residual materials, arose in the United States in the early 2000s (Jackson, 2016) and was further stimulated by the decline in sawmill activity from 2005 to 2008, potentially leading to reductions in the supply of bark for horticultural use (Bilderback et al., 2013; IEE, 2016). Pine tree substrates are derived from pine trees that are chipped and ground (with or without bark, limbs, or needles) in a hammer mill, or from clean-chip residual ($\approx$ 40% pinewood, 50% bark, and 10% needles) (Boyer et al., 2007; Jackson and Wright, 2009).

8.5.1 PHYSICAL PROPERTIES

Wood-based materials are characterized by good aeration but low water-holding capacity; however, this depends to a large extent on particle shape and size. Sawdust has the advantage of low bulk density (Prasad, 1979) and is often composed of fine particles with mean values typically 1–5 mm (Table 8.13), although for crops such as strawberries, fractions sieved to 6 mm have been used (Depardieu et al., 2016). The low available water content (Prasad, 1979) can result in water stress during active plant growth (Allaire et al., 2005; Dorais et al., 2005), and so irrigation must be applied frequently in small quantities (Favaro et al., 2002).

Wood fiber for use in growing media is characterized by low bulk density, high total porosity, and very high air content (Gruda and Schnitzler, 2004a; Domeno et al., 2010) with a consequent higher oxygen diffusion rate compared to peat (Clemmenson, 2004). Physical properties are determined to some extent by processing (Table 8.13). Apart from bagged modules for glasshouse vegetables, wood fiber is generally not used as a stand-alone growing media component because it retains insufficient plant available water and has a tendency to become compressed during use (Barrett et al., 2016). As with bark, the physical properties of whole tree substrates depend on the degree of processing, and in particular, the sieve size used to manufacture the end product (Table 8.13).

8.5.2 CHEMICAL AND MICROBIOLOGICAL CHARACTERISTICS: STABILITY

The pH of sawdust varies according to source with reports of neutral, 6.3–7.7 (Prasad, 1979), to more acidic values of 5.33 (Goh and Haynes, 1977) and 4.71 (Marinou et al., 2013). Wood fiber pH values are reported at 3.8–6.6. Sawdust and wood fiber have little buffering capacity, and their pH may be determined more by materials with which they are mixed, or by pH of the liquid fertilizer applied during plant growth. Indeed, the CEC of sawdust, wood fiber, and whole tree substrates is low: the relatively high value in Table 8.14 given by ADAS (2016) for wood fiber is probably due to grinding of samples to 2 mm. For all wood-based substrates, macro and micronutrient concentrations in water or $\text{CaCl}_2/\text{DTPA}$ (CAT) extracts, with the exception of K in some materials, are generally very low (Table 8.14).

As with some barks, plant growth is severely restricted in uncomposted sawdust (Allison, 1965). In some cases, notably with sawdust from cedar, phytotoxic substances may inhibit plant growth, but in most instances depletion of nitrogen is responsible for growth inhibition. Immobilization of nitrogen supplied to plants in the substrate solution, and associated media degradation and shrinkage, is a major problem with wood-based substrates, and may be linked to the proportions and possibly relationships between polysaccharide and lignin present in these materials. In contrast to peat, coir and bark, wood-based materials have lower lignin contents and according to one report (Domeno et al., 2010) a very high concentration of hemicelluloses. The latter polysaccharides may be readily degraded in wood-based substrates, especially those with small particle sizes, such as sawdust, fine wood fibers, and finely ground whole tree substrates that have a large surface area for microbial colonization. As with bark, microorganisms colonizing wood substrates derive their carbon principally from the polysaccharide components and nitrogen from fertilizers added to growing media: depletion of nitrogen in wood-based as well as other substrate components is referred to as nitrogen drawdown, seen in Table 8.15 (Handreck and Black, 2010), requiring supplementary N in the form of liquid feed or controlled release fertilizers.

Table 8.13 Physical Properties of Wood Substrates

Material, Origin, Particle Size (Where Noted)	Dry Bulk Density (kg/m ³)	Total Pore Space (% By Volume)	Air Content (% By Volume)	Available Water: 1–10 kPa Suction [†] (% by Volume)	Source
Sawdust (<i>Pinus radiata</i>) 0.6–3.57 mm	140	90.1	42.3	5.7	Goh and Haynes (1977)
Sawdust (<i>Pinus sylvestris</i>) <4 mm	117	89.8	nd	nd	Marinou et al. (2013)
Sawdust (<i>Eucalyptus camaldulensis</i>) 0.1–5 mm	180	86	51	15	Handreck and Black (2010)
Wood fiber: coarse over 50% >2 mm Germany	83–163 ^a	93.2	51.9	19.4	Gruda and Schnitzler (2004a)
Wood fiber: fine: over 50% <1 mm Germany	102–204 ^a	91.4	21.9	34.7	Gruda and Schnitzler (2004a)
Wood fiber, Spain	70	95.1	58	13.5	Domeno et al. (2010)
Wood fiber (United Kingdom)	60–111	nd	25.4–51.7	13–24.7 ^b	ADAS (2016)
Pine tree substrate (<i>Pinus taeda</i>)					
1.59 mm screened	150	83.3	18.2	50.4	All from Jackson and Wright (2009)
2.38 mm screened	150	85.7	23.2	42.8	
3.18 mm screened	150	86.6	31.8	33.6	
4.76 mm screened	150	79.2	31.3	25.7	
6.35 mm screened	150	82.4	39.1	19.8	
Whole pine tree (clean-chip residual) 12.7 mm	180	85	47	nd	Boyer et al. (2008)
For European samples, determinations of total pore space, air content, and available water were carried out using suction funnels with a porous plate as described by De Boodt et al. (1974). Handreck and Black (2010) outline the procedures used in Australia for physical determinations, with air content measured at 6 cm (0.6 kPa) suction. Boyer et al. (2008), also Jackson and Wright (2009) used an in-house porometry system developed at the Horticultural Substrates Lab at North Carolina State University with a mean suction of 3.8 cm (0.38 kPa). nd, Not determined.					
^a Lightly compressed to heavily compressed.					
^b Available water at 5 kPa.					

Loss in volume due to decomposition of media containing sawdust is reported as high as 50% over 1 year (Handreck and Black, 2010). Decomposition also causes slumping with an attendant decrease in air content. Thus application of nitrogen fertilizers at high rates are required to successfully raise plants in, particularly, sawdust derived from *Eucalyptus* spp. Even composted sawdusts may have a rate of nitrogen immobilization that, due to cost, precludes their use in media for bedding plants and propagation (Handreck and Black, 2010).

Prasad (1997b) reported that nitrogen immobilization associated with microbial activity was very pronounced in several wood fiber-based growing media, with reductions in available N up to 150 mg/L during 5 weeks of incubation. Release of N from organic forms of N in wood fiber-based media varied: with urea, N release was much faster than with hoof and horn fertilizer, the latter clearly being degraded more slowly by the indigenous microbial flora of the wood fiber-based

Table 8.14 pH, EC, CEC, and Extractable Inorganic Nutrients in Wood Substrates														
Substrate and Origin (Where Stated)	pH	EC (dS/m)	CEC (meq 100 mL)	NH ₄ N	NO ₃ N	P	K	Ca	Mg	Fe	Mn	Zn	Cu	
				Nutrients as mg/L ⁻¹ in water* or CaCl ₂ /DTPA extracts										
Sawdust	4.71	0.74				0.47	0.27							Marinou et al. (2013)
Wood fiber, France			10		3	3	35		7					Lemaire et al. (1989)
Wood fiber, Spain "Fibralur"	6.0	0.3	2.2		4.4	0.6	34.9	11.3	1.5					Domeno et al. (2010)
Wood fiber (United Kingdom)	5.4–8.3	0.053–0.441	9.0–14.9	0.9–9.1	0.6–2.8	1.0–43	2.7–586	0.4–19.8	0.2–5.1	0.06–1.63	<0.01–0.23	<0.02–0.09	<0.01–0.01	ADAS (2016)
Wood substrate: pine chips	5.7	0.09	0.2			2.7	17	3.5	3.0	0.12	0.004	0.016	0.004	Wright and Browder (2005)
Wood substrate: clean-chip residual: 0.48 and 0.95 cm*	5.05.5*	0.210.15*		0.20.1*	0.10.2*	4.11.4*	84.648.2*	9.15.3*	7.02.6*	3.72.1*	2.30.6*	0.280.06*	0.010.02*	Boyer et al. (2012)

Table 8.15 Changes With Time in Concentrations of Ammoniacal and Nitrate Nitrogen (mg/L^{-1}) in Mixtures of Peat and Sawdust. Nitrogen Added as Ammonium Nitrate at 1.0 g/L

Incubation Time (week)	Percentage Peat									
	100		75		50		25		0	
	NO_3	NH_4	NO_3	NH_4	NO_3	NH_4	NO_3	NH_4	NO_3	NH_4
0	58	61	90	90	97	97	90	108	90	112
1	94	68	87	68	108	54	119	54	115	90
2	64	76	58	7	39	4	14	4	0	0
3	90	76	87	7	32	0	0	0		
5	97	79	90	0	0					
10	104	0	83	0						

Reproduced with permission from Handreck, K., 1992. Rapid assessment of the rate of nitrogen immobilisation in organic components of potting media: I. Method development. Commun. Soil Sci. Plant Anal., 23, 201–215.

media. Decomposition of media containing wood fiber leads to shrinkage, this being far greater than in peat media (Prasad and Maher, 2004).

Nitrogen immobilization in wood fiber can be ameliorated to some extent by incorporation of N at the production stage. Gruda et al. (2000) found rapid nitrogen immobilization in untreated wood fiber, while in N-treated wood fiber the immobilization was similar to that in H2/H3 (white) peat. Tomato transplants cultivated on nitrogen-impregnated wood fiber showed significantly greater growth than plants cultivated in unimpregnated wood fiber (Gruda and Schnitzler, 1999). Nevertheless, Gruda et al. (2000) recommend supplying plants growing in wood fiber substrates with nutrient solutions or fertilizer from planting onward.

Boyer et al. (2012) showed that clean-chip residual wood substrates may immobilize nitrogen. They also found that immobilization was greater than in pine bark, and that microbial respiration in clean-chip residual was vastly greater than in peat. Furthermore, respiration was more pronounced in clean-chip residual substrates of smaller particle size. Thus supplementary N is recommended for these materials. Indeed, Wright et al. (2008) recommend addition of 100 mg/L N for a pine tree substrate to achieve similar quality to chrysanthemums raised in a peat-perlite medium. Despite N immobilization, Jackson et al. (2009b) found no visible degradation or shrinkage after 2 years in large containers with woody nursery crops. Boyer et al. (2009), in a year-long study with four nursery stock species in clean-chip residual, found substrate shrinkage to be equivalent to that in pine bark media for two species, and slightly greater (up to 1 cm decrease in substrate height) than pine bark for two others. Of four grades of clean-chip residual used, shrinkage proved greatest in the largest particle size (3.18 cm) which the authors attribute to settling effects. Shrinkage did not affect crop growth or commercial value.

8.5.3 PLANT GROWTH

Sawdust has been used, alone or in mixtures, for cultivation of greenhouse vegetables and strawberries in bag or module culture. Whereas Maree (1994) and Parks et al. (2004) found *P. radiata* sawdust suitable for raising tomato and cucumber plants, Allaire et al. (2005) found fresh sawdust

blends from white spruce [*Picea glauca* (Moench) Voss.] and fir [*Abies balsamea* (L.) Mill.] to be inferior to mineral wool as substrates for tomato culture. Mixtures of sawdust with other substrates are recommended by several authors. For strawberry (*Fragaria* × *ananassa* Duch.), [Marinou et al. \(2013\)](#) found sawdust mixtures (e.g., with pumice 50:50 vol:vol) to be superior to sawdust alone for cultivation of strawberry. [Depardieu et al. \(2016\)](#) found yields of strawberry in a 1:3 vol:vol mixture of sawdust and peat to be similar to those in pine bark and peat. Where negative results have been reported, they may be due to no or incomplete composting of sawdust ([Prasad, 1980](#)) or insufficient neutralization of phytotoxic effects ([Worrall, 1978](#)). [Prasad and Maher \(2006\)](#) reported that with a number of nursery stock species, compost produced from a mixture of sawdust, and spent brewers grain could successfully substitute for peat in a growing medium.

Like sawdust, wood fiber has been successfully used with a range of protected vegetable crops. [Hardgrave and Harriman \(1995\)](#) found wood fiber (Hortifibre) to be superior to a range of other organic substrates for growth of cucumber. [Gruda and Schnitzler \(2004b\)](#) found growth of young tomato plants in wood fiber to be equivalent to that of tomatoes in “white peat” (H undefined), with well-developed root systems in the former. Yield of tomatoes in bags containing wood fiber (Fibralur) was equivalent to that of coir with spring-planted tomatoes, but slightly inferior with autumn-planted tomatoes ([Domeno et al., 2010](#)).

[Bohne \(2004\)](#), working with several nursery stock species, found that plants grown in 25% and 50% (v/v) wood fiber substrates did not statistically differ in quality compared to peat, in contrast to plants grown in 75% and 100% fiber which were statistically better compared to peat. Due to the decreasing available water with increasing amounts of wood fiber, plants in substrates with >50% wood fiber had to be irrigated more often than peat, but overall the total amount of irrigation water per growing season for wood fiber was smaller than for peat. [Gerber et al. \(1999\)](#) demonstrated that geranium (*Pelargonium* × *hortorum* L.H. Bailey) could be grown in 100% wood fiber with similar growth to plants grown in peat if they were irrigated and fertilized more often than the plants grown in peat. Wood fiber is used extensively in mixture with peat to produce growing media for retail markets in the United Kingdom and Ireland, and as components of media for bedding plants ([Drury, 2015](#)).

A wide range of greenhouse crops have been produced successfully with whole pine tree/clean-chip residual substrates including chrysanthemums, poinsettia, as well as annual bedding plants (including ageratum, salvia, and impatiens), vegetables, and herbaceous perennials ([Jackson et al., 2007](#); [Boyer et al., 2008](#)). Nursery stock and tree species have been successfully raised in these media ([Wright and Browder, 2005](#); [Boyer et al., 2009](#)). Root growth of annual and woody plants grown in whole pine tree substrates is reported to be equal to, or better than in peat or pine bark ([Jackson et al., 2007](#)). However, when freshly harvested trees were used to produce substrates, growth reduction of young marigold and tomato seedlings in plug trays was occasionally observed, and thus the use of aged whole tree substrates for young transplants was recommended.

8.6 RICE HULLS

The hull (or husk) is the outermost layer of the rice (*Oryza sativa* L.) grain and acts as a protective cover to the grain. Rice hulls are separated from the rice grains by milling following

parboiling (soaking, steaming, and drying the grain). Worldwide, rice hulls are available in large volumes as waste from rice processing operations: rice production in Asia produces about 770 Mt of hulls annually (IRRI, 2017). Much of this is burnt as a fuel, and the ash used in construction materials. In the United States, rice processing may lead to as much as 1.9 Mt of hulls annually (Sambo et al., 2008), and here rice hulls are incorporated by several manufacturers into growing media (Anon, 2014a). About 12,000 m³ of rice hulls were used by professional growers in the Netherlands in 2013 (Schmilewski, 2017). Aged, composted, or charred (carbonized) rice hulls have been considered for use in growing media (Einert, 1972; Kämpf and Jung, 1991). Biochar produced from rice hulls has been considered for use in growing media (Kim et al., 2017).

Rice hulls have an elongated convex shape with typical dimensions of 3–4 by 5–7 mm (Evans, 2008), and dry bulk densities of uncompressed, non-ground material at around 100 g/L (Mansaray and Ghaly, 1997; Gómez and Robbins, 2011), rising to 400 g/L depending on the extent of milling/grinding/sieving (Sambo et al., 2008; Buck and Evans, 2010). Organic constituents include cellulose (30%–40% by weight), hemicelluloses (15%–25%), and lignin (15%–25%), but hulls also contain about 15% by weight of silica (López et al., 2010; Kumar et al., 2010): the latter two components confer a high degree of stability to rice hulls, which degrade only slowly (Evans and Gachukia, 2004).

Evans and Gachukia (2007) reported that the large particles of parboiled rice hulls incorporated at up to 50% (v/v) enhanced drainage and aeration in peat-based substrates. Hanan (1998) and Gómez and Robbins (2011) report water-holding capacities for fresh parboiled rice hulls respectively of 20% and 23% (v/v), total pore space of 89% and 93% (v/v), and an air-filled pore space of 69% and 70% (Table 8.16). Gómez and Robbins (2011) also showed that only small decreases in total porosity occurred in media containing pine bark and parboiled rice hulls after more than a year in storage: air space declined in all mixtures of rice hulls and peat, whilst water-holding capacities correspondingly rose. Sambo et al. (2008) and Buck and Evans (2010) using milled or ground rice hulls produced a range of particle sizes: all ground products had higher bulk densities than fresh hulls. As particle size decreased, total pore space decreased and water-holding capacity generally increased (Table 8.16). Rice hulls are reported to have a low CEC (Evans, 2008).

pH values of fresh rice hulls are reported as 4.6–4.8 by Evans (2011) rising to 6.5–7.0 in storage, values also reported by Zanin et al. (2011). Both authors reported moderate EC values (Table 8.17). Whereas Evans (2011) and Zanin et al. (2011) both reported relatively high concentrations of K in rice hulls, Evans (2011), using ground rice hulls, found relatively high concentrations of P (Table 8.17), being considered sufficiently high to cause adverse effects on plant growth if ground hulls were used as the sole component of growing media. The water extraction procedure used by Zanin et al. (2011) may have resulted in their much lower P concentration. Evans and Gachukia (2004) found little immobilization of N in media containing rice hulls and attributed this to the stability of the substrate.

Rice hulls are rarely used as the sole component of growing media. Evans and Gachukia (2004) demonstrated growth of a range of bedding plants including impatiens (*Impatiens walleriana* Hook, f) and pansy (*Viola* × *wittrockiana* Gams) in peat-based media containing up to 40% (v/v) parboiled rice hulls without adverse effect on root or shoot development compared to growth in a pure peat medium. Evans (2011) further demonstrated equivalent growth of bedding

Table 8.16 Physical Properties of Rice Hulls

Origin, Treatment, Particle Size (Where Noted)	Dry Bulk Density (kg/m ³)	Total Pore Space (% By Volume)	Air Content (% By Volume)	Available Water: 1–10 kPa suction [†] (% By Volume)	Source
Ground rice hulls, 2 mm screened, Italy	390	77.7	31.3	30.6	Sambo et al. (2008)
Ground rice hulls, 6 mm screened, Italy	280	83.9	47.0	22.1	Sambo et al. (2008)
Parboiled fresh rice hulls, milled once. Arkansas, United States	200	86.9	39.6	25.8	Buck and Evans (2010)
Parboiled fresh rice hulls, ground and screened to 0–18 –1.8 mm. Arkansas, United States	300	74.9	16.2	31.6	Buck and Evans (2010)
Fresh parboiled rice hulls, Arkansas, United States	100	93.7	70.6	nd	Gomez and Robbins (2011)

Determinations of total pore space, air content, and available water were carried out using suction funnels with a porous plate as described by De Boodt et al. (1974). nd, Not determined.

Table 8.17 pH, EC, and Extractable Inorganic Nutrients (mg.L⁻¹) in Parboiled Rice Hulls

Rice Hulls and Origin (Where Stated)	pH	EC (dS/m)	NH ₄ N	NO ₃ N	P	K	
Arkansas, United States. Ground screened 1–2 mm	4.6	0.7	12	0.5	130	375	Evans (2011)
Arkansas, United States. Ground screened 0.5–1 mm	4.8	1.2	10	0.5	60	300	Evans (2011)
Rice hulls (origin not stated). Ground to 2 mm	6.8	0.4	nd	0.47	7.7	146	Zanin et al. (2011)

Zanin et al. (2011) used water extraction; Evans (2011) used a saturated paste extract.

plants such as vinca (*Catharanthus roseus* L.G. Don), impatiens (*I. walleriana*), and geranium (*Pelargonium × hortorum* L.H. Bailey) in peat media, and media amended with up to 30% rice hulls. As a component of container media, Gómez and Robbins (2011) found that amending pine bark with up to 40% (v/v) parboiled rice hulls did not result in a significant decrease in plant growth or increase the volume or frequency of irrigation for container-grown spirea (*Spiraea × bumalda* L.) and that the physical properties of media did not deteriorate markedly over two growing seasons.

8.7 COMPOSTED MATERIALS

Materials such as coir, rice hulls and especially bark may be composted prior to use in growing media. Worldwide one of the principal aims of composting is to deal with organic wastes and by-products. Volume reductions, elimination of toxic organic substances as well as freely available carbon sources may result in end-products that can be considered for use in growing media. Composted materials per se are available in large quantities in many countries, and many composted wastes and by-products have been investigated for their potential as components of growing media (Table 8.18). However, of all candidate materials for growing media, composts are the most variable; physically with respect to bulk density, airspace, and water retention; chemically in terms of pH and nutrition, and also microbiologically (Raviv, 2011, 2013). For these reasons, composted materials (other than bark) are almost always used in combination with other materials. Composts produced from the so-called green materials (prunings, shredded branches, plant debris, and waste from gardens and nurseries) are widely used as components of growing media in the Netherlands, United Kingdom, Italy, and Germany, primarily in media for the hobby market. Compost use in growing media in Germany increased from 0.25 M m³ in 2005 to 0.7 M m³ in 2013; in the Netherlands from 10,000 m³ in 2001 to 95,000 m³ in 2013; in Italy, over 0.29 M m³ was used in 2013 (Schmilewski, 2017). In the United Kingdom, 0.31 M m³ of compost was incorporated into growing media in 2013 (DEFRA, 2014). Municipal wastes and sewage sludges are incorporated

Table 8.18 Examples of Composts Investigated as Components of Growing Media

Compost Feedstock ^a	Crop	Region	Growing System	Compost Fraction (%)	Reference
Municipal solid waste (MSC)	Various ornamentals	North America	Containers	15	Calkins et al. (1997)
Biosolids and yard trimmings	Various ornamentals	North America	Containers	100	Wilson et al. (2004)
Sewage sludge	<i>Bougainvillea</i> spp.	Mediterranean	Containers	Up to 70	Vecchiotti et al. (2013)
Sewage sludge	Broccoli	Mediterranean	Growing bed	30	Perez-Murcia et al. (2006)
Dairy manure	Bedding	North America	Pots	60	Schober et al. (2010)
Grape marc + cow manure	Vegetable transplants	Mediterranean	Trays	50	Bustamente et al. (2008)
Leached SMC ^b	Bedding plants	North America	Pots	50	Young et al. (2002)
Olive mill waste	Foliage pot plants	Mediterranean	Pots	25	Papafotiou et al. (2005)
Green brewery wastes	Tomatoes, bedding plants	Ireland	Pots	Up to 40%	Ní Chualáin et al. (2011)
Peanut	<i>Dracaena marginata</i>	Iran	Pots	Up to 100%	Torkashvand et al. (2015)
Seaweed (<i>Posidonia</i>)	Tomatoes	Mediterranean	Pots	20	Montesano et al. (2014)
Cotton gin compost	Ornamental shrubs	North America	Containers	60	Jackson et al. (2005)
Orange peels + cow manure	Tomato	Mediterranean	Containers	100	Raviv et al. (2005)
Newspaper + horse manure	Tomato	Northern Europe	Containers	100	Ball et al. (2000)

^aMain component.

^bSpent mushroom compost.

into growing media in some southern European countries, such as Spain (Moral et al., 2013) and also in Canada and the United States, where composts are often used as components of media for production of nursery stock (Chong, 2005; Bilderback et al., 2013). Strict quality control procedures are essential in preparing composts for use in growing media, with, for example, the Waste and Resources Action Programme (WRAP) in the United Kingdom and the principal body monitoring growing media standards in Europe, the Netherlands-based RHP foundation, outlining acceptable parameters for composts destined for use in growing media (WRAP, 2011; Wever and Scholman, 2011). The subject of compost production for horticultural purposes was reviewed by Raviv (2005) and Morales et al. (2016). Composting for growing media was reviewed by Raviv (2013).

8.7.1 THE COMPOSTING PROCESS

The composting process is the biological decomposition of organic materials under controlled conditions into a stable humus-like product (Golueke, 1972). Thorough composting including a prolonged thermophilic phase may reduce phytotoxicity, render pests, pathogens and weed seeds unviable, and stabilize the end product with respect to N and oxygen demand.

The C-to-N ratio is an important variable in composting and the desired range for the source material is between 25 and 35 (Day and Shaw, 2001). If the value is much higher than this, then the rate of decomposition will be very slow and the resulting compost may not be stable. At lower C:N ratios, loss of N in the form of ammonia (NH_3) to the air is often high. The desired C:N ratio can be achieved by analyzing the feedstock and correcting its C or N contents by supplementing with C rich materials such as wood or pruning waste or N rich substances such as poultry manure. The type of C and N present may also need to be taken into account: lignin-rich materials may break down slowly in composting systems, as may nitrogen within protein-based wastes from municipal solid wastes as well as cheese whey and other dairy operations. The C:N ratio falls during composting as CO_2 is liberated and a final value in the range 15–20 is generally aimed for (Kayhanian and Tchobanoglous, 1993).

Materials to be composted are first shredded to reduce particle size and to promote microbial invasion into the substrate to accelerate the composting process. Shredding also improves handling of the material and facilitates prescreening and mixing (Rynk and Richard, 2001). After shredding, adding any materials such as N and water, and typically forming into compost piles, the temperature will begin to rise as microbiological activity increases. After a few days the temperature will typically be over 60°C : the thermophilic stage that lasts until the easily decomposable organic matter has been broken down. It is important for uniform composting that all of the material be exposed to similar temperatures. This is accomplished in many systems by regular turning of the composting pile to ensure sufficient oxygen is available for aerobic degradative processes to occur. Where turning is not done, as in static piles, temperature uniformity throughout the pile is an important consideration in the design of the composting system. Turns and/or forced aeration should prevent temperatures rising above $60\text{--}65^\circ\text{C}$, since ashing or loss of organic matter may occur. Once the easily decomposable matter is degraded by microbes, temperature will start to decrease and as it falls below 40°C the compost will be reinvaded by mesophilic microorganisms that will minimize ammonia loss due to nitrification of ammonium ions. In the last stage, the active composting phase gives way to the curing period where the compost matures and becomes ready

for use as a component of a growing medium. For most materials in regularly turned piles, this will take 12–20 weeks. Some of the microorganisms that invade the compost during the curing phase are responsible for compost suppressiveness against a wide variety of soil-borne pathogens (Neher et al., 2017).

Maturation and compost stability are important characteristics in relation to use as a growing medium component. If active decomposition continues after incorporation into a growing medium, then this will adversely affect plant growth due to reduced levels of available oxygen and nitrogen and/or the presence of phytotoxic compounds. Consequently, tests have been developed to evaluate the stability/maturity/phytotoxicity of composts. Common stability and maturity tests are respirometric tests based on O₂ consumption and CO₂ evolution (Lasaridi and Stentiford, 1998; Scaglia et al., 2000; Ní Chualáin and Prasad, 2009), or self-heating tests (Kehre and Pohle, 1994). Rate of heat production using microcalorimetry has been suggested as a very sensitive measure of compost maturity (Laor et al., 2004; Medina et al., 2009). Chemical tests such as the nitrate/ammonium ratio, C:N ratio, and content of volatile organic acids are used in various countries to assess compost maturity (Hogg et al., 2002). Inbar et al. (1989) used solid state ¹³C nuclear magnetic resonance and infrared spectroscopy to determine the degree of composting. Phytotoxicity tests generally involve a bioassay to screen for negative impacts on seed germination and plant growth (Prasad et al., 2010).

8.7.2 COMPOSTS: PHYSICAL AND CHEMICAL PROPERTIES

The nature of source materials and their processing generally determines the physical and chemical properties of composted materials. Tables 8.19–8.21 indicate this well (Noble, 2005). Pore space is high, for example, in spent mushroom compost (SMC) as well as in many, but not all green composts, with materials containing animal manures having lower values. Air contents also vary, being high with materials derived from grape marc and olive wastes, with water contents being correspondingly lower. Most materials had residual lignin values of 15%–25%, lower than peat or coir. All the materials with the exception of some SMCs had much higher bulk densities than typical peats (Table 8.19), and this is a common problem with most composted materials (Surrage and Carlile, 2009).

pH levels are generally high, and all materials contained relatively high ionic concentrations, notably K, with correspondingly high EC values (Table 8.21), effectively precluding their use as single media components and indeed restricting their levels in growing media. Nutrient ions (NH₄, NO₃, P, K) and micronutrients present in some composts may enable lower rates of fertilization in growing media, but composted materials may also contain unacceptably high concentrations of Na and Cl (Cai and Gao, 2011), as well as high concentrations of heavy metals (López-López and López-Fabal, 2013). In this respect, both the WRAP in the United Kingdom and the Netherlands-based RHP have designated acceptable concentrations for ions within composted materials destined for use in growing media (WRAP, 2011; Wever and Scholman, 2011).

8.7.3 PATHOGEN SURVIVAL IN COMPOST

Plant, human, or animal pathogens may contaminate waste material destined for composting, with concerns that such organisms may survive composting processes. In a review of human and animal

Table 8.19 Physical Properties of Some Composted Materials Found by Researchers From Various Countries Participating in an International Project on the Disease Suppressive Properties of Composts (Noble, 2005)

Composted Material	Country	Dry Bulk Density (g/L)	Pore	Air	Water	Particle Size		
			Space (%)	Content (%)	Content (%)	% w/w		
						>10 mm	4–10 mm	<4 mm
SMC	United Kingdom (F)	171	91	33	58	39	42	19
	United Kingdom (M)	263	87	20	68	33	31	36
	Ireland	149	92	42	50	21	53	26
	France	275	87	31	56	5	75	20
	Greece	197	90	53	37	14	40	46
Green waste	United Kingdom	554	73	9	65	3	18	79
	Ireland	419	81	31	50	13	31	56
	France	410	81	24	57	5	75	20
	The Netherlands	216	90	26	53	9	46	45
Biowaste	United Kingdom	439	78	30	48	3	81	16
	Ireland	244	87	47	40	—	—	—
Onion waste	United Kingdom	328	85	17	68	77	13	0
Grape marc	Greece	444	82	45	37	1	15	84
OPC ^a + OLW	Greece	232	85	48	37	18	27	55
SCM ^b + orange peels + straw	Israel	461	76	35	40	24	34	42
SCM + pepper plants	Israel	329	82	48	35	16	28	56

SMC, *Spent mushroom compost*.

^aOPC, *Olive press cake*; OLW, *olive leaf waste*.

^bSCM, *Separated cattle manure*.

Table 8.20 Chemical Properties of Some Composted Materials Found by Researchers From Various Countries Participating in an International Project on the Disease Suppressive Properties of Composts (Noble, 2005)

Composted	Country	Dry Matter	N	P	K	WSC ^b	Lignin (% of Dry Matter)
Materials		(%)	g/kg Dry Matter				
SMC	United Kingdom (fresh)	41.0	26.7	9.6	26.0	1.8	22
	United Kingdom (mature)	37.3	20.8	7.2	12.1	1.3	22
	Ireland	32.7	24.5	7.7	29.7	2.2	19
	France	48.7	17.8	1.8	18.1	1.2	15
	Greece	83.4	20.3	4.0	12.2	1.7	21
Green waste	United Kingdom	72.8	—	—	—		24
	Ireland	61.3	11.2	1.9	9.8	0.9	20
	France	56.9	17.1	4.1	10.8	1.0	21
	The Netherlands	60.7	—	—	—		20
Biowaste	Ireland	74.6	24.6	5.9	11.0	1.9	28
	United Kingdom	68.3	—	—	—	—	—
Onion waste	United Kingdom	25.5	18.4	3.3	14.0	30.4	12
Olive leaves	Greece	48.7	21.6	1.4	5.6	1.0	17
Grape marc	Greece	73.4	—	—	—	0.7	39
SCM ^a + orange peels + straw	Israel	55.0	23.0	2.1	10.0	—	—
SCM ^a + pepper plants	Israel	55.0	20.0	3.6	33.1	—	—

SMC, Spent mushroom compost.
^aSCM, Separated cattle manure.
^bWSC, Water-soluble carbohydrate.

pathogen survival in composted green waste (CGW), Jones and Martin (2003) concluded that most pathogens are inactivated by a composting procedure with a residence time of 3 days at a temperature higher than 55°C. Noble and Roberts (2003, 2004) reviewed temperature–time effects and other sanitizing factors of composting on 60 plant pathogen and nematode species. For all of the bacterial plant pathogens and nematodes, the majority of fungal plant pathogens, and a number of plant viruses, a compost temperature of 55°C for 21 days was sufficient to ensure eradication. However, some pathogens with highly resistant spore and sclerotial resting bodies proved more difficult to eradicate. *Plasmodiophora brassicae* (clubroot of *Brassica* spp.), *Fusarium oxysporum* f. sp. *lycopersici* (tomato wilt), and *Macrophomina phaseolina* (dry root rot) were more temperature-tolerant, as they survived a peak compost temperature of at least 62°C (maximum 74°C) and a composting duration of 21 days. However, Fayolle et al. (2006) showed a complete eradication of *P. brassicae* in both bench- and large-scale composting procedures at temperatures

Table 8.21 Water-Soluble Nutrient Levels in Composted Materials (mg/L⁻¹) Found by Researchers From Various Countries Participating in an International Project on the Disease Suppressive Properties of Composts (Noble, 2005)

Composted Material	Country	pH	EC ^a	P	K	NH ₄ N	NO ₃ N
SMC	United Kingdom (F)	7.4	549	75	2300	0	440
SMC	United Kingdom (M)	6.9	209	35	1105	0	250
SMC	Ireland	7.1	348	61	2961	42	7
SMC	France	7.8	361	55	2260	350	0
SMC	Greece	8.0	359	82	2145	275	70
Green waste	Ireland	8.0	151	14	1009	0	119
Green waste	France	7.8	122	61	780	0	90
Biowaste	Ireland	8.4	232	36	1234	28	67
Onion waste	United Kingdom	7.1	94	70	2560	105	10
Grape marc	Greece	7.1	213	105	280	15	0
OPC + OLW ^b	Greece	6.7	319	140	561	21	0
SCM ^c + orange peel + straw	Israel	5.7	685	6	780	3	125
SCM + pepper plants	Israel	7.1	713	2	1296	2	149

SMC, Spent mushroom compost.
^amS/m in a 1:5 water extract.
^bOPC, Olive press cake; OLW, olive leaf waste.
^cSCM, Separated cattle manure.

as low as 50°C provided that the moisture content was sufficient (60% w:w). Composting of infected plant residues resulted with complete eradication of *Clavibacter michiganensis* (the causal agent of a brown rot of potato and bacterial canker of tomato) (Yogev et al., 2009). Moreover, the resulted compost was suppressive against the pathogen. Some plant viruses, notably tobacco mosaic virus are resistant to composting processes with a high temperature tolerance (> 70°C) and may only be completely inactivated in closed bioreactors (Ghaly et al., 2006). The eradication conditions of some other viruses (e.g., *Potato spindle tuber viroid*) were too difficult to define in a recent research (Kerins et al., 2018). Cyst-forming plant parasitic nematodes of the genus *Globodera* may be eliminated by 8 days of composting where all materials have reached a temperature of minimum 50°C during the period (Bøen et al., 2006).

Particular studies have been carried out on survival of *Escherichia coli* (notably strain 0157) *Salmonella enterica*, *Listeria monocytogenes*, and *Campylobacter* spp. in composts, especially those based on animal by-products (Berry et al., 2013; Erickson et al., 2015). Berry et al. (2013) found that some minimally managed composting processes can reduce *E. coli* O157:H7 and other pathogens in bovine manure but detected both *E. coli* O157:H7 and *Listeria* spp. from the edge and top of static piles after 84 days of composting, and highlighted the importance of adequate C:N formulation of initial mixtures for the generation of high temperatures and rapid composting, and the need for periodic turning of the piles to increase the likelihood that all parts of the mass are subjected to high temperatures. Inactivation of a wide range of human and other animal viruses,

including bovine and porcine viruses, Newcastle disease of poultry, and polio virus, appears to occur on exposure to 55°C for 10 minutes in composting systems (Elving et al., 2012).

Compost temperatures in excess of 60°C can be achieved in a range of composting systems using a wide range of source materials. However, there is insufficient information on the survival risks of pathogens in cooler zones of the compost, particularly with in-vessel systems in cases where the compost is not turned.

The demonstration by Steele et al. in Australia (Steele et al., 1990a,b; Hughes and Steele, 1994) that composted bark, largely based on *P. radiata*, is a source of *Legionella* infections prompted safety concerns about the use of composted materials as growing media. *Legionella longbeachae* was the principal organism of concern, with infections by this species outnumbering infections by *Legionella pneumophila*. Both of these organisms may cause pneumonia-like symptoms. Indeed, growing media are now included in the risk assessment for pneumonia within Australia. Steele et al. (1990b) found that 33 out of 45 samples of commercial potting mixes in Australia contained *Legionellae*. Studies of multiplication in these media did not show increases in numbers of *Legionellae* in the manufactured products, but the bacteria were able to survive in media for 6–10 months at temperatures between 4°C and 35°C.

Although Steele et al. (1990b) examined two samples of peat-based growing media from Europe for *Legionellae*, and could not detect these, *L. longbeachae* has been associated with growing media in Scotland and elsewhere (Lindsay et al., 2012; Potts et al., 2013), linked to the use of peat diluents, possibly exacerbated by long storage periods allowing these amoeba-borne pathogens to multiply. However, the disease incidence in Scotland is considered very low, the presence of *Legionella* species in growing media does not correlate with the number of cases confirmed there, and thus the risk to users of growing media is similarly held to be low (Anon, 2013; Currie and Beattie, 2015).

8.7.4 PLANT GROWTH

The use of compost as a component of growing media has been reviewed for nursery stock by Fitzpatrick (2001) and for vegetable transplant production by Sterret (2001). Pronk (1995) found that composted vegetable, fruit, and garden waste could be substituted for up to 15% of the peat in the potting medium, if fertilization is adjusted. Larger percentages of compost resulted to high pH levels. In a wide-ranging review of research over 10 years into amending peat media with composted materials based on green wastes and brewery by-products, Prasad and Carlile (2009) reported good growth of many species including tomatoes, pot plants as well as a range of ornamentals at amendment levels up to 40% (v/v). Germination of test species such as tomato and antirrhinum was largely unaffected at compost inclusion levels up to 25%. Enhanced growth of many plants was observed at higher rates of inclusion, attributed to elevated concentrations of macro- and micronutrients arising from composted materials. N immobilization was noted as a problem with CGW by Jauch and Fischer (1993) who also reported a reduction in the fresh weight of pot plants when it was used at a rate of 50%. Hartz et al. (1996) also noted N immobilization by CGW. However, studies undertaken by Ní Chualáin et al. (2011) using N- and P-rich dairy wastes as a compost accelerant were found to further enhance the nutrient content of compost/peat media, leading to plant growth with no signs of N deficiency, even after storage of media for several months.

(Ní Chualáin, 2016). Such amended media are marketed in for hobby gardeners in the United Kingdom (Ní Chualáin et al., 2017).

The main problems in substituting compost for peat are phytotoxicity due to incomplete composting, insufficient available N if the source materials are not properly formulated and elevated pH and salt levels. Water-holding capacity is also affected as the proportion of compost in a growing medium rises. In spite of the above, it should be noted that based on current know-how, it is possible to produce compost that in some cases can fully substitute peat moss in growing media, with similar yield results (Rogers, 2017).

8.8 ANAEROBIC DIGESTATES

Anaerobic digestion (AD) involves microbial breakdown of biodegradable materials in the absence of oxygen within a controlled environment. Suitable biodegradable materials for use in AD include leftover food, livestock manures, green compost as well as maize and grass silage. AD produces biogas which can be used to generate renewable heat and electricity, and residual digestate that has been considered as a potential component of growing media after appropriate composting (Raviv et al., 1987; Tarre et al., 1987). In most countries, the production and use of these digestates is governed by animal by-products regulations (e.g., Anon, 2014b), which require all materials to be pasteurized. This is especially important with slurries that may contain zoonotic pathogens.

Fibrous solid waste from anaerobically digested materials can be composted to form a potential component for growing media. Ridout and Tripepi (2011) investigated the use of composted digestates as a component of growing media. High values of pH (>7) and EC (>5.0 dS/m) were recorded, along with similarly high concentrations of ions (K, P, Na, and Cl). Digestates can contain high concentrations of nitrogen. The ammonium nitrogen (N) fraction (mineral N) in whole digestate or in the separated fiber of anaerobic digestate is as much as 50%–75% of the total Kjeldahl nitrogen (Möller and Müller, 2012). The remaining nitrogen is classified as organic N. The ammonium fraction is susceptible to volatilization if not stored properly, for example covered by a membrane. Crippa et al. (2013) working with noncomposted digestates from cattle and pig slurry/manures, also recorded high pH values (7.7–7.8), although ECs proved lower at 1.1–1.3 dS/m. Physical properties included dry bulk densities of 78–105 kg m³ and air volumes of 43%–50% with moderate water-holding capacities. Digestates alone were found to strongly inhibit cress (*Lepidium sativum* L.) germination, serving as a bioassay for phytotoxicity, as well as growth of lettuce (*Lactuca sativa* L.), but at 25%–30% (v/v) in mixture with sphagnum peat, both germination of cress and growth of lettuce proved satisfactory. MacConnell and Collins (2009) described satisfactory growth of many species of nursery stock in peat mixed at 50% with anaerobically digested fiber. Lamont and Elliott (2016) reported that anaerobically digested dairy fiber could replace peat at levels up to 50% for growth of poinsettias, with the fiber containing significant amounts of nutrients. Ponchia et al. (2017) found noncomposted anaerobic digestates from fruit and wine distillery wastes to have a pH of 6.7 and an EC of 1.41 dS/m, and improved growth of *Rosa* and *Abelia* × *grandiflora* Rehd. was noted when digestates were added at 20% (v/v) to mixtures of green compost and peat.

Bustamante et al. (2012, 2013) produced composts by mixing anaerobic digestates with bulking materials including wheat straw and vine prunings. The composts showed a suitable degree of stability and maturity and adequate physical properties for their potential use as growing media. The bulking agents had a positive effect, reducing EC and the N losses during composting and also diluting the heavy metal contents of the end-products. Furthermore, the composts suppressed growth of fusarium wilt of melon, *F. oxysporum* f. sp. *melonis*.

8.9 BIOCHAR

Biochar, essentially charcoal used as a soil or growing media amendment, is normally produced from organic waste using pyrolysis technology under temperatures ranging from 400°C to 700°C where oxygen is either absent or depleted. The pyrolysis process produces biochar as well as two additional materials, syngas and bio-oil that may have commercial value as an energy source. Biochars differ depending on the feedstock, temperature, and residence time and have been effective tools of waste management, soil remediation, and may also offer control of greenhouse gas (GHG) emissions through carbon sequestration (Lehmann and Joseph, 2015; Kern et al., 2017). Due to the large variability of biochars, one type of biochar may not be suitable for all growing conditions and crops. Materials closely related to biochar, such as charred rice husks, are used extensively in South America (Colombia) for growth of cut flower species (Quintero et al., 2013).

8.9.1 PHYSICAL PROPERTIES

Many biochars have a total porosity similar to peat at 90%–95%. Air space and EAW depends mostly on the particle size of the material. Bulk density is similar to H2–H4 peat, varying from 100 to 300 g/L (Blok et al., 2017). Pelletized biochar was tested as a component of a growing medium and was found to improve the physical properties of hydraulic conductivity and water availability but had a variable effect on porosity (Drumroese et al., 2011). Zaccheo et al. (2014) found that when biochar is mixed with peat they found benefits on physical properties of growing media such as increased air content and lower shrinkage. Zhang et al. (2014) found that the addition of biochar reduced bulk density, improved the water-holding capacity, and porosity in relation to CGW alone.

8.9.2 CHEMICAL PROPERTIES

In many cases the pH of biochars is very high (Table 8.22). However, this may allow replacement of some or all of the dolomitic limestone that is normally used to neutralize peat. Zaccheo et al. (2014) investigated the potential of a biochar prepared from pinewood with a pH of 10.2 to neutralize peat with a pH of 4.2. Proportions of 30% and 40% were most effective in adjusting the pH to a moderate acidic value. The fine fraction of <3 mm had the highest neutralizing power. Steiner and Harttung (2014) emphasized that acidic peat-based growing media, which are usually limed to adjust the pH for different demands of crops, may work without lime when biochar is used as an additive.

Table 8.22 Selected Physical and Chemical Properties of Biochars for Potential Use in Growing Media

Origin and Nature	pH	EC (dS/m)	Dry Bulk Density (kg m ⁻³)	K (mg/L ⁻¹)	Author
Hardwood (poplar)	9.7	0.23	131	192	Bedussi et al. (2015)
Softwood (spruce)	11.1	0.33			Bedussi et al. (2015)
80% Beechwood + 20% tomato leaves	11	6.8			Blok et al. (2017)
80% Wood chips + 20% tomato leaves	12	13	113	367	Blok et al. (2017)
Wood chips	9.4	0.53	102	14	Blok et al. (2017)
Wood chips	6.6	0.05		25*	Prasad et al. (2018)
Forest residues	9.5	0.64		990*	Prasad et al. (2018)

pH and EC as EN13037 and EN 13038; K determined in 1:1.5 water or DTPA extracts.*

When compared to other organic substrates, biochar EC is often high to very high (Table 8.22). Feedstocks such as wood chips and other wood wastes may be more suitable as candidate substrates than those derived from manures and sewage sludge waste as the former have relatively lower ECs (Table 8.22). Furthermore, biochars made from glasshouse crop wastes would not be suitable as a component of growing media due to extremely high ECs (Blok et al., 2017). All biochars have low contents of extractable macronutrients except for K (Table 8.22), but some possess moderate concentrations of Cl and SO₄ (Blok et al., 2017).

Prasad et al. (2018) evaluated four commercial biochars produced from several organic sources (mostly wood) with varying nutrient concentrations as peat diluents. The biochars contained low concentrations of N and P and generally high levels of K. The OUR values for biochars were low, varying from 1.1 to 4.4 mmol O₂/kg OM/h, indicating exceptional structural stability against microbial decay. When these materials were mixed with peat at 10%, 25%, and 50% and nutrients were added to bring them to the same level of nutrients as in fertilized peat, it was found that in biochar mixtures considerably reduced the levels of CAT extractable N (including nitrate) and P, whereas EC, pH and K levels increased with biochar addition. The ability of biochars to retain and release K and NH₄ is reported by Bedussi et al. (2015). A biochar-mediated decrease in EC has important implications regarding the use of other materials used to dilute peat, for example, CGW. The rate of dilution of peat with CGW is limited due to the high EC; biochar addition gives the potential for higher rates of peat dilution with green compost.

8.9.3 PLANT GROWTH

Blok et al. (2017) found that biochar produced from 100% beechwood chips was slightly phytotoxic. However, Prasad et al. (2018) found root development using cress was in a number of cases better than peat. Tomato plants growing in containers with biochar addition had increased height and plant biomass was in a number of cases better than peat alone indicating that biochar could be used successfully to dilute peat. Greenhouse experiments using tomato (*Solanum*

lycopersicum L.) and marigold (*Tagetes erecta* L.) plants grown in 3.0-L pots showed that addition of either biochar had no effect on dry weights of tomato plants after 4 weeks of growth, but significantly increased plant heights in all of the biochar treatments. Marigold plant heights were greater than the control in all but the 5% wood biochar, with little effect on plant dry weights. These results indicate that both pelletized straw and wood biochars would be suitable as replacements for peat in soilless substrates at these levels (Vaughn et al., 2013).

Tian et al. (2012) reported that biochar derived from green waste could be used as partial substitute for peat in the cultivation of *Calathea rotundifolia* Poepp. & Endl. cv. Fasciata, and Prasad et al. (2018) observed superior growth of tomato plants in peat: biochar mixtures to that in peat alone. Zhang et al. (2014) studied the effect of biochar and humic acid amendments added to CGW (particle size <10 mm) on the growth of *Calathea insignis* W.Bull nom.illeg. The addition of biochar and humic acid improved the growth of *Calathea* including root volume with the best treatment being a combination of biochar at 20% and humic acid at 0.7% mixed with CGW. Thus CGW could replace peat when used in combination with biochar and humic acid. Steiner and Hartung (2014) used rates of 0%, 25%, 50%, 75%, and 100% (v/v) biochar mixed with peat as a growing medium for *Helianthus annuus* L. cv Teddy Bear. Other ingredients tested were perlite, clay granules. *H. annuus* grew well in all substrates and mixtures under study. Dispenza et al. (2017) identified mixtures of 40% “brown” peat and 60% conifer wood biochar as suitable for production of the ornamental shrub *Euphorbia × lomi*.

Biochar may influence components of the microbial flora of growing media, and in doing so, protect plants against diseases, either by direct protection or by induced plant resistance (Grabner et al., 2010). Recently Frenkel et al. (2017) showed that while low concentrations ($\leq 1\%$) of biochar often suppressed several soil-borne diseases, higher concentrations ($\geq 3\%$) were mostly ineffective or induced plant disease. They suggest that for use as horticultural peat replacement, biochar feedstocks and concentrations should be standardized and the potential effect of biochar on plant disease be considered, so that growers can rely on consistent and reproducible biochars for desired effects.

In certain plant growth media, biochar amendment results in chemical responses in the plant as well as shifts in the rhizosphere microbiome (De Tender et al., 2016). Grabner et al. (2010) suggested two innovative mechanism for the effects on growth: (1) the biochar stimulates shifts in microbial populations toward beneficial plant growth promoting rhizobacteria or fungi, due to either chemical or physical attributes of the biochar or (2) low doses of biochar chemicals, many of which are phytotoxic or biocidal at high concentrations, stimulate plant growth at low doses (hormesis).

8.10 OTHER MATERIALS

The solid fraction from animal manure can serve after composting as an alternative to peat in container media. In addition to areas where peat is scarce or expensive, the Nitrates Directive (Council of the European Communities, 1991) now restricts the application of animal manure to farmland. Composts prepared from leached separated cattle manure after a process of methanogenic fermentation can serve as a peat substitute: such composts have a bulk density, air, and

water content similar to peat. Seedling growth in these media was as good as in peat mix (Chen et al., 1983). The water release curve of separated cattle manure compost shows porosity similar to that of peat, but the volumetric water content within the range 1–10 kPa tension is considerably lower than peat. This is due to high air-filled porosity and higher content of residual water held at tensions >10 kPa. This has implications for scheduling of irrigation (Raviv and Medina, 1997). Data on storage of composted material made from slurry fiber plus peat, slurry fiber, and sawdust and on its own showed that it is quite stable as evidenced by lack of changes in N content after storage of over 1 year and by maintaining its suppressive capacity against several soil-borne pathogens (Saadi et al., 2010).

Chipboard waste can also be milled, composted, and used as a substitute for peat in growing media (Molitor et al., 2004). This material can contain up to 10% urea formaldehyde. It has been found to have poor storage properties due to release of N and subsequent low pH and high levels of NH_4N and NO_3N (Dickinson and Carlile, 1995).

8.11 STABILITY

In soilless production, plants are grown over lengths of time that range from a few weeks to several years. Media sold through retail outlets may be stored outside, often in areas where wide temperature variations may occur (Carlile, 2004). In general, it is desired that the substrate remains intact during use or indeed in storage, so as to not modify the growing environment in a manner that will negatively affect productivity. It is, however, important to note that the root zone is a microcosm where organisms other than the plant also prevail. Organic growing media are essentially living systems within which there may be considerable biological activity with both negative and positive impacts. The predominant negative impacts of this biological activity are nitrogen immobilization and the physical breakdown of the substrate (i.e., lack of physical stability): positive effects include release of nutrient ions from slow-release fertilizers as well as other complex nutrient sources, and disease suppression.

Physical breakdown of the growing medium can have a detrimental effect on crops which have a long production period such as woody nursery stock. Crops that have a high aeration requirement cannot be grown well under wet conditions (e.g., ebb and flood systems). In this situation, the physical stability of the growing medium becomes important in maintaining favorable growing conditions for the whole period (Aendekerk, 1997, 2001; Maher and Prasad, 2004). The various peat types vary considerably in their capacity to resist physical degradation. Peats with higher degree of decomposition (H5–H6) are more structurally stable than younger peats (H2–H3) (Aendekerk, 1997). Prasad and O'Shea (1999) studied peats of varying degrees of decomposition from bogs in Ireland, Finland, Germany, Baltic States, and Sweden. Particle sizes of the peat had a major effect on the breakdown, with fine peat breaking down more than coarse peat. The degree of breakdown was inversely related to decomposition (von Post scale) and particle size. The degree of volume loss varied from 47% for H2 (Finnish) peat to 5% for coarse H5 (Irish) peat. Airspace decreased from 21% to as low as 3.5% over the 15-month incubation period for fine textured H2 Peat. Reduction in air-filled porosity was related to volume reduction. In another trial reported by Prasad and O'Shea (1999), peat

from bogs in Baltic States, Sweden, Ireland, and Germany of two sizes (0–3 and 5–10 mm) were compared, and the loss in volume after nearly 2-year incubation was in the order of 13%–20% for H4 and H5 peats and 29%–40% for H2 and H3 peats. Wood fibers showed an even greater volume loss of the order of 50%. Steaming of H2 or H5 peats did not appear to have a significant effect on break down, as evidenced by changes in particle size. FT-IR analysis of peat taken before and after incubation, showed a trend toward enrichment of lignin and decrease in cellulose.

Prasad (1997a) found that volume reduction of coir as a result of biodegradation was 24% over 24 months, indicating it is a relatively stable material. Shrinkage of 19%–21% was reported by Nelson et al. (2004) just after 8–10 weeks under cropping. Differences in biodegradation of coir batches could be related to differences in the original lignin content which could again be related to the age of the coir and fiber content.

Volume reduction of between 36% and 54% over a 2-year period was found with various types of wood fiber (Prasad and O'Shea, 1999; Fischer et al., 1993). In a study with coir and three wood fibers on their own or mixed with H5 peat, Prasad and Maher (2004) found that the wood fibers generally break down rapidly over time compared with peat. Generally, the shrinkage was much less in Toresa (Gumy, 2001) than in other wood fiber products (Prasad and Maher, 2004) probably due to differences in the lignin content of the materials (Table 8.19). Coir was found to be relatively stable in this study, as well. The addition of H5 peat to all the nonpeat materials led to a reduction in the rate of breakdown of the resulted mix while the addition of lime accelerated its breakdown. Breakdown of the materials was strongly and negatively related to the initial lignin content. Samples were analyzed by FT-IR spectroscopic analysis at the start of the experiment. A strong negative relationship was found between the FT-IR peak ratio $1600\text{ cm}^{-1}/1056\text{ cm}^{-1}$ (lignin to cellulose) and substrate shrinkage (Prasad and O'Shea, 1999). Fischer et al. (1993) also found volume loss of similar magnitude with wood fibers. In a review comparing peat-based and peat-free media mixes.

Fitzpatrick and Verkade (1991) have reported that some compost products used in growing media start with satisfactory porosity but undergo compaction during the growing period. Maher and Prasad (2002) studied the effect of adding CGW to two types of peat, respectively, undecomposed (H2) and moderately decomposed peat (H5), on physical properties and on physical degradation of the growing medium. A young CGW, composted for 12 weeks with no addition of N, and a mature CGW, composted for 20 weeks with the addition of 500 g/m^3 of N prior to composting were compared. Addition of CGW at 30% by volume increased bulk density and air content and reduced pore space and easily EAW content. Physical degradation was measured by monitoring height loss of the peat/CGW mixes over a 2-year period. Alone, H2 peat degraded at a much greater rate than the H5 peat. Addition of young CGW increased degradation of both peats, whereas the mature CGW only increased degradation in the H5 peat. Also in this experiment, FT-IR spectroscopic analysis of the mixes was carried out and the peak ratio $1600\text{ cm}^{-1}/1056\text{ cm}^{-1}$, which represents the aromatic/aliphatic ratio (lignin-to-carbohydrate ratio), was negatively correlated to physical degradation of the mixes.

Harrelson et al. (2004) studied the physical properties of fresh pine bark and aged pine bark over 1 year in an unprotected location in the presence and absence of coarse builders sand (bark: sand = 8:1; 11% sand by volume). Container capacity and available water in the aged pine bark–sand substrate were significantly greater than in fresh pine bark–sand substrate. The authors

concluded that fresh pine bark–sand substrate would require frequent irrigation with small quantities of water due to limited available water content. Over the course of the year, available water in the fresh pine bark–sand substrate increased from 15.8% to 22.3% while air-filled porosity decreased from 36.3% to 24.9%. Similarly, the aged pine bark–sand substrate showed an available water increase from 22.7% to 30.0% with an air-filled porosity decrease from 25.9% to 17.0%. The decrease in air-filled porosity for both mixes over time can present challenges to the growers for their long-term use in production.

Biological stability is an important consideration as part of N nutrition management. Some coir has been found to immobilize nitrogen (Prasad, 1997b). In some cases, immobilization over 5 weeks was around 50–60 mg N/L, while in others, the N immobilization was negligible. In growing media derived from composted materials, if the composting process has not been thorough, the resulting compost may have high cellulose or hemicellulose contents which are readily degraded by microorganisms. Thus many peat-free media mixes have an inherently higher microbial population and activity than peat-based media mixes (Dickinson, 1995). This often results in a very quick depletion of inorganic nutrients (microbial immobilization), particularly nitrogen (Dickinson and Carlile, 1995), with a simultaneous change in pH. Such changes can be ameliorated by the addition of a buffering agent such as expanded clay or other clay minerals in the media mixes (Bragg, 1998). Changes in the nitrogen status are common with organic media destined for retail markets: in some soluble N release from complex inorganic and organic N sources may occur; in others, especially wood and bark-based media, N immobilization may occur (Carlile, 2004).

8.12 DISEASE SUPPRESSION BY ORGANIC GROWING MEDIA

8.12.1 THE PHENOMENON AND ITS DESCRIPTION

Although most growing media are initially pathogen-free, they are frequently infested with disease-causing pathogens soon after planting. Low biological activity and particularly lack of microbial population diversity, typical of most unused media, was suggested as one of the main reasons for this potentially destructive phenomenon (Chapter 5: Root Zone Microbiology of Soilless Cropping Systems, as well as Postma et al., 2000; Menzies et al., 2005). Although several peat moss types suppressed soil-borne diseases such as *Alternaria brassicicola*, *Leptosphaeria maculans*, *Rhizoctonia solani*, and *F. oxysporum* f. sp. *lycopersici* (Tahvonon, 1982; Pane et al., 2013), peat moss is still used in many cases as a control in compost suppressiveness studies (Gaag et al., 2004; Noble and Coventry, 2005; Raviv et al., 2005; Cohen et al., 2008) due to its low microbial activity (Dickinson, 1995) leading to conduciveness to rapid colonization by a variety of pathogens (Hoitink et al., 1977; Borrero et al., 2004; Yogev et al., 2006).

Compost has been shown to provide a significant level of disease suppression in several production systems (Table 8.23). As discussed earlier, many organic substrates undergo some sort of composting as part of the processing of the raw materials into growing media components. While the focus of this has largely been on the beneficial changes of physical and chemical properties, there is also evidence of disease suppressiveness as a result of the composting process. Indeed, a

Table 8.23 Typical Examples of Commercially Feasible Uses of Composts for Soil-Borne Disease Mitigation

Pathogen	Host	Compost Feedstock	Reference
<i>Fusarium oxysporum</i> f. sp. <i>radicis-cucumerinum</i>	Cucumber (<i>Cucumis sativus</i>)	Greenhouse waste	Yogev et al. (2006)
<i>F. oxysporum</i> f. sp. <i>radicis-lycopersici</i>	Tomato (<i>Lycopersicon esculentum</i> Mill.)	Greenhouse waste	Cheuk et al. (2005), Raviv et al. (2005)
<i>F. oxysporum</i> f. sp. <i>basilici</i>	Sweet basil (<i>Ocimum basilicum</i>)	Separated cattle manure	Reuveni et al. (2002)
<i>F. oxysporum</i> f. sp. <i>chrysanthemi</i>	<i>Dendranthema grandiflorum</i>	Hardwood bark	Chef et al. (1983)
<i>F. oxysporum</i> f. sp. <i>melonis</i>	Melon (<i>Cucumis melo</i>)	Greenhouse waste; prunings	Raviv et al. (2005), Ros et al. (2005)
<i>F. oxysporum</i> f. sp. <i>dianthi</i>	Carnation (<i>Dianthus caryophyllus</i>)	Hardwood bark, olive pomace	Pera and Filippi (1987), Pera and Calvet (1989)
<i>Meloidogyne incognita</i>	Tomato (<i>L. esculentum</i> Mill.)	MSW	Ryckeboer and Coosemans (1996)
<i>Meloidogyne javanica</i>	Tomato (<i>L. esculentum</i> Mill.)	Cattle manure; greenhouse waste	Oka and Yermiyahu (2002), Raviv et al. (2005)
<i>Pythium ultimum</i>	Cucumber (<i>C. sativus</i>)	Hardwood bark	Chen et al. (1988)
<i>P. ultimum</i>	Poinsettia (<i>Euphorbia pulcherrima</i>)	Hardwood bark	Daft et al. (1979)
<i>Pythium aphanidermatum</i>	Cucumber (<i>C. sativus</i>)	Licorice compost, sugarcane residues	Hadar and Mandelbaum (1986), Theodore and Toribio (1995)
<i>Phytophthora nicotianae</i>	Various citrus at the nursery stage	MSW	Widmer et al. (1998)
<i>Phytophthora cinnamomi</i>	<i>Rhododendron catawbiense</i>	Hardwood bark	Hoitink et al. (1977)
<i>P. cinnamomi</i>	<i>Aucuba japonica</i>	Soft and hardwood bark	Spencer and Benson (1981)
<i>Phytophthora citricola</i>	<i>A. japonica</i>	Soft and hardwood bark	Spencer and Benson (1981)
<i>Rhizoctonia solani</i>	Poinsettia (<i>E. pulcherrima</i>)	Hardwood bark	Daft et al. (1979)
<i>R. solani</i>	Radish (<i>Raphanus sativus</i>)	Separated cattle manure, grape marc	Gorodecki and Hadar (1990)
<i>R. solani</i>	Cucumber (<i>C. sativus</i>)	MSW	Tuitert et al. (1998)
<i>Sclerotinia minor</i>	Lettuce (<i>Lactuca sativa</i>)	Sewage sludge	Lumsden et al. (1986)
<i>Sclerotium rolfsii</i>	Radish (<i>R. sativus</i>)	Cattle manure, grape marc	Gorodecki and Hadar (1990)

substantial body of research exists on suppression of fungal diseases, as well as inhibition of disease-causing nematodes (Oka and Yermiyahu, 2002; Raviv et al., 2005; Oka, 2010) and bacteria (Schonfeld et al., 2003; Yogev et al., 2009).

8.12.2 SUGGESTED MECHANISMS FOR SUPPRESSIVENESS OF COMPOST AGAINST ROOT DISEASES

The most dramatic and immediate effect of amending an unused growing medium with compost is the large and rapid increase in microbial and fungal activities and microbial population diversity (Diab et al., 2003). In many cases, compost sterilization considerably reduces or eliminates its suppressiveness, indicating that the mechanism of disease suppression is predominantly biological (Hadar and Mandelbaum, 1986; Trillas-Gay et al., 1986; Reuveni et al., 2002; Noble and Roberts, 2003; Postma and Schilder, 2015). However, in some cases, a swift reintroduction of diverse microbial populations soon after sterilization leads to a resumption of suppressiveness (Serra-Wittling et al., 1996; Yogev et al., 2006; Aviles et al., 2011).

The suppressiveness phenomenon consists of a complex and intricate set of mechanisms. Understanding these is important since it is desirable in soilless production to make use of this phenomenon and to maximize its effectiveness. Given the withdrawal of many plant protection chemicals, including insecticides and fungicides (Carlile and Coules, 2012), alternative strategies such as disease suppression may play an increasing role in pest and disease control. Hoitink and Boehm (1999) suggested several mechanisms, which are likely to be factors in disease suppression:

- Competition for nutrients, space, and occupation of infection sites by other microorganisms. The beneficial effects of competition for nutrients and space have been shown for a variety of pathosystems (Serra-Wittling et al., 1996; Chen et al., 1988). Another example is the antagonistic effects toward *F. oxysporum* f. sp. *radicis-lycopersici*, *Pyrenochaeta lycopersici*, *Pythium ultimum*, and *R. solani*, all known pathogens of tomato, observed after compost was added to soil. These effects were associated with marked increases in the percentage of siderophore producers within the root zone of tomato (Brito-Alvarez et al., 1995). Siderophore producers such as fluorescent pseudomonads are known to compete with fusarium through suppression of their chlamydospores' germination (Elad and Baker, 1985). Massive production of siderophores leads to reduced levels of iron, which is essential for successful germination of the pathogen and penetration of the host. In soil-grown plants, the proliferation of pseudomonads is dependent on continued supply of carbon through root exudation; when compost is applied, ample supply of carbon is provided, along with diverse communities of such beneficial microbes. Another type of competition among fungi is for colonization sites on the root surface (Benhamou and Garand, 2001; Olivain and Alabouvette, 1999; Pantelides et al., 2009). For example, Pharand et al. (2002) showed that compost-mediated resistance of tomato to crown and root rot caused by *F. oxysporum* f. sp. *radicis-lycopersici* was associated with nonpathogenic strains of *F. oxysporum* and that this was induced through the formation of physical barriers on the roots to fungal penetration, at sites of attempted penetration by the pathogen (Pharand et al., 2002).
- Hyperparasitism followed by lysis was shown to be the main mechanism of soil suppression of *R. solani*, caused by *Trichoderma harzianum* (Chet and Baker, 1980). *T. harzianum* is found frequently in suppressive composts (Kuter et al., 1983). However, Nelson et al. (1983) showed that amending immature compost with high doses of *T. harzianum* did not confer suppressiveness to this medium, thus concluding that the production of container media that were consistently suppressive to *R. solani* requires both the existence of antagonists and an

environment that favors their activity. Direct hyperparasitism of *Sclerotium rolfsii* propagules was shown by Gorodecki and Hadar (1990) in composts based on grape marc and separated cattle manure but not in peat-based media. Another example of lysis was shown for hyphae of *Pythium aphanidermatum* (Mandelbaum and Hadar, 1990). They demonstrated that hyphae of *P. aphanidermatum* grown on nylon fabric and buried in container media were rapidly lysed in a compost medium based on separated cattle manure as compared with peat-based media. Light and SEM microscopic observations of hyphae retrieved from container media showed no evidence of direct parasitism by other soil fungi; however, bacteria were associated with the lysing hyphae. In general, it was concluded that pathogen propagules can be destroyed after incubation in suppressive organic substrates. This process may be relatively fast, as shown by Yogeve et al. (2006) for microconidia of *F. oxysporum* f. sp. *radicis-lycopersici* and for *F. oxysporum* f. sp. *basilici* in different compost types, while the same conidia from these pathogens survived in peat.

- Antibiosis: The role of production of antibiotics within composts is well established (Pugliese et al., 2010; St. Martin et al., 2014). *Gliocladium virens* produced detectable levels of gliotoxin in composted mineral soil, populated with natural microbiota, and inoculated with *G. virens*. This medium effectively controlled damping-off of zinnia (*Zinnia elegans*) seedlings caused by *P. ultimum* and *R. solani*. Aqueous extracts containing gliotoxin were as effective for control of damping-off of zinnias as the intact *G. virens*—amended medium (Lumsden et al., 1992). *Trichoderma/Gliocladium* and related spp. are known to produce a variety of metabolites effective against soil/growing media pathogens, and preparations of *Trichoderma* spp. are marketed commercially as biocontrol agents (Mukherjee et al., 2013).
- Futile pathogen germination: One potential mechanism for compost-related disease suppression is the compost's role in mimicking root exudates. Normally, a pathogen propagule will not germinate in the absence of a host as signaled by root or seed exudates (Lockwood, 1990) since these chemical signals are required for the host identification by the pathogen (Chen et al., 1988). Induction of germination of pathogens in the absence of a host may occur, however, in composted material. In compost-containing medium, germination of pathogens may be triggered before the pathogen comes in contact with living plants, so that the existing inoculum is spent. This is probably the reason why plant-derived composts are usually so effective in suppressing soil-borne diseases (Cheuk et al., 2005) and especially in reducing the inoculum level, even without the existence of plants (Yogeve et al., 2006).
- Adsorption of chemical signals that enhance the germination and root colonization by pathogens on the high specific surface area of composts is another putative mechanism for disease suppression. Adsorption reduces the amount of free signal, so the germination of a smaller number of propagules can be realized. Later, part of these adsorbed molecules can be decomposed by compost microorganisms, leading to further decrease of the signal's concentration. However, conclusive evidence to support this mechanism is lacking.
- Induced systemic resistance (ISR; or systemic acquired resistance) is “a state of enhanced defensive capacity developed by a plant when appropriately stimulated” (Bakker et al., 2003). The resistance responses are usually systemic, but localized types also exist, and may be effective against a broad range of pathogens. ISR is probably one of the most important mechanisms through which compost induces disease resistance to plants. Many rhizosphere bacterial and fungal isolates have been reported to turn on ISR in plants (van Loon et al., 1998).

In addition, microbial metabolites such as salicylic acid and lipopolysaccharides have been implicated in microbially mediated ISR (Bakker et al., 2003). Some of the changes triggered by composts and its microbial communities were of anatomical nature (as described above, Pharand et al., 2002). Others are related to enzymatic activity. For example, composted pine bark was suppressive to *Pythium* root rot of cucumber (Zhang et al., 1996), whereas dark peat was not suppressive to the disease. Cucumber and *Arabidopsis* plants grown in composted pine bark expressed higher levels of β -1,3-glucanase (Zhang et al., 1998) and peroxidase (Zhang et al., 1996) than those grown in peat. Split root experiments suggested that the resistance mechanism in cucumber and melon was systemic (Zhang et al., 1996; Yogeve et al., 2010). Compost-induced expression of pathogenesis-related genes was shown by Kavroulakis et al. (2005). They concluded that suppressive composts are able to elicit consistent expression of certain PR genes in the roots of tomato plants, even in the absence of any pathogen. The expression of the PR genes may be triggered by the microflora of the compost or could be associated with abiotic factors of the compost.

In this context, it is important to note that the ISR phenomenon is not limited to the root system and inhibition of foliar diseases has also been detected. Although Krause et al. (2003) and Hoitink et al. (2006) reported that systemic suppression of foliar diseases induced by compost amendments is a rare phenomenon and is mainly related to the effect of the level of compost inoculation with *Trichoderma hamatum* strain 382, other examples for foliar disease suppression are known (Bobev et al., 2004; Horst et al., 2005; Kavroulakis et al., 2005), suggesting that this phenomenon is far less rare than previously thought. Indeed, the ability of *Trichoderma* spp., common colonists of composts, to stimulate plant defense mechanisms is now well-proven (Mukherjee et al., 2013). Yogeve et al. (2010) working with cucumber plants grown in either peat or compost based on orange peels and separated cattle manure deliberately inoculated the leaves with spores of *Botrytis cinerea*, the causal agent of gray mold—a very severe greenhouse foliar disease of many crops. After germination, the fungus grows initially as hyphae, spreading within leaves. The growth rate of the hyphae in compost-grown plants was 93% slower as compared to their peat-grown counterparts. Similar results were obtained with melon (*Cucumis melo*).

The ability of *Trichoderma* species in particular to attack soil-borne pathogens as well as stimulating defense mechanisms within plants may lead to encouragement of its development within growing media by either direct inoculation or production of naturally suppressive growing media containing the fungus (Mukherjee et al., 2013). In this respect, the organisms responsible for disease suppression colonize the compost mainly during the curing stage when the temperature drops $<40^{\circ}\text{C}$ (Scheuerell and Mahaffee, 2005). *Bacillus* spp., *Enterobacter* spp., *Flavobacterium balustinum*, *Pseudomonas* spp., other bacterial genera and *Streptomyces* spp. as well as *Penicillium* spp., several *Trichoderma* spp., *G. virens*, and other fungi have been identified as biocontrol agents in compost-amended substrates (Hoitink et al., 2001).

8.12.3 HORTICULTURAL CONSIDERATIONS OF USE OF COMPOST AS A SOILLESS SUBSTRATE

In commercial soilless production, it is important to find economical and reliable methods to reduce the risk of yield loss due to root diseases. Inclusion of compost in growing media as a method to

suppress a wide variety of root zone plant pathogens was first suggested by Hoitink et al. (1975) and is now a well-established commercial practice, corroborated by a large body of scientific evidence. This subject has been reviewed by Hoitink and Fahy (1986), Hoitink and Boehm (1999), Hoitink et al. (2001), Noble and Roberts (2003), Noble and Coventry (2005), Termorshuizen et al. (2006), and Moral et al. (2013). Some representative examples for this use are shown in Table 8.18.

Hadar and Mandelbaum (1986) found that the degree of decomposition of compost has a strong effect on the rate of disease suppressiveness. They showed that immature compost could not suppress damping-off (*P. aphanidermatum*) of cucumber seedlings, while mature compost could. Another example can be found in *R. solani*, a pathogen which is highly competitive as a saprophyte and *Trichoderma*, an effective biocontrol agent of *R. solani*. In fresh, undecomposed organic matter, biological control of *R. solani* does not occur because both organisms grow as saprophytes and *R. solani* remains capable of causing disease. The synthesis of lytic enzymes involved in parasitism of pathogens by *Trichoderma* is repressed in fresh organic matter due to high glucose concentrations. In mature compost, where concentrations of nutrients such as glucose are low, sclerotia of *R. solani* are killed by the parasite and biological control prevails (Hoitink et al., 2001). Similar results were obtained by Kanaan et al. (2015).

On the other hand, if the compost is excessively stabilized, then it will not support microbiological activity so that disease suppression is lost (Widmer et al., 1998; Danon et al., 2007). Similarly, Tahvonen (1993) found lightly decomposed peats (H2–H3) to be suppressive of soil-borne diseases. Darker, more decomposed peats are low in microbial activity and are conducive to *Pythium* and *Phytophthora* root rots (Boehm and Hoitink, 1992).

An important practical consideration of the use of compost for disease prevention is the longevity of the suppressiveness capacity. It was shown that effectiveness of several compost types against the dollar spot disease of creeping bentgrass (*Agrostis palustris* P. encross), caused by *Sclerotinia homoeocarpa* was maintained for 1 year, under shaded, uncontrolled storage conditions (Boulter et al., 2002). This subject is now under investigation and other results also suggest a full capacity for 1 year, with subsequent gradual decline, with only 50% of the original capacity after 2 years (Saadi et al., 2010).

In conjunction with the extensive European compost research described above, the effect of 18 composts was tested in seven pathosystems [*Verticillium dahliae*—eggplant (*Solanum melongena*), *R. solani*—cauliflower (*Brassica oleracea* var. *botrytis*), *R. solani*—pine, *Phytophthora nicotianae*—tomato, *Phytophthora cinnamomi*—lupin (*Lupinus* spp.), *Cylindrocladium spathiphylli*—*Spathiphyllum*, *F. oxysporum*—flax (*Linum usitatissimum*)] (Termorshuizen et al., 2006). Applying 20% of compost into potting soil or sand, 54% of the tested combinations were significantly more disease suppressive, 3% showed significant enhancement of the disease, and 43% of the tested combinations did not result in significant differences compared to the control without compost. The mean disease suppressiveness per compost ranged from 14% to 61% (Termorshuizen et al., 2006). Thus generally speaking, compost has a positive effect on suppressiveness of potting soil. However, none of the compost types could improve the suppressiveness of the potting soil for all tested pathosystems. Another disadvantage is the variability and unpredictability of compost: apparently similar compost types may give different results, and compost quality is often not stable in time and among batches of the same producer. This can be changed if strict production parameters are adopted by producers, as suggested by Raviv (2005).

Consequently, it may be feasible to partially replace peat in potting mixes with well-prepared, mature compost, whenever disease suppression is desired. Normally values of 20%–30% of the total resulting mix are quoted (Termorshuizen et al., 2006; Prasad and Carlile, 2009). Higher rates can also be used, subjected to testing with the target crop.

8.13 SUSTAINABILITY OF GROWING MEDIA AND THEIR COMPONENTS

Growers and producers of agricultural and horticultural products face increasing challenges with respect to the sustainability of their operations. Sustainability involves economic, social as well as environmental considerations (Anon, 2005) including factors such as release of greenhouse gases (GHGs) that may contribute to global warming. In this respect, extraction and use of peat, with its ability in situ to store carbon almost permanently, has attracted particular attention. Reductions in the extent of peatland habitats are also of concern in some countries (Carlile and Coules, 2013).

8.13.1 GREENHOUSE GAS EMISSIONS

Natural (undrained) peatlands function as long-term carbon (C) stores as the sequestration of CO₂ over time is greater than the amount of C that is emitted from the peatland as methane (CH₄) and leached in waterborne exports (Roulet et al., 2007; Nilsson et al., 2008). Globally, from 2000 to 2009, the agriculture, forestry, and ‘other’ land use sector accounted for 24% of all global GHG emissions (around 10 Gt CO₂-eq/year), with emissions from peatland drainage and burning alone estimated at around 0.9 Gt CO₂-eq/year (Wilson et al., 2015). Most of the emissions associated with burning come from “reclamation” of tropical peatlands for agricultural and other purposes (Brahic, 2007). Wilson et al. (2015) estimate CO₂ emissions from peatland extraction sites in the Republic of Ireland/United Kingdom to be of the order of 1.9 tonnes CO₂-C/ha/year, from Scandinavian sites at 2.3 tonnes CO₂-C/ha/year, and in Canada around 4.3 tonnes CO₂-C/ha/year, the last-mentioned figure due to greater oxidation of younger peat in that country. In contrast, Jauhiainen et al. (2012) found daytime mean annual CO₂ emissions from peat oxidation alone of 80–94 tonnes CO₂-C/ha/year on peatland in Sumatra, Indonesia. Jauhiainen et al. (2016) describe similarly high CO₂ emissions from peatlands cleared and used for rubber and oil-palm cultivation. The amount of GHG released due to peat extraction and use in growing media in the United Kingdom was reported by DEFRA at 6 Mt CO₂-eq in 2010 (Anon, 2009): about 1.3% of the United Kingdom total and although low, this figure has been criticized by Rieley (2011) and others, who indicate that peat extraction in the United Kingdom in 2010 may account for around 0.2 Mt CO₂-eq or 0.04% of GHG emissions. Since 2010, peat extraction has markedly decreased in the United Kingdom.

Life cycle analyses (LCAs) of growing media and their constituents have been undertaken (Cleary et al., 2005; Verhagen and Boon, 2008; Quantis, 2012). Conclusions from LCA studies vary according to both the units of measurement and whether interpretation follows either an overall or offset LCA (Anon, 2009). Offset figures exclude the CO₂ emitted during composting of materials well as the carbon calculated as released during breakdown of these and other biogenic

materials—wood fiber and coir—during and after use. These materials are defined as renewable resources: most authorities, including the EU do not regard peat as a renewable resource when calculating LCAs. Although the rate of carbon release from peat may be much slower than from materials such as green compost, bark and wood fiber, GHG release is measured over a 100 year time span. However, short-term release of CO₂ from composting operations can be high: for example, during composting of paper waste, mixed yard waste and food waste compost 150, 229, and 370 g CO₂-C and 2.0, 4.4 and 34 g NH₃-N/kg initial substrate were evolved respectively (Komilis and Ham, 2006): the amount of CO₂ and NH₃ evolved during the first 21 days of composting of tobacco and grape wastes per mass of volatile matter was 504.81 and 122.45 mg/kg (Kucic and Briski, 2017).

A UK DEFRA report (Anon, 2009), as well as those of the RHP foundation in the Netherlands (Verhagen and Boon, 2007) and Cleary et al. (2005) indicate that over 70% of GHG emissions from organic growing media come from carbon release after use, disposal, and ultimate degradation. EPAGMA—The European Peat and Growing Media Association—found that the highest impact on climate change resulted from use of growing media containing “white peat”: peat with a low humification index and thus subject to more rapid oxidation than more humified peat (Quantis, 2012). Transport was found to be a significant factor in GHG emissions with some substrates used in UK growing media, notably peat imported from Scandinavia and coir imported from the Indian subcontinent (Anon, 2009). In this respect, composts which in theory can be produced anywhere on the globe (Raviv, 2017), may need little or no transportation to the site of manufacture, thus mitigating GHG emissions.

8.13.2 HABITATS AND SPECIES DIVERSITY

Removal of peat clearly alters the landscape and has a detrimental effect on native fauna and flora at the site of extraction, much of which is unique to peatlands (Rotherham, 2013). In some countries where peatlands were formerly widespread, such as the Netherlands, they have all but disappeared. In the Netherlands, peatland habitats only amount to 5000 ha (Montanarella et al., 2006) and are justly regarded of high conservation status. Elsewhere, peat extraction for horticulture forms a very small part of total peat reserves, for example, in Canada which has over 100 M hectares of peatland, and where peat extraction for horticulture occupies <0.02% of total reserves (CSPMA, 2011). In Canada, peatland habitats are widespread and current extraction is unlikely to influence the biodiversity of such habitats. Nevertheless, much research has been undertaken under the auspices of the Canadian Sphagnum Peat Moss Association on restoration of peatland habitats on cutaway bog areas (CSPMA, 2011).

In between these extremes, there are countries, notably in Northern Europe, where peatland habitats and their associated biodiversity are held to be threatened by extraction for fuel and horticulture. Peat has been extracted by industrial means on >30,000 ha in northwest Germany, and since the mid-1980s rewetting and restoration is required by law on abandoned cutover peatlands (Höper et al., 2008). Since the late 20th century, much attention has been drawn to raised bog peatland habitats in the United Kingdom that have been heavily exploited for use in growing media (Rotherham, 2013). The UK Biodiversity Action Plan database (Anon, 2007) reports the area of lowland raised bog peat habitats to be 24,500 ha, of which just over 7800 ha are considered active (Alexander et al., 2008). Peat extraction for horticultural use has declined markedly

in England during the period 2000–15, but in other countries with more extensive raised bog habitats such as Scotland, some planning permissions given decades ago have been activated, with consequent peat extraction; the costs of revoking such planning permissions are prohibitive to local authorities (Anon, 2017). In the Republic of Ireland, peatland habitats include 23,000 ha of conservation grade raised bog (IPCC—Irish Peatlands Conservation Council, 2011). The IPCC indicates that Irish bogs are under threat not only from extraction by large companies for fuel and horticultural use but also the traditional practice within local communities of digging peat for fuel under turbary rights. In 2016 the principal company responsible for peat extraction in Ireland announced that it would no longer extract peat for fuel use after 2030.

Coir, wood fiber, and bark are derived from coconut palm and conifer trees, respectively, neither forming part of scarce habitats, although conifer plantations are often regarded as species-poor.

8.13.3 SPHAGNUM FARMING

Peat grown from fragments of sphagnum moss used to “seed” cutover bogs from which extraction has taken place is viewed as a renewable and thus sustainable alternative to extracted peat (Gaudig et al., 2014). Such sphagnum biomass has similar physical and chemical properties to H1–H3 peat, and cultivation of sphagnum (“*Sphagnum* farming”), the technology of using sphagnum biomass in industrial substrate production, and the use of growing media based on sphagnum biomass in professional horticulture have been extensively studied, particularly in Germany (Gaudig et al., 2008, 2014; Pouliot et al., 2014). Accumulation of sphagnum biomass reached as high as $6.9 \text{ t}^{-1} \text{ ha}^{-1}$ in 1 year in small scale field studies. Growing media made from sphagnum biomass have been tested on a wide variety of crops (Emmel, 2008; Oberpaur et al., 2010) and have generally performed at least as well as standard peat-based media (Reinikainen et al., 2012). However, the cost of producing sphagnum biomass far exceeds that of peat obtained for horticultural purposes by conventional extraction, and the lignin content is very low (<15%) indicating that it would not hold its structure if longer term crops were grown in it (Gaudig et al., 2014).

8.13.4 ECONOMIC AND SOCIAL SUSTAINABILITY

The economic sustainability of producers and manufacturers of growing media depends on several factors including the ability to source raw materials, their quality, and cost of processing in addition to longevity of media after sale. Social sustainability is directly related to the extraction and processing of raw materials for use in growing media, which often provides employment in rural areas in many parts of the world: areas where jobs are often scarce. The employment created by growing media manufacturers, and indeed many of their customers, frequently also present in rural locations, contributes much to local economies, helping to sustain, and maintain the communities there. In Ireland for example, extraction and processing of peat was established to provide employment in rural areas (Clarke, 2010). Peat production in England (until recently), Scotland, Germany, the Baltic States, and Scandinavia is predominantly carried out in rural areas. Production of coir in Sri Lanka and India is focused in coastal rural areas. Sawmills that produce bark are located in forested areas. Composting operations worldwide are often performed on the fringes of urban areas or more commonly in rural situations.

8.13.5 GROWING MEDIA: END OF LIFE

Conservation of resources and mitigation of adverse environmental effects involves the reduction, reuse, and recycling of materials. In terms of growing media, direct reduction is not a viable option since yield and plant resistance to stress are positively correlated with the volume of the root zone (Snyder and Bauerle, 1985; Xu et al., 2001). However, reductions can be achieved indirectly through use of media over several seasons with the same crop or switching crops while using the same medium. For example, sterilization and replanting in media is now common with strawberries and other fruit raised in modules, particularly with coir (Zucchi et al., 2017).

Organic media may be reused or recycled as a diluent in peat or other media, particularly for nursery stock species, many of which are tolerant of a range of physical conditions. However, the physical and chemical nature of used media may differ markedly from fresh components (Jackson et al., 2009b; Verhagen, 2009), leading to compaction, reduced air-filled porosity and increases in bulk density (Fonteno et al., 1981; Yoon et al., 2007). Nevertheless, remediation steps including substrate disinfestation and careful choice of bulking materials may allow reuse. Some media constituents (e.g., composted bark, composted almond shells, and rice hulls) are resistant enough to allow their repeated use during several seasons (Raviv, 2017).

Pesticides, particularly herbicides and insecticides, applied to organic growing media may pose a problem in reuse or recycling. For example, with nursery stock, herbicides of moderate persistence may be applied to containers in order to provide control of weeds over several months. Typical examples include isoxaben, napropamide, and lenacil in Europe; several other active ingredients are permitted in the United States including oxadiazon, flumioxazin, and oxyfluorfen (Atwood, 2016; Neal, 2016). Residues of pesticides may be carried in reused or recycled media, but information following reuse/recycling is lacking. Drakes et al. (2001) consider environmental consequences arising from land spread with used/discarded media are likely to be minimal due to the spread of waste over a relatively large area. However, growing media, both in retail and professional situations may be drenched with neonicotinoid insecticides such as thiacloprid and clothianidin for control of vine weevil (*Otiorhynchus sulcatus* L.) and other root-damaging pests. In view of their long persistence, discarded media is highly likely to contain residues of these compounds. The adverse effects of these compounds on bees is well established, and there is current concern that deleterious effects may also occur to key ground and soil inhabiting insects such as staphylinid and carabid Coleoptera, as well as other phyla including annelids (Pisa et al., 2015).

Unlike inorganic materials, organic growing media are biodegradable: once added to soil—their ultimate destination—they form part of the organic matter fraction there. Indeed, the principal routes for disposal by professional growers of organic media, as well as unwanted or substandard plants in pots or other containers, is directly to agricultural land. Unwanted material may first be placed in dumps or piles, and later spread on land (Drakes et al., 2001). Most hobby gardeners probably dispose of media locally to soil, or via organic “brown waste” bin collections. Application of spent organic media to soils may improve fertility there, this being especially important for soils in arid and semiarid regions (Raviv, 2017). A special case is roof gardens. An LCA conducted on discarding this type of substrate revealed that their end-of-life scenario is of major importance to their environmental effect (Peri et al., 2012).

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PART

PRACTICES OF
SOILLESS PLANT
PRODUCTION

III

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IRRIGATION IN SOILLESS PRODUCTION

9

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9.1 INTRODUCTION

Irrigation is the process of delivering water to plants so as to meet their needs for several important resources. It is known that providing too little or too much water can reduce crop productivity or, when extreme, can lead to plant death. Improper water management can also lead to excessive run-off. While providing water is certainly a major facet of irrigation, various other materials that are dissolved in the water are also provided to the plant at the same time. These include an array of nutrients as well as oxygen.

Although fertilization is not a specific aspect of this chapter, it should be noted that in soilless plant production, soluble fertilizers are frequently dissolved in the irrigation water using injection equipment. This type of irrigation with nutrient solution is called “fertigation.” In many soilless systems, fertigation is the preferred approach to supplying nutrients, especially where the substrate is not capable of holding (bonding) nutrient ions (i.e., due to low cation-exchange capacity (CEC). Specific technologies involved in combining fertilizers with irrigation water are covered elsewhere in the book. In this chapter we assume that this type of irrigation is being practiced. So here, the terms “fertigation” and “irrigation” are used interchangeably.

In general, the plant’s fresh weight typically consists of more than 90% water and less than 10% everything else. There are, of course, many exceptions reflecting organ function and plant adaptation to the native environment. When comparing the amount of water held in fresh matter of a particular plant with the total amount of water that the plant has taken into the roots, only a small percentage (about 1%) of the total water actually ends up as fresh matter (Raviv and Blom, 2001). The rest of the water is transpired and most of this water use is necessary as part of the plant’s need to cool itself and to transport materials from the root zone to the aerial parts of the plant.

The total amount of water that flows through a particular plant system is generally called “the transpiration stream” and includes all the water that enters the roots and thus also accounts for the fraction of water that ends up tied up in biomass. While transpiration cools the leaves, this flux of the transpiration stream has several other consequences that are very important. One of these is the fact that when this flux is positive, it has to match an equivalent flux in the root zone of liquid moving toward the roots. This liquid, called the root zone solution, carries with it nutrients and gases, and the motion of this liquid is driven by the transpiration stream of the plant.

It should be noted that dissolved elements can move by either diffusion or bulk (mass) flow. Diffusion is the process whereby an element moves due to a gradient in concentration of that element. Bulk flow (or mass flow) is when the liquid in which the material is dissolved carries the element to another location. Diffusion is a relatively slow process so that movement of an element over 1 cm distance can take hours or days (depending on the magnitude of the gradient); bulk flow can carry elements over this distance in a matter of seconds. The biggest implication for this is that when the transpiration stream is actively moving water, or when water is applied through irrigation, bulk flow of nutrients and oxygen is carrying these elements toward the root surface.

Of all resources that the plant needs, water is needed in greatest quantities and needs to be plentiful and readily available so as to sustain a transpiration stream that is actively carrying the nutrients needed for plant growth or production to the surface of the roots.

9.1.1 WATER MOVEMENT IN PLANTS

Plants are able to capture light energy and use it to transform water and CO_2 into a useful chemical form (carbohydrates) through the photosynthetic reactions that occur in the leaves. While water is mainly absorbed by the plants through its roots, CO_2 enters the plant through pores in the leaves called “stomata.” These pores must be open for the CO_2 to enter and at the same time water inside the leaves is able to evaporate into the atmosphere. This movement of water into the atmosphere is called “transpiration” and this process drives the movement of water from the soil, through the plant, and into the air surrounding the leaves.

At the leaf surface there is a relatively still, thin layer of air, called the “boundary layer.” Even with wind or air current flowing past the leaf, this boundary layer persists but becomes thinner as air movement increases. The movement of water from the stomata through this boundary layer is driven by diffusion (which is relatively slow) and only works if the water molecules are then carried away by air currents (a type of bulk flow which can be relatively fast). If the air surrounding the leaf is saturated with moisture (i.e., $\text{RH} > 95\%$), then water movement is slow since the diffusion gradients are small.

Most of the evaporation process at the leaves occurs through the stomata. The plant has the ability to control the degree to which the stomata are open and this mechanism allows the plant to exert control over the amount of water evaporated from the leaves. It should be noted that the surfaces of the plant are generally covered with a waxy layer (cuticle) which limits water loss. This water loss is generally negligible compared to the rate of water loss through the stomata. However, young leaves may not have a substantial cuticle and water loss control through these leaves is not as effective. Under normal conditions the total rate of water loss through evaporation is slightly less than (but very close to) the rate of water uptake from the root zone. The total water loss from plants is also affected by the number of stomata that are present on the foliage. Thus, the more leaf area the plant has, the greater the potential rate of transpiration.

The potential rate of removal of water by plants from the root zone is driven by the following factors: temperature, light, relative humidity, wind speed, leaf area, and the degree to which the plant’s stomata are open. The potential amount of water removal is a matter of integrating this rate over time. Various irrigation control strategies (described below) use this information as part of the control strategy.

9.1.2 WATER POTENTIAL

The hydraulic forces of water within both plants and substrate play an important role in how water is managed. These forces can generally be expressed in terms of energy potentials such as matric, osmotic, and gravimetric potentials. When dealing with irrigation issues, such potentials are expressed as negative numbers with units of kilopascal (kPa). “Matric potential” or “moisture potential” is synonymous; they have positive values in conditions of positive pressure and negative values for suction. In contrast, we use the term moisture “tension” as being the absolute force of suction. As a consequence “high tension” in substrates means “dry conditions.” In relation to the types of values that are typically recorded in soilless production, a relatively dry condition in most substrates corresponds to a moisture tension of between 10 and 20 kPa (or a matric potential between -10 and -20 kPa). In general, all irrigation practices that explicitly attempt to avoid water stress in soilless production are confined to the range of 0 – 8 kPa of tension. In some substrates, such as stone wool, the range is even narrower, with the onset of water stress problems occurring if the production system is allowed to exceed 5 kPa. In contrast, the types of tensions encountered in field soils may well range as high as 75 kPa; in such systems, we rarely see tensions below 10 kPa, except during or just after an irrigation event. The reasons for this difference in tension range are mainly due to the substrate texture: the size of the particles making up the soil and substrate. More correctly, it is the size of the pore spaces between the particles and the number of contact points between particles, to where water is attracted, per volume unit. In general, the texture of field soils is much finer with smaller particulates with many more contact points between them and smaller pore spaces compared to soilless substrates.

Matric potential, Ψ_m , results from the interaction of water and solid materials. It can be thought of as the pressure that would need to be exerted to remove water from a substrate. For example, we understand that a sponge can hold water and we need to squeeze it (impose a pressure) to make the sponge release the water. If we could maintain a constant pressure on the sponge, at some point no more water would be released while some would be retained in the sponge. Matric potential is holding the remainder of the water in the sponge against the squeezing pressure.

Gravimetric potential, Ψ_g , is related to the height of water above some point of reference. This force of gravity translates to approximately 1 kPa per 10 cm of water column (note that the horizontal size of the layer has no influence on this). For example, take a sponge saturated with water, stand it on one end, and allow it to drain. At some point water will stop draining from it. If the height of the sponge is 10 cm, then the gravimetric potential at the top of the sponge would be 1 kPa. The force that is holding water in that sponge can be measured as matric potential. Since energy needs to be balanced and we know that the gravimetric potential of the 10 cm sponge is 1 kPa, then the matric potential must be -1 kPa so that there is a net of 0 kPa.

Solute or osmotic potential, Ψ_s , is related to salinity [and thus electrical conductivity (EC)] and can be described mathematically with the formula:

$$\Psi_s = -RT \left(\frac{n_s}{v_w} \right)$$

where R is the gas constant, T is the temperature, n_s is the number of moles of solutes, and v_w is the volume of water. Note that the n_s/v_w portion is a concentration, so solute potential is directly related to the concentration of salt in the substrate solution. In the root zone, this can increase

through the addition of salts or through the removal of water. This second process can occur in plant systems whenever water is removed from the growing substrate at a greater rate than the solutes within it. The empirical relationship between EC (dS/m) and osmotic potential (kPa) of

$$\Psi_s = -36EC$$

has been suggested to estimate how salinity, at levels that plant growth will occur, accounts for the contribution to the overall potential (Richards, 1954). So, as the concentration of solutes in a solution, and thus EC, increases, the solute potential decreases.

Understanding the concept of potential energy can provide insight to how management practices affect plant water status. Water will move passively from a state of high energy to a relatively lower energy state, and the difference of these two energy states is related to the rate of water movement. So, increasing the gradient will increase the movement of water down that gradient. When an irrigation event occurs, the matric potential of the substrate increases and the water therein is of higher energy. This facilitates water movement into the roots that might be present since the water in them contains solutes and is lower in energy. If the water that is applied happens to contain a large amount of solutes, which reduces its energy potential, the gradient is reduced and the “ease” that water can move into the roots decreases.

The moisture content of any substrate is also affected by the physical characteristics of the substrate (particle size distribution, particle surface properties, pore space, etc.), each affecting the various potentials. For any substrate, there is a relationship between the moisture tension and moisture content. This relationship can be thought of as a “signature” for a particular substrate.

This curve has various names such as “moisture release curve,” “moisture retention curve,” etc.; the diagram (Fig. 9.1) shows the relationship between moisture tension and moisture content and

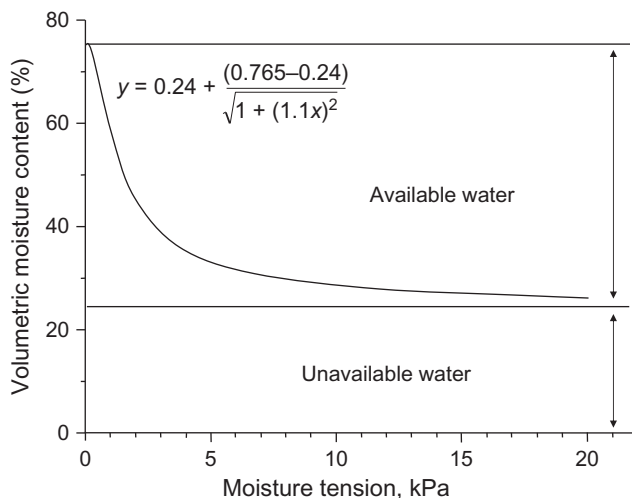


FIGURE 9.1

Moisture release curve of UC mix. The moisture content at saturation is 76.5% and the unavailable portion is 24% (Kiehl et al., 1992).

indicates a container capacity of 75% for the substrate University of California (UC) mix (see [Chapter 3: Physical Characteristics of Soilless Media](#), and [Chapter 7: Inorganic and Synthetic Organic Components of Soilless Culture and Potting Mixtures](#), for other examples of this curve). In addition to the various potentials, there are also forces due to gravity that affect the moisture status of the root zone and [Fig. 9.1](#) can be used to interpret how a root zone might behave with regard to moisture dynamics by considering the typical root zone as consisting of multiple layers of substrate “stacked” on top of each other and each affecting the layers above and below as the water is held in place by internal forces (capillary forces), with the weight of the water in all the layers below a particular layer pulling down on the water in the layers above. Thus the top layers have more weight pulling on them than the bottom layer so that the moisture tension varies with depth. Because field soils generally have finer textures than soilless substrates, these variations are relatively small per cm depth. But with soilless substrates, the moisture tension changes more dramatically with centimeters of depth.

With the moisture release curve ([Fig. 9.1](#)), it is possible to estimate the water content of such a containerized root zone immediately after an irrigation event. Taking into account the gravimetric potential, when a steady-state is reached at the bottom of the root zone of a container with a depth of 20 cm, it will have a moisture tension of 0 kPa while the top of the root zone will be at 2 kPa (=20 cm depth). Reading off the curve shows that moisture content at the bottom will be 75%, while the substrate at the top will be at about 50%. The differences are even greater as the dry-down progresses. Many growers typically think that the entire root zone is the same, but that is clearly not the case and it can be deceptive since the top layer (which the grower can readily observe) is always drier than the bottom layer.

The moisture release curve also illustrates some general points about irrigating. The wettest condition (i.e., after a thorough irrigation) is represented on the curve at tensions near zero. The general shape for most potting substrates is similar, but they do vary enough to affect final set-point recommendations for use in making irrigation decisions. As water is removed from the substrate by the plant (and by evaporation), the status progresses along the curve to higher tensions (lower water content). The pattern is approximately asymptotic to some volumetric water content with increasing high tensions. Most plants will show wilting when they are subjected to moisture content that approaches the levels represented by this line. Thus, in the literature, the water that remains in the substrate at this point is termed “unavailable water,” while the difference between container capacity and this level is called “available water.” Also, the term “easily available water” is generally used to refer to water released between 1 and 5 kPa, while the range 5–10 kPa is generally referred to as “water buffering capacity.” During rewetting, typically a much faster process than dry-down in soilless substrates, the pattern is always similar, even if slightly different (a phenomenon called “hysteresis”). However, with regard to the precision needed to make irrigation decisions, this aspect can be ignored.

9.1.3 THE ROOT ZONE

Within this book we use the term “root zone” to represent the space (generally filled with substrate) which has been made available in a production system to be occupied by plant roots. There may be areas within the root zone where few roots are present because of adverse conditions at various times or because the plant has not yet filled the root zone with roots. In root zones that are designed and assembled well, such “voids” are present only in early phases of plant production; in poorly

designed systems and in systems where irrigation is not controlled well, there may be many such pockets. But even in the most uniform systems, there will be areas with fewer roots. This heterogeneity in root distribution can cause problems.

One factor contributing to this heterogeneity is that root growth is typically geotropic, meaning that they grow downward. In constrained root systems where the bottom is an impervious layer with access to oxygen through drainage holes, there will generally be more roots there. In the areas in the root zone where many roots are present, water can be removed at a more rapid rate than in areas where fewer roots are present. At the same time, water is removed through evaporation from the parts of the root zone that are exposed directly to the atmosphere. The result is that the removal of water is not uniform throughout the entire root zone.

One important facet in soilless production is a substrate characteristic known as “hydraulic conductivity.” This refers to the rate at which water can move within the root zone. As discussed elsewhere in this book ([Chapter 3: Physical Characteristics of Soilless Media](#)), hydraulic conductivity is generally much higher in soilless media than soil-based media. For all substrates, there is a large difference between the hydraulic conductivity when the substrate is moist, compared to when it is dry. With regard to irrigation schemes, it is the unsaturated hydraulic conductivity (UHC, i.e., the value when moisture is present, but not saturating) that is of particular relevance. The higher the UHC, the faster water will move laterally during, and immediately after, an irrigation event. With drip irrigation systems, this rate dictates the flow rate of the emitters that should be used (if UHC is low, then emitters must be used that have a low flow rate).

It should be noted that if the rate at which water moves in the root zone is slow, then it may be possible for the plants to extract some of the dissolved elements at a rate higher than can be replenished with future irrigation pulses. This can result in problems with nutrient deficiency as well as oxygen deficiency. For this reason, soilless production systems always consist of substrates that have higher UHC than soil.

In intermittent irrigation approaches there may be the risk of drying out portions of the root zone to the extent that the hydraulic conductivity is so poor that water being applied to the top or bottom of the root zone cannot distribute itself throughout the root zone during the interval between irrigation events. This is especially of concern with substrates containing compost or sphagnum peat which become hydrophobic in dry conditions and then are very difficult to rewet. The regions in the root zone that are most prone to drying out are those that contain a lot of roots. Thus, when this happens, it becomes very difficult to rewet the entire root zone uniformly and especially those areas where the roots are most prevalent. This can lead to situations where the plant appears to be water stressed despite the fact that the container is heavy with water. In such instances, the plant is responding to the fact that there is little water around the roots and as this is further depleted, the low hydraulic conductivity makes replenishing the substrate surrounding the roots a slow process. This can be particularly problematic if the plants are exposed to high temperature, high light levels, and dry winds, representing a need for urgent action. The best approach to dealing with such an emergency is to apply water to the entire plant directly to provide water for cooling and increase the humidity to reduce the water loss rate. At the same time the root zone may need to be irrigated several times within a few hours so as to attempt to elevate the hydraulic conductivity throughout the root zone and to replenish oxygen through bulk flow.

If pockets of the root zone are allowed to dry out, then this can result in root death in those pockets. Also, as the water is removed preferentially from such areas, the concentration of solutes

increases to the point where some of them may be toxic to the plant. Historically, the main technique for preventing such problems has been to provide an excess amount of water (i.e., more water than the root zone can hold) and to apply the water slowly so that it has time to distribute itself laterally in the root zone even as it slowly leaches from the bottom of the root zone.

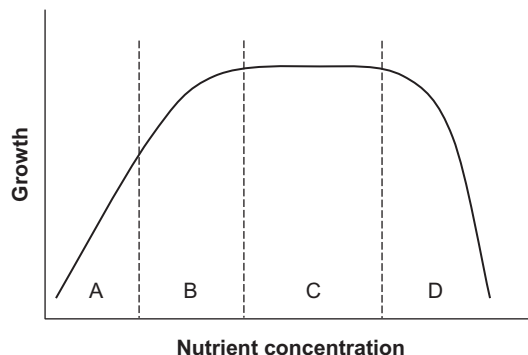
There is a facet to irrigation that is particularly counterintuitive. There is widespread knowledge that oxygen starvation can occur and that flooding can cause this so that some practitioners associate the application of excessive amounts of water with oxygen starvation. Many irrigators are careful to not irrigate too often but do try to keep the root zone as moist as possible; and it is exactly this scenario that is likely to result in oxygen deficiency. When new water is applied, it carries dissolved oxygen into the root zone and this mass flow of oxygen can resolve the problem very quickly. But if that oxygen is consumed by the plant at a faster rate than water use and this stagnant water is not replaced, then oxygen starvation can occur, leading to poor growth and plant diseases.

9.1.4 WATER QUALITY

Traditionally most water sources used in commercial soilless production did not consist of pure H_2O , having some minerals and/or salts dissolved in the water. For the practitioner it is important to know that materials dissolved in the water do not change the clarity of the water although the color may change. Salty water appears just as clear as pure water to the naked eye. If the water appears cloudy, then it is said to be turbid and this is due to materials or biological entities (microbes, algae) which are suspended but not dissolved in the water. These cause turbidity which can typically be filtered out of the water, while dissolved materials will pass through fine filters unimpeded. Filtration is an expense that growers must evaluate and expense increases as the size of the pollutants decreases. For example, removal of sediment might simply be accomplished by moving water through a settling pond to slow the movement of water to allow the particulates to fall out of suspension. In contrast, extraction of dissolved materials can be accomplished with membranes, but this is typically relatively expensive as it requires considerable energy. Use of water that is not of high quality nearly always results in problems for the grower so many growers have systems on hand to generate high-quality water. While there are various methods for desalination, the reverse osmosis becomes the most widely used approach. Unfortunately, those systems typically function to separate the water into two streams, one of higher quality and another of even lower quality than the original. This low-quality water is generally of no use and must be discarded. The obvious consequence is that the total amount of water available for irrigation is reduced. This then may be problematic if water is in short supply. The cost of discarding the low-quality water may be high and must be taken into account when planning the system.

Water is typically evaluated for EC, pH, and alkalinity. EC is a measure of the amount of salt that is dissolved in the water. There are various salts that could be involved, some are harmful; some may not be toxic, but are undesirable; while others are nutrients that the plant uses as part of its metabolism. Most water sources will contain some of each of these types of materials. As with virtually all environmental variables, there is a response curve for each nutrient for any plant.

Such a curve generally has a range where the plant will have inadequate amounts (deficiencies), a range where the plant can tolerate the nutrient, and a range of concentrations where the plant will respond in a negative way, possibly showing foliar and/or root damage (Fig. 9.2).

**FIGURE 9.2**

A nutrient response curve. At low concentrations, small increases in availability result in large changes in growth (A). Further increases in nutrient concentrations have smaller affects as nutrient levels approach optimal levels (B). At some point, additional amounts do not increase growth. This is the range of luxury consumption (C). At high levels, toxicity is reached and growth diminishes. Eventually, plant death occurs.

Adapted from Nelson, P., 2003. Greenhouse Operation and Management, sixth ed. Prentiss Hall, Upper Saddle River, NJ (Nelson, 2003).

EC, generally measured in units of dS/m, is typically seen in soilless production from 0 to 5 dS/m. Historically, growers have been advised to keep the EC below 3 dS/m to assure rapid growth of plants. Clearly this is impossible if the source water already is high in dissolved salts and the addition of nutrients will raise the EC to greater than 3 dS/m. If the source water has too high an EC, then the only options the grower has is to remove some of the salts, blend with low-EC water such as stored rain water, or to grow plants that can tolerate the elevated salt content. It should be noted that with development of complex hydroponic systems that include recirculation, it has become apparent that higher levels of EC might be infeasible.

pH is a measure of the hydrogen ion concentration ($-\log_{10}$ of mol/L), describing the extent that the fluid is an acid or a base. Water is said to be “neutral” if it has a pH of 7. For soilless production, however, this term is deceptive because for virtually all crops in such production systems, water that has a pH significantly lower than 7.0 is preferred. In soilless production, a range of irrigation solution pH of between 5.6 and 6.8 is required. For some crops, even lower pH levels have been found to be ideal. In general, pH levels of the solution around the roots should not be below 5 or above 7. Note that this is a wide range of the hydrogen ion concentration because the pH is a logarithmic scale.

One important aspect of water quality is that dissolved minerals in the water may build-up in the root zone. This is due to the fact that plants can preferentially remove water over dissolved minerals. Thus as the water is extracted from the root zone, they become more concentrated as water is depleted. At some point the concentration may reach a point where the material comes out of solution as mineral deposits. This condition should be avoided.

Alkalinity is a measure of the irrigation water to raise the pH of the root zone solution. This is measured in units of milliequivalents (meq) or ppm of CaCO_3 . In soilless production, low alkalinity (less than 3 meq) is generally acceptable, while a level greater than 8 meq is so problematic for

growers that such water is not suitable for crop production. When the source water has alkalinity below 8 meq, it can be used if treated with acid injection.

On the opposite end of the water quality spectrum is water that is much purer than the typical sources described above. As fresh water supplies on the planet have declined due to overuse, it has become necessary in some countries to desalinate seawater or brackish water. Such high-quality water is essentially very low in solutes. Another side of soilless culture is the increase in indoor production without sunlight, where light is generated with lamps, and air conditioning systems are used for cooling. One by-product of air conditioning is that water is condensed out of the air that is being cooled, and this water is available to be used in irrigation. Here the water is basically distilled water, and thus mostly free of solutes.

While it is generally assumed that such high-quality water is perfect for agriculture, this is technically not the case. There are elements (ions) in such water at very low concentrations. In reporting on the use of desalinated water in Israel from the Ashkelon facility, [Yermiyahu et al. \(2007\)](#) reported that agricultural crops produced with such water suffered from deficiencies of various nutrients which are typically found in ground water used in agriculture in the past. They suggested that irrigation with desalinated water may also lead to deficiencies in such essential elements as calcium and sulfur. [Bar-Tal et al. \(2017\)](#) showed that desalinated water with Ca and Mg levels significantly below optimal rates resulted in significantly greater incidence of blossom-end rot in tomatoes. Thus growers shifting to use of desalinated water need to modify their fertilization program. Also boron is generally not removed through reverse osmosis (RO) systems. At the Ashkelon facility, where the source water contains considerable amounts of boron, the resulting high-quality water contained levels of boron that are known to be a problem for plants, while being safe for human consumption ([Yermiyahu et al., 2007](#)).

Condensate water from air conditioning systems makes contact with the cooling coils and these are typically made of metal (copper) and soldered together using solder. Some small amount of metals in such systems can thus end up dissolved in such condensate. In most horticultural situations, these concentrations may be so low that it is not relevant, but in recirculated systems, these metals can build up to where toxicity symptoms can occur.

Both of these “high-quality” water supplies have another property that is a potential problem: they tend to corrode metal pipes. Consequently, irrigation systems transporting such high-quality water should be made of plastic. Furthermore, due to the absence of ions that provide the water with buffering capacity, pH fluctuations must be anticipated and materials must be included in the fertilization program to address this.

Recycled or reclaimed municipal water, secondary level treated municipal waste water, is becoming more common as a source for agriculture. In California, this water contains moderate levels of salts with ECs around 1.0 dS/m; however, they consist mostly of sodium and chloride ions (Na^+ and Cl^- , respectively) and may be in concentrations high enough to cause plant damage ([Nackley et al., 2015](#); [Tanji et al., 2008](#)). In Israel, irrigation of rose plants with secondary level treated municipal waste water with ECs around 2.0–2.5 dS/m increased the Cl concentration in leaves by 47% in perlite and 73% in coir grown plants ([Bernstein et al., 2006](#)). So, the sodium and chloride sensitivity and tolerance to salinity of plants to be irrigated with this water need to be considered before applying the water. In some cases, blending the reclaimed with other higher quality water to dilute the salts or with water that has appropriate levels of calcium and magnesium to lower the sodium absorption ratio can reduce toxicities.

In California, the Irrigated Lands Regulatory Program requires that runoff water from irrigated agriculture must meet water quality standards (see https://www.waterboards.ca.gov/water_issues/programs/agriculture/). To comply with this regulation, runoff water is captured for reuse. Concerns of using this water for irrigation not only include all of the quality issues discussed previously but also include plant pathogens. As irrigation water washes over infected plant tissues or leaches through soils containing soil-borne diseases and then accumulates as surface run off, inoculum can travel in the water. Applying this water that is infested with pathogenic inoculum as irrigation is, obviously, not desirable. To eliminate the inoculum, the water must be treated to remove pathogens and the potential to spread diseases. Various and numerous pathogens have been documented in irrigation water sources and include *Phytophthora* spp., *Pythium* spp., *Fusarium* spp., plant pathogenic viruses, and nematodes (Hong and Moorman, 2007; Hong et al., 2014; Koenig, 1986; Stewart-Wade, 2011).

There are several methods to remove pathogens from the water that fall into three categories of treatment: (1) chemical, (2) physical, and (3) biological. Chemical treatments frequently used in horticulture include chlorination (using sodium and calcium hypochlorite, chlorine gas, or other chlorine source), chlorine dioxide, ozonation, hydrogen peroxide, and copper ionization. Physical methods include reverse osmosis, heat treatment/pasteurization, and UV radiation. Biological methods include constructed wetlands and slow sand filtration. Slow sand filters have been demonstrated to be effective in removing *Phytophthora capsici* (Lee and Oki, 2013), *Phytophthora* spp. (Ufer et al., 2008), tobacco mosaic virus (Oki et al., 2017), and *Fusarium* (Prenafeta-Boldu et al., 2017). There are numerous references to these water treatment methods since the largest interest is in the production of drinking quality water. For growers, there are excellent online sources of information at the University of Florida (<http://www.watereducationalalliance.org>) and the University of Guelph (<http://www.ces.uoguelph.ca/water/pathogen.shtml>) comparing different treatment methods for specific pathogens, effectiveness, and costs.

9.2 ROOT ZONE MOISTURE DYNAMICS

The moisture dynamics in the root zone during irrigation are typically much faster than during the time between irrigation events. There are significant complexities related to this that affect irrigation decisions.

9.2.1 DURING AN IRRIGATION EVENT

The irrigation event is generally triggered by a human or automated (preprogrammed) decision-making process. The onset of the event is characterized by the opening of a valve which leads to water or dilute nutrient solution, flowing under pressure. An irrigation event lasts a particular period of time (typically minutes) and the end of the event is specifically controlled to achieve a desired effect. Usually the objective is to bring the root zone up to full capacity, although deviations from this mission are possible and discussed below as part of irrigation decisions.

While the focus during irrigation is generally on the delivery of water, dissolved substances and gasses are also provided to the plant. These other substances may be present because they were

part of the source water or were purposely added to the irrigation water as part of a crop production strategy.

It is also noteworthy that during the time of an irrigation event, the spatial distribution of water in the root zone is heterogeneous and dynamic (changing from second to second). Water moves downward through the substrate fast (in seconds) and slows laterally (in minutes or hours). As the water enters the area surrounding the roots, it carries with it substances that were dissolved in the irrigation solution and substances that were already in the root zone. Thus the bulk movement of water moves materials to the root surface.

9.2.2 BETWEEN IRRIGATION EVENTS

Immediately after an irrigation event where the volume of water applied exceeds the amount needed to achieve container capacity, there is a period of time where water continues to leach from the root zone. The leachate has to be replaced with air, which is pulled into the root zone at any point where the substrate is exposed to air. This action is generally overlooked, but it should be noted that air entering this way consists of over 21% oxygen so that this can account for a substantial amount of oxygen.

Once the leaching has stopped, there is a period of gradual decline in water content due to evapotranspiration (ET): water being removed by the plant (transpiration) plus evaporation of water from the substrate directly to the air. It should be noted that water moves down an energy gradient. A number of variables affect this gradient between the leaf and the atmosphere and, consequently, the rate at which this removal occurs.

Ion (including nutrients) uptake by the roots may be at rates that are not the same as the rate of water uptake. Thus ions could be depleted more rapidly or less rapidly. In the latter case, this could lead to a concentrating effect which can manifest itself as an increase in root zone salinity. An important part of irrigation management is salinity control and between irrigations it is very likely that salinity is increasing in the liquid that surrounds the roots. There may well be times when an irrigation event is needed to flush out salts that have accumulated in this way.

Sensors that monitor soil moisture may also be able to monitor EC. Time domain reflectometry (TDR) may be able to measure EC in substrates, but these sensors are affected by organic matter content. In addition, when EC is high, the TDR signal is also affected so that these measurements are difficult to make. [Oki and Lieth \(2004\)](#) used a suction lysimeter with an embedded EC cell to measure substrate solution EC. There are several commercially available sensors available that measure substrate moisture, temperature, and EC.

9.2.3 PRIOR TO AN IRRIGATION EVENT

One of the most notable characteristics evidenced by moisture release curves of soilless substrates is that the dry-down process is not linear with regard to moisture tension and moisture content. Thus as water is depleted past some threshold (5 kPa for UC mix, [Fig. 9.1](#)), the tensions increase more and more rapidly with each unit of water depleted. This acceleration forces irrigation managers to schedule irrigations well in advance of critical high-tension conditions.

If the plant is not able to adequately control water loss as the moisture tension rises toward a critical point, then the plant will begin to dehydrate. This typically results in a loss of turgor

pressure in the leaves, resulting in the visible symptoms of wilting leaves and flagging stem tips. Within a very short time, the leaves lose the ability to cool themselves since the amount of water evaporating from them is inadequate to meet the cooling demand of the tissue. If this persists, then the temperature of the leaf will rise in the presence of high light, possibly resulting in temperatures high enough to cause damage to cells in the leaf. Plants have the capacity to adapt somewhat to this. If a plant experiences such high tensions frequently, then it adapts to close stomata at lower moisture tensions so as to avoid wilting. One additional consequence of stomatal closure is that it also prevents CO_2 from entering the leaves, thus shutting down photosynthesis, possibly reducing biomass accumulation. Thus it is best to use irrigation strategies that keep stomata open so as to maximize plant growth.

Even before wilting occurs during a dry-down, there are other deleterious effects which may not be easy to detect. Leaf and stem expansion are very sensitive to water stress (Hsiao, 1990) and can be reduced even under moderate water stress. Maximizing turgor pressure within young cells is necessary to maximize cell enlargement and stem and leaf growth. A reduced transpiration stream in the plant may also result in fewer mineral nutrients moving within the plant to the sites where they are needed as part of building biomass. There may also be less water available for the essential metabolic processes within the plant, resulting in suboptimal growth.

While the same phenomena and processes occur in soil-based and soilless systems, there are some significant differences. In general, plants growing in soil-based systems experience slower movement of water in the root zone so that the plant has to grow larger root systems to mine a larger soil volume for water so as to meet its transpiration demand. The smaller and confined root systems in soilless systems require greater attention by the irrigation manager as the plant has less protection from catastrophic irrigation system failure. Also, with smaller root systems, the plant may be less capable of responding to climatic extremes that increase water demand.

9.3 IRRIGATION OBJECTIVES AND DESIGN CHARACTERISTICS

An optimally designed irrigation system will deliver water to the plants to maximize water use efficiency (WUE). This means that water is provided in time just as the plants require it and in a manner so that all the delivered water is utilized by the plants and little is wasted. This requires that the delivery and control systems are properly designed for the application that they are to service.

Irrigation delivery systems are always designed to optimize various specific characteristics. This includes system capacity as well as system uniformity. The degree to which these characteristics are optimized has a greater impact on how irrigation is controlled rather than on the water utilization of the crop.

9.3.1 CAPACITY

The capacity of an irrigation system relates to the number of plants that can be irrigated at any one point in time. This is governed by the maximum rate that water can be extracted from its source. In some cases, production facilities have to install tanks to store water and the capacity of these tanks

and the dynamics of how they are filled and emptied governs the maximum rate of water use during production. For example, a grower needs 16,000 L/day to maintain the crop. The well that is available can produce only 1000 L/h. If an 8000 L water tank is available, it can be filled when there is no water demand, for example, at night. Both the well and the storage tank will be accessed to provide water for the crop during the day.

Irrigation systems should group plants so that this source capacity is distributed over time. As such, each irrigation circuit is separate and needs to have its own decision-making. Most large soil-less production operations have considerably more plants than could be watered with new source water all at once. Each circuit must be queued (scheduled) for irrigation and care must be taken to only irrigate as many circuits at any one time as can be supplied by the fertigation equipment (pumps and injectors).

9.3.2 UNIFORMITY

The grouping of plants is generally done so as to combine plants of similar water and nutrient use patterns into the same irrigation circuit. Typically there is an expectation by the grower that the entire crop stays uniform in water and nutrient use and that each plant reaches the production goal at the same date. Without temporal and spatial uniformity, the grower may face economic problems. Since the availability of water and nutrients directly impacts plant growth, the need for this uniformity dictates that the irrigation system delivers water in as uniform a manner as possible. This means that within an irrigation circuit, water is delivered to each plant at nearly the same rate to provide nearly the same amount of water and nutrients as possible.

Absolute uniformity is generally not feasible for any crop (Seginer, 1987). Crop response to water, cost of the irrigation solution, type of irrigation system all affect the level of practicability of taking special measures to achieve greater uniformity (e.g., pressure compensation at each emitter, respacing plants as they grow, etc.). For any situation, there is a level of uniformity that will result in the greatest level of economic feasibility (Chen and Wallender, 1984; Seginer, 1987).

Precise evaluation of irrigation system uniformity is difficult. There are a number of sources of variation that can be considered and measured. For any one irrigation system, the distribution of water from an irrigation system after the system is fully pressurized is typically different from the distribution during the time that it is pressurizing. While many practitioners consider the latter to be insignificantly short, in practice it is almost always a significant percentage of the duration of the irrigation event.

Uniformity in an irrigation system is generally measured by placing a container at each emitter to collect the water that is delivered during an irrigation event. The water collected determines the delivery rate of each emitter. Uniformity has been quantified in two differing approaches: (1) by summing the absolute difference of each rate from the mean and (2) by calculating variances or standard deviations. Both methods are equally effective for optimization purposes (Seginer, 1987), although the first of these is somewhat easier for practitioners, while the second is easier to implement in mathematical treatments of the problem. Also, if the uniformity measure is used within economic analyses, then it is important to note that the use of variance tends to give

disproportionately greater weight to larger deviations. Thus for a specific set of data collected as above, a “coefficient of uniformity” (U) can be calculated as a percentage value:

$$U = 100 \times \left[1.0 - \frac{\sum |x_{\text{mean}} - x_i|}{n \times x_{\text{mean}}} \right]$$

where x_i is the individual amount of water collected at a particular location, x_{mean} is the mean of these values, and n is the total number of these samples (Lieth and Burger, 1993). The alternative would be

$$U = 100 \times \left\{ 1.0 - \frac{[\sum ((x_{\text{mean}} - x_i)^2)]^{1/2}}{n \times x_{\text{mean}}} \right\}$$

Other methods have also been presented, but, as Seginer (1987) points out, the high degree of correlation between the methods suggests that the choice of method is not particularly important. Seginer (1987) concludes that while both approaches are useful under various conditions, the second of these two methods should generally be used.

The duration of a uniformity test can have an effect on the outcome. Water delivery to the plants close to the irrigation valve begins very quickly after the valve has opened, while the emitter farthest away from the valve begins delivering water later and at a lower rate due to pressure loss over the length of the flow of water. Thus, while conducting such tests, it is important to operate the system for durations that are similar to the durations that will be used in practice. Some emitters are pressure compensated and will minimize or eliminate fluctuations in flow rate due to pressure variations.

It should be noted that subirrigation systems are difficult, if not impossible, to assess for uniformity. Some practitioners have the notion that subirrigation systems are highly uniform, but this is not necessarily true. Subirrigation systems must be completely level to be absolutely uniform. At the same time, the ability to drain the system in a timely manner is very important to prevent the stagnation problems described above. Ebb-and-flood tables and trays with channels molded into the bottoms to facilitate water flow and distribution can achieve this in a way that still allows the system to be filled uniformly and drained fairly rapidly. Other subirrigation systems (troughs, flooded floors, and capillary mats) typically are not uniform since the bases of the plants sit at different levels along a drainage gradient. The troughs and flooded floors need to be sloped to provide drainage. Capillary mats installed on long, sloping benches can be problematic. If the height difference from one end of the bench to the other is large, it may create a pressure head in the mat adequate to suck nearly all water out of the root zones of plants sitting at the highest part of the bench.

The only type of system that has the potential to be completely uniform is liquid culture such as hydroponics, or NFT (Nutrient Film Technique) (Hurd, 1978) where the roots are always submerged or wetted and have access to water at very low tensions at all times.

9.4 IRRIGATION DELIVERY SYSTEMS

All irrigation systems consist of tubing and/or pipe to transport the irrigation solution from the source to the individual plants or to a group of plants. Generally, the irrigation solution is prepared

from water which is under pressure and injectors are used to inject various soluble fertilizers into the water. Mixing or blending tanks assure that the dissolved materials are distributed uniformly within the water, and filters remove any insoluble materials. Pumps, pressure tanks, and pressure regulators may be used to increase or stabilize the pressure at which the solution arrives at the valve to the irrigation circuit.

In all irrigation systems, the flow of water requires some form of pressurization to facilitate moving the water. The higher the pressure, the more rapidly the liquid can flow in the system. Friction within the pipes imposes resistance to flow, causing the pressure to drop over the length of the pipe. The diameter of the pipe has a significant effect on this phenomenon, with pipes of smaller diameter having greater pressure drop along the length of pipe. Tables are available that indicate the pressure drop for various lengths of pipes of various diameters and materials. It is important to properly design irrigation systems so that all of the parts (main lines, valves, pressure regulators, laterals, and emitters) are correctly sized to deliver water at the desired pressure and flow rate.

When a valve is opened in a system where the irrigation water is under pressure, this allows the irrigation solution to flow. While there is active flow in the system, there will be a dynamic pressure gradient with the highest pressure at the source and the lowest pressure values at the emitters. Any one irrigation circuit must be designed so that this pressure at each emitter is at or near the optimal pressure for that emitter. If too many emitters are included, then some emitters experience inadequate pressure and consequently deliver little or no water, while other emitters (closer to the valve) experience higher pressure and thus deliver water at a higher rate. If the total potential rate of flow out of all emitters is less than the capacity of the system, then pressure-compensating emitters can be used to assure that each plant receives water. But if the total rate of water delivery through all emitters in the circuit is greater than the source can provide, then there is no way that this problem can be solved without increasing the source pressure or installing pipes of larger diameter to increase the flow rate within the system. Under such a circumstance, replacing all of the emitters with ones that have a lower delivery rate (e.g., slower drip) would be a solution. Obviously the duration of operation of the irrigation circuit would have to increase to assure that each plant receives the appropriate amount of irrigation solution.

In commercial soilless production, one of the most important crop production factors is uniformity. This is particularly important if the entire crop needs to be shipped (harvested) at once. Since the availability of water and fertilizer is a major factor for growth, variations in available water and nutrients will result in variation in the size and quality of the plants. For this reason, the uniform delivery of irrigation water is important in such systems. Minimizing the pressure drop within the system, and especially the variation in pressure among the emitters, maximizes irrigation uniformity. Along with general upkeep of the irrigation system, it is advisable to periodically conduct an assessment of distribution uniformity (DU) of both overhead and drip systems. There are numerous methods published to do this and include those from the University of California ([Shaw and Pitteneger, 2009](#)) and the [Irrigation Association \(2011\)](#). Test results will inform the need for maintenance and repair.

The most common errors in commercial production consist of having irrigation circuits too large (attempting to irrigate too many plants through one control valve) and minimizing the plumbing costs by using the smallest diameter pipe that is feasible. Generally, this is an economic decision designed to minimize installation costs. However, a price will be paid during the years of operation. If each circuit is large, then it may be impossible to irrigate all plants during a hot dry

day because of inadequate flow rates. By having many smaller circuits, it is possible to make better use of the available pumping capacity, while providing better uniformity.

Various types of irrigation systems are in use in soilless production in greenhouses or nurseries. These include various approaches to delivering the water to the plant: overhead systems (which apply irrigation solution “over the top” of the foliage), drip or microspray systems that apply water to the substrate surface (while keeping the foliage from getting wet), and subsurface systems (which apply water to the base of the root zone and rely on capillary action or flooding to bring water into the root zone from below).

9.4.1 OVERHEAD SYSTEMS

Overhead irrigation systems (Fig. 9.3) deliver water to the space above and surrounding the foliage of the plants with the expectation that this water will disperse through the canopy of the plant to the top of the root zone. While some water will evaporate in the process, effecting some cooling,



FIGURE 9.3

Overhead irrigation systems: (A) hand watering, (B) boom system, (C) mist system, (D) sprinkler system in an outdoor nursery.

much will penetrate the canopy and end up at the top of the root zone. If the plants are grown in containers which are spaced apart on the ground or bench, these systems are very inefficient and have a low WUE. Most of the water applied will not reach the substrate surface, wetting the bench, floor, or ground and become unused runoff.

Overhead irrigation systems are widely used in commercial nurseries because they tend to be relatively inexpensive to install and can irrigate large areas. However, if materials are dissolved in the water (e.g., minerals or fertilizer), then this method can result in unsightly residues on flowers and leaves. This is generally only a problem for ornamental plants and many times such residues can be washed off with clear water irrigation or with rain. Water on the foliage may also cause disease problems if water droplets stay long enough to allow germination and growth of fungi and bacteria. Such situations are most likely to occur when relative humidity is high (e.g., at night or during rainy or foggy weather).

While it is frequently undesirable to wet the foliage, overhead irrigation can be used to cool plants directly on hot days by wetting the plants and allowing the water to evaporate. This type of water application is called “syringing” and is generally done with clear water and scheduled to allow the foliage to dry completely before the night so as to avoid disease problems.

The following are specific types of overhead irrigation systems that are of use in soilless plant production.

9.4.1.1 Sprinkler systems

Overhead sprinkler irrigation (Fig. 9.3D) is one of the least expensive water distribution methods since each emitter serves many plants. With potted plants, the major drawback to this type of irrigation is the large quantity of wasted irrigation solution due to water that misses the root zone. Also, making such systems uniform requires that the circular distribution patterns of each sprinkler head be overlapped extensively; even with that precaution in the design, it is very difficult to create a uniform distribution pattern. Thus in such systems, extensive overirrigation of many plants is needed to assure that those plants in the locations receiving the least amount of water get an adequate amount of irrigation solution. Since crops are typically irrigated based on the driest location, maximizing the uniformity of the application of water will reduce the duration that water needs to be applied and will increase WUE.

Sprinkler heads typically throw a stream of water while rotating in a circular pattern (e.g., gear driven, multi-stream rotors, or impulse sprinklers) or spray water in all directions simultaneously (spray heads) through a particular nozzle design. They are available in full-circle or partial-circle patterns and at various rates of water delivery. Sprinkler systems lend themselves to automated control of irrigation in potted outdoor nursery production. Selecting the most appropriate sprinklers requires considering the size of the area to be irrigated, available water pressure and flow, type and growth stage of plants receiving the water, and other factors.

9.4.1.2 Mist and fog systems

Mist and fog systems dispense water into the air surrounding the plants. Generally, these systems are primarily cooling or humidification systems rather than delivering water to the root zone. On propagation benches where cuttings are rooting, mist systems can provide adequate water to meet the plants irrigation needs, but this is largely due to the fact that the transpiration rate is significantly reduced. Care needs to be taken in such situations to supplement the water to the newly

forming roots if the mist system does not deliver enough water to wet the entire root zone. Mist systems (Fig. 9.3C) generally produce smaller water droplets than sprinkler systems. Their objective is to wet the foliage, either for cooling the crop or to prevent or reduce transpiration (as in the rooting of cuttings). Fog systems generate even smaller water droplets with the objective of cooling or humidification. Both fog and mist systems must be controlled very precisely to avoid disease and algae problems. They must be used in concert with ventilation to remove the moist air before more water can be dispensed.

Controlling irrigation in greenhouses with mist or fog systems requires more care since it is possible for the grower or installed sensors to be deceived by water in the air, on the foliage, and on the root zone surface. It is possible, under such conditions, for the root zone to be too dry, with very low hydraulic conductivity while visible surfaces are wet. In this setting, sensor-based irrigation that measures the moisture deeper in the root zone is very useful to determine when the root zone is becoming too dry. It should be noted that sensor inputs into model-based irrigation control systems may also be deceived by the plethora of water surrounding the plants. Properly managed fog systems should not result in wet surfaces.

9.4.1.3 Boom systems

A boom system (Fig. 9.3B) consists of a machine which travels above the plants on rails, dispensing irrigation solution to the plants below through nozzles that are mounted on a boom suspended over the plants. In protected cultivation, the travel of such a device is linear. In some cases, the nozzles generate a spray similar to a sprinkler system, while in other cases, water is allowed to flow in a stream to plants below. In the latter case, the plants need to be positioned very precisely under the path of the boom and the substrate needs to have adequate bulk density and consistency so that the water stream does not wash it out of the pot or knock plants over.

Some boom systems can be programmed to modify the delivery rate of the irrigation solution along the path and to even skip over sections. In many nurseries, boom systems are installed so that each section of plants, or perhaps an entire greenhouse consisting of many benches of plants, has a dedicated boom. Such systems can operate fully automated.

Control of such systems commonly is through manual operation, especially if intervention is needed to start and stop the irrigation or to reposition the boom equipment. The duration of the irrigation event for each plant is dictated by the travel speed of the boom across the plants.

Boom systems allow for uniform overhead irrigation and customized delivery to each plant. The main drawback is the installation expense and (perhaps) the need for manual intervention (labor costs). In highly automated (and capital intensive) moving-tray systems, it is possible to mount one boom permanently and have the trays with the plants move underneath.

9.4.2 SURFACE SYSTEMS

Surface systems (Fig. 9.4) come in a wide range of schemes, delivering water directly to the top surface of the root zone with minimal wetting of the foliage.

9.4.2.1 Hand-watering

Irrigating by hand with a hose and wand affords the possibility of customizing the delivery of water to the top of the plant (Fig. 9.3A) or to just the root zone of each plant. Some sort of diffuser is

**FIGURE 9.4**

Surface irrigation systems: (A) spaghetti system, (B) micro-spray emitter, (C) drip irrigation monitoring, (D) perimeter micro-spray irrigation, (E) drip tape, and (F) custom-spaced in-line emitters.

generally used to prevent washing substrate out of the pot and to disperse the water over the entire top surface of the root zone. Potted plants should have adequate headspace at the top of the root zone since enough water needs to be placed there quickly to allow replenishment of a considerable amount of readily available water (perhaps as much as 50% of the container volume). Hand watering also dictates that the infiltration rate must be fast enough to accept water quickly while the pot is being watered, but not so fast as to have the applied water run through the pot without being retained.

Hand-watered plants should always be watered to bring each pot to container capacity. Partial irrigation can cause a portion of the root zone to dry out. With most container mixes it is difficult to know whether the substrate has become completely rewetted. Water draining out of the bottom of a pot is not necessarily a good indicator as to whether the substrate has been uniformly rewetted to container capacity, since in porous substrates, water can rush (channel) through the root zone and not be uniformly adsorbed by the substrate.

Although hand watering can be very effective, this is entirely dependent on the competency of the irrigator and very skilled and knowledgeable ones are rare. More typically hand watering is the

most inefficient method of irrigation. The applications are usually made based on the schedule of the irrigator and not necessarily when it is best for the plants. Usually volumes of water applied are much greater and application rates are much faster than the substrate can accept the water. As the stream of water is moved from plant to plant, water continues to flow between containers, so it is applied where it is not effective. This results in a very low WUE and a significant amount of runoff containing nutrients and other pollutants.

9.4.2.2 Drip irrigation systems

Drip irrigation systems (Fig. 9.4A) deliver a spray, drip, or slow flow of irrigation solution directly to the top of the root zone. While some of these types of systems do not technically “drip,” they all have water flowing from an emitter at a relatively slow rate.

Drip systems deliver water directly to the base of each individual plant slowly as to allow water to move laterally in the root zone before water starts coming out the bottom of the root zone. Water that is applied too fast will channel straight through the substrate and out the drainage holes, where it is lost from the root zone. Other ways to facilitate getting water to all parts of the root zone include the use of more emitters per plant or pot or to use emitters which deliver a spray of water (e.g., spray stakes or “spitters”) (Fig. 9.4B) so as to wet a larger area of the substrate surface.

Drip systems should always be equipped with filters to prevent clogging and pressure regulators to assure proper pressure (25–40 psi, see emitter manufacturer’s recommendations). Fluctuations in pressure can result in erratic delivery volumes and poor system uniformity. Also, excessive pressure (e.g., during surges when valves are opened or closed or due to water hammer at the end of an irrigation cycle) can blow out or destroy emitters.

Since this type of irrigation places an emitter at each plant, considerably more supply pipe and/or distribution tubing is needed than with other irrigation systems. Near the plants, this pipe is generally a soft plastic tubing (typically polyethylene) to allow emitters to be attached directly to them or with drip tubes.

Tube or “spaghetti” systems (Fig. 9.4A) are those which use smaller diameter tubes that are connected to laterals to deliver water to each plant. Spaghetti systems can be used with or without emitters. In fact, some emitters actually function only as weights attached at the end of a tube. The tube itself has a small inner diameter, slowing the flow of water. Each tube must be exactly the same length since the flow rate from the tube varies with its length. Another function of the weight is to diffuse the water emerging from the tube. Some such emitters have also been developed that allow the tube to be shut off when not in use. Other emitters also have pressure regulation as a built-in feature to ensure that the dynamic pressure at each emitter is identical so that the flow rates delivered from them are also similar.

In a spaghetti system, the rate of water delivered to each pot will depend on the distance of the end of each tube from the main valve with the further away from the valve, the lower the delivery rate at the emitter. Furthermore, this rate will vary with pressure fluctuation. Pressure-compensating emitters can be used to help make such systems more uniform and are necessary to maximize uniformity and WUE.

Drip systems allow the highest degree of precision and uniformity but the substantial costs of acquisition, installation labor, and maintenances mean that they may not be feasible for smaller pots or crops with short production cycles. Drip systems can easily be monitored by measuring the performance of a particular emitter at a representative position within the irrigation circuit (Fig. 9.4C).

In-line systems have emitters embedded at regular distances directly in tubing. These can either be in the form of drip tape (Fig. 9.4E), tubing that has emitters molded into the tubing during the manufacturing process at regular intervals, or tubing to which emitters are attached directly (Fig. 9.4F). Such a system is precise in how it places water near the plants, but it reduces the flexibility a grower has in spacing options. For example, tubing with drippers built in at specific intervals dictates the spacing for the crop.

Drip irrigation systems are also used in conjunction with bag or slab crop production systems using various materials such as stone wool. In such production systems, the irrigation system delivers a slow drip to the substrate and this water moves relatively quickly through the substrate to the base of the bag, pot, or tray. Holes or slits cut into the base of the container allow water to drain, but some puddling inside the container is unavoidable (and desirable) assuring that the substrate does not dry out as long as irrigation occurs frequently. Irrigation control in such systems is typically by timer, but it is also possible to have one bag mounted in a tray that is designed to trigger irrigation whenever the water content in this tray drops below a certain level.

9.4.2.3 *Perimeter and microspray*

Similar to overhead sprinkler systems, irrigation systems can also be constructed to deliver a spray directly to the base of the plants (Fig. 9.4D). Such systems are sometimes characterized as drip systems due to the relatively slow rate of water delivery. In cut flower production, this type of system generally is installed around the perimeter of a raised bed of substrate or amended soil.

9.4.3 SUBSURFACE

Subsurface systems (Fig. 9.5) bring irrigation solution into the root zone from below. These systems include capillary mats (Fig. 9.5A), troughs (Fig. 9.5B), flooded trays, benches (Fig. 9.5C), and floors (Fig. 9.5D). When irrigating with these, the bottom of the substrate of the root zone is put in contact with water and capillary action in the growing medium carries the water into and throughout the root zone. Substrates that have poor UHC will not distribute water adequately above the level to which the base of the root zone is flooded. Thus with subirrigation, it is imperative that (1) root zones cannot be allowed to dry out to any extent and (2) only shallow root zones (less than 10 cm) can be used unless the substrate is specifically designed to have high water-holding capacity, high hydraulic conductivity, and excellent capillary rise.

While there are advantages to using subirrigation, all such methods share a drawback: there is a tendency for salts to build up in the upper portion of the root zone. This occurs because irrigation solution (containing water and fertilizer salts) enters at the bottom. While some salts are taken up by the plant, water evaporating from the soil surface will leave salts behind, concentrating them at that surface. Monitoring of salt concentrations and leaching by occasional irrigation from above are essential with subsurface irrigation, unless the source water has very low EC and the crop is grown for a relatively short time.

9.4.3.1 *Capillary mats*

Capillary action can be used by setting the pots on a wet fibrous mat and allowing irrigation solution applied to the mat to be absorbed into the bottom of the pot (Fig. 9.5A). The mat must be on a level, waterproof surface and the base of the pots must have holes so that the potting medium can

**FIGURE 9.5**

Subirrigation systems: (A) capillary mat, (B) trough system, (C) flooded tray or table, (D) flooded floor system.

come into contact with the mat. The basic idea is to saturate the mat with irrigation solution and to keep the rooting medium in every pot at the same moisture tension (and thus the same moisture content).

Improper leveling can result in problems with capillary mat systems. If there is just a 10 cm drop over the length of a bench, for example, then the plants at the lower level will have moisture tensions 1 kPa lower than those of higher levels. For typical container media (e.g., represented in Fig. 9.1), this can result in substantial differences in water content and a loss of crop uniformity.

Fibrous mats are available bonded to a layer of perforated plastic film which is designed to inhibit evaporation and algae growth and to reduce the incidence of roots growing into the mat.

The edges of the mat should not be draped over the edge of the bench, since the wicking action will cause water to be drawn off the mat, draining most of the water away from the plants.

Three sides of the perimeter of the bench can have the edge of the mat and water-tight plastic under the mat slightly elevated so that all runoff occurs at the end opposite the water supply (Fig. 9.5A). If water is applied to the mat slowly, then the water movement across the bench will be through the mat and runoff from the bench will be slow until the mat is saturated. If water is

applied rapidly, then water will travel through, as well as on top of, the mat. In closed systems the drained water is recycled and hence can be applied frequently. In this case, fast application is possible. If the edges of the mat are not elevated slightly, then water may run off the bench, near the source, long before the remainder of the mat is saturated. Some growers also elevate the edge on the fourth side during the irrigation to raise the water level above the surface of the mat. Once all plants have absorbed water, the excess water is allowed to drain off by lowering one edge or folding down a flap of the mat.

A manufactured mat product is available that integrates tubing within the fibrous portion between two plastic layers. The tubing is connected to the laterals of an irrigation system and facilitates the distribution of water throughout the mat.

9.4.3.2 Troughs

Irrigation troughs consist of flat-bottomed metal or plastic channels, mounted at a slight slope. Pots are set into such channels and may be filled with substrate (Fig. 9.5B). Irrigation solution is applied at the elevated end and runs slowly down the length of the trough. Warping or twisting of the trough will create poor distribution causing water to miss pots. While it may be tempting to line the trough with matting material, this will essentially convert the trough to a capillary mat system. The configuration of the pots within the trough, particularly the location of holes in the base of the pot, is very important since water has to migrate to these holes. Water which reaches these holes is drawn into the pots via capillary action. During an irrigation event, water runs in the channel continuously and much of it may miss pots entirely. Unless this water is recycled, there is a potential to produce a large amount of wasted water.

9.4.3.3 Flooded trays or benches

Flooded trays or benches (also called “ebb-and-flood,” “ebb-and-flow,” or “flood-flow” systems) are designed to be flooded with irrigation solution for 5–10 minutes, submerging the bottom of the pots. Each of the pots should be submerged to the same depth and for the same length of time to achieve maximum uniformity. Modern trays are equipped with grooves (Fig. 9.5C) to facilitate drainage and to move water rapidly to all parts of the tray at the start of an irrigation event as well as promoting drainage from the pots at the end of the irrigation.

A typical irrigation cycle starts by flooding the bench or tray. The duration of this phase is dependent on the rate at which water comes onto the tray and the size of the tray. Care should be taken that water coming onto the tray is not flowing so fast as to push pots over. The water level in the tray is raised so that the bottom 2–5 cm of each pot is submerged. The optimal level would be one where the holes in base of all of the pots are submerged simultaneously at the start and the water level is maintained to keep the pots in water throughout the flooding period. The depth to which the water must be raised is also dictated by the hydraulic conductivity of the substrate. The lower the UHC, the deeper the water must be and the longer the duration that the plant must be allowed to stand in this water. The duration needs to be long enough to allow water to be drawn up into the top portions of the root zone by capillary action. It should be noted that during this time a significant portion of the root zone is above container capacity. As the tray is drained, the excess water in the pots also leaves the pots. This can have significant consequences if some pots are infested with water-borne pathogens since these pathogens are then distributed to the table or tray

and subsequently delivered to all nearby pots with the next irrigation. The duration of this flooded state should be minimized to avoid predisposing the roots to disease infection.

Usually there will be a lot of water that needs to be drained from such a system. This water can be either recycled in a separate recycling system or pumped directly onto the next bench. There is, however, a risk that pathogens from diseased plants, or from plant material lying on the bench, may be transmitted from bench to bench if the irrigation solution is not disinfected between applications. This also means that growers must typically be more diligent about rogueing-infected plants and proper sanitation of benches.

9.4.3.4 Flooded floors

Subirrigation can be carried out on an even larger scale by lining large sections of the floor with concrete or plastic, contoured so as to allow all water to drain into a central drain (Fig. 9.5D). Water supply lines and drainage pipe are incorporated directly into the concrete. Low walls (curbing or rubber bumpers) are used to section the system into functional circuits. Operation is the same as with flooded tables.

Proper design and construction is critical since even small improper variations in grade can have extensive consequences that are virtually impossible to repair. Flooded floors are sloped to allow drainage. Incorporation of sloped channels into a perfectly level pad (i.e., akin to trays) would be prohibitively expensive and/or weaken the concrete pad to where it could easily crack. Also, since the surface is shaped by human hands, the concrete is likely to have some areas where puddles will form. Drainage grooves need to be cut to drain these. Otherwise the plants in these areas will be at saturation much longer than others and the continual moisture can cause disease problems.

Another potential problem is that on most soils it is very difficult to pour a large pad of concrete without some cracking occurring in the concrete sometime after it has been put into use. It is extremely important during the design phase to verify how large a pad can be poured so that the floor will not crack later. Once a crack has occurred, allowing water to seep through it will exacerbate the situation. Driving forklifts or other heavy equipment onto the pad may also contribute to cracking. Obviously, flooded floors should be designed by an experienced engineer and installed by an experienced concrete contractor, and only after the soil has been tested and properly prepared.

Sanitation on flooded floors is very important since workers can carry pathogens on the soles of their shoes directly into the irrigation system. Once disease symptoms appear in one area, it is likely that workers walking on the floor will already have transmitted the problem throughout the nursery unless proper sanitation measures have been utilized.

9.5 IRRIGATION SYSTEM CONTROL METHODS

For all irrigation systems, decisions must be made when to irrigate and for how long. With regard to irrigation, we refer to irrigation “events” and “schemes.” An irrigation event is a single application of water or nutrient solution. With soilless growing systems, an irrigation event lasts from a few seconds to several minutes (rarely longer than 1 hour unless the irrigation scheme is

continuous). Irrigation schemes are management strategies developed to attain specific crop production goals utilizing various delivery and monitoring methods and refer to the overall plan of managing the irrigation water for the duration of the crop.

9.5.1 OCCASIONAL IRRIGATION

Conventional irrigation schemes can best be termed “occasional” irrigation. Many times in the horticulture literature, irrigation is described as being on an “as needed” basis. This vague terminology refers to the use of occasional irrigation such that the root zone does not dry out to an extent that might result in “water stress.” In this approach, the objective is to deliver water at intervals dictated by the plants’ removal of water from the root zone, but applications usually occur based on the timing of the observations of the irrigation manager and secondarily on the plant’s need.

In general, this involves an interval of time between irrigation events that ranges from a few hours to several days. This irrigation scheme generally involves bringing the root zone to full capacity of water (i.e., the “water-holding capacity”) and then providing additional water that will result in a percentage of the applied water leaching from the root zone.

9.5.2 PULSE IRRIGATION

Typically, overhead irrigation systems apply water at rates greater than can be absorbed and held by the substrate. This can generate substantial amounts of runoff. “Pulse irrigation” is an approach that utilizes several frequent irrigations with shorter durations. The objective of “pulse” irrigation is to match the application rate of the irrigation system with the absorption rates of the substrate (Zur, 1976). By applying the water in cycles of a short duration irrigation followed by a “rest” period, the substrate is allowed to absorb water that is applied before additional water is added (Lamack and Niemiera, 1993). The duration of such a rest period is dictated by the hydraulic conductivity of the substrate as this determines how well water diffuses laterally in the root zone.

To implement this method, irrigations that are normally applied in a single relatively long duration application are divided into several shorter intervals or “pulses.” For example, an irrigation duration of 40 minutes may be divided into four individual 10 minute intervals separated by a period of 50 minutes. Pulsing may result in a shorter total irrigation duration that may translate into water savings and reductions in runoff. Using the previous example, the pulses might be reduced to 8 minutes and still provide adequate water to the plants (Levin et al., 1979; Fare et al., 1994). On the other hand, dividing water applications over time lengthens the periods of high water availability. This usually results not only with increased transpiration rate but also with increased WUE of the marketable part of the crop (Fernández et al., 2005; Katsoulas et al., 2006).

With pulse irrigation, there is a danger of pockets within the root zone drying out. For instance, if the roots within the root zone are much denser in a particular layer within the root zone, then this layer will dry out more rapidly. If the pulse is too short to replenish this zone completely, then over time this zone will become hydrophobic and prevent water movement into this substrate. Furthermore, this barrier to flow may prevent water from flowing to other areas

of the root zone, which then also dry out. Generally, these pockets are not visible to the grower so that the grower will not recognize this until the killed roots allow entry of pathogens into the plants, causing plant diseases to appear. It is generally best to avoid pulse irrigation unless (1) the roots are distributed uniformly throughout the root zone and (2) the UHC of the substrate is such that water still moves rapidly within the root zone when it is somewhat dry. Hence, pulsing is more effective when the substrate is not allowed to dry to a state at which hydraulic conductance is limiting.

9.5.3 HIGH-FREQUENCY IRRIGATION

In some types of soilless production systems, irrigation events may occur several times per day and perhaps per hour and each or at least some of the irrigations results in bringing the root zone to its full water-holding capacity. In such situations, the system is dependent on the bulk flow of water, nutrients, and oxygen. This approach can only be used in systems where the substrate allows complete drainage of all excess water between irrigation events. The advantages of high frequency irrigation for maintaining moisture and nutrients at low fluctuation level close to the optimal status are described in [Chapter 6](#), Nutrition of Substrate-Grown Plants.

9.5.4 CONTINUOUS IRRIGATION

A number of soilless production systems are best referred to as “continuously irrigated” systems. In such systems there is no irrigation per se, but rather the root zone is managed in a manner that the roots are constantly submerged and bulk flow is imposed on a continual basis either by moving the water with pumps or by bubbling air into the water to induce water currents. This irrigation approach is appropriate in systems where the root zone consists of substrates that have little or no water-holding capacity, or possibly have no substrate at all. NFT and other “water culture” systems are of this type. Another system of this type is aeroponics where the irrigation solution is continually sprayed onto the roots ([Zobel et al., 1976](#); [Feigin et al., 1984](#); [Soffer, 1986](#); [Soffer et al., 1991](#)).

Continuous irrigation can also be applied in the form of microirrigation. In this case, water is applied at a rate that coincides with the transpiration, so roots are never submerged in water. In this case an additional amount of water is applied periodically so as to discharge accumulated salts.

9.6 IRRIGATION DECISIONS

Irrigation control involves two facets: making the decision to irrigate and implementing the decision through a particular irrigation control approach. The main irrigation decisions are (1) when to initiate the irrigation event (how frequently to open a valve) and (2) how much irrigation solution to deliver during the irrigation event (duration of irrigation events).

9.6.1 IRRIGATION FREQUENCY

Plants must never be subjected to conditions that cause stress and reduce plant growth if the objective is to optimize production. These stresses can be by both the lack and the abundance of water. Plants that have run out of readily available water may be delayed and suffer death of roots or even the entire plant. So, irrigation needs to occur before the water content of the substrate is reduced to a growth-limiting condition, but not so often so as to maintain water-logged and anaerobic conditions that favor root diseases.

The management allowable depletion (MAD) amount can be used to trigger irrigation. MAD is the maximum amount of water than can be removed from the soil, expressed as a fraction of total plant available water (PAW), before plant stress occurs (Snyder, 2014) and is used to determine when irrigation is needed to replace the volume of water removed from the soil and any additional amounts for leaching purposes. The substrate water depletion rate is linked to the water use of the plant and evaporation from the substrate surface. So the environmental conditions around the plant directly affect plant water use. Irrigations that substantially increase the humidity should be done so that the relative humidity does not stay near 100% for more than a few hours. In situations where the irrigation is likely to be followed by a period where there is no ventilation (e.g., late afternoon, evening, or cloudy conditions), relative humidity can rise to levels where condensation will occur. Since the leaves and greenhouse glazing are generally slightly cooler than the air in the greenhouse, those are the surfaces where condensation will occur first. For many plant species, liquid water on the foliage should not be allowed to remain for any extended period of time since this is likely to allow disease organisms to proliferate, infect the plants, resulting in disease problems.

Another consideration is the availability of water pressure in the supply lines. In most production systems, the water supply capacity is limited so that not all irrigation circuits can be irrigated at the same time. If all plants need to be irrigated during a short period of time (as may happen on hot dry summer days), it may be necessary to use short pulses of irrigation during the day to minimize water stress effects, followed by irrigation events later in the day that are more complete.

In automated irrigation control systems, two parameters need to be explicitly set to prevent the system from irrigating too frequently or too infrequently:

Maximum duration between irrigation events—This parameter identifies the longest tolerable interval between irrigations (in hours or days). It is set in an irrigation control program as a safety measure to prevent damage to plants due to failure of other control decisions or equipment. System failures can happen in various ways: the solenoid valve leaks and allows enough water to get to the pot that is being monitored (the representative plant) so that it never dries out. This flow rate would be inadequate for the entire crop so many plants would die without use of this parameter. Failure can also occur if water gets into the electronics associated with sensors causing faulty measurements.

Minimum duration between irrigation events—This parameter sets the shortest allowed interval between irrigations (in minutes, hours, or days). It is needed to prevent crop damage by overwatering in case the sensor fails so that the controller continually sees a signal that represents a dry condition that does not change. This could occur if the emitter to the representative plant has been inadvertently pulled out of the pot. It could also be caused by a signal problem with the sensor or controller or a break in the cable between them.

9.6.2 DURATION OF IRRIGATION EVENT

The duration of an irrigation event is a factor of the uniformity of the crop, the uniformity and design of the water distribution system, the water-holding capacity of the substrate, and the expected level of substrate water depletion at the start of an irrigation event. Typically within a uniform crop, plants that are exposed to more sunlight or air movement, or to low vapor pressure deficit (VPD), deplete more water. In crops that are less uniform, the plants with the greatest total leaf area will use water at a faster rate than other plants. Also, the plants that are closest to the irrigation valve may receive the highest rate of water application if the irrigation system is improperly designed and significant pressure differences exist among the sprinklers or emitters. The design of the irrigation system and the source water pressure and capacity will influence the length of time between the opening of the irrigation valve and the time when the last plant to get water begins to receive it. In large irrigation circuits, this can take tens of seconds or longer. Typically, the plants furthest from the irrigation valve will see the slowest water application. Plants in this area that are large and have plenty of light and air movement will require the longest time before the substrate reaches container capacity.

The duration of an irrigation event is mainly dictated by the amount of water that needs to be supplied to the plant to rewet the substrate, plus any amount of leachate that is needed. The percentage of leachate in relation to the total amount of water applied is called the “leaching fraction.” In conventional irrigation approaches, the leaching fraction can range from 0% to 20% under well-controlled conditions. In some instances, greater percentages are necessary in order to drain excessive accumulated salts. Leaching fraction can be measured in real-time using, for example, an electronic tipping bucket located beneath the drainage holes of a representative plant.

The leaching fraction is typically a function of the degree to which salt build-up is likely to be a problem. If the source water is somewhat saline and fertilizer salts are added to this water, then the salinity may well rise to a level where it is problematic to the plants. If this is the case, then the amount of leaching should be relatively high. Leaching will be desirable in cases where the water used for irrigation is low in quality as it will ensure that salts that may accumulate are moved out of the root zone. If the water is of high quality and fertigation results in little build-up of salts in the root zone, then the leachate volume can be minimized. However, if the quality of the irrigation water is marginal, then leaching fraction will need to be greater. The leaching fraction can be calculated by dividing the EC of the irrigation water by the maximum tolerable EC of the leachate (Marshall et al., 1996). That is, if the EC of the irrigation water is one-fifth of the EC of the maximum desired EC of the leachate, then the leaching fraction should be one-fifth of the volume applied (Table 9.1).

The purpose of an irrigation event is to replace substrate water that has been removed by the plant through transpiration and evaporated from the substrate surface thus to provide fresh water and nutrients to the plants. So the duration of an irrigation event needs to be long enough to replenish the used water and also provide a leaching fraction, if necessary. The amount of water to be replaced, then, is dependent on the total amount of water depleted from the soil. This can be measured using various methods or estimated knowing the total PAW of the substrate and the MAD. The product of these two factors is an estimate of the volume of water to be replaced through irrigation. The volume of water needed for leaching is necessary to determine the total volume of water to be provided.

Table 9.1 Recommended Leaching Fractions (Evans, 2014)

EC Applied (dS/m)	EC Leached (dS/m)			
	3	6	9	12
0.50	0.17	0.08	0.06	0.04
0.75	0.26	0.12	0.09	0.06
1.00	0.33	0.17	0.11	0.08
1.25	0.43	0.20	0.15	0.10
1.50	0.50	0.25	0.17	0.12
1.745	0.60	0.28	0.21	0.14
2.00	0.67	0.33	0.22	0.17
2.25	—	0.36	0.27	0.18
2.50	—	0.42	0.28	0.21
3.00	—	0.50	0.33	0.25
5.00	—	—	0.56	0.42

The volume of irrigation required to obtain the desired leaching fraction EC based on the EC of the irrigation water. Higher irrigation water EC and lower plant tolerance to salinity requires greater leaching fractions. EC, Electrical conductivity.

Knowing the volume of water to be provided, information of the irrigation systems is needed. The application rate (aka, precipitation rate), usually expressed as cm/min or in/min, is needed to calculate the length of time the water should be applied, through a perfectly performing irrigation system. Since there are no perfect systems, the DU is necessary to adjust the run time to accommodate the driest areas in the irrigation zone. To irrigate plants grown in containers, the diameter and volume of the containers are also needed. The irrigation duration can be calculated as shown in the example of a crop grown in a no. 1 size container of 2.7 L volume:

Substrate PAW = 0.52

MAD = 0.4

Leaching fraction (L_f) = 0.2

Container opening (d) = 15 cm

Container volume (V_C) = 2.7 L

Application rate (A_r) = 1.8 cm/h

Distribution uniformity (DU) = 0.84

Amount of water to apply (V_w) in liters = $V_C \times \text{PAW} \times \text{MAD} \times (1 + L_f) = 0.67 \text{ L}$

Application rate per container (A_C) in L/hour = $(d/2)^2 \times \pi \times A_r / 1000 = 0.32 \text{ L/hour}$

Run time (irrigation duration, hours) = $V_w / (A_C \times \text{DU}) = 2.5 \text{ hours} = \mathbf{150 \text{ minutes}}$

The plants that are the last to receive water in the irrigation zone are typically also getting the lowest application rate due to pressure losses or inadequate system performance; thus they are also the last to reach container capacity. It is these plants that should be used to determine the desired leaching fraction. By measuring the leachate volume and timing the duration required to achieve the minimal amount of leaching, it is possible to identify the length of time to add to the irrigation duration to achieve a minimum leaching fraction. It should be noted that the volume of the actual

leaching fraction for each plant will vary due to factors such as the nonuniformity of the delivery system and various differences among the plants.

9.7 APPROACHES TO MAKING IRRIGATION DECISIONS

There are several approaches to making irrigation decisions: (1) look-and-feel, (2) gravimetric, (3) timer-based, (4) sensor-based, and (5) model-based methods. Some of these methods involve explicit considerations of the plants while others may not and are based on approximations of information about the plants.

9.7.1 “LOOK-AND-FEEL” METHOD

The look-and-feel method involves close inspection of the plant, paying particular attention to the color of the substrate surface, the occurrence of flagging foliage, or slight color changes in the foliage which occur in some crops just prior to wilting. In containerized production, this might be accompanied with lifting up one or two pots to gauge their weight or sticking a finger into the growing medium to attempt to feel whether a significant amount of water has been removed from the pot. Doing this accurately requires a substantial amount of experience of relating these “human sensor readings” to moisture content and moisture tension and is very difficult to do reliably and consistently.

While this approach can be fine-tuned to be superior to a time-based approach, there are numerous problems. One is that human touch is not a particularly good measure of moisture content. As long as there is any amount of readily available water, moisture can be felt by the skin surface. The absence of moisture is generally felt only at fairly dry conditions that are suboptimal for plant growth. In general, moisture levels much greater than those coinciding with water stress should be maintained for optimal growth. Thus, if irrigation is based solely on the appearance of the plant or the feel of the soil, the plants will be consistently subject to stress. As the plants adapt to this stress, it will become less noticeable while still being suboptimal for growth.

Judging the weight of pots by lifting a few representative pots is a better approach than looking at or feeling the substrate. However, it is subject to problems if the grower is not aware of the moisture release characteristics of the medium and has not learned how various levels of water content relate to how heavy the pot feels. Also, when doing this, the grower has to somehow subtract the weight of the plant and substrate. The fresh weight of the plant also varies dramatically with its water status and as it grows. (A quick check can be done by irrigating a pot to container capacity. Then the weight of that pot can be compared to other pots. The difference would be the amount of water that had been retained in the irrigated pot.)

Many practitioners (and some scientists) consider the concept of “water stress” to be an important consideration in irrigation management. However, it should be noted that this term has no specific, quantitative definition and thus means different things to different persons. It is purely a qualitative term. Many persons feel that “water stress” represents a condition in the plants that results in the appearance of observable symptoms (e.g., wilting). However, it has been shown that such conditions always lead to reductions in plant growth. Even cacti, which are known to be

extremely resistant to water stress, only grow at appreciable rates if ample water is present. In fact, the reduction in productivity can be observed even under conditions that might not qualify as “water stress.” Thus the term “water stress” is relatively useless in irrigation management, except to say that if water stress is occurring, then there has been a failure in either the irrigation system or the irrigation scheme. In general, “water stress” should not be the deciding factor for starting an irrigation event; doing so assures that the irrigation event will be too late.

9.7.2 GRAVIMETRIC METHOD

It is possible to use weighing as a tool in irrigation scheduling. Generally, weight changes due to plant growth are much smaller and less rapid than weight changes due to loss of water from the pot. Recording the weight of the container after the previous irrigation can enable the irrigation control system to determine the amount of water lost since that irrigation. Thus, it is possible to weigh the pot day to day and determine the cumulative amount of water that has been lost to ET. Since the container volume and water-holding capacity of media are known, it is possible to approximately track water use. There are, however, possibilities for inaccuracies. For example, as the roots fill in the pot, they take over some of the space which would otherwise be occupied by water. Thus, the volume of readily available water decreases as the plant grows. In root-bound plants (where the root mass has filled most of the root zone), this loss can be substantial.

Devices specifically designed for use with irrigation control are available. These are, however, not commonly used as they require frequent adjustments and tend to be inaccurate for indicating how much readily available water remains. Most growers find the labor needed for this approach to not be cost-effective. Still, with proper care, weighing the plant can be a very good research tool (Graaf et al., 2004; Wallach and Raviv, 2005).

9.7.3 TIME-BASED METHOD

The simplest and least costly approach to automation of irrigation is to use timers to turn irrigation systems on and off. This approach is feasible since soilless media, although having high water-holding capacity, typically have high infiltration rates and high porosity, allowing excess water to drain away readily. Under these circumstances it is possible to apply irrigation solution on a fixed schedule, so watering always occurs sooner and for longer than is presumably needed by the crop. While this wastes some irrigation solution, many growers currently consider this a small price to pay for the insurance that the plants will always be watered and fertilized adequately. This price, however, is likely to rise steeply as economics, environmental considerations, government regulations, and public pressure force growers to eliminate this waste. The combination of time-based irrigation and a closed irrigation system is, therefore, a natural solution.

A purely time-based approach can also lead to suboptimal moisture conditions in the root zone if the substrate cannot provide adequate oxygen for the roots under saturated conditions. This can easily happen especially if the substrate depth is shallow and the plants use water relatively slowly.

Timed irrigation is easily done with the use of electrical or electronic timers specifically designed for controlling irrigation valves. Adjustments to irrigation timing should be made to reflect changes in climate or season or other factors that would affect plant water use. Timers can

also be used where the decision to irrigate is made by other methods (e.g., look-and-feel or sensors) and a manually started timer is then used to control the duration of the irrigation.

9.7.4 SENSOR-BASED METHODS

Sensors can be used to register some facets of the environment surrounding the crop and to use the resulting signal as part of irrigation decisions. “Sensor-based” irrigation control involves a sensor to directly measure some aspect of the moisture content of the root zone or water demand of the plant.

It is possible to use sensors to measure the moisture content (or related characteristics) in the root zone and then use this information to control moisture levels. A number of different sensor types exist. Currently the most feasible sensor for this is the tensiometer, due to its relatively low cost among the sensors which are not affected by salts (fertilizer) in the irrigation solution. Generally, the sensors that are used in this type of irrigation attempt to measure moisture content of the substrate of the root zone. It is very important that there is an understanding of the measurement a sensor is providing since there are two measurements related to soil moisture and they are very different.

9.7.4.1 *Tensiometers*

A tensiometer is a device which measures moisture tension or matric potential. It consists of a tube fitted with a porous ceramic tip on one end and a pressure/suction gauge on the other end. In automated systems, the gauge is supplemented with or replaced by a transducer to convert the tension (suction) to an electrical signal that can be sensed by a computer or electronic controller. The tube is filled with water and the device is sealed and inserted in the substrate so the ceramic tip is in the middle of the root zone. As the substrate dries, it will “pull” water out of the tensiometer and create or change the vacuum inside of the tube. The level of the vacuum is indicated on the gauge or converted to an electrical signal by the transducer. Tensiometers may be installed oriented vertically, horizontally, or any other angle. Considerations in determining the angle include the effect of the water column within the tensiometer on the pressure measurement, the ability of applied water to channel along the tensiometer, and the size of the container or pot. Tensiometers with “high-flow” ceramic tips respond rapidly to changes in moisture and are thus better suited for use in soilless substrates than tensiometers designed for use in soils.

The basic operation is to have one tensiometer, or other sensor, coincide with each irrigation valve. This sensor needs to be in the root zone of a plant in a location that is representative of the whole crop. The irrigation system should be designed and operated to apply irrigation water as uniformly as feasible. One approach to using tensiometers in irrigation control is to use the signal to merely override irrigation systems (using the rain detection cut-out circuit, for example) to prevent timer-controlled solenoid valves from coming on unless a specific level of dryness has been reached. Another approach involves two set-points: (1) a high-tension set-point, representing the level of dryness (e.g., where 80% of the readily available water has been removed) when irrigation is initiated and (2) a low-tension set-point representing the set-point at which an ongoing irrigation event will stop. These set-points are used as follows.

As the tension in the root zone rises with water depletion, the tensiometer is monitored. Once the high-tension set-point is reached, the irrigation is initiated (in large-scale operations, this may

mean that this particular irrigation circuit is scheduled for irrigation by placing it in an irrigation queue). Once the irrigation is in progress, the tensions will drop (usually over seconds or minutes) as water is applied. Hopefully the hydraulic conductivity and application method adequately disperses water laterally in the root zone. The rate of water application should be slow enough to allow this to occur and also to allow the tensiometer to follow the changes in moisture condition. Applications may be applied in pulses to match substrate infiltration and water application rates. When the low-tension set-point is reached, the irrigation is stopped (or allowed to continue for a specific length of time to obtain the desired leaching fraction).

Kiehl et al. (1992) investigated the use of tensiometers in UC mix and found that in potted plant production in this substrate, much of the readily available water in the substrate is exhausted by the time the dry-down has reached 7 kPa. At this point, extraction of a little more water sends the tension over 10 kPa. They observed that at tensions over 10 kPa, a plant that is accustomed to fairly moist conditions will start to wilt. Unless irrigation occurs at that point, the plant will be exposed to damaging conditions as the substrate dries out further. Thus it is generally wise to irrigate when the tension is around 5 kPa; after that the urgency increases with increasing tensions. They also investigated a number of scenarios involving lower levels of high tension set-points and found that an irrigation regime that involves a significant draw-down to about 5 kPa is best when working with UC mix. Raviv and Blom (2001) found that UC mix, despite having relatively good aeration, does subject the plants to oxygen deficiency if the tension is not allowed to rise so as to deplete some of the water in the root zone.

Kiehl et al. (1992) also investigated the suitability of various low-tension set-points. They found that a low-tension set-point fairly close to the high-tension set-point (attempting to maintain relatively constant nonsaturating moisture conditions at all times) was not conducive to optimal growth, even if such an approach allowed them to nearly eliminate all runoff from the crop. They also found that a low leaching percentage (close to 0%) could be achieved with much lower low-tension set-points of 0.5–1 kPa. While low- and high-tension set-points of 1 and 5 kPa, respectively, appear to be suitable for UC mix or other similar potting mixes, other substrates (e.g., peat, coir, stone wool, and perlite) may require different set-points. It has not been shown whether tensiometer-based irrigation control is feasible with these systems.

The system parameters described earlier in the chapter for irrigation control are all pertinent when using tensiometers to control irrigation. As described above, high- and low-tension set-points can be used to trigger irrigation and to determine when to stop an irrigation. As described earlier, various scheduling overrides based on time or other factors should be included as safety features. Such systems can also include alarms that alert growers that something is potentially wrong and that human intervention is necessary to resolve a conflict. For example, with tensiometers in soil-less substrates, a high-tension alarm should be implemented so that the operator can be made aware of tensions that have risen to levels that represent dangerous conditions. This can happen if the irrigation system fails (due to, e.g., defective pump, solenoid valve, or emitter in representative pot). It could also signal defective signal wiring (depending on design of signal wiring/processing). It might also be an indicator that the tensiometer has been accidentally removed from the root zone and is drying out.

There may be other reasons for overriding sensor-based irrigation control. The grower may have other motives for the crop that would suggest more or less frequent irrigation. For instance, if plants are relatively small and are not able to remove water from the entire root zone, then the

substrate solution immediately around the roots may become depleted of nutrients and oxygen without becoming depleted of water. Under these conditions, the grower should schedule extra irrigations so as to provide the plants with needed oxygen or fertilizers.

9.7.4.2 Electrical conductance sensors

In field soils, a common type of moisture sensor is one that has electrodes separated by some distance and actually measures the electrical conductance of the substrate. The main problem with these types of sensors is that the conductance of pure water is quite different from water that has salts dissolved in it. Commercial plant production implies explicitly that there will be fertilizer salts in the root zone. If the salinity is relatively constant, then these types of sensors may be feasible as part of sensor-based irrigation control. However, if there will be times where irrigation is with just clear water or when salinity of the root zone solution fluctuates differently from moisture status, then this type of sensor is not ideal for controlling irrigation.

In granular matrix sensors, the electrodes are embedded in a material such as ceramic, gypsum, or other porous material to minimize problems associated with salinity. This type of sensor, although the measurement is conductance, actually measures matric potential since the moisture level within the matrix changes with soil matric potential. In gypsum block sensors, the water within the block is saturated with Ca^{2+} and SO_4^{2-} ions making it less sensitive to changes in the salinity of the water in the substrate. However, this solubility means that the blocks slowly dissolve and have a finite life. Soil Moisture Equipment Corp. (Santa Barbara, CA), for example, rates the life of their gypsum block at 2–5 years.

The sensor needs to be in good contact with the substrate to facilitate water movement between the substrate and the gypsum as moisture conditions change. If the structure of the substrate is inconsistent, as moisture content changes, the contact with the substrate may be affected. Imbedding the gypsum blocks into a granular matrix (e.g., Watermark, Irrrometer Co., Riverside, CA) reduces this effect and improves precision, although not to the level of other sensors such as tensiometers.

Since the movement between the gypsum block and the substrate is usually very slow, so is the response of these types of sensors to changes in moisture conditions (Munoz-Carpena et al., 2005). For example, after a rapid change in moisture content due to irrigation, a gypsum block may take up to an hour to equilibrate to the change.

Implementation of this type of sensor (specifically granular matrix sensors) can be to sense an alarm or a high set point. For example, a system can be set to initiate an irrigation event upon reaching a high set-point (dry) condition and an application of a set duration would deliver water to the crop. Since these sensors are slow to respond to changes, a timed application is required.

Sensors that specifically measure substrate electrical conductance should not be used for irrigation management. Conductance is related to substrate water content, but these sensors do not measure moisture levels.

9.7.4.3 Dielectric capacitance (time domain reflectometry, frequency domain)

Methods to measure the moisture content of substrate by measuring dielectric capacitance or dielectric permittivity of substrates have recently become more prevalent since they are fast (fractions of seconds) and some methods also have the capability to measure salinity (EC) of the substrate (Dalton et al., 1984). TDR utilizes parallel metal rods (wave guides) to carry an electromagnetic

pulse into the substrate. An oscilloscope is used to analyze the electrical pulse as it enters the substrate, travels the length of the wave guides, and then returns. The shape of the waveform can be analyzed to determine the moisture content and salinity of the substrate. The time that it takes for the pulse to travel down the waveguides is related to water content and the dissipation of the signal is indicative of the EC of the substrate (Dalton and Poss, 1990).

TDR probes have either two or three parallel bare metal rods ranging in length from 10 and 50 cm, are relatively inexpensive, and are very robust due to their simplicity. The cost of the electronics to generate the electromagnetic pulse and to detect and analyze the signal, however, can be expensive for research grade equipment. Variations in sensors include coiling the waveguides to reduce the overall length of the probes (Nissen et al., 1998) and embedding in gypsum to measure matric potential (Persson et al., 2006).

A newer adaptation of TDR is TDT (time domain transmissometry) and, as indicated in the name, measures the time to transmit an electrical pulse along a closed loop in contrast to the pulse being reflected in a linear circuit in TDR. TDT has been demonstrated to effectively manage irrigation in potted flowering crops (Miralles-Crespo and van Iersel, 2011).

Frequency domain (FD) reflectometry is similar to TDR. However, as TDR measures changes in time characteristics of the electromagnetic pulse as it travels down the waveguides, FD analyzes changes in the frequency characteristics of the pulse (Lin, 2003). The advantage over TDR is the ability to use much shorter (less than 10 cm) waveguides and is much simpler in implementation as it does not require a complex set of electronics. The sensors can be connected directly to a data logger without the need for the pulse generator as required by a TDR probe. A sensor variant imbeds the waveguide in a circuit board (fiber-reinforced epoxy) material (EC-5, Decagon Devices, Inc., Pullman, WA) that makes them very easy to install and even more durable. As with TDR, these sensors require no regular maintenance.

Although there are advantages of TDR and FD such as sensor durability and rapid measurement, there are drawbacks. These measurement systems need to be calibrated for each substrate type as texture, structure, and organic matter content affect the accuracy of measurements (Morel and Michel, 2004; Baas and Straver, 2001). They are also affected by high ECs (~ 5 dS/m) as the electromagnetic pulse is unable to be reflected at the end of the waveguides as the conductance of the substrate increases.

The use of TDR (Murray et al., 2004) and FD systems can be similar to a system using tensiometers: they can sense when a dry set point is reached. Since they are very rapid, they can be used to monitor soil moisture changes as water is applied to terminate the irrigation event. A major difference from tensiometers is what is being measured. Tensiometers, as their name reflects, and granular matrix sensors measure matric tension or potential, the x -axis on a soil moisture curve (Fig. 9.1). Capacitance sensors, TDR, and FD measure volumetric water content, which would be the y -axis of the curve.

Various plant-based sensors, such as stem flow gauges, dendrometers, leaf thickness sensors, stem dielectric sensor, are used in research related to plant water status but are generally not used in commercial production settings.

9.7.4.4 Pros and cons of various types of sensor-based control

Tensiometers tend to be immune to issues regarding salinity. They respond purely to moisture tension and thus measure the force that the plants have to overcome to extract water from the root

zone. Tensiometers are also not sensitive to variations in soil texture. A reading of -5 kPa, for example, is the same for any soil texture. On the other hand, most other sensors measure volumetric water content (VWC) and a 15% VWC is a wet state in a sandy soil but is a very dry state for a loam. However, the tensiometer contains water within the tube and the porous ceramic tip needs to be wet to be airtight. Under high tension conditions the ceramic may dry to the point that the vacuum in the tube pulls air into it. The resulting air pocket in the tensiometer will affect the readings and the response time. Thus when high tensions are allowed to occur (above about 80 kPa depending on the characteristics of the ceramic tip), the sensor will require service to replace water lost from the tensiometer tube.

Electrical conductance sensors are generally less expensive than tensiometers and require less service while in use. However, since the sensor reading is affected by salt content even at normal fertilizer concentrations, such sensors are not suitable for use in situations where precision is desired. Granular matrix sensors and gypsum blocks are less sensitive to salinity since the electrodes are embedded in gypsum.

Dielectric capacitance sensors generally require considerably more electronics to be feasible as sensors in irrigation control. This means that they cost considerably more to use than the other sensors. However, with miniaturization and electronic customization, this is changing and these types of sensors are becoming more cost-effective in commercial production settings.

9.7.4.5 Sap flow methods

Sensors can be installed on plant to measure the flow of sap in the stems. Stem sap flows in both the upward, acropetal, and downward, basipetal directions and the volume of water translocated upward in the plant is usually much greater than the downward flow of plant sap; in normal day-time conditions there is a net upward flow. As water availability declines and water uptake rates follow, then this rate also declines. One way this rate can be measured is with stem flux sensors. These sensors consist of a heat source and two temperature sensors in close contact with the stem. The temperature sensors are both above and below and equidistant from the heat source. When a pulse of heat is generated and transfers to the stem, the liquids inside the stem are warmed and the temperature change is detected by the temperature sensors. The time difference is an indication of the net rate of liquid flow in the stem. For scientific studies, these generate some very interesting data. However for control of irrigation in production systems, these sensors are not practical due to their fragility, sensitivity to improper installation, and impacts of external factors including wetting of the stem and sensor by irrigation, for example.

9.8 SENSOR INTERFACES

With the development and use of sensors, there have been improvements in how the signals generated by the sensors are communicated to the instruments where these are turned into data to be monitored and used in making decisions. Previously, sensors needed to be wired to an interface. Improvements in radio and wireless network communications have eliminated the need for cables, reducing cost of data collection and facilitating deployment of many more sensors than was possible in the past. Wireless sensor networks have been demonstrated to be effective in managing

irrigation (Coates et al., 2012; Lea-Cox et al., 2013) by optimizing irrigation and improving WUE. New radio technology is also being used for individual sensors and control systems that are low power requiring and still have long range.

9.8.1 MODEL-BASED IRRIGATION

It is possible, through the use of mathematical models, to compute how much water a crop has used and to then irrigate so as to replace this water. One way to accomplish this is to set out a pan of water and track the water level in it as it evaporates. The evaporation of water from this pan is dependent on local weather which can be related to the ET of the crop and thus irrigation schemes can be used that are based on this measurement. Evaporation pans can be obtained for this specific use. A device has also been developed that uses this same principle to estimate ET without using an open pan. It protects that water uses a wicking process to facilitate evaporation and automates the measurement process so that a data signal is generated which can be used as part of input to a control device. These types of devices are frequently used in outdoors production of field crops and orchards.

Another similar approach is to use a reference plant with known water-use characteristics measure its ET rate by weighing it and tracking its water loss. This is basically identical with the gravimetric approach, described previously.

In all these cases, calculations have to be made to relate the measured evaporation rate to the ET of the crop and to also integrate this rate over time so as to determine a cumulative amount of water that has been lost from the root zone of the plants. Such calculations are made with mathematical models and methods based on such calculations are known as model-based irrigation control.

By relating the total volume of water removed from the root zone to the volume of the root zone, it is possible to calculate when a particular level of depletion has occurred. As discussed previously, the moisture release curve for the substrate can be used to identify the level of depletion that should be used to trigger or schedule an irrigation event or to elevate the priority of a previously scheduled event.

In general, all model-based irrigation control methods involve the use of inputs from sensors in the environment. Calculations are typically made by computers, but it is also feasible to program controllers to do this, although controllers are generally limited in the level of complexity that can be implemented.

Greenhouse environmental control computers are used to control many characteristics of the crop environment. Many schemes for irrigating crops have been devised to use data being recorded by these computers. As scientists develop models for water-use in crops, the various manufacturers of these computer systems implement these models in their systems, providing growers with a continually wider array of choices of model-based irrigation strategies. These methods attempt to relate the amount of water that has been lost by the plant and from the root zone to one or more environmental variable (light, temperature, VPD, and wind speed) so that irrigations can be scheduled to replace this lost water before stressful situations occur.

Various methods exist for approximating how much ET has occurred since the last irrigation. Implementation of these methods in irrigation control in soilless production is only feasible with computerized irrigation systems, since this estimation requires extensive computation using sensor

data. The most common method for estimating ET is the Penman–Monteith model (Penman, 1948; Monteith, 1965) although other methods can also be found in the scientific literature (Sharma, 1985). These approaches generally involved the abovementioned four environmental variables as well as a variable related to the size of the plant's foliage surface area. It should be noted that these models were developed for plants growing in an outdoor environment so that certain considerations are needed when dealing with protected cultivation. In virtually all cases, implementation of such models involves simplification so as to remove variables and parameters that typically are not easy to monitor or evaluate in a practical situation. Estimations of plant water use are available to assist in irrigation management from online computer programs such as the California Irrigation Management Information System (www.cimis.water.ca.gov). The worldwide standards for measuring reference ET are detailed in Allen et al. (1998) and are available on the web at <http://www.fao.org/docrep/X0490E/X0490E00.htm>.

One irrigation control method that focuses on a single environmental parameter sums light measurements at regular intervals. Once a specified light-sum level is reached, an irrigation is initiated. The light-sum-level set-points are refined by the individual grower for each crop. Another method estimates the VPD of the air and relates this to the rate of ET. VPD essentially measures how much water can be taken up by the air and, in conjunction with other variables, can be used to compute the cumulative amount of ET over time. However, when implemented without the use of light, temperature, wind speed, and leaf area, the calculations can only serve as a rough approximation requiring frequent set-point modifications or irrigation system interventions.

When using a particular method, it is important to understand how calculations are being made and which variables are being used in the computations. Using a method that makes ET calculations based on measurements of temperature, light, relative humidity, and wind speed, such as the Penman–Monteith model, has the greatest potential for success but still requires frequent grower intervention as the crop grows, is harvested, affected by insects and diseases, etc. As models for plant growth become available, this need for grower intervention may well decrease.

There are generally circumstances under which any of the irrigation systems and control strategies are appropriate. However, for any one crop and cropping system, there is generally one system which is best. The main variables determining which is best include labor and installation costs as well as the degree to which each is able to keep the crop at optimal moisture conditions.

9.9 FUTURE RESEARCH DIRECTIONS

In soilless production, growers typically pay much closer individualized attention to the plants than in field production. Current research into methods that are broadly termed “precision agriculture” focus on striving toward greater individualized attention to specific plants or groups of plants. As such, soilless crop production leads the way, with various implementations already in practice or close to adoption. For example, wireless irrigation control has already been developed so as to eliminate wiring associated with sensors and valves. Some companies are developing commercial products based on this concept. Further innovations are certain to come first to soilless production agriculture, striving for specific identification of the irrigation and fertility needs

of particular plants to groups of plants using sensor technologies and controlling delivery of water to each plant on an individualized basis. Further advances can be anticipated to also control other root zone variable in this way.

Another area where more research and development are needed is in the integration of more sensors into irrigation control to dynamically account for more variables that affect plant growth. As such, soilless production could benefit significantly from a control strategy to simultaneously optimize root zone temperature, oxygen and nutrient concentrations (van Iersel et al., 2013), salinity, pH, as well as water content. Currently the most advanced control strategies use a few environmental variables (estimated integrated VPD or light) and water content. Work has been done demonstrating crop-specific fertigation adjustments based on in-line conductance monitoring (Coates et al., 2014), but ultimately it may be possible to control fertigation based on all measurable root zone variables. It may, in fact, be possible to involve measurements of nonroot-zone variables. For example, one could even integrate irrigation system hydraulic data (e.g., flow rates or water pressure) so as to optimize flow characteristics and optimize water pressure at each emitter. Ultimately, it may also be feasible to involve sensor data for various plant processes (e.g., transpiration rate), as this information can be useful in diagnosing unexpected conditions and plant problems. For example, if the root zone sensors indicate that there is adequate water, but the transpiration rate and environmental data suggest that the plant is not using enough water to compensate for the atmospheric demand, then this will indicate particular conditions that may well require a change in the irrigation control or another intervention by the grower. It has been observed that optimizing irrigation and reducing the volume of water delivered to plants may also compromise plant quality since the amount of nutrients provided is also reduced if irrigation water also contains soluble fertilizers. More research is needed to understand how to compensate for the reduction in delivered water by increasing the concentration of nutrients in the nutrient solution.

Another area where research is needed in the future is in queuing irrigation events subject to sensor input or model calculations. For example, if irrigation for a particular irrigation circuit is planned when the representative plant reaches 5 kPa of moisture tension, then in most commercial nurseries, it is not feasible to turn on the corresponding irrigation valve if other plants are already being irrigated due to inadequate total system capacity. In general, this is handled through irrigation scheduling. On days when the plants consume a lot of water, it needs to be anticipated that many plants will be scheduled for irrigation. As the plants wait in the queue for their turn at irrigation, they continue to consume water so that the tension continues to rise. Clearly some optimization strategy and research are needed to identify how to best juggle the rising moisture tension, irrigation capacity, and risk assessment to minimize impacts on plant quality and potential financial losses in the event that some plants may end up seeing water stress. At the same time, each group of plants will have a different value on the market so that a grower may want to prioritize the irrigations, particularly if there is no way to give all the plants all the water that they need.

In the same way, modeling approaches can also be integrated with queuing methods to provide anticipatory control. Using all the root zone variables, as well as plant water needs and stress tolerances, environment measurements, weather forecasts, and irrigation system capabilities, it should be feasible to forecast when harmful conditions can be anticipated. With such an approach, models would be used to forecast when an irrigation will be needed for each group of plants by accounting

for the aforementioned queuing approach and using simulation modeling to calculate whether the currently selected high-tension set-point will result in a situation where not all plants can be irrigated as needed. If that situation occurs, then the control algorithm would either lower this set-point or reduce the irrigation duration so as to eliminate leaching, perhaps do partial irrigations throughout the day, and prioritize crop water needs, to avoid or at least minimize financial losses due to wilting of plants. Research is needed to develop such models, sensors, and integration strategies. Engineering research is needed to develop decision support tools and control algorithms that encompass these concepts.

Emitters could be developed to do more than just deliver water in a particular fashion. In addition to pressure regulation, which is already common, emitters could be developed to include sensor technologies so as to dynamically adjust flow through the emitter based on sensed root zone conditions. In particular, if a sensor could be developed to assess hydraulic conductivity of the substrate, then the emitter flow rate could be dynamically adjusted to match the ability of the substrate to carry irrigation water to the roots of the plants since substrate hydraulic conductance will increase as water content increases.

Perhaps the most important facet of future developments in irrigation in soilless production is to recognize that current approaches are primitive in comparison with what they could be. We know most of the variables that are important as part of optimal root zone moisture management. Yet the vast majority of commercial soilless plant production does not make use of these tools, relying instead on substrates that can be overirrigated without a significant penalty and plants that can tolerate the resulting conditions. Clearly there is a lot of room for advancement toward optimization, especially as the cost of electronics and sensors continues to decline.

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WATER AND NUTRIENT MANAGEMENT AND CROPS RESPONSE TO NUTRIENT SOLUTION RECYCLING IN SOILLESS GROWING SYSTEMS IN GREENHOUSES

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10.1 INTRODUCTION

The use of soilless culture is expanding worldwide, and in many countries with water scarcity or where there are problems with the emissions of nutrients and pesticides, the interest in (semi)closed systems is increasing (van Os, 2017). The main reasons for recycling greenhouse effluents are to avoid soil and underground water contamination and achieve high water and nutrients use efficiencies (WUE and NUE, respectively). Greenhouse crops grown in substrate systems with free drainage systems are irrigated with an overall surplus of water, to ensure sufficient irrigation to all plants in the greenhouse, avoid salts accumulation and maintain sufficient nutrients concentrations (Sonneveld and Voogt, 2009). As a result, leaching from soilless culture growing systems can be substantial (Voogt, 2004; Teunissen, 2005). Moreover, when the discharged N reaches underground water and dissolves uniformly in a 50 m deep water body $\times$ 2000 ha (twice the greenhouse area), it can increase the $\text{NO}_3\text{--N}$ concentration in that water volume by $1.5 \text{ g/m}^3/\text{year}$, and within 7 years surpass the EPA permitted $\text{NO}_3\text{--N}$ concentration in drinking water (10 g/m^3). In situ denitrification, ex situ ion exchange, and reverse osmosis (RO) methods to remove the dissolved nitrates are expensive.

The potential reductions in water and fertilizer use in closed loop irrigation systems (IRs) are significant at a national level but insufficient to persuade most growers to adopt them. The incentive for growers to reduce N-leaching by adaptive fertigation are low since the percentage of the costs of fertilizers from the total yearly costs are relatively low. The cost savings for water and irrigation are estimated at $<2\%$ of the total production cost for Dutch conditions (Vermeulen, 2016). It can be assumed that this will not be much different for growers in other circumstances. Charging growers for nitrate pollution clean-up might change their attitude toward recirculation; however,

this is hard to implement. Therefore governments have imposed regulations to enforce farmers and growers to take measures. In the European Union the Water Framework Directive (WFD) and the Nitrate Directive act as a framework for the member states to implement regulations to reduce nutrient—emissions to ground—and surface waters (WFD, 2000). In some EU countries, regulations have been implemented to reduce the emission of nutrients and plant protection products from greenhouses (Vermeulen et al., 2017). Like in the Netherlands and Belgium, closed systems are obligatory, and in Netherlands, even discharge is regulated. In other countries, such as in Germany and Scandinavia, growers are limited in the discharge of nutrient leaching from soilless systems. Recently, the Dutch authorities require complete purification of the water to be discharged to the surface water or the sewer (Van Ruijven et al., 2017).

Several factors constrain the potential for recycling of nutrient solution. The main constraint is the potential risk of rapid spreading of soil-borne pathogens from infected spots. Another important factor is related to the chemical water quality (salinity and ionic composition), determining volumes of greenhouse effluent discharge. In particular in arid zones, characterized by high evapotranspiration (ET) and salt concentration in fresh water (~ 1 dS/m), the main problem is the rise in osmotic potential (OP) and accumulation of Cl and Na^+ to toxic levels. In temperate climates (low ET and better water quality), the main concern is Na^+ accumulation in solution and resulting toxicity in certain crops (Voogt and Sonneveld, 1996; Baas and van der Berg, 1999; Sonneveld, 2000; Voogt and van Os, 2012). Removal of solutes and microbes from water with any of the available methods is very costly financially as considerable energy is required as well as expensive equipment. Moreover, with current of technologies, it is hardly possible to discriminate at the removal of solutes from the effluent water between the valuable and nonvaluable ions at reasonable costs.

Another factor is the accumulation of root exudates in the recycled solutions. This pertains mainly to carboxylate release (Marschner, 1995) as well as root debris, secreted sugars, and amino acids. Economic and environmental constraints limit the daily quantity of water and fertilizer that can be added via open IRs and the irrigation frequency. Consequently, salt concentrations at the root surface increase between successive irrigations, while nutrient concentrations decrease, thus reducing water and nutrients uptake efficiency. In well-planned closed loop IRs, crop development and yield are not limited by temporary stresses due to reduced water and nutrient potentials at the root surface. It is expected therefore that transition from open to closed loop irrigation results not only in safer environment and reduced input costs but may also enhance yield and produce quality (Bar-Yosef et al., 2003a,b, 2005). In general, for the majority of current closed systems, reuse of the drainage water results in higher NUE and WUE.

The objectives of this chapter are to (1) describe principles related to nutrient solution recycling; (2) evaluate the horticultural consequences of reduced solution emission from greenhouses and the impact of salinity build-up and ample and high-frequency fertigation on yield and quality of important greenhouse crops; and (3) describe and assess efforts to simulate salt accumulation and crop response to recirculation under different environmental conditions.

10.2 SYSTEM DESCRIPTION

There are various agricultural production systems which are described as involving nutrient solution recycling. In this chapter, it refers to soilless production systems where irrigation nutrient solution

that passes through the root zone is collected and reused in a closed loop. Sometimes these systems are called “closed systems” or “closed growing systems” (van Os, 2017). This can be confusing as this terminology is used also for closed aboveground environments. In the context of this chapter, we use these terms in relation to the “below ground” environment. This description applies for many systems widely used in commercial practice, for example, nutrient film technique (NFT), deep flow technique (DFT), aeroponics, ebb-flood systems, and systems with substrate with collection and reuse of the drainage water (see Chapter 12: Liquid Culture Hydroponic System Operation, and Chapter 13: Technical Equipment in Soilless Production Systems, for details on these systems). In this chapter, the systems mentioned first (NFT, DFT, aeroponics) are referred to as “hydroponics” as this is the term commonly used for this type of systems, where the root environment consist mainly of water (i.e., nutrient solution). We refer to systems where a growing medium is used to hold the roots, as “substrate” systems. The description is general and covers systems with either manual addition of fertilizers and water, or fully automated systems with nutrient solution mixing tanks and automatic electrical conductivity (EC) and pH control. The term “root environment” is the part of the system in where plants’ roots are located.

10.2.1 ESSENTIAL COMPONENTS

A closed loop IR is presented schematically in Fig. 10.1. The essential components of this system are (1) root environment (substrate, or nutrient solution) (S); (2) drainage system collecting substrate effluents by gravity flow and delivering it to drainage tank (D); (3) submerged pump (P) pumping solution from D to either the nutrient solution mixing tank (R) or to a storage tank for buffering drainage water (V_{out}); (4) booster (B) boosting solution from R into the IR defined by emitters geometry and discharge rate (Q_{in}); and (5) fresh water and fertilizer replenishment device (W, NPK). In some cases, a disinfection device is added to treat the recycling water. Disinfection methods aimed at root pathogens are discussed in Chapter 5, Root Zone Microbiology of Soilless Cropping Systems.

10.2.2 PROCESSES AND SYSTEM VARIABLES AND PARAMETERS

10.2.2.1 The system and its state variables

The closed loop IR can be described as a model consisting of various state variables which quantify the water and nutrient quantities in various parts of the system, as well as rates at which these flow within the system dynamically. Furthermore, the various elements dissolved in the water are quantified as both quantity and concentration variables; both are needed when applying the model to soilless culture since the system operator provides them in quantity, while the plant needs to be exposed to particular concentrations within min/max limits.

Some of these variables are illustrated in Fig. 10.1. The state variables are the instantaneous water volume in the operational tank and in substrate (V_{rw} and V_{sw} , respectively) and the element concentration in the tank and substrate solutions (C_r and C_g , respectively). The term “element” covers all the pertinent nutrient ions in the system. The variables related to the substrate are the volume (V_s) and the substrate volumetric water content (θ). Other variables are operational water tank volume (V_r), IR discharge rate (Q_n), and element concentrations in the replenishment solution (C_{ad}).

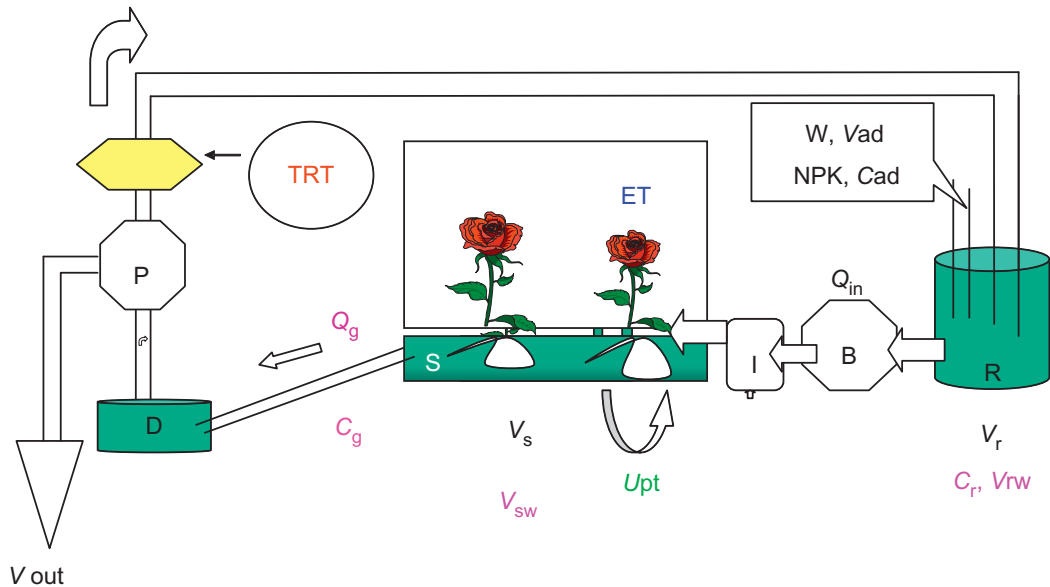


FIGURE 10.1

Schematic description of a closed loop irrigation system. System components: D—drainage tank; P—pump (normal flow—toward operational (also mixing) water reservoir R; when receiving command to dispose—toward V_{out}); S—substrate; B—booster; I—irrigation system, ET—evapotranspiration rate $L/m^2/h$ discharge rate Q_{in} (L/m^2 substrate/h); TRT—slow filtration (biofilter) effluent treatment (disinfection by chemical means is done at I); W, NPK—fresh water (quantity V_{ad}) and fertilizer (concentration C_{ad}) replenishment. Other terms are explained in the text.

Additional variables that define the system are the quantity of a specific nutrient in the solution refilling the supply tank (Q_{fi}), the concentration of a specific nutrient in the refill solution (C_{fi}), the volume of the water supplied (V_w), the specific ion concentration in the fresh water (water source) (C_{wi}). If there are more sources used, V_w and C_w will be V_{w1} , V_{w2} , etc., the concentration of a specific nutrient in the nutrient solution supplied (drip irrigation or replenishing solution) (S_{si}), the uptake concentration of a specific ion by the crop (further explained below) (C_{ui}), the specific ion concentration in the root environment (C_{gi}), and the ion concentration in the drainage (C_{di}).

The volume of the operational tank per unit greenhouse area V_r (L/m^2) is designed according to the capacity of the IR and flow rates. The flow rate capacity should be able to cover the peak crop water demand for 1 hour of a full grown crop in midsummer conditions, which is for example—as a rule of thumb—1.5 mm/h in the Netherlands and ~ 3 mm/h in Israel. To ensure unrestricted supply of water, the water volume of the operational tank (V_r) should at least contain 1 day of water consumption by the crop. This requirement stems from the small substrate volume and limited water buffering capacity in soilless systems relative to growth conditions in soil. As an example, a crop having $ET_{max} = 5 L/m^2$ ground/day and occupying $1000 m^{-2}$ greenhouse ground should have V_r of $5 m^3$. The added water volume reduces the rate of EC increase and allows application of

various sources of fertilizers via the replenishment solution without affecting considerably the nutrient composition and concentrations of the recirculated solution (see Sections 10.2.6 and 10.2.7).

The target nutrient concentration in recirculated solution is crop specific and also depends on some characteristics such as the substrate type, the fertigation management, and source water quality aspects (e.g., salinity level).

The permitted EC in recirculated solutions, which is a measure of the total ion concentration in the solution is called threshold EC (EC_{rec}). It is crop specific and varies with season, climate, and crop growth stage. It depends on the expected trade-off between yield and quality effect as well as savings in water and nutrients input and emission. If the EC or components making up the EC (including fertilizer ions as well as Na and Cl) accumulate beyond a certain threshold level, part of the effluent has to be discharged rather than transferred to R. The amount of discharge requires grower insight and use of proper measuring devices to balance the IR discharge rate (Q_{in}) with ET and uptake.

Substrate, ion, and crop parameters including medium's hydraulic properties, effective ion diffusion coefficients, and crop uptake and dry matter (DM) and N partitioning parameters, all affect the system processes.

10.2.2.2 Processes and rates

The main processes taking place in the system are ET (ET, $L/m^2/h$); water and ions uptake ($kg/m^2/h$); ions partitioning between substrate solution and solid phase; water and ions transport in the substrate; crop growth and DM production ($kg/m^2/h$), and root extension within the substrate.

It should be noted that solution volume in the drainage collection tank D (Fig. 10.1) is small in comparison with the total drainage volume and therefore will not be taken into account in following estimations. The container can be small since pump P is operated automatically when a particular water level is reached and drainage volume D is maintained at that level. Water replenishment, irrigation, drainage, and discharge volumes are continuously measured by water meters. ET is estimated by the following equation:

$$ET = W - V_{out} \quad (10.1)$$

where W is the replenishment water volume and V_{out} is discharge volume. Note that (1) irrigation (Irr) is the amount of water applied to plants and could be severalfold greater than W , and (2) ET also equals to Irr—Drainage, considering that the water uptake for growth is assumed being part of ET; hence, ET in this case stands for “total plant water uptake + evaporation” rather than just ET. Moreover, this estimate is prone to considerable measurement error when using commercial water meters (Bar-Yosef, 2003). In today's modern greenhouse horticulture, often accurate and precise drainage measuring devices are used, such as weighing gutters, drainage gauges, and the like, by which dynamic determination of ET is possible even per minute (Dieleman and Marcelis, 2006; Tsafaras and de Koning, 2017).

10.2.3 THE WATER AND SALT MASS BALANCE

For soilless culture, the quality of the water sources used is crucial, though for closed systems it is of even greater importance, since undesired ions will accumulate, necessitating frequent discharge of part of the recirculating solution. However, this is very unproductive as one of the reasons for

using closed systems is just to reduce water and nutrient losses; moreover, in some countries, discharge is regulated quite strictly. Then it thus inadequate to use water of inferior quality, though in many areas around the globe there are no alternatives. Since the suitability of water sources for closed system depends on the mass balances of salts and nutrients, it is a matter of input and output balances.

For optimal system operation, salt concentration in recirculated solution (C_g) and water reservoir (C_r) and their water volumes (V_g and V_r , respectively) must be known. For a simplified system (with no biofilter, with known ion uptake and ET rates, and without a drainage container), these variables can be estimated by solving the rate Eqs. (10.2)–(10.5) for a single ion, for example, Cl.

$$\frac{dC_r}{dt} = \frac{Q_{ad}C_{ad} - Q_{in}C_r + (Q_g - Q_{out})C_g}{V_r} \quad (10.2)$$

$$\frac{dC_g}{dt} = \frac{Q_{in}C_r - Q_gC_g - U_{pt}}{V_g} \quad (10.3)$$

$$\frac{dV_r}{dt} = Q_{ad} + Q_g - Q_{in} \quad (10.4)$$

$$\frac{dV_g}{dt} = Q_{in} - Q_g - ET \quad (10.5)$$

Here U_{pt} = Cl uptake rate by plants, g/m²/h; Q_{ad} = time-averaged rate of fresh water addition to system, L/m²/h; C_{ad} = Cl concentration in added water, ET, Q_{in} , the irrigation system discharge rate (L/m² ground); and Q_g , the drainage rate (L/m² ground), where $Q_{out} = Q_{ad} - ET$. When Q_{ad} , Q_{in} , Q_g , and ET are constant rates, the equations can be solved by ordinary differential equation solvers (e.g., Cutlip et al., 2005) and used to plan the recirculation system. The mathematical description of the system processes becomes more complex when temporal and spatial solute and water distributions in the substrate are included. This is taken into account in the comprehensive models described later in this review.

Another approach was proposed by Voogt and Sonneveld (1996), based on ionic concentrations mainly for the purpose of evaluation of the required water quality for closed systems and to assess the overall nutrient requirements in terms of nutrient solutions (see Section 10.3.9). They considered a steady state situation where the system is adjusted to the main input and output flows. In this approach, all ionic inputs and outputs are normalized by dividing them by the quantity of added water to the system (Q_{ad}), thus transforming them to concentration, either in mmol/L or mg/L. Transferring this approach to the system description from Fig. 10.1 leads to the following for each individual ion: C_{Wi} refers to the concentration of ions in the source water. Since there can be more than one source at a time or during a growth cycle it will be $C_{Wi,a}, \dots, C_{Wi,n}$ for the concentration in the various sources ($a \dots n$). C_{Ui} is the uptake concentration, being the ion uptake in mmol or mg divided by Q_{ad} :

$$C_{Ui} = \frac{U_{pt}}{Q_{ad}} \quad (10.6)$$

Note that for realistic evaluations, this should not be used for short time steps, but rather for time steps of days or longer. From this approach, it is understandable that if $C_{Wi} > C_{Ui}$ there will be accumulation resulting in an increase in C_{gi} . This description is valid for both nutrients, as well as for residual salts (Na and Cl). It even holds for the total concentration of ions expressed in terms

of the EC value (dS/m). There is a strong relationship between the EC and the total concentration of ions. True enough, each ion has its own specific contribution to the EC (McNeal et al., 1979). The contribution of a specific salt to the EC depends on the valence of the constituent ions, the dissociation constant, the ion activity, and ion pair formation. The temperature of the solution has a strong effect too, and therefore, the EC is expressed at a standardized temperature, usually 25°C. For mixed salt solutions McNeal et al. (1979) showed a linear segment method with which the contribution of various concentrations of different salts to the EC can be calculated. The method is suitable for concentrations up to 50 and 25 mmol/L for mono- and bivalent ions, respectively. The low intercepts given with these linear relationships point out that they are suitable for calculations until rather low concentrations. Another fundamental approach to relate EC to the ionic strength is given by Griffin and Jurinak (1973) and Kramer et al. (2002). For rough estimations of the EC of a mixed nutrient solution the formula given by Sonneveld et al. (1999a,b) can be used for mixed salt solutions.

$$EC \approx 0.095 \times C^+ + 0.19 \quad (10.7)$$

where EC is the electrical conductivity of the solution in dS/m and C^+ is the sum of charges of the cations in mmol/L.

However, for a precise calculation of the EC from the ion composition the already mentioned method is presented by McNeal et al. (1979).

The accumulation of an ion has repercussions on the physiology of the plant, both by itself and in synergy with other ions. A good example is the effect of K:Ca:Mg ratios as will be further discussed in Section 10.3.11. Here it is important to note that the accumulation results in a higher concentration in the root zone, and this may increase the uptake C_{Ui} . If C_{Wi} is not exceeding a certain threshold value of C_{Ui} , over time, an equilibrium between the two parameters will be established and the result will be a certain C_{gi} . Certainly these equilibria will be constrained by plant physiological boundaries, since too high C_{ui} as well as C_{gi} may cause antagonistic or even toxic effects, but most likely it will constrain yield or quality.

If $C_{Wi} < C_{Ui}$, depletion will occur. Logically, this can only occur if plants can actively take up the ion considered, which is the case for example for K^+ , NO_3^- , NH_4^+ , and orthophosphate ions ($H_2PO_4^-$, HPO_4^{2-}). For ions absorbed passively, like Ca^{2+} , BO_3^{3-} , that is, by mass flow with water uptake, depletion is not likely to occur. For Cl, this is crop specific and is depending on the situation. Usually Cl is high and accumulation is most likely to happen; however, there are clear examples where Cl can be depleted to rather low concentrations. If Cl is too low for sufficient supply by mass flow most plants will take up Cl actively (Xu et al., 2000) (see Section 10.3.9).

This system approach provides two important parameters which can be used for establishing crop-specific criteria for water quality (i.e., the maximum concentration of undesired ions such as Na and Cl) for closed systems as well as for making operational decisions for discharge of some of the drainage water. The first parameter is the maximum permitted concentration for an ion in the root environment ($C_{gi,max}$) and the second is the uptake rate of the same ion by the plant. The $C_{gi,max}$ is C_{gi} above which there will be damage to the crop, either by reduction in growth (yield) or quality depreciation. This parameter is analog to the threshold value from the Maas–Hofmann equation (Maas and Hoffmann, 1977), which relates however to total salt concentration. For the nutrient ions, these $C_{gi,max}$ values are naturally part of the overall composition of the nutrient solution and should be related to other parameters as will be discussed in Section 10.2.7. For Na and

Cl, these $C_{gi,max}$ values should be derived from experimental data. This has been carried out for a number of crops (Voogt and van Os, 2012). The other parameter is the specific uptake rate C_{ui} , at $C_{gi,max}$. Within the context of the issue of salinity, this can be considered also as “uptake capacity” of the crop. Voogt and van Os (2012) showed that for Na and Cl, there is a linear relationship between the average concentration in the root environment and the uptake concentration. From this follows that for each crop, there will be also a maximum intake concentration ($C_{ui,max}$), that corresponds with the $C_{gi,max}$. Hence, to avoid accumulation and salinity, for a closed systems, C_{wi} may on average not exceed $C_{gi,max}$, in other words, for Na and Cl, $C_{gi,max}$ is also the maximum acceptable concentration in water sources. In view of the fact that this uptake has no direct relation to a physiological crop demand, the uptake capacity can be used as indicator for the ability of crops to deal with certain concentrations in the input water sources without concessions on yield or quality in closed growing systems (Voogt and van Os, 2012). Mathematically, this is $C_{ui,max}$, corresponding with $C_{gi,max}$, that is, the uptake concentration at the maximum acceptable concentration in the root environment.

Solution culture (without substrate) or hydroponic systems constitute a special case of the aforementioned recirculation systems (Maloupa, 2002). The water volume in the growth container is maintained constant (e.g., via an external constant water head tank with feed solution) and solution is discharged from the growth container when $EC \geq EC_{thr}$ (e.g., the Ein Gedi system, Brooke, 1990). An interesting modification of deep flow (DFT) hydroponics is aero-hydroponics (Soffer and Burger, 1988; Feigin et al., 1984), where solution is sprayed on part of the root system in the container, thus improving oxygen availability. Another hydroponic system is NFT and semi-DFTs (see Chapter 13: Technical Equipment in Soilless Production Systems) where oxygen is supplied by diffusion from air and mixing occurs due to the relatively rapid solution flow rate.

10.2.4 SUBSTRATE CONSIDERATIONS

10.2.4.1 Physical properties

The substrate is defined in terms of its volume (V_s , length $\times$ width $\times$ height, L/m^2 ground), water, and nutrients concentrations (θ_s , L/L, and C_{si} , g/L), and hydraulic conductivity (K_h , m/h). To describe water transport in substrate, one needs to know its K_h versus θ and θ versus water matric potential (φ , m) functions. The theoretical basis of the hydraulic conductivity and water retention functions and how they should be used to predict temporal and spatial θ in substrates are reviewed in Chapter 3, Physical Characteristics of Soilless Media. Here more applicative aspects related to the system dynamics will be discussed. The first aspect is the substrate’s capillary fringe, or capillary rise, that can be derived from carefully constructed water retention curves (h_c , m , Caron et al., 2002, 2005; Hillel, 2003). To avoid anoxic conditions when the bottom of the substrate is submerged in water (e.g., when rate of water supply exceeds the drainage rate), the substrate’s height should equal h_c + [air buffer layer], the buffer layer being usually >5 cm. Data on the capillary fringe of several substrates is summarized in Table 10.1. For a given height, the substrate width is determined by the desired substrate volume per plant (see later) and number of plants per surface area.

The second property is substrate bulk density (ρ , kg/L) from which the medium total porosity (V_a , L/L) can be estimated ($V_a = 1 - \rho/\rho_r$, where ρ_r is the particle density). In soil, ρ_r is ~ 2.65 kg/L,

Table 10.1 Hydraulic Properties^a of Representative Substrates^b

	K_{hs}	$\theta_{95\%}$	$\theta_{0.5\text{ h}}$	$\theta_{12\text{ h}}$	$\Theta (-1\text{cb})$	$J(\theta_s)$	J_{fd}	ρ	h_c
Substrate	cm/h	v/v				cm/h		kg/L	cm
Perlite	>1000 ^c	0.90	0.66	0.43	0.30	450	10	0.10	6.0 ^d
Stone wool	276 ^e	0.90	0.85	—	0.42	3.5	0.75	0.12	3.5 ^d
Tuff	180 ^f	0.70	0.34	—	—	5.0	0.9	1.1	5
Pumice	—	0.38	0.21	0.19	—	360	10	0.75	—
Coir	200 ^g	0.92 ^d	—	—	0.80	—	—	0.16	4.7 ^d
Peat	360 ^h	0.89 ⁱ	—	—	—	—	—	0.10 ⁱ	3.6–5 ^h

^a K_{hs} = saturated hydraulic conductivity; $\theta_{95\%}$ = volumetric water content at 95% saturation; $\theta_{0.5\text{ h}}$ = θ after $\frac{1}{2}$ h drainage starting at saturation; $\Theta(-1\text{cb})$ = θ under 1 centibar suction; $J(\theta_s)$ = flux of drainage between 100% and 95% saturation; J_{fd} = mean J between θ_s and $\theta_{0.5\text{ h}}$; ρ = bulk density; h_c = capillary fringe.

^bIf source is absent the parameter was derived from unpublished data by the author. All drainage data were obtained in $0.5 \times 0.2 \times 0.2\text{ m}^3$ containers (see text).

^cMarfa and Orozco (1995), average of their coarse and fine perlite data.

^dCaron et al. (2002).

^eda Silva et al. (1995).

^fWallach and da Silva (1992).

^gRaviv et al. (2001).

^hCaron et al. (2005), 60% sphagnum peat, 30% composted pine bark, 10% coarse sand.

ⁱWeiss et al. (1998).

in organic substrates ρ_r is $\sim 1\text{ kg/L}$, and in perlite $\rho_r = 1.16\text{ kg/L}$ (perlite producers product leaflet). Available data of ρ is summarized in Table 10.1.

Attempts to simulate solute transport and water drainage in substrates by employing the convection–diffusion and Richard’s equations (see Hillel, 2003) were mostly unsuccessful (e.g., Wallach and Raviv, 2006). The main reason for this shortcoming is the strong dependency of hydraulic parameters on packing uniformity (Caron et al., 2002; Fermino and Kampf, 2005), variations in bulk density (Fonteno, 1993), and substrate matrix stability during crop production (Nelson et al., 2004; Wallach and Raviv, 2006). To avoid this uncertainty in simulation, Bar-Yosef et al. (2004c) described the water flow in two steps: (1) during irrigation by the one dimensional Darcy equation (or optionally by the plates theory) and (2) during drainage by using empirical drainage functions obtained in real growth containers to estimate outflow fluxes from substrate after irrigation termination. They employed a 50–60 cm long plexiglas container with adjustable side walls; drainage holes at the bottom (optionally attached to electric valves to accurately control initial water content) deliver the effluent into a container placed on electronic scale. They (1) saturated the dry substrate and recorded the volume of water consumed; (2) open the drainage holes, recorded effluent weight (W) as a function of time, and calculated drainage fluxes (J_{ef} , L/m^2 substrate/s); and (3) transform $W(t)$ to entire-substrate mean water content (θ_m) as a function of time, and then to J_{ef} as a function of θ_m . Example of functions obtained by the described tool with pumice is shown in Fig. 10.2.

It can be seen that after approximately $\frac{1}{2}$ hour the fast drainage stops and outflow during additional 10 hours comprised less than $\sim 2\%$ of the $\frac{1}{2}$ hour discharge (not presented). Drainage studies in stone wool, perlite, and tuff yielded similar results, and consequently $\theta_{0.5\text{ hour}}$ can be used as an indicator of

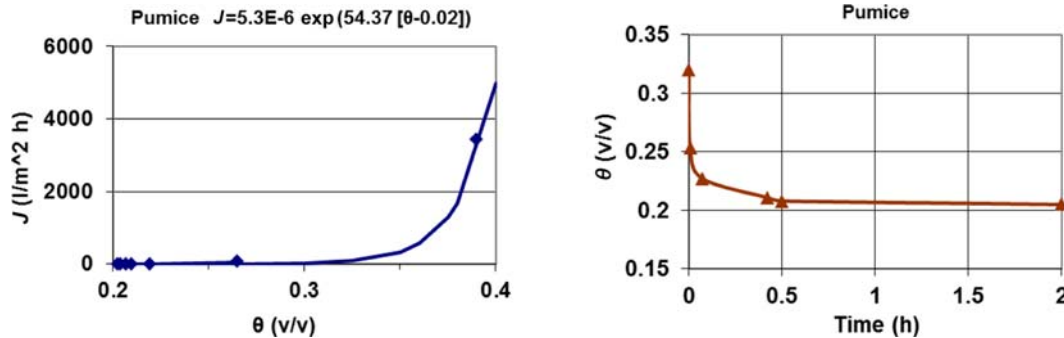


FIGURE 10.2

Obtaining the empirical pumice drainage function (for details see Bar-Yosef et al., 1999a,b). The substrate in the experimental container is saturated then water is allowed to drain while weighing the container. The derived water volume in substrate is divided by its volume to give average θ and drainage flux (J) as a function of time. The corresponding J and θ along the drainage process are plotted to give the $J(\theta)$ function. Water flow during irrigation is described by employing the Darcy law or plate theory.

From Bar-Yosef, B., Markovich, T. and Levkovich, I. (2001). Pepper Response to Leachate Recycling in a Greenhouse in Israel. *Acta Hort.* 548, 357–364, <https://doi.org/10.17660/ActaHortic.2001.548.41>.

the substrate's moisture content when it is ready to receive a new irrigation. Data of $\theta_{0.5 \text{ hour}}$ for the abovementioned substrates, and their saturated hydraulic conductivity (K_{hs}) are summarized in Table 10.1. For a 1000 m² greenhouse, substrate dimensions of $0.4 \times 0.2 \times 600 \text{ m}^3$, and $\theta_{0.5 \text{ hour}} = 0.21$ or 0.66 (pumice and perlite, respectively, Table 10.1), the water quantity at the end of fast drainage (V_{swbl}) is 10.1 and 31.7 m³, respectively. The data show that K_{hc} is considerably greater than $J(\theta_s)$. This is not surprising in view of the fact that θ in the system described above could not be maintained at saturation for a sufficiently long time once drainage began, θ at a depth of 10 cm immediately dropped [see, $\theta(-1\text{cb})$, Table 10.1] and the decline in average θ caused a decrease in J . Another empirical parameter is the average drainage flux between θ_s and $\theta_{0.5 \text{ hour}}$ (J_{fd} in Table 10.1) which is more stable than $J(\theta_s)$.

When planning a drip IR, it is essential to choose emitter discharge rate and distance between emitters to promote complete and uniform salt leaching and avoidance of salt accumulation in pockets. During irrigation, a water saturated zone is formed around the emitter. In clayey soils, the saturated volume is a hemisphere, but in sandy soil, the depth of the saturated volume exceeds its lateral distance (Bar-Yosef and Sheikolslami, 1976). In substrates, which are characterized by weaker water retention than sand, the saturated zone below emitters can be approximated as cylinder like. The relationship between emitter discharge rate (Q , L/h), irrigation time (t , hour), and the saturated cylinder radius (r , cm) is given in Eq. (10.8), where $\theta_{0.5}$ is θ at irrigation commencement and h_s is the cylinder height (=substrate height, cm):

$$Q t = \pi r^2 h_s (\theta_s - \theta_{0.5}) \quad (10.8)$$

Note that θ_s cannot be accurately determined, but it can be estimated from the 95% θ_s ($\theta_{s95\%}$) presented in Table 10.1. In the case of stone wool ($\theta_{s95\%} = 0.90$ and $\theta_{0.5} = 0.85$, Table 10.1), $h_s = 10 \text{ cm}$, $t = 0.1 \text{ hour}$ and $Q = 2.4 \text{ L/h}$, r is calculated from Eq. (10.8) to be 12.3 cm. This means that a distance of 20 cm between adjacent 2.4 L/h emitters should provide uniform salt leaching in

Table 10.2 Distribution of EC in a Stone Wool Slab in Two Treatments Differing in Threshold Electrical Conductivity (EC) at the End of Experiment

EC _{thr}	123 Days After Planting			
	Distance ^a (cm)		Depth ^b (cm)	
dS/m	0	10	0–2	0–10
EC (dS/m) in slab				
2.5	2.5	2.6	3.8	2.7
4.5	4.4	4.4	6.2	4.4

Test crop was pepper grown at Bet Dagan, Israel.
^aFrom emitter. Distance between emitters on lateral = 20 cm. Substrate sampled at depth of 4–5 cm. 7–10 irrigations per day.
^bFrom top. Mean of substrate samples taken at distances of 0 and 10 cm from emitter.
 Data derived from Bar-Yosef, B., Levkovich, I., Markovich, T., 1999a. Pepper response to leachate recycling in a greenhouse in Israel. *Acta Hort.* 548, 357–364; Bar-Yosef, B., Raviv, M., Meiri, A., 1999b. *Biological and Chemical Aspects of Water and Fertilizers Recycling in Greenhouses: Pepper, Muskmelon and Gypsophila*. Final Report 1996–1998 Submitted to the Chief Scientist, Ministry of Agriculture, Israel (Hebrew).

this substrate and prevent salt pockets formation. Plant water uptake may reduce substrate θ below $\theta_{0.5}$, but under common irrigation frequencies, this can be neglected. For example, ET of 1 L/m² ground/h is a very high value in greenhouses. When irrigating every hour at $\theta_{0.5}$, the plant extracts water during the 30 minutes between fast drainage termination and new irrigation. The stone wool volume is 24 L/m² ground and at $\theta_{0.5}$ it holds 20.4 L water. Correcting for the water uptake gives $\theta = (20.4 - 0.5)/24 = 0.83$ (instead of 0.85).

Experimental data of lateral EC distribution in stone wool at the end of a pepper (*Capsicum annuum*) growth period indeed indicates that no lateral salt pockets existed at the end of the season at the aforementioned drippers discharge rate and geometry (Table 10.2).

However in both presented treatments in Table 10.2, the EC in the top 2 cm was significantly higher than in the 0–8 cm layer. This could stem from evaporation-mediated salt accumulation in the top few mm that could not be effectively leached by drip irrigation and where roots are practically absent due to the effect of light. Experimental results of Cl distribution in perlite at the end of gypsophila (*Gypsophila paniculata*) growth season in closed loop IR also show uniform leaching at similar irrigation regime, but when irrigation rate was reduced by ~45%, the Cl concentration at a distance of 10 cm was higher than near the emitter (Table 10.3). As in stone wool, the Cl concentration at the top layer exceeded the concentration at the bottom.

In pumice ($\theta_{s95\%} = 0.38$ and $\theta_{1/2} = 0.21$, Table 10.1), $h_s = 20$ cm, and $Q = 2.4$ L/h an irrigation period of 0.1 hour can saturate a cylinder of $r = 4.7$ cm, indicating possible salt pockets between emitters located 20 cm apart. Experimental data with this substrate indeed show salt accumulation at distance of 10 cm from the emitter (Table 10.3). To obtain a saturated $r \geq 10$ cm, a higher discharge rate and/or longer irrigation period are needed. To avoid salt pockets without changing Q or irrigation time, a smaller distance between emitters should be used.

Table 10.3 Distribution of Cl in Pumice and Perlite^a at the End of an Experiment With Gypsophila in Closed Loop Irrigation System at Bet Dagan, Israel (EC_{thr} 3.3 dS/m)

Depth	Distance From Emitter (cm)					
	0	10	0	10	0	10
Cm	Pumice		Perlite 2		Perlite 2 (reduced)	
Cl concentration (mg/kg substrate)						
5	400	690	3240	3190	3080	3690
10	250	540	2420	2640	2660	2810
20	270	440	2120	1990	1740	1970

EC, Electrical conductivity.

^a8–10 irrigations per day, 6 min per irrigation; in the “perlite 2 reduced” treatment 5–6 irrigations per day, 6 min per irrigation. Distance between emitters was 20 cm.

Data derived from Bar-Yosef, B., Levkovich, I., Markovich, T., 1999a. Pepper response to leachate recycling in a greenhouse in Israel. *Acta Hortic.* 548, 357–364; Bar-Yosef, B., Raviv, M., Meiri, A., 1999b. *Biological and Chemical Aspects of Water and Fertilizers Recycling in Greenhouses: Pepper, Muskmelon and Gypsophila*. Final Report 1996–1998 Submitted to the Chief Scientist, Ministry of Agriculture, Israel (Hebrew).

Microbes' migration in closed loop IRs depends, among other things, on the substrate's filtration capability. Microbes' migration was studied in soils and soil columns (for review see, e.g., [Friedman, 1999](#); [Or et al., 2007](#)), but meager information can be found on the mobility in substrates per se. Comparison between representative microbes concentration in the leachate of pumice and stone wool that received the same recycled solution via drip irrigation for 218 days indicates that pumice was considerably more effective at filtering of bacteria (total and *Pseudomonas fluorescens*), and probably also of *Pythium* ([Table 10.4](#)), out of the recirculating solution.

These results were for pepper. Similar data were obtained with gypsophila, with the exception that pumice was shown to be a better filter than perlite ([Table 10.4](#)).

Another substrate-related question is whether to grow a crop in a long, continuous trough or in individual containers at the same plant density and substrate volume per plant. This is related to the spread of pathogens and unimpeded root growth. In trough-like systems, the runoff of one plant enters the adjacent roots environment of neighboring plants and certainly may infect those plants easily in case of diseased plants. Stone wool, perlite, or coir bags are examples of this, although they seldomly measure more than 1 or 2 m in length. The distribution of pathogens is similar in long trough/bag or container systems if leachate from the system is mixed and reapplied over the entire greenhouse area without a disinfection step. The main difference between container and trough is that lateral root growth in the latter persists without interference, while in containers root contact with walls inhibits elongation (e.g., [Romano et al., 2003](#)) and promotes formation of more root tips per unit root weight. Implications of this effect are further discussed in [Section 14.1.4](#). Another problem with a reduced root zone volume is the reduced oxygen supply and lower root respiration rate ([Peterson et al., 1991](#)). In conclusion, there appears to be no reason for different fertigation management for container- and trough-grown crops as long as the substrate volume per plant and substrate height are identical in both systems.

Table 10.4 Comparison Between Representative Microbe Concentrations in Leachates of Pumice, Stone Wool, and Perlite Under Pepper^a and Gypsophila^b

Substrate	Crop	Total Bacteria	<i>Pseudomonas fluorescens</i>	<i>Xanthomonas</i>	<i>Pythium</i>
		cfu/mL			per mL
Pumice	Pepper	6.20E4	3.20E3	4.0E3	0
Stone wool	Pepper	2.36E6	1.24E5	3.0E5	0.25
		$P < .05$	$P < .05$	n.s.	n.s.
		Total bacteria	<i>P. fluorescens</i>	CVO <i>Erwinia</i>	PDA ^c
		cfu/mL		per mL	
Pumice	Gypsophila	1.3E4	1.0E4	0	26
Perlite	Gypsophila	4.0E6	3.0E6	0.4	97
		$P < .01$	$P = .10$	n.s.	n.s.

PDA, Potato dextrose agar. In both experiments the two test substrates shared the same recirculation system, namely, the inflowing solutions were identical but the outflow solutions differed due to different substrate reactivity.

^aMicrobiological analyses were carried out by Dr. G. Kritzman, Agricultural Research Organization, Bet Dagan, Israel. Data were derived from Bar-Yosef et al. (1999a,b).

^bAs footnote "a" but data derived from Bar-Yosef et al. (2001a,b).

^cTotal fungi onto PDA.

10.2.5 IRRIGATION SYSTEMS

Irrigation solution can be applied from the bottom, top, or sides of the substrate (Chapter 9: Irrigation in Soilless Production). Bottom irrigation is a common practice for potted ornamentals as well as young plant nurseries, when grown on tables or (concrete) floors using flooded bench (eb and flood) irrigation. When irrigating from below, the bottom of the substrate is submerged in irrigation water or is in liquid contact with a wet mat. If upward transport is fast enough (depending on substrate's hydraulic properties) and diffusion downward into the saturated zone is sufficiently slow, then this results in a system where the average EC of the liquid in the root zone increases slowly over days and weeks (de Kreij and Straver, 1988; Otten, 1994). The advantage of this system is that there are virtually no water and nutrients losses and the energy requirement is minimal. The disadvantage is that the salts accumulating at the top substrate layer must be actively managed. This is mainly a problem for potted plants with a long growing period. As long as the plants are flooded frequently, the plant development is less affected by a partially high EC value in the root environment, since plants escape from spots with high salinity (Voogt and Sonneveld, 1999). Redistribution of these ions within the system is limited, especially Ca, K, and P accumulate in the top layer (Otten, 1994). Therefore the concentration of nutrients in the irrigation water should be closely attuned to the crop demand (see Section 10.3.8). The problem can be managed by prevention of evaporation of the top layer of pots, which can be achieved by specific substrate mixtures and management of the flood height and frequency. The upward capillary movement depends on the substrate's matrix and evaporation rate, and the inability to leach the unpredictable salt

accumulation from the top of the root zone represents a risk of developing a layer in the root zone (at the top) that is toxic to the plant. Such a salt accumulation on the top of pots of ornamentals can cause serious problems in the postharvest phase at the consumer, when the plants are irrigated commonly from the top and salts will be moved down. High salt concentrations from top reach the active roots developed at low concentration at the bottom of the pot. These roots are not yet adjusted to such high concentrations and the plant can seriously suffer by osmotic shock as has been demonstrated by Verberkt and Van den Berg (1993). This suggests that subirrigation be undertaken only if the supply water is of low salinity.

The most common method in closed loop IRs is irrigation from the top of the root zone. The substrate is in containers (slabs or bags) perforated at the bottom and placed on a rigid plastic tray (steel) gutter at adequate slope in the direction of flow; the effluent is captured and available for recirculation. This type of irrigation is generally applied via mini-sprinklers or drip irrigation.

IRs are characterized by their effective discharge rate (J_{irs} , L/m² substrate/h or J_{irg} , L/m² ground/h). Under drip irrigation, J_{irs} is the product of emitter discharge rate (J_{tr} , L/h) multiplied by the number of emitters/m² substrate (n_{tr}). The substrate-to-greenhouse area ratio (f_s) is necessary to transform J_{irs} into J_{irg} :

$$J_{irs} = n_{tr} J_{tr} \quad J_{irg} = f_s J_{irs} \quad (10.9)$$

10.2.5.1 Disinfection

While disinfection of the drainage water in closed loop systems is discussed in Chapter 5, Root Zone Microbiology of Soilless Cropping Systems, it is important here to note that some disinfection methods have an effect on the nutrient concentration. Runia (1994) showed that UV sterilization and ozone treatment reduces the Fe concentrations, particularly if Fe-diethylenetriamine penta-acetic acid (DTPA) is part of the nutrients. The same effect is observed in systems with chlorine disinfection (not used in soilless culture but used in drinking water treatment). Fe-polyphosphate however is not affected (Voogt, Unpublished data).

10.2.6 FERTILIZER ADDITION

Fertilizer nutrients must be in the system to meet the requirements of the plants. Fertilizers usually are supplied in dissolved form within the irrigation water (fertigation). There can also be an application of basic level into the growing media and some nutrients may inherently be part of organic substrates. A base fertilizer charge may be either readily available crystalline fertilizers or controlled release fertilizers. For fertigation there are various fertilizers available, which can be divided into two groups: (1) compound fertilizers or blends with a predefined composition and (2) fertilizers consisting of a single salt. Both can be available in liquid or in crystalline form. A prerequisite for all fertilizers to be applied in closed systems is that they should be low in residual salts content, to prevent Na or Cl accumulation. Today there are several ways to administer the concentrated fertilizer solution into the system, either directly into the water flow or by using a buffer (mixing tank) or operational water reservoir (W, NPK in Fig. 10.1 also see Chapter 13: Technical Equipment in Soilless Production Systems). The most important prerequisite is that the dosage is controlled by continuous measuring of the EC and pH.

10.2.7 MONITORING AND CONTROL OF NUTRIENTS

Circulated solutions should be monitored regularly to ensure that nutrient concentrations, EC, and pH are kept within predefined limits and salinity threshold levels are not exceeded.

10.2.7.1 Electrical conductivity and pH

EC and pH measuring in closed systems should be done at regular intervals using electronic control systems. If measured manually then measurements should be preferably made around the same time of the day, or collected over a 24-hour time period each day. The reason is that the EC and pH will vary considerably over the day depending on water and nutrients replenishment volume and timing. Typical variations in EC and pH over the day are shown for roses (*Rosa hybrida* L.) in closed system in Israel in Fig. 10.3.

For substrate systems operated with a targeted leaching fraction, this timing effect is of less importance, given the time delay and the large buffer of the system. However, apart from temporal effects, there are local differences, for instance there is a significant difference between ionic composition of emitter and drainage solutions (Fig. 10.3), generated by the water and ion uptake from substrate and resulting in higher EC in the drainage than in the irrigation solution. Since the

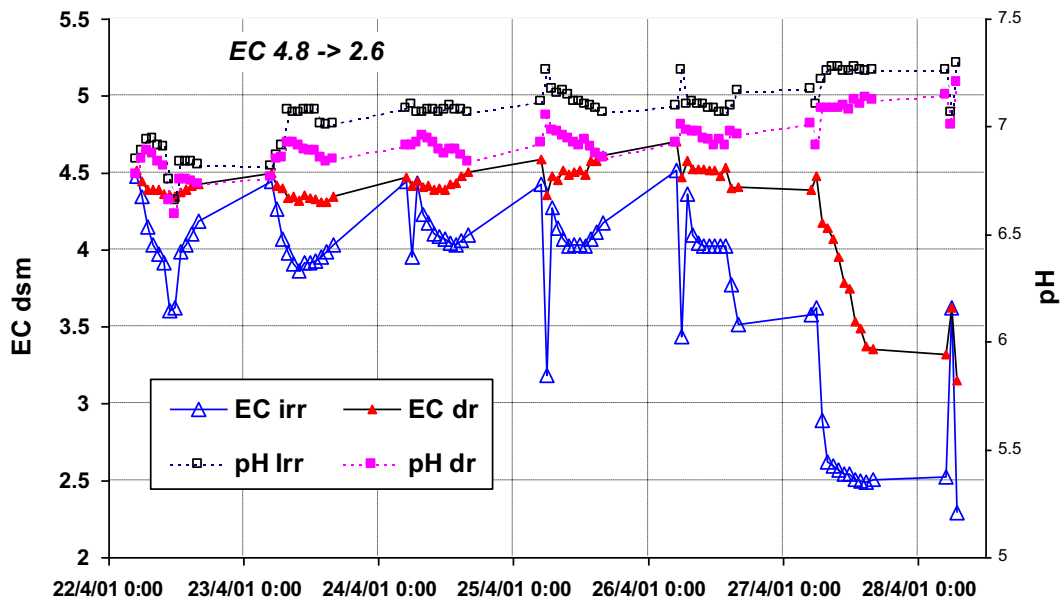


FIGURE 10.3

Hourly variations in recycled solution EC and pH at the substrate inflow (irr) and outflow (dr) points. The daily replenishment of fresh solution into the operational water reservoir (Fig. 10.1) occurred at 0600 hours. One-year-old rose plants, Arava, Israel.

Derived from Kramer, S., Bar-Yosef, B., Tzabari, G., Kritzman, G., Oshervitch, A., 2002. Rose Response to Water and Fertilizer Recycling in the Arava. Annual Report Submitted to the Research and Development Authority, Arava, Israel (Hebrew).

resulting EC difference is an important indicator of irrigation sufficiency, it is therefore recommended that both emitter and drainage solutions be monitored regularly and simultaneously. Moreover, this method not only gives an acceptable estimation of the average concentrations in the root environment, but it also informs about lowest and highest concentrations in the root environment, generally the supply and drainage in a substrate system, respectively. Especially for interpretation of the EC value knowledge about lowest and highest values are very useful (Sonneveld and Voogt, 1990). The pH difference between irrigation and drainage solutions is caused by plant root and microbiological activity. In addition, the variation in pH within the root environment can be considerable. It is therefore advisable to measure pH in both solutions, since information is needed for controlling the NH_4/NO_3 ratio in the replenishment solution (discussed in detail later and in Chapter 6: Nutrition of Substrate-Grown Plants). Moreover, it is essential to control the pH of the irrigation solution to keep it below levels of pH 6.2, in order to prevent precipitation of salts (such as CaHPO_4) which will otherwise cause clogging of the drip irrigation.

10.2.7.2 Nutrients

For adequate control over nutrients and salinity build-up in the root environment, the nutrient solution should be sampled and analyzed regularly, to ensure that they are close to the target concentrations.

Much research has been carried out in hydroponic systems and in substrate systems, results are usually related to the nutrient solution in which the crops have been grown. These nutrient solutions therefore can be considered as soil and substrate solutions. It has been established that crop reactions to nutrient levels and salinity in hydroponic solutions are in agreement with reactions in soil and substrate solutions. However, if the soil or substrate matrix are involved in the reaction, such as rhizosphere or adsorption–desorption, the reactions will differ from those in hydroponics. This, for example, is the case for the availability of micronutrients. Cu, for example, can be strongly bound by organic compounds of peat substrates (Verloo, 1980). Even Cu deficiency may occur in plants, despite high concentrations analyzed in the material as the roots have to compete for Cu strongly bound to organic complexes (De Kreijl et al., 1985). The choice of what should be sampled depends largely on the growing system, the buffering capacity of the system as well on the rate of replenishment (hydroponic systems) or drainage fraction (substrate systems). In case of hydroponic systems with bulky volumes of the nutrient solution in the root environment, such as in DFT systems, the buffering capacity is huge and allows just sampling of the ponds.

Systems with continuous circulation such as NFT and aeroponic systems, the sampling directly from the root environment is not adequate and is also prone to spatial variations; therefore it can be derived better from the collection or circulation tank (D, Fig. 10.1), which have a much larger buffer and alleviate the spatial and temporal variations.

For substrate systems, the so-called substrate solution can be used for routine testing. This especially is the case for substrates with high water contents at a high pressure head, such as mineral fibers and foams (Sonneveld, 1995). During cultivation the crops are irrigated frequently in such substrates, sometimes dozens of times a day, which ensures a stable and high water content in the substrate. Under such condition, the substrate solution easily can be gathered, withdrawing it from the substrate by suction with the aid of a simple syringe. Substrate solution can be extracted also

with the aid of ceramic cups placed in the substrate. A drawback of this method is the accidental placement in the root zone. Therefore in view of the great variation of concentrations of salts and nutrients in greenhouse soils and substrates, a substantial number of different subsamples are required, distributed both horizontally as well vertically. It is to be noted that certain ions can be adsorbed by the material of the cups; therefore the right choice of cups is important to avoid contamination by the material (Shen and Hoffland, 2007).

The composition of the substrate solution also can be estimated from the composition of the drainage water. This method seemed to be very suitable, because it not only gives an acceptable estimation of the average concentrations in the root environment, but it also informs about lowest and highest concentrations in the root environment, generally supply and drainage in a substrate system, respectively. Especially for interpretation of the EC value knowledge about lowest and highest values are very useful, as discussed in Chapter 6, Nutrition of Substrate-Grown Plants.

Ebb and flood systems need specific treatment. Sampling of the collection tank is the obvious though it is questionable to what extent the composition of the buffering tank will be influenced by the nutrients within pots, since the flooded nutrient solution will be sucked by the pots to saturation and the volume drained off to the floor into the returning solution is relatively limited compared to the huge volume of the flooded solution (Otten, 1994). In those cases, it is more appropriate to collect samples from the pots, although it is also advisable to monitor the composition of the supply solution regularly. The frequency of analyses depends on the permitted deviation from the objective values and the daily replenishments. Chemical analysis of plant leaves can have an additional value, though it cannot replace the monitoring of the nutrient solution. Nutrient contents in plant tissue will vary tremendously depending on the age of the leaves, the position on the plant and environmental conditions. Adams and Ho (1993) found differences in Ca concentrations within a tomato plant that ranged from 1 to 600 mmol/kg DM; the lowest at the distal ends of fruits and the highest in laminae of young leaves, respectively. The differences should even be more striking when the comparison was made with old leaves, because the Ca concentrations in old leaves can be double of those in young leaves (Sonneveld, 1980). Measured tissue analysis results are compared with target nutrient concentrations so that deviations can be used to identify the needed fertilizer management modifications. However, reference values have to be established for specific leaves and specific conditions, for individual crops and likely for different varieties as well.

Another monitoring approach is to determine crop characteristics not directly comparable with target values, but yet indicating time variations that can be interpreted as positive or adverse plant response to prevailing growth conditions. The so-called phyto-monitoring includes chlorophyll fluorescence imaging (Chaerle and Van Der Straeten, 2001) on the one hand and simpler devices such as stem and fruit diameter, leaf temperature, and sap flow, on the other hand. For review of commercially available methods and data interpretation see, for example, Ehret et al. (2001), De Swaef et al. (2012), Hanssens et al. (2013), and Van de Wal et al. (2017).

In substrate systems, the substrate water status can be monitored by using water content sensors. Today this is commonly used in commercial crops, as an additional value to the gauging of drainage. Although the reuse of drainage overrules the need for irrigation control to a certain extent, growers like to have more control tools over irrigation and seem to look for advantages in playing around with irrigation, drainage, and substrate moisture throughout the day (Heuvelink and Kierkels, 2012).

10.2.8 CONTROL

10.2.8.1 Water replenishment

In hydroponic systems, the reservoir with nutrient solution needs frequent replenishment, depending on its volume relative to water consumption by plants. During replenishment the water reservoir is brought to full capacity. If this procedure is repeated every day then the recharged volume is equal to $ET + [\text{disposed volume}]$ of the previous day. The entire water volume in the system is $V_{wT} = V_r + V_{wse}$, where V_{wse} is the water volume after the end of the rapid drainage (L/m^2). In modern systems, the replenishment is fully automated, using float valves for the water volume and dosage of fertilizer solution and acid/base dosage controlled by EC and pH sensors.

When the recirculation system includes a biofilter, the reference condition for replenishment should be the end of the entire drainage filtration, and return of the treated solution to the water reservoir. The biofilter may add a considerable volume of water to the system depending on its technology and design.

In substrate systems, the replenishment of the water in the system is by irrigation and a certain surplus irrigation is deliberately required to avoid drought, nutrient imbalances, and salinity build-up. The required overirrigation is due to a combination of the lack of water and nutrient buffer in these systems and heterogeneities in crop transpiration and the water supply. The transpiration of individual plants will differ as this is affected by the plant and canopy characteristics as well as local variations in shadowing and microclimate. This is illustrated well by an investigation from Van der Burg and Hamaker (1987) (cited in [Sonneveld and Voogt, 2009](#)), who measured irrigation and drainage on individual plants at various spots in a commercial stone wool grown greenhouse tomato from March until September. The average coefficient of variation of the water supply increased during the season from 4.8% till 13.2%, while the coefficient of variation of the water uptake was not increased during the season and on average amounted to 10.2%. The required drainage water is affected by the variation of the uptake by the plant as well as by the variation of the supply. Those variations are independent, and thus the variation of the drainage water will be calculated as

$$s_d = \sqrt{s_s^2 + s_u^2} \quad (10.10)$$

where s_d is the standard deviation of the drainage water, s_s is the standard deviation of the water supply, and s_u is the standard deviation of the water uptake of the plant.

In practice, the root system from individual plants cover a larger area than only the reach of the irrigation nozzle at the spot alleviating heterogeneities partly. For example, in case of three plants per stone wool slab, the coefficient of variation will be divided by a factor $\sqrt{3}$, resulting from an average reaction of three plants together. Furthermore, the drainage should be always ≥ 0 and will be approximated with a confidence limit of for example 1% ($P \leq .01$) (not more than one on a hundred plants will suffer from drought). This agrees with a standard normal distributed unit T of 2.33 then the drainage will be calculated in relation to the water uptake by the following equation:

$$d \geq T \frac{s_d}{\sqrt{p_n}} \quad (10.11)$$

where T is the standard normal distributed unit agreeing with a determined confidence limit; d is the quantity of drainage water, required as oversupply; p_n is the number of plants that can equalize

the water mutually and the value mostly will vary between 1 and 3; and s_d is the standard deviation of the drainage water as given in Eq. (10.10).

In this formula, s_d will be calculated from Eq. (10.9) in which s_s will be found by iteration, when the water uptake is determined. The required oversupply can be expressed on the supply, and this fraction is independent of the level of the supply.

10.2.8.2 Solution discharge

Any closed loop IR must support an option for solution discharge when a predetermined discharge criterion is met. Beerling et al. (2014) reviewed the criteria that play a role to justify discharge from closed systems and is discussed in Section 10.2.4.

10.2.8.3 Nutrients replenishment

Continuous recording of daily application of nutrients is one of the most important control tools, as it allows for imposition of an application rate that is at least equal to the projected consumption rate of the crop. If cumulative supply exceeds the sum of the projected cumulative uptake of nutrients, the amount of discharged nutrients, plus the nutrients found in the recirculated solution, then the difference is likely to be located in substrate pockets located between wetting fronts of adjacent emitters. Existence of such pockets requires alteration of the substrate composition or irrigation regime to induce greater lateral water flow and more efficient solute leaching (see Chapter 3: Physical Characteristics of Soilless Media).

In the previously mentioned example of a 1000 m² greenhouse, the water volume $V_r + V_{wsb}$ (perlite) was 43.4 (=5 + 38.4) m³. At an N solution concentration of 100 g N/m³, the quantity of N in the greenhouse is 4.34 kg, in comparison with a daily N uptake rate of ~0.2 kg/day. When replenishing the consumed 0.2 kg by adding it in one dose into the water reservoir (R, Fig. 10.1), the N concentration there would increase to 140 g N/m³, which will have small impact, and after the first irrigation, the solution will gradually dilute to 105 g N/m³ (uptake neglected). If instead the 0.2 kg N is injected into the irrigation water and diluted in a water dose of 1 m³, the N concentration in the water pulse would be 200 g N/m³, which is less desirable than 140 g N/m³ due to the resulting elevated EC, but in subsequent irrigations, it will also be diluted to 105 g N/m³. Such considerable fluctuations in N concentration suggest that dividing the daily N rate into —three to five portions seems more appropriate when injecting the nutrients directly into the irrigation water.

10.3 MANAGEMENT

10.3.1 INORGANIC ION ACCUMULATION AND DEPLETION

One of the important factors in closed systems is the dynamics of ions over time. Ion accumulation or depletion in closed IRs depends on the balance between the rates of application and uptake by plants. The rate of accumulation stems from the difference between C_{wi} and C_{ui} (Section 10.2.1) and is different for each ion and differs per crop. Accumulation of ions may occur for nutrients as well as for nonnutrients, whereas for some nutrients, a certain accumulation is permitted deliberately to reach certain target levels. These targets are needed to effectuate a required uptake for optimum production or for product quality parameters. A good example of this is the different

requirements for Ca, Mg, and K in fruit vegetables. For Ca, the concentration in the root environment has to be increased by a factor of 2, whereas K can be reduced by a factor 0.5, compared to the supply concentrations and based on the same ionic sum. See the example and discussion in Section 10.3.8.2. Another example is the rapid uptake of NH_4 ions relative to the uptake of NO_3 . As a consequence, there are specific target values for all ions, which then clearly have to be different from the concentration in the supply. This is subject to the whole set of nutrient management and will be discussed further on. In this topic we will discuss undesired accumulation of ions due to water quality issues. An example of the differences in accumulation rate of the ions Na^+ , Ca^{2+} , Cl , and SO_4 for some important greenhouse crops are presented in Table 10.5.

In this example, the chloride accumulation is greatest (mg/L basis) but did not result in a problem in crops having short growth period and low transpiration rate, for example, lettuce (*Lactuca sativa*). Calcium and Mg^{2+} accumulate in solution of all presented crops except lettuce and muskmelon (*Cucumis melo*). Notably the greater Ca and Mg consumption by these two crops in comparison with supply (Table 10.5) stems from the fact that the actual fresh water Ca and Mg concentrations were higher than specified in the table (see footnote).

In temperate zones such as in Western Europe, water application rates are lower than in semi-arid and arid zones, and the fresh water salt concentration is usually lower than cited in Table 10.5. In The Netherlands for example Cl accumulation is not regarded as a problem and the main concern is Na concentration build-up (Baas and van der Berg, 1999). However, due to the tight regulations regarding discharge of collected drainage, accumulation of Na and Cl is of concern and intensive research has been carried out to deal with salinity in closed growing systems.

It must be noted that the accumulation of Na or Cl in the root environment coincides strongly with the EC. Principally, the accumulation will cause increase of the EC and that eventually will affect yield or quality of the products (Sonneveld, 2000). However, in modern greenhouse crops, the EC in the root environment is controlled more or less automatically by managing the EC of the nutrient solution supplied. Increasing EC of the drainage as a result of increasing Na or Cl will then result in lower supply of fertilizer solution to match the target EC of the irrigation solution. Then Na or Cl accumulation eventually causes reduction in the nutrient concentrations (Voogt and Sonneveld, 1996). For many crops, maximum acceptable Na and Cl concentrations have been established ($C_{\text{gNa}}^{\text{max}}$, $C_{\text{gCl}}^{\text{max}}$), derived from experiments in closed systems (Baas et al., 1994; De Kreijl and Van den Berg, 1990; Sonneveld and Voogt, 1983; Sonneveld and Van der Burg, 1991; Sonneveld et al., 1999a,b). Corresponding with these values, uptake capacities $C_{\text{UNa}}^{\text{max}}$ and $C_{\text{UCI}}^{\text{max}}$ have been established (see Section 10.2.3). Data of the outcome of these trials are summarized in Table 10.6.

The data clearly shows the great variability in uptake concentrations of various crops, which implies for instance that cucumber can be grown in closed systems without discharge better than roses which take up Na or Cl at much lower rates. Without exception $C_{\text{UNa}}^{\text{max}}$ is lower than $C_{\text{UCI}}^{\text{max}}$, so consequently, Na is generally the bottleneck element for closed systems. These uptake concentrations could be used as parameters for judging the suitability of water sources for closed growing systems. The different behavior of both elements could be caused by the fact that Cl is an essential nutrient for plants (Chapter 6: Nutrition of Substrate-Grown Plants) and the uptake is even metabolically controlled, which cause a preferred uptake at low levels in the root environment (Mengel and Kirkby, 2001). The rather low uptake capacities for Na for Rose and Bouvardia crop are

Table 10.5 Evaluating Potential Cl, SO₄, Na, Ca, and Mg Accumulation in Closed Loop Irrigation System of Representative Greenhouse Crops [Tomato, Pepper, Lettuce, Muskmelon (M. melon), Strawberry (Strbry), Roses, and Lisianthus (Lis)] Grown in Semiarid Zone

	Tomato	Pepper	Lettuce	M. melon	Strbry	Roses ^c	Lis
Fresh Water (mm)	480	490	55	350	330	1293	610
C1 (conc. in fresh water 175 mg/L)							
Addition (kg/ha)	840	857	96	612	577	2263	1067
Uptake (kg/ha)	371	125	23	154	85	103	301
Pot. conc. (mg/L)	1172	1830	182	1145	1230	5400	1915
S (SO₄–S conc. in fresh water 50 mg/L)							
Addition (kg/ha)	240	245	27	175	165	646	305
Uptake ^c (kg/ha)	50	45	5	30	80	176	—
Pot. conc. (mg/L)	475	500	55	362	212	1175	—
Na (conc. in fresh water 110 mg/L)							
Addition (kg/ha)	528	539	61	385	363	1422	671
Uptake (kg/ha)	291	172	24	85	80	78	214
Pot. conc. (mg/L)	593	917	92	750	712	3360	1142
Ca (conc. in fresh water 85 mg/L)							
Addition (kg/ha)	408	416	47	298	280	1099	518
Uptake (kg/ha)	302	132	18	340	79	212	70
Pot. conc. (mg/L)	265	710	72	—	502	2217	1120
Mg (conc. in fresh water 24 mg/L)							
Addition (kg/ha)	115	118	13	84	79	310	146
Uptake (kg/ha)	13	72	9	202	34	61	75
Pot. conc. (mg/L)	255	115	10	—	112	622	177
The accumulation is expressed as potential concentration (Pot. conc.)^a of elements in the recirculated solution at the end of the growing season. For simplicity, all crops were assigned with the same element concentrations in fresh water (in parenthesis) even though there were some differences between experiments. Where deviations from the indicated concentrations were significant they were mentioned in text. Uptake data were obtained experimentally^b in treatments owning EC_{thr} > 4 dS/m; under no circumstances had the presented uptake rates caused yield reductions >20% of the control (EC_{thr} = 2.7 dS/m). ^a(Addition – uptake)/total solution volume. ^bReferences for reported crops, all in recycled solutions, are tomato—Reshef et al. (2006); pepper—Bar-Yosef et al. (2005); lettuce—Bar-Yosef et al. (2002); Muskmelon—Bar-Yosef et al. (2000a,b,c,d); strawberry—Matan et al. (2003); Roses—Bar-Yosef et al. (2003a,b); Lisianthus—Dori et al. (2005). ^cEstimate based on prevalent %S in plant as derived from Hochmuth et al. (2004) and plant DM in above mentioned references.							

particularly striking. For sweet pepper, differences in the Na uptake have been found between the vegetative stage and periods with heavy fruit load (Sonneveld and Van der Burg, 1991).

The increase in ion concentrations is equal to the accumulated quantity over time divided by the total water volume (V_{wT}). Representative examples depicting concomitant temporal increases in EC

Table 10.6 Uptake Concentrations (C_{Ui}^{max}) of Na and Cl for Some Greenhouse Crops, Expressed in mmol/L, as Derived From Trials With Closed Systems and Long-Time Cropping Periods of at Least 6 Months at Continuous Low (<5 mmol/L) and High (10 mmol/L) Na and Cl Concentrations in the Root Environment

Crop	Na Root Environment		Cl Root Environment	
	<5 mmol/L	10 mmol/L	<5 mmol/L	10 mmol/L
Gerbera	0.2	0.8	1.7	2.4
Carnation	0.1	0.3	0.5	0.8
Lily	0.4	1.0	0.6	0.9
Aster	0.2	1.5	0.4	2.0
Rose	0.0	<0.1	0.1	0.2
Bouvardia	<0.1	0.1	0.2	0.3
Tomato	0.4	0.8	0.6	1.0
Cucumber	0.3	1.0	0.3	1.5
Sweet pepper vegetative stage	0.2	0.6	0.3	0.7
Sweet pepper heavy fruit load	<0.1	0.3	0.3	0.7

Data from Sonneveld, C., Voogt, W. 1983. Studies on the salt tolerance of some flower crops grown under glass. Plant Soil 74, 41–52; De Kreijl and Van den Berg, 1990; Sonneveld and van der Burg, 1991; Baas, R., Nijssen, H.M.C., Van der Berg, T.J.M., Warmenhoven, M.G., 1994. Yield and quality of carnation (Dianthus caryophyllus) and gerbera (Gerbera jamesonii L.) in a closed nutrient system as affected by sodium chloride. Sci. Hort. 61, 273–284; Sonneveld, C., Voogt, W., Spaans, L., 1999a. A universal algorithm for calculation of nutrient solution. Acta Hortic. 481, 331–339; Sonneveld, C., Baas, R., Nijssen, H.M.S., de Hoog, J., 1999b. Salt tolerance of flower crops grown in soilless culture. J. Plant Nutr. 22, 1033–1048.

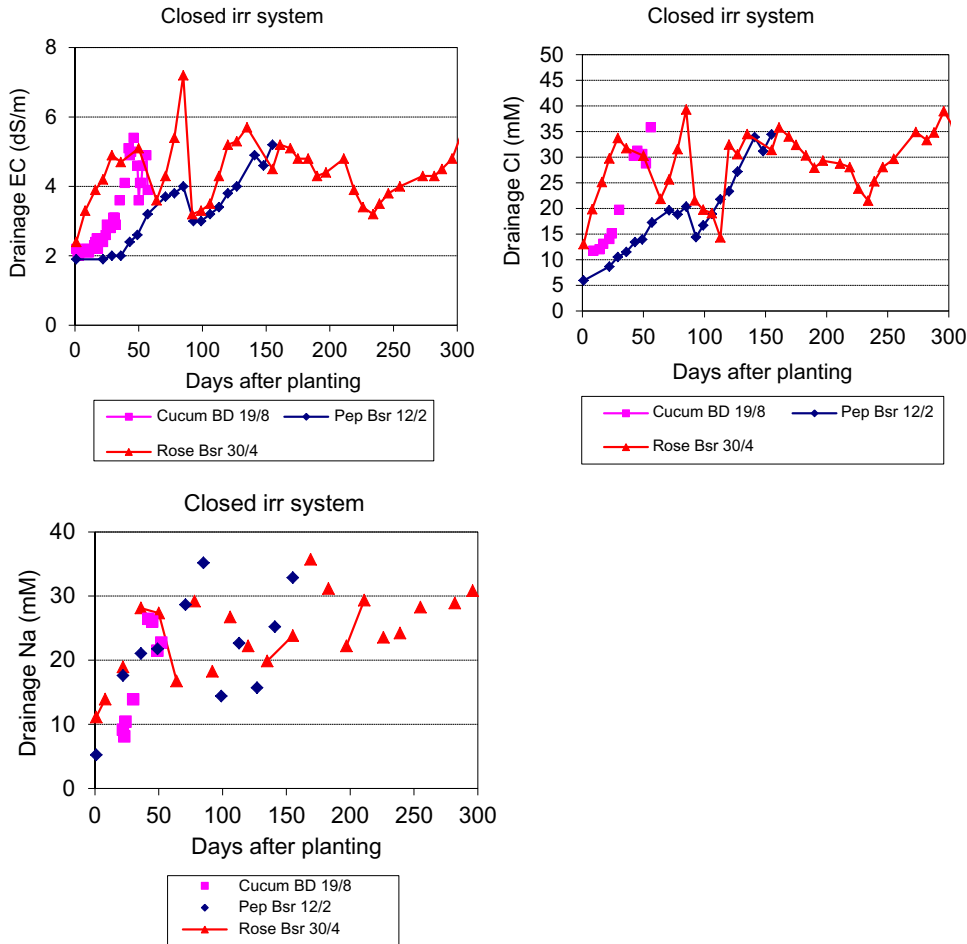
and Cl and Na^+ concentration in recycled solution of three crops grown under threshold EC of ~ 5 dS/m is presented in Fig. 10.4.

Pepper planted in winter showed the slowest rate of EC, and Cl and Na concentrations increase during the first 3 months of growth until EC_{thr} was reached. Rose (planted in spring) had a faster salt accumulation rate and cucumber (planted in autumn) had the fastest rate due to its highest growth rate. The cumulative transpiration (ET_{cum}) which is required to obtain the threshold Cl concentration in solution (C_{thr} , corresponding to EC_{thr}) depends on the Cl concentration in fresh water (Cl_{fw}), initial Cl concentration in recycled solution (Cl_{in}), Cl uptake by the crop (U_{Cl}), and total recycled solution volume (V_T):

$$ET_{cum} = \frac{V_T(Cl_{thr} - Cl_{in}) + U_{Cl}}{Cl_{fw}} \quad (10.12)$$

A similar mass balance approach was used by Carmassi et al. (2003) and Kempkes and Staghellini (2003).

After EC_{thr} was obtained (Fig. 10.4), the solution EC of the three crops continued to fluctuate, sometimes in excess of the permitted lower and upper deviations. The fluctuations stemmed from inadequate solution discharge control stemming mainly from automation failure. The impact of such fluctuations on yield, and quality is hard to evaluate and is further discussed later in this chapter.

**FIGURE 10.4**

Increase in EC and Cl and Na concentration of recycled greenhouse solutions in the presence of cucumber (planted on 19/8/2002 at Bet Dagan, Bar-Yosef et al., 2003a,b), pepper (planted on 12/2/2001 at Besor, Reshef et al., 2002a,b), and roses (planted on 30/4/2002 at Besor, Bar-Yosef et al., 2003a,b). Greenhouse water volumes were 29, 50, 30 m³/1000 m² greenhouse and plant stands 2500, 3500, 6000 pL/1000 m² greenhouse, respectively. NaCl concentration in fresh water was 6.6 mM in Besor and 6.0 mM in Bet Dagan. Threshold EC for discharge was ~5 dS/m. EC, Electrical conductivity.

While there was a clear similarity between EC and Cl build-up pattern and fluctuations, Na behaved differently: its accumulation was not crop specific, and it was subject to larger concentration fluctuations over time. This result agrees with the weaker correlation between EC and Na concentration in comparison with EC versus Cl concentration (Fig. 10.4).

Temporal Na^+ concentration in solution (C_t) was satisfactorily simulated by Carmassi et al. (2003) and Kempkes and Staghellini (2003) who used the following ion mass balance equations:

$$\begin{aligned} V_T C_t &= V_T C_{\text{in}} + V_{\text{in}} C_{\text{fw}} - U - \sum V_{\text{dis}} (C_{\text{thr}} + \Delta C_{\text{up}}) \\ C_t &= C_{\text{in}} + C_{\text{fw}} V_{\text{in}}/V_T - (C_{\text{thr}} + \Delta C_{\text{up}}) \sum V_{\text{dis}}/V_T - U/V_T \end{aligned} \quad (10.13)$$

where V_{in} is the cumulative fresh water application ($= \sum \text{ET} + \sum V_{\text{dis}}$), C_{in} is the initial ion concentration in the system solution and the other variables were defined as part of Eqs. (10.10) and (10.11).

EC variation over time cannot be calculated because (1) plant uptake is undefined, and (2) occasional variations in nutrient concentrations cause variations in EC not accounted for by Eq. (10.12). However, as shown earlier (Fig. 10.4), the empirical relationship between C_{Cl} and EC is highly correlated, so predictions of C_{Cl} can be used as estimators of EC.

The rapid accumulation of Na and Cl in this trial for pepper and rose is probably caused by the low uptake concentrations of these ions (Table 10.6).

Accumulation of both Ca^{2+} and SO_4^{2-} may result in gypsum (CaSO_4) precipitation and consequently emitters clogging. From solubility product (K_{sp}) considerations, precipitation will not occur as long as $(\text{Ca}^{2+})(\text{SO}_4^{2-}) < K_{\text{sp}}$, where parentheses denote activity and $K_{\text{sp}} = 10^{-4.5}$ (Lindsay, 1979). For $(\text{SO}_4^{2-}) = 2 \text{ mM}$ (concentration $\sim 2.2 \text{ mM}$) which is attainable in closed loop IRs (see Table 10.5) (Ca^{2+}) should not exceed 15.8 mM , otherwise gypsum precipitation is likely to occur.

Two less common elements that may accumulate in recycled solutions are Si and Al. Si is sometimes found in well-water sources; both may be released from substrates containing aluminosilicates or their by-products, for example, stone wool, perlite, pumice, and tuff. The solubility and activity of free Al^{3+} ions or one of the complex hydroxides is subject to the prevailing pH level (Lindsay, 1979). Usually above pH 5, there is hardly Al available for uptake (Wright, 1989). An example of what can be found of these ions under practical conditions is shown in Table 10.7. At

Table 10.7 Concentrations of Al, Si, and Mn Found in Recycled Solution Under Pepper, Muskmelon, and Gypsophila Grown in Perlite, Stone Wool (RW), and Pumice

Ion	Pepper (68 DAP)			Muskmelon (67 DAP)				Gypsophila (68 DAP)	
	RW		Pumice	Perlite			RW	Perlite	Pumice
	Solution pH								
	5.5	6.9	6.4	4.2	4.9	6.0	5.1	4.5	3.8
Solution ion concentration (mg/L)									
Al	0.13	0.010	0.038	0.66	0.32	0.10	0.35	0.17	1.3
Si	13.9	16.8	26.9	5.4	6.6	5.2	10.0	21	41
Mn	0.37	0.08	0.26	0.30	0.46	0.13	0.22	0.43	0.47

Fresh water Al, Si, and Mn concentrations were $5 \mu\text{g/L}$, $6 \pm 2 \text{ mg/L}$, and 0, respectively. Solution sampling time is given as number of days after planting (DAP).

Data derived from Bar-Yosef et al. (pepper—1999, muskmelon—2000, gypsophila—1999).

pH 6.4, Al concentration in pumice is similar to its concentration in fresh water (Table 10.7). It increases to 1.3 mg/L as pH dropped to 3.8 due to increased NH_4 supply.

The data indicate (Table 10.7) that transition from pH 4.9 to 4.2 caused a sharp increase in Al concentration from 0.32 to 0.66 mg/L. According to Adams and Moore (1983), a concentration of 1 mg/L Al^{3+} is potentially toxic to most plants, but complexation as hydroxides as well as by various organic anions present in plant rhizosphere may reduce the activity of the free ion, thus ameliorating its potential phytotoxicity (Grauer and Horst, 1992). According to Mengel and Kirkby (2001), the toxicity of Al declines rapidly if the pH of a solution is increased from pH 4 to 5. At similar pH, perlite, pumice, and stone wool did not differ considerably in their Al release to solution.

The chemistry of Si is rather complex, depending on the type of aluminosilicate, weathering, and age, silicates in substrates will become water-soluble silicon. The compounds, most likely formed are monosilicic acid, (H_4SiO_4 or $\text{Si}(\text{OH})_4$) (Iler, 1979). It may reach saturation at 100–200 ppm of SiO_2 , from where polymerization will start. Depending on pH and salt concentration, the properties of polysilicates differ from amorphous silica at low pH, forming a gel with increasing concentration, to stable colloids, with spherical particles at high pH and without salts (Iler, 1979). In an effort to estimate the release of Si from substrates, Voogt and Sonneveld (2001) tested several types of growing media, which were exposed to constant moisture under greenhouse conditions. Stone wool, in particular used slabs and slabs exposed to low pH, resulted in the highest Si concentrations, whereas perlite and pumice were quite stable (Table 10.8). Lindsay (1979) found that Si concentration in treatment solutions surpassed expected Si concentration in equilibrium with quartz (2.8 mg/L). Except in pumice the Si concentrations were lower than expected in soil solution

Table 10.8 Si Concentrations in the Nutrient Solution Released From Several Growing Media, After 6 Months of Immersion in Three Times the Water Volume at Saturation With a Standard Nutrient Solution for Cucumber (Voogt and Sonneveld, 2001)

Substrate		Si
		mg/L
Stone wool new slab		19.7
Stone wool used slab		30.9
Glass wool new slab		5.6
Peat substrate		8.4
Peat substrate + 50% rice hulls		30.9
Perlite		1.4
Pumice		4.2
Stone wool new slab	High pH (7.5)	11.2
Stone wool new slab	Low pH (4.0)	39.3
Stone wool used slab	High pH (7.5)	21.1
Stone wool used slab	Low pH (4.0)	53.4

From Voogt, W. and Sonneveld, C. (2004). Interactions Between Nitrate (NO_3) and Chloride (Cl) in Nutrient Solutions for Substrate Grown Tomato. *Acta Hort.* 644, 359–368, <https://doi.org/10.17660/ActaHortic.2004.644.48>.

containing aluminosilicates (22 mg/L); in pumice the Si concentration (27 or 41 mg/L depending on pH, Table 10.8) slightly exceeded that value.

Silica concentration in perlite fluctuated with pH in the range of pH 4.2–6.9, and was more crop than pH dependent. The evidence of beneficial or harmful effects of Si in the root environment is still debated. Positive effects of Si have been reported for cucurbits such as cucumbers and melons (Adatia and Besford, 1986; Belang r et al., 1998; Voogt and Sonneveld, 2001), whereas for other crops, no effect were found (Voogt and Sonneveld, 2001), or even negative effects (Lieten et al., 2002). The crop-specific concentrations reflect plausible differences in Si uptake by various plants (Table 10.7). Stamatakis et al. (2003) reported that soluble Si at 2 mM (56 mg/L) in hydroponic solution ameliorated salinity induced yield reduction in tomato (*Lycopersicon esculentum*) and alleviated incidence of blossom end rot (BER) under nonsaline conditions; unfortunately the effect of lower concentrations was not investigated. On the other hand, Lieten et al. (2002) found that irrigation solution Si concentration >0.55 mM (as K₂SiO₃) was associated with enhanced strawberry fruit albinism. The direct mechanism of this disorder is still unexplored.

Some of the microelements are prone to accumulation in recycled solutions. These ions can be present in water sources. Boron can be present in too high concentrations in groundwater or in desalinated water by RO, since the membranes used for RO cannot prevent HBO₃ going through, except at very high pH. For example, in two closed loop IR experiments in Israel, B was added continuously in the replenishment solution at concentration of 0.2 mg/L, and its concentration in the recycled solution increased depending on the threshold EC (which determined discharge) and time after planting (only one date is presented in Table 10.9). In lisianthus (*Eustoma grandiflorum*), the concentration at the end of the second flush reached values of 0.4–0.8 mg/L and in cucumber the concentration varied between 0.5 and 1.1 mg/L.

Zinc, Copper, and Mn can be present in water sources too, sometimes from natural source or due to contamination from the technology used in the water infrastructure. When supplied in excess of plant consumption, that is, in concentrations above C_{Ut}^{max}, this can be harmful. For the metal ions, the problem can be a specific, for instance, Zn can be exchanged for Fe at the chelating complex

Table 10.9 Boron Accumulation in Recycled Solution as a Function of Threshold EC (EC_{thr})^a in Two Crops

EC _{thr}	Lisianthus ^b	Cucumber ^c
	Harvest Second Flush	45 Days After Planting
dS/m	B (mg/L)	
2.3–2.4	0.4	0.5
3.5–3.6	0.6	0.9
4.3–4.6	0.8	1.1
Note that solution sampling time was different in the two experiments. Boron concentration in the feed solution was 0.2 mg/L. ^aAs EC_{thr} increases discharge volume decreased. ^bBar-Yosef (2003, Unpublished data). ^cDerived from Bar-Yosef et al. (2003a,b).		

and causing Fe to be less available for the plant (Bugter and Reichwein, 2007). For all microelements, the risk of toxicity may occur (Sonneveld and De Bes, 1984).

Similarly, Yermiyahu et al. (2003) found no yield reduction in greenhouse pepper exposed to B concentrations of 0.5, 1.0, 2.0, and 4.0 mg/L, no toxicity symptoms were observed in the crops in Table 10.9 either.

Another inorganic toxicant that may accumulate in recirculated solutions is nitrite (NO_2^-). The problem can be acute in new synthetic substrates lacking nitrifying bacteria where NH_4 is a significant N source. Under such conditions, oxidation of NH_4 to NO_2 occurs at a faster rate than conversion of NO_2 to NO_3 , so that nitrite accumulates in the solution. An example of this phenomenon is presented for cucumber growing in perlite in Fig. 10.5.

At the end of the first week of growth, $\text{NO}_2\text{-N}$ concentration was ~ 1 mg/L regardless of N total concentration, and it sharply increased within 2 weeks to the peak which varied between 9 and 13 mg $\text{NO}_2\text{-N}$ in the 2–8 mM N treatments (30:70 $\text{NH}_4\text{:NO}_3$ ratio). In the highest N treatment (16 mM), the same trend was seen, but NO_2 concentrations were lower probably due to lower solution pH which may have inhibited the NH_4 oxidation to NO_2 . The nitrite concentration returned to ~ 1 mg/L value after ~ 6 weeks of growth. When plants are young and their root systems small, the exposure to such high nitrite concentrations render them susceptible to root pathogens (e.g., *Pythium*). Nitrite may be formed and accumulate under anoxic conditions too, stemming from

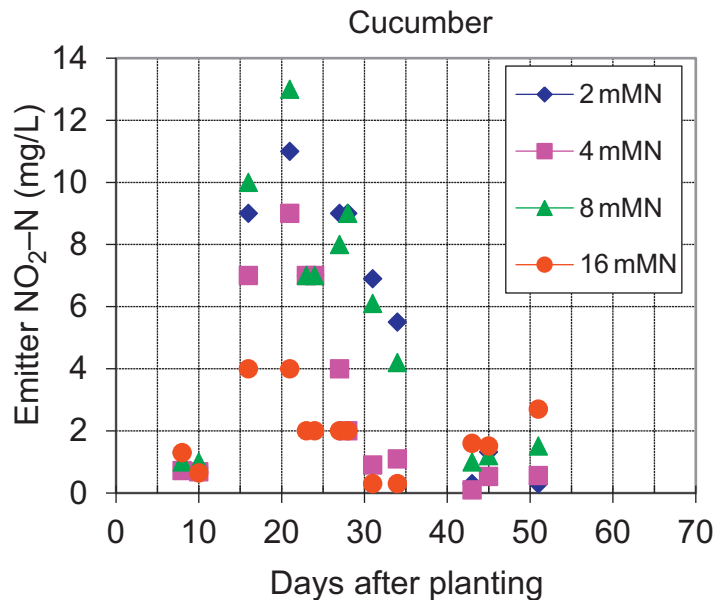


FIGURE 10.5

Nitrite (NO_2^-) concentration in recycled solution of cucumber plants grown in perlite 2 as a function of time and concentration of N in replenishment solution ($\text{NH}_4\text{:NO}_3$ ratio = 30:70). Planting date 1/3/04.

Derived from Bar-Yosef, B., Cohen, Y., Kritzman, G., Matan, E., 2005. Integrated Crop Response to Solution Recycling, Greenhouse Climate and Root Disease: Pepper, Roses and Cucumber. Report Submitted to the Chief Scientist, Ministry of Agriculture, Israel.

nitrate reduction or incomplete nitrification. Some authors studied nitrite dynamics in growing media and found sometimes considerable concentrations (Daum and Schenk, 1998). Given the conditions of frequent periods of stagnant water (between irrigation events) and the unavoidable formation of biofilm in the IR, it is highly presumable that denitrification is taking place and nitrite is formed. Kierkels (2010) cited growers and consultants who repeatedly analyzed NO_2 in drip irrigation water, resulting in problems with crop growth. However, given the abundant presence of microorganism, nitrite is rapidly oxidized to nitrate. Nevertheless, both the presence as effect of this ion might be highly underestimated.

10.3.2 DISSOLVED ORGANIC CARBON

Organic carbon (OC) accumulation is important as it serves as substrate for various microorganisms and affects cations speciation due to complexation with organic acids. Microorganisms consume oxygen and thus compete with plants on dissolved O_2 , but well adapted beneficial populations compete with root pathogens on the available OC and thus suppress their potency (see Chapter 5: Root Zone Microbiology of Soilless Cropping Systems). The development of the OC accumulation was studied in various crops and substrates in Israel. It can be seen that in systems with EC_{thr} around 4 dS/m discharge was small, and therefore, OC concentration exceeded the one in lower EC_{thr} (e.g., 2.1 dS/m) which did not differ meaningfully from fresh water OC concentration (Table 10.10).

Note that the values reported are the mean OC concentration during the main growing season, as fluctuations with time after the first month of growth were small and within the experimental error ($\pm 20\%$ of the mean). The steady state OC concentration reflects the crop-specific balance between root excretion and microbial consumption of C. Measurements in pepper indicated that OC concentration in winter (greenhouse heated to 16°C) surpassed the concentration in spring

Table 10.10 Steady-State^a Organic Carbon (OC) Concentration in Recycled Solution of Various Crops and Substrates at Different Growth Seasons in Israel

Crop	Substrate	EC_{thr}^b (dS/m)	Season	OC mg/L	Season	OC mg/L
Pepper	Stone wool	4.0	Winter	38	Spring	11
Muskmelon	Perlite	2.5	Spring	11		
		4.5	Spring	21		
Gypsophila	Perlite	2.3	Spring	14	Summer	13
		2.9	Spring	18		
Lettuce	Stone wool	2.1	Autumn	8		
		2.8	Autumn	16		
Cucumber	Stone wool	2.1	Spring	15		
		4.8	Spring	33		

EC_{thr} , Threshold electrical conductivity. OC concentration in fresh water = 10–13 mg/L C.

^aIn all presented systems solution samples were taken in 2 weeks' interval along the growth season. Between 1 month and harvest the OC concentrations fluctuated with time by approximately $\pm 20\%$ of the presented mean (Bar-Yosef, Unpublished data).

^b EC_{thr} is the threshold EC at which recycled solution is discharged to avoid exceeding this value.

(mean daily temperature $\sim 27^{\circ}\text{C}$) probably due to reduced microbial activity in winter (Table 10.10). The crop effects can be evaluated only for the same substrate and growing season, so meaningful crop effects from the presented results are hard to obtain.

Data on OC constituents in recycled solutions is meager. Original data for pepper (same crop as in Table 10.10) show that citrate and oxalate constitute more than 90% of analyzed organic anions and that concentrations in solution flowing into the substrate and out of it are practically equal (Fig. 10.6).

The concentration of citrate ($\sim 5\text{ }\mu\text{M}$) is similar to the molar concentration of Zn and Mn in the same solution; therefore complexation might influence the mobility and adsorption of these microelements in the system. The OC concentration in the recycled solution of pepper was shown to be $\sim 20\text{ mg/L C}$ (Table 10.10) which exceeds the $\sim 0.5\text{ mg/L C}$ contributed by the organic acids presented in Fig. 10.6. This means that the OC in recycled solutions is composed mainly of more stable carbon compounds, probably stemming from microbe and root debris.

It is noted that in all analyzed recycled solutions (Table 10.10), dissolved O_2 concentration was higher than 90% of the equilibrium concentration with air at the given solution temperature ($\sim 8\text{ mg/L O}_2$, data not presented). This was possible under regime of 8–16 irrigations per day exercised in the reported studies, which prevent formation of stagnant solutions in the substrate.

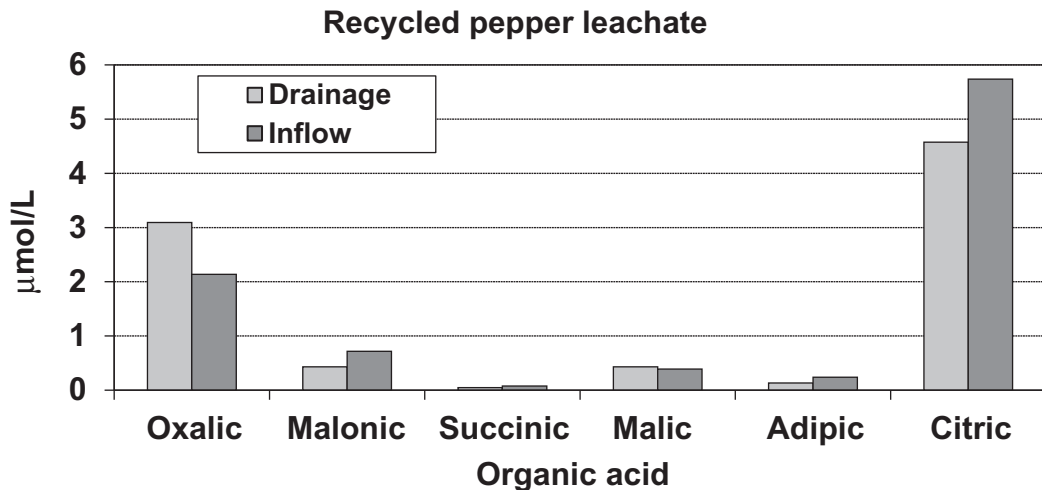


FIGURE 10.6

Concentration of organic acids found in recycled solution of pepper grown on stone wool at Bet Dagan, Israel. The total organic carbon concentration is presented in Table 10.10. Solutions were sampled in emitter (inflow) and in substrate drainage in late autumn (30 days after planting). Formulae of acids yielding concentration $>0.5\text{ }\mu\text{M}$ are $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$ (oxalic), $\text{CH}_2(\text{COOH})_2$ (malonic), and $\text{H}_3\text{C}_6\text{H}_5\text{O}_7\text{H}_2\text{O}$ (citric).

Schwarz, D., Erner, Y., Bar-Yosef, B. (1999, Unpublished data).

10.3.2.1 Microflora accumulation

In pepper, the concentrations of total bacteria and *P. fluorescens* in circulated solutions were two orders of magnitude smaller in pumice than in stone wool and in gypsophila two orders of magnitude smaller in pumice than in perlite (Table 10.4). In muskmelon ($EC_{thr} = 4.5$ dS/m), the difference in microbe concentration between stone wool and perlite was relatively small and inconsistent (Table 10.11).

The substrate effect plausibly stemmed from different pore size distributions, determining the filtration capacity of the medium but this effect should be further investigated before practical conclusions can be drawn. Increasing threshold EC had no clear effect on microbe population, but in muskmelon on perlite, the total bacteria concentration was lower than in pepper and gypsophila (Table 10.11). Comparison between microbe accumulation in summer and spring (muskmelon on perlite, Table 10.11) showed small, insignificant differences. No disinfection was used in any of the aforementioned experiments; despite this, the incidence of plant mortality was negligible in all three crops.

10.3.3 DISCHARGE STRATEGIES

In open drainage systems, problems connected to the accumulations of substances are simply handled by implementing some specific leaching fraction. In closed systems there is a risk associated with accumulations and circulation of undesired substances. The considerations to discharge or not to discharge is a matter of considering on the one hand data the crop response to Na and Cl near threshold values, while on the other hand considering alleged effects from accumulation of metabolites or root exudates. Beerling et al. (2017) showed in a series of experiments with cucumber and sweet pepper that long-term crop growing in a completely closed system, with adequate nutrient supply, good water quality, and disinfection with UV treatment, did not show any yield reduction.

Table 10.11 Concentration of Total Bacteria, *Pseudomonas fluorescens*, and Total Fungi [on Potato Dextrose Agar (PDA)] in Circulating Solution of Given Crops Grown in Perlite and Stone Wool (RW)^a

Crop	EC_{thr} (dS/m)	Substrate	Season	Total Bacteria (cfu/mL)	<i>Pseudomonas fluorescens</i> (cfu/mL)	PDA Fungus (cfu/mL)
Pepper	4.5	Perlite	Spring	1.9×10^6	2.3×10^5	—
	2.5	Perlite	Spring	5.4×10^6	2.9×10^5	—
Gypsophila	2.3	Perlite	Summer	0.9×10^6	1.0×10^5	160
Muskmelon	4.5	RW	Spring	4.0×10^4	2.6×10^4	260
	4.5	Perlite	Spring	7.0×10^4	1.1×10^4	66
	2.0	Perlite	Spring	2.0×10^5	1.8×10^4	310
	4.5	Perlite	Summer	2.4×10^5	1.2×10^3	2080

^aUnless otherwise stated solutions were sampled 2–3 months after planting. Analyses were done by Dr. Kritzman (see Bar-Yosef et al., 2003a,b, 2004a, 2005). Total bacteria count in fresh water was 6.2×10^4 cfu/mL. No disinfection treatments took place in presented experiments.

Nevertheless, it is common practice that discharge of solution is nearly always done in practice to minimize risk. Raaphorst et al. (2014) found that the quantity of discharge in Dutch greenhouse growers differs largely among crops and greenhouses with the same cultivation system: the discarded amount varied from almost zero to almost 1200 m³/ha/year (Table 10.12). Beerling et al. (2014) assessed in a joint fact-finding process with Dutch growers and other stakeholders the obstacles for reusing nutrient solutions and reported that these are (1) problems with filter back-flush, (2) salinity build-up, (3) risk of disease spreading, (4) nutrient imbalances, (5) the fear for growth inhibition.

10.3.3.1 Filter back-flush water

Discharge of filter back-flush water is one of the main emission routes; for the frequent cleaning of sand filters, usually 100–150 L/ha/day recirculation water is used, which adds up to a yearly discharge of ~30–50 m³/ha/year, depending on crop and cultivation system. The prevailing reason for discharging back-flush water is ignorance of the importance of this emission route. Filter back-flush water can safely be reused, as has been shown by several commercial producers, and could be regarded as a typical “quick win” measure, with high impact and minimal effort and costs (Raaphorst et al., 2014). Another way of decreasing the amount of discharged flush water is installing automatic, self-cleaning filters; these need only 10% of the water for back-flushing compared to sand filters. The costs involved for reusing back-flush water (pipework, clearing out accumulated sediment in return water storage) or for installing an automatic filter, can at least partially be recouped by saving on fertilizers and water (Table 10.12).

Table 10.12 Total Amount of Discharged Nutrient Solution, and the Amount of Back-Flush Water Contributing to the Discharge, for 11 Commercial Greenhouses in 2011, and the Estimated Quantity of N and P Losses by This Discharge and the Costs

Greenhouse	Crop	Total Discharged	Discharged Back-Flush	Discharged N	Discharged P	Total Costs
		m ³ /ha/year	m ³ /ha/year	kg/h/year	kg/ha/year	€/ha/year
1	Rose	1160	484	244	43	1209
2	Rose	1126	0	236	42	0
3	Gerbera	32	7	8	1	18
4	Gerbera	310	172	78	14	431
5	Gerbera	167	56	42	8	139
6	Gerbera	94	28	24	4	69
7	Cucumber	1012	173	326	56	432
8	Pepper	489	0	144	27	0
9	Pepper	414	406	122	23	1014
10	Pepper	26	0	8	1	0
11	Tomato	37	0	16	2	0

Extrapolated from data from Raaphorst et al. (2014).

10.3.3.2 Salt accumulation

The accumulation of Na or Cl in the recirculation water is an often mentioned reason for discharge to prevent salinity stress. The issue of accumulation has been treated in [Section 10.2.3](#) and is closely connected with the water quality used. It is to be noted that although mentioned as main driver for decisions for discharge, the growers' decision to discharge is often quite below $C_{\text{gNa}}^{\text{max}}$, as was found in Na concentrations in discharged water ([Baltus and Volkers-Verboom, 2005](#)).

10.3.3.3 Diseases

An often mentioned reason to stop reusing the nutrient solution is the occurrence of certain root zone diseases or viruses, such as *Clavibacter michiganensis* in tomato and green mottle mosaic virus in cucumber ([Van Paassen and Welles, 2010](#) cited in [Beerling et al., 2014](#)). Occasionally, fear of the spread and accumulation of pathogens is mentioned as a reason for not reusing the condensation water, because in this water, spores of airborne diseases can be found.

Disinfection of the recirculating water can be done effectively by heat treatment or UV radiation ([Van Os, 2010](#)), so fear for spreading diseases is generally not justified, provided that enough attention is paid to maintenance and adequate capacity of the disinfectant. A failing disinfectant usually causes significant emission because a grower will not use return water that is not disinfected, and storage capacities are usually not large enough.

10.3.3.4 Imbalances in nutrient composition

The prevailing practice is that a maximum of 50% of the nutrients (expressed as EC) originates from the return water, resulting from the growers maximize the set point for mixing drainwater at an EC of 1.5 mS/cm. This leaves room for adjustment of imbalances between nutrients. In practice, much lower percentages are often applied because many growers use fixed settings for the amount of return water, irrespective of the crop need. As a result, the return water storage is often not large enough and therefore return water has to be discharged. A solution to this problem is to increase the fraction of nutrients from return water to a higher percentage, or even better, to apply “dynamic settings” for the composition of irrigation water. With dynamic settings, the fraction of nutrients from the return water depends on the total amount of nutrients in the irrigation water, so when the EC of the irrigation water is increased, also more return water can be used. For this strategy, regular nutrient analyses are indispensable. [De Boer-Tersteeg et al. \(2012\)](#) have shown that this strategy is successful and safe.

10.3.3.5 Growth inhibitors

There is a prevailing opinion among grower's that long-term recycling of the nutrient solutions will affect crop development negatively. This growers experience and feelings are paramount and often result in the decision to refresh the nutrient solution. [Van Os et al. \(2012\)](#) demonstrated that indeed growth inhibition in rose could be caused by the accumulation of not-yet identified organic substances, which can be eliminated successfully by using advanced oxidation (UV with peroxide) of the recirculating water.

It is also common practice to not recirculate at the start of a new cultivation, in fruit vegetables in particular, because of fear for growth inhibition ([Lee, 2011](#)), though there is no scientific ground for this concern.

Very little is known about effects of permitted deviation from the mean threshold EC ($\Delta C_{up} + \Delta C_{low}$, Eq. 10.9) on plant performance in closed loop IRs. Salinity stress (obtained by a large permitted deviation, or amplitude) has been shown to improve fruit quality (Mizrahi et al., 1988; Plaut and Meiri, 1997) but possible toxicity and uptake inhibition effects might delineate this advantage and minimal deviations around EC_{thr} may be preferred. Kramer et al. (2002) investigated the EC amplitude effect in roses in the Arava valley in Israel where fresh water EC was 2.6 dS/m. Four treatments were investigated: DR4 and DR3 had EC_{thr} of 3.7 dS/m and amplitudes of 1.7 and 0.3 dS/m, respectively. Treatment Dr1 had EC_{thr} of 2.8 dS/m and amplitude of 0.3 and treatment Dr2 had EC that fluctuated between the maximum EC of treatment 3 and the minimum EC of treatment 1 (Fig. 10.7).

The treatments were applied for a year and results showed no significant difference in cumulative fresh flower weight and ET between low and high amplitude treatments (Dr3 and Dr4). However, water and N discharge were 35% smaller in the high than in the low amplitude (see the table below Fig. 10.7). The reason is that in order to dispose a certain quantity of salts, a smaller volume is needed when salt concentration is higher.

When reducing the mean salinity level in solution (treatments Dr1 and Dr2), the yields and ET significantly increased but again the high amplitude treatment resulted in significantly smaller discharge in comparison with low amplitude. The “large saw tooth” (high amplitude) approach requires fewer discharge events (points of sharp decline in EC, Fig. 10.7); therefore its operation is expected to be cheaper than “small saw tooth” systems.

10.3.4 SUBSTRATE AND SOLUTION VOLUME PER PLANT

The substrate and solution volumes per plant affect root growth (see earlier discussion and Chapter 14: Growing Plants in Soilless Culture: Operational Conclusions), determine the buffering capacity of the closed IR to water, nutrients, and oxygen, and influence the salt accumulation rate in the substrate. Reduced root mass (R , g fw root/m² substrate) decreases the water and nutrients uptake rate (Q_W , Q_N , respectively, Eq. 10.14, g/m² substrate/h) and may also reduce the root cytokinin and abscisic acid synthesis and supply to the canopy (Aiken and Smucker, 1996).

$$Q_N = F_N R; \quad Q_W = J_W R \quad (10.14)$$

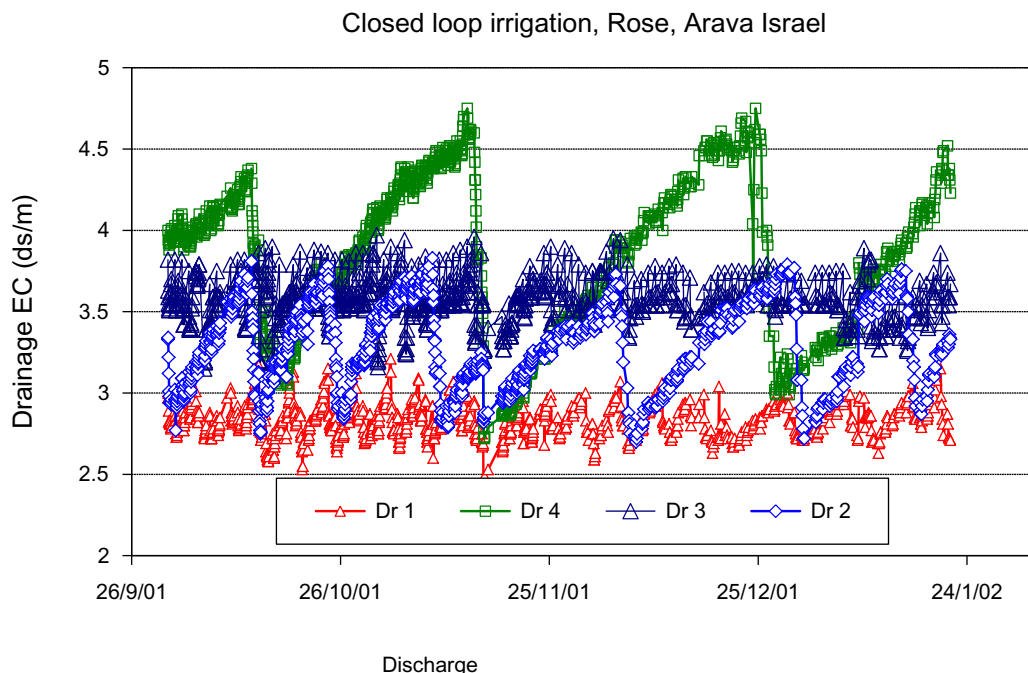
The flux terms (F_N and J_W) are commonly described by the Michaelis–Menten equation (10.15) and the water flux of uptake equations (10.16):

$$F_N = \frac{F_{maxN} C}{K_{mN} + C} \quad (10.15)$$

$$J_W = K_{sr}(\varphi_s - \varphi_r) \quad (10.16)$$

where F_{maxN} is the maximum F value for N, K_{mN} is the uptake efficiency constant (L/g N), K_{sr} is the combined soil–root resistance to water permeability into root (g H₂O/bar/h), and φ_s and φ_r (bar) are water potential in soil at the root surface and inside the root, respectively.

When the target uptake rate (Q_{target}) is known, the minimal root mass which is required to sustain it under maximum flux conditions ($F = F_{max}$) is $R_{min} = Q_{target}/F_{max}$. If actual root weight is less than R_{min} , modifying the fertigation regime will not facilitate an increase in uptake rate and the



Tr	FWt	ET	Water	N
	g/m ²	L/m ²	L/m ²	g/m ²
Dr1	9070	1311	1750	131
Dr2	9284	1260	791	79
Dr3	8111	1239	678	68
Dr4	7997	1221	428	43

FIGURE 10.7

Effect of EC saw tooth amplitude ($\Delta C_{up} + \Delta C_{low}$, Eq. 10.10) on total rose cut flower (var. Long Jaguar) weight (FWt), cumulative ET and discharge of water and N during the growth period 31/12/99–31/8/01. The substrate was tuff, fresh water EC was 2.6 dS/m, and irrigation frequency 8–12 per day. ET, Evapotranspiration; EC, electrical conductivity.

only option to enhance uptake is to increase root mass, for example, by increasing substrate volume per plant. Michaelis–Menten constants for some ions and crops are presented in Table 10.13.

For example, in order to satisfy a nitrate Q_{target} (Q_{NO_3}) of 500 mg N/m² substrate/day (=50 mg N/m²/h for 10 hours of uptake per day) by a tomato plant with $F_{max} = 25 \mu\text{g N/g fw root/h}$, $R_{min} = 50/0.025 = 2000 \text{ g fw root/m}^2 \text{ substrate}$. If the substrate is 0.2 m deep this root volume occupies $\sim 2 \text{ L}$ (assuming root density of 1 g/mL) out of 200 L substrate. Estimating the minimal root

Table 10.13 Michaelis–Menten Constants^a of Selected Ions and Crops Grown in Stirred or Unstirred Solution Culture

Crop	Ion	$F_{\max}$ (mol/cm/s)	K_m (μ M)	System
Maize	NO ₃	1.16	10	Well stirred
	H ₂ PO ₄	0.50	3.0	
	K	5.02	16.	
Soybean	H ₂ PO ₄	0.10	2.0	Well stirred
Tomato ^a	NO ₃	5.1	258	Well stirred
Tomato ^b	NO ₃	4.6	400	Stirred
	NH ₄	20.3	3000	
	H ₂ PO ₄	2.2	350	
Tomato	NO ₃ –NH ₄	11.0	3000	Sand
	H ₂ PO ₄	1.8	320	Rock wool
	K	3.0	1000	Aeroponics
Pepper	NO ₃ –NH ₄	14.0	550	Unstirred
	H ₂ PO ₄	17.0	25	

Michaelis–Menten constants of plants grown in stirred and unstirred cultures (Scaife and Bar-Yosef, 1996).

^aUnits conversion: for root owning radius of 0.025 cm and bulk density of 1 g fw/cm³ root, $F_{\max}$ of NO₃ = 10×10^{-13} mol N/cm root/s is identical to 25.8 μ g N/g fw root/h; K_m = 500 μ M NO₃ is identical to 7 mg/L NO₃–N.

^bScaife and Bar-Yosef (1995).

weight which is necessary to sustain a given transpiration rate is more difficult because the J_w parameters (Eq. 10.19) are not available in the literature. The role of the Michaelis–Menten parameter K_m in planning recirculation regimes will be discussed below.

Water volume per plant determines the rate by which a nutrient concentration in solution varies with time. For the aforementioned 200 L substrate volume and predawn θ of 0.6 v/v, daily uptake of 500 mg N/m² substrate reduces C_N by ~ 4 mg/L. This is true of course if no N is added via the water replenishing ET. For $f_s = 0.3$, this uptake rate is equivalent to 150 mg N/m² ground. For tomato consumption rate in the high range (~ 350 mg N/m² ground, Reshef et al., 2003), the daily reduction in C_N is 9.7 mg/L N. By applying Eqs. (10.17) and (10.18) one can estimate the expected reduction in N uptake rate due to this decline in C_N and decide what should be the frequency of N replenishment prior to exceeding the permitted deviation from Q_{target} . If f_c is 0.125 (20 cm substrate width in a bed of 160 cm, instead of 48 cm substrate above) the daily change in C_N would be 23.3 mg/L N.

The faster decline in C_N (and faster increase in salinity) associated with decreased root volume suggest that greater substrate and water volumes per plant would be better. Other factors however suggest smaller substrate and water reservoir volumes, among them: (1) economic reasons such as substrate acquisition and disposal expenses; (2) reduced disinfection costs; and (3) smaller drip IRs are required.

10.3.5 EXPERIMENTAL DATA OF CROP RESPONSE TO SUBSTRATE VOLUME

Pepper (*cv.* Cuby) grown in a closed IR in Besor, Israel, was subjected to EC_{thr} of 2.7 or 4.0 dS/m. Total and export quality yields were highest in 0.50 m wide and 0.2 m high perlite beds (16.6 L/pL substrate), followed by 0.2 and 0.1 m wide beds (6.7 and 3.3 L/pL). The 0.2 and 0.1 m wide treatments did not differ significantly from each other (Fig. 10.8).

Xu and Kafkafi (2001) compared pepper growth response to container volumes of 9, 18, and 33 L/plant and in agreement with the above result reported that growth and fruiting were maximal in the intermediate volume. Substrate width in the Besor experiment had no significant effect on

Width	Exp yld	BER	Fruit wt
cm	kg/pL	%	g/frt
50	2.68 a	39	164
20	2.08 b	41	162
10	2.10 b	41	164

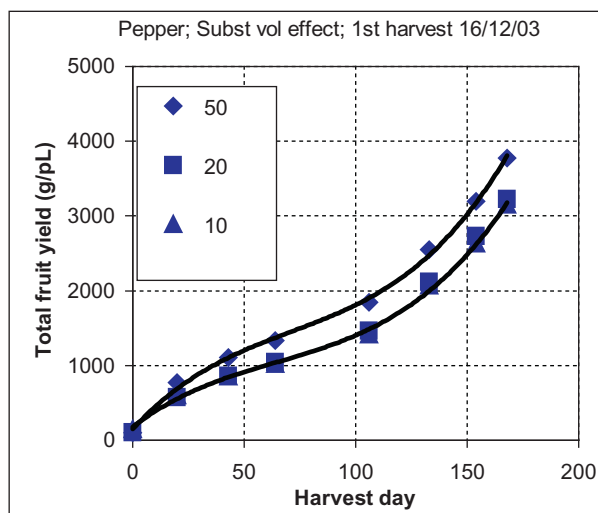


FIGURE 10.8

The effect of perlite width (10, 20, or 50 cm) on total pepper (*cv.* Cuby) fruit yield as a function of time (figure), and high-quality fruit yield (Exp yld), BER incidence and average single fruit weight (Fruit wt) (Box). The substrate volumes per plant were 3.3, 6.7, and 16.6 L/pL, respectively; substrate height was 0.20 m, EC_{thr} = 4.0 dS/m and irrigation frequency was 10–16 per day, depending on season. First harvest day was 16/12/03. Experiment was conducted in Besor, Israel (Bar-Yosef et al., 2004a). BER, Blossom end rot; EC_{thr} , threshold electrical conductivity.

the incidence of BER or the average fruit weight at the beginning of June. The response was similar under the two studied EC_{thr} levels, therefore only the higher EC_{thr} is presented. The data (Fig. 10.8) indicate that the advantage of the 50 cm wide bed became apparent ~ 60 days after start of harvest.

Root excavation carried out in the Besor experiment on June 10 showed that root weight (excluding >4 mm diameter roots) in 0.5, 0.2, and 0.1 m wide substrate decreased from 21 to 17 and 9 g DM/pL, respectively. This order is compatible with the treatments effect on fruit yield thus indicating that root growth restriction might have limited fruit production.

The substrate volume effect on crop performance in closed IRs was also studied in tomato. As above, yield in the 0.5 m wide perlite exceeded the yield in 0.2 or 0.1 m substrates, the latter two not differing significantly from each other (Table 10.14).

The yield increase stemmed from more and larger fruits, and lower BER incidence. Increasing the substrate width elevated P and K (but no other nutrient) concentrations in leaves in late November (0.97%, 1.06%, and 1.12% P and 5.0%, 5.2%, and 5.5% K, respectively), and increased transpiration rate (ET) during the spring (1.6, 2.8, 3.8 mm/day, respectively). The effect of substrate width on leaf nutrient concentrations in the spring was negligible, so it may be concluded that the yield reduction in the 0.1 and 0.2 m wide substrates resulted from water stress in the spring.

In the cited pepper experiment, the maximum ET was 2 L/pL/day. During the noon hours the corresponding ET was 0.5 L/pL/h. In the 10 cm substrate (3.3 L/pL), water quantity after the rapid drainage was ~ 1.5 L/pL ($\theta_{1/2 \text{ hour}} = 0.45$ v/v). Removal of additional 0.5 L water by the plant reduces θ to a value which strongly decreases the perlite's unsaturated hydraulic conductivity, thus jeopardizing water and nutrients transport to roots. In 0.2 m substrates, this problem did probably not occur.

10.3.5.1 Effect of substrate type

Substrates and their physical and chemical properties are reviewed in various chapters. The most important substrate properties in closed loop IRs are physical characteristics such as water infiltration rate, drainage rate, and, where applicable, microbes filtration capacity and chemical properties such as the input of unwanted compounds and fixation. In particular, the use of organic substrates

Table 10.14 Effect of Substrate Width¹ on Tomato Total and Marketable Yields, Blossom End Rot (BER) Incidence, and Large Fruit Yield (Besor, Israel)²

Substrate Width cm	Accumulated Yield		BER	Fruit Size
	Total	Marketable		>67 mm
	kg/pL			kg/pL
10	9.55 ^b	6.00 ^b	1.68 ^a	3.12 ^b
20	9.46 ^b	6.11 ^b	1.58 ^{ab}	3.22 ^b
50	10.10 ^a	6.74 ^a	1.50 ^b	3.63 ^a

¹ $EC_{thr} = 4.5$ dS/m in all treatments; fertigation frequency 10–16 per day, depending on season.

²Reshef et al. (2006).

^{a,b} $p = 0.05$

can be of concern here. Verhagen (1999) showed that the input of coir as growing medium in a closed system may add considerable quantities of Na if not properly treated before use. And, as mentioned in Section 10.2.7.2 Cu, can be strongly bound by organic compounds of peat substrates.

10.3.5.2 Water and nutrients replenishment

Nutrient concentrations in fill solutions (C_f , mg/L) are designed to replenish the net loss due to crop nutrient consumption (U) and system disposal (D). For systems with a large buffer reservoir, this replenishment can be once per time (day), though for the majority of the systems with frequent irrigation throughout the day and the whole cropping period, there will be continuous replenishment of the irrigation tank, which will be made up partly by (whether or not) disinfected drainage water and partly fresh water. The tank will be topped up constantly to a predefined EC and pH set point, whereby the rate of drainage to be mixed is either fixed at a certain volume or at a certain EC value. The refill solution should be such as to reach and maintain a predefined target concentration C_g in the root environment. The added water volume (V_f , L/m² ground) multiplied by C should equal $U + D$ (mg/m² ground) (Eq. 10.17)

$$C_f V_f = U + D; \quad C_f = \frac{U}{V_f} + \frac{D}{V_f} \quad (10.17)$$

When $D = 0$, the quotient U/V_f is the time-averaged nutrient concentration in the transpiration stream, which can be considered as the uptake concentration of each individual nutrient i (Section 10.3.5.2) (C_{Ui}). In practice, U and D are not measured daily; instead U is estimated from a crop-specific target nutrient consumption curve, and D is evaluated using the estimated nutrient concentration in the drainage and the measured disposed drainage volume. Over- and underestimation of C_f may cause accumulation or depletion of nutrients in the recirculated solution, respectively; therefore chemical analysis of the solution must be performed at least every 2 weeks. Under conditions of (1) $D = 0$ and (2) real $C_{Ui} = C_f$, the concentration of the nutrient in the recirculated solution is maintained at C_g .

The target nutrient concentrations in solution (C_{gi}) for each nutrient are defined as a concentration providing an uptake rate equaling the target nutrient consumption rate by the crop at the specific growth stage. The target nutrient consumption curve is defined as a unique time function coinciding with optimal yield and product quality. Currently, these functions for individual ions are to be obtained experimentally (Bar-Yosef, 1999). There are striking differences in the requirements for individual nutrients that stem from basic plant physiological uptake and distribution processes in plants, which have relevant practical consequences. The most optimal nutrition of the plant is determined by the required concentrations in the root environment. It has been demonstrated that some elements, require accumulation (relative to the supplied concentration) at the root surface to achieve sufficient uptake (Voogt, 1988a,b; Voogt and Bloemhard, 2013), while other nutrients can be depleted almost completely (Voogt and Sonneveld, 2004). The demand of a certain crop cannot be deducted from the uptake concentrations only. Some crops could deplete a nutrient solution down to very low concentrations for a specific nutrient. This phenomenon not necessarily means a demand of the crop. This can be illustrated by studies with Boron in rose, strawberry, and sweet pepper. Rose and strawberry could deplete B from the nutrient solutions in the root environment strongly and accumulate B in the transpiring older leaves, [De Kreij et al., 1985; van Bastelaere

(personal communication)]. This however may result in toxic levels as can be illustrated by the results of an experiment with Rose “Motrea” (Fig. 10.9).

Surprisingly, this was affected by grafting. With rootstock, the uptake and toxicity were much more a problem than without. The opposite occurs with sweet pepper, as for this crop relatively high B levels in the root environment are needed, to achieve sufficient uptake (Voogt and Bloemhard, 2013). Even at the required concentrations in the root environment, deficiency symptoms often occur.

The accumulation factor (AF_i) describes the rate of accumulation in the recirculating solution over time in relation to the supply (Eq. 10.18)

$$AF_i = \frac{C_{gi}}{C_{fi}} \quad (10.18)$$

AF_i can be used to characterize the differences in nutrient uptake and demands of crops. If AF_i is <1 , plants can easily absorb this nutrient. If this is the case for micronutrients B, Mn, Cu, or Zn then this could lead to toxicity. If AF_i is >1 , the uptake is more difficult and higher concentrations in the root zone than supplied are necessary to enforce sufficient uptake.

This effect is clearly demonstrated by Voogt and Sonneveld (1996) for K, Ca, and Mg with tomatoes and with carnations both grown in stone wool in full closed systems (Table 10.15). AF_i is higher for Ca and Mg than for K. For K, the AF_i increases with increasing K supply, while for Ca and Mg, this factor decreases with the supply, except for Ca with tomato.

This is also demonstrated in Fig. 10.10, showing that for K at 6.5 mmol/L C_f and C_g are equal, hence $AF = 1$. However, for Ca and Mg, even at the lowest supply in this trial, AF is already 2. The results can be explained by the different uptake mechanism of these elements, as K uptake is characterized by a strong active mechanism, in contrast to Ca which is absorbed only passively by the transpiration flow (Marschner, 1995). It should be noted that these trials were carried out at EC levels of 3.0 mS/cm in the root environment with supply of K, Ca, and Mg in the form of nitrate fertilizers accordingly to keep the EC at target and not affecting other nutrients.

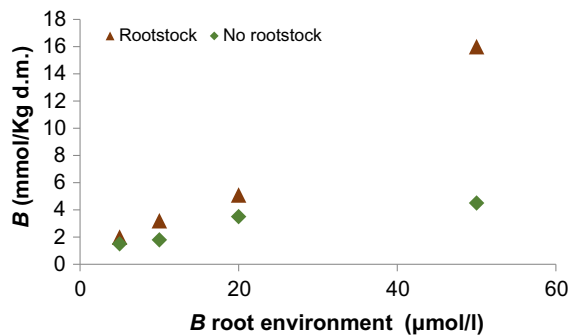


FIGURE 10.9

Effect of B concentration in the root environment and the B content in the foliage of rose “Motrea” on rootstock “Natal Briar” and without rootstock, grown in hydroponics for 12 weeks exposed to a range of constant B concentrations.

Voogt, W., 1995. Unpublished data.

Table 10.15 Average Concentrations Supplied (C_{ad}) and in the Root Environment (C_g) Both in mmol/L, and the Accumulation Factor (AF_i), of K, Ca, and Mg, Recorded in Trials With K:Ca:Mg Ratios With Tomato and Carnation in Recirculating Nutrient Solutions

Treatment	K			Ca			Mg		
Tomato	C_{ad}	C_g	AF	C_{ad}	C_g	AF	C_{ad}	C_g	AF
1	3.2	1.5	0.5	2.9	13.6	5.1	0.6	2.1	3.5
2	4.7	4.1	0.9	2.3	10.9	5.0	0.7	2.2	3.6
3	6.3	10.2	1.7	1.7	6.0	3.6	0.7	1.6	2.4
4	7.5	16.9	2.3	1.0	2.1	2.1	0.7	1.1	1.7
Carnation									
1	2.7	4.8	1.8	2.4	4.2	1.8	0.4	1.5	3.8
2	3.2	6.0	1.9	1.9	3.2	1.7	0.5	1.6	3.2
3	4.3	8.8	2.0	1.0	1.9	1.9	0.7	1.6	2.3
4	5.1	10.0	2.0	0.6	1.6	2.7	0.8	1.6	2.0

Data from Voogt, W., Sonneveld, C., 1996. Nutrient management in closed growing systems for greenhouse production. In: *Plant Production in Closed Ecosystems & Automation, Culture and Environment*. Kluwer Academic Press, Dordrecht, pp. 83–102.

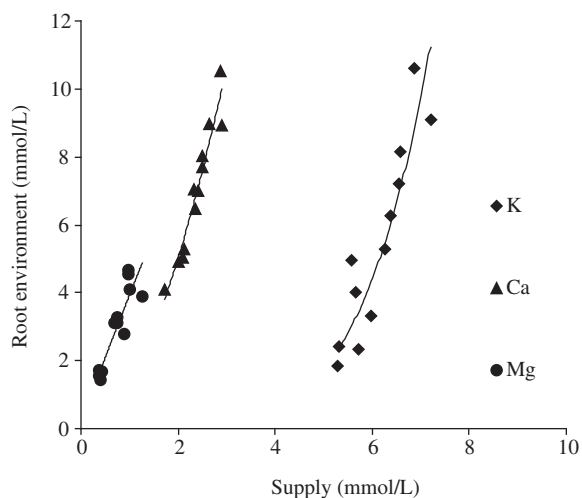


FIGURE 10.10

Relationship between the average K, Ca, and Mg concentrations supplied and the average concentrations in the root environment (stone wool slab) of a long-term tomato crop grown in stone wool with reuse of drainage water

Derived from Voogt, (2002).

The importance of choosing the right cation ratio for the nutrient solution is illustrated by the results of trial with a wide range of K/Ca ratios at different Mg concentrations (Fig. 10.11). The results show that not only the absolute concentration of a nutrient is important, but also the mutual ratios. Increasing K/Ca ratio resulted in a clear improvement of shelf life as well as factors affecting taste: sugar and acid content (not shown). However, BER was increased at the same time and—as opposite to BER—the incidence of gold speck on tomato fruits. This phenomenon is connected

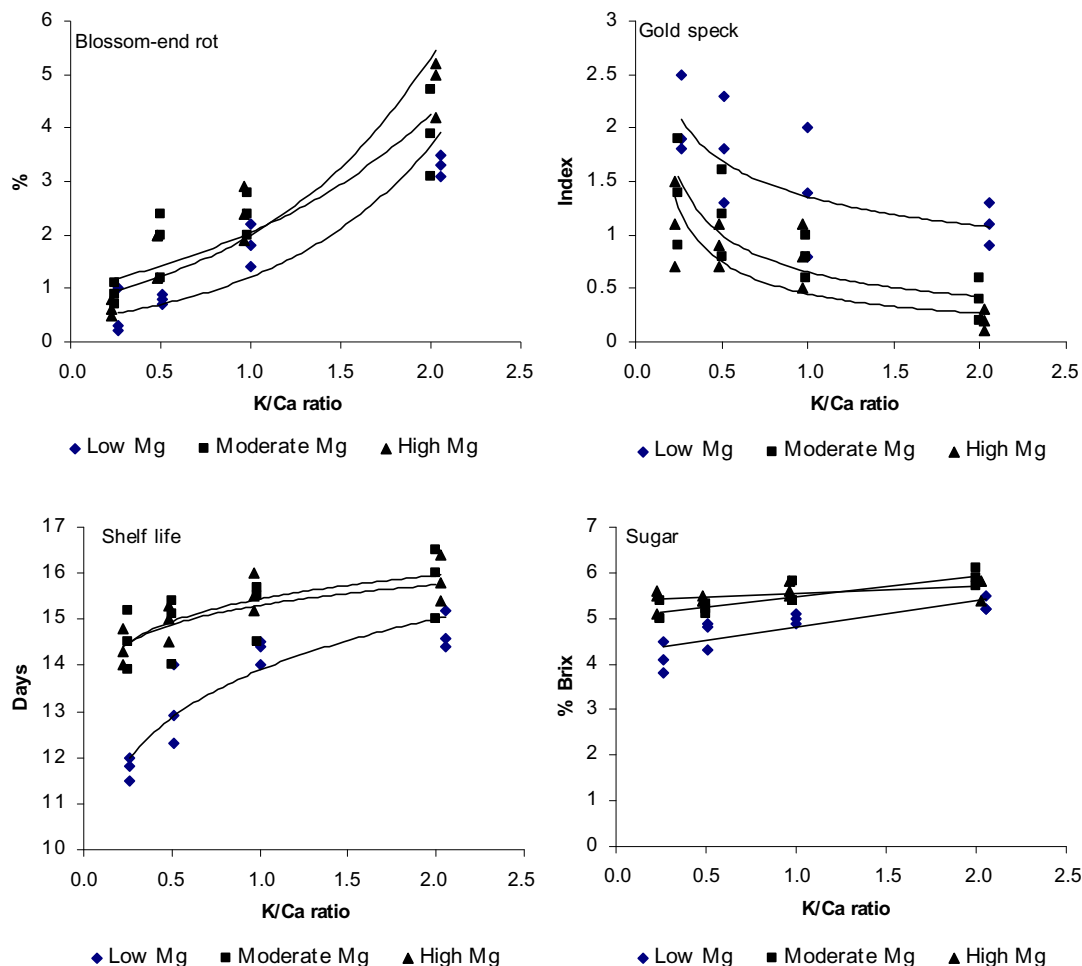


FIGURE 10.11

Effect of the K/Ca ratio at three Mg levels in the root environment on the percentage of BER fruits, the gold speck index, shelf life, and sugar content of tomato fruits grown in stone wool grown tomato.

Adapted from Voogt, W., 2002. Potassium management of vegetables under intensive growth conditions. In: Pasricha, N.S., Bansal S.K. (Eds.), *Potassium for Sustainable Crop Production*. International Potash Institute, Bern, pp. 347–362.

with Ca accumulation and Ca-oxalate crystallization (De Kreijl et al., 1992). As in this experiment, the increase in K is connected with decrease in Ca, it is complicated to distinguish whether K or Ca caused the effects mentioned. On the other hand, the Mg concentration interferes with the K/Ca effect, increasing Mg concentration enhances the incidence of BER and reduces gold speck incidence. For BER this was confirmed recently by the results published by Bar-Tal et al. (2017).

Rather than experimental evidence, the optimum concentrations and supply might be deducted from theoretical crop models. When inserting C_{trg} in the Michaelis–Menten equation (10.13), a unique flux is obtained. Multiplying this flux by the current given (or estimated) crop root mass yields an uptake rate value that should match the target uptake rate by the plants at the appropriate growth stage. Apparently C_f (derived from quantity factors) is not necessarily equal to C_{trg} (derived from intensity factors) but the closer these values are to each other, the more balanced and stable is the recirculation system. The target concentration C_{trg} can be used as long as the nutrient quantity in the system ($C_{\text{trg}} \times V_T$) exceeds the current daily demand by the crop multiplied by a confidence factor f_c ($f_c \geq 1$). When this restriction is not met, and V_T satisfies the ET demand and cannot be increased, the nutrient concentration in solution should be elevated above C_{trg} despite possible adverse effects on plant development. In practice, C_{trg} can seldom be calculated accurately due to scant Michaelis–Menten parameter estimates (Table 10.13) and root mass data. Consequently, it is approximated as C_f and corrected during production to a steady state concentration.

Fertigation frequency may affect the choice of C_{trg} because it is one of the factors determining the time-averaged nutrient concentration at the root surface (see Chapter 6: Nutrition of Substrate-Grown Plants). At higher frequency the nutrient depletion zones around roots are more often replenished by fresh solution thus the time-averaged concentration in the root zone increases. Depletion zones are formed if the flux of uptake exceeds the rate of element transport from the bulk solution to the root. The transport is driven by diffusion and convection. The first is determined by the effective ion diffusion coefficient in the substrate, θ , and the nutrient concentration gradient along the pathway from the bulk solution to the root. Convection is caused by transpiration and the nutrient concentration in solution. Under similar weather and substrate conditions, a higher C_{trg} is required under low irrigation frequency than high irrigation frequency because a greater concentration gradient toward the root is needed in order to provide a predefined flux of uptake.

In the Michaelis–Menten equation (10.14) the concentration term C is defined at the root surface. When adopting Michaelis–Menten parameter estimates from the literature, it is imperative that the experimental C indeed represents conditions at the root surface and not in the bulk solution (e.g., unstirred nutrient solution) or bulk substrate solution (Table 10.13).

Olsen and Kemper (1968) applied the steady state solution of the general convection–diffusion equation in cylindrical coordinates to describe nutrient concentration at the root surface (C_a , g/L) as a function of its concentration in the bulk substrate solution (C_b , g/L), steady water uptake rate per unit root length (W , mL/h/cm root), the system's effective diffusion coefficient (D_p , cm²/h), root radius (a , cm), and substrate volumetric water content (θ , v/v). Simple algebraic transformation of their original solution yields:

$$C_a = -\beta + \sqrt{(\beta^2 + K_m C_b \alpha)} \quad (10.19)$$

where $\beta = 0.5[K_m - (F_{\text{max}}/W)(1 - \alpha) - C_b \alpha]$ and $\alpha = a_p^{W/2\pi\theta D}$ and $D_p = D_o \exp(s\theta)$. D_o is the diffusion coefficient (cm²/h) in water, s = substrate constant (units 1/ θ), and K_m and F_{max} are the

Michaelis–Menten parameters in Eq. (10.14). Note that $F_{\max}$ is expressed per unit root length rather than per unit root weight.

Salts accumulate at the root surface between successive irrigations if transport rate to root exceeds the rate of uptake by the root. In this case fresh irrigation solution displaces the salts from the root surface and reduces the OP in the root zone.

Salinity is affecting the choice of C_{trg} because it reduces mass flow toward roots and decreases the flux of nutrients uptake due to competition on ion uptake sites, or transporters (Marschner, 1995). A semiempirical function representing inhibited uptake is the competitive Michaelis–Menten equation (Fried and Broashard, 1967):

$$F_{\text{NO}_3} = \frac{F_{\max} \text{NO}_3 \text{ CNO}_3}{K_m \text{NO}_3 + \text{CNO}_3 + C_{\text{Cl}} K_m \text{NO}_3 / K_{m\text{Cl}}} \quad (10.20)$$

The function describes the flux of nitrate uptake, F_{NO_3} , in the presence of a given Cl concentration (C_{Cl} , mg/L) by taking into account the affinity of Cl to the uptake site ($K_{m\text{Cl}}$, L/mg). Other variables in Eq. (10.20) were defined above at Eq. (10.20). According to Eq. (10.i) adding 150 mg/L Cl to a solution of 100 mg/L NO_3^- reduces F_{NO_3} by tomato roots from 8 to 1.3 $\mu\text{g/g}$ root $\text{NO}_3\text{--N}$ ($K_m \text{NO}_3 = 3.6 \times 10^{-3}$ L/g and $K_{m\text{Cl}} = 1.0 \times 10^3$ L/g). Eq. (10.19) has been used in modeling recirculation (Bar-Yosef et al., 2004c) but has yet to be calibrated or validated for a wide range of practical applications.

Competitive uptake is also known between NH_4^+ and Mg^{2+} and K^+ . For background information on this subject see Marschner (1995) and Chapter 6, Nutrition of Substrate-Grown Plants; specific information for hydroponics roses can be found in Lorenzo et al. (2001) and Bar-Tal et al. (2001).

Substrate volume is affecting chosen C_{trg} because reduced root mass calls for increase in flux of uptake if the absorption rate is to be unaltered.

10.3.6 EFFECT OF TARGET N CONCENTRATION ON CROP PERFORMANCE

Bar-Yosef et al. (2000a,b,c,d) found that pepper fertigated 18 times per day gave similar total yield, large fruit yield and unmarketable yield under target N concentration of 70 and 140 mg/L. In this experiment, however, P and K concentrations varied concomitantly with N (Table 10.16). The expected effect of the higher NH_4 concentration in the 140 mg/L N treatment on higher incidence of unmarketable fruits (mainly BER) due to Ca uptake inhibition (Kirkby and Mengel, 1967) could not be confirmed in this study (Table 10.16).

Tomato response to N target concentration was stronger than pepper (Table 10.17) and showed a clear decline in total, marketable, and large fruit yields as N concentration increased from 50 to 150 and 250 mg/L N. The decline in yield occurred despite a small but significant increase in N and P concentration in leaves (data not presented), and decrease in Cl leaf concentration from 4.4 to 2.3 and 2.0 g/100 g. The yield reduction may be related to the observed decline in Ca and Mg concentration in leaves stemming from the increased NH_4 in the fill solutions (fixed $\text{NH}_4\text{:NO}_3$ ratio) (Table 10.17). The reduction in Ca concentration was accompanied by an increase in the incidence of BER (Table 10.17). Even though the EC was identical in all treatments ($\text{EC}_{\text{thr}} = 4.5$ dS/m), the ET in the 250 mg/L N treatment was lower than in the 150 mg/L N (data not presented), compatible with the observed decline in yield in the two treatments.

Table 10.16 Effect of Macronutrient Concentration in Solution (C_{NPK}) (Same Irrigation Regime) on Pepper¹ Yield in a Closed Loop Irrigation System² at Bet Dagan, Israel³

C_{NPK} ⁴	Yield		
	Total	Large	Unmarketable
mg/L N, P, K	kg/m ² ground	kg/m ² ground	%
70, 15, 30	10.9 ^a	5.3 ^a	17.5 ^b
140, 75, 150	11.0 ^a	4.9 ^{ab}	19.1 ^{ab}

¹ var. Cuby; planting 30.8.99 in perlite; first and last harvests 20.12.99 and 15.5.00, respectively.

² The irrigation regime was identical in all treatments.

³ Derived from a larger table by Bar-Yosef *et al.* (2000a,b,c,d).

⁴ $\text{NH}_4\text{:NO}_3$ ratio in fill solution = 1:4; $\text{EC}_{\text{thr}} = 3.5 \text{ dS/m}$; 18 irrigations per day.

^{a,b} $p = 0.05$.

Table 10.17 Effect of Target N Concentrations in Solution on Tomato¹ Yield and Ca and Mg Concentration in Plant Leaves

C_{N} ²	Yield				Content in Leaves	
	Total	Marketable	Large	BER ³	Ca	Mg
mg/L N	kg/m ² ground				g/100 g	
50	10.8 ^a	7.7 ^a	4.2 ^a	1.2 ^c	1.01 ^a	2.78 ^a
150	9.4 ^b	5.7 ^b	2.8 ^b	1.7 ^b	0.81 ^b	2.01 ^b
250	7.6 ^c	3.6 ^c	2.2 ^c	2.2 ^a	0.72 ^b	1.64 ^c

¹ var. 870; planting 07/09/04; 9–18 irrigations per day, depending on season.

² Plants grew in closed irrigation system.

³ greenhouse at Besor, Israel.

^{a,b,c} $p = 0.05$.

Nukaya *et al.* (1991) in a study on the interaction of anions in tomato in NFT found that at equal EC levels, the addition of Cl and proportional reduction of NO_3 enhanced the Ca uptake and reduced BER. These effects were confirmed in a more extensive study with two successive tomato crops in stone wool with circulating nutrient solution, with a range of $\text{NO}_3\text{:Cl}$ ratios from 22.8 to 0.8 mmol/L and 0.5 to 23.6 mmol/L for NO_3 and Cl, respectively, which were kept as target values for the root environment (Vooft and Sonneveld, 2004). BER was clearly reduced, though gold speck index was increased with the decrease of NO_3 and increase of Cl. As both ions were entangled in the treatments, it is impossible to conclude whether the effect on Ca transport to the fruits was related to the decrease in NO_3 or the increase in Cl. In the experiment the target NO_3 concentration, C_g could be reduced to 3 mmol/L without any yield loss. Strikingly, the N uptake did not differ that much between the treatments. C_u was reduced by 20% when C_g was reduced 85%. For NFT systems, a minimum concentration of 1.5 mmol $\text{NO}_3\text{/L}$ was found (Massey and Winsor, 1980). It is to be expected that depletion zones in the root could occur easily; therefore the

Table 10.18 Effect of N Concentration in Solution (C_N) and N Source (NH_4NO_3 vs Urea) in Fill Solution on Rose (Long Jaguar)¹ Yield in a Closed Loop Irrigation System² in Arava (Hazeva), Israel

N Source and Target Concentration		Cut Flower Yield		
$\text{NH}_4\text{NO}_3\text{-N}$	Urea-N	Total	50 cm	70 cm
mg/L		Number/m ² greenhouse		
28	—	1080 ^a	260 ^{ab}	34 ^a
56	—	1075 ^a	260 ^{ab}	32 ^{ab}
112	—	1075 ^a	267 ^a	32 ^{ab}
168	—	1000 ^b	231 ^b	29 ^b
—	112	1020 ^{ab}	244 ^{ab}	33 ^a

¹Growth period—30/05/02–31/12/03.
²Tuff; total water volume in the system 30 L/m² greenhouse; 9 irrigations per day, ET~4–5 mm/day; minimal night temperature: 18°C.
^{a,b} $p = 0.05$.

refreshment or replenishment rate is of importance to provide for sufficient N for the crop at these low target N values. In common stone wool systems the rate of replenishment of the solution is ~1.3 times the water uptake (L/L) (Sonneveld, 2002). In the experiment described, it was between 4 and 5 and in NFT it often is between 40 and 50 (Winsor et al., 1979).

The concentration supplied with such a low circulation rate needs to be close to the uptake concentration, and this concentration is often around 10 mmol/L (Sonneveld and Voogt, 2001). With a low circulation rate as mentioned, plants can absorb easily 80% of the N from the circulating solution (Sonneveld, 2002).

A study of rose response to N target concentration ($\text{EC}_{\text{thr}} = 3.5$ dS/m, 8–10 irrigations per day) showed that N concentration could be reduced to 28 mg/L without impairing total yield or increasing the number of excessively short or long stems in comparison with concentration of 56 or 112 mg/L N (Table 10.18). It is shown that a target N concentration of 168 mg/L N caused a considerable decline in yield relative to the lower concentrations (Table 10.18). It could not be resolved in this experiment whether the yield decline stemmed from excess N uptake or NH_4 mediated reduced cation uptake.

As discussed above, the demonstrated crop preference for low N concentration is characteristic of irrigation involving high fertigation frequency with ample amounts of irrigation solution. It does not contradict current N concentration recommendations in open IRs which are considerably higher (e.g., de Kreij et al., 1999) because irrigation frequency and dose volume are appreciably lower.

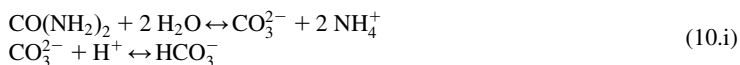
10.3.7 EFFECTS OF THE TYPE OF NITROGEN

10.3.7.1 $\text{NH}_4\text{:NO}_3$ ratio

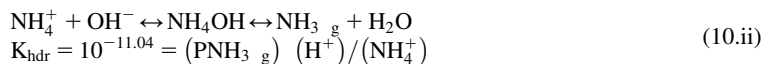
The ammonium to nitrate ratio is an important factor in fertigation management (Bar-Yosef, 1999 and Chapter 6: Nutrition of Substrate-Grown Plants). For a given N dose and irrigation regime, the

supply ratio affects several solution and crop factors: (1) N uptake efficiency; (2) Ca, Mg, and K absorption rates decline due to competitive NH_4 uptake; (3) NH_4 supply decreases solution pH while NO_3 tends to increase it (see, for example, [Imas et al., 1997](#)); and (4) Excess NH_4 uptake particularly at root temperature $>28^\circ\text{C}$ is deleterious to root development. It is superfluous to ascertain that the NH_4/NO_3 ratio is an important factor in pH control in soilless culture, and this subject is further discussed in [Chapter 6](#), Nutrition of Substrate-Grown Plants.

Another potential source of nitrogen is urea [$\text{CO}(\text{NH}_2)_2$] which is the cheapest source of nitrogen per unit N. Due to its zero charge, it is unaccompanied by ions that accumulate in closed loop IRs, it does not contribute to solution EC, and its impact on solution OP is weakest among all N fertilizers. For illustration—the osmolality (O_s/kg , where O_s = freezing point depression/1.86) of 5 g/L solution of KNO_3 , $(\text{NH}_4)_2\text{SO}_4$, NH_4Cl , NaNO_3 , and urea is 0.092, 0.093, 0.173, 0.11, and 0.083, respectively ([Weast, 1977](#)). Per g of N the smaller effect of urea on freezing point depression and OP is much more pronounced. The disadvantage of urea is that its uptake efficiency before hydrolysis is significantly lower than ammonium or nitrate ([Kirkby and Mengel, 1967](#); [Herndon and Cochlan, 2007](#)), and therefore, it is not used in greenhouses having open IRs where the residence time in the substrate is too short for hydrolysis. The hydrolysis consumes H^+ and therefore increases pH as per this equation:



According to [Ikeda et al. \(2001\)](#), 60% of the urea applied to soilless medium is subjected to hydrolysis at greenhouse normal temperature in 1 day. This rate is similar to NH_4 nitrification ($\sim 50\%$ in 1 day, [Ikeda et al., 2001](#); [Lang and Elliott, 1991](#)); therefore the hydrolysis end product is ammonium plus nitrate. The presence of plants may change the hydrolysis rate and ratio between the ammonium and nitrate species. There are two important constraints to using urea: (1) It should be free of biurate which can accumulate in the closed loop IR and become toxic to plants. (2) The urea–N concentration should not exceed a certain fraction of the total N in the system due to (1) its slow uptake and (2) urea might induce toxic ammonia (NH_3 g) release (Eq. 10.ii) due to the pH increase during hydrolysis and the presence of NH_4^+ . According to



with $\text{PNH}_3 \text{ g}$ in bars, a PNH_3/NH_4 ratio of 1/1000 is expected at pH 8.04 so addition of urea at cf. pH > 7.5 is not advised.

The optimal fraction urea:total N in recirculated solutions has not yet been determined, so it is arbitrarily recommended to be $<15 \pm 5\%$ in order to minimize the above mentioned concerns. If the total daily N uptake plus discharge is also $15 \pm 5\%$ of the total N then the entire daily depletion can be replenished by urea. [Kramer et al. \(2003\)](#) compared rose response to ammonium nitrate versus urea supply in the fill solution at a target N concentration of 112 mg/L N in the recirculated solution. The cumulative number of cut flowers over 18 months of harvest and their quality were the same in both sources ([Table 10.18](#)).

10.3.7.2 Effect of N sources and concentration on root disease incidence

The nutritional status of plants has a major impact on their disease susceptibility to various pathogens (see collection of papers in [Engelhard, 1989](#)). The exact mechanism underlying this

relationship is unknown, but there are indications that it is associated with N concentration and $\text{NH}_4\text{:NO}_3$ ratio in soil or substrate solutions. Duffy and Defago (1999) reported that the severity of *Fusarium* crown and root rot in tomato seedlings grown in stone wool was increased by NH_4 and reduced by NO_3 ; intermediate NH_4NO_3 concentration (40–80 mg/L N) reduced disease, but rates > 100 mg/L N increased it.

Voogt and Paternotte (1997) studied the effect of pH and $\text{NH}_4\text{:NO}_3$ ratios on the incidence of *Humicola fuscoatra*, causing corky root in tomato. The phenomenon of corky root was aggravated when pH in the solution was decreased; however, this was not the case when NH_4 was supplied. Strikingly, root browning was found in all treatments, no matter whether inoculated or not, but the development of corky roots—which is considered to be a secondary effect of infection with *Humicola*—was mainly found at low pH, with one strange exception at high pH (without NH_4). But notably there was hardly sporulation of this fungus at high pH (Table 10.19) (Voogt and Paternotte, 1997).

Bar-Yosef et al. (2005) investigated the solution N–root disease relationship in a closed loop IR with *Pythium* inoculated cucumber as test crop. The N variables were $\text{NH}_4\text{:NO}_3$:urea ratio at equal total N concentration (126 ± 14 mg/L N) and N concentration at constant $\text{NH}_4\text{:NO}_3$ ratio of 30:70 in the fill solution. They reported that 40 days after planting the mortality incidence increased as the $\text{NH}_4\text{:NO}_3$ ratio rose from 20:80 (70%) to 80:20 (94% mortality), and in a treatment of 50% NO_3 :50% urea the mortality dropped to 22% (Table 10.20). The NH_4 concentration in the recirculated solution of the urea treatment was similar to the concentration in the 50 NO_3 :50 NH_4 treatment and the concentration of free urea-N was <10% of the total N. At the end of the growing

Table 10.19 Effect of pH and NH_4 Concentration in the Nutrient Solution Supplied and Inoculation of Spores of *Humicola fuscoatra* on the Incidence of Brown Coloring of Roots, the Incidence of Corky Roots, and the Number of Conidia on Root Samples of *H. fuscoatra* on Tomato Grown in Stone Wool With a High Recirculation Rate

Treatment				Brown Roots	Corky Root	Conidia	Yield
	pH	NH_4 mmol/L	Inoculation	Index ^a	Index ^b	Nr ^c	kg/m ²
1	4.5	0	Yes	6	1.5	10	42
2	4.5	0	No	4	0.5	10	46
3	4.5	1.5	Yes	5	1	6	42
4	4.5	1.5	No	5	1	10	39
5	6.3	0	Yes	8	6	3	37
6	6.3	0	No	7	0.5	0	43
7	6.3	1.5	Yes	6	0	1.5	43
8	6.3	1.5	No	7	0	0	42

Experiment carried out in Naaldwijk, NL (Voogt and Paternotte, 1997).

^aIndex by visual judgment of the severity of brown coloring of roots from 0 = white 10 = dark black-brown roots.

^bIndex by visual judgment of the incidence of corky root: 0 = no corky; 10 = completely corky.

^cCounting results of conidia of *H. fuscoatra* on samples of roots taken from beneath the pits in the rock wool slabs.

Table 10.20 Effect of $\text{NH}_4\text{:NO}_3\text{:Urea}$ Ratio (NH:NO:U) at Constant Total N Concentration in the Fill Solution of Recirculation System (Experiment 1) and Increasing Total N Concentration at Constant $\text{NH}_4\text{:NO}_3\text{:Urea}$ Ratio (Experiment 2) on % Cucumber Plants Mortality on Day 40 and at the End of Experiment

Experiment	Treatment				Mortality %		Num Days
	NH	NO	U		End	40 Days	60% Mort.
1 8–10 mM N	50	50	00	E1	94	86	32
	20	80	00	E2	91	70	38
	80	20	00	E3	94	94	27
	00	50	50	E4	88	22	51
2 30:70:0	2			W1	0	0	—
	4			W2	78	30	48
	8			W3	69	18	51
	16			W4	82	74	35

Also presented is the number of days required in each treatment to obtain 60% mortality. Plants were inoculated by *Pythium* on planting day. Mortality was due to *Pythium* and *Fusarium* (spontaneous infection). Plants were planted on 01/04/04, in closed loop irrigation system at Bet Dagan, Israel.^a

^aDerived from Bar-Yosef et al. (2005).

season (day number 75), more than 90% of the plants died in all treatments as a result of *Fusarium* root rot (spontaneous infection) and/or *Pythium*.

When maintaining a 30:70 $\text{NH}_4\text{:NO}_3$ ratio and increasing total N concentration from 2 to 4–8 and 16 mM the mortality incidence after 40 days increased from 0 to 18–30 and 74%, respectively, and at 2 mM N, no plant died until the end of the experiment. A similar conclusion can be drawn from the results of number of days that were required to kill 60% of the initial plant population (Table 10.21). Enhanced NH_4 concentration in solution decreased pH which was linearly correlated with the % mortality and number of days required to kill 60% of the plants (Fig. 10.12).

It is unknown if the low NH_4 per se caused the reduced mortality (Duffy and Defago, 1999) or it was due to the higher pH, but it is clearly indicated that reducing the N concentration ($\text{NH}_4\text{:NO}_3$ ratio = 30:70) to 2 mM N in a closed IR considerably abated cucumber susceptibility to *Fusarium*. The implications of such a low concentration on crop yield and quality must be studied in detail, but previous data indicated that at least in tomato grown in closed IR C_N reduction from the conventional 10 mM N to 4 mM was feasible.

10.3.8 MACRO NUTRIENTS OTHER THAN N

10.3.8.1 Phosphorus

P is usually supplied in the form of orthophosphates. Polyphosphates are used nowadays widely in commercial practice, usually these P-source is promoted to prevent clogging of drip irrigation, but also growth enhancement and better crop performance have been claimed, although these effects could not be confirmed in research (De Kreij et al., 2003). More recently polyphosphates were

Table 10.21 Effect of Irrigation Frequency (Same Threshold Electrical Conductivity and Irrigation Rate) on Pepper¹ Yield in a Closed Loop Irrigation System² at Bet Dagan, Israel³

Fertigation Frequency	Yield		
	Total	Marketable	Unmarketable
Num/day	kg/m ² ground	kg/m ² ground	(%)
18	11.0	4.9 ^a	19.1 ^{ab}
12	10.8	4.8 ^a	17.8 ^b
6	10.5	4.3 ^b	20.0 ^a

¹ var. Cuby; planting 30.8.99, first harvest 20.12.99, last 15.5.00.
² Threshold EC in all cases 3.5 dS/m; perlite substrate 50 cm wide; NH₄-N, NO₃-N, P, K concentrations 2, 8, 1, 4 mM, respectively
³ Derived from Bar-Yosef *et al.* (2000a,b,c,d).
^{a,b} *p* = 0.05.

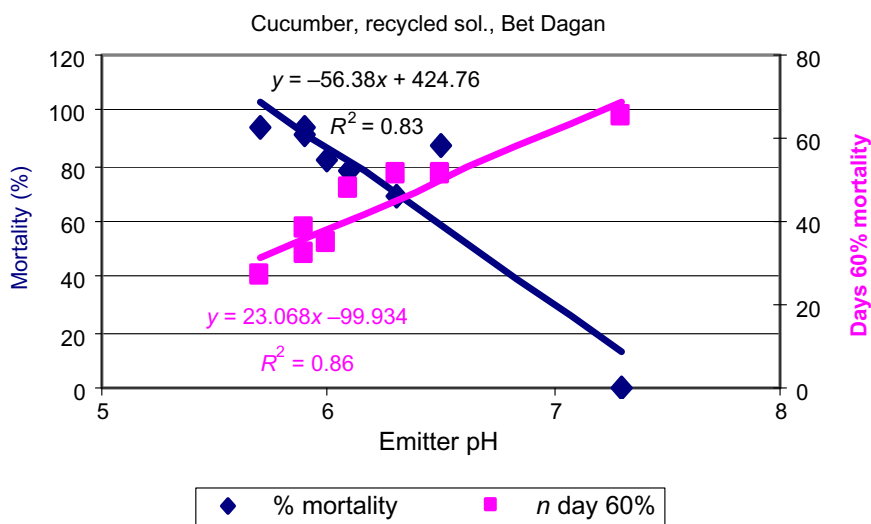


FIGURE 10.12

The effect of irrigation solution pH in closed loop irrigation system (emitter pH) on % cucumber plant mortality 30 days after planting, and on number of days (n) that were required to kill 60% of the plants. Variations in pH were caused by N treatments (4 NH₄:NO₃ ratios at constant total N, and 4 N concentrations at constant ratio); plants were inoculated by *Pythium*, but mortality was due to *Pythium* and *Fusarium* (spontaneous infection). Experiments were conducted in Bet Dagan, Israel (Bar-Yosef *et al.*, 2005).

introduced as alternative for Fe-chelates (Voogt and IJdo, 2015). The authors showed that the stability of these products is such that the Fe-uptake was comparable with the plants treated with traditional Fe-DTPA, though the concentrations in the recirculation solution are low, due to the high pH of the drainage water.

In open IRs, P concentration in irrigation solution is ~ 30 mg/L P (Hoagland and Arnon, 1950; de Kreij et al., 1999). The appropriate concentration in closed loop IRs depends on the interrelationship between P uptake and salinity (e.g., Feigin, 1985), fertigation frequency, and substrate's P adsorption characteristics (empirical results are presented later).

10.3.8.2 Potassium

The K demand of crops is relatively high, in particular with fruit vegetables crops, and show often peaks during periods with heavy fruit load (Voogt, 1988a, 1993), although also in crops such as butterhead lettuce a strong increase in K demand in the period of head-filling is observed (Voogt, 1988b) as well as for rose during periods of flower stem elongation when grown in shifts. The dynamics in the uptake of the cations K, Ca, and Mg is illustrated in Fig. 10.13, as monitored in a typical long-term tomato crop in a Dutch greenhouse, with stone wool and reuse of drainage water. The uptake of K peaks with the more or less exponential increase of the fruit load during the first eight to nine clusters. This peak in K demand is obviously related to the relatively high demand of K for fruit growth and development. This stems from the observations that the distribution of K is more or less equal to the total DM partitioning over fruits and vegetation (Voogt, 2002). Strikingly the uptake of Ca, and to a lesser extent Mg, decreases in the same period. It has been suggested that this could be due to the restriction in root development as a result of the sink/source competition between developing fruits and roots, causing less new root tips and less sites along the root for the passive uptake pathway for Ca ions. As a result, the K/Ca ratio of C_u changes dramatically

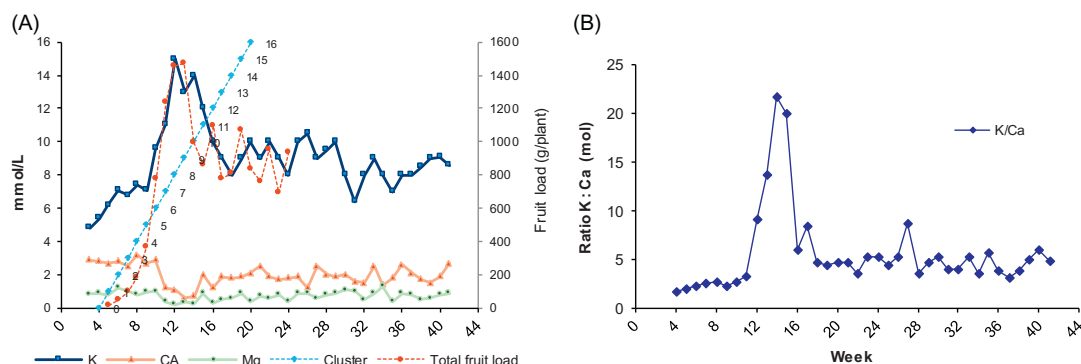


FIGURE 10.13

The course of the uptake concentration C_u of K, Ca, and Mg during a long-term tomato crop grown in stone wool with reuse of drainage water, for comparison the development of the cluster flowering (number), and the total fruit load (g/plant) is shown for the first 20 weeks (A). The uptake of K and Ca expressed as K/Ca ratio (B).

From Voogt, W. (1993). Nutrient Uptake of Year Round Tomato Crops. *Acta Hortic.* 339, 99–112, <https://doi.org/10.17660/ActaHortic.1993.339.9>.

(Fig. 10.13). It is highly recommendable to counteract these expected shifts in uptake to prevent strong K-depletion, which easily occurs otherwise. For tomato, Voogt (1988a,b) developed guidelines for adjustment of nutrient solution corresponding with the number of the flowering cluster, which has been implemented in the Dutch fertilizer recommendation system (de Kreij et al., 1999). Since the shift in K coincides with a reduction in the Ca and Mg uptake, this is effectively a change in the mutual K:Ca:Mg ratios, leaving the total EC and the other nutrients unchanged.

Potassium concentration in open loop IRs varies between 150 and 240 mg/L, depending on crop (Hoagland and Arnon, 1950; de Kreij et al., 1999). In recirculated solutions, where significant part of the N is added as NH_4 the K uptake rate might be inhibited, leading to the need for higher K target concentrations. Due to the similar N and K transport mechanism in nonadsorbing substrates, it is expected that increased fertigation frequency will allow reducing target K concentrations similarly to N.

10.3.8.3 Microelements

When Fe, Zn, and Mn are found in chelated form [usually ethylenediaminetetraacetic acid (EDTA) or DTPA] their target concentrations in open IRs are ~ 1.0 , 0.25 and 0.5 mg/L, respectively (de Kreij et al., 1999). Their availability to plants is pH dependent (see later). As discussed in the next section, fertigation frequency affects recirculated solution pH, so that this, in turn can be expected to affect microelements availability.

10.3.9 FERTIGATION FREQUENCY

10.3.9.1 Principles

Maximum irrigation frequency depends on the substrate's water holding capacity, the substrate drainage rate and the quantity of water, which is applied to meet plant's demand and leaching salts from the substrate. The water quantity/ m^2 substrate per irrigation (V_p) should be $\leq [\theta_{95\% \text{ sat}} - \theta_{0.5 \text{ hour}}] \times [\text{substrate volume per } \text{m}^2 \text{ substrate}]$. In perlite ($h = 0.25 \text{ m}$), for example, V_p should be $< 60 \text{ L/m}^2$ (see Table 10.1). The actual water dose (V_{irr}) should be smaller ($\sim 30\%$ of V_p , or 18 L/m^2) to prevent temporary water saturation and to slow down the drainage rate. Exceeding V_p results in water flow on top of the substrate rather than through it, thus impairing effective nutrients and oxygen recharge and root flushing with fresh solution. The irrigation time (t_{irr}) needed to supply this dose is $t_{\text{irr}} = V_{\text{irr}}/I_{\text{irr}}$, where I_{irr} is the IR discharge rate ($\text{L/m}^2 \text{ subst/h}$). For a conventional $I_{\text{irr}} = 50$, each irrigation event should last $(60 \text{ minutes} \times 18/50) \sim 20 \text{ minutes}$. The time between successive irrigations (t_{si}) should exceed $t_{\text{irr}} + t_d$, where t_d is drainage time, calculated as V_{irr}/I_d and I_d is the drainage rate in $\text{L/m}^2 \text{ subst/h}$. The I_d for perlite is approximated from its J_{fd} (cm/h , Table 10.1) using the transformation $10 \text{ cm/h} = 100 \text{ L/m}^2 \text{ subst/h}$. The t_d is therefore $(60 \times 18/100) \sim 11 \text{ minutes}$, and $t_{\text{si}} (= 11 + 20) \sim 30 \text{ minutes}$. This result indicates that 2 irrigations per hour, each comprising 18 L/m^2 substrate, is a safe irrigation frequency in perlite. In order to compare with ET (ET, $\text{L/m}^2 \text{ ground/h}$, or mm/h), the above V_{irr} is transformed into $\text{L/m}^2 \text{ ground}$. For this, we need to know the fraction of ground area covered by substrate. For a conventional fraction of 0.25 (0.4 m wide substrate per 1.6 m wide bed), the above 18 L/m^2 substrate becomes 4.5 L/m^2 ground, and the permitted rate of $4.5 \times 2 \text{ mm/h}$ is about four- to fivefold greater than the ET during summer middays in arid zones. The recommended V_{irr} for perlite in Israel at the aforementioned I_{irr}

is 2–4 mm/irrigation event, compatible with 15%–30% of V_p . The recommended number of irrigations per day, n_I , is

$$n_I = \frac{[\text{daily irrigation rate, } V_d]}{V_{\text{irr}}} \quad (10.21)$$

For a conventional $V_d = 5 \times [\text{daily ET}]$, $[\text{daily ET}] = 3 \text{ mm}$ and $V_{\text{irr}} = 2 \text{ mm}$, the number of irrigations per day is seven to eight, which is within the safe range discussed above.

10.3.9.2 Experimental evidence

The effect of irrigation frequency in closed loop IRs was investigated in pepper at Bet Dagan, Israel, and for tomato at Besor, Israel (Tables 10.21 and 10.22).

In pepper, 12 irrigations per day (n_I) gave better yield and quality than $n_I = 6$ and did not differ from 18 irrigations per day. In tomato, the irrigation frequency effect was very similar to pepper and the intermediate frequency (8–16 per day) gave best results (Table 10.22).

The range in frequency in a given treatment stems from seasonal differences in ET. Baas and van der Berg (2004) studied rose (cv. Frisco) response to irrigation frequency under normal, $\times 2$, and $\times 4$ irrigations per day (water rates increasing proportionally to frequency) using closed loop IR in the Netherlands. Under the prevailing growth conditions (3 L substrate per plant and EC of 0.9 or 1.9 dS/m), irrigation frequency had no significant effect on cut flower production.

As noted earlier, the irrigation frequency effect is interrelated with the nutrient concentration in the recycled solution. In the case of tomato, $n_I = 16$, the yield at N concentration of 50 mg/L exceeded that at 150 mg/L, the latter being the concentration at which the frequency effect was studied (see Table 10.22).

An important constraint to irrigation dose and frequency (at fixed V_p) in recirculated systems are the devices used for disinfection treatments. These systems will be designed for a certain capacity (m^3/day). Increasing the capacity of these systems to account for greater daily irrigation and the handling of the larger drainage volume are expensive; therefore it is essential to minimize solution

Table 10.22 Effect of Irrigation Frequency (Same Threshold Electrical Conductivity and Irrigation Rate) at Two N Concentrations on Tomato¹ Yield and Incidence of Blossom End Rot (BER) in a Closed Loop Irrigation System² in a Greenhouse at Besor, Israel³

Irrigation Frequency	Solution N Concentration (mg N/L)					
	50	150	50	150	50	150
Num/day	Total Yield (kg/pL)		Marketable Yield (kg/pL)		BER (num/pL)	
4–8	—	8.06 ^b	—	4.35 ^b	—	42.5 ^a
8–16	10.57	9.87 ^a	7.11	5.94 ^a	24.4	23.5 ^b
16–32	10.10	—	7.07	—	23.0	—

¹var. 870; planting 1/9/04 and 15/7/05.

²Threshold EC in all cases 4.5 dS/m; perlite, mean of 10, 20, and 50 cm wide substrate; P, K concentrations identical in all treatments (1 and 4.5 mM, respectively).

³Derived from Reshef et al. (2006).

^{a,b} $p = 0.05$.

recirculation. Even in case of a buffering system such as the so-called daily storage tanks, which allow to do treatments 24/7, have their limitations. For NFT systems, the rate of circulation is as such that disinfection is hardly applicable.

10.3.10 pH CONTROL: PROTONS AND CARBOXYLATES EXCRETION BY ROOTS

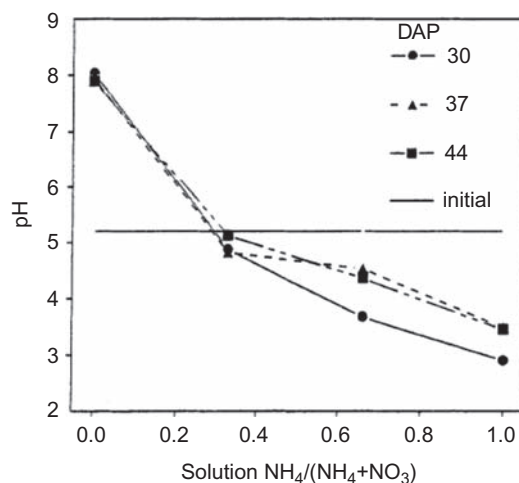
10.3.10.1 NH_4/NO_3 ratio-induced effects

Regarding pH in nutrient solutions, we have to distinguish between the different flows. For substrate-based systems, the pH of the irrigation water should be kept below 6.2 to avoid precipitation of phosphates and blocking of the IR (Sonneveld and Voogt, 2001; see also Chapter 6: Nutrition of Substrate-Grown Plants). The pH in the root environment depends on specific requirement of the crop but as a rule should be kept between 5 and 6.5 (Sonneveld and Voogt, 2009). The solution pH can be controlled by two methods: (1) modifying the NH_4/NO_3 ratio in recycled solution and (2) applying acid for reducing pH (HNO_3 , H_3PO_4 , or HCl where water Cl is low) or base for increasing pH (K_2CO_3 , KHCO_3 , KOH). The advantage of method 1 over 2 is that it eliminates the safety measures associated with handling strong acids, and it does not add ions not consumed by the plants. However, its disadvantage is that NH_4 might inhibit Ca, Mg, and K uptake, and it will only change the pH indirectly as it requires activity of roots and microorganisms, and there will be a time delay between application and effect in the entire solution volume of hours or even days. So this method can only be used for controlling the pH in the root environment. The advantage of method 2 is that the pH of the irrigation water can be controlled instantaneously. In closed systems, the pH of the irrigation water is affected highly by the pH of the drainage water; therefore an adequate automatic pH control, with both an acid and low dosage pump is essential.

The choice of the right $\text{NH}_4:\text{NO}_3$ ratio in the nutrient solution supplied (C_f) is important, since in substrate systems the rooting volume is restricted and the buffer capacity of the substrate itself is small or even nonexistent. Therefore the pH of the substrate can be affected easily by the addition of small quantities of NH_4 (Voogt, 1995).

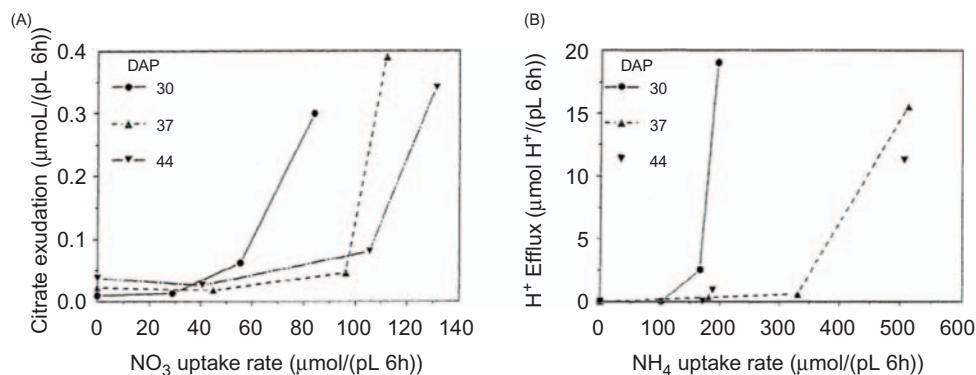
Method 2 for adjustment of the pH in the root environment is only possible in hydroponic systems when the nutrient solution has a high flow rate. Because of this high flow rate, it is frequently possible to add small quantities of alkalis or acids for adjustment. However, in substrate systems, the refreshing rate of the root environment is too low to correct directly by addition of sufficient alkalis or acids. The quantities to be added to the solution supplied are restricted, because extreme low or high values of pH must be prevented. Values below 4.5 have risks of damage on roots (see further), but also the matrix of mineral wool substrates are easily damaged at low pH values. In order to adopt method 1, it is necessary to evaluate the following three factors: (1) the potential proton excretion by roots of a given crop, (2) the nitrification rate in the recirculation systems, and (3) the effect of nitrate uptake on carboxylates release into solution. Insight into factors 1 and 3 can be derived from the work published by Imas et al. (1997).

They measured pH (Fig. 10.14) and short-term (6 hours) NH_4 and NO_3 uptake and proton and citrate release by roots in solutions varying in $\text{NH}_4:\text{NO}_3$ ratio in the presence of young tomato plants (30–44 days after planting) (Fig. 10.15). The data indicate that a certain threshold uptake rate must be surpassed before exudation starts and then it proceeds at a constant rate which declines with root age. The potential accumulation of protons and citrate in real recycled solutions was

**FIGURE 10.14**

Solution pH after 6 h of uptake by tomato plants from solutions varying in $\text{NH}_4\text{:NO}_3$ ratio (total N = 7.5 mM).

Derived from Imas, P., Bar-Yosef, B., Kafkafi, U., Ganmore-Neumann, R., 1997. Release of carboxylic anions and protons by tomato roots in response to ammonium/nitrate ratio and pH in nutrient solution. *Plant Soil* 191, 27–34.

**FIGURE 10.15**

(A) Citrate exudation versus NO_3^- uptake and (B) H^+ excretion versus NH_4^+ uptake in young tomato plants.

Derived from Imas, P., Bar-Yosef, B., Kafkafi, U., Ganmore-Neumann, R., 1997. Release of carboxylic anions and protons by tomato roots in response to ammonium/nitrate ratio and pH in nutrient solution. *Plant Soil*. 191, 27–34.

estimated from data in [Figs. 10.14 and 10.15](#) and is summarized in [Table 10.23](#). The calculations indicate that during 30 days of recirculation, tomato plants absorbing NH_4^+ from a solution of $\sim 5 \text{ mM } \text{NH}_4^+$ may reduce an unbuffered recycled solution pH from ~ 7 to 3.1 (50 L solution/ m^2 ground). The effect of $\text{NH}_4\text{:NO}_3$ ratio on carboxylates exudation is evaluated from the data in

Table 10.23 Estimated Potential H^+ Release by Tomato Plants (40 Days Old) and Resulting pH in Recycled Solution of 5 mM NH_4 , Zero pH Buffering Capacity, and 2.5 Plants/ m^2 Ground

Threshold Uptake ^a	H^+ Release/ NH_4 Uptake ^b	Effective NH_4 Uptake Rate ^c	Estimated Daily H^+ Release ^d		
mol NH_4 /pL/day	mol H^+ /mol NH_4	mol N/ m^2 ground/day	mol/ m^2 ground	Estimated Daily pH	30 Days pH
0.0012	3/10	0.004	0.0012	4.6	3.1

^a NH_4 uptake rate below which no H^+ is excreted.^bThe ratio is the slope of the line in Fig. 10.15 subject to NH_4 uptake rate > threshold (average of three plant ages).^cEquals to [Total uptake rate] – [threshold uptake rate] $\times$ number of plants/ m^2 ground. Total uptake rate = 0.007 mol N/ m^2 /day.^dColumn 2 $\times$ column 3.**Table 10.24 Estimated Potential Citrate Release by Tomato Plants (40 Days Old) Stemming From Nitrate Uptake From Recycled Solution of 5 mM NO_3 and 2.5 Plants/ m^2 Ground**

Threshold Uptake ^a	Citrate Release/ NO_3 Uptake ^b	Effective NO_3 Uptake Rate ^c	Estimated Daily Citrate Release ^d	Estimated Citrate Conc. ^e	30 Days Citrate Release	30 Days Solution Citrate
mol NO_3 /pL/day	mol Citrate/mol NO_3	mol N/ m^2 ground/day	mol/ m^2 ground	μ M	mol/ m^2 ground	μ M
0.0004	0.4/10	0.006	0.00024	4.6	0.0072	144

^a NO_3 uptake rate below which no citrate is excreted.^bThe ratio is the slope of the line in Fig. 10.15 subject to NO_3 uptake rate > threshold (average of three plant ages).^cEquals to [Total uptake rate] – [threshold uptake rate] $\times$ number of plants/ m^2 ground. Total uptake rate = 0.007 mol N/ m^2 /day.^dEquals to column 2 $\times$ column 3.^eSolution volume = 50 L/ m^2 ground.

Fig. 10.7. When all N is absorbed as nitrate, and assuming that no consumption by microbes takes place, the estimated potential citrate concentration after 30 days is 144 μ M (Table 10.24).

This concentration is $\sim$ 25-fold greater than the C concentration found in actual greenhouse recycled solution of 30 days old pepper plants receiving 100 mg/L N at NH_4/NO_3 ratio of 15:85 (Table 10.10). The difference indicates the extent of organic acid consumption by microbes in recycled solutions (Table 10.5). The equivalent C concentration in the 144 μ M citrate solution is $\sim$ 10 mg/L C, which is similar to the Total Organic Carbon (TOC) values found in recycled solutions reported above (Fig. 10.6). This indicates that the TOC consists mainly of microbial mass which consumed the exuded organic acids, minus CO_2 released in respiration.

As mentioned earlier, the rate of pH evolution stemming from nitrification must also be known for pH management. In reality, this will be quite impossible to quantify as it will depend on the microbiological activity which depends on quite some unknown and unpredictable variables and

Table 10.25 The Daily NH_4 Concentration and pH in Open and Closed Loop Irrigation Systems in a Pepper Greenhouse (cv. Celica) in Besor, Israel.

	Open Irrigation System			Closed Irrigation System $\text{EC}_{\text{thr}} = 3.2 \text{ dS/m}$			
	Emitter	Drang + pl	Drang – pl	Fill	Emitter	Drang + pl	Drang – pl
Date	$\text{NH}_4\text{--N (mg/L) (and pH)}$						
12 February	59 (7.6)	27 (7.3)	—	—	29 (8.3)	12 (6.3)	—
19 March	19 (7.0)	10 (6.6)	19 (6.5)	44	4 (6.8)	0.2 (6.6)	1.9 (6.3)
30 April	47 (7.0)	15 (5.8)	17 (6.2)	49	12 (6.2)	5.3 (5.1)	8.2 (5.6)
28 May	46 (7.1)	17 (5.7)	18 (6.1)	30	2 (6.3)	0.3 (6.3)	0.4 (6.1)
18 June	46 (7.0)	18 (5.5)	21 (6.0)	46	10 (6.7)	0.3 (6.4)	2.3 (5.6)
16 July	48 (7.0)	18 (5.6)	24 (6.4)	33	9 (6.9)	0.1 (6.3)	0.8 (6.1)

The ionic composition of the fill solution (recirculation treatment) was equal to the irrigation solution ionic composition in the open irrigation system. Both the emitter and effluent solutions (with or without plants) were collected for 24 h then analyzed.^a
 EC_{thr} , Threshold electrical conductivity.
^aDerived from Reshef et al. (2003).

parameters. In a control open loop irrigation treatment, Reshef et al. (2002a,b) reported that 72% of added NH_4 disappeared from substrate solution within 24 hours in the presence of plants, compared with 57% in the absence of plants (mean of 6 dates in winter and spring months, Table 10.25).

In recirculated solution with fill solution identical to the emitter solution in the open system, the emitter NH_4 concentration was considerably lower (~ 10 vs 45 mg/L N) and the NH_4 disappearance rate was 82% in 24 hours in the presence of plants and 67% in their absence (Table 10.25).

In the open IR, the NH_4 and NO_3 concentrations were ~ 45 and 95 mg/L N , respectively, and the pH indeed declined in the presence of plants from ~ 7 to ~ 5.8 in 24 hours. In the absence of plants (nitrification only), the pH dropped to ~ 6.2 . Notably the root excretion effect on pH was considerably stronger in the spring and summer than in winter and beginning of spring.

In the closed loop IR, the NH_4 and NO_3 concentrations were ~ 10 and $\sim 130 \text{ mg/L N}$. Under these conditions, the drainage pH in the presence of plants exceeded the pH in the absence of plants. The reason was that in the absence of plants basidification stemming from NO_3 uptake did not occur (Table 10.25). In order to obtain a pH comparable to the pH in open system, the fill solution NH_4 concentration should exceed the one in the open system.

10.3.11 SPATIAL DIFFERENCES OF pH IN THE ROOT ZONE

NH_4 is absorbed by plants preferentially (van Beusichem, 1988; Marschner, 1995). Thus when drip irrigation is used in substrate systems, the majority of the NH_4 ions supplied will be absorbed by the roots close to the dripper, reducing the pH in the surrounding root environment. The further away in time and distance from the drip irrigation flow, the nutrient solution will be depleted of NH_4 relative to NO_3 causing an increase in pH, and this will result in considerable spatial differences of pH values. An indication of this phenomenon was found during monitoring pH in a

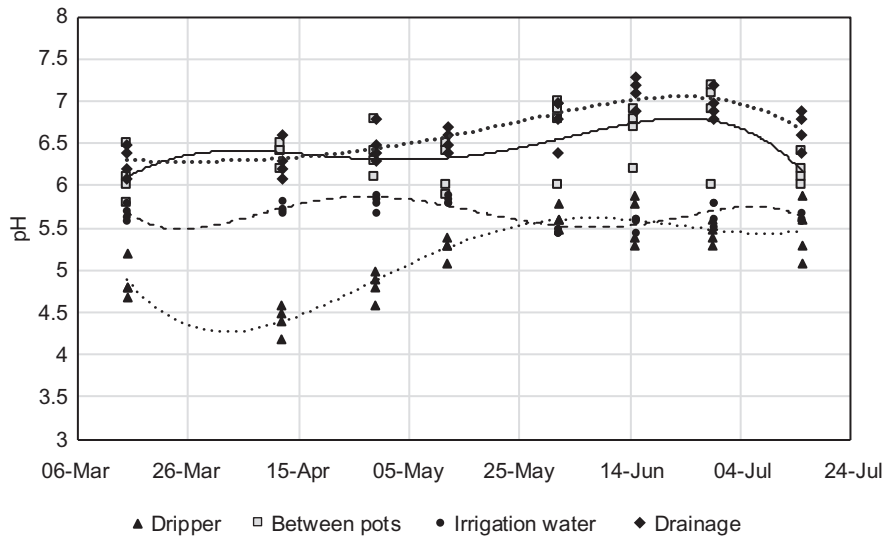


FIGURE 10.16

Course of the pH measured in samples taken by a syringe from spots close by the dripper and in between the pots and from the irrigation and drainage solution, at four replicates in a commercial stone wool grown greenhouse cucumber crop.

Voogt (Unpublished data).

commercial greenhouse where pH was measured at samples collected from various spots (Voogt et al., 1989).

A more detailed survey on the pH gradient has been described by Sonneveld and Voogt (2001). Obviously this pH gradient can be affected also by nitrification of NO_3 ; however, for the result it makes no difference, whether the pH increase stems from NO_3 uptake or NO_3 reduction. The tendency of higher values pH in the root environment can be explained by the larger anions than cation uptake in the presence of high NO_3 concentration and neutralizing the proton exchange (see also Section 10.3.13). If mixed samples are collected from stone wool slabs, or if drainage is collected, the HCO_3 concentration in part of the subsamples with a high pH nullifies the H_3O^+ concentration in the other part of the subsamples with a low pH as can be seen in the pH values from the drainage in Fig. 10.16.

Then the high average value in a randomized sample is more or less misleading and for adequate information about the development of the pH value the spots with high and low pH should be sampled separately throughout the cropping cycle.

10.3.12 CROP RESPONSE TO pH

As discussed in Chapter 6, Nutrition of Substrate-Grown Plants, the pH is important for the availability of plant nutrients as well as some other root functions for the plant. The uptake of micronutrients is affected largely by the pH, as well known from the deficiency of some microelements,

such as Fe, Mn in calcareous soils (Mengel and Kirkby, 2001). On the other hand, the uptake of Mn can be increased almost uncontrollable at pH levels below 5 (Sonneveld and De Bes, 1984). The pH range for optimum nutrient availability is between 5 and 6 for more or less all crops, but there is a great difference among the sensitivity of crops for high pH values. Gerbera, cucumber, and rose for example easily show chlorosis and yield reduction when the pH rises above 6. Sometimes with these crops, the micronutrient concentrations in the plant were reduced, which could be traced as being the cause for these effects. However, this supposition was not always fully confirmed by the data of the tissue analysis (Sonneveld and Voogt, 1994; Voogt and Sonneveld, 2009). Nevertheless, a positive yield response was found when crops were grown at pH values of ~ 5.5 , comparison with higher values, for cucumber (Sonneveld and Voogt, 1994), gerbera (Sonneveld and Voogt, 1997), carnation (Voogt, 1995), and rose (Voogt and Sonneveld, 2009). Many other crops can survive quite well with values between 6 and 7. Values below 4.5 are not advisable, for either root damage or an increased risk of root diseases. Voogt and Paternotte (1997) found an increase in the incidence of corky roots with tomato when the pH was decreased from value of 5.5 (see the next section).

The question whether uptake or plant performance is related to the pH or to the NH_4 supply has been investigated in experiments where the effects of pH levels and $\text{NH}_4\text{:NO}_3$ ratios were investigated independently, using cucumber, gerbera, and roses as test crops. The crops were grown in glass wool substrate as being pH neutral and inert, with reuse of drainage water. The circulation rate was kept high and the pH of the circulating solution was controlled automatically with acids (mixture of HNO_3 , H_2SO_4 , and H_3PO_4) and alkaline solution (KHCO_3) and was kept at constant levels of 7.0, 5.8, and 4.5. At each pH level, three $\text{NH}_4\text{:NO}_3$ ratios were kept: 0, 11.7% and 22.8% of total N. The yield response showed a clear interaction, and decreased with increasing pH, though when NH_4 was added, the yield reduction was nullified (Table 10.26).

The chlorosis index increased—as likely the result of micronutrient stress and part of the explanation of the yield response—also aggravated with increasing pH but increased less with NH_4 supply. However, at the high pH levels the high NH_4 supply resulted in yield decrease, which could be caused by NH_3 toxicity, which starts to develop for the NH_4OH – NH_3 equilibrium at H levels above 7. In the absence of NH_4 , a small surplus of acid was required in the balance of supplied acid and alkaline to keep the pH at the set points. When NH_4 was added, the balance turned into a surplus of required alkaline, which was almost equivalent with the concentration of NH_4 supplied.

10.3.12.1 pH and fertigation frequency

Recirculated solution pH is expected to be affected by fertigation frequency as well. This stems from the aforementioned observation that increasing frequency enhances ion uptake rate by plants. In solutions consisting predominantly of NO_3^- , higher frequency should increase the pH, while under relatively high NH_4^+ concentrations, the change in pH should depend on the relative efficiency of protons release due to NH_4^+ uptake and nitrification. In the experiment where pepper response to fertigation frequency was studied (Table 10.20), the time-averaged pH in recirculated solution slightly declined (6.4–6.2) when the frequency was increased from 6 to 12 fertigations per day, while increasing the frequency to 18 fertigations per day had no further effect on pH. In tomato (see Table 10.21), raising the frequency from 4–8 to 8–16 (at total N concentration of 150 mg/L) or from 8–16 to 16–32 (at 50 mg/L N) had no measurable effect on pH. In open IRs, the pH response to fertigation frequency was reported to be much stronger. For example,

Table 10.26 Effects of the pH Level Combined With Three NH_4 Rates Yield (Number of Fruits/m² and Total Yield kg/m²), Index for Chlorosis (Visually Judgment: 0 = Green 10 Complete Yellow Leaves) on Cucumber Grown on Glass Wool Substrate With a High Circulation Rate ($Q_g/Q_{in} > 0.8$)

Treatments			Yield		Chlorosis Index	pH Corrections		
pH	NH_4 mmol/L	NH_4 % of N total	Fruits/m ²	kg/m ²		H^+	OH^-	Balance
4.5	0	0	137 ^a	51.1 ^a	4.7	0.81	0.01	0.79
5.5	0	0	129 ^b	47.3 ^b	5.0	0.40	0.04	0.36
6.5	0	0	126 ^b	45.8 ^c	7.3	0.10	0.40	− 0.31
4.5	1.5	11.7	131 ^a	48.9 ^a	4.9	0.30	0.70	− 0.40
5.5	1.5	11.7	126 ^b	49.7 ^a	5.0	0.03	1.85	− 1.82
6.5	1.5	11.7	135 ^a	49.4 ^a	6.4	0.03	2.24	− 2.21
4.5	3	22.5	130 ^a	47.5 ^b	4.7	0.23	1.28	− 1.05
5.5	3	22.5	124 ^b	48.6 ^a	5.4	0.04	3.18	− 3.13
6.5	3	22.5	118 ^c	45.5 ^c	6.2	0.01	3.37	− 3.36

The pH were setpoints for the circulating solution, automatically controlled by acid, and base dosage; the NH_4 rates were applied in the NS supplied (C_{ad}). The quantities of H^+ and OH^- and the acid balance are expressed in mmol/L water supplied (V_{ad}). Voogt, (unpublished data).

Bar-Yosef et al. (2000a,b,c,d) reported that fertigrating waxflower (*Chamelaucium uncinatum* L.) in sandy soil with 7 mM N (70% NH_4^+) at three irrigations per day, one irrigation every 3 day, and one irrigation every 5 day (identical water and N quantities) resulted after 1 year in soil–root-volume pH of 5.8, 6.4, and 6.7, respectively. The interpretation was that the time-averaged NH_4 concentration in the soil increased as irrigation frequency increased, nitrate concentration and uptake decreased, and consequently the pH declined. This mechanism cannot operate effectively in closed loop IRs because fresh N solution is applied only via the fill solution which comprises ~25% of the total N in the system, most of which is in the form of nitrate.

10.3.12.1.1 pH-dependent nutrient concentrations in solution

Nutrients whose solubility is strongly pH-dependent are Ca, P, Fe, Zn, and Mn. The relationship between pH, CO_2 partial pressure (PCO_2), and Ca concentration (C_{Ca} , M) in calcite (CaCO_3) saturated solution is (Lindsay, 1979)

$$0.5 \log(C_{\text{Ca}}^{2+}) = 4.87 - 0.5 \log(\text{PCO}_2) - \text{pH} \quad (10.22)$$

Under $\text{PCO}_2 = 3 \times 10^{-4}$ bar (300 ppm CO_2) and pH 7, $C_{\text{Ca}}^{2+} = 183$ mM; at pH 8.0 and 8.5 and same PCO_2 , C_{Ca}^{2+} drops to 1.83 and 0.183 mM, respectively. When C_{Ca}^{2+} is controlled by gypsum, its concentration was shown to be 4.8 mM Ca^{2+} regardless of pH which means that at $\text{pH} \leq 7$ precipitated gypsum (if present) controls Ca solution concentration.

In addition to Ca^{2+} , two common Ca species should be mentioned: CaHCO_3^+ and CaCO_3^0 . At total carbonates concentration of 10 mM, total Ca 2 mM and pH 8, the mole fractions of Ca^{2+} ,

CaHCO_3^+ , and CaCO_3^0 are 0.92, 0.05, and 0.03, respectively. At common carbonates concentration of 3–5 mM, the concentration of CaHCO_3^+ and CaCO_3^0 becomes negligible.

At a fixed Ca concentration, the pH determines the solution P concentration in equilibrium with Ca–P minerals. The solubility decreases in the order $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ (DCPD) $>$ $\text{Ca}_4\text{H}(\text{PO}_4)_3$ (OCP) $>$ $\text{Ca}_3(\text{PO}_4)_2$ (β TCP) $>$ $\text{Ca}_5(\text{PO}_4)_3\text{OH}$ (HA). If the activity product of Ca^{2+} and HPO_4^- exceeds the DCPD solubility product, the mineral precipitates within minutes-to-hours, and if the solution is maintained at $C_{\text{Ca}}^{2+} = 4.5$ mM and pH 7.0 an equilibrium P concentration of 3.5 mg/L is obtained. At pH 6.4, the concentration increases to 14.0 mg/L P (Table 10.27).

OCP precipitates slower than DCPD (within hours) and its corresponding equilibrium P concentrations are 0.9 and 9.0 mg/L P; β TCP precipitates within months and is therefore not formed within one growing season; its equilibrium concentrations at $C_{\text{Ca}}^{2+} = 4.5$ mM and pH 7.0 and 6.4 are 0.15 and 2.4 mg/L P, respectively (Table 10.27).

When the 4.5 mM Ca includes Ca^{2+} , CaHCO_3^+ , and CaCO_3^0 , the total-P equilibrium concentration can be $\sim 20\%$ – 50% greater than shown in Table 10.27, depending on specific conditions. Under high-frequency irrigation, the substrate solution is usually supersaturated with respect to the concentrations shown in Table 10.27, as the experimental results of P concentration in recycled solution versus pH indicate (Fig. 10.17). The data clearly show a decline in P concentration as pH rises above 6.5, as theory predicts.

Iron, Zn, and Mn have limited solubilities due to formation of insoluble hydroxides ($\text{Fe}^{3+} + 3(\text{OH}^-) \leftrightarrow \text{Fe}(\text{OH})_{3(s)}$, $\text{Zn}^{2+} + 2(\text{OH}^-) \leftrightarrow \text{Zn}(\text{OH})_{2(s)}$ and $\text{Mn}^{3+} + 3(\text{OH}^-) \leftrightarrow \text{MnOOH}_{(s)} + \text{H}_2\text{O}$) and oxides ($\text{MnO}_2_{(s)}$) at elevated pHs. These reactions decrease Fe^{3+} , Mn^{3+} , and Mn^{4+} concentration to practically zero at pH $>$ 5. The three microelements are added to fill solution as salts of the divalent cations. To precipitate as hydroxides the Fe^{2+} and Mn^{2+} must be oxidized first. The kinetics of oxidation and precipitation in closed or open IRs is as yet unknown. Zaw and Chiswell (1999) reported that the oxidation of Mn^{2+} to Mn^{4+} (the Mn^{3+} is an intermediate in this reaction) in water with pH $<$ 8.5 is slow and takes days to weeks to complete; Scott et al. (2002) found on the other

Table 10.27 P Concentration in Equilibrium With Dicalcium Phosphate Dihydrate (DCPD), Octa Ca Phosphate (OCP), and β -Tricalcium Phosphate (β TCP) at Two pHs and One Ca Concentration^a (C_{Ca}) Representing Conditions in Arid Zone Recycled Solutions (Ionic Strength = 10 mM)

Mineral	pH	Equilibrium Concentration
		mg/L P
$\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ (DCPD)	6.4	14.0
	7.0	3.5
$\text{Ca}_4\text{H}(\text{PO}_4)_3$ (OCP)	6.4	9.1
	7.0	0.9
$\text{Ca}_3(\text{PO}_4)_2$ (β TCP)	6.4	2.4
	7.0	0.15

^a $C_{\text{Ca}} = 4.35$ mM [$= 174$ mg/L]. The calculation assumes that the only Ca species was Ca^{2+} .

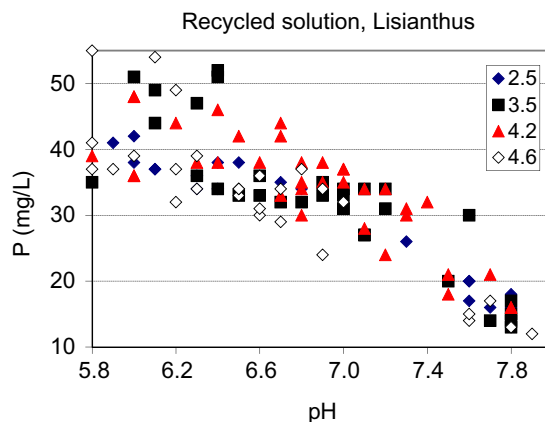


FIGURE 10.17

Inorganic P concentration in recycled solution as a function of pH. The crop was lisianthus grown in perlite in Besor. P was replenished in all threshold EC treatments (inset) to maintain 35 mg/L P in the recycled solution. EC, Electrical conductivity.

hand that Mn^{2+} oxidation and precipitation in a slightly acidic mountain stream was a matter of hours to days. The contrasting results can be explained by the fact that Mn^{2+} is oxidized both abiotically and biotically (Morgan, 2005). The first process is slow, as described by Zaw and Chiswell (1999). The second process is mediated by a wide variety of bacteria and is rapid (half-life of ~ 1 –10 hours, Morgan, 2005) compatible with rates reported by Scott et al. (2002).

The redox potentials at which Mn^{4+} and Fe^{3+} begin their reduction to Mn^{2+} and Fe^{2+} are 225 and 120 mV, respectively (redox potential decreases under anoxic conditions). These reduction reactions cannot occur before all NO_3^- in the system is depleted as nitrate reduction starts at redox potential of 250 mV (for review of this subject matter, see, for example, Cronk and Fennessy, 2001).

Minerals that may control Mn^{2+} concentration in solution are MnCO_3 (Schwab and Lindsay, 1983; Luo and Millero, 2003) and $\text{Mn}_5\text{H}_2(\text{PO}_4)_4$ or $\text{Mn}_3(\text{PO}_4)_2$ (Boyle and Lindsay, 1986). Zinc concentration in solution is controlled by $\text{Zn}(\text{OH})_2$ which is very insoluble at $\text{pH} > 6$ and $\text{Zn}_3(\text{PO}_4)_2$ (Lindsay, 1979). At pH 7 and H_2PO_4^- of 1.0 mM, the equilibrium Mn^{2+} and Zn^{2+} concentrations are $10^{-3.4}$ and $10^{-6.1}$ M, respectively (solubility products taken from Lindsay, 1979; no carbonates or Ca^{2+} were assumed to be present in the calculation). The Mn^{2+} concentration (21 mg/L) is appreciably higher than target Mn in solution (~ 0.5 mg/L) while the Zn^{2+} (0.05 mg/L) is approximately fivefold smaller than needed. When Mn^{2+} is in equilibrium with 5 mM HCO_3^- (pH 7, no P or Ca present) its concentration is 2 mg/L (solubility product of MnCO_3 taken from Lindsay, 1979), which is still higher than the target concentration. At pH 8, the Mn^{2+} concentration in equilibrium with the above PO_4 or CO_3 minerals is 1.0 and 0.2 mg/L, respectively.

10.3.12.1.2 pH and chelates

The precipitation and immobilization problem of some micronutrients is solved by using chelates (Table 10.28) that can sustain desired microelement concentrations in solution even at high pHs.

Table 10.28 Stability Constants (K_{sc}) of Several Ca, Mg, Fe, Zn, and Mn Chelates and Complexes Pertinent to Recirculated Solutions

Reaction ^a	Product	EDTA ⁴⁻	DTPA ⁴⁻	EDDHA ⁴⁻	Citrate ³⁻	Oxalate ²⁻
		log K_{sc}				
Ca ²⁺ + L	CaL	11.6	12.02	8.20	4.2	2.0
Mg ²⁺ + L	MgL	9.8	10.6	9.0	—	—
Fe ²⁺ + L	FeL	15.27	17.67	15.3	—	—
Fe ³⁺ + L	FeL	26.5	29.2	35.4	12.5	8.9
Zn ²⁺ + L	ZnL	17.44	19.6	17.8	5.5	4.6
Mn ²⁺ + L	MnL	14.5	16.7	—	4.5	3.7

^a $zM + L = M_nL$; $K_c = (M_nL)/[(M)^z (L)]$.
 Lindsay, W.L., 1979. *Chemical Equilibria in Soils*. Wiley-Interscience, New York; Norvel, W.A., 1972. *Equilibria of metal chelates in soil solution*. In: Mortvedt, J.J., Giordano, P.M., Lindsay, W.L. (Eds.), *Micronutrients in Agriculture*. Soil Sci. Soc. Am., Madison, WI, pp. 115–138 (Norvel, 1972).

Among the aforementioned cations, Fe³⁺ forms the most stable chelates with common natural and synthetic agents. For all greenhouse fertigation purposes, the stability constants (K_{sc}) of Fe³⁺, Fe²⁺, and Zn²⁺ with EDTA (Table 10.28) are sufficient to warrant solubility in recirculated solution even at pH ~7 (Table 10.29). The K_{sc} of Mn²⁺ is weaker; therefore its concentration in the recycled solution decreased sharply as pH approached 6.5 (Table 10.29).

Information on K_{sc} of Mn³⁺ or Mn⁴⁺ could not be found in the literature and it is also not known what the Mn²⁺/(Mn³⁺ + Mn⁴⁺) ratio was in the recycled solution. The natural complexing agents citrate and oxalate are not adequate substitution to EDTA (Table 10.28). The aforementioned K_{sc} constants and the equilibria between the free and complexed ions depend on the pH (Norvell, 1991). A clear demonstration of the practical significance of this phenomenon is given by Bugter and Reichwein (2007). They studied the interaction of metal-micronutrients with EDTA, DTPA, and EDDHA Fe-chelates under a wide range of pH values in mixed nutrient solutions, where Cu, Zn, and Mn were added as sulfates. They concluded that under conditions of the prevalent pH values in nutrient solutions when using Fe-EDTA and Fe-DTPA, virtually all supplied Cu²⁺ will replace equivalent Fe³⁺ from these chelates due to the higher stability. For Zn²⁺, the exchange with Fe-EDTA chelate becomes significant from pH values above 5.5 and for Fe-DTPA above 6. For Mn, this complexation starts from pH values of 7 and higher. This either partial or complete replacement of Fe from the chelating complex is less of a problem for Fe, as this ion is supplied in much higher concentrations than Cu and Zn. However, complexation of Cu and Zn is not necessarily an advantage as has been shown by Voogt and Sonneveld (2009). In an experiment with Fe-DTPA and Fe-EDDHA and continuous pH in the root environment of 4.5, 5.8, and 7.0 with roses in a closed system, they found strong accumulation of Zn in the circulating solution with DTPA and high pH, in contrast to EDDHA treatments. At the same time, the Zn uptake was lower compared to the lower pH treatments. The complexation of Zn-DTPA might explain this Zn accumulation. It is suggested that chelated Zn is less or even unavailable for plants (Parker et al., 1992). This phenomenon has been found in previous experiments with pH and DTPA (Voogt and Sonneveld, 2017).

Table 10.29 Microelement Concentrations as a Function of pH in Lettuce and Pepper Grown Using Recirculated Solutions in Besor, Israel

pH	Mn	Zn (mg/L)	Fe	Source
5.6	0.12	0.30	1.9	<i>Lettuce</i>
6.1	0.12	0.30	1.9	Bar-Yosef et al. (2001a,b)
6.5	0.035	0.30	2.1	
6.8	0.04	0.16	2.0	
5.0	0.46	0.24	2.5	<i>Pepper</i>
5.9	0.40	0.33	3.5	Bar-Yosef et al. (1999a,b)
6.7	0.03	0.16	1.5	
7.0	0.02	0.16	1.5	

Concentrations were determined by atomic absorption; thus they comprise the sum of all species in solution.

The above discussion emphasizes the fact that maintaining the recirculated solution pH between 5 and 6, which is essential to avoid Mn deficiency (without resorting to foliage Mn application) is one of the main and most difficult tasks in closed loop IRs management. The available knowledge and efficient monitoring of solution pH warrant that this problem will not become a growth limiting factor in closed loop IRs.

10.3.13 ROOT ZONE TEMPERATURE

The effect of root temperature on water and ion uptake has been intensively studied during the last three decades (Clarkson et al., 1986; Bowen, 1991; Macduff et al., 1994; Chung et al., 1998) and will not be reviewed in detail in this chapter. The subject is relevant because solutions in closed IRs are stored for one-to-several days in reservoirs which are usually exposed to outside air temperatures. It has been shown that a root temperature decline from $20 \pm 2^\circ\text{C}$ to $14 \pm 2^\circ\text{C}$ is associated with reduced water (Ho and Adams, 1995; Adams, 2002; Urrestarazu et al., 2003) and nutrients uptake rates (Engels, 1993; Engels et al., 1993; Adams, 2002; Urrestarazu et al., 2003), and the magnitude of the effect depends on the crop's physiological stage. Bravo and Uribe (1981) used a Q_{10} value of 2–2.5 to account for temperature effect on active nutrients uptake and stated that the Q_{10} for nutrient transport processes was considerably lower. Benoit and Ceustermans (2001) reported that stone wool cooling to $17\text{--}22^\circ\text{C}$ increased Ca concentration in paprika fruit relative to temperature of $23\text{--}33^\circ\text{C}$ and attributed this effect to higher oxygen concentration in the root zone at the lower temperature range. Root zone temperature affects root growth too. For example, decreasing the maize roots temperature from 20°C to 5°C reduced the root growth rate from 1.2 to 0.02 mm/h (Pritchard et al., 1990). Gannmore and Kafkafi (1980, 1983) found that tomato and strawberry root temperature exceeding $28\text{--}30^\circ\text{C}$ was detrimental to root development, particularly at high $\text{NH}_4\text{:NO}_3$ ratio in the feed solution. Heller et al. (2015) reported that the temperature in the substrate increased with the height of the pots and the frequency of irrigation (the maximum

daily increased from 23°C to 28°C) leading to higher occurrence of tip burned leaves. It may be concluded that root zone temperatures below ~18°C and above 28°C may seriously impair nutrients uptake and root growth. This implies that water reservoirs should be insulated to avoid temperature fluctuations exceeding this range.

10.3.14 INTERRELATIONSHIP BETWEEN CLIMATE AND SOLUTION RECYCLING

Increasing the relative humidity (RH) in greenhouses lowers ET and thereby decreases salt accumulation in closed loop IRs by reducing fresh water addition (Cohen et al., 2003). Another unexplored question is the crop response to RH under varying EC conditions mediated by solution recirculation (Sonneveld and Welles, 1988). The aforementioned questions were studied by Bar-Yosef et al. (2003a,b, 2004a) with roses and pepper as test crops, presented here as study cases. In roses, three climate regimes were imposed (roof ventilation, RV, wet pad, WP, and shading, SH), each at threshold EC of 2.7 or 4.0 dS/m. The whole-season mean RH in WP, RV, and SH were 64%, 59%, and 54%, respectively. The whole-season mean air temperature in WP was 1°C lower than in RV or SH, albeit during warm hours the difference was 2–6°C, depending on season. Global radiation in SH was 30% lower than in WP or RV, which significantly decreased DM production and yield (Table 10.30).

The two other climates gave similar total flowering stem yield, but RV increased the fraction of shorter (40–50 cm) stems in comparison with WP. A similar effect was obtained by increasing the threshold EC from 2.7 to 4.0 dS/m (same table). The advantage of WP over RV became apparent at EC 4.0 dS/m where it gave a significantly higher yield and reduced discharge of water (Table 10.30) and N. Total water savings could not be ascertained since water consumption by the

Table 10.30 Effect of Three Climate Regimes (Wet Pad, WP; Roof Ventilation, RV; and Sliding Screen Shading, SH) Coupled With Two Threshold EC (EC_{thr}) Values for Replacing Recycles Solutions on Rose Yield (cv. Mercedes) (Total¹ and 40–50 cm Long, Expressed per m² greenhouse), Evapotranspiration, ET, Dry Matter (DM) Production, and Water Discharge²

Climate ³	EC_{thr}	Flowering Stems Yield		Cum	Total	Discharged
		Total	40–50 cm	ET	DM	Water
	dS/m	Num/m ²		mm	kg/m ²	mm
WP	2.7	256 ^b	54 ^c	877	2.85	524
	4.0	298 ^a	73 ^b	947	3.11	162
RV	2.7	274 ^{ab}	88 ^{ab}	1091	2.66	596
	4.0	268 ^{ab}	100 ^a	1053	2.40	241
SH	2.7	233 ^c	60 ^c	949	2.42	556
	4.0	239 ^c	75 ^b	852	2.14	213

¹Total yield comprised the following length groups: 40–50, 50–60, 60–70, >70 cm.

²Concentration of N in discharged solution was in all treatments 140–160 mg/L N.

³The experiment took place at Besor, Israel (Bar-Yosef et al., 2003a,b).

^{a,b} $p = 0.05$.

WP per se was not measured. However, the reduced N disposal ($\sim 24 \text{ g/m}^2$ ground in WP and ~ 36 in RV) is significant. Two more facts should be emphasized: (1) the water use per kg DM was $306 \pm 2 \text{ L}$ in the WP and $425 \pm 15 \text{ L}$ in the RV treatment (calculated based on data in Table 10.30). This is compatible with the hypothesis that increasing RH reduces ET without affecting DM production. (2) Yield in the WP treatment was higher at $\text{EC}_{\text{thr}} 4.0$ than at $\text{EC}_{\text{thr}} 2.7 \text{ dS/m}$. This indicates that in recirculation one or more of the accumulating ions was beneficial to roses and under WP conditions its benefit outweighed its adverse effect via increased salinity. Under RV the salinity impact was stronger and no yield increase was recorded as EC_{thr} rose from 2.7 to 4.0 dS/m (Table 10.30). The accumulated ions in this experiment were Ca, Mg, and B, and their time-averaged concentration during the last 4 months of growth (when discharge controlled the solution EC) were 125, 100, and 0.26 mg/L, respectively, at $\text{EC}_{\text{thr}} 4.0 \text{ dS/m}$, and 100, 65, and 0.16 mg/L at $\text{EC}_{\text{thr}} 2.7 \text{ dS/m}$. The concentrations in fresh water were 57, 28, and 0.1 mg/L. All three ions and interrelationships among them could contribute to the beneficial effect of the $\text{EC}_{\text{thr}} 4.0 \text{ dS/m}$ treatment under WP cooling.

The effect of climate–EC interaction on pepper is summarized in Table 10.31.

Total fruit yield was significantly higher in WP than in RV or fogging (FG) with tap water while the effect of EC was insignificant. FG caused leaf torch which reduced the photosynthetic leaf area and thus yield. BER incidence was significantly higher in RV than in WP or FG, and in all treatments it was lower at EC 4.0 than 2.7. Like roses, the water consumption per kg DM was lower in WP than in RV and similar to FG (218 ± 1 , 244 ± 7 , and $220 \pm 4 \text{ L/kg}$, respectively); water discharge at EC 2.7 was considerably lower in WP than in RV or FG but at EC 4.0 dS/m the differences diminished because less water was discharged in all cooling treatments (Table 10.31).

Table 10.31 Effect of Three Climate Regimes (Wet Pad, WP; Roof Ventilation, RV; and Fogging, FG) Coupled With Two Threshold Salinities (Threshold Electrical Conductivity, EC_{thr}) for Replacing Recycled Solutions on Total Pepper (cv Celica) Yield (per m^2 greenhouse), Blossom End Rot (BER) Incidence (% of Total Fruit Number), Cumulative (Cum), Evapotranspiration (ET), Dry Matter (DM) Production, and Water Discharge¹

Climate ²	EC_{thr}	Fruit Yield		Cum	Total	Discharged
		Total	BER	ET	DM	Water
	dS/m	kg/m ²	% Num	Mm	kg/m ²	mm
WP	2.7	5.50 ^a	19.4 ^b	520	2.19	304
	4.0	5.26 ^a	15.8 ^c	531	2.28	125
RV	2.7	4.87 ^b	25.7 ^a	559	2.07	459
	4.0	4.83 ^b	24.4 ^a	521	1.98	146
FG	2.7	4.52 ^c	24.2 ^a	387	1.79	451
	4.0	4.44 ^c	14.5 ^c	407	1.83	105

¹Concentration of N in discharged solution was in all treatments 140–160 mg/L N.

²The experiment took place at Besor, Southern Israel (Bar-Yosef et al., 2004a,b).

^{a,b} $p = 0.05$.

The accumulating ions in solution in the pepper experiment were Ca, Mg, SO_4 , and B, and their concentrations at the end of the experiment were 133, 75, 207, and 0.31 mg/L, respectively, at $\text{EC}_{\text{thr}} = 4.0$ dS/m, and 75, 45, 96, and 0.26 mg/L at $\text{EC}_{\text{thr}} = 2.7$ dS/m. The concentrations in fresh water were 48, 26, 57, and 0.1 mg/L. Improved supply and utilization of Ca is usually associated with decline in the incidence of BER (for review see, for example, [Saure, 2001](#)). Since BER incidence declined at EC 4.0 in comparison with $\text{EC}_{\text{thr}} 2.7$, it is deduced that the accumulating beneficial ion was Ca. Indeed, for pepper the recommended Ca concentration in the irrigation solution in inert substrates in The Netherlands is ~ 120 mg/L Ca ([de Kreij et al., 1999](#)). The fact that there was a response to higher Ca concentration in solution (at EC 4.0 vs 2.7 dS/m) even though Ca concentration in treatment EC = 2.7 dS/m increased with time is not surprising as uptake is affected by competition with other cations, for example, NH_4 and K (see, for example, [Terada et al., 1996](#)) and enhanced Ca concentration increases its capacity to compete on common absorption sites.

Final decision whether RV or WP should be employed in closed loop IRs should be subjected to economic analysis taking into account WP installation, operation, and maintenance cost relative to RV. Certainly the effectiveness of WP rises as the recycled solution EC increases and in regions and seasons where high greenhouse temperatures limit production.

10.4 MODELING THE CROP-RECIRCULATION SYSTEM

Dynamic models provide a powerful tool for representing dynamic systems where many variable change in relation to driving variables such as climate, affecting growth and development of crops. As described earlier, models of closed loop fertigation systems for soilless culture systems can be used to describe the systems so as to improve our understanding of the system but can also be implemented as control models and subjected to optimization procedures.

10.4.1 REVIEW OF EXISTING MODELS

Any comprehensive model describing greenhouse crop growth in a climate-controlled closed loop IR should include the following components:

1. A simulator of crop DM production and partitioning between plant organs as a function of greenhouse climate and crop water and N status (for review, see [Marcelis et al., 1998](#)). Nitrogen is given higher priority than other nutrients due to its importance in plant development and specific effects of NH_4 and NO_3 on root growth, excretion of protons and carboxylates, and uptake of certain ions.
2. A simulator of water and individual nutrient uptake by the crop as a function of their concentration and distribution in the substrate. The specific problem of salt accumulation in closed IRs requires that competitive uptake phenomena (e.g., Na–K and Cl– NO_3) and OP effects on water uptake and nutrients mass flow to roots be accounted for in the simulation. Another concern is the salinity effect on plant characteristics, for example, DM content or specific leaf area ([Heuvelink et al., 2003](#)).

3. A simulator of water, solute, and nutrient transport in substrate, including ion partitioning between solid and liquid phases, mineralization of organic N, and nitrification as a function of root zone conditions.

Comprehensive crop–soil–atmosphere models (e.g., [DSSAT, 2004](#)) fulfill the above requirements but are not calibrated to greenhouse crops, detached substrates and constrained root volumes, nor are they programmed to account for solution recirculation and salinity build-up effects.

Two models specifically designed to simulate crop growth and uptake in closed loop IR greenhouses and to account for the aforementioned requirements are Greenman by [Bar-Yosef et al. \(2004a,b\)](#) and CLOSYS by [Marcelis et al. \(2003, 2005\)](#). The models differ in the simulation of root controlled processes and transport phenomena but are quite similar in their approach to calculate crop net photosynthesis and DM and N partitioning. Both models still lack the capability to predict produce quality and the effect of tissue elements other than N, on DM production.

Other greenhouse crop models simulating only part of the entire system have also been published. [Heinen \(1997\)](#) presented a detailed model of nutrients transport, root growth, and uptake but it lacks the interaction with DM production and does not account for the various components of recirculation systems. [Giuffrida et al. \(2003\)](#), [Staghellini et al. \(2004\)](#), and [Carmassi et al. \(2003\)](#) published models capable of predicting salt accumulation in closed loop IRs, but these models include no feedback relations with crop response to rising EC values. More recently [Voogt et al. \(2012\)](#) developed the model WATERSTREAMS that predict mainly all water flows in a closed system, including transpiration and discharge for filter flushing and describes the accumulation of Na (and Cl) in closed loops and predicts the discharge moments and discharge rates. Although this model lacks a comprehensive description of the Na and Cl dynamics in the substrate and has a simple input–output approach of these ions in the system compartments, validation of the model in situ produced a good match between predicted and real Na accumulation ([Voogt, unpublished data](#)).

10.4.2 EXAMPLES OF CLOSED LOOP IRRIGATION SYSTEM SIMULATIONS

10.4.2.1 Chloride accumulation and impact on N and water uptake

The threshold Cl concentration for replacing recycled solution determines the time it takes under given climatic conditions and irrigation regime until solution discharge must take place. As described above, threshold EC is simpler and less expensive to monitor but cannot be calculated directly; therefore it is being approximated from simulated Cl concentrations. Simulation results from the Greenman model ([Bar-Yosef et al., 2004a,b](#)) show that in order to maintain an average Cl concentration of 500 mg/L while growing tomato, the recycled solution must be partly discharged after day 27. If a concentration of 1750 mg/L is tolerated, then the discharge starts after 126 days ([Fig. 10.18](#)). A similar trend was obtained in a pepper experiment reported earlier ([Fig. 10.18](#)) where ~140 days were required to obtain an average Cl concentration of ~1050 mg/L in recycled solution at Besor, Israel. The model also predicts that Cl concentration at the bottom of the substrate is considerably higher than in the water reservoir (=irrigation solution), and the difference is widening with increasing threshold concentration ([Fig. 10.18](#)). The simulated daily fluctuations are greater at the bottom of the substrate than in the water reservoir, which is replenished with fresh solution in order to maintain constant water volume in the system.

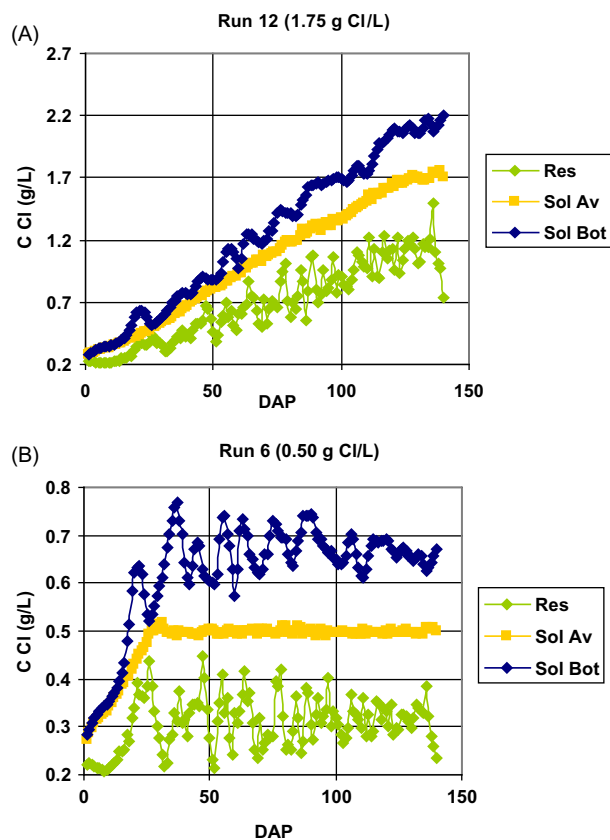


FIGURE 10.18

Simulated concentration of Cl in recirculated solution in closed loop irrigation system as a function of time and threshold Cl concentration. Simulated crop was tomato; growth conditions and fertigation management were typical to the Besor area in southern Israel. Threshold Cl concentration = 1750 mg/L (A) or 500 mg/L (B). *Res*, Water reservoir; *Sol Av*, average concentration in entire water volume; and *Sol Bot*, concentration in solution at the bottom of the substrate. Calculated by the Greenman model (Bar-Yosef et al., 2004c).

From Bar-Yosef, B., Fishman, S. and Kläring, H.-P. (2004). A Model-Based Decision Support System for Closed Irrigation Loop Greenhouses. *Acta Hortic.* 654, 107–122, <https://doi.org/10.17660/ActaHortic.2004.654.11>.

The accumulated Cl affects plant growth via several mechanisms: (1) inhibiting nitrate absorption by competitive uptake, (2) reducing water uptake and therefore mass flow of nutrients to roots due to OP, (3) decreasing the specific leaf area and increasing the %DM, particularly in fruits (Heuvelink et al., 2003). The Greenman model distinguishes between effects (1) and (2) while keeping (3) constant. Based on calibrated parameters for tomato, the model predicts that fruit yield is affected more by N uptake inhibition than water uptake. When excluding competitive uptake, the simulated N quantity in plant was $\sim 35 \text{ g N/m}^2$ ground (day 80), while when accounted for with a threshold Cl concentration of 500 mg/L the quantity is reduced to 10 g N/m^2 ground (Fig. 10.19).

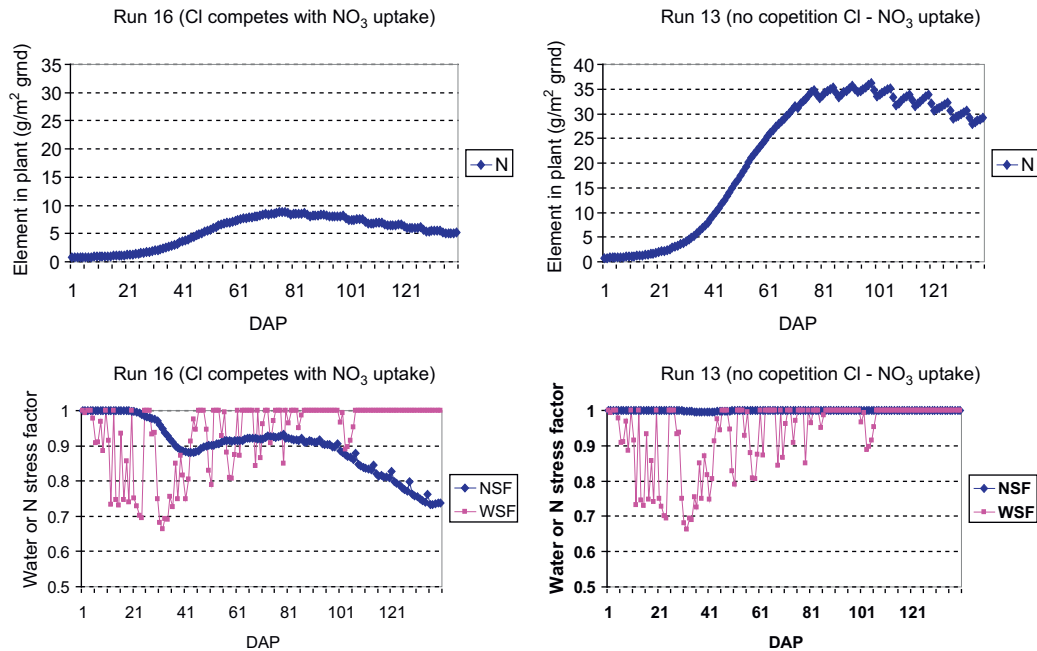


FIGURE 10.19

The effect of competitive Cl–NO₃ uptake on N absorption by tomato plants and NSF ($Cl_{\text{thresh}} = 500 \text{ mg/L}$). The effect is visualized by comparing results when competitive uptake is accounted for and when it is disregarded. The WSF was determined mainly by the assigned irrigation regime and weather conditions. Plants grew in a closed loop irrigation system in the Besor, southern Israel. Calculations by the Greenman model (Bar-Yosef et al., 2004c). NSF, N stress factor and WSF, water stress factor.

The N deficiency thus created induced a reduced DM production rate (expressed as “N stress factor”) that increased with time. When the Cl effect on N uptake was disregarded the N stress was negligible (stress factor = 1) (Fig. 10.19). The water stress factor (Fig. 10.19) was determined by the fertigation regime and weather conditions, and was negligibly affected by the reduced N uptake.

The solution OP effect on water and N uptake and disposal was simulated for threshold Cl concentration of 500 or 1225 mg/L (Fig. 10.20). The higher concentration created an OP of $\sim 8 \text{ MPa}$. The tomato plant “water retention curve” (water content θ vs water potential φ in plant) which is used in Greenman allows very low values of φ ($\sim 30 \text{ MPa}$, Bar-Yosef et al., 1980); therefore water uptake (which is proportional to the total water potential difference inside and outside the root) could be sustained at a rate not appreciably smaller than at 500 mg/L Cl. Experimental ET and N uptake data indeed show that both values did not differ considerably when EC_{thr} was 2.5 or 4.0 dS/m ($EC_{\text{thr}} 4.0$ corresponds to $\sim 40 \text{ mM}$ of monovalent electrolyte which is close to the 1225 mg/L Cl concentration). The model further predicts that in order to maintain a threshold Cl of 500 instead of 1250 mg/L, $\sim 200 \text{ L}$ water and 20 g N/m^2 ground must be discarded (Fig. 10.20).

Both the model and experimental results showed that maintaining the recirculated solution EC at 4.0 dS/m caused insignificant reduction in tomato yield (data not presented).

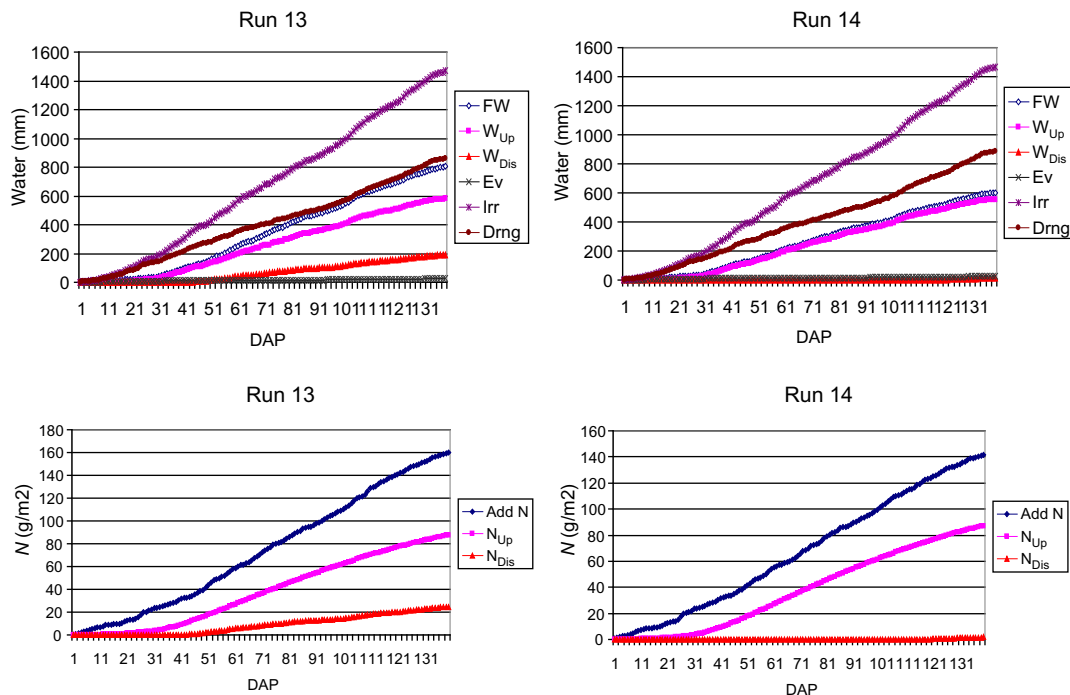


FIGURE 10.20

The simulated effect of threshold Cl concentration (run 13 = 500, run 14 = 1225 mg/LCl) on W_{up} and N_{up} and W_{dis} and N_{dis} by accounting for the osmotic potential mechanism and disregarding Cl–NO₃ competitive uptake. The applied Irr and simulated Ev and Drng are also included in the top figures. *Drng*, Drainage; *Ev*, evaporation; *Irr*, irrigation; N_{dis} , N disposal; N_{up} , N uptake; W_{dis} , water disposal; and W_{up} , water uptake.

10.4.2.2 Predicting solution pH

Controlling pH of recycled solution was mentioned earlier as one of the major problems in managing closed loop IRs. Based on principles discussed earlier, the Greenman model simulates both the nitrification and ammonium uptake effects on recirculated solution pH (Fig. 10.21). It is shown that for NH₄:NO₃ ratio 1:4 and 140 mg/L N, the NH₄ uptake is more effective in decreasing solution pH than nitrification. The model predicts an increase in solution pH starting ~40 days after planting coinciding with the commencement of fruit filling and reduced DM allocation to roots. The short-term fluctuations in pH stem from the difference between fresh water and recirculated solution pH. Predicted variations in pH along the season resemble experimental observations from the Besor area, Israel (Fig. 10.21B), so it is expected that when better calibrated the model might be helpful in choosing ammonium nitrate ratios and concentrations facilitating not only target N uptake rates but also target solution pH.

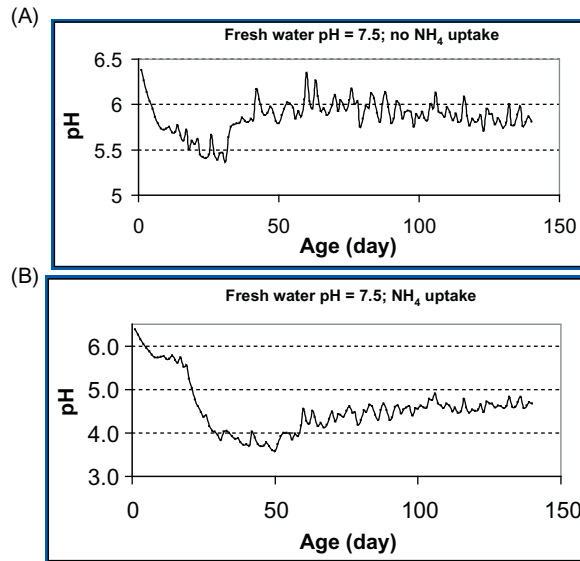


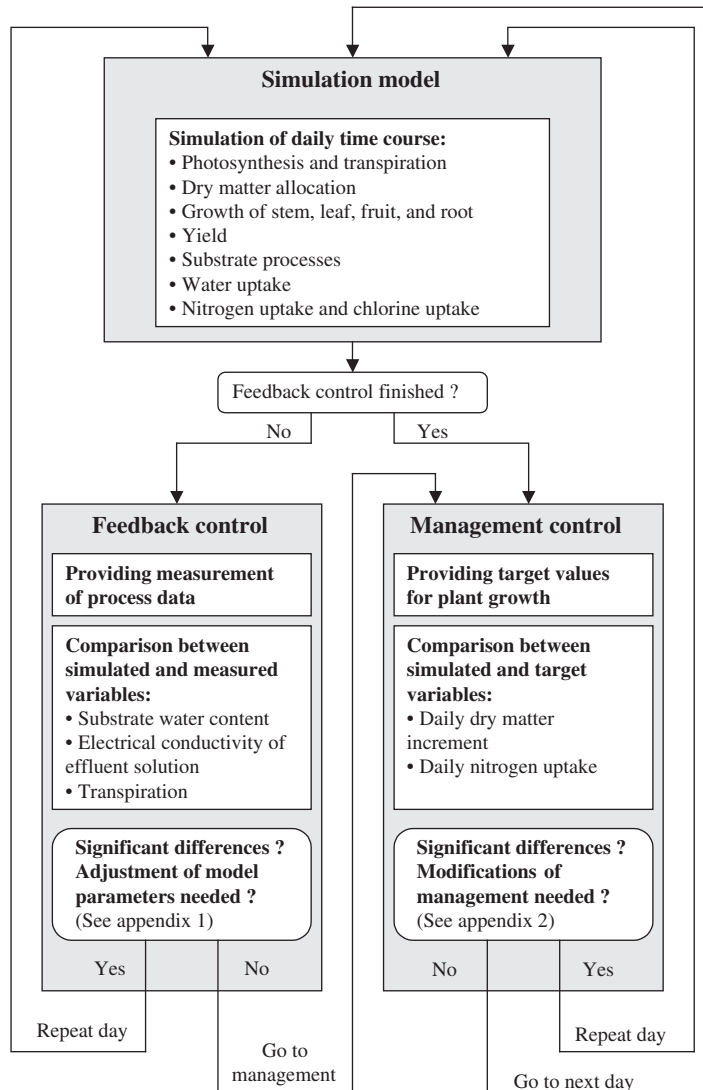
FIGURE 10.21

Simulation of recirculated solution pH as a function of days after planting (age). Initial pH 6.5, fresh water pH 7.5; $\text{NH}_4:\text{NO}_3 = 1:4$. (A) Disregarding NH_4 uptake effect on H^+ excretion by roots (nitrification driven pH change only). (B) Disregarding nitrification effect (NH_4 uptake driven pH change only).

10.4.3 OUTLOOK: MODEL-BASED DECISION SUPPORT TOOLS FOR (SEMI)CLOSED SYSTEMS

One of the major potential contributions of models is in their use for deriving management recommendations, and in the context of this chapter, optimizing the control of (semi)closed greenhouses. A decision support system (DSS) for drainage management in semiclosed hydroponic systems, based on a Na^+ mass balance model (Savvas et al., 2005, 2007; Varlagas et al., 2010) and measurements of plant water consumption was developed by Katsoulas et al. (2012). The DSS was tested and evaluated during several experimental periods and was capable to maintain a predefined level of Na^+ concentration in the nutrient solution and minimize water drainage and nitrate emission in tomato crops grown in semiclosed hydroponic systems.

The state-of-the-art DSS models for soilless systems used today by growers are part of modern day climate computers. These models mainly consist of algorithms for the prediction of ET, using actual and predicted weather and have a feedback on the real-time measured drainage (de Koning and Tsafaras, 2017). More advanced systems use not only the overall drainage flow but use sophisticated electronic weighing scales for the substrate, the drainage, and (in case of hanging crops such as tomato and cucumber) even plant weight (Tsafaras and de Koning, 2017). Usually simple algorithms are included for automatic EC correction of the irrigation, mainly driven by irradiation. So far individual nutrients or pH are not part in these DSS.

**FIGURE 10.22**

The DSS used by Bar-Yosef et al. (2004a,b). DSS, Decision support system.

To transform a model into a DSS, two essential components must be incorporated in it: (1) DM and uptake target functions according to which crop must develop in order to give an anticipated yield. (2) An “expert driven interface” that analyzes deviations between model predictions and target values and recommends modifications in management rules to minimize the gap. Two secondary prerequisites are (1) developing a hardware–software package allowing real-time acquisition of

weather and EC and pH data to the DSS. (2) A feedback mechanism that evaluates model's predictions based on real-time comparison between calculated and measured data, for example, solution pH or daily transpiration. An example of such a system is presented in Fig. 10.22 (Bar-Yosef et al., 2004a,b). A DSS by Marcelis et al. (2003) is driven by similar considerations, but it relies more on plant-related variables, for example, plant height and stem expansion. Greenhouse DSSs, such as the above, are designed to achieve automatic control of cooling or heating the greenhouse, CO₂ enrichment, RH, harvest and leaf pruning timing, and irrigation–fertilization management, including quantities and frequency of application and permitted threshold salinity and disposal. The DSS also allows evaluating preplanting decisions such as substrate type and volume and volume of water reservoir.

It is hard to assess how much time will be needed until currently developed DSS will become operative under real world conditions. Some engineering problems still exist in executing in real-time even simple model recommendations, for example, “increase irrigation frequency by 50% and report back.” Another difficulty in applying DSS is lack of validation under commercial production and feedback from growers, extension personnel, and associated scientists.

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ANALYTICAL METHODS USED WITH SOILLESS SUBSTRATES

11

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11.1 INTRODUCTION

Growing media constituents include processed raw materials (e.g., peat, pumice, tuff); waste products from agricultural, forest, and food industries (e.g., bark, rice hulls, coir); recycled materials (e.g., from mattresses, tires, paper waste); processed materials (e.g., poly phenol foam, urea formaldehyde foam, stone wool); or composted materials (Urrestarazu Gavilán, 2004; Zaccheo and Cattivello, 2009). A nearly unlimited number of possible growing media are obtained by mixing these diverse materials. Ideally each potential growing medium should be tested under commercial growing conditions in field trials in order to verify its suitability for crop production. However, besides the costs and duration of these “biotests,” even these tests cannot give a complete guarantee, since commercial growing conditions such as temperature or fertigation are never identical with the experimental conditions. Moreover, the performance of a product may be crop specific, which would considerably increase the number of required field trials. While manufacturers of new growing media test their products in plant production under commercial conditions, laboratory tests of the growing media are also used since these provide a reliable indication of the performance of the medium with respect to water, nutrient, and oxygen availabilities in both the short and long term. Such laboratory tests are normally more cost effective and can be performed under standardized conditions with reference samples.

The analytical methods for such tests—mostly adopted from soil science—target four areas of application: growing media selection, quality control, fertilization advice, and growing media usage/management recommendations (Table 11.1).

11.1.1 GROWING MEDIA SELECTION

Selecting soilless growing media for particular applications is one of the extensive uses of these analytical methods. In particular, it allows for benchmarking growing media against each other and for recommending ranges for acceptable values for particular applications.

Table 11.1 Overview of Parameters From Physical, Chemical, and Biological Analyses Used for Growing Media and Their Relevance and Use for the Applications

	Application of Growing Medium Analysis				
Parameters	Selection	Quality Control	Fertilization Advice	Recommendation	Remarks
Physical analysis					
Bulk density	+	++	− / +	++	Little experience Little experience
Porosity	++	++	−	++	
Water retention including rewetting	++ +	++	−	++ +	
Particle size	+	+	−	+	
Shrinkage	++	+	−	++	
Hardness, stickiness	+	−	−	+	
Penetrability	+	−	−	+	
Hydrophobicity	+	−	−	+	
Hydraulic conductivity	+	−	−	+	
Oxygen diffusion	+	−	−	+	
Chemical analysis					
Direct available elements in the rhizosphere	+	++ +	++ +	++	Depending on growing medium Depending on growing medium Depending on growing medium
Potentially available elements, EC	++	++	++	++	
Exchangeable ions, CEC, AEC	++	++	++	+	
Total analysis	+	+	+	+	
N fixation/P fixation	++	++	++	++	
CaCO ₃	− / ++	−	+	+	
pH	++	++	++	+	
Biological analysis					
Stability	++	− / +	−	++	Depending on growing medium
Phytotoxicity	++ +	− / +	−	++	May depend on crop
The columns show applications for selection of growing media, routine quality control, routine fertilization advice, and growing media usage/management recommendations. Note: Scale increases from − = parameter not relevant, to ++ + = very relevant. AEC, Anion exchange capacity; CEC, cation exchange capacity.					

11.1.2 QUALITY CONTROL

Another reason for the use of reliable analytical methods is that producers of growing media and/or laboratories should be able to use these methods for quality control by themselves or by independent standardization organizations for production of described quality in order to provide growers or retailers with media with prescribed physical and/or chemical characteristics. Among the most prominent of these organizations are ISO (International Organization for Standardization), ASTM (American Society for Testing and Materials, www.astm.org), CEN (European Committee for Standardization, www.cenorm.eu), DIN (Deutsches Institut für Normung e.V., www.din.de), VDLUFA (Verband Deutscher Landwirtschaftlicher Untersuchungs- und Forschungsanstalten www.vdlufa.de), AFNOR (Association Française de normalization, www.afnor.org), and RHP (Regeling Handelspotgronden, www.rhp.nl).

11.1.3 FERTILIZATION ADVICE

The analysis of growing media is done routinely for advice on fertilization of the plants being grown. Systems for fertilization advice based on various nutrient analysis techniques have been used for many years in many countries and are indispensable in hydroponics. Frequent (weekly) sampling with analysis and advice within 2 days has become a standard procedure.

11.1.4 GROWING MEDIA USAGE/MANAGEMENT RECOMMENDATION

Analytical methods are used extensively by end-users such as growers to decide on how to use particular growing media based on the physical, chemical, and biological characteristics of the media, in combination with the plant or crop requirements. These requirements will vary according to the crop, the growing system (slabs, containers), the water supply system (e.g., ebb-flood, overhead irrigation, drippers), water quality, and the growing period. Each combination of crop, growing system, and water supply system thus may require specific characteristics of a growing medium. Attempts to systemize the recommendation of particular growing media based on crop parameters (such as the fertilization advice systems mentioned) are not yet generally accepted (e.g., Kipp et al., 2000).

Obviously, not all parameters which are determined by means of physical, chemical, and biological methods are equally important for the abovementioned applications of the analysis methods. The overview table (Table 11.1) shows that some of the parameters such as directly available nutrients and electrical conductivity (EC) are rather important and frequently used for all applications. For physical analysis, water retention characteristics including rewetting are essential for screening, quality control, irrigation management, container volume, and container height recommendations. This does not mean that other analysis methods are less relevant. For example, the CaCO_3 content can be very important in screening rooting media such as composts. Hydraulic conductivity has been argued to be an essential tool for assessing water availability for plants (Raviv et al., 1999; Wallach and Raviv, 2008). However, a routine method for (unsaturated) hydraulic conductivity is not yet used due to technical difficulties (Wever et al., 2004). For an updated discussion of this subject see Chapter 3, Physical Characteristics of Soilless Media. Measurement of some parameters suffer from large variation (e.g., oxygen diffusion), and/or the high costs involved, which makes it

less suitable to use it as a method for routine measurements. Table 11.1 gives an overview of the physical, chemical, and biological parameters that are measured in growing media and the relevance of their use in specific stages along the decision-making process.

11.1.5 VARIATION

Since there is a large variety of physical and chemical analytical methods used by laboratories, efforts have been made to compare and standardize the laboratory methods. As an example, for physical analysis not all laboratories use the same sample preparation procedures. In the method used in service labs for physical parameter determination in the Netherlands, the potting mix sample is pressed and not moistened, whereas in the European Norm (CEN method), the sample is not pressed and is moistened. In the 1:1.5 vol. extract method (Sonneveld et al., 1974; Australian Standard, 2003a,b), the sample is moistened until pF 1.5 and pressed. Analysis reports can bring confusion to users familiar with their own analytical methods. Naturally, labs and countries are reluctant to change their analytical methods. The ISHS “working group on substrates in horticulture other than soils in situ” of the ISHS Commission of plant substrates should be mentioned for their efforts to encourage standardization (e.g., Gabriëls et al., 1991; Gabriëls, 1995). Research has shown that in many cases analysis results can be converted from one method into another (e.g., Sonneveld and van Elderen, 1994; De Kreij et al., 2001; Wever et al., 2005). Moreover, the variation among labs using the same methods on the same samples, both for physical (Gabriëls et al., 1991; Wever and Van Winkel, 2004), and chemical analyses (Baumgarten, 2004; De Kreij and Wever, 2005), should be improved in order to increase the reliability (Baumgarten, 2013).

11.1.6 INTERRELATIONSHIPS

To assess the suitability of a growing medium for a well-defined purpose, usually several interrelated measurements are interpreted. A common example is the relationship between stability on the one hand and water and air transport on the other. The underlying mutual cause is the breakdown of structural elements which results in a loss of pore volume. Another example is the relationship between loss of stability and the fixation of nitrogen and phosphorous. Microbiological activity is the mutual cause underlying both. Both examples show the importance of interpretation and the available room for new, more direct measurements.

In this chapter, the physical, chemical, and biological parameters as listed in Table 11.1 are described including the units used. Special cases, common values, and relation to crop growth are given too. In case a standardized European method is used (a so-called CEN method), more information can be found at www.cenorm.eu and this is referenced with the method.

11.1.7 TRENDS IN METHODS

Most laboratory methods are designed to deliver answers under static conditions, that is, time independent. The most striking example is the water retention curve which by definition shows the relation between water tension and water content at equilibrium, even if it takes 48 hours to reach that equilibrium. As the plant uptake of water, nutrients, and oxygen can be highly variable over the course of minutes to hours, such equilibria are in practice not often reached. For this reason there is a growing interest in methods which are more dynamic (Fonteno et al., 2013; Giuffrida and

Consoli, 2016). Examples are improved hydraulic conductivity and diffusivity methods and the water uptake rate.

A second development is the supply of materials from other industries, such as composts, digestates, biochars, and spent mushroom casing which come with quality parameters used in those supplying industries. It is often remarkably difficult to agree on the use (and interpretation) of methods developed for growing media. Well-known sources of confusion are the choice of extraction methods, the expression of water content in % V/V and the expression of organic matter in % V/V: (1) Extraction methods must be well defined to allow recalculation to EC, pH, and nutrient contents per unit growing medium volume. Plant roots and microbes will both react predictably to the amounts per unit volume but not at all to amounts per unit weight. Values derived from different extraction methods may be compared using a conversion factor. The conversion factor can be found by comparing two extraction method values which share the same nutrient content per unit growing medium volume. (2) The expression of water content in % V/V instead of % W/W is needed to assess the air content in % V/V in the various materials at various dry bulk densities. Air content is a factor which is often very critical for plant growth in the growing media mentioned. (3) The organic matter conversion into % V/V is needed to assess its importance to plant roots and microbes which both react to the amount in % V/V and not to the amount in % W/W.

Examples of misinterpretations by using weight instead of volume are to be found in Tables 11.3 and 11.4. Using the correct reference to units per volume of growing medium is especially important when comparisons are made between potting soil components with different dry bulk densities.

A third development is the interest in the evolution of growing media properties during cultivation. Over time, properties may change due to the actions of root growth and bacterial activity but also a large number of other physical, chemical, and biological processes (Caron et al., 2015; Judd et al., 2015; Kerloch and Michel, 2015).

11.2 SAMPLE PREPARATION

Soilless substrates manufactured for plant production come in various forms, such as loose material, pellets, compacted bulk materials, and slabs. Some soilless substrates or growing media are shipped to the end user in a compacted form; these materials are converted into loose growing media or a growing media mix using industrial methods (e.g., compressed coir bricks, various peats, and wood fibers). Samples for tests are always taken from the final growing medium. A representative bulk sample of the material must be prepared in a standardized fashion to allow the results to be representative for the whole batch of material investigated, and samples for the final individual measurements are prepared, usually by subsampling into test cylinders, such as rings. Preformed materials such as plugs or slabs are measured, without breaking them up.

11.2.1 BULK SAMPLING OF PREFORMED GROWING MEDIA

Preformed growing media, such as stone wool slabs, coir boards, peat boards, and polyurethane slabs, have a variation in properties within and between the units they are sold in, and within and between the batches they are produced in. Furthermore, most preformed growing media are pressed

into some shape during production and usually show a density profile in the direction perpendicular to the compression force. The pattern of cutting the material into the final units determines in what way these density differences may manifest themselves in the end product. Finally preformed fiber materials, such as stone wool, coir pith, and some peats, may show differences in fiber orientation in one or two directions. If it is desirable, samples should be taken in different directions of the original product. The sample preparation of preformed growing media is usually confined to cutting a sample of specific dimensions in such a way that the material is not disturbed (CEN 15761, 2009). To get a representative sample, 2 or more samples from different units are necessary (CEN 12579, 2013).

11.2.2 BULK SAMPLING LOOSE MATERIAL

Big differences among samples of the same material may exist among and within heaps or rows of loose growing media, and more importantly among different production dates of the same product. For most products like peat, coir dust, and compost, the weather conditions from the time of harvest till the day of transport influence the properties of the end product. Examples are peat excavated during a wet or a dry summer, composts made during a hot or cold period, coir dust from a lot stored for up to 5 years and composts made from summer grown plants versus winter grown plants.

Samples are taken from the heap or heaps with a hollow headed agricultural hand auger. The auger diameter should be larger than three times the maximum particle size of the sampled material. When no auger with a large enough diameter is at hand, a scoop may be used. The samples should be taken from a depth deep enough to avoid any material from the outer layer that has dried or otherwise obviously changed. This is usually deeper than 50 cm. Twenty or more samples from different places are necessary to get a representative sample. From bags (20–100 L), the whole bulk sample is taken from randomly chosen bags. Care should be taken to avoid fractionation of the sample, especially when using a scoop. The total sample size should not exceed what is needed in the laboratory to minimize the need for subsampling and the risk of segregation (CEN 12579, 2013).

11.2.3 SUBSAMPLING PREFORMED GROWING MEDIA

The sample preparation is confined to cutting a sample of specific dimensions in such a way that the material is not disturbed. Special attention should be paid to avoid compression by the knife or saw blade involved. When the material is preformed but the form is not stable, for example, boards of peat, it can be necessary to put the material in a holder.

11.2.4 SUBSAMPLING LOOSE GROWING MEDIA

The density of the materials may be stable because of the rigid nature of the individual particles. One might think of clay pellets, tuff (synonyms are pouzzolane, pozzolana, tezontle, porous volcanic material), pumice, perlite, or sand. These materials may be poured gently into a test cylinder of the right dimensions (e.g., diameter 5 cm and height 5 cm or diameter 10 cm and height 7.5 cm). No compression by weight or by tapping is applied.

For softer materials, which might be compressed by their own weight and moisture content, a laboratory compacted bulk density in its “as received state” has to be estimated. Based on that

laboratory compacted bulk density the weight of sample to be taken for subsamples is calculated. Different procedures exist in which the samples are brought to a standard density and/or moisture content. A specific example is given in CEN 13040 (2007). The medium is manually homogenized and passed through a screen beforehand. It is then sieved into a 1 L test cylinder with a removable collar until overflowing. The excess material is wiped off and the sample is then compacted with a plunger of specific weight. After a standard compaction interval, the collar is removed and the sample is wiped off at the level of the test cylinder. The test cylinder is then ready for weighing and the laboratory compacted bulk density is calculated.

11.3 PHYSICAL ANALYSIS

11.3.1 BULK DENSITY

11.3.1.1 Definition and units

In general, the bulk density (kg/m^3 , g/L) of a material is defined as the ratio of the mass of solids to the bulk volume of the growing medium. Depending on the purpose of the determination, the growing medium may be referred to in a dry, moistened, compacted, or loose consistence.

11.3.1.2 General principle of determination

The bulk density is measured by determining the weight of a known volume of material after a specific preparation procedure. For trade purposes, first the filling volume of packages is of major importance for the customer. Thus the material is taken from the package and loosened before the measurement (determination of quantity). Second, nutrient or salt contents of the material are usually declared on the basis of content per volume. In this case, the growing medium is compacted before measurement using an external weight (laboratory compacted bulk density). Third, for advisory work it is necessary to simulate the situation as it can be found at the grower during cultivation. In this case, a “self-compaction method” is used, receiving the “potting density.” For the first two cases, the moisture of the samples is taken as it occurs. In the latter case, the moisture may be adjusted to a level as used for growing purposes.

Finally, the bulk density may also be referred to the material in a dry state (dry bulk density). This parameter is determined after drying a certain volume of material following the determination of the water holding capacity (WHC) of a material.

11.3.1.3 Determination of quantity

The bulk density measurement for the determination of quantity (CEN 12580: 2013) is measured using a container with a known capacity of about 20 L and a height to diameter ratio between 0.9:1 and 1:1. A 7.5 cm high collar and a fall controller (a sieve with a defined mesh aperture) with the same diameter as the container are attached. The container and collar are filled to the top after which the collar is removed and the material is leveled off. The container is weighed and the bulk density is then calculated according to the following equation:

$$\rho_b = \frac{M}{V} \quad (11.1)$$

where M is the mass of the material in the container (fresh, kg), ρ_b is the bulk density (fresh, kg/m³), and V is the volume of the empty container (m³).

11.3.1.4 Determination of laboratory compacted bulk density

The laboratory compacted bulk density mostly serves as a basis for the display of nutrient or salt contents. A cylinder with a volume of 1 L (e.g., diameter $d = 100$ mm, height $h = 127$ mm) is filled with the growing medium, using an upper ring, a funnel, and a sieve (mesh size 20 mm). After filling, funnel and sieve are removed and a weight (650 g) is placed on top for 180 seconds. After this time, both weight and upper ring are removed and the surface is leveled using a blade or ruler. Then the mass of the cylinder is determined. The laboratory bulk density is calculated according to the following equation:

$$\text{CBD} = m_2 - m_1 \quad (11.2)$$

where CBD is the laboratory compacted bulk density (g/L), m_1 is the mass of the cylinder (g), and m_2 is the mass of the cylinder including the sample (g).

11.3.1.5 Determination of potting density

Especially for advisory work, a special bulk density determination may be used to simulate the situation as it is found at the grower after the potting process.

Before determination, the sample is prepared to reach a moisture content representing an optimal status for potting (fist test): the sample is pressed in the fist. If water beads escape between the fingers, the sample is too wet. If the sample crumbles in the hand when the fist is opened, without further action, the sample is too dry. Suitable moisture content is present if the pressed sample forms an aggregate which crumbles under mild pressure, after the fist has been opened; if, on the contrary, it only deforms, it is too wet. The optimal moisture content may be described as “moist as a well-squeezed sponge.” When moistening excessively dry sample material, the water shall be mixed into the sample material in such a manner that it is evenly absorbed. In the case of very dry samples, this process requires thorough mixing at intervals. This procedure shall last no >8 hours. Excessively moist samples shall be air-dried (<30°C) carefully and thoroughly mixed thereafter.

For material passing a 5 mm sieve, about 300 mL of the material at a moisture content that is suitable for plant growth is loosely filled in graduated cylinder with of the following dimensions: inner height 170 ± 5 mm, inner diameter 46 ± 2 mm, pedestal height of 10 ± 2 mm, and pedestal diameter of 70 ± 5 mm. Graduated up to 250 mL with a scaling of at least 5 mL. Before filling, the tare of the empty cylinder is recorded and the cylinder is placed on the compaction device (Fig. 11.1).

Then a hopper fitting to the cylinder is placed about 10 mm above the upper edge of the cylinder and the cylinder is loosely filled with an excessive amount of the moistened and sieved material through the hopper. Excessive material is removed gently with a ruler. The filled cylinder is placed in the ring of the compaction device and dropped 10 times on the rubber mat after lifting the pedestal to the lower edge of the ring. It has to be assured that the cylinder drops vertically and free without touching the ring. If necessary, the surface of the compacted sample is leveled gently without further compaction. The graduated cylinder is weighed with an accuracy of 0.1 g and the volume is read with an accuracy of 5 mL.

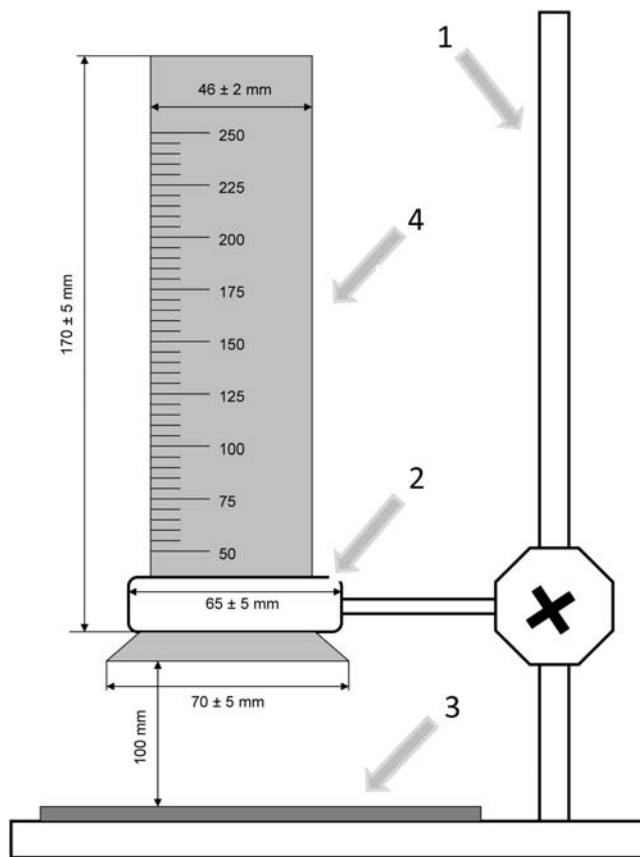


FIGURE 11.1

Compaction device for materials passing a 5 mm sieve. 1 = laboratory stand; 2 = ring; 3 = rubber mat; 4 = graduated cylinder.

The potting density is calculated according to the following equation:

$$PD = \frac{m_L - m_o}{V} \cdot 1000 \quad (11.3)$$

where PD is the potting density (g/L), m_o is the mass of the empty cylinder (g), m_L is the mass of the compacted sample and the cylinder (g), and V is the volume of the compacted sample (mL).

For material not passing through a 5 mm sieve, the dimensions of the cylinder are the following: inner height 270 ± 5 mm, inner diameter 78 ± 4 mm, pedestal height of 15 ± 5 mm, and pedestal diameter of 120 ± 5 mm. Graduated up to 1000 mL with a scaling of at least 25. The compaction device has to be changed; accordingly, the procedure of the determination stays the same.

11.3.1.6 Determination of dry bulk density

The dry bulk density (kg/m^3) is defined as the ratio of the mass of dry solids to the bulk volume of the growing medium (after Blake and Hartge, 1986). It is measured by determining the dry weight

**FIGURE 11.2**

Two test cylinders prepared for the determination of the dry bulk density.

of a known volume of material after the determination of the WHC at -10 cm water head (-1 kPa) and subsequent drying at 105°C . The determination is carried out using two test cylinders with approximately 100 mm diameter and 50 mm height each (Fig. 11.2). One test cylinder is extended to double its height by securing a second test cylinder on top. The test cylinder and rim (second holder) are then filled with the moist and homogenized sample, avoiding compaction or artificial air pores. Then the sample is saturated with water for 24 hours. Afterward, the WHC at -1 kPa is determined by equilibrating the water content on a suction table for 48–72 hours. To determine the WHC, the two cylinders are removed, separated, and the surface of the lower cylinder is leveled by wiping or scraping off the excess material with a rigid blade. Then the mass is determined. The mass is determined a second time after drying at 105°C , and the dry bulk density is calculated according to the following equation:

$$D_{\text{BD}} = \frac{(m_2 - m_1)}{V} \times 1000 \quad (11.4)$$

where D_{BD} is the dry bulk density (g/L), m_1 is the mass of the cylinder (g), m_2 is the mass of the cylinder including the sample after drying (g), and V is the volume of the cylinder (mL).

11.3.1.7 Special cases

Nonrigid growing media such as peat and composts have to be handled with special care as described in “sampling.” During transport, filling and weighing procedures vibrations might alter bulk density considerably.

More than one figure for bulk density is necessary to sufficiently characterize media with a density profile as in some peat products, or with layers of different density as in some types of stone wool (Bullens, 2001).

11.3.1.8 Common values for media

The dry bulk densities for most growing media are 3–20 times lower than for most soils (soils are up to 1500 kg/m³, for growing media see Table 11.2).

11.3.1.9 Influence on plant growth

For a given material, an increase in bulk density is associated with decrease in total pore space (TPS) and thus affects plant growth mainly through the effects of reduced free pore space. A decrease in TPS will often decrease both water and air (including oxygen) contents, and their transport. A decrease in TPS will also decrease root penetration. Water retention, hydraulic conductivity, and gas transport are however also strongly affected by pore diameter distribution. Therefore bulk density and total porosity alone are not enough to estimate water retention, hydraulic conductivity, and gas transport nor root growth.

11.3.2 ORGANIC MATTER CONTENT

11.3.2.1 Definition and units

The organic matter content is the amount of organic components expressed as percentage of weight over the weight of the dry sample (% W/W). Organic components include plant and animal residues at various stages of decomposition, cells and tissues of microorganisms, and substances synthesized by those microorganisms.

Table 11.2 Bulk Density, Volume Fractions of Solids, Water, and Air at a Suction Head of –10 cm and Total Pore Space as Measured in Some Growing Media

	Dry Bulk	Volume (% V/V)			
	Density	Solids	Water	Air	Total Pore Space
Units	kg/m ³	% V/V	% V/V	% V/V	% V/V
Glass wool	—	2	59	39	98
Stone wool	49	3	69	28	97
Perlite	105	4	35	61	96
Polyurethane	78	5	18	77	95
Peat	113	9	54	37	91
Pumice	431	17	32	51	83
Clay granules	489	24	21	55	76
Tuff	700–850	47	20	33	53

After Kipp, J.A., Wever, G., De Kreij, C., 2000. International Substrate Manual. Elsevier, Amsterdam, The Netherlands.

11.3.2.2 General principle of determination

The organic matter content is measured by bringing an oven-dried sample slowly to a temperature of 550°C for 24 hours (CEN 13039, 2011). The loss on weight is reported as the percentage of organic matter in % W/W of the original oven dry sample. The weight of remaining minerals and ash is reported as the percentage mineral matter in % W/W of the original oven dry sample.

11.3.2.3 Special cases

The temperature of ignition must be reported with the method as many different temperatures and durations of ignition are used. Temperatures reported range from 375°C to 650°C and duration ranges from 2 to 24 hours. An extensive comparison by Matthiessen et al. (2005) showed 550°C for 2 hours to be an acceptable general solution. Reasons to use the lower temperatures with less complete oxidation are samples with high levels of organic material leading to an open flame which throws out particles from the sample holders. Samples are usually only 2 g which requires very careful presampling. The usually small weight loss by oxidation of nitrate is thus counted as organic matter.

11.3.2.4 Common values for media

In soils values between 0% and 5% W/W are common with values 0%–2% W/W common in arid areas. Growing media can vary from 0% to 100% W/W with many potting soil mixes having values of >70% W/W. Some values for well-known growing media: perlite 0% W/W, stone wool 2% W/W, composts 40%–80% W/W, coir 70%–90% W/W, peat 80%–100% W/W (Kipp et al., 2000).

11.3.2.5 Influence on growth

Indirect effects of organic matter are the influence on water retention, the partial adsorption of nutrient cations to the cation exchange capacity and the support of a population of microorganisms which may stimulate plant growth.

11.3.2.6 % V/V and % W/W units

The quantity is reported in % W/W. As long as the organic material is mixed in mineral soil with densities of 1000–1250 kg/m³, this is usually not confusing. Once composts are mixed with growing media which are considerably lighter and possibly lighter than the compost itself, % W/W becomes a source of error which even affects scientific literature. Tables 11.3 and 11.4 illustrate the case for mixing with mineral soil and mixing with perlite.

To reach a level of 80% V/V organic matter in sand 33% W/W is needed. To reach 80% V/V in perlite, 89% W/W is needed. Of course the amount of organic matter is exactly the same in both cases! The reason for the apparent difference in % W/W is entirely caused by the difference in density of the receiving growing media. The use of % W/W in this comparison is wrong and prone to lead to serious error! An underlying reason to prefer % V/V is that roots and soil microbes both experience the amount of organic matter in their environment on a volume basis (distance to the next organic particle). Disease resilience is correlated with the amount of organic matter in % V/V not by the amount in % W/W (Veeken et al., 2005).

Table 11.3 Soil Organic Matter Content in Sand-Compost Mixes, Expressed as % V/V and % W/W

DBD Sand	DBD Compost	Minerals	OM	Minerals	OM	DBD	OM
kg/m ³	kg/m ³	% V/V	% V/V	kg	kg	kg/m ³	% W/W
A	B	C	D	E	F	G	H
			100% – D	A × C	B × D	E + F	F/G
1600	200	0	100	0	200	200	100.0
1600	200	10	90	160	180	340	52.9
1600	200	20	80	320	160	480	33.3
1600	200	30	70	480	140	620	22.6
1600	200	40	60	640	120	760	15.8
1600	200	50	50	800	100	900	11.1
1600	200	60	40	960	80	1040	7.7
1600	200	70	30	1120	60	1180	5.1
1600	200	80	20	1280	40	1320	3.0
1600	200	90	10	1440	20	1460	1.4
1600	200	100	0	1600	0	1600	0.0

DBD, Dry bulk density; OM, organic matter.

Table 11.4 Soil Organic Matter Content in Perlite-Compost Mixes, Expressed as % V/V and % W/W

DBD Perlite	DBD Compost	Minerals	OM	Minerals	OM	DBD	OM
kg/m ³	kg/m ³	% V/V	% V/V	kg	kg	kg/m ³	% W/W
A	B	C	D	E	F	G	H
			100% – D	A × C	B × D	E + F	F/G
100	200	0	100	0	200	200	100.0
100	200	10	90	10	180	190	94.7
100	200	20	80	20	160	180	88.9
100	200	30	70	30	140	170	82.4
100	200	40	60	40	120	160	75.0
100	200	50	50	50	100	150	66.7
100	200	60	40	60	80	140	57.1
100	200	70	30	70	60	130	46.2
100	200	80	20	80	40	120	33.3
100	200	90	10	90	20	110	18.2
100	200	100	0	100	0	100	0.0

DBD, Dry bulk density; OM, organic matter.

11.3.3 POROSITY

11.3.3.1 Definition and units

The porosity is the total space that is not occupied by solid material, including space where water cannot readily enter. The porosity or TPS (% V/V) is calculated from the total volume minus the volume occupied by the solids. Often, only the maximum volume of water which will enter the material on immersion (effective pore volume) is reported as this is the volume which affects several other properties such as water and air transport. Effective pore volume can be measured more readily.

11.3.3.2 General principle of determination

Most siliceous materials have a density of the solid phase (a.k.a. particle density) of about 2650 g/L and most organic materials have a density of the solid phase of about 1550 g/L. When the dry bulk density of a sample with single constituent, for example, siliceous sand is known, it is assumed that the density of the solid phase of that sand is 2650 g/L. Thus a common dry bulk density of 1500 g/L would mean that $1500/2650 = 0.57$, that is, 57% V/V of the volume is occupied by the sand solids and 43% V/V of the sample is the total pore volume or porosity.

The effective pore volume may be found by measuring the amount of water entering a material upon immersion. Care should be taken to allow the sample to saturate from the bottom up by raising the water table slowly, taking 30 minutes or more. This is necessary to prevent air from being trapped in the pores.

A more accurate method to find the effective pore volume, using air instead of water, is the pycnometer. In a pycnometer measurement, a sample is put into a vacuum chamber and the volume of air entering the chamber upon breaking the vacuum is measured.

11.3.3.3 Special cases

Some growing media are composed of solids with a different density, for example, some stone wool mineral melts yield a density of about 2900 g/L. If the density is not known, matters can become increasingly complicated when mixtures of two or more materials are present.

The porosity calculated from the dry bulk density and the specific weights may have to be corrected for the amount of nonconnected pores. These pores are embedded in the material and remain air filled during immersion as in perlite, volcanic products like tuff, and extruded plastics like phenols and polyurethane. There is no fixed method to measure the total porosity, but a fair estimate is usually found by grinding the material to a fine, compressed powder. Measuring the volume of the sample before and after grinding yields a volume lost by grinding. Then the effective pore volume of the powder is measured by submersion or preferably a pycnometer. Finally, the total pore volume of the original material can be calculated by adding the volume lost upon grinding and the effective pore volume of the powder.

11.3.3.4 Common values for media

TPS for most growing media is 1.5–2.8 times higher than the values found for common soils (about 35% V/V, other values in Table 11.2).

11.3.3.5 Influence on growth

An increase in TPS by an increase in large pores will decrease the water retention, increase oxygen transport, and increase root penetration. These, in turn, will have a positive influence on plant growth. If the increase in TPS is brought about by an increase in small pores, the water retention will increase and oxygen transport and root penetration may decrease or increase. The effect of pore space is therefore complex and may affect plants in opposite ways also depending on pore size distribution.

11.3.4 PARTICLE SIZE DISTRIBUTION

11.3.4.1 Definition and units

A particle size class is a fraction (%W/W) of a material with particles with a diameter larger than a given lower limit and smaller than a given upper limit, for example, 2–4 mm.

11.3.4.2 General principle of determination

The material is air-dried and either gently ground or put directly on a set of sieves with decreasing mesh size. Thus the material is fractionated into classes which pass through the sieve above but not through the sieve underneath. A common sieve set in square mesh has the following sizes: 16.00, 8.00, 4.00, 2.00, 1.00, 0.500, 0.125, and 0.063 mm (CEN 15428, 2007).

11.3.4.3 Special cases

Sieve analysis is very sensitive to effects of drying, breaking prior to sieving, and the sieving process itself. Therefore detailed specifications are needed to ensure reproducible results. Samples with a lot of fine organic material, like green composts, may form rigid “cakes” upon drying. When this happens and the resulting cake does not readily disintegrate, it is better to follow a procedure using lower drying temperatures and periodic gentle agitation.

Breaking prior to sieving is only done to avoid starting with largely oversized clods. There is a maximum amount of material for a given sieve size to prevent clogging the mesh. The amount of energy in moving the sieves and the time of sieving is usually standardized, for example, to 7 minutes duration with 10 second bursts with a rest interval of 1 second between them and an amplitude of 1 mm.

The fraction passing one sieve size and not another is not a true measure of the particle diameter. It means that there is at least one diameter of the particle which enables it to pass the upper sieve. Particles with largely different dimensions for length, width, and height in particular, like reed compost, tend to orientate horizontally and not pass sieves which they can easily pass if handled manually. Finally, the sieves may have square or round holes which slightly influence the result (round holes retaining very slightly more material, depending on particle shape).

11.3.4.4 Common values for media

Media used in horticulture generally have particles ranging, arbitrarily, from 0.125 to 2 mm to reach an optimum between available air and available water (Abad et al., 2005). Much coarser materials, for example, many barks, are in common use to increase aeration for larger plants. Much finer materials, for example, clays, are in common use to enhance water retention characteristics.

Many materials are offered in different size ranges, for example, perlite may be delivered in fractions of 0–1, 0.6–2.5, and 1–7 mm. The vast variety of peat products may be offered as a single product, for example, fractions of 6–18 mm, but are quite often mixed. A common mix being 70% V/V peat <2 mm with 30% V/V of fractions 20 mm.

11.3.4.5 Influence on growth

The particle size distribution (in combination with the total pore volume) indicates the amount of small and large pores. This, in turn, is an important indication of the percentage of water and air at different suction levels and the ease of rewetting. In some cases, it gives information on the ease of handling of the material by machinery, for example, drilling holes for container plants or filling sowing trays. It is difficult to use the particle size on its own to estimate water characteristics, because the different size classes may show interstitial filling, which means a smaller size class can fill the holes among larger size classes, thus creating an unexpectedly high bulk density. In addition, water retention is also affected by surface characteristics such as the affinity between water molecules and the surface of the particles. The relation between particle size distribution and water characteristic is still being researched (Naveed et al., 2012).

11.3.5 WATER RETENTION

11.3.5.1 Definition and units

The WHC or retention is a function of the TPS (% V/V) and a suction force applied (either in cm water pressure or kPa). The more suction is applied, the drier the material gets (this is represented by the main drainage curve in Klute and Dirksen, 1986).

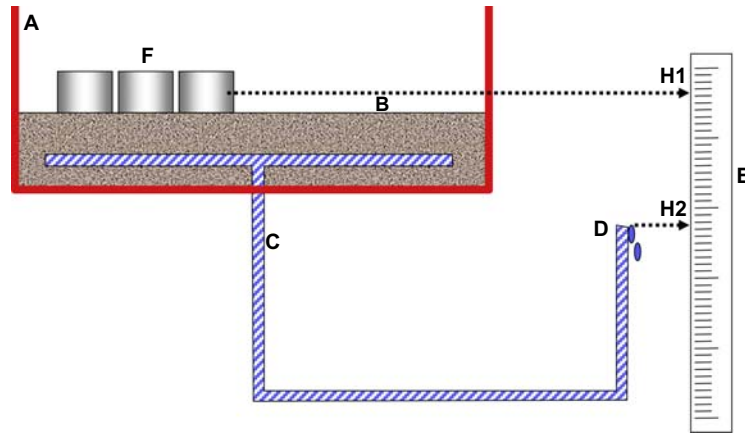
11.3.5.2 General principle of determination

For media with suctions up to 10 or even 20 kPa suction tables are in common use. The suction is usually applied by placing the samples on a bed of very fine sand or clay in a watertight container connected to an adjustable overflow (Fig. 11.3).

Water retention measurements are time consuming, because equilibrium between the applied suction and the water-filled pore space is essential. Usually each measured point on a curve with 3–10 points requires 1–2 days to reach equilibrium. Frequently the term water retention is used for the measurement of both the main drainage curve and the main wetting (=rewetting) curve. Common measuring points include the water content at saturation, and at suctions of –2.5, –10, –20, –50, –100, and –500 cm suction head. Usually the main drainage curve and the main wetting curve share the wettest and the driest points but differ in between. This effect is called hysteresis. Its causes are discussed in the paragraph on hydrophobicity. Hysteresis also influences the rewetting, rehydration rate, and unsaturated hydraulic conductivity. Further discussion appears in Chapter 3, Physical Characteristics of Soilless Media.

11.3.5.3 Special cases

Various variants exist, for example, using air pressure instead of water columns. Especially at higher suction levels air will enter the sand bed of the suction table. Strict procedures for sample preparation, saturation, and measuring time have to be observed. The method is derived from soil

**FIGURE 11.3**

Schematic suction table with a water tight container (A), filled with moist fine graded and packed sand in a flat layer of ± 10 cm (B). In the layer of sand, there is a drainage system with multiple well-distributed entries (C) connected via water-filled tubes to an adjustable overflow (D). The overflow height corresponding to the suction ($H_2 - H_1$) can be read on a scale in mm (E). The suction is applied to the samples in sample holders (F) and is usually defined as the difference in cm between the overflow and half the height of the samples.

science and its application for media creates some interpretation problems. One interpretation problem is related to the sample height of 5 cm, which is treated as a point sample. In fact, for small suction forces applied (from 0 to 15 cm suction force, i.e., 0–1.5 kPa), the sample height does influence the reading to a large degree.

Another interpretation problem concerns the point of 0.0 cm suction, which is defined at a free water table up to half of the sample's height. The point of 0.0 cm suction is thus changing with the sample height. Sample heights other than 5.0 cm are common, for example, 3.5 cm for propagation plugs.

A third common interpretation problem is that many growing media used in horticulture are in practice subjected to only a part of the measuring range mentioned. There is no point in measuring, for example, polyurethane foam at suctions above 20 cm of water column since at these suctions there is no longer water in interconnected pores.

11.3.5.4 Common values for media

Water retention forces in growing media are usually 10–100 times lower than the common values for soil (10–100 kPa). The plant reaction to high suction forces is highly plant specific and the common permanent wilting point defined at 1585 kPa has to be treated as an indication at best (Fonteno and Jackson, 2014).

11.3.5.5 Influence on growth

The results are indicative of the ease of the uptake of water—and nutrition—by plants as well as the wetness in various growing systems. Growth is highest at low water retention forces, but very

low water retention forces are sometimes avoided, for example, when the amount of air-filled pores becomes too low for proper oxygen transport (Nichols and Duggan-Jones, 2013). Water retention forces high enough to decrease the fresh weight growth may actually be desirable, for example, to create denser, that is, better quality pot plants and in transplant production, when hardy plants are preferred by the growers. The term “easily available water” is sometimes used to indicate water available for unhindered growth, defined as the % V/V released between suction forces of -1.0 to -5.0 kPa. This quantity is indicative for peat-like growing media but not so much for many other growing media.

11.3.5.6 Graphic presentation

Fig. 11.4A–D shows examples of water retention (or drainage or moisture release) curves for sand, peat, coir pith, and a stone wool. In the figures, the water content drops with increasing suction force but much faster so for stone wool and sand. At the same time, the air content increases in proportion to the water content loss. The dark bar in the top represents the volume taken by the solids of the growing media, for example, about 69% for the sand and about 3% for the stone wool. Note that a linear scale is used, not a logarithmic scale which is in use in soil science, and which is called a pF curve. Soil scientists need a logarithmic scale to fit in the permanent wilting point at 15,850 cm suction force, that is, pF 4.2 (Fonteno and Jackson, 2014). Growing media analysis on the other hand requires a scale of 100 cm suction force at most, usually much less, for example.

11.3.6 REWETTING

11.3.6.1 Definition and units

The rewetting curve (main wetting curve in Klute and Dirksen, 1986) shows the ability of a material to take up water against gravity from a well-defined point of dryness (Fields et al., 2014). The rewetting curve is the water retention curve during rewetting. Ample time is allowed for the changing moisture content to reach equilibrium with the suction force applied.

11.3.6.2 General principle of determination

The rewetting curve is measured with the same apparatus and in the same units as described for measuring the water retention (CEN 13041, 2011). Common measuring points include -500 , -100 , -50 , -20 , -10 , -2.5 cm, and saturation. Each point shows an equilibrium between the water content in the sample and the suction force applied. The time to reach equilibrium is 24 hours or longer if required.

11.3.6.3 Special cases

Growing media are very different in their ability to rewet. For example, coir dust after drying is known to rewet to almost the level at saturation, but some stone wool may only rewet to half of the level at saturation. First, growing media differ in the amount of hysteresis they display. Hysteresis is that most growing media remain considerably drier when rewetting to a suction value than when drying to the same suction value (the drying curve and rewetting curve are not similar). Second, some growing media take minutes to rewet to equilibrium while others require days before equilibrium is reached. Third, the ability of growing media to rewet depends on the starting point of the rewetting process, that is, each starting point results in a different curve for each growing medium.

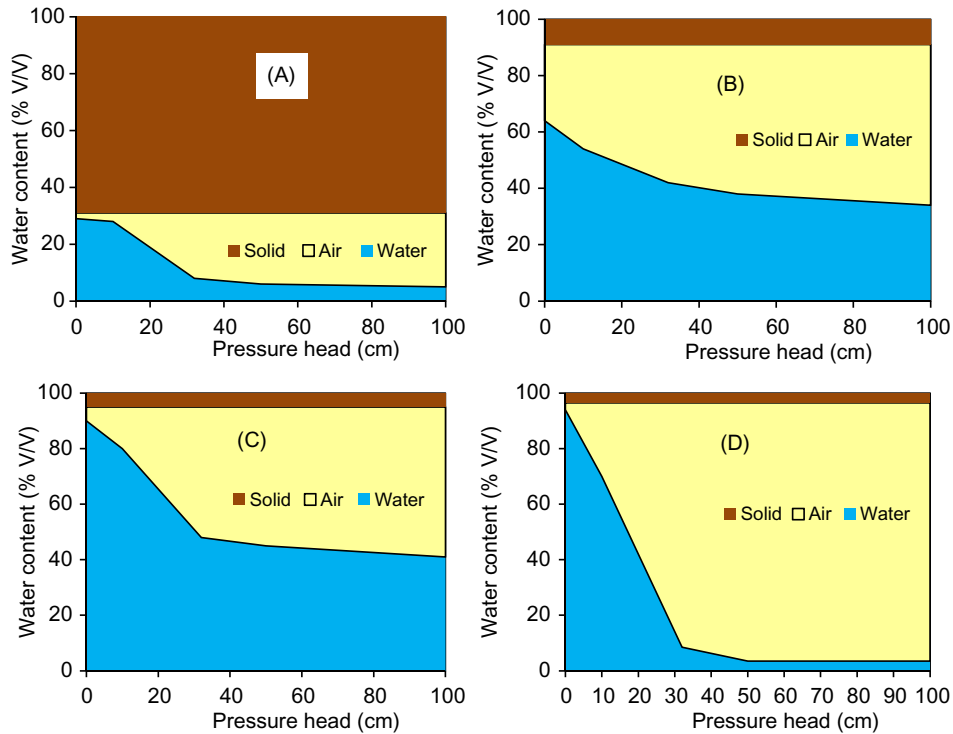


FIGURE 11.4

(A) Water retention curve for coarse sand growing medium, showing almost 70% V/V of solid matter, and a water content ranging from 30% to 5% V/V. (B) Water retention curve for (rather dry) peat growing medium, showing 9% V/V of solid matter, and a water content ranging from 64% to 34% V/V. (C) Water retention curve for coir pith growing medium, showing 5% V/V of solid matter, and a water content ranging from 90% to 41% V/V. (D) Water retention curve for stone wool growing medium, showing 3% V/V of solid matter, a water content ranging from 95% to 3% V/V. All for suction forces ranging from 0 to 10.0 kPa (0–100 cm).

Unfortunately, no one method at present describes the rewetting ability of growing media satisfactorily. The present rewetting curve from soil science requires standardized large pressure heads before rewetting. The problem is in choosing the point of defined dryness (Fonteno et al., 2013). The point of dryness has to be related to dry, but practical circumstances which are different for different growing media. The actual situation during growing is even more complicated as many drying and rewetting cycles are following one another, ending and starting from different points of dryness (Qi et al., 2011; Fields et al., 2014). Therefore at least in practice, an alternative method on the rewetting ability of the material, the rehydration rate, is increasingly used.

11.3.6.4 Common values for media

Values have to be regarded in relation to the point of dryness used as well as the particular growing medium. If stone wool is dried from a water content of 88% V/V (at -0.1 kPa) to 30% V/V (at

–2 kPa), rewetting to 60% V/V is normal. Had the point of maximum dryness been –10 kPa, rewetting to 60% V/V would indicate superb rewetting.

11.3.6.5 Influence on growth

Plants do not react directly to differences in rewetting ability, but imperfect rewetting will amplify any unevenness in crop water use of a given area. In practice, this will force growers to use more frequent irrigation cycles. Indeed, many published results suggest that plants respond favorably to more frequent irrigation (Xu et al., 2004; De Swaef et al., 2012).

11.3.7 REHYDRATION RATE (REWETTING RATE METHOD)

11.3.7.1 Definition and units

The rehydration rate is defined as the increase in moisture content of a dried sample over a set period of time (% V/V over minutes).

11.3.7.2 General principle of determination

A growing media sample of standard dimensions (such as a cylinder of 5 or 10 cm diameter and a height of 5 or 7.5 cm) is oven-dried at 105°C for 24 hours. After weighing, the sample is placed on a thin layer of water on a coarse mesh. The weight of the sample is measured after 0, 1, 2, 4, 8, 15, 30, 60, 120, 240, and 360 minutes and 1, 2, and 7 days. Special care should be devoted to the drying as small differences in drying may result in surprisingly large deviations in the rewetting rate. Second, care should be devoted to the contact with the free water surface. There should be a permanent full contact of 0 to 1 mm.

11.3.7.3 Special cases

Some growing media may develop hydrophobicity, notably materials containing nonfrozen black peat as well as some composts. Hydrophobicity may be caused by specific molecules either formed or deposited. It is difficult to distinguish effects of hydrophobicity from effects of the pore size distribution or of changes in pore geometry upon shrinking (Kerloch and Michel, 2015). Rehydration rate is closely related to unsaturated hydraulic conductivity. Some work has been devoted to relating rehydration rate and hydraulic conductivity with each other (Londra, 2010). This line of work can account for pore size distribution effects and even the effects of shrinkage but cannot account for hydrophobicity caused by specific molecules.

11.3.7.4 Common values for media

Fig. 11.5 shows the excellent rewetting rates of coir pith, a sowing soil (fine milled peat and compost 85:15% V/V), peat (H1–3), and a sandy medium used for outdoor ornamentals in large containers. In comparison to the others, coir pith shows the best rewetting rate, reaching over 70% V/V in 15 minutes.

11.3.7.5 Influence on growth

A slow rewetting rate, for example, <40% V/V in 15 minutes, increases the risk of too dry growing medium in the upper layer of the containers. It also increases the risk on hydrophobicity in places

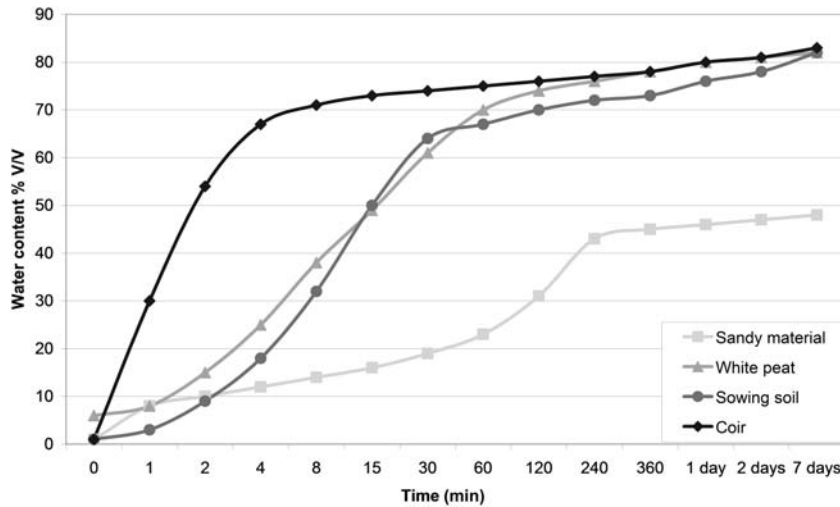


FIGURE 11.5

Rehydration rate of dry samples of four growing media from a free water surface.

where there is more evapotranspiration such as borders. Some growing media, such as the sandy material in Fig. 11.5, rewet too slow to be used in combination with sub irrigation.

11.3.8 HYDROPHOBICITY (OR WATER REPELLENCE)

11.3.8.1 Definition and units

Hydrophobicity is the irreversible loss of water retention upon drying caused by the change during drying in molecular organization of organic matter, and as a consequence, deposition of hydrophobic molecules on pore walls.

11.3.8.2 General principle of determination

It can be measured as a change in water contact angle with the pore wall. The direct measurement requires many microscopic observations and is laborious. A method based on capillary rise is more often used to estimate the contact angle (Michel et al., 2001, 2008; Naasz et al., 2008; Caron et al., 2015; Goebel et al., 2004).

11.3.8.3 Special cases

It is difficult to discern between incomplete rewetting and pore wall hydrophobicity. Incomplete rewetting is also caused by air entrapment and static pore diameter variations (which stands for different diameter sections in one continuous pore). These are the classic explanation of hysteresis, that is, the difference between a drainage curve and the subsequent wetting curve. Pore wall hydrophobicity is caused by the deposition of hydrophobic organic substances on soil particles (Ahmed et al., 2015; Ellerbrock et al., 2005). The difference is of practical importance as the rewetting

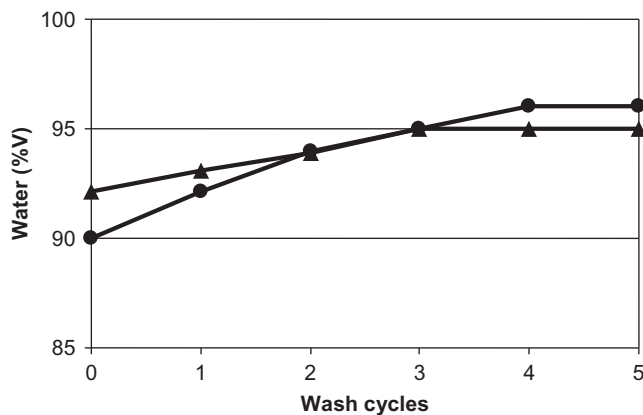


FIGURE 11.6

Volume of water at container capacity after repeatedly “washing” two brands of stone wool treated with a wetting agent before testing.

After Kipp, J.A., Wever, G., 1998. Characterisation of the Hydrophysical Behaviour of Stone Wool. Applied Plant Research. Naaldwijk, The Netherlands (Kipp and Wever, 1998).

method, based on equilibrium between suction and moisture content, shows mainly effects of hysteresis and the rehydration rate shows combined effects of hysteresis and pore wall hydrophobicity. Care should be taken in defining the terms explicitly in publications, as there seem to exist several interpretations of hydrophobicity, air entrapment, static pore diameter variations, and shrinkage (dynamic pore diameter variations). Wetting agents are sometimes used to facilitate rewetting of hydrophobic growing media like unfrozen black peat, mineral fibers, composts, and growing media in which decomposing roots and algae are present (Fig. 11.6). When using wetting agents, the initial maximum water content reached is lower than after a few rewetting cycles because of the reduced surface tension of the water (Yang, 2008; Fields et al., 2014).

11.3.8.4 Common values for media

It is recommended to measure against one or more well-known reference samples as figures from literature may be difficult to reproduce due to variations caused by temperature, ions from the growing medium and wetting agent–medium interactions.

11.3.8.5 Influence on growth

Hydrophobicity is a common and unwanted property. By nature hydrophobicity will be more pronounced when the material gets drier. The results are often seen in nurseries, where the boundary rows dry out faster, develop more hydrophobicity and consequently take up less water and get even more hydrophobic. This will reduce yield or quality of the plants from the boundary rows or will cost extra labor to manually rewet the growing medium.

11.3.9 SHRINKAGE

11.3.9.1 Definition and units

Shrinkage is the volume loss (% V/V) of a sample of standard dimensions after drying to 105°C as compared to the volume at standard moisture (usually –10 cm suction force) (CEN 13041, 2011).

11.3.9.2 General principle of determination

The material is moistened to standard moisture and put in a test cylinder of known dimensions in a standardized way. The material is then oven-dried. The dimensions of the remaining samples are measured by hand. The use of image analysis methods to measure shrinkage is emerging (Michel et al., 2004; Michel, 2015).

11.3.9.3 Special cases

Some growing media stick to the walls of the test cylinder, form cracks and even break in several pieces. The resulting volume may be measured after sealing the sample pieces with wax and measuring the underwater volume of the pieces.

11.3.9.4 Common values for media

In laboratories 5%–10% V/V is a common value. Some growing media like transplant media may shrink over 20% V/V (Jackson et al., 2008; Nowak, 2010).

11.3.9.5 Influence on growth

Shrinkage is a problem for outside ornamentals, such as trees, shrubs, and flowering plants, which may move in their containers or even be blown by wind. It is often related to hydrophobic effects caused by drying. In these cases irrigation may be problematic as any excess water drains very fast through the cracks or the void between container wall and the medium (this is called channeling). Also the evaporation from the medium is much faster at the surface area to the surrounding air, including the area of the cracks and voids, which then is much larger than the container surface alone. Shrinkage may be desirable in some transplant mixes used in tray cells as the plugs are to shrink loose from the surrounding tray which facilitates transplanting.

11.3.10 SATURATED HYDRAULIC CONDUCTIVITY

11.3.10.1 Definition and units

The mass of water which passes through a unit area in time ($\text{m}^3/\text{m}^2/\text{s}$) in a saturated growing medium.

11.3.10.2 General principle of determination

The saturated hydraulic conductivity is characterized by Darcy's equation (11.5):

$$Q = K_{\text{sat}} \cdot \frac{dH}{dx} \quad (11.5)$$

where Q is the flux (m/s), K_{sat} is the saturated hydraulic conductivity (m/s), dH is the head difference (m), and dx is the distance (m).

According to this equation it is possible to calculate or model water transport rates for any head difference over any distance in a given growing medium. Apparatus to measure K_{sat} with the constant head method usually consist of a supply container from which water flows to a sample of known dimensions. The saturated sample is kept under a constant head of water. The water passing through the sample is collected in a drain collection container and measured per unit time. Apparatus for the running head method are similar in construction, but the containers are directly connected to the sample and the head is allowed to change during the test. Care must be taken to fully saturate the sample and to prevent air bubbles from appearing in between the growing medium particles while running the tests.

11.3.10.3 Special cases

The application of standard measurement devices from soil science for the measurement of saturated hydraulic conductivity in growing media has proved difficult as the saturated hydraulic conductivities are 100–1500 times larger than in soils (1–100 cm/day). At all times it should be checked and reported that the apparatus without sample allows flows of at least one order of magnitude larger than the K_{sat} measured. This check is necessary to rule out the possibility that the hydraulic conductivity of a part of the apparatus such as the sample support or a low diameter valve or bend is limiting, and the measured value reported as if it were that of the test material. Several designs of apparatus have been proposed (Da Silva et al., 1995; Wever et al., 2004; Naasz et al., 2005).

11.3.10.4 Common values for media

Unvalidated measurements suggest 1 m/min for stone wool and coarse perlite (1–7.5 mm) and 0.01 m/min for fine perlite (0–1 mm).

11.3.10.5 Influence on growth

K_{sat} is usually not of direct importance for water and nutrient uptake by plants (but see unsaturated hydraulic conductivity). It is however relevant for salt leaching and changing the composition of the substrate solution.

11.3.11 UNSATURATED HYDRAULIC CONDUCTIVITY

11.3.11.1 Definition and units

The mass of water which passes through a unit area in time ($\text{m}^3/\text{m}^2/\text{s}$) under unsaturated conditions.

11.3.11.2 General principle of determination

The theoretical background of unsaturated hydraulic conductivity is thoroughly discussed in Chapter 3, Physical Characteristics of Soilless Media. There are methods for direct measurements of K_h , but these are tedious methods and they are not described here. An alternative method described in Chapter 3, Physical Characteristics of Soilless Media, is using a model based on the measured K_{sat} and the water retention curve. The model requires three constants to be determined

per growing medium. To further reduce the amount of parameters to be determined, here Eqs. (11.6) and (11.7) are offered.

The formula after Darcy is, with one adapted parameter, used to characterize unsaturated water transport as shown in the following equation:

$$Q = K_h \cdot \frac{dH}{dx} \quad (11.6)$$

where Q is the flux (m/s), K_h is the unsaturated hydraulic conductivity (m/s), dH is the head difference (m), and dx is the distance (m).

According to Eq. (11.6) it is possible to calculate or model water transport rates for any head difference over any distance for a given growing medium. K_h may be found using K_{sat} with a series of formulae according to the structure in the following equation:

$$K_h = K_{sat} \cdot \left(\frac{WFP}{TPS} \right)^m \quad (11.7)$$

where K_h is the unsaturated hydraulic conductivity (m/s), K_{sat} is the saturated hydraulic conductivity (m/s), WFP is the water-filled pore fraction (V/V), TPS is the total pore space (V/V), and m is the constant.

TPS is a constant which is usually known, leaving only the constant m to be determined. The only variable now is the water-filled pore space. The influence of water-filled pore on K_h is very large as can be seen from the exponential function. The relation water-filled pore over TPS is thought to represent the tortuosity of the transport path (Allaire et al., 1996). The measurement of either K_{sat} or K_h requires careful pretreatment of the samples and the apparatus used (Da Silva et al., 1995; Wever et al., 2004; Londra, 2010).

11.3.11.3 Special cases

Several designs of apparatus have been proposed (Da Silva et al., 1995; Wever et al., 2004; Naasz et al., 2005). Increasingly measurements are performed in the field, using on line sensors for highly localized water and oxygen content measurements (Londra et al., 2012; Caron et al., 2014).

When growing media have been properly characterized for K_h , or the relation K_h-K_{sat} , further work may be based upon model calculations for specific growing media dimensions and specific cultivation and supply techniques. It is common to link the water retention characteristics and unsaturated hydraulic conductivity (models using the work of Mualem and work of van Genuchten). These models are now being expanded to include dynamic rewetting and oxygen diffusion (Caron et al., 2002; Wallach and Raviv, 2008; de Swaef et al., 2012; Caron, 2004). There is also research into the evolution of K_h-K_{sat} over time in cultivation as influenced by structure loss and root growth (Kerloch and Michel, 2015).

11.3.11.4 Common values for media

The hydraulic conductivities at higher water contents are 100–1500 times higher than the values found for soils (1–100 cm/day, Fig. 11.7).

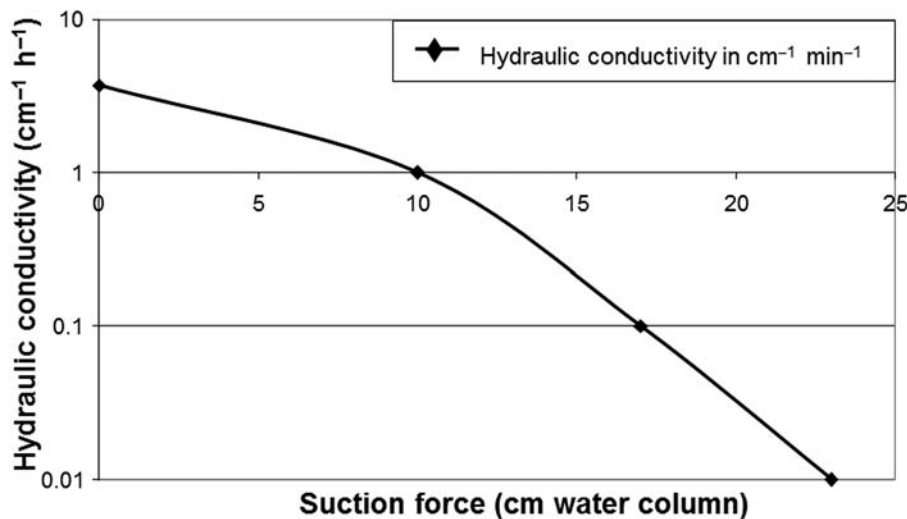


FIGURE 11.7

Unsaturated hydraulic conductivity of stone wool in relation to the suction force applied.

After Da Silva, F.F., Wallach, R., Chen, Y., 1995. Hydraulic properties of rockwool slabs used as substrates in horticulture. *Acta Hortic.* 401, 71–75.

11.3.11.5 Influence on growth

Despite the very high initial hydraulic conductivities, water transport rates may be growth limiting. The transport rate drops very rapidly with a decrease in water content, water uptake by horticultural crops is very high and the growing volume is usually small (Raviv et al., 1999). Growth may also be limited by low water content and transport rates in the rhizosphere, while the overall water content still seems acceptable (Kerloch and Michel, 2015; Londra et al., 2012; Caron and Nkongolo, 1999; Caron et al., 2001).

11.3.12 OXYGEN DIFFUSION

11.3.12.1 Definition and units

The mass of oxygen which passes through a unit area in time ($\text{g/m}^2/\text{s}$) in a growing medium.

11.3.12.2 General principle of determination

The oxygen transport by diffusion through air is characterized by the following equation:

$$Q = D_h \cdot \frac{dC}{dx} \quad (11.8)$$

where Q is the flux ($\text{g/m}^2/\text{s}$), D_h is the unsaturated oxygen diffusivity (m^2/s), dC is the concentration difference (g/m^3), and dx is the distance (m).

According to this equation it is possible to calculate or model oxygen transport rates for any concentration difference over any distance in a given growing medium. The real difficulty is in finding D_h . D_h may be found with a series of formulas with a structure as in the following equation:

$$D_h = D_o \cdot (\text{AFP}^a \cdot \text{TPS}^b) \quad (11.9)$$

where D_o is the oxygen diffusivity in air, AFP is the air-filled pores (V/V), TPS is the total pore space (V/V), and a and b are the constants.

The only variable is the air-filled pore space. The influence of air-filled pores on D_h is extremely large as indicated by the exponential function (Wever et al., 2001).

The measurement of either D_h or D_o requires careful pretreatment of the samples and the apparatus used (Klute and Dirksen, 1986). It is usually done by measuring the oxygen concentration with a fiber optic electrode. Just as for hydraulic conductivity, increasingly measurements are performed in the field, using on line sensors for highly localized water and oxygen content measurements (Londra et al., 2012; Caron et al., 2014).

11.3.12.3 Special cases

A basic assumption is that diffusion through air-filled space is the main transport mechanism for oxygen to the roots. In some cases, mass transport of oxygen in either air or water will be larger than diffusion. This may be the case when there is a temperature difference in the growing medium or when water is frequently or continuously moving as in nutrient flow techniques or aeroponics.

An alternative technique for these measurements is the use of air tight containers around the roots of a growing plant (Holtman et al., 2005). In such systems with substrate, an oxygen gradient from top to bottom arises which may be used as input for improving models (Blok and Gérard, 2013). The measurement of oxygen in air or water has become much easier with the introduction of fiber optic methods (Blok and Gérard, 2013). The introduction of the sensors in the material still requires great care and a control procedure to be sure that the results are reliable, that is, no air leakage is introduced by the measurement.

Another approach is the installation of air chambers with local contact to the atmosphere in the growing media examined (Londra et al., 2012; Wever et al., 2001). The oxygen content in the air chambers can be measured either by entering the sensor through a suitable septum or by using an air tight syringe to take samples for gas chromatography. Thus both oxygen and carbon dioxide may be measured simultaneously, showing a typical rapid decrease respectively increase with depth (Fig. 11.8).

11.3.12.4 Common values for media

Actual measurements of diffusivity against air-filled pores are becoming common (Caron, 2004; Dresbøll, 2010; De Swaef et al., 2012; Chamindu Deepagoda et al., 2013). The results show an exponential decrease of D_h with air-filled pores (Fig. 11.9). An important consequence of the exponential decrease of D_h with air-filled pores is that oxygen diffusion rates in all growing media with interrelated pores may, for lower air-filled porosities, that is, under 40% V/V, be simplified to one relation with air-filled porosity without much extra error in the air-filled pore volume found (Wever et al., 2001).

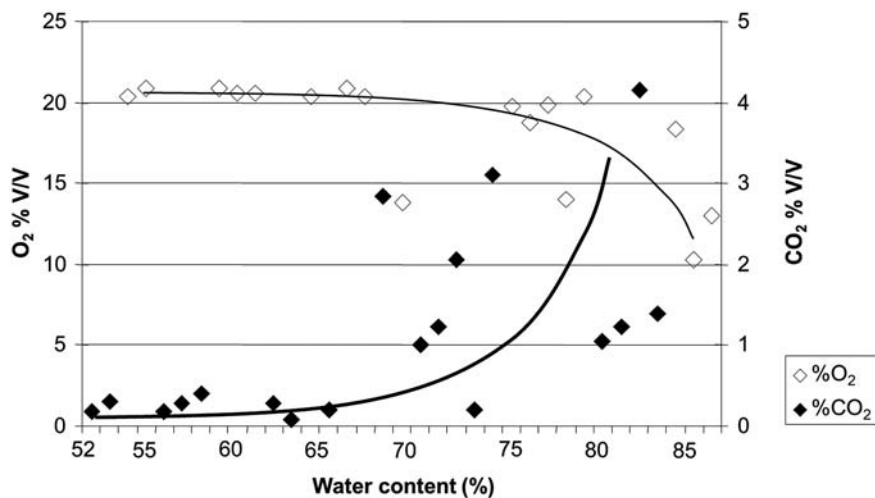


FIGURE 11.8

Microscale measurements of oxygen content and carbon dioxide content in the air of pores in stone wool against the local water-filled pore volume in % V/V.

After Baas, R., Wever, G., Koolen, A.J., 2001. Oxygen supply and consumption in soilless culture: evaluation of an oxygen simulation model for cucumber. *Acta Hortic.* 554, 157–164 (Baas et al., 2001).

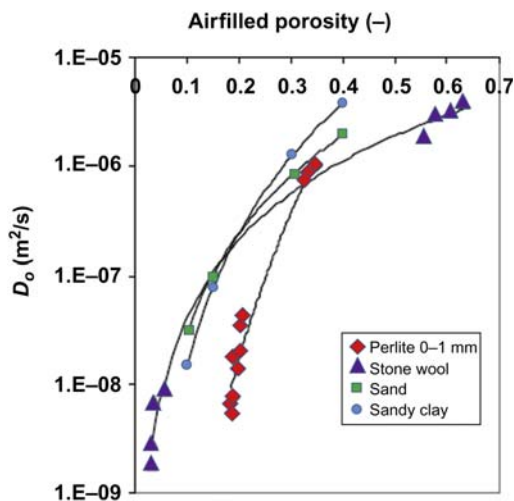


FIGURE 11.9

Measurements of the diffusivity of oxygen in various growing media against the air-filled pore volume in % V/V.

After Baas, R., Wever, G., Koolen, A.J., 2001. Oxygen supply and consumption in soilless culture: evaluation of an oxygen simulation model for cucumber. *Acta Hortic.* 554, 157–164.

For horticultural evaluations, flux calculations must be adapted to the dimensions and properties of the system under evaluation. For example, plant containers are open at the top and parts of the bottom, but impermeable at the sides. Furthermore the shape has to be included to be able to calculate the minimum pathway of oxygen to a particular point, given the changing water content and the oxygen consumption along the path. Based on flux calculations for diffusion only, and with container dimensions of <20 cm for both, width and height, oxygen levels in horticultural substrate systems may drop way below 20% in layers with <30% V/V air-filled pores (Wever et al., 2001).

11.3.12.5 Influence on growth

Plant oxygen use may be as high as 0.2 mg/g (FW-roots)/h (Bar-Yosef and Lieth, 2013; Blok and Gérard, 2013). At 20°C, water contains a maximum oxygen content of 8 mg/L which allows the conclusion that oxygen supply from the ambient atmosphere is essential. Anoxia may be expected in layers in horticultural substrate systems with <10% V/V air-filled pores. The exception in Fig. 11.9 is perlite which had to be corrected for the amount of closed not interrelated pores before it fitted the common line.

11.3.13 PENETRABILITY

11.3.13.1 Definition and units

The resistance to root growth (kPa) can be measured with a cone penetrometer (Fig. 11.10).

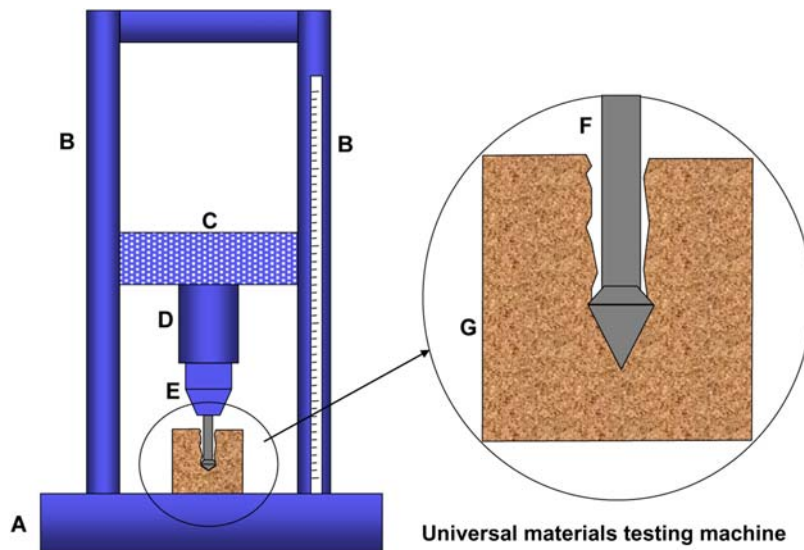
11.3.13.2 General principle of determination

Penetrometers are devices to measure the force needed to push a metal rod of known diameter into a growing medium. They may be hand operated and portable or machine driven and stationary.

The cone penetrometer is supposed to represent a root growing through the material. The force needed to insert the cone in the material is measured. Ideally, the penetrometer reading is then correlated to measured pressures of growing roots, but more often it is used alone. Before the reading becomes stable, some compression of the material takes place. A larger cone angle and a larger cone diameter increase the distance over which compression takes place before a stable reading is found (Bullens, 2001). To avoid the influence of friction along the shaft, it is necessary that the shaft driving the conical head is smaller in diameter than the head (Fig. 11.10).

11.3.13.3 Special cases

Fibrous growing media, such as stone wool and coarse peat, may be compacted over several centimeters before a stable reading is found unless an adapted cone is used with a maximum diameter of 2 mm and a cone angle of 30 degrees. Rigid growing media like stone wool should be measured in three directions as the fibers may be deposited in planes and roots will prefer certain directions. Rigid growing media, such as stone wool and pressed peat/soil plugs and cubes, may also show a marked compaction profile. Material with coarse particles may occasionally show very high readings which are better reported separately. This type of high reading is caused by the probe trying to press such a particle down whereas roots will grow around it.

**FIGURE 11.10**

Schematic representation of a universal testing machine used for the determination of mechanical properties as penetrability, hardness, and elasticity. A is a solid support to suppress vibrations, B are supporting shafts with a traction mechanism, C is a carrier beam, D is housing for the load cell, E is the mounting head, F is a shaft with a conical head, and G is the sample.

11.3.13.4 Common values for media

The ease of root growth is determined by the material density and the way the particles are interconnected. Typical penetrometer values for growing media are 2–20 times lower than the results found for soil (approximately 1200 kPa for soil).

11.3.13.5 Influence on growth

Root and shoot development react to differences in growing resistance over the whole range of values found (Fig. 11.11). In comparative research, threshold values for inhibition of fresh weight growth of roots and shoots were different for different growing media, for example, 250 kPa for stone wool and 450 kPa for perlite (Bullens, 2001). This seems to indicate that the method still needs to be improved.

11.3.14 HARDNESS, STICKINESS

11.3.14.1 Definition and units

Hardness is the maximum resistance value (kPa) measured when pushing a defined shape through a body of growing medium, just before the body gives way and splits into fragments. Hardness is used to characterize the handling forces some preformed products as plugs can withstand.

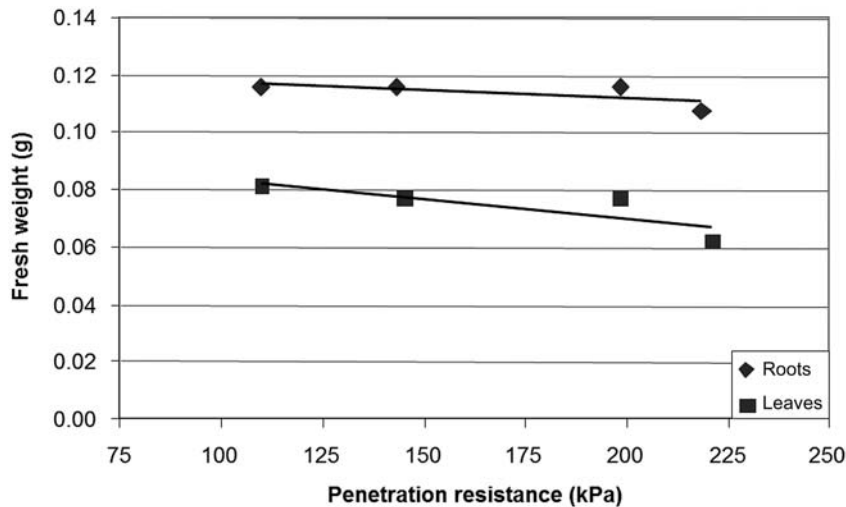


FIGURE 11.11

Decreasing growth with increasing penetration resistance of Irish white peat <15 mm.

After Bullens, H.P.G., 2001. *Mechanische eigenschappen van tuinbouwkundige groeimmedia (Mechanical properties of horticultural growing media)*. In: *Soil Technology*. Wageningen University, Wageningen, The Netherlands.

Stickiness is the maximum negative force (kPa) measured when lifting a defined surface from a compressed body of growing medium. Stickiness characterizes the cohesion between particles, a quality which determines whether the material can be used to be pressed into coherent growing medium forms, such as plugs or blocks, mainly for transplanting.

11.3.14.2 General principle of determination

Both hardness and stickiness can be measured with a universal material testing machine (Zwick, Instron, many others). These machines measure pressure against replacement and use load cells and displacement meters.

For a specific hardness test, a cylindrical plunger with an accurately measured maximum diameter and a conical head with an angle of 45 degrees is pressed into the material until 75% of the original height of the sample is reached. As with the measurement of penetration resistance, it is important that the shaft driving the conical head is of a markedly smaller diameter than the diameter of the conical head to be sure there is no influence of friction of the material along the shaft (Fig. 11.12). Appropriate values for the material testing machine are a downward speed of 50 mm/min for the plunger and a 100 N load cell to measure the force and calculate the pressure in kPa.

The maximum peak load registered during insertion is used to characterize hardness (Wever and Eymar, 1999). Recently results for a method to push a defined blade through peat blocks were reported in the trade press but no method was published yet (van Staalduinen, 2017).

The energy necessary to withdraw a plunger after insertion to 75% of the original height has been used to characterize stickiness or adhesiveness (Wever and Eymar, 1999).

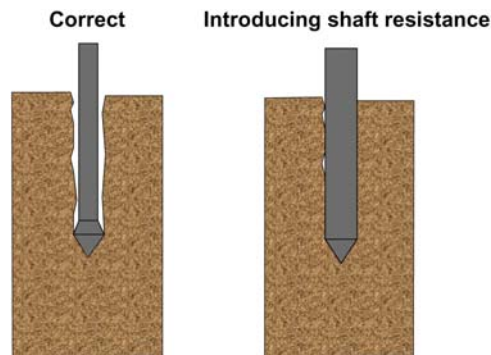


FIGURE 11.12

The shaft driving the plunger head should have a markedly smaller diameter than the plunger head.

For both hardness and stickiness one has to be aware of the extreme sensitivity of the properties to moisture content. It is therefore necessary to report results together with the moisture contents used. Hardness can be measured at saturation and for oven dry material, but stickiness can only be measured on wet material, for example, subjected to a suction force of -10 cm water column, after saturation.

11.3.14.3 Special cases

The measurement strongly resembles the penetration resistance measurement. Again fibrous growing media, such as stone wool and coarse peat, may be compacted over several centimeters before a stable reading is found, reason to use an adapted cone with a cone angle of 30 degrees. Rigid growing media, such as stone wool and peat blocks, should be measured in three directions as the fibers may be deposited differently in some directions and roots will prefer particular directions.

Many other mechanical properties have been measured and procedures have been suggested. Among these are: hardness and elasticity (Wever and Van Leeuwen, 1995), cohesiveness (Wever and Eymar, 1999), brittleness, stickiness, and firmness. Many more mechanical properties are known from descriptive work on building materials and food materials. There is, however, no agreement on which properties are essential for growing medium evaluation and which methods are appropriate. There is certainly room for a comparative study on this subject!

11.3.14.4 Common values for media

Hardness of pressed growing media may vary from 400 to 2500 kPa.

11.3.14.5 Influence on growth

Data from the penetrability readings indicate growth reduction >250 –450 kPa. Hardness is however often measured to characterize or compare structures, such as peat fractions, which are used to bring stability in an otherwise easily penetrable matrix or to characterize the suitability for automated handling of blocks and plugs. The readings have therefore usually to be set against a standard acceptable for a specific handling system.

11.4 CHEMICAL ANALYSIS

One single extraction method in growing medium analysis is not sufficient for a balanced characterization due to the enormous diversity of raw materials as well as the large number of elements and compounds involved. Every extraction method treated below is designed for a specific group of elements/compounds or a particular fraction of such a group. In the first place the methods are aimed at investigating nutrient element contents available to plants or whether the growing medium contains high concentrations of elements which may be damaging to the plant, for example, sodium. Second, growing media must not contain high concentrations of elements which may be dangerous for food crops, such as heavy metals. One of the relevant factors is how rapidly or easily the element is released from the growing medium. There are, for example, “mobile compounds” which are water-soluble, so that after analysis an impression may be gained of the readily available amount in the growing medium. In addition to this group, there are “semimobile” compounds which are soluble in an aqueous solution of ammonium acetate or a complex forming reagent [e.g., diethylenetriamine penta-acetic acid (DTPA)]. Mostly, the ammonium acetate extraction is applied for the determination of cations and the DTPA for the determination of trace elements. In addition, there is the “overall determination” which uses concentrated acids to provide pseudo-total concentrations of inorganic components (CEN 13650, 2001). With this method both the organic and the inorganic matrices are accessed. In addition, several other extraction methods are being used for specific aims such as the determination of a single element only (CEN 13654-1, 2001; CEN 13654-2, 2001).

In the instructions and descriptions given in this chapter, the term “substrate solution” is used similarly to the term “soil solution” which is used in soil science. Substrate solution in growing media should represent the solution present in a medium under growing conditions. Thus the concentrations of dissolved substances in the substrate solution are closely related to their actual concentration in the rhizosphere during most of the growing period. Under growing conditions, the moisture content is certainly not constant; consequently, a definition for moisture condition has to be chosen which is achieved artificially but is sufficiently close to the moisture condition under realistic growing conditions. Fluctuations occurring due to irregularities in the moisture content are obviously not represented in this definition, so that the definition relates to the circumstances in the growing medium briefly after irrigation and the termination of free drainage. This moisture condition is called “container capacity.” However, this is close to saturation. In substrate-grown crops water is often supplied frequently, so that substantial fluctuations in the moisture condition during the growing period are limited. Under growing conditions in general terms, two situations occur: (1) growing media of which a low negative pressure height is built up, for example, peaty materials and coir and (2) growing media which are not suitable for the build-up of any negative pressure height in the matrix and are continuously kept more or less saturated. Examples are stone wool, expanded clay granules, and artificial foam. So, in principle two moisture conditions are defined: for the growing media under (1) a moisture content at a pressure head of -32 cm (pF 1.5) and for the growing media under (2) a moisture content at nearly saturation, called “container capacity.” The pressure head at which the substrate solution is being defined must be indicated individually for every growing medium and extraction method executed.

For growing media analysis in glasshouse horticulture, aqueous extracts are used. The reason for this is the fact that in growing medium analysis for glasshouse horticulture great value is attached to the composition of the substrate solution. The most important parameter of the substrate solution is the osmotic pressure and estimates of it can only be obtained by using aqueous extracts (CEN 13038, 2011). The osmotic pressure of the substrate solution is an important factor in growth regulation in glasshouse crops. High values cause growth inhibition, while low values may lead to insufficient product quality and, particularly under light-deficient conditions, to vegetative crop development. Osmotic pressure in substrate solution can be built up by letting salts accumulate or by excessive application of soluble fertilizers. In this way the osmotic pressure, the salt content, and the nutrition are intertwined. It is therefore important that the various parameters for salt and nutrition are determined in the same aqueous extract, in order to be able to regulate them and their interrelations.

The best method to evaluate the ionic composition of the substrate solution is by analyzing the substrate solution extracted immediately from the growing medium. In stone wool crops, where the substrate solution can be obtained by means of a simple suction device, this is a standard procedure. With many other growing media, this is more difficult to realize, because the water is bound much stronger to the matrix and can then only be obtained by way of pressing out or displacement. Such methods are less suitable for routine analysis. For routine analysis in such cases, a known amount of extra water is added to the sample to prepare an extract more easily. On the basis of the concentrations of salt and nutrients found in the extract, the concentrations of salt and nutrient elements in the substrate solution may then be calculated. As a rule, the less water is added to the sample, the more accurate is the calculation, since in that case disturbances due to dilution and valence effects are reduced to the minimum.

The reason so many methods exist is the higher dilutions are easier and faster while the lower dilutions require more care but are more accurate. Furthermore various rooting materials cause application problems with one or more of the methods described.

11.4.1 WATER-SOLUBLE ELEMENTS

11.4.1.1 *The 1:1.5 volume method*

The 1:1.5 volume extract is meant to estimate the composition of the substrate solution of peaty growing media and comparable media. The growing medium sample is mixed manually in such a way that its natural structure is not disturbed. Water is added to the “fresh” (as received) nondried material until water can be pinched though one’s fingers during gentle pressing a ball-shaped sample in one’s hand. In this way, the water content corresponds reasonably well with a pressure head of -32 cm. Different operators should check this regularly by using a standard sample and determine the water content after adjustment to the desired pressure head. If growing media are wetter than a pressure head of -32 cm, they have to be air-dried a little in advance and the procedure commences with the “visual” wetting. The material is put into two rings placed one upon the other, the lower having a minimum content of 60 mL at a height of 5 cm. The material is pressed for 10 seconds with a pressure of 10 kPa. The upper ring is removed by cutting the growing medium between the two rings carefully. The content of the lower ring is mixed with 1.5 times its volume of demineralized water. Subsequently the suspension is shaken firmly for about

15 minutes and after a waiting period of 1 hour the suspension is filtered using a filter paper which should not contaminate the extract with the elements to be determined. For some elements (mainly trace elements) filtration is done with a $0.45\text{ }\mu\text{m}$ ceramic filter. The method can be applied for all growing media which consist of at least 50% peat or other organic materials such as coir, bark and wood fiber. EC, NH_4 , Na, K, Ca, Mg, NO_3 , Cl, SO_4 , HCO_3 , P, Mn, Fe, Zn, B, Cu, and Mo are measured in the extract. Element contents are expressed per volume of extract. To convert the contents per volume of extract to contents per volume of growing medium Eq. (11.10) should be used:

$$C_s = A \cdot C_e \cdot (1.5 + \theta) \quad (11.10)$$

where C_s is the element content per volume unit of growing medium (mg/L substrate); C_e is the element content per volume unit of extract (mmol/L extract); θ is the water content on volume basis at pressure head -32 cm (m^3/m^3); and A is the atomic weight (g/mol).

Typically, peaty growing media have water contents at pressure head of -32 cm of about $0.4\text{ m}^3/\text{m}^3$, which means that (for simplicity) the element contents per volume unit of extract should be multiplied with 1.9 to reach the element content per volume unit of growing medium.

The pH is measured in slurry, the semisolid settled matter below the supernatant, after 6 hours waiting time.

For coarse growing media it is recommended to use a wider ring for measuring out the growing medium. As a general rule, it is recommended to use a ring with a diameter 2.5 times bigger than the growing medium's biggest particles. The height of the lower ring, however, remains 5 cm . The method is described by [Sonneveld et al. \(1974\)](#) and [Australian Standard \(2003a,b\)](#).

Element contents are expressed in mmol/L per volume unit of extract or could be converted to mg per liter fresh growing medium (see [Eq. 11.10](#)). Expressing results in moles rather than weight is more rational from both scientific and practical points of view. EC and pH are expressed on extract basis.

For peaty growing media, coir dust, and comparable media the 1:1.5 volume extract is used in Denmark, South Africa, Australia, and The Netherlands. Note the volume is measured out in a standardized way after the moisture content is standardized at a pressure height of -32 cm (pF 1.5). In this method substrate solution is diluted about four times ([Table 11.5](#)).

The 1:1.5 volume method, however, is not suitable for some kinds of other growing media or field circumstances, when (1) the volume cannot be measured as in the 1:1.5 volume method as growing media are rigid (tuff) or too compressible (glass wool); (2) it is too laborious to estimate the moisture content at the pressure height of -32 cm ; (3) in practice growing media are used at moisture contents nearby saturation (synthetic foam). In those cases or circumstances other methods have to be applied.

11.4.1.2 The 1:2 soil solution method

The 1:2 solution method is developed for the evaluation of mineral soils and should not be used for growing media. Reason to mention it is that the field measurement of EC and pH is easier than for the other methods and is therefore used when time is important. The method starts with saturation the soil. Then a volume of demineralized water is taken, usually 80 mL , and saturated soil is added to this volume until a volume of 150% V/V is reached which is 120 mL in the chosen example.

Table 11.5 Influence of Extraction Methods on Dilution of a Sample of 1000 mL

	Unit	Code	Formula	Soil Moisture	Saturation Extract	1:1.5	1:2	1:3	1:5
Sample volume	mL	SV		1000	1000	1000	1000	1000	1000
FMC	mL	A		300	300	300	300	300	300
Squeezed wetness (pF 1.5)	mL	B		500	500	500	500	500	500
Saturated	mL	C		600	600	600	600	600	600
Total pore space	mL	D		700	700	700	700	700	700
Volume of solids	mL	E	100-D	300	300	300	300	300	300
EC soil solution	dS/m	F		1.0	1.0	1.0	1.0	1.0	1.0
Water added to standardize sample	mL	G		0	300	200	300	0	0
Formula for line above					C – A	B – A	C – A		
Water added to standardized sample	mL	H		0	0	1500	2000	1800	5000
Formula for line above						SV × 1.5	SV × 2	C × 3	SV × 5
End volume all water	mL	I	A + G + H	300	600	2000	2600	2100	5300
EC in extract	dS/m	J	A × F	1	0.50	0.15	0.12	0.14	0.06
Dilution compared to FMC	Factor	K	I/A	1.0	2.0	6.7	8.7	7.0	17.7
Dilution compared to pF 1.5	Factor	L	I/B	0.6	1.2	4.0	5.2	4.2	10.6
Dilution compared to saturation	Factor	M	I/C	0.5	1.0	3.3	4.3	3.5	8.8
Dilution compared to sample volume	Factor	N	I/SV	0.3	0.6	2	2.6	2.1	5.3

The use of an absolute volume is needed to compare methods defined in different volumes and % V/V. The table is specific for one growing medium. Each growing medium will bring specific values for the values A–F in the “Code” column. Furthermore the value for “field moisture content” may differ between batches of the same material and will influence the outcome of some of the methods too.

Soil Moisture: in some systems it is possible to use extracts squeezed or sucked out of the sample.

Saturation extract: extracts squeezed or sucked out of the sample after saturation (like the Pour Thru method). 1:1.5: moist the sample up to squeezed wetness (pF 1.5); add 150% V/V water to the sample volume. 1:2: moist the sample up to saturation; take 100% V/V water volume and add sample until the total volume is 150% V/V. 1:3: measure moisture content at saturation; add three times the water content at saturation to a fresh sample. 1:5: a laboratory compacted sample is prepared; 500% V/V water is added (five times the sample volume). FMC, Field moisture content.

The volume of demineralized water added to the sample material is thus two times the volume of saturated soil, hence the name 1:2 method. In this method substrate solution is diluted about four times (Table 11.5).

11.4.1.3 The 1:3 substrate solution method

The aim is to determine the composition of the substrate solution for other than peaty growing media which cannot easily be hand-pressed. The dilution of the substrate solution is comparable with the 1:1.5 method. In this way, the element contents of the extracts of both methods can be used in the same recommendation system. The moisture content of the fresh growing medium is determined. A ring with a diameter of about 10 and 5 cm height is filled loosely with fresh growing medium with a pressure of 1 kPa on top. Subsequently it is saturated for 2.5 hours and then leached out on a grid for 2.5 hours. To a newly prepared volume of growing medium subsequently add sufficient demineralized water so that it contains three times the amount of moisture at container capacity after leaching out. The measured volume of the growing medium chosen should depend on the coarseness of the growing medium. For fine growing media with a particle size <20 mm a minimum volume of 100 mL is sufficient, while for growing media with a particle size of >20 mm a volume of 400 mL is recommended. The suspension is rotated (using an end-over-end shaker) for 2 hours and subsequently remains standing for one night. The rotation rate must be low enough not to damage the texture of the growing medium. The suspension is filtered through a filter that does not contaminate the extract with elements that must be determined. For some elements (especially trace elements) a 0.45 μm filter is used. The method is designed for all granular growing media for which the 1:1.5 method is impractical, such as expanded clay granules, perlite, pumice stone, granulated stone wool, wood fiber, rice husk, and bark. The following determinations are carried out on the extracts: EC, pH, NH_4 , Na, K, Ca, Mg, NO_3 , Cl, SO_4 , HCO_3 , P, F, Mn, Fe, Zn, B, Cu, and Mo. Element contents are expressed per volume unit of extract. The method is described by De Kreij et al. (1995, 2001).

11.4.1.4 The 1:5 volume method

The 1:5 volume method has been developed by CEN in order to extract growing media and soil improvers with a wide variety of characteristics in an unequivocal way with demineralized water (CEN 13040, 2007; CEN 13652, 2001). The growing medium is first measured into a 1 L cylinder having a height of 10 cm and a loose collar extension of 5 cm. The cylinder and collar are filled via a screen which functions as a fall-breaking device (to minimize operator sensitivity) and is subsequently pressed with a pressure of 0.9 kPa. After 3 minutes the collar is removed and the cylinder is leveled off using a straight edge. The weight of the sample is determined—thereby providing the “laboratory compacted” bulk density of the sample in g/L. For the preparation of the extract, a certain volume necessary for preparing the extract is determined through weighing, based on the calculated bulk density. In materials in which the particles are smaller than 20 mm a weight equivalent of 60 mL is weighed out. Of the other materials with particles smaller than 40 mm a weight equivalent of 250 mL is weighed out. The sample is mixed with five times the volume in demineralized water. After shaking for 60 minutes the suspension is filtered through filter paper which does not contaminate the extract with the elements to be determined. For some elements (trace elements) filtering is done with a 0.45 μm filter. The method is designed for all granular growing media. In the extracts the following determinations are carried out: EC, NH_4 , Na, K, Ca, Mg, NO_3 , Cl, SO_4 ,

HCO₃, P, F, Mn, Fe, Zn, B, and Cu. Element contents are expressed per volume unit of growing medium. EC is expressed on the basis of the extract in mS/m.

The method poses problems when measuring out very coarse and very wet, cohesive materials. Materials which compact very easily also give problems when the weight is mounted. Due to the high dilution of the substrate solution it gives rise to valance/dilution effects; some elements reach the detection limits of the instruments routinely used for all other extraction methods as well; growing media containing gypsum show high EC values due to gypsum dissolving, which in insoluble form does not pose a risk for crop growth. Proficiency tests have been executed by [De Kreij and Wever \(2005\)](#).

11.4.1.5 *The pour-through method*

The method was developed in the United States for on-farm measurements by [Yeager et al. \(1983\)](#). It needs low-cost equipment and instruction of people is easy. The procedure is as follows: the crop is irrigated thoroughly. After approximately 1 hour for equilibration a saucer is placed under each container to be tested. A sufficient number of containers have to be sampled to ensure reliability. A sufficient amount of distilled water is poured evenly on top of each pot to yield ~50 mL leachate out of the bottom of the container. The demineralized water will replace the substrate solution during drainage of the container (via the “piston effect”). The container has to drain completely in about 5 minutes. The solution on the saucers is poured into beakers and its EC and pH values are measured. Some variation due to the amount of water poured onto the container can lead to variation of the data. When channeling occurs the water does not displace the substrate solution.

The results obtained using the pour-through (PT) method must be carefully interpreted, and calibration curves were developed for comparing PT and saturated media extract nutrient values ([Cavins et al., 2004](#)). Yet, pH data derived from the PT methods differ significantly from those of other methods and they require careful interpretation ([Handreck, 1994](#); [Rippy and Nelson, 2005](#)).

11.4.1.6 *Comparing extraction methods*

In [Table 11.5](#), some methods are compared with respect to the dilution they create in the sample. In the last four rows, the dilution with respect to four different points of reference is shown. It is clear the 1:1.5 and 1:3 methods are usually very close and results can be compared to each other. Do notice however that the quantities of water added to reach a standardized moisture content (respectively squeezed moisture content and saturation) differ depending on the material chosen and the field moisture content of the sample.

To compare results of different methods with each other, a conversion factor as given in the last four rows can be used. If enough properties of the samples are known a more exact factor can be calculated following the structure of [Table 11.5](#). But even then caution is needed as elements unavailable at ambient moisture conditions will become available when the sample is further diluted. Obviously especially the 1:5 method will overestimate the amount of normally precipitated elements such as calcium, magnesium, iron, sulfate, bicarbonate, and phosphate (from precipitates such as gypsum, CaSO₄, lime, CaCO₃, and others).

11.4.2 EXCHANGEABLE, SEMI- AND NONWATER-SOLUBLE ELEMENTS

11.4.2.1 *The calcium chloride/diethylenetriamine penta-acetic acid method*

One of the characteristics of calcium chloride as an extraction liquid is that it offers a certain salt concentration during the extraction. This results in the release of weakly adsorbed reserves, mostly concerning main elements, N, P, K, and Mg. This corresponds to what happens in reality when fertilizers are supplied. The DTPA is a strong complex forming agent, binding the trace elements Fe, Mn, Cu, and Zn. The combination of calcium chloride and DTPA is therefore meant for the extraction of both main and trace elements. Moreover, by using these extractants the activity of the roots is simulated to some extent. The method is described in [CEN 13651 \(2001\)](#). The extraction liquid is 0.01 M CaCl_2 /0.002 M DTPA with an average pH of 2.6. NH_4 , NO_3 , P, SO_4 , K, Na, Mg, Fe, Cu, Zn, Mo, Mn, Pb, Cd, and B are determined in the extract. Contents are converted and presented in mass element (g or mg) per unit volume of the growing medium. Determination of B may give rise to problems, depending on the detection method used, due to the strong coloring of the extract. The elements are expressed in mg/L growing medium.

11.4.2.2 *The 0.1 M barium chloride 1:1.5 volume method*

The extraction with barium chloride is meant to determine the amount of exchangeable cations, displaced from adhesion sites by the barium ion. Since it is not naturally present in growing media, barium does not give rise to disturbances of the measurement of individual cations. The results of the determination give an estimate of the quantity of cations “available” to the plant. The sample is wetted to a state equivalent to that arising from a pressure head of -32 cm and a volume of, for example, 100 mL is measured (see 1:1.5 method).

This volume of growing medium is mixed with 1.5 times the volume of a 0.1 M barium chloride solution, after which the suspension is shaken intensively for 15 minutes. After a waiting period of 1 hour, the suspension is filtered through a paper filter which gives no contamination of the extract with the elements to be determined. For some elements (particularly trace elements) filtration is done with a $0.45\text{ }\mu\text{m}$ filter. The method is being applied for coir pith, coir fiber and coir chips ([Verhagen, 1999](#)), and biochar ([Silber et al., 2010](#); [Kloss et al., 2014](#)). In the extract pH, NH_4 , K, Na, Ca, Mg, Fe, Mn, Zn, and Cu are measured.

11.4.2.3 *Active manganese method*

Manganese is a micro nutrient which may become present in toxic quantities ([Silber et al., 2009](#)). Mn may occur in various forms in a growing medium: (1) water soluble, (2) adsorbed or exchangeable, (3) easily reducible, and (4) inert ([Leeper, 1947](#)). Acute toxicity may occur when less available forms of manganese suddenly convert into more available forms. The conversion may be triggered, for example, by a drop in pH or steam sterilization. The aim of the determination of “active” manganese is to investigate how much Mn can theoretically be released. Mn-active is the sum of the three Mn forms a, b, and c mentioned above. On the basis of these a prediction can be made of the possible toxicity of Mn ([Sonneveld, 1979](#)).

To a weighed amount of air-dry medium, 20 times as much (by weight) hydroquinone (1,4-benzenediol) extractant containing 1 M ammonium acetate solution with a nominal pH of 7 is added. After 1 hour of shaking, the solution is filtered and the Mn is measured in the filtrate. The

determination is recommended for materials which are expected to contain significant quantities of Mn (such as some barks). Mn is expressed in mg per kg dry material.

11.4.3 THE pH IN LOOSE MEDIA

The method is described in CEN 13037 (2011) and the volume is determined according to CEN 13040 (2007). The pH can also be measured in the abovementioned water extracts.

According to CEN 13040 (2007) a weighed amount of growing medium, corresponding to a volume of 60 mL (for materials consisting of >80% particles <20 mm) or 250 mL (for all other materials) is put in a plastic bottle, and the necessary amount of water (300 or 1250 mL) is added. It is shaken for 1 hour on a horizontal shaking machine. Within 1 hour after shaking the pH is measured. A reading is taken if the value changes no more than 0.1 U during 15 seconds. The method is suitable for all loose, nonpreformed media. Note, that also pH, EC, and nutrients can all be determined on the same extract.

The same procedure is being used with 1 M KCl or 0.01 M CaCl₂ as extractant.

11.4.4 NITROGEN IMMOBILIZATION

Some organic growing media can immobilize a substantial amount of nitrogen during the course of the cropping and especially during the first weeks of use as a growing medium or as a component in potting soils. For these media it is important to estimate the potential for nitrogen immobilization by the medium prior to their use. Several methods and variants have been proposed.

The Nitrogen Draw Down Index (NDI) (Handreck, 1993) reports the percentage of the nitrogen present in a growing medium over the quantity initially applied. The initially applied quantity was defined as 75 mg/L nitrate in growing medium. This initial quantity was later increased to 150 and 300 mg/L (Sharman and Whitehouse, 1993; Bragg and Whiteley, 1995). The adapted NDI then was reported with the initial dose, for example, NDI₁₅₀ or NDI₃₀₀.

The Nitrogen Fixation Index (NFI) is based on the NDI and also reports the percentage of the nitrogen present in a growing medium over the quantity initially applied as nitrate. The NFI is reported for day 0 and after 4 and 20 days, respectively. The initially applied quantity is defined as 0.1 mmol NO₃/g dry matter. For materials with a bulk density close to 200 kg/m³, this content is equal to 20 mmol NO₃/L growing medium. In the 1:1.5 extraction solution, this will be close to 10 mmol/L. Only limited experience has been gained with this method, so that no estimate can be given as to the accuracy which can be obtained with the method.

VDLUFA (2000) developed a similar method which has been suggested for CEN as well. In this method, the N is added as ammonium nitrate. The addition of nitrogen is not calculated per unit dry weight but per unit volume of sample. The quantity applied is 500 mg/L growing medium (about 37 mmol/L). As in other recent methods, nitrogen fixation is reported as the original amount present minus the amount actually present, both in mmol/L, based on extraction according to CEN 13651 (2001). The differences between the methods are in the initial amount of nitrogen applied, the nitrogen source, and the extraction method. As growing media differ in their pattern of fixation over time, three or even more observations over time may be necessary.

11.4.5 CALCIUM CARBONATE CONTENT

The aim is to determine the amount of carbonates. The contents of the carbonates are expressed as calcium carbonate. A certain weight of air-dry medium, for example, 3 g, is placed into a closed container and an excess of 4 M HCl is added so that all carbonates in the sample react to produce CO₂. On the basis of the increase of the amount of gas above the sample, the amount of CaCO₃ can be calculated. The method is meant for all growing media (NEN-ISO 10693, 2004).

The sum of the carbonates is expressed as calcium carbonate equivalent; the mode of expression is in the fraction of CaCO₃ on a weight basis. There may be other substances (for instance organic aromatic acids) which may give off CO₂, or another gas may be formed (e.g., H₂S and/or SO₂). FeCO₃ does not completely dissolve in hydrochloric acid, which results in a faulty representation of the carbonate content. In most situations this does not lead to significant practical errors in determining CaCO₃ content.

11.4.6 BUFFER CAPACITY

Apart from pH and calcium carbonate content, it is sometimes of interest to know the amount of acid or base required to compensate for a too high or too low substrate pH, respectively. Such methods also allow liming of acid rooting materials and the mixing of two or more materials which counteract each other, resulting in a desired neutral or slightly acidic pH (Huang et al., 2007; Blok and Kaarsemaker, 2013; Sullivan et al., 2014).

11.5 BIOLOGICAL ANALYSIS

The determination of the following characteristics will be discussed: stability/breakdown (respiration), the presence of weeds and phytotoxicity (growing tests). Different methods will be applicable, depending on the nature of the material, such as its mineral and organic content, whether it is preformed material or not and its EC.

11.5.1 STABILITY AND RATE OF BIODEGRADATION

11.5.1.1 Definition and units

There are different ways to look at stability (Page, 1982; Pagga, 2000). There is physical, chemical, and biological stability. Physical stability has to do with mechanical strength, chemical with the breakdown of certain carbohydrate structures, and biological stability has to do with microbiological activity. A common approach is to define stability as the microbial degradation rate of the organic matter under aerobic conditions (Haug, 1993). A lower degradation rate corresponds to a higher degree of stability. The microbial degradation of the organic matter (C_aH_bO_cN_d) can be represented by the following equation:



where ΔH is the heat production (kJ), X is the biomass (g/L), and O₂ and CO₂ are the oxygen and carbon dioxide (mg/g/h), respectively.

Compost stability and maturity are terms which indicate the degree of organic matter decomposition and potential of phytotoxicity caused by insufficient composting (Chapter 12: Liquid Culture Hydroponic System Operation; Wu and Ma, 2001; Alburquerque et al., 2006; Van Der Wurff et al., 2016). When unstable materials are added to growing media they may have a negative impact on plant growth due to reduced oxygen content and/or available nitrogen and/or the presence of phytotoxic compounds (Brodie et al., 1994; He et al., 1995; Keeling et al., 1994). Phytotoxic substances of organic origin can be broken down by microorganisms during the composting process into non-toxic organic compounds. Because of the mentioned concerns, extensive research has been conducted to study the composting process and to develop methods to evaluate the stability/maturity of compost prior to its agricultural use (Hue and Liu, 1995; Iannotti et al., 1994; Jimenez and Garcia, 1992; Mathur et al., 1993). CEN has issued two methods for characterizing the stability: the OUR (oxygen uptake rate) method (CEN 16087-1, 2011) and the “self-heating test” (CEN 16087-2, 2011). However, a number of different other tests are still available (Cooper, 2004), some which are simple to operate, others requiring quite sophisticated and expensive apparatuses. Some workers argue that more than one test may be required to sufficiently characterize stability. A relatively simple test was suggested by Caceres et al. (2015), which simultaneously assesses the hygienization degree. Many volatile organic compounds are emitted during and after composting which can provide information about the progress of composting process (Lopez et al., 2016). Most methods lock a sample in a fixed volume of air, accepting a change in gas composition during the measurement, while others advocate measuring with a stable level of oxygen (Adani et al., 2005).

11.5.1.2 Determination of potential biodegradability

Potential biodegradability is used as an indication of field stability. Knowledge of the potential biodegradability is a valuable tool, especially for compost and material intended for landfill (Godley et al., 2004). The question which remains to be answered is how potential degradability and field degradability relate to each other.

Van der Sloot et al. (2003) suggested that the measurement of water-soluble organic matter is a simple and rapid procedure giving valuable information on the potential biodegradability of the material under test. The determination of cellulose, hemicellulose, and lignin, that is, fractionation of organic matter, gives additional information but is not often used in the routine analysis of growing media of compost. It is performed mostly by animal nutrition laboratories (Van Soest, 1963; Van Soest and Wine, 1967). Another method of carbohydrate fractionation, as a measure of estimating the biological stability, is using near infrared reflectance to determine the degree of mineralization of wastes (Lohr et al., 2017). The technique has potential but is material-specific (Albrecht et al., 2011). Thus large databases for calibration of the method are required.

11.5.1.2.1 General principle of methods used for growing media (constituents)

During incubation of the sample, the self-heating, the evolution of CO₂ or the consumption of O₂ is recorded. For the latter two, the sample is incubated at a certain moisture content and temperature.

11.5.1.3 Determination of heat evolution (“Dewar” test)

Heat evolved during the mineralization of organic material, for example, the composting process can be measured and used as an indicator of the aerobic microbiological activity and hence the “biological stability” of an organic material. The test method has been standardized for compost

(CEN 16087-2, 2011). The sample is placed in a heat-retaining flask (e.g., “Dewar”—flask or similar) and the initial and the maximum temperature (after 2–5 days) are recorded either by a thermocouple or thermometer. The ambient temperature has to be kept at $22 \pm 2^\circ\text{C}$. The test is limited in the sense that it best distinguishes very immature from mature compost; it cannot distinguish moderate maturity from high maturity which may be important for the use of composts in potting soil (Brinton, 2000). A more sensitive test is based on the very high sensitivity of modern micro calorimeters (Laor et al., 2004; Medina et al., 2009).

11.5.1.4 Determination of Solvita test

The Solvita test is a commercially available field test kit. It employs gel–colorimetric technology in which respiration gases from composts (CO_2 and NH_3) are captured, and the amounts are indicated by the color change on test sticks and calibrated to a wide range of known conditions. This easy to use system can, under certain circumstances, give an indication of the stability. Various overviews doubt the general applicability though (Brinton, 2000; Cooper, 2004).

11.5.1.5 Determination of respiration rate by CO_2 production

In this method the rate of biodegradation is estimated by measuring the rate of CO_2 production under specific circumstances (Cooper, 2004). The method was initially proposed as a European standard but the work was closed.

11.5.1.6 Determination of respiration rate by O_2 consumption (oxygen uptake rate)

In this method the rate of biodegradation is estimated by measuring the rate of O_2 consumption under specific circumstances (Veeken et al., 2003). Based on the method formerly known as the OxiTop, CEN has developed a standard (CEN 16087-1, 2011). The pressure drop in a vessel, caused by oxygen consumption, is measured. A sample with an equivalent of 2 g organic matter is incubated in a buffered nutrient solution containing a carbon dioxide scrubber and a nitrification inhibitor for a maximum 7 days at room temperature. The carbon dioxide is scrubbed to avoid interference with the pressure in the vessel. The nitrification inhibitor has to be added as the conversion of ammonia to nitrate also consumes oxygen, which would influence the result. The OUR is calculated over an interval with constant pressure drop (see Fig. 11.13).

The abovementioned procedure can also be used for the determination of N-mineralization and N-immobilization (Grigattia et al., 2007).

11.5.1.7 Common values for media

Some common values are found in Table 11.6. Criteria based on the results of O_2 uptake method are presented in Table 11.7.

11.5.1.8 Influence of the property on growth

In several studies, OUR has proved its suitability for characterizing the stability of a material. Silva (2004) shows the correlation between the growth of Kohlrabi and the OUR of two different composts-peat mixes (Fig. 11.14).

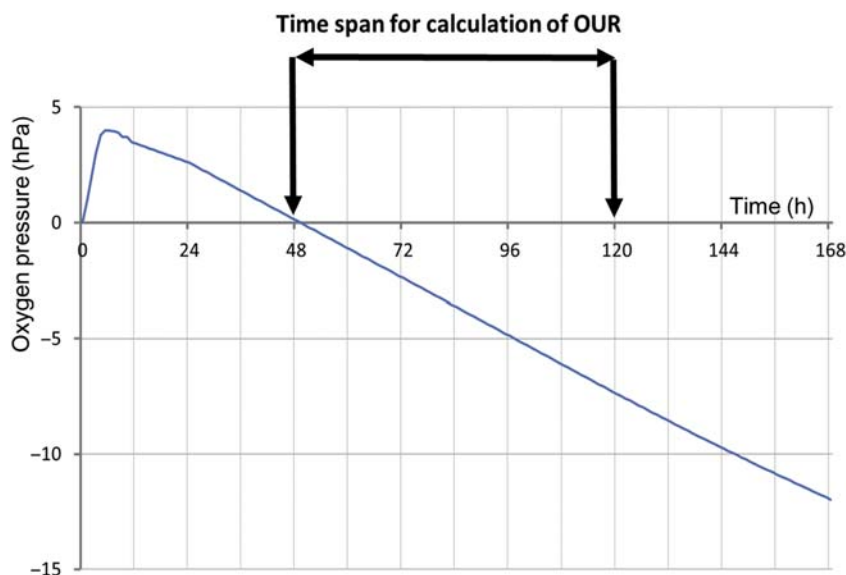


FIGURE 11.13

Typical development of the pressure drop during OUR measurements. *OUR*, Oxygen uptake rate.

Table 11.6 Respiration Rates of Various Types of Organic Matter Under Standardized Conditions

Type of Organic Matter	O ₂ Respiration Rate	N-Mineralization
	mmol/kg/h	mg/kg/h
Biowaste compost	17.0 ± 7.8	6.6 ± 3.4
Green waste compost	10.4 ± 3.3	5.1 ± 2.3
Spent mushroom compost	28.1 ± 3.8	38.8 ± 23.5
Peat	2.1 ± 0.9	5.6 ± 3.3
Bark and composted bark	13.4 ± 9.6	− 2.9 ± 2.5

After Veeken, A.H.M., De Wilde, V., Hamelers, H.V.M., Moolenaar, S.W., Postma, R., 2003. OxiTop[®] measuring system for standardised determination of the respiration rate and N-mineralisation rate of organic matter in waste material. NMI MeststoffenInstituut and Wageningen University, Agrotechnology and Food Sciences. Wageningen University, Wageningen, The Netherlands.

11.5.2 WEED TESTS

11.5.2.1 Definition and units

The contamination of horticultural growing media with viable seeds and sprouting plant parts (propagules) is of high practical importance as there are high costs of eradicating them in horticultural practice. With the “weed” test the number of seedlings and propagules emerging under specified

Table 11.7 Classification of Stability of Composted Materials With Four Stability Classes, Based on the OUR Method

Stability Class	Respiration Rate	
	mmol O ₂ /kg OM/h	mg O ₂ /g OM/day
Very stable	< 5	< 4
Stable	5–15	4–11
Unstable	15–30	11–23
Very unstable	> 30	> 23

OM, Organic matter; OUR, oxygen uptake rate.
 After Veeken, A.H.M., De Wilde, V., Hamelers, H.V.M., Moolenaar, S.W., Postma, R., 2003. OxiTop[®] measuring system for standardised determination of the respiration rate and N-mineralisation rate of organic matter in waste material. In: *Environmental Technology*. Wageningen University, Wageningen, The Netherlands.

circumstances is assessed (CEN/TS 16201, 2013). The number of emerged weeds is given as the number per liter of original sample or per area (m²) at a given growing medium height (e.g., 20 mm for CEN/TS 16201, 40 mm for the RHP certificate). The EU eco-label requires that composts used for growing media will be free of weed seeds, but this requirement surpasses the limits of practical control systems (Lopez-Lopez and Lopez-Fabal, 2016).

11.5.2.2 General principle

To investigate whether a growing medium is infected with weeds, pH and electric conductivity (salt content) of the sample are optimized either by diluting with peat or liming.

The growing medium is wetted to optimal moisture with clean water. For CEN/TS 16201 optimal moisture is according to the “fist test” (optimal moisture is reached, when the sample feels moist but no water emerges while squeezing gently). At least 3 L of the original sample are filled in the respective number of seed trays (50–100 mm height) to a maximum height of 20 mm and kept at 18–30°C without direct sunlight and high humidity (the trays are covered with plastic film). During the test the growing medium is kept sufficiently moist, as under normal growing conditions. Care must be taken to ensure no secondary infection arises, for example, from seeds blown in by the wind. The number of emerging weeds is monitored and recorded for 3 weeks. An overview of the available weed tests is given by Baumgarten and Dersch (2004).

11.5.2.3 Special cases

To ensure the germination of all present seeds, the physiological phenomenon of seed dormancy may be broken by different physical treatments. Several methods are proposed by the CEN/TS 16201, for example, stratification (low temperate stimulus under wet cool conditions at 4°C for 3 days), gibberellic acid treatment or KNO₃ treatment. To reduce the concentration of substances inhibiting sprouting, the growing medium may also be “washed,” following a process also described in CEN/TS 16201.

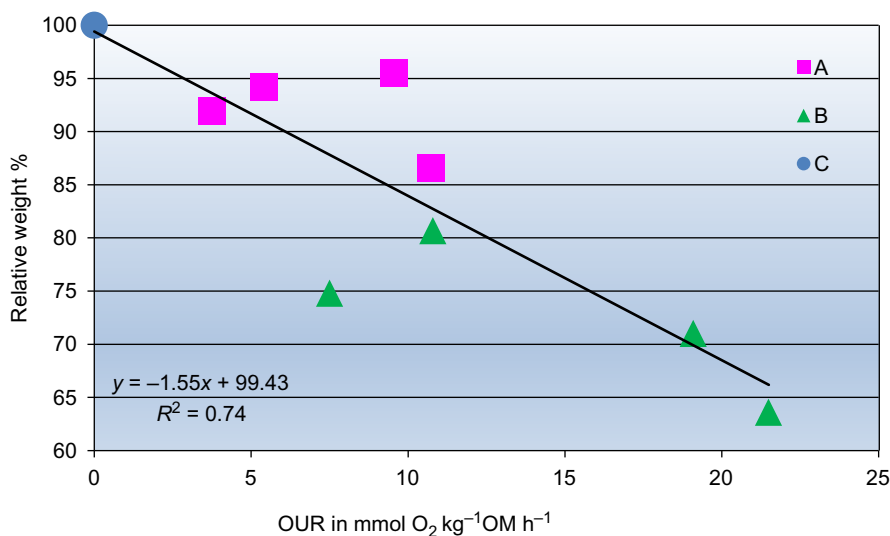


FIGURE 11.14

Relation between relative growth of kohlrabi and stability measurements using the OUR method. A = a mix of peat/compost A 80/20% V/V (Compost A: 70% trimmings, 30% grass). B = a mix of peat/compost B 60%/40% V/V (Compost B: 30% trimmings, 30% grass, 40% leaves). C = no addition. *OUR*, Oxygen uptake rate.

Based on Silva, M., 2004. *Compost Stability. Relation Between Respiration Measurements of Compost and Plant Growth*. PPO Report. Praktijk Onderzoek Plant en Omgeving, Naaldwijk, The Netherlands.

11.5.2.4 Common values for media

Depending on the source of the materials, different levels can be expected. With respect to the use or legal requirements, different levels may be acceptable. For peaty growing media, standards set for RHP certification are <15 per m² for plants typical for bog vegetation, corresponding to 0.375 plants/L, and <4 per m² for nonpeatbog-vegetation plants, corresponding to 0.1 plants/L (Bos et al., 2002; Zevenhoven et al., 1997; Verhagen, 2016, personal communication). For other constituents, such as coir, bark, pumice, compost, etc., requirements are based on individual risk assessment. In Austrian legislation, a limit for growing media is set with 3 plants/L. For compost used in growing media production, the German “Bundesgütegemeinschaft Kompost” demands a limit of 0.5 plants/L (<http://www.kompost.de/guetesicherung/guetesicherung-kompost/produkte-anforderungen/>).

11.5.3 PLANT RESPONSE TEST (“PHYTOTOXICITY” TEST)

11.5.3.1 Definition and units

Phytotoxicity is defined as a delay of seed germination, inhibition of plant growth or any adverse effect on plants caused by specific substances (phytotoxins) or growing conditions (REAL CCS, 2014).

In chemical research it is possible to determine a number of important parameters and also to indicate certain dangers with respect to plant growth. However, it is practically impossible to screen a material for all substances which might cause damage. In order to assess whether there are harmful substances present in a material, a plant response test may be helpful. Such a test does not give definitive answers as not all crops react the same and the reason for a possible effect can only be assumed, but it does give more confidence about the suitability of a growing medium or a growing medium constituent.

Plant response tests may be used, if, for example, chemical analysis shows unexpectedly high concentrations of substances, problems of a certain crop with a growing medium are reported or harmful substances (e.g., phenolics in bark) are suspected to be present. In some cases, it may be necessary to carry out a plant response test with the material diluted and formulated as a growing medium. This may, for example, be the result of chemical research in which unexpectedly high concentrations of certain substances were found. To test whether these substances affect plant growth, a plant response test should be undertaken. Also, when it is already known that the use of a growing medium with a certain crop gives problems and the cause is unknown, a cropping test can be informative, for example, phenolics in bark (Machrafi et al., 2006; Cui et al., 2017). In some growing media also phytosanitary problems may be expected, for example, with nematodes or certain virus diseases. Specific tests for such cases are available. Occasionally it may also be necessary to test whether a process in growing medium manufacturing (e.g., composting or sterilization) gives an adequate elimination of certain pathogens. Currently, several methods for examining plant response are available (ISO 11269:2012, 2012; CEN 16086-1, 2011; CEN 16086-2, 2011; Blok et al., 2014).

11.5.3.2 General principle

A material can be tested as it is, in diluted form or by means of a watery extract from the material. Testing the material per se offers the advantage that the plant is in direct contact with the medium. However, if factors, such as physical characteristics, influence growth significantly, the extract method may be preferable. The choice of a test crop should reflect the purpose of the growing medium. There is no general advice for a good indicator crop. Based on the work by Gossow et al. (1995), who tested different crops for sensitivity to different toxic substances, Chinese cabbage (*Brassica chinensis*) is widely used. However, another indicator crop may be used due to specific questions. Sometimes, two or three test crops from different families are used, for example, Chinese cabbage, lettuce (*Lactuca sativa*), and barley (*Hordeum vulgare*). For short-term root development tests, mostly cress (*Lepidium sativum*) is used (Del Buono et al., 2011). When testing for contamination with herbicides it is advised to test both monocots and dicots as many herbicides are targeting specifically on monocots or dicots.

11.5.3.3 Determination on loose growing media

The material to be tested is mixed in various ratios with a reference growing medium, for example, sphagnum peat. Recommended ratios for individual materials are listed in CEN 16086-1 (2011). Furthermore, a sufficient supply of nutrients and water has to be secured. The test plant is sown on or planted in the prepared sample, the sample has to be kept under suitable conditions (temperature, light intensity, moisture). Germination and growth are recorded over a certain time. For the

**FIGURE 11.15**

Plant response test using Chinese cabbage, various materials.

Photo: N. Schlatter, AGES.

germination rate, CEN 16086-1 (2011) with Chinese cabbage suggests a duration of 5 days, plant biomass shall be determined as soon as at least 50% of the plants show more than five foliage leaves (Fig. 11.15).

11.5.3.4 Determination on an extract

Especially for coarse media or growing medium constituents, direct germination on the material is not possible. In these cases, a sufficient amount of material (12 L volume) is extracted for 4 hours using a specific nutrient solution. Consequently, the extract is added to pots filled with perlite. Then, Chinese cabbage is sown on the surface and covered with a small amount of perlite.

Alternatively, the extract can be used in hydroculture. In this latter case, oxygen levels in the rooting solution must be maintained. If nutrient fixation is known to take place (e.g., nitrogen), necessary corrections must be carried out. Like for the test using the material directly, germination and plant biomass are determined after a given time.

11.5.3.4.1 Short-term root development test (“petri dish test,” CEN 16086-2, 2011)

If an immediate evaluation of a material is necessary, a root development test using a certain petri dish or test plate is possible. In this case, the medium is placed in square petri dishes, 10 seeds of cress are put on top. The petri dishes are set in a way that the emerging roots grow along the cover of the petri dish (Figs. 11.16 and 11.17).

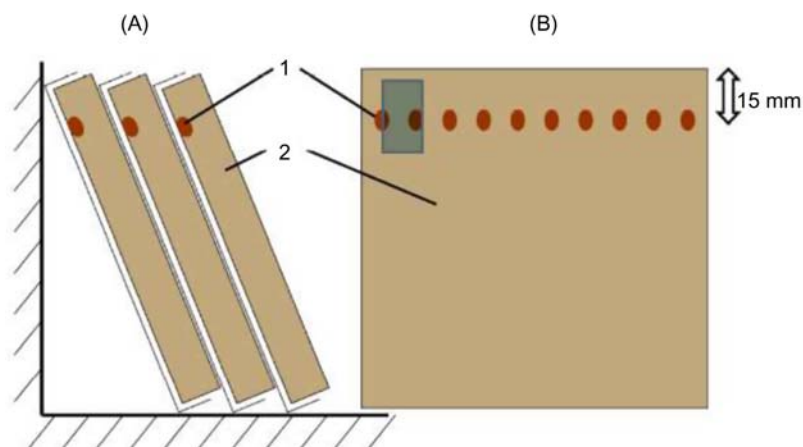


FIGURE 11.16

Situation of the petri dish for germination—(A) side view and (B) front view; (1) cress seeds; (2) test growing medium.

After CEN 16086-2, 2011. Soil Improvers and Growing Media—Determination of Plant Response—Part 2: Petri Dish Test Using Cress.



FIGURE 11.17

Petri dish test with germinated cress.

Photo: N. Schlatter, AGES.

Germination Rate and root length are measured after 72 hours at $25 \pm 5^\circ\text{C}$ and can be used for calculating the Munoo-Lisa Vitality Index, which sets the tested material in comparison to a reference. The petri dish test can be performed using the material directly or again perlite saturated with an extract.

11.5.3.5 Special cases

Dilution of the tested material is necessary if chemical properties, especially the salt content, are unfavorable. Respective criteria are listed in CEN 16086-1 (2011). An overview of available growing tests is given by Baumgarten and Spiegel (2004).

11.5.3.6 Common values for media

Common values given for media phytotoxicity are % germination and root elongation. Comparisons of different test methods can be found in Blok et al. (2008), Emino and Warman (2004), and Wever (2004).

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PART

SOILLESS PLANT
GROWING
SYSTEMS
DESIGN AND
MANAGEMENT

IV

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LIQUID CULTURE HYDROPONIC
SYSTEM OPERATION

12

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Growing plants in “liquid culture” or “water culture,” without the use of any solid substrates, has been around for decades. The key in this type of plant production is that the water has nutrients and oxygen dissolved in the water at levels that lead to high rates of plant growth. The roots of the plants are grown within this solution.

Initially, water culture was used to better understand the role of soil in plant growth. In 1699, John Woodward published an experiment with spearmint, whereby plants grown in water containing a soil extract had much greater growth than plants growing in distilled water. [von Liebig \(1840\)](#) promoted the concept that soil was not necessarily required for plant growth but that it contributed mineral elements necessary for plant growth, specifically potassium, phosphate, and ammonium. During the intervening decades, several scientists employed the water culture method to determine the necessity of various mineral elements in plant growth which led to the development of various nutrient solution recipes. One of these early recipes was proposed by Johannes Sachs in 1860 and contained potassium nitrate, calcium phosphate, magnesium sulfate, calcium sulfate, sodium chloride, and iron sulfate ([Hoagland and Arnon, 1950](#)). Nutrient solution recipes were further refined as scientists became aware of additional elements essential for plant growth.

The term “hydroponics” was coined by [Gericke \(1937\)](#) to mean water culture without employing any substrate. Over time the term hydroponics has evolved to encompass a broader meaning to include any of the soilless culture methods where the nutrient solution moves past the roots of the plants either continuously or many times per day regardless of whether there are solids in the root zone, as long as the substrates do not restrict the flow of the nutrient solution. Some sort of substrate may well provide stability to the plant and to provide an inert matrix which may provide some benefits such as providing air porosity or increased water retention. In addition to including systems where the roots are within a substrate matrix, systems where relatively small amounts of water are continuously or frequently move to the roots within a volume that includes mostly air, and without any substrates at all, are also considered hydroponic systems.

Strictly speaking, however, hydroponics is the practice of growing plants in nutrient solutions where the roots are exposed to nutrients in a dynamic manner. Any nutrient molecule dissolved in the water has an opportunity to interact with a root surface many times in a short time, perhaps

even as a continuous flow. Furthermore, such nutrients are the only source of nutrients. This is in contrast with conventional potted plant production in organic substrate where the organic matter contributes to the nutrients which the plant takes up. Also in conventional potted plant production, irrigation is provided only when a considerable amount of the water has been taken up by the plants. Such conventional irrigation practices place water into the root zone less often than with hydroponic systems. Between irrigation events, the transport of nutrients from the root zone to the root surface is accomplished through diffusion along the concentration gradient set up by the root taking up nutrients. Ion transport from roots to shoots then occurs both passively (through the transpiration stream as plants take up water) and actively (through channels and pumps which can allow for uptake against a concentration gradient).

In hydroponics, water flows through the root zone during irrigation events. During such movement of water, the nutrient molecules dissolved in this water move at speeds that are several orders of magnitude greater than the diffusion processes. This type of movement is termed “bulk flow.” A key facet of all hydroponic systems is that bulk flow is used much more extensively to move nutrients and as such the root surfaces are exposed to more nutrient molecules at greater rates. Furthermore, these rates are so much larger that typical hydroponic nutrient solutions can be much more dilute than conventional irrigation nutrient solutions yet still give the plant as much or more opportunity to take up nutrients. Such low concentrations of nutrient solutions, as measured by lower electrical conductivity (EC) level (less salinity), can be very beneficial to plant growth as described in [Chapter 6](#), Nutrition of Substrate-Grown Plants.

Prior to the advent of hydroponic plant production, fertilizers were provided to the root zone of plants through a combination of organic materials, nonsoluble substrates, slowly dissolving fertilizer products as well as ionic formulations where the fertilizers were salts which dissolved to provide the nutrients which the plant needs. Invariably such salts included some anions or cations which were not particularly useful to the plant or which the plant needs only in smaller amounts (e.g., Na^+ , Cl^-). When organic compounds are used to provide nutrients, microbes are generally depended on to release the nutrients. By the mid-1930s [Gericke \(1937\)](#) reported that agricultural crops (vegetables and flowers) had been produced for market employing hydroponic techniques. The publication of Hoagland’s nutrient formulation ([Hoagland and Arnon, 1938](#) and revised in [Hoagland and Arnon, 1950](#)) led to increased commercial plant production using hydroponics either in true solution culture or with an inert substrate such as gravel or sand.

One logical conclusion that might be drawn from the above is that one could simply increase the concentrations of desired nutrient ions in the nutrient solution and to thus make the bulk flow less of an imperative. When one tries to increase the nutrient load in a nutrient solution above certain levels, the result is less growth by the plants. This is due to the fact that total salinity has an effect on the osmotic potential. Thus there is a maximum concentration for the total nutrient load and once this maximum is exceeded, the plant experiences reductions in growth. So there is a trade-off between more nutrient availability and negative effect due to salinity. The latter can be measured with EC. A general guide in soilless culture has been to avoid EC levels above 3 dS/m but this is only a general guide, as the threshold EC value is species and cultivar dependent and is affected by the environmental conditions.

Over time, several types of liquid culture systems have been used commercially. The earliest systems we identify today as “Deep Flow” or “Deep Water Culture” are basically a relatively large vessel filled with nutrient solution with the roots dispersed freely in this liquid ([Fig. 12.1](#)).



FIGURE 12.1

Deep water culture.

It is well known that oxygen has to be forced into this water and that without it, plants would suffer from “water logging” which is basically oxygen starvation. To avoid oxygen depletion, air is introduced at a rate required to maintain the appropriate level of dissolved oxygen (normally, around 8 ppm). Water can also pick up oxygen from the surface as long as there is adequate head space above the water and oxygen can freely diffuse into this head space. As such, the concept of aeration is a key feature of hydroponics and can be accomplished with either bubbling with an air/aquarium pump, by causing the nutrient solution to be splashed or frequently moved to the surface with a water pump, or use of a pump with a Venturi air intake to aerate the return water. Commercially, some deep flow systems also inject oxygen from a tank, which can allow for supersaturated dissolved oxygen (Goto et al., 1996), but more often is used as alternative to mechanical pumping and to ensure adequate dissolved oxygen in large commercial installations.

One facet to liquid culture systems is that it is important to minimize the amount of nutrient solution that is exposed directly to light. Growth of algae in the system will proliferate if light is not restricted since the nutrient solution is just as optimal for algae as the crops being grown. It is unavoidable that some algal growth will be present in nearly all systems. The antagonistic effect of algae is twofold: (1) the algae take nutrients and dissolved oxygen that are destined for the crop and thus compete with the crop and (2) the proliferating cells will cause unattractive sludge and perhaps clog components in the system, such as filters and drip emitters. There is an entire industry that uses the algae as the crop for energy, biomass, etc.; in that domain, the nutrient solution containing algae is exposed explicitly to high light and dissolved oxygen to maximize their production.

The other reason why light is avoided is because blue and ultraviolet light degrades chelating molecules which can then lead to iron/micronutrient deficiency (Albano and Miller, 2001).

12.1 TYPES OF HYDROPONIC SYSTEMS

Many hydroponic systems are described with technical detail in Chapter 13, Technical Equipment in Soilless Production Systems. So here we define terms for later use in this chapter where the discussion focuses on crops and operations. As such, specifics about system design are handled in Chapter 13, Technical Equipment in Soilless Production Systems. It is relevant that the popularity of hydroponics is increasing steadily as both commercial and noncommercial (hobby; home-use) installations are increasing. This has resulted in hydroponics supply stores springing up in many communities, even in small towns. This itself is encouraging further expansion of the market because the technical elements (fertilizers, pumps, systems) are easy to obtain.

Deep Flow systems, as mentioned above, involve a relatively large body of water in which the roots are suspended (Fig. 12.1). The plant is held above the water surface and all fertilizer nutrients are dissolved in the water. These types of systems can range in size from buckets to ponds. Generally commercial installations are aerated with both air pumps (or by injecting oxygen from tanks) and water pumps. Water movement is important to ensure proper mixing/replenishment of nutrients and dissolved oxygen. The flow rate and water movement are customized by the grower, increasing either if oxygen becomes deficient or the plant shows signs of nutrient deficiency.

One type of implementation, sometime referred to as a “raceway,” has the vessel consisting of a long trough where water is pumped in at one end and allowed to flow to the other end where the pump intake is located. The system is level so that the motion of the water is imposed by the pump.

Deep Flow systems can have the plants mounted by holders that hold the plants above the water and only the roots and end of the stems are in the water. Another approach is to mount the plants on floats (often insulation boards) which have adequate buoyancy to keep the crown of the plant above water at all times. This is also sometimes referred to as a “raft/pond” system.

Nutrient film technique (NFT) uses an approach whereby roots are suspended in a trough whereby a thin layer of nutrient solution is continually recirculated. The NFT system was developed beginning in the late 1960s by Cooper (1975). Since only a thin film of water is used at a time, the system has less water during operation than deep water systems. However, that does not mean that it uses less water. Water use is a function of transpiration of the plant and that could easily be the same for all hydroponic systems.

NFT systems (Fig. 12.2) consist of long white channels with holes in the top side for mounting the plants. The plants grow through these holes while the roots stay in the channel where the relative humidity is 100% and the layer of water is flowing from one end to the other using the force of gravity. A slope of 1%–4% is typically maintained from the end of the channel where water flows in, to the drain end so as to obtain adequate water flow and drainage. The drainage water is captured and reused directly to prevent the NFT root film from drying out.



FIGURE 12.2

NFT system. *NFT*, Nutrient film technique.

Aeroponics involves spraying the nutrient solution onto the roots by pushing the nutrient solution through misters either continuously or many times per hour within the root zone. The root zone is primarily air and the roots, the water quantity is similar to NFT in that only the floor of the root zone is wet and it is constantly draining back into a reservoir. The water is filtered as part of the continual reuse, so as to avoid clogging the misters in the root zone.

“Aerohydroponics” was developed as the Ein Gedi system (Fig. 12.3) in the 1980s (Soffer, 1986; Soffer et al., 1991). It is a hybrid between Deep Flow and Aeroponics. It was commercialized by General Hydroponics and, as such, accounts for a very large installed base in the United States and Europe, mostly in the hands of urban growers. As with Deep Flow there is a relatively large amount of nutrient solution in the root zone, and as with aeroponics, there is a relatively large amount of head space above the nutrient solution which is continually misted onto the roots. Here some roots are suspended in humid air and other roots are suspended in the reservoir. The frequent misting is responsible for saturated dissolved oxygen concentration of the nutrient solution so that both nutrient and oxygen availability are maximized. The original Ein Gedi system also had a speed control for the spraying so that it was possible to fine-tune the oxygen concentration and even control it down to near zero oxygen for research purposes (Soffer et al., 1991). While this system consisted of a large round tub equipped with spinner for creating the mist, another version consisted of a long channel with holes in the top to accommodate a plant.

The above hydroponic systems often use small amounts of substrate to anchor the plant into the system as well as for early development of seedlings/cuttings prior to transplanting into the system. Inorganic substrates such as rockwool or phenolic foam (e.g., oasis) are most frequently used; however, it is possible to get excellent plant production results with some organic substrates as long as the organic substrate is slow to decompose. The decomposition process releases nutrients and changes the substrate physical properties. Coconut coir, peat, and various organic/inorganic mixes

**FIGURE 12.3**

Ein Gedi aeroponic system seen with banana tree growing within it. Note spinner motor (black) on top with control for rate of aeration.

have been used successfully and many growers prefer using organic media rather than inorganic media in such systems. When used in these solution culture hydroponic systems, these organic substrates may be glued together or held by a wrapper to avoid the components being lost into the solution which can clog pumps and emitters.

The selection of which system to use for a given application often relates more to logistical considerations than to differences in plant performance between the different systems. Some of the practical considerations include installation and maintenance cost, labor efficiency and automation, ability to respace plants (more common in deep flow vs NFT) to improve space efficiency, and ability to control root zone conditions. For example, deep water ponds are often 20–30 cm in water height. This large volume of water reduces fluctuations in temperature, dissolved oxygen, pH, and nutrition as compared with NFT which typically utilizes a much smaller volume of water in the reservoir. As stated earlier, actual water consumption of the two systems is typically similar, depending primarily on plant transpiration. Because of the large surface area between channels and the aerial environment and the relatively smaller volume reservoir, solution temperature is more difficult to control for NFT as compared to deep flow systems. Thompson et al. (1998) reported that control of solution temperature to 24°C for lettuce

(*Lactuca sativa*) “Ostinata” in deep flow allowed for acceptable lettuce production under otherwise unacceptably high air temperatures (31/26°C day/night). Ikeda et al. (1994) grew several spinach varieties in deep flow and NFT. Interestingly they found greater yield in deep flow during the winter but with NFT during the summer. When comparing deep flow to NFT for basil (*Ocimum basilicum*), Walters and Currey (2015) found only subtle differences in yield between the two systems when water temperature, aerial environment, and nutrient solution were held similar.

12.2 WATER QUALITY

In closed hydroponic systems, such as deep flow, NFT, and aeroponics, the irrigation water is captured and reused whereas in open systems it is not. EC of source water should ideally be less than 0.25 dS/m for closed systems. Many hydroponic growers have found it necessary to filter source water and perhaps use reverse osmosis to remove some solutes so that it is pure enough for closed hydroponic systems. It is relevant that any solute in the water can only leave the root zone by either entering the plants or by having some of the nutrient solution discarded. These solutes consist of whatever is dissolved in the source water (e.g., calcium carbonate, etc.) and fertilizers put there by the manager. Water that evaporates from the nutrient solution does not carry these solutes out, leading to a gradual concentration buildup. Furthermore, then plants take up water and nutrients somewhat independently, and especially take up more water in response to a need to cool the foliage through evapotranspiration. All this points to the need for nutrient solutions to be as light in these solutes as possible.

The way to counter salt buildup in closed systems is to blend in fresh water with little or no solutes, or, if this is not possible, then purposely drain off and discard some proportion of the nutrient solution and replace this with fresh water. In open systems the buildup of salts can be managed by applying excess water to leach out soluble salts. Therefore, source water EC can be higher than in closed systems; ideally less than 1.0 dS/m. The issue of the effects and management of EC in opened and closed systems is further discussed in Chapter 10, Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses.

12.3 CONTROLLING THE NUTRIENTS IN HYDROPONIC SYSTEMS

Perhaps the most difficult facet of operating a hydroponic system is to manage, over days and weeks, the many nutrients which the plants needs to take up through the roots. The key facet of hydroponics is to capture all leachate and to reuse it as much as possible by pumping the water through a process system (filtration, disinfestation, injection of nutrients, etc.). The technical and scientific details of this are covered in Chapter 10, Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses. Theoretically, if we can have a system where the injected nutrients are in perfect balance with what

the plants take out of the nutrient solution, then we can avoid discharging water from the system and the only water lost is through transpiration. While this is easy to consider, it is not at all easy to do, particularly if the environment around the plant is changing, thus affecting transpiration and growth differently in relation to the targeted growth.

The key problem is that there are many nutrients and the plants use each at different rates, at different times (see [Chapter 6: Nutrition of Substrate-Grown Plants](#)). To provide exactly what the plant needs requires a method for predicting this for the particular plants. Models can be used but most are not as yet adequately accurate to account for all the possibilities.

Ideally we would have a sensor for every nutrient that is dissolved in the water (N, P, K, Fe, Ca, etc.) as well as the compounds which are undesirable (e.g., sodium) so as to be able to identify whether any one of these is insufficient (i.e., causing deficiency in the plant) or in excess (causing toxicity), preferably as a continuous variable with sensor systems at adequate resolution to allow for anticipation of undesirable conditions.

Achieving better control is indeed possible by obtaining more accurate measures of each nutrient in the nutrient solution. Many such in-line sensors exist but these are fairly expensive and may need frequent calibration and replacement so that growers generally seek a simpler, less expensive solution. It is also possible for growers to use in-house ion-specific probes or test kits to periodically measure many of the nutrients of interest (e.g., CleanGrow technology). This can still be time consuming and expensive, and accuracy depends again on frequent calibration. Another approach is to periodically send nutrient reservoir samples to analytical labs. However, the turn-around time tends to be slow and the expense is significant. Nevertheless, commercial producers using deep flow ponds often send nutrient solution for analysis every week or two and use the results to adjust the pond solution over time.

Ultimately, because exact nutrition composition is difficult to monitor, EC, a measure of total salinity (both fertilizer and nonfertilizer ions) is more practical for day-to-day monitoring and adjustment. While EC values do not tell us the nutrient composition, they give an indication as to the salinity and therefore provide some information as to the relative strength of the nutrient solution. Typically, EC is monitored not only to ensure there is adequate fertility but also to make sure salinity is below the level where it affects the growth of the plant through osmotic pressure. EC targets vary by crop but are often in the range of 1.0–2.5 dS/m from the fertilizer plus the EC contribution from the water source. A low EC indicates that not enough fertilizer is being supplied to meet plant needs. It is relevant that this approach to control is only possible through the use of nutrient solutions which contain ionic fertilizers. This makes it possible to use the measurement of salinity to account for changes in the nutrient solution. Urea is generally not used in recirculated hydroponics, in part for this reason, and partly due to potential for phytotoxicity.

Beyond EC, it is important to closely monitor and adjust nutrient solution pH. In some hydroponic systems, pH can swing more than one pH unit in the course of a day as the plants selectively remove certain ions during active growth. As such, production management may require testing more frequently than daily. Generally, the more water is in the system, the slower the pH fluctuations. When the principle source of nitrogen is nitrate, its consumption by the plant results in rising pH, while nitrogen consumed as ammonium tends to lower the pH. As such, growers may try to balance the ratio of these two N sources so as to achieve a stable pH. The exact ratio required to

achieve stable pH is not readily predictable as it depends upon several factors: plant's differential uptake of nitrogen form over another, the rate that ammonium is converted to nitrate through nitrification (a process which also decreases pH), and the relative buffering of the water source due to its alkalinity. A study with hydroponic cut flower roses in which ammonium:nitrate:urea ratio was adjusted as the sole method of pH control in a high alkalinity water source found that in winter 25:75:0 resulted in optimum growth, while in summer 12:88:0, 25:75:0, or 0:50:50 resulted in optimum growth (Bar-Yosef et al., 2009). Despite the use of different N forms, pH was still very difficult to manage in the experiment. Adjusting N form is an efficient tool for hydroponic growers; however acids (such as nitric, muriatic, or sulfuric acid) or bases (such as potassium hydroxide) are also used to adjust pH.

In commercial facilities, pH and EC are monitored and adjusted at least daily. If done manually, common practice is to measure and adjust 1–3 times daily.

The minimum set of sensors that a professional grower must use is two: one for pH, resolving to 0.1 dS/m over the range from 3 to 8 dS/m with targets between pH of 5 and 7 and another for EC over the range from 0 to 5 with most measurements falling between 0.1 dS/m and 3 dS/m. The wider range for the meters is needed to be able to detect when the nutrient solution is very far from optimal. At such times an accurate measurement is needed to allow calculating blending percentages with either clear water or nutrient solution. In-line pH and EC sensors are commercially practical and so these attributes may be measured continuously and used to inject fertilizer stock solution or acid/base (e.g., BlueLab controllers).

While the control via pH and EC can achieve excellent results, there will invariably come a point where an existing nutrient solution cannot be used anymore as indicated by the measurements. This may be due to nutrient imbalances, buildup of unwanted ions (e.g., Na, Cl), or lack of knowledge as to the nutrient composition. Ultimately the solution may need to be discarded (in part or wholly) and be replaced with a fresh solution where the exact concentrations of each nutrient are known and optimized.

12.3.1 PREPARING NUTRIENT SOLUTIONS

Plants require 14 essential elements that must be supplied in the root zone including the macronutrients (required in relatively large quantities) of nitrogen, phosphorus, potassium, sulfur, calcium, and magnesium; and the micronutrients (needed in smaller amounts) of iron, manganese, zinc, boron, copper, molybdenum, chloride, and nickel (Chapter 6: Nutrition of Substrate-Grown Plants). All of these nutrients must be supplied by either the hydroponic nutrient solution, the substrate, or the water source. Many hydroponic fertilizers provide 12 of the essential elements but don't specifically provide chloride and nickel as they are available in sufficient quantities as impurities with the fertilizer. For fruiting crops, chloride (or saline conditions/elevated EC) is sometimes provided at high concentrations to increase flavor/soluble solids but at the expense of crop yield as in the case of tomato (Pascale et al., 2001).

Numerous recipes exist for preparing nutrient solutions of hydroponic crops (Resh, 2001; Sonneveld and Voogt, 2009). Because many factors influence plant nutrient uptake (genotype, crop growth stage, weather), the selection of a nutrient solution recipe (i.e., target values for macro- and micronutrients) is not an absolute science but refined over time based on empirical observations.

That is, a nutrient recipe is initially selected based on past experience or from conversations with a crop consultant or cooperative extension educator. Then this recipe is refined over time based on periodic tissue and nutrient solution analyses. A couple of different strategies are typically used when preparing hydroponic nutrient solutions: (1) Small and hobby growers often prefer to purchase preformulated fertilizers with fertilizer salts already preblended. In this case typically 1–3 liquid concentrates or powdered solids are mixed together to prepare a complete nutrient solution. (2) Medium-to-large commercial growers often prepare their own fertilizer solutions on site using bulk ingredients (i.e., individual fertilizer salts, concentrated liquid stock solutions, etc.). This then uses precise quantities of raw fertilizer materials to be dissolved into stock tanks dosed into source water to create the desired nutrient solution. Details of such systems and their operation are described in [Chapter 13](#), Technical Equipment in Soilless Production Systems. The first approach requires less background knowledge and labor; however, it is typically more expensive and does not allow much flexibility with adjusting the composition of the nutrient solution over time. The latter approach is cheaper when large quantities are used and allow greater flexibility to adjust individual fertilizer salts if particular elements are used by the plant in different quantities than the original target values.

When fertilizer ingredients are purchased for hydroponic use, it is important that they are “greenhouse/solution grade” not “field grade,” they have less impurities, and also they do not have waxy coatings. For example, field grade calcium nitrate has a greasy coating around the prills so that the fertilizer which is hygroscopic does not absorb too much water from the air. However, such a coating is undesirable for dissolving into stock tanks for greenhouse use. It is possible to obtain premade dry or liquid nutrient solution concentrates from various vendors. Regarding strategy no. 1 (using premade preparations), these are formulated to group various fertilizer nutrients together in such a way that (1) the various chemicals do not react with each other, (2) are completely soluble, (3) every nutrient stays in solution until it is used. For small/hobby growers, hydroponics shops sell an array of products from various manufacturers, and some are series that some measured amount is taken from each concentrate and stirred into a fixed amount of water. Generally there are two-part solutions and three-part solutions. With the three-part solutions, the grower can also make some adjustments during the phases of crop development (e.g., adjusting nitrogen during the crop cycle).

Most commercial hydroponic growers prepare nutrients in concentrated stock tanks and then dilute these either manually or with a fertilizer injector to prepare them at the final desired concentration with the deep flow pond or NFT reservoir. When preparing concentrated stock tanks, which are often at 100–200 times greater concentration than the nutrient solution target concentration, it is important to be aware of the solubility limits for each specific fertilizer salt at room temperature. If a fertilizer salt is not fully soluble in water, a precipitate will form, which means that a fraction of the fertilizer is not available. Also, such precipitate can clog fertilizer injectors and hydroponic drip lines. In addition, when multiple fertilizer salts are combined in the same stock tanks, one must be aware of compatibility. Some fertilizer salts are incompatibility meaning they rapidly combine to form an insoluble salt. The most common incompatibility issue is that calcium is incompatible with sulfates and phosphates. Compatibility of common fertilizer salts is presented in [Table 12.1](#).

The various issues with fertilizer compatibility dictate that several concentrated stock tanks (2, 3, or more) are used. Growers who need to be able to deliver many varied recipes to crops at

Table 12.1 Compatibility of Common Fertilizer Salts Used in Concentrated Stock Tanks for Fertigation

	Urea	Ammonium Nitrate	Ammonium Sulfate	Calcium Nitrate	Potassium Nitrate	Potassium Chloride	Potassium Sulfate	Ammonium Phosphate	Fe, Zn, Cu, Mn Sulfate	Fe, Zn, Cu, Mn Chelate	Magnesium Sulfate	Phosphoric Acid	Sulfuric Acid	Nitric Acid
Urea	C													
Ammonium nitrate	C	C												
Ammonium sulfate	C	C	C											
Calcium nitrate	C	C	I	C										
Potassium nitrate	C	C	C	C	C									
Potassium chloride	C	C	C	C	C	C								
Potassium sulfate	C	C	R	I	C	R	C							
Ammonium phosphate	C	C	C	I	C	C	C	C						
Fe, Zn, Cu, Mn sulfate	C	C	C	I	C	C	R	I	C					
Fe, Zn, Cu, Mn chelate	C	C	C	R	C	C	C	R	C	C				
Magnesium sulfate	C	C	C	I	C	C	R	I	C	C	C			
Phosphoric acid	C	C	C	I	C	C	C	C	C	R	C	C		
Sulfuric acid	C	C	C	I	C	C	R	C	C	C	C	C	C	
Nitric acid	C	C	C	C	C	C	C	C	C	I	C	C	C	C

C, Compatible; I, incompatible; R, reduced solubility.

Adapted from Smart Fertilizer Management <https://www.smart-fertilizer.com/articles/fertilizer-solubility>.

various stages of growth may have each of the fertilizers that is used in largest quantity in a separate stock tank. The most common approach is to consider the use of two tanks and then increase the number of stock tanks depending on the need. In this approach the two tanks are identified as A and B tanks, with the A tank holding most of the nitrogen fertilizer.

The following identifies a common strategy for using three stock tanks whereby stock tank A is used for calcium nitrate and other compatible salts; tank B is used for sulfates, phosphates, and other compatible salts; and tank C is used for either an acid or base to adjust pH (Adapted from Resh, 2011):

- Tank A
 - Calcium nitrate
 - Half of the potassium nitrate
 - Iron chelate
- Tank B
 - Half potassium nitrate
 - Potassium sulfate
 - Monopotassium phosphate
 - Magnesium sulfate
 - All other micronutrients
 - Monoammonium phosphate
 - Ammonium nitrate
- Tank C
 - Acid, used to drive down pH (or base if very pure waters, acidic N source)
 - Sulfuric/Phosphoric
 - Nitric

This strategy can also be implemented with two tanks (A and B) if the tendency for pH adjustment tends to be very similar week after week. In that case, sulfuric and phosphoric acids would be included in the B tank, while other acids, such as the nitric acid, would be included in the A tank.

This, however, takes away direct control over pH. This might not be a big issue for some growers, but having the ability to control pH in hydroponic production is very important for most growers.

There may also be reason to increase the number of stock tanks and injectors. For example, some growers find that they prefer to have better control over calcium to cope with instances of leaf margin burn, blossom end rot of fruits, or similar calcium disorders.

12.4 NUTRIENT SOLUTION RECIPES FOR LEAFY GREENS AND HERBS

In this section we present two nutrient solution recipes for growing leafy greens and herbs that have been successfully used for many years by the Cornell University Controlled Environment Agriculture group in deep flow and NFT. These examples are for 1000 L of dilute fertilizer

solution. If using stock tanks (i.e., a $100\times$ concentration), then the quantities are the amount to use per 10 L in a $100\times$ concentrated stock tank. The Tank A and Tank B components *Must* be prepared separately so a precipitate does not form.

The first is an approach that uses 750 g of calcium nitrate (15.5-0-0) per 10 L of stock solution in one stock tank. In the second stock tank, 750 g of preblended semicomplete hydroponic fertilizer with analysis of 5-12-26 (“Hydroponic,” J.R. Peters Inc., Allentown, NJ) is dissolved in each 10 L of water.

The second approach uses individual fertilizer salts to create a modified Sonneveld’s solution for leafy greens as described by [Mattson and Peters \(2014\)](#). Note that the items marked with an asterisk (*) must be weighed out with an analytical balance to achieve the desired accuracy:

- Tank A
 - 486 g $\text{Ca}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$
 - 38.0 g NH_4NO_3
 - 442 g KNO_3
 - *10.0 g 10% iron-DTPA
 - Sprint 330 or Sequestrene 330
- Tank B
 - 136.1 g KH_2PO_4
 - 245.9 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
 - *0.777 g $\text{MnSO}_4 \cdot \text{H}_2\text{O}$
 - *0.930 g H_3BO_3
 - *0.061 g $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$
 - *0.573 g $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
 - *0.092 g $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

These two nutrient solution recipes result in final nutrient composition as follows:

	Calcium Nitrate + Hydroponic Base (5-12-26)	Modified Sonneveld’s Solution
Macronutrients (mM)		
Nitrogen (N)	10.7	10.7
Phosphorus (P)	1.3	1.0
Potassium (K)	4.1	5.4
Calcium (Ca)	3.5	2.3
Magnesium (Mg)	1.9	1.0
Micronutrients (μM)		
Iron (Fe)	41.2	17.9
Manganese (Mn)	6.9	4.6
Zinc (Zn)	1.7	2.0
Boron (B)	35.2	14.8
Copper (Cu)	1.8	0.4
Molybdenum (Mo)	0.8	0.3

The above recipes can be adjusted to suit specific crops. It is relevant that in the above the primary target is to grow leaf materials along with a small amount of stem and roots. Clearly the two recipes are not identical in what is delivered to the plant, this demonstrates that many nutrient solution recipes can be used fairly effectively; however the grower may adjust over time as indicated by laboratory analysis.

These two approaches are presented since many practitioners will begin hydroponic production with off-the-shelf solutions (such as approach 1 above) and begin to make adjustments with further experience, ultimately ending up with an approach similar to no. 2. The most common facet of this experience is a notion that premixed fertilizers are more expensive relative to ones that must be mixed from the raw materials.

It is also relevant to understand how the nutrient solution recipes have evolved in the hands of firms marketing fertilizers. Many target some particular crop for growers that share a particular set of conditions (climate, water sources, substrate mixes, etc.). Most notable in any set of these conditions is that it can be expected that the pH of the nutrient solution will range between some specific minimum and maximum and that the source water or blending of water will result in buffering capacity which will cause the root zone to be at some pH level fairly frequently. This pH level affects nutrient availability just as much as that actual amount of nutrient present in the solution. For hydroponic systems the range of 5.5–6.0 is fairly commonly used so that nutrient concentrations in recipes assume this range.

Growers who find their pH fairly different from the midpoint of this range will need to account for the reduced micronutrient availability by supplying some nutrients at a greater rate; nutrients which are more available at such pH levels would need to be reduced in the nutrient solution. Clearly this is not feasible with off-the-shelf fertilizers so that pH control must be practiced to keep pH stable (within ± 0.2 range) and at a level that works well.

As part of pH control it is important to know where fluctuations come from. The most dominant factor shifting pH is the nitrogen form. Nitrate taken up by plants results in increasing the pH while ammonium use reduces the pH. So, it makes sense to use both nitrate and ammonium in such a way that the pH changes very little or not at all.

12.5 NUTRIENT SOLUTION RECIPES FOR FRUITING CROPS

With crops where the target is flowers or fruit, or even tubers or roots such as carrots, the recipes need to be different from those for growing primary leaf material. Providing fertilizer nutrients can be viewed simplistically as providing the building blocks of the biomass. The proportion of N, P, and K, and other nutrients in flowers and fruits, is very different from the leaves; so it makes sense that adjustments are needed in rates of each.

At the same time, once enough of any one nutrient is available, the plant can easily cope with some excess, as long as the total salinity stays low. This provides some flexibility to the practitioner since it is not imperative to get every nutrient concentration perfect.

When fruiting and flowering plants are young, they can initially be provided fertilizer in the same approach as the leafy-greens recipe above. As the plants grow, the proportion of N in relation to Ca and K needs to be changed during the production period, and the timing needs to be commensurate with observations of flowers, fruit, or tubers.

Below is an example of a fertilizer solution adjusted according to crop stage (from the University of Arizona CEA Center, [Mattson and Peters, 2014](#)).

Macronutrients (mM)			
NO ₃ -N	6.4	8.6	13.6
NH ₄ -N	0.0	0.0	0.0
P	1.5	1.5	1.5
K	3.7	9.0	9.0
Ca	3.6	4.0	5.0
Mg	2.5	2.5	2.5
S	3.6	3.6	3.6
Cl	2.5	2.5	2.5
Micronutrients (μM)			
Fe (DTPA/EDTA)	35.8	35.8	35.8
Mn	10.0	10.0	10.0
Zn	5.0	5.0	5.0
Cu	0.79	0.79	0.79
B	31.5	31.5	31.5
Mo	0.52	0.52	0.52

As above we can take two approaches to getting a recipe that works well. As before one strategy is to buy off-the-shelf fertilizers from a store marketing specifically hydroponics fertilizer. These products come with “feed charts” (i.e., labeled rates) that identify roughly how much of each purchased stock solution to dissolve in water to create a nutrient solution. For flowering and fruiting plants, these feed charts will be subdivided into the various growth phases of the crops. As above, the firms have taken the task of separating the stock nutrients into parts which are compatible (not forming precipitates in the bottle) and also providing enough flexibility to allow growers to easily make adjustments as the crop grows.

For tomatoes (and other fruiting crops) we present two approaches for developing the Stage 3 (multi-crop) recipe. A simple approach for 1000 L of dilute fertilizer solution (or 10 L of 100 × concentrated stock solution) is to use 950 g of calcium nitrate (15.5-0-0) dissolved in water in one stock tank. In the second stock 10 L tank uses 950 g of preblended semicomplete hydroponic fertilizer with analysis of 5-12-26 (“Hydroponic”, J.R. Peters Inc., Allentown, NJ).

The second approach uses individual fertilizer salts to create the Stage 3 (multi-crop recipe) as used by the University of Arizona CEA Center ([Kroggel and Kubota, 2018](#)). Note that the items marked with asterisk must be weighted out with an analytical balance to achieve the desired accuracy.

- Tank A
 - 919 g $\text{Ca}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$
 - 403 g KNO_3
 - *20.0 g 10% iron-DTPA (Sprint 330 or Sequestrene 330)
- Tank B
 - 171.4 g KH_2PO_4
 - 486.9 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
 - 303.0 g K_2SO_4
 - *1.693 g $\text{MnSO}_4 \cdot \text{H}_2\text{O}$
 - *1.600 g H_3BO_3
 - *0.127 g $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$
 - *1.450 g $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
 - *0.195 g $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

These two nutrient solution recipes result in final nutrient composition as follows:

	Calcium Nitrate + Hydroponic Base (5-12-26)	University of Arizona, CEAC Recipe
Macronutrients (mM)		
Nitrogen (N)	13.6	13.5
Phosphorus (P)	1.6	1.3
Potassium (K)	5.2	8.7
Calcium (Ca)	4.4	4.3
Magnesium (Mg)	2.5	2.0
Micronutrients (μM)		
Iron (Fe)	51.1	35.8
Manganese (Mn)	8.7	10.0
Zinc (Zn)	2.14	5.05
Boron (B)	44.4	25.9
Copper (Cu)	2.20	0.79
Molybdenum (Mo)	1.04	0.52

12.6 DAY-TO-DAY MANAGEMENT

It is relevant that hydroponic plant production involves daily, perhaps even hourly, changes within the system. Changes over a 2-day span can easily be very detrimental. The best advice is to inspect everything twice per day for anomalies. The most obvious of things the grower must attend to is the total amount of water that the plants are exposed to every day. Every reservoir needs to be inspected daily to verify that it holds enough resource until the next inspection. An injector stuck in open position can drain an entire stock tank and apply the resulting salty solution to the plants.

It cannot be taken for granted that all hydroponic systems are totally dependent on electricity. Power failures create huge problems. Failure to aerate for 20–30 minutes in a deep flow system can result in all plants wilting (depends on reservoir size). Aeroponic or NFT systems without power for 30 minutes will likely result in dead plants. For many systems the critical duration may well be much shorter.

Growers should have working pH and EC meters and be using each at least once per day. Data should be recorded so that trends can be observed. As a general guideline, the pH for hydroponic crops should be maintained at 5.5–6.0 and the EC at 1.4–2.0 dS/m (for leafy greens) and 2.0–2.6 for fruiting crops. Any changes in the system should be recorded as part of the data collection. Each pH and EC meter will have specifications that identify how often to calibrate this test equipment. It is imperative that a standard calibration practice be implemented by the grower. Every time a problem occurs, then test equipment should be recalibrated to assure that the error is not an artifact.

Another element of the system is the fact that oxygen is explicitly managed. As such any aeration equipment should be inspected with each crop inspection. It is common that air hoses have come out. Less common is that air pumps are failing, but this too is possible. Where air pumps are bubbling into nutrient solution, each such instance should be inspected to assure it is happening and that the air hose is not dangling above the water level. Many times scanning the crop and seeing a few plants wilting is due to inadequate aeration so that the wilted plants are sitting in stagnant water, deprived of oxygen.

An important facet of crop production is to have an eye on the target: selling the plant/crop for profit. This means that there will be key stages of crop development and this will include the first date on which the product can be shipped and the last day on which it might be marketable. During any routine inspection, the grower should check the crop production charts to verify that the crop is on track for the marketing objective. If the fertilizer injection, humidity control, pH, EC, etc. are suboptimal, then the product will be delayed or damaged. It is important to spot signs of this as early as possible so as to mitigate any damage.

12.7 OTHER ADJUNCTS

Since we recommend starting with stock solutions from stores, we must also provide a bit of advice about the many adjuncts that can be purchased. Are these needed? Most often the answer is “no.” Can they do harm? Generally not, as long as they are used according to the label. Some plants may well have a greater requirement for some nutrient (e.g., silicon) than is typically present in most nutrient solutions, but this is rare and so crop specific that a practitioner can find out whether a product is likely to be of some benefit or not (further discussion on silicon in [Chapter 6: Nutrition of Substrate-Grown Plants](#)). A grower who does not have high-quality fresh water available will need to be very judicious with inclusion of compounds that might have some elusive effect because anything added to the nutrient solution may well build up and handicap the grower in prolonged use of the irrigation water.

In this discussion we have not included the use of plant growth regulators (PGRs). Some adjuncts or amendments include PGRs and these should only be used with a full understanding of what these materials will do to the plants and what consequence might be on the final marketed horticultural product. If a material does not disclose its constituent materials, then it should not be used.

12.8 ORGANIC HYDROPONICS

With the continued growth of hydroponic plant production, there is considerable interest in being able to certify the resulting products as “organic.” This is, however, problematic with the above approach since there are items in each of the fertilizer recipes which are synthetic rather than mined or derived from plant or soils. In each country where foods in stores are marketed as “certified organic,” this means that the producer has been inspected and found to be using only materials and methods which are on a list of approved “organic” materials. Furthermore, the crop cannot have been produced with any material that is not on the list. The most basic requirement in this domain has been the requirement that crops be grown in soil and that soil health and sustainability is of paramount consideration. Thus “soilless” culture would appear to be a clear violation. Mineral wool or synthetic foams are not allowed. However, there are substrates which are obviously organic or could be certified as such. So that it may be technically possible to overcome some of the hurdles to getting certification. Plastic mulches are allowed and there may well be some latitude where some new materials could be placed on the list. Where plastic vessels are not allowed, these could be replaced with ceramics (at greater expense, of course). The most relevant omission from the “organic certified” list is calcium nitrate, a mainstay in all hydroponic systems. Without it, the fertilization approaches described above get very complicated and frustrated. Sodium nitrate (a mined product) is technically usable but in recirculated hydroponics, the sodium builds up very quickly to unacceptable levels. Many other fertilizers derived from plant or animal products tend to be much more expensive per amount of fertilizer and also typically include particles which clog emitters and filters.

As such there are commercial operations which claim to be practicing organic hydroponic production. These generally do not recirculate their irrigation water, instead growing with approved products in a flow-through approach using substrates derived from plants (peat, coir, sawdust, etc.). These are not hydroponic systems as defined in this book.

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TECHNICAL EQUIPMENT IN SOILLESS PRODUCTION SYSTEMS

13

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13.1 INTRODUCTION

The introduction of soilless culture on a commercial scale (Steiner, 1967) was motivated by a potential for increased crop productivity and efficiency. As part of this technical developments were made related to problems with root diseases, root zone oxygen deficiency, fertility control, and increased complexity in irrigation strategy. Technical solutions to these problems and opportunities resulted in widespread adoption of soilless container plant production in outdoor nurseries in the 1950s and 1960s. In the early 1970s, production of greenhouse crops in stone wool dramatically expanded commercially viable soilless crop production (Cooper, 1979; Verwer, 1978). Technical innovations in fertilization and irrigation resulted in adoption of fertigation technologies wherein completely soluble fertilizers are dissolved in irrigation water so as to deliver to plants the nutrients they need for optimal growth.

In all modern production systems, fertilization and irrigation have been integrated into a system that the grower seeks to optimize. Traditionally growers applied dry fertilizer products to the top of the root zone, resulting in fertilizer nutrients being carried to the root surfaces with the applied irrigation water. Once it became evident that all essential fertilizers could be supplied through completely soluble fertilizer salts, systems were developed where such salts are dissolved at relatively high concentrations in special stock solutions. By using one or more injectors, such concentrated solutions could be injected into the irrigation water (fertigation). This chapter focuses on the technical equipment associated with fertigation and irrigation in greenhouse production. In most such soilless production systems, fertilization is accomplished through injection of soluble fertilizer into irrigation water. We also discuss which system is best suited to which crop and where continued innovation is likely to result in future technical advances. Irrigation control methods are the subject of the previous chapter.

13.2 WATER AND IRRIGATION

13.2.1 WATER SUPPLY

An adequate supply of high-quality water is essential in soilless crop production, regardless of whether in outdoor nursery production or in greenhouse crop production. Potential water sources include rain water, surface water, groundwater, and desalinated sea water. The latter three can be secured either directly or through municipalities as part of their drinking water supply. In in-door production systems using artificial light, condensate for air conditioning systems is another possible source of water for irrigation.

The quality of irrigation water is typically evaluated through consideration of the dissolved minerals and salts in the water. Salinity is typically measured as electrical conductivity (EC) and it is known that source water with high salinity ($EC > 2 \text{ mS/cm}$) can result in growth suppression (and perhaps also other problematic manifestations) for many plants. In some areas of the world, the available supply of water has EC of nearly 2 mS/cm (or perhaps higher) so that this presents the grower with serious challenges.

The EC of the irrigation water results from a combination of the dissolved materials in the supply water plus any fertilizers that are dissolved in the water. It is desirable for the supply water to have an EC of less than 0.5 mS/cm and the sodium concentration less than 0.5 mmol/L (Sonneveld, 2000). This then allows for addition of ample fertilizer ions so that the irrigation solution seen by the plants is below a level where problems might result. In recirculating irrigation systems, where some soluble materials will build up over time, EC management can be particularly challenging if the EC of the source water is higher than 1 mS/cm . Management of this type of situation is discussed in Chapter 10, Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses, and is an area of continued innovation in soilless production because water quality in many areas of the world is low, containing higher levels of sodium (Na^+) and other problematic elements such as SO_4^{2-} , Fe^{2+} , or Mg^{2+} .

In addition to causing problems for plants, it is also possible for materials dissolved in the water to cause problems to the irrigation system. For example, iron and carbonates can deposit in the pipes causing blockages in pipes and filters.

13.2.1.1 Rain water

Since the advent of agriculture, rain has always played a role in crop production. This source of water has the advantage that it is generally very clean (with low EC) but has the disadvantage that its availability is generally sporadic. Many soilless production systems include full protection of the plants from precipitation because of the uncertainties and potential production problems associated with rain events imposing uncertainty on the nutritional status of the root zone. Thus, the use of rain water in soilless production involves capture of precipitation, storage in reservoirs, and pumps (along with filters) to put the water under pressure for use in the nursery.

In general, rain water has very low EC levels relative to the needs and limitations discussed above. Sometimes Na^+ concentration in rainfall is increased by rain and wind near oceans. Rain water can be collected from the roofs of the greenhouses. In the Netherlands, where rain throughout the year is plentiful, growers are required to have a storage capacity of at least 500 m^3 of rain water per ha of production facility. Rainfall averages may range from 50 to 90 mm/month so that with a

storage capacity of 500 m³/ha can provide for about 60% of the irrigation water needs (Van Os and Stanghellini, 2002); if the storage capacity is increased to 1500 or 4000 m³/ha, then 75% or 95%, respectively, of the necessary water can be secured this way. This rain water is stored in basins or tanks depending on the desired capacity. Earthen basins are used for capacities of more than 1500 m³ while tanks (Fig. 13.1) are used for smaller volumes since this is less expensive and less production space is needed to house such tanks.

Collection of rain water is not used in some production areas for various reasons. If the quality of other water sources is high and it can be obtained at low cost, then it may not be economically feasible to make the investment in collection equipment and water storage, since the space lost to this equipment can remove a substantial amount of production space from the nursery. Also, in areas where low-cost greenhouse structures are in use, rain water collection may not be feasible if the greenhouses are not designed with this in mind (lacking gutters or drainage plumbing). Also, in many areas (e.g., Northern California, Mediterranean countries), precipitation patterns are more uneven, requiring that the collection reservoirs be substantially larger than those identified above for Northwest Europe.

In some regions, rain water can be stored underground in aquifers. This saves using expensive space, and water quality is maintained. In fact, many groundwater sources are such sources, although on a much larger scale.

One salient issue related to the use of collected rain water is that it is never adequate for 100% of the needs of most nurseries. Thus, irrigation systems must always include another water supply. This must be considered as part of the investment cost.



FIGURE 13.1

Rain water storage in metal tanks.

13.2.1.2 *Municipal tap water*

Tap water may be unsuitable for use in closed soilless systems if it has been treated for human consumption to improve its flavor and kill bacteria. If calcium and chlorine have been added during processing, then it is particularly important that chlorides be present at less than 1.5 mmol/L. In arid and semiarid regions of the world, chloride levels are typically higher than this so that growers have to manage this by using salinity-resistant cultivars and other means.

In many areas of the world, municipal drinking water is relatively cheap compared to creating such water quality and capacity on site. Where this is not the case, the greenhouse operation must use groundwater or surface water, perhaps in combination with stored rain water and captured irrigation run-off.

13.2.1.3 *Surface water*

Water from surface sources (creeks, rivers, and lakes) is sometimes available in large quantities at a low price. While such water can be of high quality if relatively remote from human impact, the quality can also be low, especially if the water has been inadequately treated after municipal use or due to other human environmental impacts (e.g., canal water). The water may be polluted by salts, agricultural chemicals, inadequately treated sewage, or storm water run-off from nearby urban areas.

Even when surface water is generally of good quality, it should be noted that water quality might not be uniform throughout the year, requiring frequent testing by the grower to assure its continued suitability for crop production. If such water quality deteriorates at particular times of the year, then the grower will need water treatment equipment to improve the water quality. Alternately, a second source of high-quality water can be used to blend with poorer quality water, as is done by many growers in the Netherlands with captured rain water. Reverse osmosis is a commonly used treatment approach; while this process is effective, it increases the net price of supply water dramatically.

Desalination of sea water has been tested in an attempt to develop water resources for greenhouse horticulture. The price of freshwater resulting from this is relatively high, without government subsidies. During the last decade, the technology of desalinization has been improved and large and more efficient desalinization plants have been constructed in arid areas, producing and supplying water at a relatively reasonable price of about \$0.50–0.60/m³. Thus, desalinated water became a major source of water for households and agriculture in Southern Spain, Cyprus, Saudi Arabia, the Gulf countries, and others. This new innovation is discussed in [Chapter 14](#), Growing Plants in Soilless Culture: Operational Conclusions.

13.2.1.4 *Groundwater*

In regions of the world with aquifers, groundwater can be an excellent source of water. Typically, such aquifers are large and well buffered so that the supply quality is relatively stable. This is important for growers since it means that the fertigation systems design does not need to change over time. Since variations in water quality can, nonetheless, occur, it is wise for the grower to have the source water tested every 6–12 months to monitor water quality. Many domestic groundwater sources have been unchanged for centuries but are now experiencing changes as downward leaching of water carries nutrients from agricultural operations into the aquifer (a process that can take decades).

Also, in many areas where intensive agriculture is practiced, groundwater depletion is a problem. Along coastal areas, where agricultural production is frequently intense due to the milder

climates, the sea water intrusion into the aquifer can result in groundwater too high in salt for use in plant production (e.g., EC 10–15 mS/cm). In the region of Spain around Almería, groundwater, which mainly originates from the nearby mountains, is often used (EC ranges between 0.4 and 3.5 mS/cm). Water is pumped from wells between 150 and 600 m deep and, consequently, the groundwater table is decreasing, raising environmental and political concerns (Heuvelink and da Costa, 2000). The same phenomenon has been observed in the Central Coast area of California, particularly around Watsonville and Salinas where salt water intrusion has forced some growers to find alternate sources of water or close down.

13.2.2 IRRIGATION APPROACHES

The equipment used to convert the raw supply water into water that is suitable for use in soilless cultivation should focus on improving water quality and to put the water under pressure so that it will flow through the plumbing. Sand filtration is frequently used to clean up water to make it suitable for use. In addition to filters, pumps are needed to create water pressure. All soilless production systems require water to be pressurized so as to make it feasible to have uniform irrigation. Water delivery systems typically involve the use of pipes that are of largest inner diameter near the pumps. The further the water in the pipe is from the source, the lower the pressure. This problem can partly be mitigated by decreasing the pipe cross section as the water travels through the pipe. But ultimately the irrigation water supply system has to be divided into circuits in such a way that the pressure in each circuit is adequate to assure that the plants are irrigated uniformly.

As described in Chapter 9, Irrigation in Soilless Production, irrigation can be from above the plant canopy, to the top of the root zone, or from below the root zone. Each of these irrigation approaches requires different technical equipment. A schematic overview of irrigation equipment in soilless production is illustrated in Fig. 13.2. Generally, in greenhouse soilless production, irrigation and fertilization methods are integrated.

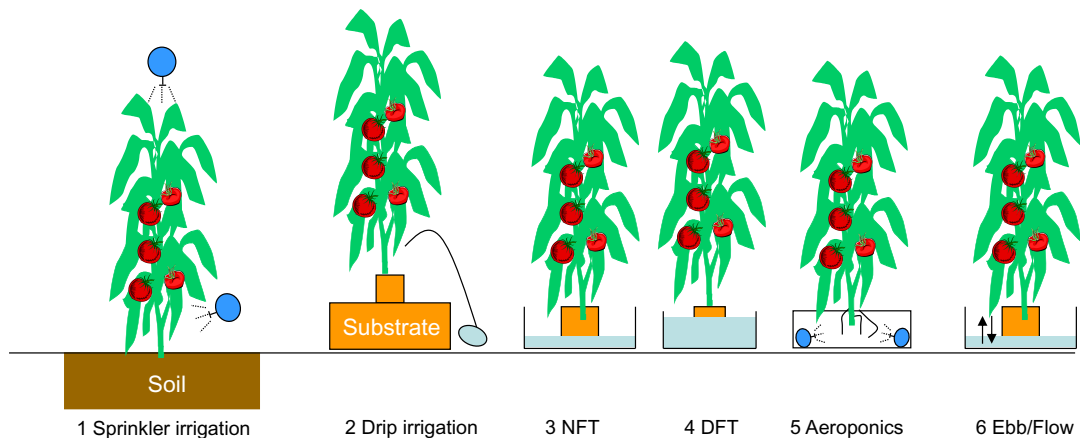


FIGURE 13.2

Various types of watering methods.

13.2.2.1 Sprinkler irrigation

Sprinkler irrigation systems apply irrigation water to the plants from above (Fig. 13.2). In some greenhouse and shade structures, plumbing is mounted overhead. Overhead installation does have the advantage that the plumbing is protected from mechanical damage by vehicles and persons. Installations can also have supply lines buried in the soil below the crop or mounted to the infrastructure. Movable sprinkler systems are also in use in various parts of the world. Crops for which it is undesirable to wet the foliage (e.g., some ornamental crops) can be irrigated with microsprinklers to the base of the plants. One positive feature of sprinkler irrigation from above is that it supplies water to most of the top of the root zone with relatively low investment and low maintenance costs (Heemskerk et al., 1997). Each sprinkler head type has a particular circular water distribution pattern; multiple sprinkler heads with short distances between emitters can be used to create a more uniform distribution pattern, but inherently sprinkler systems have uneven water distribution. For some crops, wetting of the foliage introduces a higher risk of plant disease development.

Microsprinkler irrigation systems, which deliver water to the base of the plants, overcome some of the drawbacks of overhead supply of water: there is no light interception due to overhead pipes and the crop becomes only partly wet. However, the plants themselves interfere with the even distribution of the water and prevent irrigation of large surface areas so that many more sprinkler heads are required, each delivering smaller amounts of water, to provide water to all plants.

Modern sprinkler installations are typically designed for the type of cropping system and the specific demands of the grower. In outdoor nursery production, sprinkler emitters may include rotating impact sprinklers and gear-driven ones which throw a stream of water 2–20 m, to ones with no moving parts which provide a circular spray pattern with a radius of 1–5 m. In greenhouse production, sprinkler systems are less common and emitters are frequently microsprinklers which deliver water at lower rates with greater precision to the base of the plants. Emitters have also been designed with other features that improve performance; pressure compensation assures that emitters deliver water only while the pressure is between specified minimum and maximum levels to avoid leaking of the emitters.

Due to the low expense of sprinkler systems relative to other systems, it is the most widely used system in outdoor container production. But even in these types of operations, growers typically use sprinkler irrigation only with tightly spaced plants. Plants which are spaced far apart are typically irrigated with drip or microsprinkler systems to reduce the amount of irrigation that misses the plants entirely. In greenhouse production, sprinkler irrigation is less commonly used due to the drawbacks mentioned above.

13.2.2.2 Drip irrigation

Drip irrigation is currently the most common irrigation approach in soilless culture in greenhouses. Drip systems are of two types: microtube systems or in-line systems. The former generally involves many emitters and supply tubing to allow an emitter to place an irrigation solution at a specific location, generally near the base of a plant. The latter consists of supply tubing with emitters attached directly to, or embedded in, the supply pipe.

Drip irrigation with microtubing (capillaries) and drip emitters (Fig. 13.2) allow the system to operate with relatively high pressure (at the pump or source), reduced to uniform lower pressure at the emitters. This approach allows the grower to maximize uniformity of delivery of irrigation

solution to every plant. This is important for crops with a low planting density or where each plant is of high value, making it financially feasible to optimize fertigation of each plant. Each emitter is typically connected either to a small stake (Chapter 9: Irrigation in Soilless Production, Fig. 9.4B) or a weight (Chapter 9: Irrigation in Soilless Production, Fig. 9.4A) to assure that they release the irrigation solution at the desired spot. Emitters or nozzles with various flow rates and working pressures are available and chosen according to the requirements of the crop and substrate. Emitters are typically matched to particular types of substrates so that emitters with low flow rates are typically used with substrates that have high infiltration and drainage rates. Pressure compensated emitters can be used to obtain even water distribution. Leakage-protected emitters assure that water is only supplied during irrigation events (Fig. 13.3) and prevent dripping at various points in the system; by keeping the irrigation system from draining between irrigation events, the system also starts up faster during an irrigation, leading to greater irrigation uniformity especially for short-duration irrigation events.



FIGURE 13.3

Leakage-protected emitters to water plants in soilless cultivation.

When applied to crops in beds where plants are growing at a specified spacing, in-line drip irrigation can be used by matching the planting density to the fixed in-line emitter spacing. In-line emitters can have the same properties as the standard emitters.

13.2.2.3 Nutrient film technique

Nutrient film technique (NFT) (Fig. 13.2) involves growing plants by maintaining a coating of nutrient solution around the roots, without the use of a substrate. When NFT first appeared, it seemed to be an ideal growing system because it seemed to offer optimal control over the watering of the roots without the expense of a substrate (Cooper, 1979; Graves, 1983). Optimization schemes were difficult to develop so that today NFT is used only for a few specific crops. Technically most crops could be grown in an NFT system (examples amongst others: Hortiplan, 2005; Morgan, 1985; Lataster et al., 1993; Ito et al., 1994) but widespread adoption has not occurred probably because such systems lack the ability to buffer even the slightest interruption in water and nutrient supply and there is a considerable risk of spreading root-borne diseases.

The system consists of a trough on a slope of 0.3%–2%; the roots of the plant lie inside the bottom of this trough. Nutrient solution is continually applied at the elevated end so that the solution flows down through the trough at exactly the rate required to keep the roots completely wet. At the bottom end of the trough, the solution is allowed to drain. The nutrient solution layer should be as thin as possible, almost as a film. The width of the trough varies according to the crop; troughs of 4–8 cm are sufficient for crops such as lettuce (*Lactuca sativa* spp.) and chrysanthemums (*Dendranthema grandiflora*), while for tomato (*Lycopersicum esculentum*) and sweet pepper (*Capsicum annuum*), a trough of 15 cm is needed. The length of the trough varies from 1 to 20 m. Depending on the crop and the sizes of the troughs, various types of materials have been used: polyethylene liner, polyvinyl chloride (PVC), polypropylene, and coated metal. Ideal water flow rates have been identified to be between 3 and 8 L/m²/h for crops such as chrysanthemums and lettuce (Benoit and Ceustermans, 1989b; Ruijs et al., 1990a,b; Benoit and Ceustermans, 1994). Fruiting vegetables benefit from a faster flow rate. The slowest flow rate that is adequate to keep the roots coated with water may not be adequate in an NFT system; if the flow rate is too low, the problem is not lack of water but lack of nutrients, especially for plants whose roots are downstream in the trough and are exposed to water from which many other plants have already extracted some nutrients. The last plants in the row get the least nutrients, especially potassium. Sometimes a distinction is made in flow rates needed for a young crop (2 L/m²/h) versus a mature crop (5 L/m²/h).

Plants destined for use in NFT systems are raised in small pots which are placed in the trough when a substantial root system has formed. In situations where the flow of water meanders in the trough, thus bypassing some plants, a lining of tissue in the bottom of the trough can be used to minimize this problem (Van Os and Kuiken, 1984). Shaped troughs (e.g., V shaped) also prevent this problem (Formflex, 2005).

13.2.2.4 Deep flow technique

Deep flow technique (DFT) refers to a method that also attempts to keep the roots continuously exposed to moving water and nutrients. While with NFT the water film is as thin as possible, with

DFT the continuously flowing nutrient solution has a depth of about 5–15 cm (Fig. 13.2). The large buffer of water and nutrients make it considerably simpler to control the nutrient solution. Only a relatively small fraction of the water and nutrients are actually taken up by the plants. The large volume of water also buffers the temperature, making the system practical in regions where nutrient solutions temperature fluctuations can be a problem (Ikeda, 1985; Both, 2005; Ito et al., 1994; Park et al., 2001). The width of the troughs in a DFT system is typically about 100–130 cm. Plants are secured in holes on polystyrene panels by means of a polyurethane foam; the panels float on the water or rest on the troughs sidewalls. The system is often installed at working height so that crops such as the lettuce or herbs can be easily planted and harvested.

13.2.2.5 Aeroponics and aerohydroponics

In aeroponic growing systems, the roots of plants are suspended in a volume where emitters continually or intermittently spray the roots with nutrient solution. The construction is similar to DFT (closed square box of about 1.2 m wide and 5–10 m long) but there is no water layer, rather there is a constant misting of the roots. A similar enclosed space is created in a triangle construct of polystyrene boards, as in a Japanese commercial design (Greenhouse Manager, 1988) and in a research arrangement (Leoni et al., 1994). In the triangle construct, the relative humidity is 100% and oxygen availability is increased around the roots (Fig. 13.4). However, care should be taken that the upper plants receive enough water. Often a thin layer of water is formed on the bottom, which acts as a buffer to the plants. Kratky (2005) showed the usefulness of the additional oxygen



FIGURE 13.4

Inside look to an aeroponic watering system.

available in these systems by measuring higher lettuce yields in comparison to a continuous water layer around the roots. The amount of water supplied to the plants is not mentioned in the literature; often a timer is used, but the release rate of the nozzle is not given. Ruijs et al. (1990b) suggest a flow rate of 2 L/m²/h for chrysanthemum crops.

NFT and aeroponic systems share the disadvantages due to lack of a buffer around the roots for water, nutrients, and heat. A hybrid system was devised by Soffer and Levinger (1980) by combining aeroponics with DFT, which they called “The Ein Gedi System” and coined the term *aerohydroponics* for it. Here, the roots of the plants extend through a panel, dangling in an airspace as with aeroponics, but into a substantial quantity of flowing nutrient solution, rather than just a thin film. One facet of this system is that the nutrient solution is continuously aerated and the rate at which the bulk flow of water and nutrients bathes the roots exceeds that of any other soilless production system.

With all variations of aeroponics, electricity is used to achieve all the bulk movement of water and this movement is critical to the survival of the plants. Evaluation of the commercial feasibility of such system requires balancing this energy cost with the enhancements in nutrient uptake and oxygen availability.

13.2.2.6 Ebb and flow/Ebb and flood

In “ebb and flow” or “ebb and flood” (EF) systems, container-grown plants are flood irrigated on a water-tight tray or floor. During irrigation, water flows onto the tray or floor so that the base of each pot is submerged as the tray or floor is flooded (Fig. 13.5). The required duration of the flooding depends on the hydraulic conductivity of the substrate in the containers. The duration



FIGURE 13.5

Ebb and flood irrigation of foliage container plants on a concrete floor.

should be adequate to allow the water within the root zone to be wicked to the top of the root zone: generally 10–30 minutes if the substrate has high unsaturated hydraulic conductivity (longer if not). Then, the tray or floor is drained which also allows the substrate to drain. Sloping surfaces and drainage grooves or channels are engineered into these systems to maximize the uniformity in water contents across a floor or tray between all pot/containers while facilitating rapid drainage. It is particularly important that the water be allowed to drain away completely so as to prevent root malaeation and diseases. If wet spots remain around the plant, algae growth, root diseases, and uneven watering will occur.

13.2.3 FERTIGATION HARDWARE

Water serves two important functions in plant production: it provides a vital resource for growth and also acts as a transport system for nutrients. Irrigation practices where both of these functions are actively combined in one system through the use of completely soluble fertilizers are called fertigation. In this section, we focus on the dynamic control of the supply of nutrients in fertigation systems.

A diluter/dispenser unit combines nutrients as concentrated chemical solutes from two to eight stock tanks (Fig. 13.6) with irrigation water. A system of pumps, valves, and irrigation capillaries delivers the resulting nutrient solution to each individual plant.

The amount of irrigation is often controlled by time-duration operation of the supply or by a sequence of a controlled number of equal volumetric additions. Valves connect the nutrient diluter/dispenser unit to the various irrigation sections of the greenhouse. Approximately 50 section valves can be allocated flexibly; each section is equipped with instruments to measure and log the amount of supply and drainage and daily averages of EC, pH, and water usage.

Supporting programs manage the rain water stock storage tanks or basins, the drain storage, and the storage for cleaned drain water. When additional water is needed in the system, selection criteria determine the source of the clean water. Since clean water is a highly valuable resource, water from different sources should not be mixed beforehand.

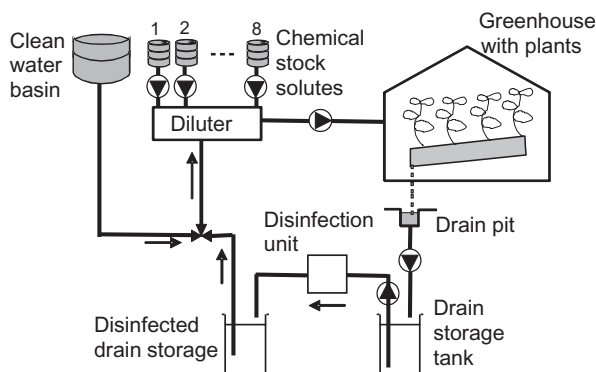


FIGURE 13.6

Schematic overview of the nutrient solutions pathway in a closed production system. The diluter/dispenser unit assures that concentrated fertilizers are diluted to a nutrient solution to be taken up by the plant.

In closed growing systems, the drain water is recycled. More data on this can be found in [Chapter 10](#), Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses. Precipitation of crystallized salts may occur when highly concentrated stock fertilizers are mixed with each other before dilution. The recycled drain water will be mixed with freshwater in a proportion selected to achieve a predefined EC value of the input water to be sent to the plants. An accurate control of the EC and pH of this input or supply water should be incorporated into the unit's controller, in combination with procedures for automatic cleansing of filters in the recirculation circuit. Comprehensive and detailed fertigation management should include these techniques to prevent failures due to unexpected EC and pH values or blockages in the system due to precipitation. EC and pH sensors are available for installation directly in control equipment. In horticultural practice, both types of sensors can be used for dynamic control and compared with the results of laboratory analyses to verify the calibration. The laboratory analysis is also used for determination of the composition of the nutrient solution.

Ion-selective electrode (ISE) sensors allow determination of specific ion dissolved in a solution. ISEs are used to measure the activity of cations and anions in the root environment. ISE sensors are available for most macronutrients, like K^+ , Ca^{2+} , NO_3^- , SO_4^{2-} , NH_4^+ , and for Na^+ and Cl^- . When used in manual test equipment, the resulting measurement provides powerful feedback to the grower as to how the nutrient solution is changing. However, this technology requires frequent calibration (daily) so that installation into dosing systems is not yet feasible.

13.2.3.1 Stock solutions: A and B solutions versus individual liquid fertilizers

A specific recipe of nutrients can be made for the particular crop in relation to the chemical composition of the source water. In 1970, when hydroponic production was introduced in The Netherlands, a system was developed by [Verwer \(1978\)](#) with one central tank to hold irrigation solution created by combining water with two stock solutions, called A and B solutions.

This type of system is still in widespread use today. The two stock solutions are prepared at concentrations 100 or 200 times higher than the average concentration of the solution to be supplied to the plants. Tank A contains all calcium compounds. In container B, all sulfates and phosphates are dissolved ([Fig. 13.7](#)). If the fertilizers in the two stock solutions were to be mixed at these high concentrations, precipitation would occur in the pipes, valves, and emitters, with disastrous results. To prevent precipitates, the solutes from the A and B tanks are diluted with freshwater in the mixing tank to achieve the right concentration and the appropriate EC level of the nutrient solution. An extra amount of an acid or base solute may be used to control pH of the batch of nutrient solution in the mixing tank.

The recipe of the A and B mixtures is selected according to the requirements of the crop to be fertigated and is based on a laboratory analysis performed twice per month. Fertilizers are usually added to the A and B tanks by hand.

Another approach is to deliver individual liquid fertilizers sourced from a supplier and stored in individual stock tanks. Depending on the supplying company, 6–10 liquid fertilizer concentrates are combined in a standard recipe for most commercial greenhouse crops ([Fig. 13.8](#)). The delivery truck is equipped with separated tanks for each concentrated fertilizer solution. A standardized discharge procedure ensures that each chemical fluid ends up in the right container, thus minimizing the chance of accidents. The need for additional manual labor is eliminated. A significant disadvantage of this method is the cost since a large quantity of water is transported.

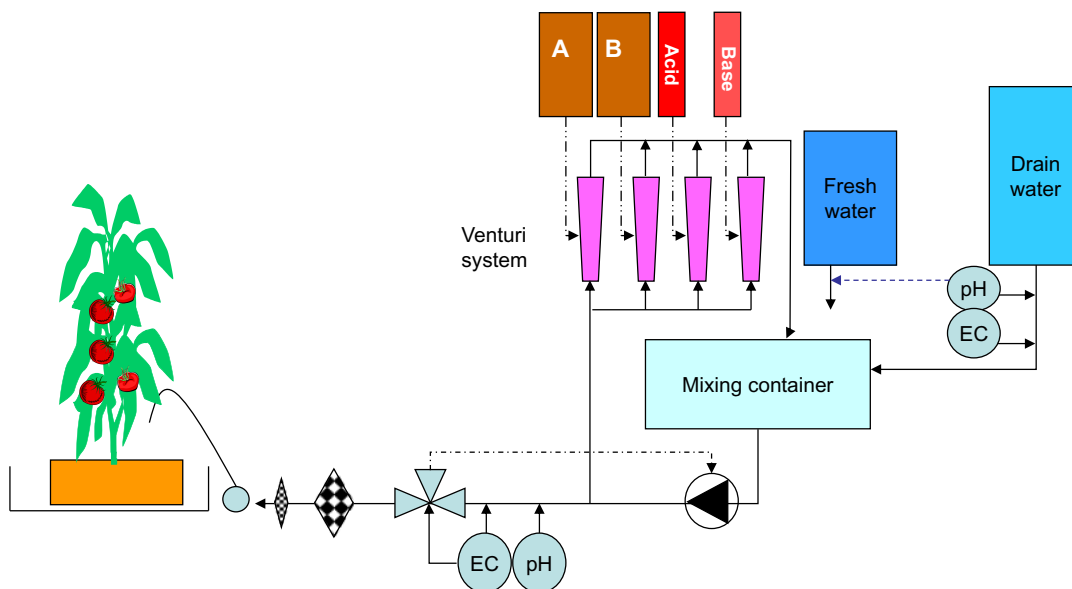


FIGURE 13.7

Example of a fertilizer diluter/dispenser unit where a Venturi fitting in the main supply pipe creates a pressure drop to transport concentrated fertilizers.

Courtesy Priva bv, De Lier, Netherlands.

This method is only economically feasible in regions where there are many greenhouse operations, such as in The Netherlands, the region around Almería, Spain, or the Brittany region in France.

Another option is to manually mix precisely weighed soluble fertilizers in the A and B stock tanks from bags of standard fixed mixtures. The amount of labor can be decreased by the use of large plastic bags of 1 or 2 m³, where the solid fertilizers are premixed by the supplier in an A bag and a B bag, in accordance to the latest laboratory analysis. The bags are delivered on standard pallets, placed on a platform above the A and B tanks by a fork lift truck and easily emptied through an opening at the bottom of the bag. The whole content of the big bags is used at once and mixed with exactly the prescribed amount of water to create the stock solutions.

New recipes of the A and B stock solution are prepared sequentially, but the release of the new recipe in both stock tanks is done at the same time to ensure that changes in the composition of the A and B solutions happen at the same time.

The recipe of the nutrient solution must be adapted to the needs of the crop during the development of the plant from a seedling stage to a reproductive stage. Cultivation starts with a species-related standard recipe. This recipe is then adjusted in accordance with advice from a specialized laboratory after a test performed every 2 weeks on a sample taken from the root environment. The advice takes into account the composition of the different stock nutrients, whether they are solid or liquid stock fertilizers and information about the status of the crop. The appropriate adjustments to the recipe are calculated on the basis of this information. Not all the

Individual nutrients

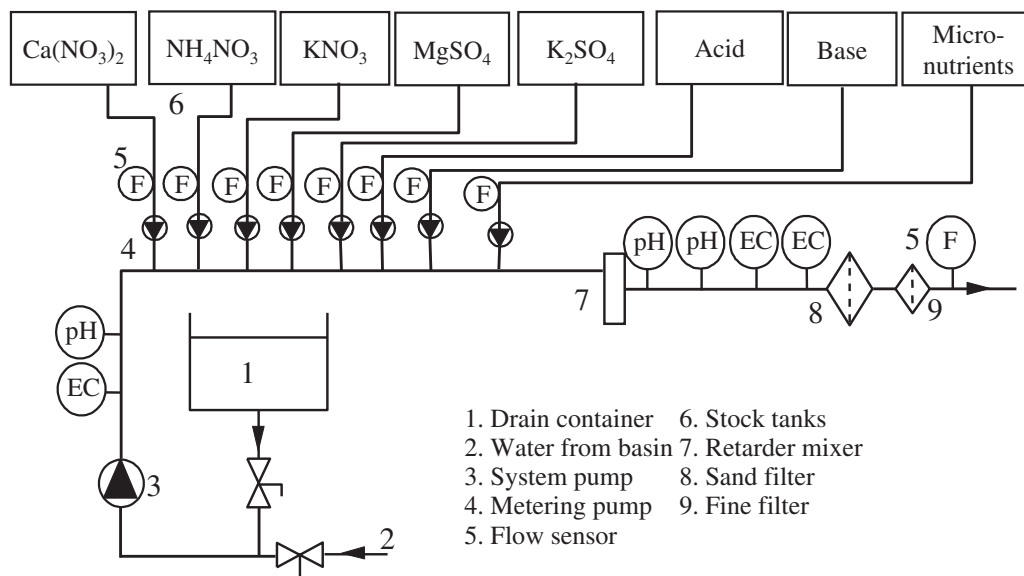


FIGURE 13.8

System with metering pumps to supply the stock liquids directly into the main supply line or into an open nonpressurized mixing tank.

possible outcomes of these calculations result in recipes that the diluter/dispenser unit is capable of producing. During recent years, fertilizers have evolved from simple single- or multielement fertilizers to more sophisticated combinations of elements, fine-tuned for a specific growth strategy. In liquid fertilizers, the option of adjusting pH levels has also been introduced (Fig. 13.8) by adding acid or basic fertilizer solutions. Manual calculation of the recipe has been replaced by computerized calculation, which has made it possible to take all kinds of constraints into account. Nowadays, these algorithms are also directly available to the grower as a spreadsheet or as a specialized calculation program related to a particular brand of fertilizer (Incrocci et al., 2012). Certain mixtures may lead to precipitation or to combinations of ions that cannot be produced from the available fertilizer stock solids or stock solutes. If the diluter/dispenser units should be forced to proceed with an “impossible recipe,” it can cause precipitations, blockages in the transport conduits, or even explosions (Schrevens and Cornel, 1990). In addition, optimization algorithms may on occasion lead to a more cost-effective choice of similar fertilizers.

13.2.3.2 Commercial injector systems for the dilution of concentrated fertilizers

13.2.3.2.1 The mixing tank system

The operating principles of mixing tank diluters are as follows: A mixing tank is connected to the various fertilizer stock tanks. Quantities of the highly concentrated fertilizer solutions are fed into

the mixing tank using nutrient injector system where the nutrients are diluted with freshwater to a specific concentration per recipe and finally pumped to the plants in the greenhouse.

There are several ways of feeding the highly concentrated stock mixtures into the mixing tank:

- Metering pumps supply the stock liquids into an open nonpressurized mixing tank or directly into the main supply line. One batch is prepared at a time in the mixing tank and then pumped to the greenhouse (Figs. 13.7 and 13.8). Examples of these systems are found in the early designs of computerized nutrient fertilizer systems (Bauerle et al., 1988; Papadopoulos and Liburdi, 1989). Presently commercial growers have a small mixing tank of about 2 m³/ha and additionally a day storage tank of 10–50 m³/ha. This allows the diluter unit to be relatively small and prepare, at its own speed, new solutions to fill the storage tank from where it is used for irrigation.
- A Venturi fitting inside the main supply line creates a pressure drop which forces the stock fertilizers (Fig. 13.7) into the mixing tank or into premixing chambers.
- Small container tubes (~5 L) are placed just above a mixing tank, one tube for each concentrated stock solution (Fig. 13.9). A valve at the bottom of the tube releases a precise quantity of stock solution into the mixing tank. A pressure sensor in the bottom of the tube measures the amount of stock solution released. The container tubes are filled by means of simple pumps or by gravity if the stock tanks are in an elevated position.



FIGURE 13.9

Small container tubes (~5 L) are placed just above a mixing tank to precisely release a quantity of stock solution into the mixing tank.

Although the three methods exist, commercial practice mostly makes use of the first one with $100 \times$ concentrated fertilizers in A and B containers with an additional day storage tank.

The EC of the supply solution is controlled by altering the extent of the dilution of nutrients and water. The simplest systems are laid out to mix a small number of stock solutions, such as systems where four nutrient stock fertilizers are used with an extra acid or base solution for pH correction. EC- and pH-sensors are installed in duplicate to ensure safe operation. A computer system controls the mixing process, the alarms, and operator communication.

Mixing tank dispensers are used in systems with overhead sprinklers, drip irrigation systems, or in ebb-flow systems. They are configurable for both in-line as well as in batched by-pass mode. The systems are available for various output capacities in the range of $1 \text{ m}^3/\text{h}$ to more than $40 \text{ m}^3/\text{h}$. A small number (four to six) of different nutrient recipes are possible. The EC set value can be changed according to changes in global radiation but is hardly useful in substrate systems. The replacement of the old solution with a higher EC by a solution with a lower EC (if more radiation) takes too much time. Steering EC is mostly done preferred vegetative or more generative growth, while daily influences of radiation are compensated with less frequent irrigation turns.

13.2.3.2.2 Direct injection system

The injection of concentrated fertilizers by means of a pressure drop in the main supply line is a modern and attractive alternative to the use of metering pumps. The pressure drop is created across a Venturi (Fig. 13.7). This pressure drop is the driving force to suck in the concentrated fertilizer fluids. It is a compact means of fertilizer transport, using fewer devices and moving parts that require maintenance than a system with metering pumps. Direct injection diluter/dispenser units are available in several sizes and capacities.

Simple direct injection systems use a mixing tank to mix and dilute the stock fertilizers from the A and B tanks by using separate premixing chambers into which the highly concentrated solutions of stock fertilizers are combined with supply water. This solution is then injected into the main supply line, where it is diluted to the concentration that is needed for the plants.

Systems that blend and dilute highly concentrated stock solutions of individual liquid fertilizers are more complex. The dispenser unit uses a fertilizer channel for each stock tank. A Venturi pressure drop in the main supply line is again the driving force behind the movement of the concentrated fertilizer fluids. The channel is closed off with a small ceramic insert with an orifice. The orifice ensures a fixed and limited flow rate. Ceramic inserts with different orifice sizes are used for the different channels. A flow meter and a fast-switching valve control precisely the accumulated flow over time. Four channels are connected to the mixing chamber. The amount of fertilizer transported into the mixing chamber is determined by pulse activating the valves. A static retarder in the main supply line mixes the highly concentrated fertilizers with the supply water (Fig. 13.10) and prevents the contact of hazardous combinations of highly concentrated stock fertilizers with residues of nutrient ions in the supply water (recirculated in case of closed systems).

The process of mixing and diluting the various individual fertilizers is controlled by feeding back the measured EC value of the supply water into a computer controller. The amount of each individual fertilizer is calculated and corrected by the controller as a function of the measured water supply flow rate and the volumetric relation between the various fertilizers, as prescribed in the recipe selected by the grower. The intended pH and EC will be realized in the solution when the



FIGURE 13.10

More sophisticated direct injection system for single-element liquid fertilizers.

Courtesy Priva bv, De Lier, Netherlands.

mixing algorithm is correctly applied. However, other factors, such as small changes in the composition of the clean input water of the system or the use of recirculated drainage water, introduce the need for fine-tuning the pH and EC value. This is carried out automatically by adjusting the output of the appropriate individual fertilizer channel in relation to its recipe value. An excess of a particular nutrient ion can be corrected in the next fertilizer supply cycle.

The concentration of the individual stock solutions is used as an input parameter in the calculation of the fertilizer recipes. The system is thus easily adjusted to accommodate all available concentrations of stock solution, such as standard industrial solutions or special stock solutions prepared for horticulture. Durable materials and a small number of moving parts ensure intrinsic reliability.

In general, eight liquid fertilizer fluids are sufficient to realize every possible recipe, including one channel for micronutrients. Injection dispensers are available at a capacity in the range of $100 \text{ m}^3/\text{h}$.

13.2.3.2.3 Safety and protection

Mixing highly concentrated ionic nutrient liquids can be hazardous. Extra precautions must be built into the system to protect the grower and the staff from hazardous situations. For this purpose, extra sensors should check the process of dilution, mixing, and dispensing for potential hazards, such as

- Excessive or insufficient flow in the individual fertilizer channels,
- Liquid fertilizer leakages,

- Intake of air into the fertilizer mixing channels,
- Unexpected drop in capacity, and
- Jamming of the channel.

General checks on the following should also be made:

- Sudden or large deviations from the expected EC and pH values,
- Leakage from the stock tanks (also when the equipment is switched off), and
- The stock tanks should be stored in a containment room with the capacity to house all the desired stock tanks.

13.3 PRODUCTION SYSTEMS

Since the advent of the use of soilless culture, various types of production systems have been developed to produce specific crops. In general, crops grown this way are “high-value” crops that have a high rate of economic return, thus ensuring that the deployed input-intensive system is profitable for the grower. Initial phases of this industry involved production in the ground in highly amended soils with above-ground soilless systems being a more recent development. In soilless production today, some systems sit directly on the ground, while others are raised above ground, with no connection between plants and soil.

For each of these types of production systems, we will discuss the advantages and disadvantages, and the suitability for specific groups of crops (crops in rows, crops in beds). Substrates will not be discussed in this section, although a distinction will be made between solid substrates (such as stone wool or polyurethane slabs), granulated substrates (such as perlite, pumice, coir, or peat), and no substrate (as in NFT and aeroponics). Most systems can be either open [all leachate flows into the subsoil without recirculation or is being used in other (field) crops] or closed (leachate is reused).

13.3.1 SYSTEMS ON THE GROUND

13.3.1.1 Ground preparation

Ground preparation is extremely important when installing a production system that will be placed directly on the ground, so as to minimize the incidence of pathogens and weeds. Steam sterilization, a chemical treatment or a solarization treatment, can be used to ensure a disease-free start and to prevent the germination of weeds. Treatment of the top 10 cm of soil is generally sufficient. Production systems which sit directly on the ground also require that the ground be prepared to accommodate the hydraulic needs of the system so that the drainage water from the production system is managed correctly. Wet and dry areas will develop if the root zone is not perfectly horizontally or on an even incline. Consequently, water availability for the plants will not be uniform resulting in oxygen deficiency in some pockets, leading to root rot. Deviations from this may also lead to variations in EC levels, leading to reduced productivity and quality.

Leveling is mostly done at a slope of 0.3%–0.5% along the length of the greenhouse span. The overall length is subdivided into sections of 20–25 m to avoid excessive differences in height

within the greenhouse. A molding machine can be used to adequately shape the subsoil after leveling. The substrate bed that will make up the root zone may be lower or higher than the aisle and small canals are dug for drain water collecting tubes. Shaping by hand is impractical for larger areas, and the accuracy of the work is likely to be inadequate. Compaction of the soil may also be needed in cases where the production system is heavy.

A plastic barrier between the production system and the ground is generally used to manage weeds and excess moisture. This can take the form of a plastic film or durable woven material with a tight mesh (antiroot foil). Selection of a liner made of a light-colored material provides the feature that it reflects light back up toward the plants. This is especially important in latitudes with prolonged dark periods and short daylight periods. A disadvantage of a light-colored (white) material is that it generally allows enough light to penetrate into and through the material to allow for some weed germination and UV degradation of the plastic. A black liner may be more advantageous than a white liner in areas where it is of greater value to have the ground below the liner attain slightly higher temperatures (cold greenhouses without possibilities to heat during the night) since it will store more heat during the day and release it during the night. A dark material is also generally more resistant to the degrading effects of UV radiation. In addition, a liner prevents the movement of plant diseases from the ground into the production system.

Woven material, rather than impervious film, should be used in cases where water needs to seep into the ground below the aisle and production system. In such cases, the ground preparation should be such as to avoid excessive compaction of the ground where the aisle will be located. Mostly the woven material gets less slippery when wet in contrast to the impervious film.

There is a significant difference in the type of layout of plants within ground-based systems. Some production systems consist of long rows of individual plants spaced at some particular spacing (i.e., “single rows”). Other systems consist of rows of plants with several plants side by side, resulting in (i.e., a “bed” of plants). In this chapter, we define “single row” to be a row of individual plants in a substrate and/or trough forming a root zone that is 4–25 cm wide. On the other hand, a “bed” is defined as consisting of several plants growing side by side in wider arrangement (>1 m) with the root zone completely filled with substrate.

13.3.1.2 Crops in single rows

Single row systems are in use for crops such as roses, cucumber, tomato, pepper, strawberry, and lettuce (Benoit and Ceustermans, 1989a). For crops in narrow rows, small quantities of substrates are normally used during propagation in the form of pressed peat pots, blocks, or plugs. Propagules are started in these, either from seed or cuttings, and later transferred to a production system. Many times, single row systems are placed so that two single rows are adjacent (forming pairs of rows); this is done when the width of a single row is such that harvesting and plant management can be accomplished from one side of the row. This effectively eliminates a large percentage of aisle space, resulting in greater overall production efficiency.

Generally, the soilless substrate that makes up the root zone of plants grown in single rows is enveloped in plastic film so as to contain the irrigation water. Preformed materials, such as stone wool slabs and coconut fibers, as well as some granulated substrates (perlite) are generally enclosed in bags which have adequate support to allow them to be placed directly on top of the liner or in the troughs without the need for a reinforcing container. Substrate products which are not in these

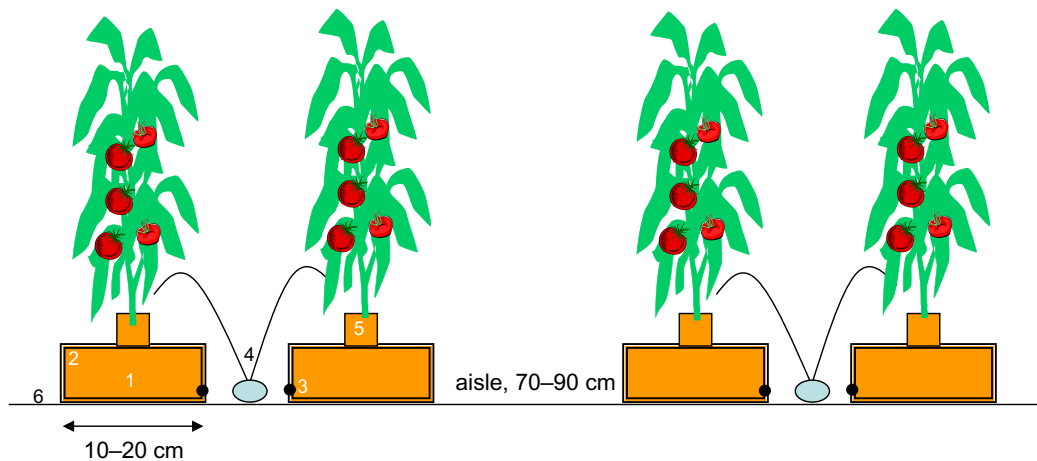
forms require some physical containment to hold the root zone in its desired form. This is either in the form of nursery containers, buckets, bags, or troughs.

Production systems which are placed directly on the ground are generally more difficult to configure as closed systems. In an open system, drain water seeps directly from the root zone through the liner into the ground below the crop (Fig. 13.11). This traditional approach has the benefit that the required investment in materials is low, but a lot of water and fertilizer is wasted while also polluting the environment (Van Os, 1994). Governmental regulations in most of Europe and North America now control the permitted amount of nitrate leaching (Nature Management and Fisheries, 1989; Planning and the Environment, 1989) so that open systems are being phased out in many such areas. This has accelerated the development toward closed systems.

A variety of materials have been used to create the troughs that collect the drainage water from slabs and bags (Lataster et al., 1993): plastic film, with or without integrated drainpipe, U-shaped polypropylene troughs, V-profiles and PVC profiles, and coated metal troughs (Fig. 13.12).

One of the least expensive closed systems consists of a plastic bag or film wrapped around a substrate slab with collection pockets on one side. Often a drainpipe is laid into the channel along the length of the row to keep the drainage channel open.

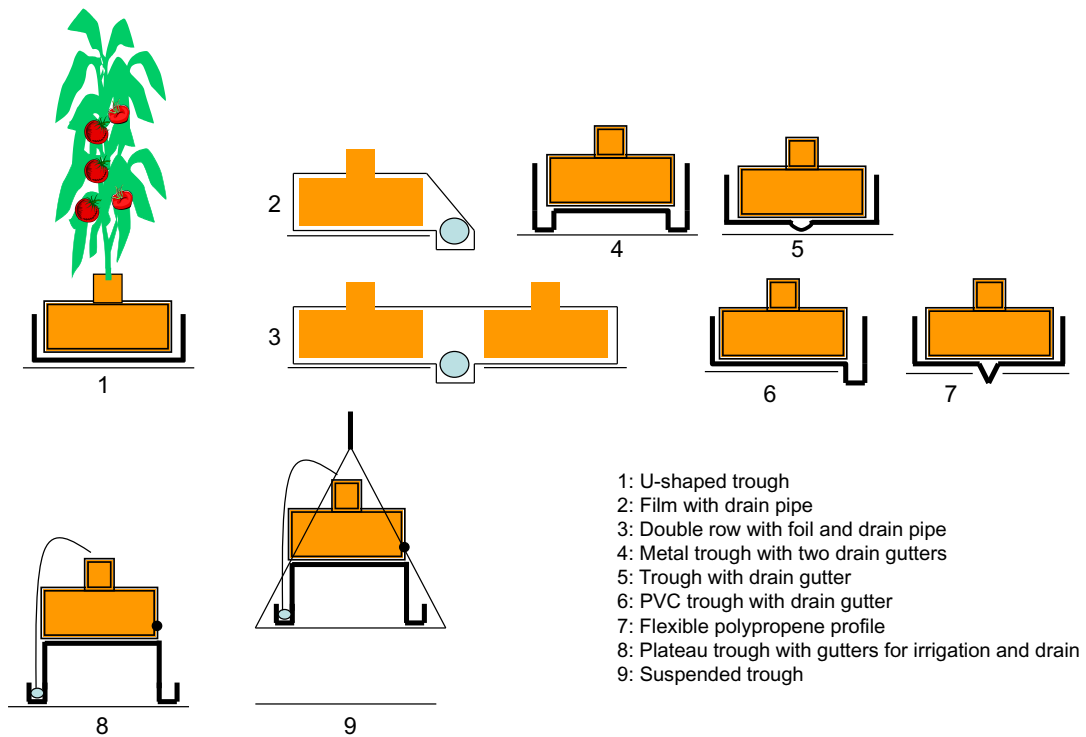
Plastic sheeting material consisting of polypropylene or polyethylene, 0.5–1.5 mm thick, delivered in rolls, can be used to create a system of drainage troughs. These are typically 15–25 cm wide and have vertical edges 5 cm high. The bottom can be flat or V shaped. The system requires careful leveling and compaction of the ground to prevent depressions where water can stagnate.



- 1: Substrate slab or granulated substrate
- 2: Foil around the substrate
- 3: Drain holes are made 1–2 cm above ground level and at the non-aisle side
- 4: Drip irrigation at the non-aisle side
- 5: Pots for plant raising, using the same substrate
- 6: Plastic liner covering the soil

FIGURE 13.11

Open systems for single row crops.

**FIGURE 13.12**

Troughs and profiles for closed single row systems.

Recirculation of the nutrient solution to the water treatment unit is easy because special fittings interconnect the troughs and the main pipeline.

Troughs have also been made of PVC, but such systems are much less flexible. This material is brought in its final shape from the factory and, consequently, cannot be delivered in roll form to the greenhouse for production of the trough on the spot. Lengths of 6–12 m have to be assembled at the greenhouse into the desired configuration. There is a risk of breakage. However, the material has the advantage that it can be easily sawed and glued to create the desired connections.

Metal troughs can also be created, either through shaping on-site off of large rolls (with a machine) or from preshaped trough segments (max 12 m, depending on truck size). While expense may be a deciding factor as to whether plastic or metal needs to be used, it is important to note that some subsoil (peat or clay soils) may not have adequate support for a plastic system so that the stronger metal systems are needed so as to prevent breakage and reduce sagging.

One consideration when selecting a material for drainage troughs is the need to clean and/or to disinfect the troughs between crops. If steam or hot water is applied (high-pressure application) to a trough made of thin plastic, then the trough is likely to deform. Solarization of these materials is possible, but temperatures must reach at least 70°C to eliminate pathogens, and this is

obviously dependent on the prevailing weather conditions. PVC is not very much in use for single row crops, but it can be cleaned rather easily because of its firm structure. It has the added disadvantage that burning during disposal releases chlorine into the atmosphere and is therefore prohibited in many countries, causing a disposal problem. Polyethylene and polypropylene can be burned completely without introducing too much pollution after greenhouse use. In the greenhouse, especially the firm material can be easily cleaned or disinfected with high-pressure equipment or chemicals.

A system consisting of both a coated metal trough and plastic film may be needed in situations where the structure of the soil of the ground is such that it results in sagging of a production system installed directly on the ground. Coated metal troughs are available in many widths and shapes (Formflex, 2005, Fig. 13.13) and can be placed directly on the ground or above the ground and even as movable systems (as described below). A metal trough system is typically more expensive than a plastic (polypropylene) trough, but it is less susceptible to settling and sinking of the ground, and may be more durable, lasting more than 10 years. Suspending the troughs improves the labor conditions during harvest tremendously but demands a strong greenhouse construction. The metal troughs can also be placed on supports independently of the greenhouse construction.

For economic reasons, it is not recommended to place the substrate directly on the ground without collecting the drain water. If the drain water is collected via troughs into a drain tank via gravity, the (expensive) fertilized water can be reused elsewhere (another greenhouse or open field crop).



FIGURE 13.13

Suspended troughs for growing crops such as tomato, cucumber, sweet pepper, and strawberry.

13.3.1.3 Vertical systems

Vertical systems are ones where some plants are on the ground while others are stacked in rows vertically. Such systems have been tested for crops with small plants like strawberry and lettuce with limited results (Fig. 13.14). The nutrient solution is applied at the top and drips through a bag filled with substrate (Massantini, 1977). Drain water is collected at the bottom. Similar systems are in use as aeroponics or as stacked containers (Liu et al., 2005) or containers fixed to a pillar (Vincenzoni, 1982). The main reason for using such an approach is to make optimal use of available floor space in an attempt to maximize the yield per m^2 . However, in a greenhouse, the available light intensity is distributed over more plants, making less light available to those at the bottom than those at the top. The consequences of this on year-round cropping should not be neglected: if harvesting of all plants begins at the moment the upper ones are ready, then the top layer crop is of good quality, but the yield from the lower layers decreases; if, on the other hand, replanting is only begun after the bottom plant is ready, each layer will yield a high-quality crop, but it will take longer before all plants are harvested and the system can be replanted.

A similar stacking approach to increase yield per m^2 greenhouse area is to use A-frames as described below.

Vertical systems for production of Belgian endive (*Cichorium endivia* L. witloof, Fig. 13.15) are feasible despite this problem (Van Kruistum, 1997). Plants grow 6 months in the open field from seed to a mature root to be lifted and stored cool (-1°C). Each week a certain amount of

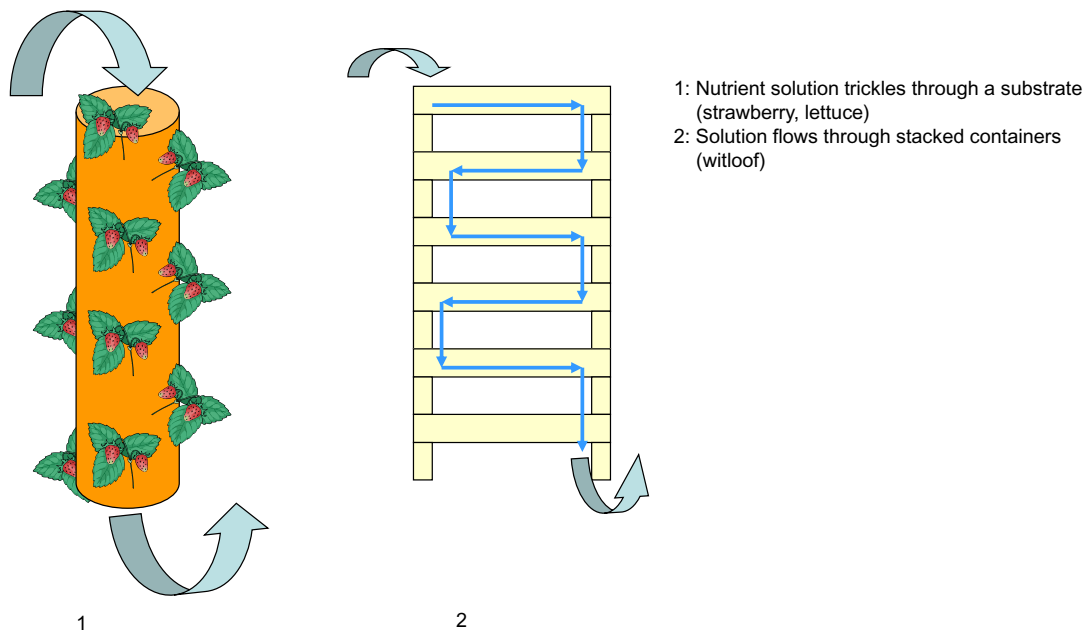


FIGURE 13.14

Vertical systems.

**FIGURE 13.15**

Belgian endive in a stacked container system (120×120 cm).

roots is set up in containers outside the growing area as close to each other as possible. These are later installed in 4–8 tiers of containers of 100×120 cm. Nutrient solution can flow through the containers in a zigzag pattern from top to bottom; the solution is then recycled. The plants are watered, kept at a temperature of approximately 22°C , and are protected from light. White heads of Belgian endive can be harvested after a forcing period of 3–4 weeks. A similar approach has been tried for asparagus (Van Os and Simonse, 1988).

Recent developments in vertical farming are resulting in plant factories with artificial light, PFALs (Kozai et al., 2015), allowing for the development of urban farming systems which use soil-less culture exclusively. Here, various types of leafy-green vegetables such as lettuce or herbs are grown in hydroponic systems in several layers above each other using only artificial lighting, with no sunlight entering the growing facility at all. The lamps are generally not the same as those used

in greenhouses; instead they are LED lamps that radiate less heat to the plant and can thus be placed in close proximity of the foliage. Since all lamps generate heat, air conditioning is needed to continuously cool the air. As a result, the greater use of energy makes this a rather expensive production method. However, the fact is that all variables can be optimized in ways, which is not possible in other production systems; the return is also greater (more biomass per unit area and time). As such, it is extremely important for the grower to calculate the economics to identify whether the extremely intensive farming operation will have a positive return on investment.

Here, it is important to have a market before the start and this market needs to have a higher market price compared to the traditional (bulk) produce. Investments levels are much higher (equipment, artificial light, and labor), which are generally not balanced with savings on transport.

Another key element in PFALs is the careful management of the air in which the plants are growing. This air is highly filtered to exclude all pest so that no pesticides need to be used. Furthermore, plant growth can be 24 hours per day while providing the maximum CO₂ concentration that the plants can use. Humidity is a key controlled variable in PFALs and this connect back to operation of the hydroponic system. The bubbling of air in the nutrient solution to sustain high levels of dissolved oxygen also results in humidity that must be removed. As such, the air conditioning system must be designed to simultaneously handle the humidity load as well as the temperature load. An important feature is that condensate water is available for reuse.

13.3.1.4 Crops in beds

Several flower crops, such as chrysanthemums, carnation (*Dianthus caryophyllus*), freesia (*Freesia* spp.), or alstroemeria (*Alstroemeria aurantiaca* L.), are grown in beds at a density of 20–60 plants/m². In this case, row systems as described above with slabs or substrates in bags are not used, because the return on investment is not adequate to support such a system for these crops (Fig. 13.16).

13.3.1.4.1 Polypropylene liner and substrate

The most widely used system for crops grown in beds consists of a 110–140 cm wide bed, with a 1–1.5 mm thick polypropylene liner between the subsoil and a layer of 5–10 cm of granulated substrate. Sidewalls are needed when the system is above the subsoil. If the aisles are raised, they can serve as sidewalls. A drainage pipe is laid at the lowest point of a bed. Depending on the type

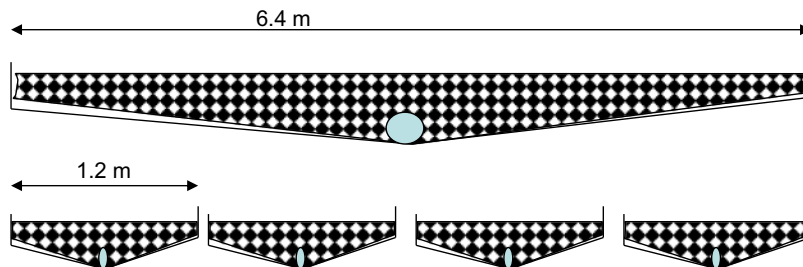


FIGURE 13.16

Systems for crops in beds.

of crop, either heating or cooling pipes can be embedded. This system is especially useful for crops such as bulbs, tubers, and rootstocks [lily (*Lilium longiflorum* Thunb.), freesia, and alstroemeria] which have to be removed entirely between cropping cycles. Still, there is no breakthrough for this type of crops. The area is limited in Europe and the variation between systems is big, it gives the impression one is still looking for the best system.

13.3.1.4.2 Polystyrene panels

Crops in liquid culture hydroponics can be grown with the plants mounted into a polystyrene panel using NFT, DFT, or aeroponic system (Ruijs et al., 1990a,b; Kratky, 2005; Ikeda, 1985; Ito et al., 1994; Both, 2005; Both et al., 1999). Young plants rooted in a synthetic substrate (mineral wool, urethane foam, etc.) are secured in polystyrene panels which have been customized specifically for use in plant production.

In an aeroponic system, structural panels form the top of a chamber where misters are spraying nutrient solution on the roots at a frequency that keeps the roots wet at all times. In this application, the polystyrene sheet must be of adequate structure to support the fully grown plants. As such, this is generally not used with large plants such as greenhouse tomatoes.

In DFT systems, the polystyrene panels, with plants installed as above, can float on top of the water so that the roots of the plants penetrated into the nutrient solution. A polystyrene material of adequate thickness is used to assure that leaves of fully grown plants do not touch the nutrient solution. The alternative approach in many DFT systems is to mount the plants at a fixed distance above the water level; in such installations, the level of the nutrient solution must be controlled carefully to assure that the roots are always wet. This application is generally in greenhouses and can also be found in plant factories.

Fully mechanized liquid culture hydroponic systems allow for a high level of automation for such systems, motivated by the desire to minimize labor costs. DFT systems consisting of large ponds constructed in greenhouses, filled with nutrient solution to a depth of >15 cm, in which the plants move through the greenhouse on the polystyrene panels floating on the nutrient solution (Fig. 13.17). The floats with young plants are placed into the nutrient solution on one side of the production area, while pushing the existing floats toward the opposite side, where panels with harvestable plants are removed by employees who pack them for storage or shipping to markets. Generally, the pond is completely covered with panels so that the number of panels removed during harvesting is equal to the number of panels inserted at the opposite end. The complete cover of the panels minimizes algae growth. The rate at which the panels move is dictated by the harvest rate.

Less expensive systems are also possible: a box with a lid. The cover has a size of about 50 × 100 cm, there is no control of the composition of the nutrient solution, just topping up if the water level decreases too much.

A recent development was a DFT system for the growth of leek (*Allium porrum*). Here, the challenge was to create a white shaft of minimal 14 cm (market demand) while growing out of the soil. Finally, a DFT system with floating panels came at which per plant hole a tube was placed in which the leek plant could grow (Fig. 13.18). The result was a faster growth, a sand-free high quality but also higher investments and running costs (Van Os, 2017).



FIGURE 13.17

Deep flow pond; lettuce growing on polystyrene panels in pool of nutrient solution with air bubbling for aeration.



FIGURE 13.18

Leek growing in hydroponics.

13.3.1.4.3 Concrete floor

One variant of the wide bed approach incorporates a concrete floor (Fig. 13.5) with EF irrigation. This type of system is used mainly for labor-extensive pot and bedding plants, raising young plants for fruiting vegetables and rooting cuttings in crates (60 × 40 cm). The concrete floor may have embedded heating tubes. The floor is divided into watering sections; irrigation water enters the section and through gutters which are also used for drainage. The floor has a slope of 0.2%–0.4% to facilitate drainage. If sections are too wide or if the slope is too steep, the difference in water height becomes too great to realize a uniform water level for all containers on the floor. The depth of water during irrigation can be as much as 5 cm depending on the container size. Containers placed on an EF floor must have features to facilitate drainage. This type of growing system is especially suited to highly mechanized methods where containers are placed or collected with specialized machines (automaton/robotics). The expected life-time of a concrete floor is the same as for a greenhouse (around 15 years). It should be considered that removing the concrete floor is just as costly as its original construction, and it is therefore hard to conclude whether the use of such a floor is economically feasible. It is also relevant that the frequent irrigation with nutrient solution will result in algal growth. Furthermore, employees walking on the floor during placement of the plants and removal, may spread pathogens into the irrigation systems. Special care for hygiene is needed: such as boots only for the greenhouse, cleaning wheels of trolleys moving from one to another compartment.

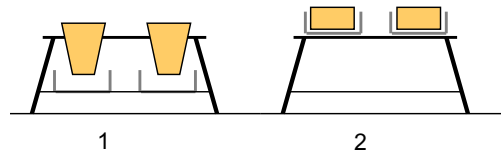
A floor should be designed with flexibility in mind. Birch (1979) designed a concrete floor system with a 10 cm trough-profile for year-round lettuce production. Transport equipment was incorporated into the design. Unfortunately, such a system excludes the possibility of changing to any other crop, forcing the grower to achieve the payback on the investment without any opportunities for changing the cropping system over the years.

13.3.2 ABOVE-GROUND PRODUCTION SYSTEMS

It has also been found that elevating the production system above the ground can have significant benefits due to better air circulation around the plants, reducing the incidence of plant diseases and improving the labor efficiency. Some plants also benefit from improved control over the temperature of the root zone. Another reason for elevating crops above ground level is to improve working conditions in the greenhouse by allowing workers to reach the plants or fruits to be harvested more easily. Raised systems can be either fixed or movable and are used for crops such as potted plants, strawberry, lettuce, rose, tomato, pepper, cucumber, and gerbera (*Gerbera hybrida*). As with production systems which are placed on the ground, crops can be in single rows or in beds. Above-ground beds are typically called “benches” and represent the majority of container grown flowering potted plant production.

13.3.2.1 Single rows

Racks constructed of metal or wood are installed on stone blocks or risers made of metal or treated lumber, so as to form a single or double row of container-grown plants from which harvests are taken on an ongoing basis. The root zone can either be formed by a set of pots or by elongated pans or troughs. Examples of crops which have been grown in such systems are roses, gerbera, and



- 1: Fixed racks for containers (gerbera, strawberry)
 2: Troughs or long-shaped containers on racks

FIGURE 13.19

Systems above the ground.

strawberry (Fig. 13.19) and various other crops are being tested by growers. The racks are placed on the ground or secured partially in the ground depending on degree to which the ground is able to support the system without subsiding.

One example of this type of system consists of 15–25 cm diameter containers placed in a metal rack, where a trough is placed just below the containers to carry away drainage water (Meteor Systems, 2009). This trough can be made of polypropylene, PVC, or coated metal. The system can be tailor-made on site for the grower.

Another example uses long trays (100–133 cm long, 15–20 cm wide, and 7.5–10 cm high) to hold bags or slabs. Each tray is supported by a rack at the point where two trays meet (every 100–133 cm). More sophisticated systems consist of an aluminum frame into which containers are hung. Two frames are placed upon a rack, the height of which can be adjusted.

Outdoor crops can be grown in a one-row system, where the supporting frame is rigidly placed in the subsoil. Special accessories are also available to fit a rain shelter over the one-row system.

13.3.2.2 Suspended troughs

Some systems have also been developed where the entire production system is suspended from the greenhouse frame by steel cables (Fig. 13.13). In this situation, there is no need for racks to support the system from below. Clearly, such an installation is only feasible if the greenhouse is engineered to support this significant weight in addition to the greenhouse covering material. Cables attached to the structural supports of the greenhouse at 2.5–4 m intervals are installed so as to give the suspended rows a smooth slope of 0.2%–0.5%. Troughs of various cross-sectional shapes are available for such application. A commonly used design is called “the plateau trough” (10–30 cm wide) where slabs or bags lie on an almost flat plateau, with small channels (4.5 cm) on both sides (about 10 cm lower) for drainage water collection. The gap between the bottom of the bag and the water is especially useful for avoiding root growth in the drain water gutter.

13.3.2.3 A-frames

The use of an A-frames or arches to increase the efficiency of space utilization has been investigated. Morgan (1985) described the principle for a trough system with 12 tiers; Leoni et al. (1994) described the use of an A-frame for the growth of tomato and lettuce with aeroponic

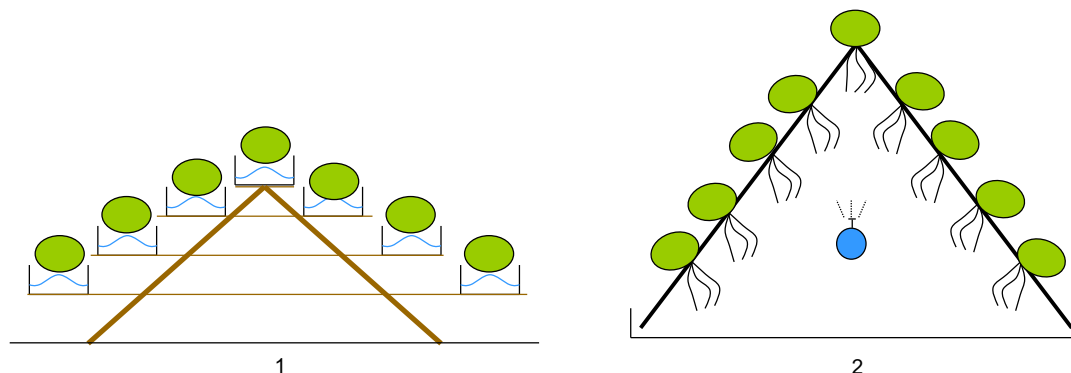


FIGURE 13.20

A-Frame systems: 1. Troughs with NFT channels and 2. Polystyrene panels with aeroponic misters.

watering (Fig. 13.20). Morgan's system was 240 cm wide at its base, 40 cm wide at the top, and 230 cm high. It accommodated 12 troughs with an NFT system for lettuce and was designed for a 3.2 m span of greenhouse space. This configuration doubled the cropping surface (up to 40 lettuce plants/m²) with 8–9 crops annually (Morgan, 1985). The system of Leoni et al. (1994) consisted of two polystyrene panels of 100 cm wide. The two panels formed a triangular A-frame shape with 120 cm at its base. The system included two A-frames in a 3.2 m span. In such A-frame systems, the lower tiers of plants receive less light than the higher ones resulting in lower productivity for the lower plants. In the format of the system by Leoni et al. (1994), there is also a risk that plants will grow in a crooked shape of the lettuce head. At demonstration sites frequently show such system, but they have not been adopted in commercial practice.

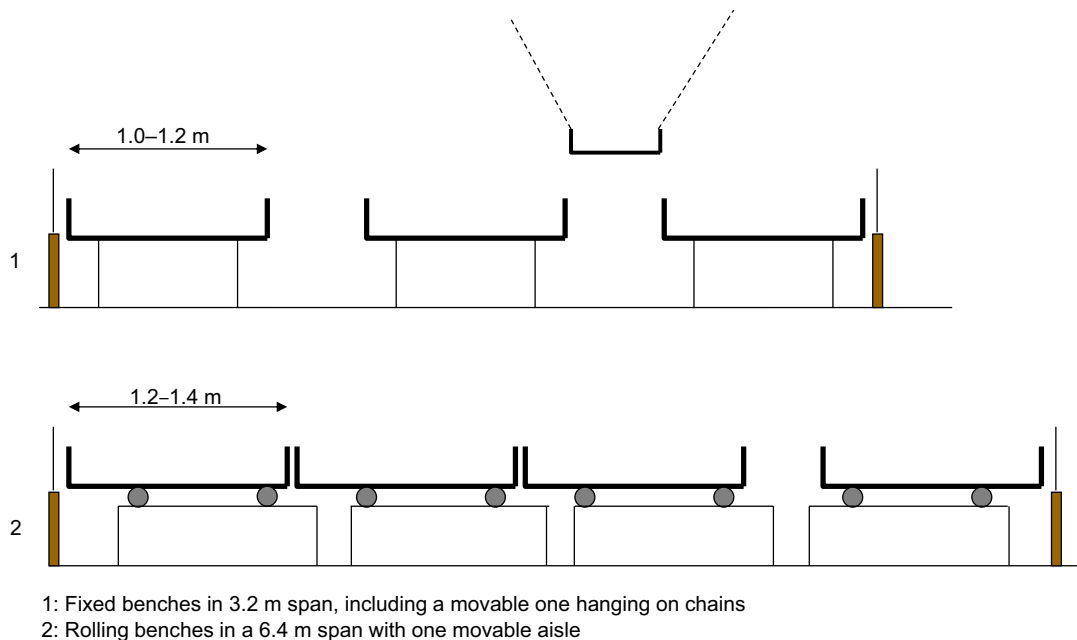
13.3.2.4 Benches

Benches are generally used to hold a large number of plants side by side so as to minimize the aisle space in the production area. The bench width is dictated by the distance that workers can reach into the crop to manipulate the plants. Benches can consist of fixed (Fig. 13.21), rolling benches or movable benches. Fixed benches cannot move and always have the same position in the greenhouse. They are mostly used for the cultivation of potted plants where aisles are needed for crop maintenance. Space utilization is relatively low (65%–75%).

Rolling benches are made of lighter weight reinforced materials, installed so that they sit on two long metal pipes which extend the full length of the bench. Cranks on the ends of the pipe allow the entire bench to be rolled side-to-side. The distance that a bench can roll is governed carefully with stops to prevent the bench from rolling off its supports. In such systems, only one aisle is needed per greenhouse span, since every bench can be rolled so that the utilization of space is increased to 85%–93%.

Another step in the development of rolling benches was the introduction of aluminum benches, with water-tight polyester profiled bottom for EF irrigation.

There has also been a trend toward wider benches, some as wide as 1.5 m. This, however, appears to be too wide for rolling benches with labor-intensive crops. For ergonomic reasons, a

**FIGURE 13.21**

Fixed and rolling benches.

bench width of less than 1.4 m is recommended, because the reach of an average Northern European worker is about 65–70 cm. If workers are of a shorter average height, then the width of the bench would need to be decreased.

13.3.2.4.1 Movable benches

Movable bench systems were developed in regions where labor and land costs are high, for use with plants that require a lot of manipulation (spacing). Special transport systems are designed to allow an entire tray of plants to be moved at one time by rolling it to a different location on this transport system (Fig. 13.22). This allows the plants to be easily moved in large batches to the worker, instead of having the workers spend time going to the plants. Such systems also allow for minimization (sometimes elimination) of aisle space.

Movable benches are custom made for the grower but mostly in standard sizes with length of 3, 4, or 6 m. The width of the bench varies between 120 and 150 cm. This results in a greenhouse space utilization of about 90%–95%. Many such movable benches are also equipped with a water-tight bottom (PVC or polyester) so that irrigation can take place via EF or capillary mats.

The concept of bringing the plants to the workers is also currently being tested for cut-flower production. For rose and gerbera, labor-intensive crops with a daily harvest from the same each plant, transporting the single rows of plants from the greenhouse to a separate harvesting area is being tested (Van Weel, 2005; Van Henten et al., 2005; Van Tuijl et al., 2005; Fig. 13.23). When

**FIGURE 13.22**

Movable benches.

**FIGURE 13.23**

Movable row system for cut-flower production: roses in stone wool on left and gerbera pots with organic substrate on the right. Entire rows of plants moving through greenhouse in a loop, toward central aisle in one greenhouse bay and in the opposite direction in adjacent bay. Harvesting and handling of plants happen only at central aisle so that no other aisles are needed.

set up in this way, all the plants in the row can be reached easily for maintenance or harvesting. Two problems have yet to be overcome with this type of system: One is the development of an optimal drip irrigation systems that can easily (and in an automated fashion) be disconnected and reconnected; this is a technical challenge that can probably be overcome. The other problem is related to the frequency with which such small rows of plants need to be moved and the rate at which they can be transported. The rows need to be transported at an interval of 4–8 hours. Rows in one span come to the main path, where maintenance or harvesting takes place; it then returns to its normal location via a separate aisle. Managing the cycles of harvests for each plant is a difficult one. It appeared that the economics are still the biggest challenge: there is little more production but much more running costs.

13.3.2.4.2 Individual plant

In intensive greenhouse production, plants in crops are being treated in an ever more individualized fashion. In the case of rolling benches, the smallest movable unit is the 1.5 m wide bench with its whole span length of 20–50 m. In case of transporting benches, the smallest movable unit is the bench of 1.5 m by 3–8 m. In case of the movable troughs, the smallest movable unit is one row of plants with a size of 0.8 by 3–8 m. Ultimately, it may happen that individual plants will be moved and treated individually.

The “walking plant system” offers the most individualized treatment (Formflex, 2005; Fig. 13.24). After potting up in pots, they are placed on a movable belt. A machine places the plants onto a movable belt in the bottom of a trough. This system is programmed by the grower to place the plants into a certain row at a particular spacing. Other machines are used to space the plants within the trough by using one automaton or machine to pull the plants out of the trough and place them onto a main transport belt; a second automation sets the plants into a trough at a specified spacing. Recently, equipment with cameras became available to judge for grading, scouting on diseases and spraying.

13.4 EXAMPLES OF SPECIFIC SOILLESS CROP PRODUCTION SYSTEMS

The list of soilless growing systems is continually expanding. Many different crops have been tried and a considerable number of them have been adopted by growers. In this section, a variety of such production systems will be described and some advantages and disadvantages will be discussed (Table 13.1).

13.4.1 FRUITING VEGETABLES

One of the largest segments of soilless production is the production of fruiting vegetables in greenhouses. These crops generally have long cropping period (4–12 months), 1–3 plantings per year, 2–5 plants/m², rows separated by 0.8–1.2 m aisles, and within the row a plant distance of about 0.3–0.6 m.

Along the length of the row, substrate slabs (stone wool, coir) or bags (perlite, pumice) are laid in different types of trough systems which are directly lying on the ground or supports



FIGURE 13.24

“Walking plant” system: automatic transport and spacing of individual potted plants.

(approx. 30–50 cm high) or suspended to the greenhouse construction (approx. 50–100 cm high). Plants are propagated in the same substrate to ensure the correct rate of drainage when planted on the slab or in the bag. Drip irrigation is used to fertigate the plants. Nearly all fruiting vegetable crops in the USA, Canada, and Europe are grown in this type of system. There are few other types of systems such as NFT (Cooper, 1979) or EF. This is mainly because of the potential for the dispersal of soil-borne pathogens via the recirculated water, the enormous quantity of circulating nutrient solution, and, in places, the relatively poor water quality which can lead to sodium accumulation.

Currently in the Netherlands, all new greenhouses are installed with a suspended trough system, developed for tomato, cucumber, and sweet pepper (Fig. 13.13). The big crop advantage is the uniform water content of the substrate and, after a few weeks, the uniform height at which the fruits hang to be harvested and, consequently, more efficient use of labor.

Table 13.1 Overview of Crop Groups

No	Group Name	Crops
1	Fruiting vegetables	Tomato (truss, cherry, beef, round), bell or sweet pepper, cucumber, melon, eggplant (aubergine), courgette, French beans, hot pepper, chili peppers
2	Single-harvest leaf vegetables	Butterhead lettuce, iceberg lettuce, kohlrabi, endive, Chinese cabbage, other salad greens
3	Single-harvest sown vegetables, drilling	Radish, spinach, (medicinal) herbs
4	Other specialty (fruit and vegetable) crops	Strawberry, witloof, asparagus
5	Cut flowers	Rose, carnation, gypsophila, bouvardia
6	Single-harvest cut flowers	Chrysanthemum, aster, lisianthus
7	Flowering bulb, tuber, and rootstocks	Freesia, amaryllis, alstroemeria, lily, tulip, iris, hyacinth
8	Other cut-flower crops	Anthurium, gerbera, cymbidium, cannabis
9	Flowering potted plants	Cyclamen, begonia, saintpaulia, pot chrysanthemum, fuchsia, kalanchoe
10	Foliage potted plants	Ficus, dracaena, monstera, schefflera, palms, ferns

In Mediterranean and African countries, bags filled with a local substrate (pouzzolane, sawdust, sand) are popular. Mostly 5–10 L bags are placed beside each other in polypropylene troughs lying on the ground. It is an easy and cheap system, mostly there is no recirculation but drain water is collected and used elsewhere.

13.4.2 SINGLE-HARVEST LEAF VEGETABLES

Many leafy vegetables are grown to size and then harvested when each plant has reached some specified size (weight). Row spacing is generally 10–30 cm depending on plant size, with similar distances between plants along the length of the row. For these mostly short-term (4–12 weeks) crops, it is generally not feasible to use drip irrigation nor substrate slabs or bags (too many plants per m²). Most systems used are based on troughs with NFT or beds with DFT.

In the United Kingdom in the 1970s, commercial hydroponic lettuces production (2 ha) was set up using 10 cm wide concrete troughs (Birch, 1979). In the Netherlands in the 1980s, a commercial system (Van Os and Kuiken, 1984) was introduced to facilitate the seasonal switch between lettuce and tomato to address the issue that economic feasibility is generally related to competition with field-grown vegetables where such systems are profitable during the winter, but not during the summer. Seven troughs of 25 cm with two rows of lettuce per trough in winter are combined with only four troughs with slabs for tomato in summer. However, Dutch growers who invested in soilless tomato systems and butterhead lettuce have disappeared from the greenhouse market. Many other systems were developed for research or on a semicommercial level (Ikeda, 1985; Morgan, 1985;

Greenhouse Manager, 1988; Ruijs et al., 1990a; Benoit and Ceustermans, 1994; Ito et al., 1994; Leoni et al., 1994).

On a large commercial scale, the movable gully system of Hortiplan, based on a Swedish design, (Hortiplan, 2005) appears to be economically feasible. However, the level of investment required is also high. It is a system in which troughs are spaced during cropping; planting and harvesting take place at the same side of the greenhouse. Another spacing system was developed in the United States by Both et al. (1999, 2005) where lettuce is produced on floating panels or floating rafts. The panels have a fixed planting density and various densities are used throughout the year. A similar large-scale system is in use in Canada for year round production of lettuce and herbs. In Australia and the United States, variants of trough systems are in use, which are more suitable for small-scale production. Special systems have been developed for special crops such as lamb's lettuce (Benoit and Ceustermans, 1989a), cabbage (Anderson, 2004), and medicinal herbs (Dorais et al., 2001). Other crops use systems developed for butterhead lettuce with some adaptations, but these systems are only economically viable under certain special circumstances.

The archway concept for lettuce production (Section 3.2.3) has progressed no further than use as a research tool. The practical problem of growing differences within one panel reduces the economic feasibility of the system. Lettuce is often used as a pilot crop in plant factories. In a plant factory, mostly of Japanese or Korean origin (Kozai et al., 2015), lettuce plants are placed in a huge growing chamber with artificial lighting without sunlight. Benches with a water buffer and without substrate are moved to a central place to be harvested.

13.4.3 SINGLE-HARVEST SOWN VEGETABLES

Crops such as radish or spinach are grown in soil in a system covering the entire area in rows 5–15 cm apart from each other and a spacing of 2–5 cm within the rows. For soilless systems, this means that the entire greenhouse area has to be covered with substrate, which is generally not economical. Alternately, a NFT or DFT system has to be used. In this case, a separate system must first be used to germinate the seeds, grow the seedlings, and then transplant them into the soilless system where they will grow to maturity. Both et al. (1996), Ikeda et al. (1995), and Maruo et al. (2001) describe such a system for spinach. In Japan, spinach is commercially grown in soilless systems, while in northwestern Europe, economic feasibility is too low because of competition from outside production. Ruijs et al. (1990a) also describe several systems for radish, but here again it appears that economic feasibility is poor.

13.4.4 OTHER SPECIALTY CROPS

13.4.4.1 Strawberry

Problems associated with soil-based production of strawberry—such as the low utilization of space, loss of yield through diseases, and limited potential for crop rotation—have prompted growers to seek soilless cultivation methods. In the 1980s, growers tried to increase the utilization of space by putting plants in large containers (25 cm diameter buckets) so that the ripening fruits were hanging along the container, resulting in less botrytis and easier harvesting. In a movable suspended trough

system, every second trough was hanging near the top of the greenhouse creating a path to pick the lower troughs, hanging at easy picking height (Van Os, 1986). This system proved too expensive and was replaced by a system where the containers were mounted higher in the greenhouse without any aisle space. Here, the fruit was hanging above the harvesters, making the harvesting of fruit uncomfortable. Many other systems have been tested for strawberry production such as a vertical wall with plant holes to collect solar heat for the night; archway concepts; DFT systems with floating panels; and vertical bags filled with substrate. Nowadays, many growers use a system of metal, plastic, or polystyrene troughs with a special profile into which bags or slabs can be laid. The strawberries to be harvested hang at eye level. A cheaper alternative is the use of containers filled with peat or coir, mostly placed in round pipes at a similar height. The 20 cm diameter white PVC pipe is a drain water collector and a container support at the same time. In tunnels and shelters, strawberries are also grown in a soilless system. These systems are less sophisticated but work well technically. Containers are often placed at a height of between 50 and 100 cm to make harvesting easier and prevent the hanging fruits coming into contact with the wet soil. Alternatively, metal troughs are placed at a height of 1.30 m, into which substrate bags are laid; the crop is watered via drip irrigation (Fig. 13.25).

13.4.4.1.1 Belgium endive (witloof)

In France, Belgium, and The Netherlands, Belgian endive (witloof) is an important hydroponically grown crop. The first stage of cultivation takes place in the open field where a root is grown from seed. Before winter, the roots are lifted and stored. Over the course of the winter, the roots are



FIGURE 13.25

Strawberry crop in suspended metal troughs.

taken in batches and forced in dark temperature-controlled chambers, where around eight containers are placed on top of each other. A 3–5 cm water layer flows from the upper to the lower container over the roots. After about 3 weeks, a yellow-white head can be cut from the roots (Fig. 13.15). This has become a year-round forcing method which has completely replaced soil forcing.

13.4.4.2 *Asparagus*

In a similar system to that used for witloof, asparagus plants can be forced for white spears (van Os and Simonse, 1988). Growers became aware that asparagus plants could also produce spears under temperature-controlled conditions during the winter, but no way has been found to exploit this commercially. The supply season was considerably extended by use of greenhouses, soil heating, and other planting methods (ridges of 1 m high).

13.4.5 CUT FLOWERS

One type of cut-flower production involves the use of the same plant for several months or years with daily harvests. The crop stays in substrate-based hydroponics at a spacing that is dictated by harvesting needs.

Crops such as rose are grown in single or double rows, while crops such as carnation, alstroemeria, and bouvardia were grown in wider beds of 1.30–1.45 m. The feasibility of producing cut flowers in soilless production is linked to the profitability of the crop. Carnation, for instance, has proved uneconomical in soilless production in northern Europe and North America and was abandoned while they are still grown in tuff beds in Israel and in above-ground beds in Colombia. Rose production in these regions also changed dramatically with nearly all rose production in these areas now being in soilless production. The change to soilless cultivation resulted in an increase of the production level and fewer diseased or dead plants. This was the main reason why the change to soilless systems for roses happened so rapidly. In other parts of the world, soil dispersion due to continuous application of monovalent ions (K^+ and NH_4^+) was another cause for shifting to soilless culture. Later, rose bushes were placed in single rows with a stone wool slab or a bag with granulated substrate. This made the system cheaper, as less substrate per square meter was needed. The introduction of the bent cane system in rose production became feasible when production was shifted to above-ground systems and this approach has become standard practice to control the quality of rose cut flowers. The introduction of the movable trough may be seen as a system for the future (Van Weel, 2005). The trough is at a slight incline to collect the drain water, and watering takes place by drip irrigation. Connecting the rigid water network to the movable drip irrigation system was a problem for a long time but has been solved (Van Zaal, 2005). During harvesting and crop maintenance, the troughs are disconnected from the watering system and pushed to the main path to perform the necessary tasks. After that, the trough moves sideways over a conveyor belt and into the following span. Each rotating unit consists of two spans 12–16 m wide and 50–100 m long. The advantages of the mobile trough system are the reduced labor requirements (the plants come to the worker) and the improvement of the quality of the labor (the roses to be harvested can be seen better; the worker does not need to reach to the middle of a bed).

Anthurium, gerbera, and cymbidium cut flowers can also be produced in a soilless production using containers, slabs, or bags. For cymbidium as cut flower and several other orchids, the

substrate is only needed for physical support. A large container filled with substrate (bark) is sufficient. Growers have successfully used polyurethane foam chunks as a substrate for cut-flower cymbidium production; the key is that the plant must be trained up on strings to prevent the flower spikes from toppling over.

Over the course of a number of years, anthurium grows into a large plant. A heavy substrate, such as clay or lava granules, is often chosen to fill large containers. Sometimes containers are placed in a bed of an EF system, otherwise drip irrigation is used. Less frequently, beds filled with granules are used; in this case, too much substrate is needed to fill the bed, and the plant needs support in height to avoid falling by imbalances.

Gerbera flower production is currently entirely done in soilless production, frequently in containers, but also in bags and slabs (Meteor Systems, 2017) which are placed in racks or in long-shaped containers filled with substrate. These rack systems are designed to have the plant at a height suitable for easy harvesting. As most flowers are pulled from the plant, it is important that the plant is fixed well in the substrate. Mobile bench systems, similar to those used for rose (Van Tuijl et al., 2005), using pots mounted on movable benches that have been developed specifically for gerbera (Fig. 13.23), allow harvesting to be done from a central aisle, as yet most production has such pots set in place with a drainage channel underneath, to allow capturing, processing, and reusing the leachate.

13.4.5.1 Single-harvest cut flowers

Crops such as chrysanthemum, lisianthus, and aster are grown from cuttings or plugs in beds 1.0–1.2 m wide and at a density of 30–70 plants/m² with all flowers programmed to be ready for harvest at the same time so that the entire bed can be harvested at once. This means that large areas of the greenhouse floor are covered with plants, so the introduction of a soilless system means that the whole greenhouse area must be covered with substrate. In the 1980s, NFT systems were introduced for research purposes (Van Os, 1980; Morgan, 1985). Flat-bottomed troughs 3–5 cm wider than the stem or the rooting medium block appear to work best. In that case, the flowing nutrient solution easily finds its way past the plants and blocks, even if the root mat grows to fill the entire trough at later stages. In the 1990s, several studies were undertaken to investigate if commercial chrysanthemum production is feasible in soilless culture (Ruijs et al., 1990b; Lataster et al., 1993). Production in soilless systems resulted in only a small increase in yield, while investment and running costs are much higher, resulting in reduced economic feasibility. Systems using beds filled with substrate, aeroponics, or movable benches appear more expensive than NFT, and even NFT is no more financially viable than soil growing (Ruijs et al., 1990b). In 2004, a new NFT system was developed using troughs: the plants are gradually spaced further and further apart during the long-day period; by the short-day period, they have reached their final spacing distance. Optimum light, temperature, and nutrient solutions are supplied during all the stages of growth (Van Henten et al. 2005; Van Os, 2006). Calculations made clear that up to 30% extra yield per m² is possible, which could pay the system costs. Still, there was no commercial breakthrough. From 2010, new attempts were undertaken with DFT and NFT trials after a literature reference (Vermeulen, 2009): what can we learn? Systems failed because of lack of knowledge on the needs of water, nutrients, and water contents of a substrate resulting in a rapid dispersal of *Pythium*. Consequently, a number of crops grew well but suddenly one went wrong. There is no solution yet.

13.4.5.2 Flowering bulbs and tubers

Flowering bulbs, tubers, and rootstocks, must be discussed in two separate groups: those which are grown in beds to produce flowers and those which are first grown in open fields to a marketable bulb size to be forced in soilless production during winter and spring. These production systems are also single-harvest systems with the planted plant material being replaced after harvest.

13.4.5.2.1 Freesia, amaryllis, alstroemeria, lily

Freesia and some varieties of alstroemeria need root cooling to obtain a perfect year-round crop. These crops need a granulated or loose substrate to facilitate the planting and digging of the bulbs and tubers. Rolling benches (four benches of 1.45 m/6.4 m span) appear to be the most economical soilless system to use (Hendrix and Ruijs, 1992), followed by 60 × 40 cm trays/boxes placed upon a plastic liner in which a drainpipe has been laid. In both systems, combined heating/cooling pipes are used, while watering takes place by drip irrigation. The increase in the utilization of space leads to an improvement in efficiency. In Blok and Kromwijk (2014), 10 substrates were compared for freesia. Growers choose for sand as the best one because of the easiness of planting tubers, low price, and cleaning afterward although yield was less as compared to coarse coir or coarse peat. Alstroemeria and lily are currently being tested in a commercial greenhouse in which 1.2 m beds are used, made of a plastic film and filled with a loose substrate.

13.4.5.2.2 Tulip, iris, hyacinth

In The Netherlands, around 2 billion tulip bulbs are produced per year in open fields. A small fraction of these bulbs are forced hydroponically for cut flowers. Here, a forcing method is employed that uses clear water, relying on the fact that the bulbs already contain the nutrients they need to grow. This fraction represents an area of around 500 ha. The main problem is keeping the bulbs in the correct position, with their tip pointing upwards, and preventing the stem from dropping during the growth. After many years of trial and error, two systems are currently in use: (1) containers with an aluminum device (90 × 120 cm, like those used in witloof forcing) with holes in which to place the bulbs and to allow them root into the water layer beneath the device; (2) bulbs are impaled on pins in the base of the case to keep them in the right position (Armstrong, 2002a). With either method, the vernalized bulbs are placed in the greenhouse at a temperature between 16°C and 21°C. Water is added via in-line drip irrigation. This approach saves on labor and avoids the expense of a substrate. Similar systems are being developed for forcing hyacinths (Armstrong, 2002b). Recent experiences show that the best results are achieved with the NFT and EF systems. Stagnant water decreases yield, probably because of the low concentration of dissolved oxygen. Disinfection against pathogens is required.

13.4.6 POTTED PLANTS

Flowering potted plants are relatively labor intensive compared with cut-flower production and grown on benches (orchids, cyclamen, begonia, saintpaulia, pot chrysanthemum, fuchsia, kalanchoe, and a host of others) (Fig. 13.22). Many are forced to flower on particular dates (e.g., holidays), requiring manipulation and grading. Green foliage plants typically have less labor input and are not scheduled as precisely. They are grown to reach a particular size, rather than a stage of

flowering. A wide range of production systems are in use, ranging from gravel beds and concrete floors to fixed or moving benches or trays. Potted plant lends itself to extensive automation since the plants are individually grown.

13.4.6.1 Flowering potted plants

Introducing movable benches is the first step to the automation of the potted plant production process. The increased utilization of space translates into an increase in yield, which is generally enough to achieve a return on the investment. High labor input and increasing labor costs are driving investment in automation. Movable trays and benches allow work to be performed in a central location: the plants come to the worker rather than the worker going to the plants. Labor conditions can be improved in the work area and the less-favorable working conditions in the greenhouse are avoided. Nearly all facets of potted plant production lend themselves to automation so that it can be expected that ultimately all tasks from the filling of pots to planting, grading, and packing of the plants can be automated. All this automation comes at a price; consequently, it is not suitable for all types of potted plants. Generally, flowering potted plants require the most labor so that it is the cultivation of these plants where the greatest savings can be derived from automated system.

Potted flowering orchids, such as phalaenopsis, are also grown with such a system and this type of automation is being used by several growers. This crop takes a relatively long time to produce and benefits from various movements of plants through various temperature regimes and pot spacing as the plants grow.

Another automation system that is seeing continued development is the process of color automated grading. Saintpaulia (African violet) is grown in production systems in Europe where the plants pass through an image analysis system which identifies how much color the plant is showing. This can be used to assess whether the plant is ready to be marketed and in conjunction with spacing machines can be used to sort plant onto several trays depending on how far from harvest each plant is. The various trays are transported to differently controlled greenhouse compartments to adjust the rate of development and force the plants to be ready on desired target dates. Similar image analysis systems are feasible for many other high-value potted flowering plant crops and have already been developed for orchids.

13.4.6.2 Foliage plants and outdoor container production

Large foliage plants (Ficus species, Monstera, Dracaena species, palm species, ferns) spend longer periods in the greenhouse without little need for maintenance except for watering and fertilizing. For this type of plants, movable systems are often too expensive and the plants are also bigger, which make transporting more difficult. Here, the process of collecting and placing of larger containers has been automated. If watering takes place by EF upon a concrete floor, special adaptations to the pot and floor need to be made.

The outdoor production of container plants is comparable with the foliage plants mentioned above. The growing period is relatively long, while labor input per plant is lower than with other potted plant production. Overhead sprinkler irrigation is generally used on gravel beds covered with a weed barrier (Bailey et al., 1999). Robotic automation is also being tested for outdoor nursery crops with considerable success. Such equipment includes automated pot filling, transplanting, placement of plants onto carts, and spacing of plants into bed using robotic fork-lifts.

13.5 DISCUSSION AND CONCLUSION

Irrigation and fertigation systems, along with all the technology related to them, are undergoing continual innovative transformations leading widespread use of automation and new developments in many areas. Irrigation approaches as covered in [Chapter 9](#), Irrigation in Soilless Production, in particular, have undergone considerable changes. Irrigation emitters improved with the introduction of pressure compensated emitters, both for drip and sprinkler irrigation. In areas where water is becoming more expensive and where pollution due to excess fertigation is being regulated, sprinkler irrigation is being replaced by drip irrigation with more precise emitters and control strategies. Improvements in filtration equipment are responsible for major advancements in soilless production technology.

There is a rising interest for NFT in crops such as lettuce in narrow troughs which are fixed (NGS system) or movable in space (Hortiplan, Viscon). There is not one system, each automation company designs its own system. Lettuce on DFT is in use to avoid excessive increase of the temperature of the recirculating nutrient solution and where control options are minimal. A few open field growers use the Dry Hydroponics system (<http://www.dryhydroponics.nl/>) at which also other crops are tried (bok choy, endive, basil; Blind, 2015; Van Os, 2017). EF is mainly in use for pot plants on benches or concrete floors and for raising young plants in stone wool blocks. Aeroponics has seen a resurgence with new approaches designed for commercial applications; Aessense Corporation has developed a fully automated system with nutrient injection from various stock solutions, with models for vertical stacking of plant production as well as for production of large plants. Good production is possible with all these methods, but each method requires a different strategy to achieve optimal growth conditions and economic viability.

In the 1980s, the first soilless systems were open or run-to-waste systems. The excess of nutrient solution flows into the soil or to surface or groundwater was polluting the environment. Research projects showed that savings in water consumption and a decrease in the eutrophication of surface water by greenhouse crop production could be accomplished through recirculation of irrigation water (Van den Boogaard et al., 2003; Lecomte et al., 2004). In many countries and states, this has led policy makers to force growers to switch to closed systems which involved reusing the nutrient solution. The basic logic behind this approach is well understood by growers: it saves water and fertilizers and it saves the environment. But the resulting systems are explicit management of the nutrient solution with sensing, blending, disinfection, etc. resulting in greater efforts and needed skills but finally yield and income for less water and fertilizer. Again, this is technically feasible but may not be universally feasible economically, unless all competing growers are forced to adopt the same expenses. With a substantial increase in government regulation in Europe and North America since 2000, many growers are now seeing the need to modify their systems into closed or semiclosed systems to avoid fines. At the same time, this area is developing rapidly as new components and technologies related to recirculation are coming on the market. Researchers are also now developing information that helps this field evolve and much of this information is covered in [Chapter 10](#), Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses. During the last 10 years,

it also became clear that within the leached water, there are remnants of pesticides used in the greenhouse. In The Netherlands, it led to much stricter legislation at which all nutrient discharges have to be purified (elimination of 95% of the concentration of plant protection products) by 2018. This legislation also leads to the question if discharge of the nutrient solution is required, should it be possible to grow without discharge, a zero-liquid emission? A joint research program of government and suppliers showed the options (Beerling et al., 2017; Van Ruijven et al., 2016; Van Os et al., 2018): zero emission is possible for sweet pepper without the loss of production and quality. It may also be that as regions legislate production practices as above, that plant factories may become the dominant production method.

Stanghellini et al. (2003) calculated the water-use efficiency of tomato grown in a closed climatized greenhouse to be 15–25 L of freshwater for 1 kg of tomato, as compared to 25–40 L for a soil-grown, nonclimatized greenhouse and 40–60 L for outdoor production. The variation between 15 and 25 L water per kg is accounted for by the difference between an open soilless system (25 L) and a closed system (15 L) in a climatized greenhouse. In a soilless system with zero-emission, tomato can be produced with 12–16 L/kg fresh product. Much lower, 4–8 L/kg tomato is possible in an airconditioned greenhouse, which is technically possible but not yet economical. This means not only that greenhouse production in dryer or hotter climates may be able to use less high-quality water, compared to outdoor production, but also that still more water (and expensive fertilizers) can be saved if a closed soilless system is used rather than an open system.

This raised the question as what the most extreme level of efficient water use might be and whether research and development can lead to this. With plant factory systems, there is the opportunity to capture all condensate which is all the water that the crop plants transpired. If this water can be continually reused and blended back into the hydroponic systems, it should then be possible to produce 1 kg of fresh produce with 1 kg (1 L) of water. In trials at UC Davis, we have already achieved this but generally do still discard water as it is yet too expensive to remove the risk of the unknown by sending samples off to analytical labs.

Finally, the water use efficiency, and also the fertilizer use, can still be increased. However, large investments might be required while the yield is not increasing very much. For the modern methods, the water price is still too low.

The commercial acceptance of soilless-grown crops in the 1980s and early 1990s raised expectation that many more crops would soon change over to soilless systems. Indeed, when one looks at all the systems developed for various crops, there are no technical limits to soilless cultivation which cannot be overcome: any crop can be grown perfectly well in soilless systems. However, to date, not all crops have been changed over to soilless systems apparently, in many cases, due to the large economic investment. The required investment and annual costs in particular may be too high. Insufficient increases in yield and quality are realized, while the price per unit of production area is declining. Those crops where the transition was made to where commercial soilless systems is now the norm achieved this because of two or more of the following benefits to the grower: an increase in production at the same plant density, an increase of the utilization of space, or a lower planting density. Upon transition from soil- to soilless production, crops such as tomato, cucumber, and sweet pepper that are characterized by a relatively low plant density (2–5 plants/m²) showed

an increase in yield. When grown in soil, these crops were subject to many soil-borne diseases which could not be eliminated. At the same time, they changed from a rather low-quality water supply to an excellent quality rain water supply. All of this has led to increased production. Almost all of these crops grown in the Netherlands now use a soilless system and are the major crops grown in soilless systems in other countries too.

On the other hand, leaf vegetables are grown in the ground at many plants/m² (about 10–25), completely covering the ground. Changing to a soilless system results in relatively little increase in yield compared to soil production. The utilization of space can only be increased in some special cases. In such systems, increases in productivity would have to be realized through more rapid accumulation of biomass, leading to shorter intervals between harvests. Consequently, soilless systems for these crops are feasible where this increase in production per unit greenhouse space per unit time is adequate to compensate for the large investment that is required. The costs in such systems may be due to the need for the entire greenhouse area to be covered with substrate or due to the need for an extensive and expensive NFT or DFT system. At the same time, competition from open-field production or from other countries is keeping the prices low, limiting the extent to which growers can recoup investment cost. Similar results can be seen in floriculture. Crops such as chrysanthemum, freesia, and alstroemeria have been widely tried in soilless systems. The technical system and the cultivation method are not the problem, but the fact remains that for a commercial grower, there is no reason to change if the crop has a high plant density and there is insufficient improvement in yield or in the utilization of greenhouse space. With rose and gerbera, on the other hand, the plant density is low and the utilization of space increases in a soilless system; consequently, it is financially beneficial for the grower to change to a soilless system. In addition, these crops suffer from soil-borne diseases, which can be avoided in soilless cultivation and, consequently, direct yield also increases.

Fertigation control involves precise control over the various fertilizer ions that are needed for plant nutrition. In the past, this has been done mostly by controlling the stock solutions and precise combination into irrigation solution. With the advent of recirculation, this has become more difficult since augmentation of nutrients to blended irrigation water requires knowledge of concentrations of the various ions. Currently, many growers collect run-off in large storage tanks to combine it with freshwater based on a preset EC value of drain water and freshwater in the diluter unit and so on. Liquid fertilizers are in use in concentrated soilless horticulture areas at which automated filling of fertilizer storage tanks is done. Direct injection is not very much in use. The use of A and B containers with an additional day storage tank gave large farmers more flexibility in dosing with a high frequency during day time. While ISE can theoretically be used to dynamically monitor recirculating water, this technology is not yet stable and thus not feasible. Further research and development are needed to improve this.

Mathematical models which describe plant nutrition-related phenomena are widely used in research (Heinen, 1997; Gieling et al., 1998a,b) and can also be adapted for use in fertigation management through implementation in computer-controlled systems. Such models can replace direct measurements with predictions based on models (LeBot et al., 1998). In some cases, this has been shown to be successful, but more research is needed. The wish to use online ion selective measurements is still there. Up to now, the sensors have not been adequately robust. On the other hand, the

automation in laboratories ensures an analysis within 24 hours after taking the sample with a relatively low price. Consequently, the sampling increased from once per 3 weeks via once per 2 weeks to once per week (Van Os, 2017) at which advice is given to optimize growth and a new nutrient solution is given for the next period.

Alternately, if ion-specific electrodes can be developed to be as stable as current in-line pH and EC sensors, then such model-based ion concentration predictions may not be needed. Ion-specific measuring principles promise much greater degree of accuracy in the supply of nutrients to the plants than is possible with existing methods. The ratio of the uptake by the plant of various ions and also the uptake-rate itself may change over time, due to variations in the plant's needs. Using ion-specific control allows faster compensation for these fluctuations by adjusting the ion content of the incoming water and opens the possibility for feedback on the actual uptake of these ions (Gieling, 2001; Gieling et al., 2005). One of the biggest advantages would be the fact that water could be recycled dynamically rather than first accumulating batches.

Application of this innovative control technology would ultimately include the following:

- ion-specific monitoring of the nutrient solution in a closed growing system;
- monitoring of plant-related variables like photosynthesis, leaf area index, water uptake, and leachate flow;
- control of fertigation based on feedback on the ion content of the drain water;
- prediction of the uptake of water and nutrients using a model-based soft sensor or observer;
- decision support systems for feed-forward control of the diurnal nutrient supply, using a plant-substrate model; and
- use of a plant-substrate model in optimal model-based control of water and nutrient supply.

The idea of controlling the supply of water and nutrient ions by means of feedback measurements is the subject of considerable research and development into new control technologies. Such controllers have been shown to operate consistent with expectations (van Straten et al., 2006). This and other research projects exemplify the role that model-based optimal control currently plays and will play in future issues related to systems engineering and automatic control of plants in protected cultivation (Van Henten, 1994; Tap et al., 1993; Van Henten and Bontsema, 1996; Van Henten et al., 1997).

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GROWING PLANTS IN SOILLESS CULTURE: OPERATIONAL CONCLUSIONS

14

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14.1 EVOLUTION OF SOILLESS PRODUCTION SYSTEMS

The previous chapters presented various specific facets of soilless plant production, representing the state of the art, and include some perspectives on the direction in which the field is moving. In some cases, gaps in knowledge were pointed out and opportunities for future research were suggested. Generally, the chapters focused on particular facets of a system whose full complexity was beyond the scope of each specific chapter. In some cases, lack of appreciation of the complexity imposed by the interaction of all the factors may lead to problems and failures for practitioners. The aim of this final chapter is to integrate various concepts from throughout the book and from other sources, to provide practitioners with practical operational tools, allowing them to optimize crop production. The focus is to develop a better understanding of the intricate processes taking place within the system along the root-zone–plant–atmosphere continuum, as affected by the interactions among the growing substrate, the liquid and gaseous phases held in its matrix, and its nutritional status. Aerial conditions, although exerting major effects on plant performance, are beyond the scope of this chapter.

One concept that was discussed in [Chapter 6](#), Nutrition of Substrate-Grown Plants, was the historical perspective on nutritional limitations to plant growth. The work by Mitscherlich suggested that plant productivity may be limited by various nutritional factors. This concept has been verified and is appropriate if all factors except one are nonlimiting. As such this concept is also true in a broader sense when considering all other factors that impact soilless crop production. However, when multiple factors are limiting, we must anticipate interacting effects which are more complex than simply attributing causality of suboptimal production to the most limiting factor. This is particularly important in practice because it is extremely rare that all production factors are simultaneously optimized.

A further consideration regarding this issue is the fact that the root zone is a dynamic microenvironment where many of the factors are in continual change, as described in [Chapters 2, 5, and 10](#). Many factors can change second-to-second or minute-by-minute, while some changes occur

over the course of a days or weeks. Management of such dynamic systems requires constant attention, decision support tools and the advent of computer technology coupled with sensor technology provides a situation where we can anticipate significant changes in the field as control technology improves.

14.1.1 MAJOR LIMITATION OF SOILLESS- VERSUS SOIL-GROWING PLANTS

In the first chapter, we described the historical reasons for the advent of soilless cultivation and its further expansion to the level it has reached to date and we briefly suggested reasons why its share in global food and plant production will continue to grow in the foreseeable future. It is particularly important to note the speed with which soilless production has changed over the past 60 years. Anyone considering soilless crop production should be aware that the field as a whole is dynamic, with horticultural practices changing continuously. One consequence of this is that in various chapters, methods and technologies are presented as being in widespread use or commonly accepted. Within this rapidly changing field, such methods may be quickly replaced by alternate methods as scientific understanding and economic feasibility change. Perhaps the most dramatic change is currently occurring in relation to management of recirculation systems for various soilless production systems, as described in [Chapter 10](#), Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses.

Various reasons persist for the continued growth of soilless production as part of agriculture. Some of these reasons can be inferred from the chapters dealing with the physical and chemical characteristics of substrates ([Chapter 3](#) and [4](#)), emphasizing their superiority over soil cultivation, ease of control of water, oxygen and nutrient availabilities, and the resulting improved crop performance. Another advantage is the relative freedom from soil-borne pathogens and improved possibility of disinfestation of the medium among growing cycles as described in [Chapter 5](#), Root Zone Microbiology of Soilless Cropping Systems. However, growing crops out-of-soil also impose some limitations which test the grower's skill to avoid losing the potential benefits offered by the advantages. The extent of these limitations and how to practically deal with them is discussed in this chapter.

14.1.2 THE EFFECTS OF RESTRICTED ROOT VOLUME ON CROP PERFORMANCE AND MANAGEMENT

One of the biggest contrasts between soilless and soil-based production is the spatial confinement of the roots into a specific, well-defined root zone. The smaller the root zone, the more intensive the production system needs to be managed within this volume.

[Sonneveld \(1981\)](#) reported that the volume of medium, available to a tomato (*Lycopersicon esculentum* L.) plant grown in a soil bed in a greenhouse, is approximately 200 L, while the corresponding volumes for production in substrates are typically an order of magnitude smaller. He calculated that the volume of water (and hence nutrients) in the root zone at any given moment to a tomato plant would be an order of magnitude greater in soil than in soilless production. Thus in soilless production involving such substrates as peat and stone wool, irrigation needs to be carried out much more frequently than in soil-based production if the nutrient and water storage aspects

were the same. In fact, with nutrient film technique (NFT) systems, where the water volume may well be an order of magnitude smaller than substrate-based soilless production, application of water needs to be continuous so as to avoid nutrient deficiencies and water stress.

Numerous experiments and simulation studies demonstrated that a continuous growth process of the root system is essential to ensure efficient water and nutrients uptake from soil (Williams and Yanai, 1996; Nye and Tinker, 1977; Sadana et al., 2005). Root growth is especially important in the case of low mobility nutrients such as P, Mn, Zn, and more so when their concentrations in the root-zone solution are low since in this case ion diffusion toward the depletion zone is slow. Plants growing in soil typically exhibit fine root growth so as to gain access to water and nutrients from less-explored regions of the root zone. In fact, active, vital growth by such plants depends on continued formation of new roots (Williams and Yanai, 1996; Nye and Tinker, 1977; Sadana et al., 2005) and root branching (Nibau et al., 2008). In frequently flushed soilless root zones, the near-absence of clear depletion zones somewhat diminishes the need by the plant for such active “foraging.” However, since the process of root growth is, by nature, a persistent one (Bengough et al., 2006), root systems of container-grown plants are usually very dense. Optimization of soilless production systems means that irrigation and fertilization must be carried out with precise timing and location and that such systems suffer from a low tolerance for error.

Under normal horticultural conditions of sufficient supply of water and nutrients, substrate volume has little or no effect on root/shoot ratio (NeSmith et al., 1992). Therefore, to satisfy the canopy needs for water and nutrients, root density must increase with decreasing substrate volume (Keever et al., 1985; Boland et al., 2000). The increased root density involves greater oxygen and nutrient consumption per unit volume of root zone, resulting in intense root-to-root competition for oxygen and nutrients, leading to more rapid declines in the concentration of dissolve oxygen (DO) and available nutrients. At the same time, the reduced DO levels can negatively affect root function, increase their susceptibility to diseases, and eventually cause their death. Shi et al. (2007) showed that forcing aeration into hydroponic system can alleviate the negative effect of the small pot volume. Lack of sufficient DO may also inhibit ammonium oxidation, leading to accumulation of toxic levels of ammonium in the liquid phase and even to toxic levels of ammonia in the gaseous phase of the root zone. This phenomenon may be aggravated by the consumption of oxygen by microorganisms that decompose organic matter. Even in inorganic media, the decomposition of dead roots and root exudates contributes to oxygen consumption. The continued mineralization of organic matter may also be accompanied by compaction of the medium, resulting in decreased oxygen diffusion rate (ODR), leading to slow flow rate of gaseous oxygen into the medium gaseous phase. Frequent solution replenishment with DO-saturated water can minimize the severity of these problems. Another method to prevent low and decreasing ODR is through the choice of medium’s components.

14.1.3 THE EFFECTS OF RESTRICTED ROOT VOLUME ON WATER UPTAKE AND PLANT NUTRITION

Two effects of restricted root volume on water uptake and plant nutrition are that (1) it can physically restrict root growth so that volume, length, and surface area of the roots are reduced and (2) limited reservoir size within the root zone and small buffer capacity of inert media may lead to

restricted water, nutrients, and oxygen supply. [Poorter et al. \(2012\)](#) found in a metaanalysis of published data of pot experiments 43% average increase in plant biomass with doubling the pot volume. It is therefore important to select the appropriate container size for each specific crop, based on its growth rate and final desired size and growing conditions ([Poorter et al., 2012](#); [Sinclair et al., 2017](#)). [Poorter et al. \(2012\)](#) suggested that the ratio of plant biomass to pot volume will not exceed 1.0 g/L.

Water deficit has been considered as the main factor for plant growth reduction under restricted root volume in many publications ([Poorter et al., 2012](#); [Sinclair et al., 2017](#); [Dambreville et al., 2017](#)). [Poorter et al. \(2012\)](#) found that reduction in photosynthesis and transpiration correlated with plant biomass and pot volume. [Sinclair et al. \(2017\)](#) obtained quantitative relation between transpiration and pot volume as described in the following equations:

$$TSW = SV \times VASW \quad (14.1)$$

where TSW is defined as the transpirable soil water (cm^3), SV is the soil volume explored by roots, and VASW is the fraction of available water (cm^3/cm^3). In soilless culture, we may assume that SV equals the whole volume of the pot.

The amount of water required for uninhibited transpiration ($AW_{\text{threshold}}$, cm^3) is calculated using [Eq. \(14.1\)](#) and a threshold value (TH) of the fraction of TSW that enables undisturbed transpiration which is derived from drying experiment:

$$AW_{\text{threshold}} = SV \times VASW \times (1 - TH) \quad (14.2)$$

The value TH is dependent of the substrate and the plant species and cultivar.

The environment conditions control the transpiration demand. [Sinclair et al. \(2017\)](#) suggested a simplified estimation of the daily transpiration amount (TA, cm^3/day):

$$TA = g_p \times LA \times VDD \quad (14.3)$$

where g_p is the plant vapor conductance (cm/day), LA is the leaf area (cm^2), and VDD is the vapor density deficit (cm^3/cm^3), which is obtained directly from air vapor pressure deficit (VPD, kPa) and temperature (T , K) such that $VDD = 0.0022VPD/T$.

Assuming daily irrigation and setting [Eqs. \(14.2\) and \(14.3\)](#) allows calculation of the pot volume (assuming that pot volume is equal to SV) for a specific combination of substrate and plant that becomes a limiting factor for transpiration under defined environmental conditions:

$$SV = \frac{g_p \times LA \times VDD}{(VASW \times (1 - TH))} \quad (14.4)$$

[Eq. \(14.4\)](#) can be rearranged to calculate the threshold LA or g_p value for existing pot volume under specific environmental conditions. Other models like the Penman–Monteith for calculation of the effects of environmental conditions on potential transpiration can be included in [Eqs. \(14.3\) and \(14.4\)](#).

The above-described publications are based on data obtained in experiments where plants in different pot volumes were irrigated at the same irrigation frequency ([Poorter et al., 2012](#); [Sinclair et al., 2017](#); [Dambreville et al., 2017](#)). A partial solution to the limited availability of water and nutrients is through their frequent supply through fertigation. The positive effects of high irrigation frequency on phosphorus uptake by lettuce grown in pots ([Silber et al., 2003](#); [Xu et al., 2004](#)) were presented and discussed in [Chapter 6](#), Nutrition of Substrate-Grown Plants. The main findings were

that increasing the irrigation frequency enhanced P uptake as well as biomass production of lettuce grown in low volume pots. However, it has been shown that high-frequency irrigations or even continuous supply (as is practical in closed systems and hydroponics) cannot overcome all the constraints posed by small root volume, as discussed in Chapter 10, Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses, Rieger and Marra (1994), Bar-Tal et al. (1990, 1995), Bar-Tal and Pressman (1996), Passioura (2006), Ronchi et al. (2006), Poorter et al. (2012), Bourgault et al. (2017), and Dambreville et al. (2017). Graham and Wheeler (2016) used a combination of high-frequency fertigation on with slow release fertilizers in pots to ensure sufficient availability of water and nutrients to pepper plants grown in a wide range of volumes. They found that the overall biomass of the shoot and roots, the leaves area, and the evapotranspiration decreased as the pot volume was smaller. However, the reduction in the fruit yield was relatively smaller, the harvest index of the pepper plants grown in small volume higher, indicating that restricting root volume can be an efficient tool to enhance the efficiency of fruit production per given volume, relevant to production in space stations.

Increased concentrations of nutrients can be used as a mean to minimize or to prevent the low buffer capacity and small reservoirs of nutrients in small volume pots. Bar-Tal et al. (1995) investigated the combined effects of root size and fertilization on the uptake of nutrients and water by tomato plants. Restriction of the root system resulted in a decrease in root volume compared with unrestricted roots. Severe and mild root restriction nearly stopped the increase in root volume 70 and 130 days after transplanting (DAT), respectively. The growth curves of the unrestricted and restricted root systems in both nutrient solutions had a smooth “S” shape, which is described well by the Gompertz growth model [Eq. (14.5)].

$$V_t = V_f \times e^{-b \times e^{-k \times t}} \quad (14.5)$$

where V_f is the maximal root volume (cm^3) and b and k are dimensionless parameters. The estimated parameters k and V_f were significantly affected by the solution nitrate concentration (Bar-Tal et al., 1995). The rate coefficient value (k) of the intact root, 0.0228 day^{-1} , was in good agreement with published values of $0.035\text{--}0.039 \text{ day}^{-1}$ (Bar-Tal et al., 1994). Root restriction significantly reduced V_f , k (the rate coefficient, day^{-1}), and b (dimensionless). The resulting V_f of the restricted root was in good agreement with the bag volume (Fig. 14.1).

Root restriction induced an increase in F_N , the nitrogen uptake rate per root unit volume. This indicated an adaptation of the smaller root system to the shoot demand. Thus, the smaller root system was more efficient in uptake per unit root weight, in agreement with published data (Jungk, 1974; Jungk and Barber, 1975; Edwards and Barber, 1976; de Willigen and van Noordwijk, 1987; Bar-Yosef et al., 1988). Eshel et al. (2001) found that the efficiency of absorption of K and P by the root increased following removal of part of the roots; however, this was not true for other nutrients. Consequently, the mass or the surface area of the root may affect nutrient uptake of plants grown in substrates flushed frequently with nutrient solution. Root restriction increased F_N ; values of 8.0, 13.5, and $25.6 \mu\text{mol/mL/day}$ were obtained at 133 DAT for the control and for mild and severe root restriction, respectively (Fig. 14.2).

Root restriction significantly reduced the potassium uptake rate per plant but it had no effect on F_K , the potassium uptake rate per root unit volume. The values of the parameter $F_{\max}$ [Michaelis–Menten uptake model, Eq. (14.4) in Chapter 6: Nutrition of Substrate-Grown Plants] (24.9 and $8.8 \mu\text{mol/g/day}$, for the young and old plants, respectively) fell within the range of

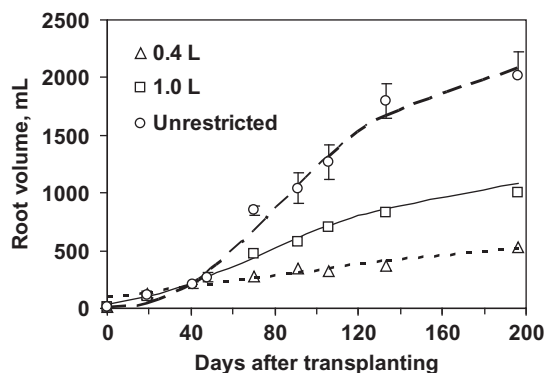


FIGURE 14.1

Root growth as affected by root restriction. Each point represents a mean of two $N-NO_3$ solution concentrations (1.0 and 9.0 mmol/L), except for the mild restriction that was studied only in 9.0 mmol/L solution. The vertical bars represent $+ LSD_{0.05}$. The curves were calculated with the Gompertz growth model [Eq. (14.5)]:

$$V_t = V_f \times e^{-b \times e^{-k \times t}}$$

where V_t is root volume at time t (cm^3), V_f is the maximal root volume (cm^3), k is a rate coefficient (day^{-1}), and b is a dimensionless parameters. The values of the parameters were fitted with the NLIN procedure of SAS.

Based on Bar-Tal, A., Feigin, A., Sheinfeld, S., Rosenberg, R., Sternbaum, B., Rylski, I., et al., 1995. Root restriction and $N-NO_3$ solution concentration effects on nutrient uptake, transpiration and dry matter production of tomato. *Sci. Hortic.* 63, 195–208.

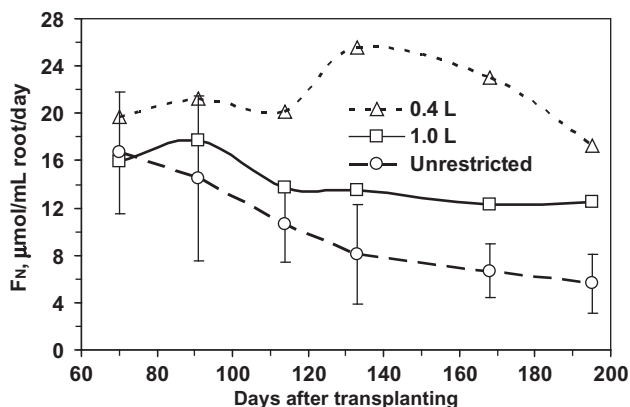


FIGURE 14.2

Nitrogen uptake rate per root unit volume (F_N) as a function of time and root volume. Each point represents a mean of two $N-NO_3$ solution concentrations (1.0 and 9.0 mmol/L), except for the mild volume that was studied only in 9.0 mmol/L solution. The vertical bars represent $+ LSD_{0.05}$.

Based on Bar-Tal, A., Feigin, A., Sheinfeld, S., Rosenberg, R., Sternbaum, B., Rylski, I., et al., 1995. Root restriction and $N-NO_3$ solution concentration effects on nutrient uptake, transpiration and dry matter production of tomato. *Sci. Hortic.* 63, 195–208.

published data for four different species (Wild et al., 1974; Chapter 10: Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses).

In the study of Bar-Tal and Pressman (1996) in the aero-hydroponics system, root restriction decreased Ca uptake rate per plant, but there was a trend of increasing F_{Ca} when root restriction was combined with low K concentration, which is consistent with the ability of the restricted-root plants to maintain Ca concentrations in plant organs similar to those in nonrestricted plants. Choi et al. (1997) reported that root restriction by small containers (190 mL) strongly suppressed transport of ^{45}Ca ions to new leaves and apices relative to the plants in the control containers (20 L). Water transport, expressed on leaf area basis, was marginally reduced by root restriction, an indication that calcium transport was more severely limited than water transport. Karni et al. (2000) also obtained a decline in Ca uptake by pepper root due to removal of three quarters of the root system, while the effect on water uptake and status in the plant was marginal.

In conclusion, root restriction reduces both nutrient uptake per plant and nutrient uptake per unit root when the buffer capacity and the rate of replenishment of the growing medium are the limiting factors. When the growing system enables maintaining a constant concentration of each nutrient at the root surface (as is the case in NFT, Deep Flow Technique (DFT), aeroponics, and with very frequent irrigation pulses), the ability of the restricted root system to meet the plant requirements is not the primary limiting factor. The minimal root size to meet plant nutritional requirement for nitrogen and other mobile nutrients is much smaller than that required for water supply, aeration, and other physiological demands. However, the supply of phosphate and calcium may be the limiting factors as a result of restricted root system. It is possible to raise nutrient supply by elevating their concentration, but this involves a potentially detrimental increase in osmotic potential, which interferes with water uptake (Sonneveld, 1981).

14.1.4 ROOT CONFINEMENT BY RIGID BARRIERS AND OTHER CONTRIBUTING FACTORS

As previously mentioned, regeneration of new roots is essential for normal plant development (Stevenson, 1967). Root growth is directed by environmental cues, including touch (thigmotropism), gravity (Oliva and Dunand, 2007), and the existence of an endogenous clock-type oscillator mechanism producing periodic pulses of gene expression in the root tip that determine the future sites of primordium initiation (Kircher and Schopfer, 2016). Gravity sensing occurs mainly in the columella cells of the root cap. Downward root growth is a natural response to gravi- and hydrotropism, typical to all active roots. A study by Massa and Gilroy (2003) suggests that normal root tip growth requires the integration of both gravity and touch stimuli. However, strong mechanical impedance such as the one that results from compact soil layer and from the walls of the container affects root growth direction (see Chapter 2: Root Growth, Physiology, and Potential Impact of Soilless Culture on Their Functioning) and causes clear anatomical and ultrastructural changes in the root apex (Wilson and Robards, 1979). These changes reflect several mechanisms that help root elongation in hard soils and even penetration through compacted layers. Root cap is generally smaller and thus cannot confer the same degree of protection to the root meristem as caps grown without mechanical impedance. Other changes include increased sloughing of border cells,

exudation from the root cap that decreases friction, and marked increase in root diameter. All these changes help to relieve stress in front of the root apex and to decrease buckling. Whole root systems may also grow preferentially in loose versus dense soil (Bengough et al., 2006). Concomitantly, mechanical impedance considerably reduces the root elongation rate and final root cell size and affects root distribution (Croser et al., 1999). Some of those energy-consuming changes may occur when roots of container-grown plant reach a rigid wall. As these roots can only change course, staying within the confined root zone, root density increases within the root zone. Eventually, root confinement within a limited volume results with reduced root growth (Hameed et al., 1987), similarly to what can usually be found in compacted soils (Hoffmann and Jungk, 1995). This reduced root growth can explain the results that were described earlier, connecting root-zone volume with plant performance even under ample supply of water and nutrients.

In container-grown plant, gravi- and hydrotropism frequently result with the accumulation of root mat on the bottom (Martinez et al., 1993). This part of the root system frequently accounts for a major part of the total root biomass and may be exposed to oxygen deficiency due both to the respiration of an extensive mass of dense roots and as a result of the existence of a perched water layer on the bottom of the container.

The main practical steps that could be applied at the grower level to ease DO deficiency are to optimize container depth, according to the crop size and growth duration and to frequently refresh the DO level using irrigation. Another useful approach is the use of hydroponics or aeroponics, where mechanical impedance is practically avoided. However, even in these conditions, increased root density leads, eventually, to growth inhibition (Peterson et al., 1991a; Bar-Tal and Pressman, 1996).

Another important limitation that sometimes occurs in soilless growing systems is the accumulation of regions of high Electrical Conductivity (EC) within the medium where irrigation water reaches only through diffusion and in the top layer experiencing a substantial amount of evaporation. This phenomenon limits the useful volume available to the plants. Salt pockets can be formed when drippers are too far from each other and hydraulic conductivity is low. Plants irrigated exclusively with subirrigation will generally experience high EC near the medium's surface as a result of evaporation-derived salt accumulation (Kent and Reed, 1996). It is important to plan both the irrigation system(s) and the water discharge rate so that a frequent downward piston-like water movement will prevent this type of salt accumulation (Schwarz et al., 1995). Thus, with subirrigation, it may be necessary to use a second irrigation system or to plan on hand-watering occasionally so as to leach such excess salts from the upper layer, and to then provide adequate irrigation solution to push these salts completely out of the root zone. The subject of selecting desirable root-zone volume is discussed in Section 14.3.

14.1.5 ROOT EXPOSURE TO AMBIENT CONDITIONS

As mentioned in Chapter 2, Root Growth, Physiology, and Potential Impact of Soilless Culture on Their Functioning, roots grown in containers are more exposed to extreme ambient temperatures than soil-grown roots, where temperature fluctuations in deep soil layers are minimal. In addition, the relative effect of ambient low VPD on water evaporation from the root zone is high, due to the typically high surface to depth ratio, as compared to this of soil-grown plants. The evaporation issue has limited horticultural significance, as normally container-grown plants are well irrigated,

the medium surface is shaded by the plant canopy, and the coarseness of the medium results in low capillary movement. It may be important, however, in outdoor nurseries and in newly planted media, where the plant cover is minimal and the effect of advective heat from the surroundings is significant (Seginer, 1994).

Moderate root-zone warming (20°C – 25°C) may have a beneficial effect in many cases and is routinely practiced in many propagation nurseries during the rooting stage of stem and leaf cuttings. Elevated temperatures, on the other hand, lead to a significant increase in root respiration rates. Since this is coupled with a sharp decrease in DO levels in the root-zone solution, the result may be detrimental. The combination of high root-zone temperature with ammonium as the source of N has a negative effect on the root system and whole plant growth and development (Kafkafi, 1990). Anaerobic processes may occur within the root tissue, significantly lowering its water and nutrient uptake and its growth rate. High temperature may, in some cases, negatively affect the activity of nitrifying bacteria which may, in turn, lead to toxic ammonium levels. Even under Northern Europe condition, temperatures within stone wool slabs in greenhouse vegetable production can reach levels of 30°C – 35°C and greatly affect their incidence of, and susceptibility to, pathogens (van der Gaag and Wever, 2005). Martin et al. (1991) showed a marked effect of the total volume of the container on the temperature in the middle of the root zone. Summer-time temperatures were 5°C – 6°C lower in 57-L than in 10-L containers. Heller et al. (2015) showed that the dimension of pots with equal volume but with different heights had strong impact on the medium temperature, up to 5°C (Fig. 14.3). They also reported that frequent irrigation with water system exposed to the sun contributed to significant increase in the medium temperature.

Practical methods that are used to mitigate high temperatures include adequate irrigation control possibility with occasional pulse irrigation, the use of mulch, a careful choice of the container color and dimensions.

14.1.6 ROOT-ZONE UNIFORMITY

One thing that should not be overlooked when contrasting in-ground field production with soilless production is the matter of uniformity. With in-ground field production, there is frequently variability in soil characteristics within the field. Many times farmers ignore this and simply treat the entire field as having the same soil, but the plants will generally grow and yield differently in the various parts of the field.

With soilless production, this is generally not the case. With a properly assembled substrate, mixed using standard best management practices, each plant experiences essentially the same growing medium as all other plants in the crop as long as supply of irrigation water and drainage are uniform. As such, in soilless production, a uniform management strategy for a particular crop is ideal, while in field production, better productivity would be achieved by customizing irrigation and fertilization for each plant depending on the variations in soil within the field, according to the concept of “precision agriculture.” With innovations in precision agriculture, it can be expected that such optimization will be feasible in the future, yet currently this type of optimization is not yet feasible in field production. With soilless production, on the other hand, such customized optimization is possible although it is questionable how economically feasible it is. As yet, in both soil-based and soilless production, the general strategy is to control irrigation and fertilization for all plants in a crop so as to avoid stress.

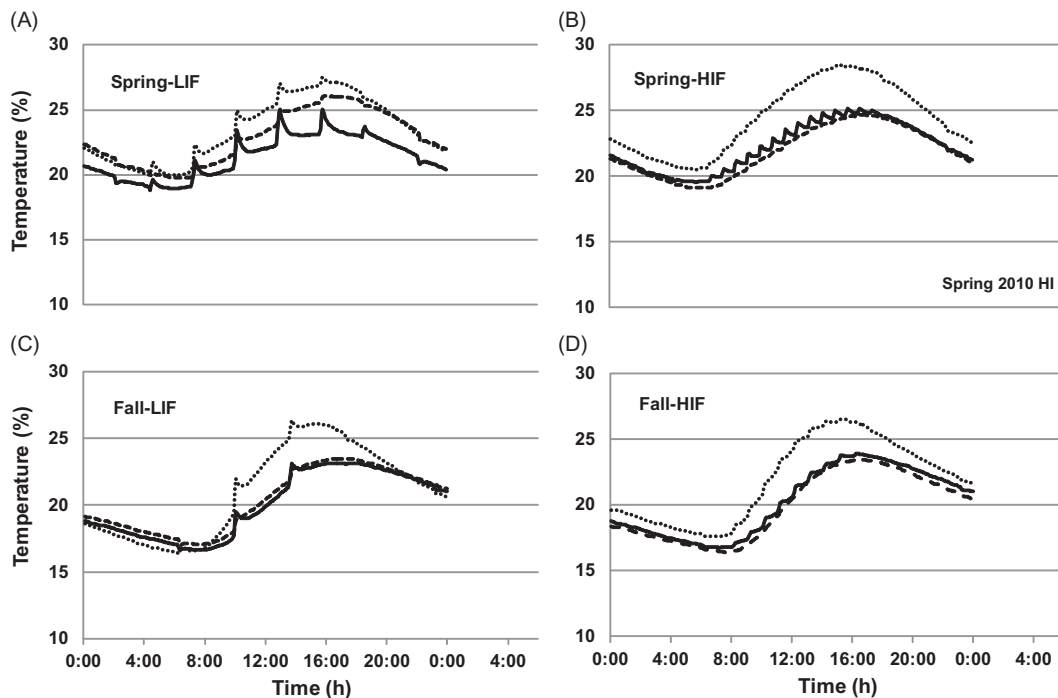


FIGURE 14.3

Representative medium temperature in LIF (A, C) and in HIF (B, D) in 10 (—), 20 (---), and 30 (...) cm container heights. On DAT 29 in spring (A, B) and DAT 30 in fall (C, D). *LIF*, Low irrigation frequency; *HIF*, high irrigation frequency; *DAT*, days after transplanting.

From Heller, H., Bar-Tal, A., Assouline, S., Narkis, K., Suryano, S., de la Forge, A., et al., 2015. The effects of container geometry on water and heat regimes in soilless culture: lettuce as a case study. *Irrig. Sci.* 33, 53–65.

14.2 DEVELOPMENT AND CHANGE OF SOILLESS PRODUCTION SYSTEMS

14.2.1 HOW NEW SUBSTRATES AND GROWING SYSTEMS EMERGE (AND DISAPPEAR)

One aspect that has had a major influence on the subject of this book is the notion by nonexperts that one can grow plants in virtually any granular or porous material. This notion is fueled by the fact that plants will grow in a variety of substrates, as long as one waters and fertilizes them according to plants' needs and the relevant characteristics of the medium. This notion has led to a nearly routine exploration of various waste materials as growing media. This has resulted in testing of many materials, some of which have been adapted as growing media. Shredded rubber tires, for instance, have been tested. Unfortunately, this material has not yielded a product suitable for widespread use as substrate in the soilless production (Khoshgoftarmansh et al., 2012). On the other hand, when testing coconut husk debris in a few tropical areas of the world, this waste material

was found to be horticulturally useful so that coconut coir is now a well-established component of some growing medium and even as a stand-alone substrate (Chapter 8: Organic Soilless Media Components). In the same way, in California, the use of redwood sawdust, a waste product from the large lumber industry in the state, was discovered to create a product that behaved quite different from other sawdust materials because its slow decomposition results in a root zone that is very slow to collapse. This makes it an excellent substrate for growing outdoor nursery products that require many months to produce. Thus, happenstance and trial-and-error have played a major role in the development of soilless growing media components. And while this approach has also led to somewhat random testing of various combinations of materials in various forms, it is today possible to engineer systems (consisting of various media components, containers, control systems, and management practices) that optimize the particular application. Horticulturalists already understand some of these principles. For instance, it is well known that if one transplants a plant from a small pot in which it is growing well, to a larger, deeper pot, then specific adjustments have to be made, either in the composition of the growing medium or by installing drainage elements, to continue to get the plant to grow. What the horticulturalist is doing is optimizing for one or two variables based on experience, even if not based on scientific knowledge of the underlying principles.

Several attempts have been made over the years to model and predict the characteristics of growing media consisting of well-defined components. Factors such as particle size distribution, bulk, and particle density were taken into account. Water release curves and rate of shrinkage were successfully modeled (Brown and Pokorny, 1975; Bures et al., 1993a,b; Pokorny and Henny, 1984a,b; Pokorny et al., 1986). Such models enable the prediction of several important properties of new combinations of ingredients.

14.2.2 ENVIRONMENTAL RESTRICTIONS AND THE USE OF CLOSED SYSTEMS

As discussed in Chapter 10, Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses, intensive greenhouse production involves several environmental risks. Environmental protection agencies in a growing number of countries require growers to capture and reuse the effluent from soilless production systems so as to minimize these risks. Simultaneously, effluent recycling leads to significant water and fertilizer savings. Moreover, the complicated issue of both temporal and spatially uniform supply of water and nutrients is not relevant if the system is closed so that it enables frequent or even constant flushing.

In view of the dwindling global reservoirs of water and nonrenewable sources of energy (oil) and as a result of environmental regulation, it appears mandatory that most above-ground systems (that allow collection of discharged drainage water) will need to be modified to capture any effluent. As described in Chapter 10, Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses, many of the problems involved with such operation have been addressed and many such systems are already in operation. There are a variety of issues that still require solutions, but the primary hurdle for many commercial growers is economic. Even this may be true only on the basis of the individual grower, and not on a national or global basis. In order to promote fast adoption of this practice, it is imperative that the true cost of water treatment be known and integrated into their final cost to the customer. In addition, the future cost for cleaning soil and water resources should be considered. Thus adoption of a

closed recirculation system involves careful study of the economic factors and a complete analysis over the long run as to handling of water when it reaches a level of salinity due to nonnutritive ions that makes it impossible to use for in commercial soilless production. As such 100% recirculation is as yet still not possible in areas where the supply water contains ions that the plant cannot use. It should be noted, however, that semiclosed systems normally reduce the amount of effluent by ~90% over a system where no recirculation is practiced. It is important that agencies regulating this issue be aware that growers generally must have the flexibility to discard water where some deleterious ion (e.g., sodium) has accumulated to a level that renders the water toxic to plants. While some argue that water can be cleaned up through water treatment, the result of such treatment is always a brine solution that must be discarded. At present, there is no easy solution to this problem and regulators that attempt to enforce 100% recirculation are imposing a condition that will ultimately put the grower out of business.

Water desalinization is probably the ideal solution to the challenges described above. Low-scale desalinization plant required for a single farm is inefficient and expensive process; however, in the last decade, the efficiency of large-scale plants gradually grows and installation of large-scale desalinization plants continues, mainly in arid and semiarid regions and expanding to less dry areas (Burton et al., 2015; International Desalinization Association, 2015). The driving forces for this rapid growth are the continuous growing demand for water due to population growth and increased water consumption per capita on one hand and the decline in fresh water supply due to overuse and contamination of natural water resources (ground water as well as rivers and lakes), and climate change that leads to reduced precipitation in arid and semiarid regions and to higher frequencies of drought events (Evans, 2009; Taikan and Kanae, 2006). At the same time, the price of desalinization is decreasing thanks to improvements in the technology (Elimelech and Phillip, 2011; Burton et al., 2015). In several regions and countries, desalinated water is becoming a major fraction of the fresh water supply. In Israel, the production of desalinated water increased from 50 MCM/year in 2004 to 650 MCM/year in 2017, whereas the supply of fresh natural water is in the range of 900–1100 MCM/year. Similar trend occurred in other arid and semiarid regions like several Mediterranean countries (Cyprus), the countries on the Persian Gulf, southern Eastern Spain (Martinez-Alvarez et al., 2016), Mexico, etc. Economical and environmental analyses indicate that irrigation of high cash crop, mainly in greenhouses, is feasible in some regions like Israel and Spain (Kaner et al., 2017; Martinez-Alvarez et al., 2017), while it is not feasible under current economical and environmental policy conditions in California (Welle et al., 2017). The supply of desalinated water to agriculture and mainly to indoor grown crops enables reduction in the quantity of water required for high yield both in open (Assouline et al., 2015; Ben-Gal et al., 2008; Silber et al., 2015) and closed systems (Cohen et al., 2003). In open system, water saving is due to the reduction in the leaching fraction required for displacing salt accumulating in the root zone (Assouline et al., 2015; Ben-Gal et al., 2008; Silber et al., 2015), whereas in closed systems, the need for discard a fraction of the water is minimized (Cohen et al., 2003). The challenge of supplying all the nutrients required for optimal growth of crops is higher with desalinated water, especially in soilless culture (Yermiyahu et al., 2007; Ben-Gal et al., 2009). Some of the macronutrients, calcium, magnesium, and sulfur, that are in sufficient concentrations in natural water become the limiting nutrients for healthy crop and yield (Bar-Tal et al., 2017; Ben-Gal et al., 2009; Martinez-Alvarez et al., 2016; Yermiyahu et al., 2007). This challenge can be overcome by two ways: either by new standards for the desalinated water composition to meet the requirements

of agriculture or including these elements in the fertilization (or fertigation) managements of the crops directly or through blending the desalinated water with natural water (Bar-Tal et al., 2017; Ben-Gal et al., 2009; Martinez-Alvarez et al., 2016; Yermiyahu et al., 2007).

The low salt content in the desalinated water, mainly the ions Na^+ and Cl^- , raises the question of the need to modify the recommended levels of several nutrients, mainly nitrogen and potassium (Yasuor et al., 2017). The basis for this question is the hypothesis of competition between the Na^+ and Cl^- ions with K^+ and NO_3^- in uptake by plants. The effect of salinity on nutrients uptake and the possible mitigation of specific toxic effects of Na^+ and Cl^- through higher concentrations of K^+ and NO_3^- are described and discussed in Chapter 6, Nutrition of Substrate-Grown Plants. The opposite question of lowering the required concentrations of K^+ and NO_3^- when high-quality water with extreme low concentrations of Na^+ and Cl^- ions were examined was raised recently in several works with different crops (Yasuor et al., 2017; Kurtzman et al., 2018). The main conclusion from these studies is that in the studied crops, significant interaction between salinity level and fertilization was obtained and therefore the required dose of fertilizers was not affected by water quality. Moreover, in opposite to the hypothesis, in some crops (potato, lettuce), irrigated with desalinated water, the potential yield was higher than under moderate salinity when combined with higher dose of nitrogen (Kurtzman et al., 2018).

The challenge of balanced plant nutrition in soilless culture becomes higher when the water supply system uses different sources of water with a wide range of water composition in different proportions that vary with time. In such conditions, there is a need for careful online monitoring of water quality, mainly through EC measurements, and a sophisticated management of the fertilization (or fertigation) with time according to the changes in the composition of the water.

The growing supply of desalinated water for crops grown in soilless culture will enable growth in closed systems without or with minor discard of water out of the system. The brine solution that is produced as a by-product of the desalination process of seawater is returned to the sea. When other inland saline or brackish water are cleaned and desalinated, the brine solution of the desalinated plant is the main environmental and economical challenge for irrigation with desalinated water.

14.2.3 SOILLESS “ORGANIC” PRODUCTION SYSTEMS

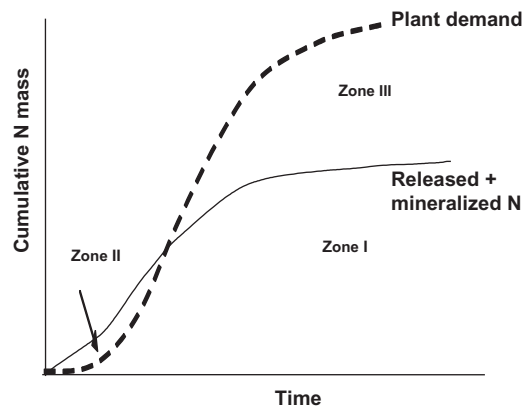
While “organic agriculture” has a very specific meaning to growers and regulators, the general public is rarely knowledgeable about the details that define this area of agriculture and what is involved in certification. The strict rules of most of the organic agriculture certification bodies in the EU and several other countries state that in order to be considered as an organically certified crop, crops should be grown in soil; yet the attractiveness of this type of agriculture to the general public is based on the desirability of having foods that were not subjected to synthetic fertilizers or pesticides. As such, it is entirely feasible to accomplish this with soilless production and this practice is acceptable in the United States, Australia, and a few other countries. A notable exception is the growing of transplants (Grunert et al., 2016) which is common worldwide. This notion became recently clear to scientists (Dresboll, 2004; Raviv, 2017) and may, eventually, affect the rules. This is especially true for areas of the world where fertile soil is limited, while food production lags behind the needs.

There is a growing desire to grow food crops in soilless culture without the use of synthetic fertilizers and pesticides. This trend, especially developed in Australia and New Zealand, is driven by the public demand for pesticide-free produce and for decreasing the use of oil-derived products (such as chemical fertilizers). Thus, while the requirement of growing crops in soil makes certification of soilless production systems as “organic” unlikely in the EU, the forces that drive the demand for organic products may well force changes to the definition or force the emergence of alternatives that are equally attractive to the general public. Such changes have already been noted with the advent of “biorational” pesticides which have changed the way integrated pest management is currently practiced in conventional field agriculture.

Discussion of the crop protection aspect of these soilless production systems is beyond the scope of the present book but it should be noted that with the advent of biological control methodologies, the certification issue may ultimately be solvable. The subject of plant nutrition as part of an “organic” soilless production system will be briefly described below.

In general, soilless “organic systems” can be based on either substrate or liquid culture. In principle, the aim of fertigation in both types is to provide completely soluble nutrients, but to have them derived from organic sources, rather than synthetic soluble salts. A variety of commercial products exist in the market, mainly based on water extraction of fermented fish, meat, compost (compost tea), or manure residues to make such liquid fertilizers available. The two main drawbacks of these materials are their extremely high cost per a unit of specific nutrient (especially nitrogen) and the fact that the ratio among the various nutrients is usually different from what plants require. Several attempts have been made to formulate an accurate combination of off-the-shelf organic fertilizers, with various degrees of success (Atkins and Nichols, 2002, 2004; Liedl et al., 2004a,b; Jarecki et al., 2005). Significant work in this area is continuing as evidenced by the frequent appearance of new products by various companies.

Development of substrates that can supply nutrients for extended periods of time is driven by the fact that some compost types can provide much of the needed nutrients for extended periods of time (Dresboll, 2004; Grunert et al., 2016) while simultaneously suppress many root pathogens (Raviv, 2005). More nutrients can be supplied once the compost’s reservoir has been exhausted in the form of guano, feather meal, or other concentrated organic fertilizers (Raviv et al., 2005). This is schematically depicted in Fig. 14.4. The difficult part of this nutritional program is, on the one hand, how to minimize nutrient leaching from the medium when the plants are young and their consumption is low (zone 2 in Fig. 14.4) while, on the other hand, accelerating the compost mineralization once the plants attained full size (zone 3), so as to maximize compost-derived nutrient uptake by the plant (zone 1). Nutrient leaching can be minimized by effluent recirculation. The decomposition process itself renders the substrate somewhat unstable (see Chapter 8: Organic Soilless Media Components) so that long-term (more than a year) crop production in such systems may not be feasible. Most concentrated organic fertilizers are very expensive, in comparison with chemical fertilizers. For example, guano-derived nitrogen costs ~\$5/kg, more than five times its cost when derived from chemical fertilizers. Compost, on the other hand, is a relatively cheap source of nutrients. However, most of the nitrogen exists within the compost in a recalcitrant form that resists mineralization. The challenge for future researchers is to find ways to accelerate nitrogen mineralization from compost, if possible, in a controlled manner, at times dictated by the grower. The work of Dresboll (2004) elegantly sets the stage for this needed effort.

**FIGURE 14.4**

Graphical presentation of plant demand for nitrogen versus compost mineralization rate.

Courtesy of Dr. Yael Laor.

One approach to developing an organic soilless nutrient source is to use a separate reactor in which compost is aerobically brewed to form liquid extract called compost tea. Such systems are already being tested in commercial production and considerable research is needed to develop methods for sustained uniform delivery of particular nutrients to the plants. The compost tea can be applied to the root system, as a drench or as a foliar spray (Pane et al., 2014).

To date, soilless “organic” production has not reached a significant share of the market, but there is considerable development in this area to attempt to solve the hurdles to making such systems economically feasible. In addition to the economical and professional obstacles mentioned above, the current regulations are the main stumbling block, effectively requiring the development of an entirely new system.

14.2.4 TAILORING PLANTS FOR SOILLESS CULTURE: A CHALLENGE FOR PLANT BREEDERS

In addition to its importance to the understanding of basic root growth processes, the study of the response of roots to various types of physical stress (such as oxygen deficiency and mechanical impedance) has clear horticultural relevance. Although some root traits such as enhanced ability to capture phosphate (Gahoonia et al., 2001; Bates and Lynch, 2002) and the nature of root biomass, length, and architecture (Fita et al., 2006; McPhee, 2005) are starting to receive increased interest, in general, the intentional selection of particular root traits has been largely neglected. The above examples were studied with field crops such as barley (*Hordeum vulgare* L.), melon (*Cucumis melo* L.), and pea (*Pisum sativum* L.). When more such responses are elucidated, especially under conditions of soilless culture, that is, ample water supply and nutrition, better manipulation of root growth responses to the specific characteristics of these growing conditions will be attainable via appropriate plant breeding or genetic engineering technologies. Some examples for root characteristics sought for soilless culture conditions are increased branching and fine roots formation, so as to

continuously provide the plant with actively absorbing root surfaces. In view of the increased importance of closed systems, better sodium and chloride exclusion efficiency is a trait of paramount importance. Root systems that can tolerate higher external concentrations of these ions allow the grower to set a higher threshold before effluent discharge becomes mandatory. As explained in [Chapter 10](#), Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses, any such small increase leads to a very significant decrease in the required leaching fraction. Similarly, root systems that can tolerate higher osmotic potential in the rhizosphere can lead to the abovementioned result, while greatly improving water and nutrient use efficiencies.

Another facet of soilless plant production that could be addressed by breeders is the plant's response to root-zone temperatures. These are typically higher than the temperatures to which plant roots have adapted through evolution. Accompanying higher temperatures is a lower level of dissolved oxygen so that selecting for plants with low susceptibility to hypoxia could be particularly advantageous in soilless production.

14.2.5 CHOOSING THE APPROPRIATE MEDIUM, ROOT VOLUME, AND GROWING SYSTEM

Choosing the appropriate medium for a specific combination of crop and environmental conditions is the subject of numerous scientific papers and several books ([Bunt, 1988](#); [Handreck and Black, 2002](#); [Savvas and Passam, 2002](#)). [Kipp et al. \(2000\)](#) provided detailed tabulated information about substrate's characteristics. Most of the papers dealing with comparisons among substrates or growing systems do not involve explicit economic consideration. Moreover, the results of most of these studies are often affected by arbitrary choices, made by the researchers. For example, if various substrates are irrigated with identical irrigation regimes ("control," usually is based on the "traditional" medium and is based on time-controlled irrigation), in spite of their differing physical characteristics, an unfair advantage is given to some of them, while others may be irrigated in a frequency that does not match their properties. Generally, researchers do not make the extra effort to compare substrates using irrigation frequencies based on physical parameters such as water tension, unsaturated hydraulic conductivity, water content, or real-time replenishment of transpired water based on weighing lysimeter (some who did were [Morel and Michel, 2004](#); [Raviv et al., 2001](#); [Wallach and Raviv, 2005](#)). The use of measurable plant parameters that reflect plant water status is even less common. Irrigation control using methods such as leaf water potential (with pressure chamber), stomatal resistance for water (with diffusion porometer), canopy temperature (with thermocouples or infrared thermometers), flow of water in the stem (with the heat pulse method), and changes in stem diameter (with dendrometer) can be found in research studies but only rarely in horticultural studies. [Caron et al. \(1998\)](#) recommended the use of plant-based measures such as xylem water potential, to assess water availability and to control irrigation. However, these methods have not gained popularity in commercial settings. As a general rule, it can be stated that reports aimed at comparing media while using uniform irrigation treatments have only limited usefulness. In addition to avoiding biased conclusions, comparative research conducted while fitting the

irrigation regime to the medium's physical characteristics also provides relevant information about optimal irrigation management.

Another factor that may mask media effect is their potential contribution (positive or negative) to nutrient availability. As described in [Chapter 4](#), Chemical Characteristics of Soilless Media, various media can adsorb or release nutrients. If this is not taken into account while comparing growing media, false conclusions may be reached. Unlike the case of irrigation regime, overcoming this heterogeneity among substrates is rather difficult since real-time monitoring of all nutrients is not practical even under experimental conditions.

Unfortunately, we cannot suggest clear answers about the optimal choice of media for specific crop/conditions combinations except to caution the readers about the nature of this kind of pitfall. Researchers should not justify such experimental design based on that fact that such flawed method is frequently found in the scientific literature. Historically, when a large number of growers adopted a specific medium (e.g., stone wool in Northern Europe), it was accompanied by a wide assortment of peripheral elements (structural, knowledge, and customized instrumentation) that constitute a substantial economic investment. This results in an objection to replace one medium by a newer one, even if proved as somewhat superior. Adoption of a new substrate is more common when a substrate must be replaced occasionally, when a new industry emerges, or when the use of a certain substrate is not free of problems. An example of the former is the replacement by many growers of peat moss by coir due to the fact that peat moss shrinks over the years requiring to be either top-dressed or replaced ([Scagel, 2003](#), and see [Chapter 8](#): Organic Soilless Media Components).

Economics and availability play a major role in substrate component selection. Tuff, for example, is readily available in Israel and in the Canary Islands (under the name Picon) and its cost in these countries and nearby is competitive with other substrates; therefore, it found a wide use in these countries ([Plaut et al., 1973](#); [Perez-Melian et al., 1977](#); [Silber et al., 1994](#); [Raviv, 2016](#)). Similarly, bark compost is widely used in temperate countries, where forestry is a major industry and a large source of wood waste ([Pokorny, 1982](#); [Depardieu et al., 2016](#)). In some cases, the need to find solutions for used media leads to the development of yet another new media by their recycling. An example is the incorporation of recycled stone wool as a component in peat-based media ([Riga et al., 2003](#)).

As mentioned above, root volume rarely affects shoot/root ratio ([NeSmith et al., 1992](#)). It can be assumed that shoot growth is regulated by root growth ([Hoffmann and Jungk, 1995](#); [Eshel et al., 2001](#)) while, in an interactive manner, root growth is regulated by the shoot through carbohydrate supply. Modern fertigation techniques enabled growers to save costly greenhouse space and to reduce the expense of substrates by growing plants in small root volumes. High growth rate and high shoot/root ratio are common under these conditions. There is a limit, however, to the shoot/root ratio, as at a certain point, the physiological ability of the root to take up water and nutrients to meet the demands of the shoot becomes a limiting factor. In well-controlled experiments, root restriction has been shown to limit plant growth in various species, as manifested in reductions of such growth parameters as leaf area, leaf number, plant height, and biomass production ([Cooper, 1972](#); [Richards and Rowe, 1977](#); [Carmi and Heuer, 1981](#); [Carmi et al., 1983](#); [Carmi and van Staden, 1983](#); [Ruff et al., 1987](#); [Robins and Pharr, 1988](#); [Peterson and Krizek, 1992](#); [Choi et al., 1997](#);

van Iersel, 1997; Karni et al., 2000; Haver and Schuch, 2001; Xu et al., 2001; Dominguez-Lerena et al., 2006). In many cases, the initial volume in which transplants are grown affects their performance after transplanting (Bar-Tal et al., 1990; di Benedetto and Klasman, 2004; Dominguez-Lerena et al., 2006). The volume in which the roots can expand may affect plant growth either through plant nutrition and transpiration (Brouwer and de Wit, 1968; Hameed et al., 1987; Bar-Tal et al., 1990; Choi et al., 1997) or via root and shoot physiology (Aung, 1974; Jackson, 1993; Haver and Schuch, 2001). In fact, in some cases, the effects of root restriction are not implemented through nutrient deficiency (Carmi and Heuer, 1981; Robbins and Pharr, 1988; Izaguirre-Mayoral and de Mallorca, 1999) or water stress (Krizek et al., 1985; Ruff et al., 1987; Izaguirre-Mayoral and de Mallorca, 1999). In such cases, it has been suggested that root restriction stress induces a reduction in the supply of growth substances from roots to shoots and an imbalance in root and shoot hormones (Carmi and Heuer, 1981; Peterson et al., 1991a,b; van Iersel, 1997; Hurley and Rowarth, 1999; Haver and Schuch, 2001). Peterson et al. (1991a,b) showed that root respiration was reduced by root confinement and concluded that the decline in root respiration capacity represents a decline in root metabolism. Haver and Schuch (2001) found that severe root restriction caused apical dominance that was associated with low ethylene production by the root. Moreover, they showed that exposure of roots to ethylene overcame the apical dominance induced by root restriction. The transfer of growth substances from roots to shoots, hormonal balance, and root metabolism are out of the scope of this chapter and of the main subject of this book and will not be discussed further. However, when the crop in question is a finished plant of well-balanced and compact dimensions, a relatively large root volume is advisable.

It has been shown that the depth of the growing container has a more significant effect than the volume on successful plant development (Dominguez-Lerena et al., 2006). Generally, the deeper the medium depth, the larger is the ratio of air to water spaces in it. In growing media of all depths, there will be a saturated zone after drainage at the bottom due to the formation of perched water table. In shallow media, this saturated zone constitutes a larger percentage of the total root-zone volume, thus affecting air/water relations, possibly leading to water logging problems. However, this is not always true. For example, when the development of a fibrous root system of tree transplants is required for subsequent optimal development after transplanting, shallow and wide containers are considered preferable (Milbocker, 1991).

The minimal desirable root-zone volume is usually determined by the grower before starting the crop. Even in cases where one or more transplanting stages are planned, it is essential to plan them in advance, to ensure the crop's success. Factors that affect final required root-zone volume include the type of crop, its growing duration and season, nature of the chosen substrate, economical consideration such as the cost of various containers, substrate, and greenhouse area ("rent"). Logistical considerations, such as transportation costs, may also affect the decision in cases where the potted plant itself is the final product. Some typical examples are listed in Table 14.1.

One common practical consideration for growers that dictates the root-zone volume is the interval between irrigation events. While it would be ideal to tailor the irrigation schedule to the needs of the plants, many growers set the irrigation schedule for convenience or to take advantage of available capacity for irrigation. In such circumstances, the root-zone volume should be matched to this interval to assure that the plant does not run out of water.

Table 14.1 Typical Examples of Required Size of Root Volumes for Various Crops

Crop Type	Growing Duration	Special Considerations	Standard Volume/Plant	References
Bedding plants and vegetable transplants	<60 days	Densely grown	<125 cm ³	Kemble et al. (1994), di Benedetto and Klasman (2004) Gysi and Allmen (1997)
Lettuce and other leafy vegetables	<60 days from transplanting		<0.5 L	
Small-size pot plants	3–6 months	Transportation required	0.5–2.0 L	
Medium-size pot plants	6–9 months	Transportation required	1–4 L	
Large-size pot plants	1–2 years	Transportation required	10–20 L	
Short-season vegetables and herbs	2–6 months		2–5 L	Baas et al. (2001)
Long season cut flowers (gerbera, carnations)	9–12 months		3–6 L	Garibaldi et al. (2004)
Long season vegetables (tomato, pepper)	9–12 months		~10 L	Tuzel et al. (2001)
Long season vegetables (tomato, pepper)	9–12 months	Hydroponics	1–2 L	Peterson et al. (1991a)
Perennial cut flowers (rose)	3–5 years		5–10 L	Raviv et al. (1999), Raviv et al. (2001)
Tree saplings	1 year	Transportation required	~5 L	Salifu et al. (2006)

14.2.6 AQUAPONICS—SYNERGISTIC CONNECTION OF HYDROPONICS AND AQUACULTURE

There is considerable interest in aquaponics by the general public. Such production operations devote part of the production space to hydroponics and other parts to aquaculture. The basic idea is that the waste products from the aquatic organisms can be used as fertilizer for the plants and that waste material from the hydroponics can be used in aquaculture. As discussed in [Chapter 12](#), Liquid Culture Hydroponic System Operation, hydroponics is a viable enterprise by itself. Similarly, aquaculture by itself is a viable enterprise. The linkage of the two systems results in aquaponics.

It is, however, clear that aquaculture waste is not by itself a suitable fertilizer solution for hydroponics. The high investments in all the tools and methods require that fertilizer, which is relatively small part of the costs, must be optimized. As such, the standard is to use fertilizer products which are completely soluble and thus do not clog up emitters, filters, pumps, etc. Furthermore, if

the water coming from the aquaculture is of a quality that is adequate for the aquatic organisms to thrive, then the fertilizer level will be far too weak for the hydroponics. Furthermore, if this is the only type of nitrogen fertilizer, then there will be too high a proportion of ammonium. Improvements can be achieved by including a bioreactor between the two systems.

There is a notion that the plants can be used to make the water cleaner so it can be returned to the aquaculture system. But this then means that either the last plants to be exposed to this water have very little fertilizer or the water will contain too much nitrate and other ions which are not optimal for most aquatic organisms, particularly the ones which are prized as food.

Thus there is considerable research interest in developing commercial sustainable systems that would solve these problems.

14.2.7 NEW FRONTIER IN SOILLESS CULTURE: PLANT FACTORIES WITH ARTIFICIAL LIGHT

Various technological advancements have combined to create an entirely new facet to soilless culture: growing plants in in-door production systems, known as plant factories. In such facilities, plants are grown in soilless culture, many times stacked on shelves with as many vertical layers as fit into the volume. All light is generated with lamps with spectra that are matched to the photosynthetic action spectrum of the crops being grown. Generally, heat builds up in such facilities so that air conditioning is used to cool the air. All CO₂ needed by the plants is pumped into the facility at concentrations that maximize biomass production. All moisture in the air is removed using the air conditioning systems and dehumidifiers. This generates condensate which is basically high-quality water that can be used in irrigation. As this is basically distilled water, it requires very careful management to avoid fluctuations in pH.

An important innovation with these technologies has been the advancement with light emitting diode lighting which allows building lamps with very specific designed spectra. While these lamps also generate heat as every lamp does, a core advantage is that much of the heat is on the back side of the lamp while high-intensity discharge lamps radiate a lot of long wave radiation onto the plants. While the newer lamp technologies tend to be more expensive, and the high energy utilization is quite costly, the benefits gained in faster, more optimal plant production, combined with production that is free of pesticides, offer marketing opportunities that create a positive return on investment. As such many of such facilities are currently starting up all over the world. They are particularly suited for countries that lack self-sufficiency in food security. The subject of plant factories was discussed by [Kozai et al. \(2015\)](#).

14.3 MANAGEMENT OF SOILLESS PRODUCTION SYSTEMS

14.3.1 INTERRELATIONSHIPS AMONG VARIOUS OPERATIONAL PARAMETERS

The large number of variables that interact to form the production system makes it nearly impossible to consider (much less to optimize) all these factors simultaneously. As such there is still a lot of room in the marketplace for new growing media formulation that are specifically suited for

particular applications. Much of what is available today is really a raw material that still requires substantial engineering by the end user to create an ideal growing situation for plants.

It is important to note that growing media of today are not simply haphazard combinations of a few convenient media components, rather they are elements of engineered horticultural systems tested by marketing forces and adapted by horticulturalists to particular purposes.

All media provide some plant anchorage, while nutrients are often provided by added fertilization. Water and air are provided in the pore spaces of the media. The principal factors affecting air and water status in the growing media are (1) the media components and ratios, (2) media depth, (3) media handling, and (4) watering practices.

Each of the generally used media components is unique in their properties pertaining to their air/water relations. For example, peat has a comparatively higher content of unavailable water at a given matric tension compared to vermiculite. This variability in the availability of water in different types of media components means no two components are exactly alike in terms of providing water to plants. Similar variability does exist among various media components with regards to wettability (i.e., the ability of dry media to rapidly absorb water when moistened) or drainage characteristics. [Airhart et al. \(1978\)](#) reported that an air dry peat-vermiculite substrate required 5 days to reach 70%–78% of moisture saturation while milled pine bark required 48 days to achieve 58%–78% saturation. Thus, the kind and ratio of the components in the media mixes have important bearings on the hydrophysical environment in the root zone, which in turn largely determines the performance of the plants grown thereon. In general, moisture retention characteristics of media mixes is a compromise of its individual component corresponding characteristics. However, even when the same components are blended in identical ratios, physical properties may vary due to difference in particle size ([Bilderback et al., 2005](#)). [Pokorny and Henny \(1984a,b\)](#) reported that substrates with the same components and ratio were not identical in their physical properties, even though they were assumed to be the same. Differences in physical properties in their study were attributed to the shrinkage and particle size differences of the components blended. Variation in identical components occurs if dry components are blended compared to blending moist components. Dry components when mixed tend to fit together tightly and increase bulk density of the substrate compared to when moist components are blended; consequently, air-filled porosity is reduced.

In handling growing media, proper care is necessary to avoid compaction; air space can be drastically reduced by compaction. The moisture content of the media prior to filling containers or bags may also be important. For example, adding water to soilless media mixes filling plug trays causes the media to swell and helps create more aeration. Moistening the media before filling larger containers does not have much benefit.

Apart from the air/water relations of the media mixes imparted by the individual and interactive effects of their components, watering practices have a tremendous influence on their air/water relations. This is due to the fact that both water and air compete for the same pore spaces in the media, where they are nearly mutually exclusive.

14.3.2 DYNAMIC NATURE OF THE SOILLESS ROOT ZONE

Soilless production has been in transition ever since it was conceived (see [Chapter 1: Significance of Soilless Culture in Agriculture](#)). Initially, this type of production was carried out by mimicking

traditional methods based on production in soil. The wide array of soils on the planet has led farmers to adapt methods so as to deal with the many physical and chemical properties that abound. The production methods in highly porous soils were assumed to be an excellent initial step for managing production in soilless growing media. However, as more and more artificial components were tested and drawn into use, the physical and chemical properties of growing media became more and more different from soils so that the methods required to be commercially successful changed dramatically. Yet even now, most practitioners view the key elements of soilless production to be driven by water in the limited root zone and avoidance of stress related to water status and plant nutrition.

While this stress-avoidance approach has been successful in attaining excellent commercial results with soilless production, the growth of the industry has resulted in strong commercial competition which continues to push growers to achieve greater efficiencies. This has resulted in the various specialized systems described in [Chapter 13](#), Technical Equipment in Soilless Production Systems, as well as a huge outdoor nursery industry. As a segment of agriculture, it continues to increase both in size and in percentage. If one adds to this the fact that virtually every home owner owns and pampers a number of potted plants, it becomes clear that there is considerable interest in managing the root zone for more than merely survival or stress avoidance.

While the irrigation and fertilization factors described above are certainly ones that cannot be ignored, there are a multitude of other factors that are of equal importance in soilless production. The ones that must be taken into account are those which have the capacity to change rapidly, and especially where the rate of change toward a suboptimal condition can occur within a few days under conditions that may readily occur.

As such it is worthwhile exploring the various root-zone factors discussed throughout this book in the context of how readily each can change from a condition that is generally optimal, to one that is suboptimal (or vice versa). Ideally, a quantitative approach would be best here, but the numerous circumstances in the real world make this impossible. For this discussion, it is important to separate out effects that occur over the course of a day versus ones that occur faster or slower. There is no need to consider a time-frame longer than 1 week in soilless production since there are no root-zone factors that do not have the capacity to change on at least a weekly basis. And therein lies an operational conclusion: while some factors in soils take multiple weeks to change significantly, every factor in soilless production has the capacity to change much more rapidly, even with substrates that are highly buffered.

The major consequence of this capacity for rapid change is that control must involve accurate and rapid sensing of all these conditions in conjunction with implementation of mitigating measures to keep the system as homeostatic as possible. In cases where accurate and effective control is not feasible at the time-scales needed, we still attempt to implement some sort of buffering to slow the rapidity of change to a manageable level. In fact, the sole reason why we seek well-aerated media is to provide a passively buffered oxygen situation by allowing oxygen to diffuse into the root zone and ethylene and CO₂ to leave. If we were to force bulk movement of these gases (as we force bulk movement of all other materials through irrigation), then the aeration buffering provided by well-aerated materials would be unnecessary. In fact, in water culture methods, we generally force oxygen into the water either by bubbling air or spraying which force the dissolved oxygen level to saturation.

A major difference between systems that are operated as open versus closed or semiclosed systems is in the moisture regime. While moisture content can change with equal rapidity in both, closed systems are generally characterized by more frequent irrigations, preventing moisture levels from ever declining to levels that would induce even hidden moisture stress in the plants. Thus in a mixture that combines an inert medium such as sand or perlite with one or two organic substrates, the moisture conditions will typically decline to levels that are dryer in open systems than in closed systems. As was pointed out in [Chapter 3](#), Physical Characteristics of Soilless Media, one consequence of this is that the unsaturated hydraulic conductivity in open systems is likely to be much lower at times than it is in closed systems. Thus irrigation in the two systems leads to more rapid change in bulk movement of water in closed systems so that during normal operation, the practitioner can count on lateral bulk movement of water to be faster in closed systems than in open systems. Since this bulk movement carries with it dissolved gasses and nutrients, the net flow of these into plants is typically faster in closed systems, even without consideration of the bulk movement due to the more frequent irrigation events.

One stark contrast between soilless and soil-based production is the rapidity of change during irrigation events in relation to the rates of change between irrigation events in soilless substrates. Clearly, it is possible to induce very significant rapid change in most of the root-zone variables with irrigation. In fact, one logical conclusion is that dissolved oxygen concentration can be raised significantly through an irrigation event. Thus noting that oxygen declines at an hourly pace between irrigation events in both open and closed systems, a logical conclusion is that irrigation on an hourly basis (practiced in many greenhouse production settings) prevents hypoxic conditions. The same is, of course, true for the availability of various nutrients, but the rate of decline to deficiency conditions typically takes longer for nutrients than for oxygen. In many soilless production techniques involving stone wool, irrigation intervals of less than 2 hours are needed even when there is still ample fertilizer content in the root-zone solution. It can be argued that the current practices have stabilized around a set of water and nutrient supply recommendations that match this time interval and that it is dictated not by water and nutrients but rather by oxygen.

In large part, there is still a lot that is not known about oxygen in the root zone. Some studies have been carried out but scientific research was hampered by lack of sensors that could be used to make in situ measurements at specific points in the root zone of plants growing in soilless growing media. Most such methods in the past have involved taking a sample (which may well result in changing the oxygen concentration of the sample) or measuring a flowing liquid as it passes a membrane. Most early methods used membrane technology that was consumptive so that very small changes could not be detected because they were smaller than the oxygen utilization of the instrumentation. Recently, new spectrophotometric devices have emerged ([Wang et al., 1999](#)) which allow nonconsumptive measurement of dissolved oxygen at the tip of a probe that can be inserted in liquid or solid substrates. This has led to the finding that the oxygen concentration in the root zone of plants in soilless substrates is quite variable within the root zone and that it changes very rapidly. We have measured root zones of roses in 3–5-year-old coir where the oxygen concentration declined in the middle of the root zone from 8 ppm (near saturation) to less than 0.5 ppm over a period of 2 hours ([Flannery, 2007](#)). The degradation level of this particular substrate was not thought to be an issue, but carrying out the same test with brand new coir resulted in diffusion of oxygen into the root zone that was fast enough to compensate for oxygen

consumption. It may also be possible that this rapid decline in old coir might have been due to the microbial activity. Regardless what the cause is, the operative conclusion was that irrigation was the only feasible way to restore oxygen to normoxic conditions in this particular root zone.

Earlier in this chapter, it was shown that the confined root zone has in and of itself a significant effect on the plants and their roots. In general, there is a notion that we can talk about the root zone as being of a particular composition that is uniform throughout its volume. And while this may well be the case to some extent, one should not jump to the conclusion that the root zone is entirely uniform. In fact, there are significant gradients from top to bottom and from inside to outside, as well as from surfaces on which the Sun is shining to those in the shade. From the standpoint of water, there may be a perched water table at the base with declining wetness levels with height. In substrates with steep moisture retention curves, this can mean that the root zone is too wet at the bottom and too dry at the top for optimal root function. And for all substrates, this dictates the minimum and maximum depth to which the substrate can be used. Temperature gradients are quite common with the surface of the substrate being involved in active evaporative cooling even as the Sun shining on the black plastic side of the container heats the nearby substrate to temperature that are too hot for humans to handle. Clearly, there is extensive heterogeneity with temperature and this will lead to substantial variation in dissolved oxygen concentration because oxygen solubility ranges from hypoxic to normoxic levels over the temperature range that roots of container grown plants experience. These temperatures are also linked to the rate of microbial activity and population growth, further contributing to the volatility of the situation with regard to any element for which these microbes compete with the growing plant. It was shown in [Chapters 6 and 10](#) that utilization of nutrients involves exchange processes that affect pH and EC. Yet despite these dramatic changes in temperature, oxygen, pH, EC, and moisture content, the status quo in the industry today does not involve any attempt to sense these variables. Clearly, technical developments are needed in this arena so as to develop sensor-based methods for real-time optimization of the root zone.

In fact, closed production systems are really only sustainable by measuring these variables dynamically. Greenhouse soilless production of vegetables is already innovating in this area (see [Chapter 13: Technical Equipment in Soilless Production Systems](#)) and there is a significant growth in the use of sensors and instrumentation, coupled with decision support software, to attempt to stabilize optimal conditions for the plants. For plants growing in liquid culture, this type of control is already in use but even here, commercial settings still rely mostly on buffering by involving a lot of liquid in the system. As such, DFT, with its large amount of water, has less rapid changes (i.e., is more stable) than aeroponics, which is in turn more stable than NFT. As was pointed out in [Chapter 13, Technical Equipment in Soilless Production Systems](#), NFT has not emerged as a widespread commercial system, due to this instability. As we develop instrumentation and control technologies that can replace buffering with active manipulation, we may well find that NFT can be feasible in commercial settings.

Another facet that is frequently ignored by the practitioner is the presence and activity of microbes. In fact, the prevailing notion in commercial production is that it is best to eradicate these as they are viewed to be in competition with or even antagonists to the plants. Thus there is nearly indiscriminate use of fungicides with the purpose of eradicating pathogens, with little consideration as to whether the effect to beneficial microorganisms might not result in greater problems.

Chapters 5 and 8 in this book touched on the concept of disease suppressiveness in substrates that is mediated or imparted by microorganisms. The fact that the complete absence of microbes is decidedly suboptimal has been well known by practitioners; many will choose to pasteurize rather than sterilize growing media for this reason; or an attempt will be made to reinoculate sterilized growing media so as to prevent the “biological vacuum.”

We now see further evidence that we must treat the microbes as partners in production, rather than tolerable antagonists. In Chapter 10, Water and Nutrient Management and Crops Response to Nutrient Solution Recycling in Soilless Growing Systems in Greenhouses, it is shown that as systems evolve to more closed systems, we must account for the activity of microbes in the conversion of nitrogen forms and that such conversion plays a major role in the dynamics of the pH. Considerably more research is needed to assist growers with methods that manage this pH. Research in this area needs to evolve so that the microbial population is not treated as a difficult unknown, but rather as a set of dynamic populations. We do not yet know what levels of these populations are optimal, exactly which species are ideal partners for soilless plant production, and how to control the root zone variables so that these subsystems are managed in an ideal fashion.

There are, of course, various other horticultural actions that subject the plant to sudden changes which translate into rapid changes in the root zone. Soilless culture generally involves conditions where the environment can be controlled dynamically around the plant (e.g., greenhouse production) or where the plant can be moved to another location without disrupting its growth (e.g., container nursery production). Many times such changes are implemented so as to improve lighting conditions for the plants. These actions do have an effect on the plants that translates to the root zone as well. Improving lighting conditions can enhance photosynthetic activity which yields more sugars in the plant for the growth of tissues such as roots. The resulting increase in exudates, microbe activity, and nutrient uptake by roots and microbes will result in a more dynamic situation where all variables change more rapidly. Thus a grower needs to be aware that such a horticultural action will have an impact and may require more attention to pH and fertigation management. Also, such a change may result in more rapid increase in leaf biomass and increase the need for air movement around the leaves. This will result with a faster transpiration stream (Chapter 9: Irrigation in Soilless Production) which will, in turn, affect water removal from the root zone.

One operational conclusion that can be drawn from this with regard to open systems is that there is a need to maximize transpiration as part of the overall production strategy. This is something that is frequently not understood because our attention to the ever increasing value of water has led us to seek ways to reduce water use by agriculture. However, in soilless production, maximizing transpiration will assure that a steady supply of nutrients reaches the surface of the roots of plants growing in substrate (as opposed to liquid culture) and thus will increase water use efficiency (Raviv and Blom, 2001). Without this, the plant would be dependent on diffusion processes or the bulk movement during irrigation events. With diffusion of nutrient molecules being very slow, and irrigation in open systems being relatively infrequent (a few times per week, driven by water use of the plant), the boundary layer surrounding the roots would be mostly in a nutrient-depleted state.

14.3.3 SENSING AND CONTROLLING ROOT-ZONE MAJOR PARAMETERS: PRESENT AND FUTURE

With the advent of the use of computers for control of many processes in agriculture came the possibility of developing production systems that combine sensor technologies and intelligent control to create very specific production systems that marry particular plants (or even specific cultivars) to particular soilless production elements. The chapters in this book dealing with irrigation and fertigation explicitly highlight such systems as being common and dynamically developing to include new sensors, mathematical models, and horticultural approaches. For example, integration of technologies to estimate the amounts of each nutrient ion in the root zone is resulting in greater dynamic control over plant nutrition, ultimately allowing greater customization of specific production systems to specific crops.

Much of the technological developments in this area have been driven by opportunity. Moisture sensors are available and significant research has been done to incorporate them into irrigation control systems. Robust specific ion electrodes for measuring ions in the root zone have been lacking so that research in this area has been limited and consequently, there are currently no widely used control systems that obtain input about specific ions from the root zone. Once robust versions of such sensors become available, they will be integrated into production systems to dynamically optimize plant nutrition. Currently, some prototype systems exist, but these measure the leachate of soilless production systems rather than the status of the root zone directly.

Similarly, sensing dissolved oxygen concentration in the root zone has been (and continues to be) very difficult so that using this variable as a control variable for manipulating the root zone is as yet not possible. As such there are currently no irrigation systems (or aeration systems) that explicitly control oxygen in the root zone. Therefore, our current recommendations involve the use of substrates with high aeration. Should it become feasible to monitor and control oxygen concentration directly, then this recommendation may well change, possibly resulting in new and revolutionary soilless production systems.

While opportunistic exploitation of sensor and controller technology is important in advancing horticultural technology, it should be noted that soilless agricultural production has advanced to such a point, with ever-increasing percentage as part of agriculture, that it is now reasonable for research agencies to explicitly set research priorities to focus on developing sensor technologies that can be incorporated specifically in soilless production systems. In fact, throughout agriculture, there are numerous opportunities for advances in “precision agriculture.” For soilless crop production, development of sensor technologies in conjunction with simultaneous control (and optimization) of moisture, nutrients, and oxygen are likely to pay large dividends, pushing production per unit production area and production costs to efficiencies never seen before. It could be argued that if human population continues to grow as it has been, and especially in areas that are prime agricultural production areas, then soilless precision agriculture will be needed to avert food shortages.

Thus, one of the recommendations that can be extracted from this information is that horticultural engineering has significant opportunities in the area of integration of sensor technology with dynamic control over the root-zone variables that are the primary constraints to productivity. Over the past 50 years, improvements in this control have resulted in a variety of production systems, each more productive than the previous generation. It is important to note that significant advances in this area are yet possible.

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Soilless Culture

Theory and Practice

Second Edition

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